

Consumers Energy Company

Electric Rate Case No. U-21389

Index of Exhibits - A# Order

Line No.	Exhibit #	Schedule #	Witness	Witness Exhibit #	Description
	(a)	(b)	(c)	(d)	(e)
Historical Year Section					
1	A-1	A-1	Aponte, Josnelly C.	JCA-1	Revenue Deficiency (Sufficiency) for the Historical Year Ended December 31, 2022
2	A-1	A-2	Aponte, Josnelly C.	JCA-2	Historical Financial Metrics - Financial Basis
3	A-2	B-1	Aponte, Josnelly C.	JCA-3	Rate Base for the Historical Year Ended December 31, 2022
4	A-2	B-2	Aponte, Josnelly C.	JCA-4	Total Utility Plant for the Historical Year Ended December 31, 2022
5	A-2	B-3	Aponte, Josnelly C.	JCA-5	Depreciation Reserve and Other Deductions for the Historical Year Ended December 31, 2022
6	A-2	B-4	Aponte, Josnelly C.	JCA-6	Working Capital for the Historical Year Ended December 31, 2022
7	A-2	B-5	Aponte, Josnelly C.	JCA-7	13-Month Average Working Capital Balance Sheet Summary for the Historical Year Ended December 31, 2022
8	A-2	B-6	Aponte, Josnelly C.	JCA-8	Point-In-Time Working Capital Balance Sheet Summary for the Historical Year Ended December 31, 2022
9	A-3	C-1	Aponte, Josnelly C.	JCA-9	Adjusted Net Operating Income for the Historical Year Ended December 31, 2022
10	A-3	C-2	Aponte, Josnelly C.	JCA-10	Calculation of Revenue Conversion Factor for the Historical Year Ended December 31, 2022
11	A-3	C-3	Aponte, Josnelly C.	JCA-11	Operating Revenue for the Historical Year Ended December 31, 2022
12	A-3	C-4	Aponte, Josnelly C.	JCA-12	Cost of Fuel and Purchased Power for the Historical Year Ended December 31, 2022
13	A-3	C-5	Aponte, Josnelly C.	JCA-13	Operation and Maintenance Expense for the Historical Year Ended December 31, 2022
14	A-3	C-6	Aponte, Josnelly C.	JCA-14	Depreciation and Amortization Expense for the Historical Year Ended December 31, 2022
15	A-3	C-7	Aponte, Josnelly C.	JCA-15	General Taxes for the Historical Year Ended December 31, 2022
16	A-3	C-8	Aponte, Josnelly C.	JCA-16	Federal Income Taxes for the Historical Year Ended December 31, 2022
17	A-3	C-9	Aponte, Josnelly C.	JCA-17	State Income Taxes for the Historical Year Ended December 31, 2022
18	A-3	C-10	Aponte, Josnelly C.	JCA-18	Other (or Local) Taxes for the Historical Year Ended December 31, 2022
19	A-3	C-11	Aponte, Josnelly C.	JCA-19	Allowance for Funds Used During Construction for the Historical Year Ended December 31, 2022
20	A-3	C-12	Aponte, Josnelly C.	JCA-20	Tax Effect of Pro-Forma Interest Adjustment for the Historical Year Ended December 31, 2022
21	A-3	C-13	Aponte, Josnelly C.	JCA-21	Tax Effect of Interest Synchronization Adjustment for the Historical Year Ended December 31, 2022
22	A-4	D-1	Aponte, Josnelly C.	JCA-22	Overall Rate of Return Summary for the Historical Year Ended December 31, 2022
23	A-4	D-2	Aponte, Josnelly C.	JCA-23	Cost of Long-Term Debt (Excluding Securitization) for the Historical Year Ended December 31, 2022
24	A-4	D-3	Aponte, Josnelly C.	JCA-24	Cost of Short-Term Debt for the Historical Year Ended December 31, 2022
25	A-4	D-4	Aponte, Josnelly C.	JCA-25	Cost of Preferred Stock for the Historical Year Ended December 31, 2022
26	A-4	D-5	Aponte, Josnelly C.	JCA-26	Cost of Common Shareholder's Equity for the Historical Year Ended December 31, 2022
27	A-5	E-1	Breuring, Eugene M.	EMB-1	Annual Service Area Calendar Sales by Major Customer Classes and System Output – 5-Year Historical
28	A-6	-	-	-	Exhibit Number Intentionally Left Blank

29	A-7	-	-	-	Exhibit Number Intentionally Left Blank
30	A-8	-	-	-	Exhibit Number Intentionally Left Blank
31	A-9	-	-	-	Exhibit Number Intentionally Left Blank
32	A-10	-	-	-	Exhibit Number Intentionally Left Blank
	Test Year Section				
33	A-11	A-1	Aponte, Josnelly C.	JCA-27	Revenue Deficiency (Sufficiency) for the Projected 12-Month Period Ending February 28, 2025
34	A-11	A-2	Aponte, Josnelly C.	JCA-28	Financial Metrics – Ratemaking Basis
35	A-11	A-3	Aponte, Josnelly C.	JCA-29	Comparison of the Electric Revenue Requirement between the Historical Year and the Projected Test Year
36	A-12	B-1	Aponte, Josnelly C.	JCA-30	Rate Base for the Projected 12-Month Period Ending February 28, 2025
37	A-12	B-1a	Aponte, Josnelly C.	JCA-31	Development of Rate Base for the Projected 12-Month Period Ending February 28, 2025
38	A-12	B-2	Aponte, Josnelly C.	JCA-32	Total Utility Plant for the Projected 12-Month Period Ending February 28, 2025
39	A-12	B-3	Aponte, Josnelly C.	JCA-33	Depreciation Reserve and Other Deductions for the Projected 12-Month Period Ending February 28, 2025
40	A-12	B-4	Aponte, Josnelly C.	JCA-34	Working Capital Summary for the Projected 12-Month Period Ending February 28, 2025
41	A-12	B-4a	Aponte, Josnelly C.	JCA-35	Development of Working Capital for the Projected 12-Month Period Ending February 28, 2025
42	A-12	B-5	Aponte, Josnelly C.	JCA-36	Projected Capital Expenditures
43	A-12	B-5.1	Blumenstock, Richard T.	RTB-3	Summary of Actual and Projected Electric Capital Expenditures for the Years 2022 through February 2025
44	A-12	B-5.2	Carveth, Adam S.	ASC-1	Summary of Actual & Projected Capital Expenditures
45	A-12	B-5.3	Cumberworth, Audra L.	ALC-1	Summary of Actual and Projected Electric Capital Expenditures
46	A-12	B-5.4	Foster, Matthew J.	MJF-6	Electric Corporate Services Capital Expenditures for the Years 2022, 2023, 2024 and the 12 Months Ending February 28, 2025
47	A-12	B-5.5	Guinn, Quentin A.	QAG-1	Summary of Actual & Projected Capital Expenditures
48	A-12	B-5.6	Hale, Cullen M.	CMH-1	Summary of Actual and Projected Capital Expenditures - Customer Experience and Operations
49	A-12	B-5.7	Kelly, Michael P.	MPK-1	Electric Distribution Summary of Actual and Projected Capital Expenditures
50	A-12	B-5.8	McLean, Steven Q.	SQM-1	Projected Capital Expenditures – Demand Response
51	A-12	B-5.9	Myrom, Jeffrey A.	JAM-1	Transportation Electrification Program Capital Expenditures
52	A-12	B-5.10	Schmoekel, Daniel A.	DAS-1	Summary of Actual and Projected Electric Capital Expenditures for the Years 2022 through 2025
53	A-12	B-5.11	Weller, Heather N.	HNW-5	Projected Capital Expenditures Information Technology Summary of Actual and Projected Electric and Common Capital Expenditures
54	A-13	C-1	Aponte, Josnelly C.	JCA-37	Adjusted Net Operating Income for the Projected 12-Month Period Ending February 28, 2025
55	A-13	C-2	Aponte, Josnelly C.	JCA-38	Calculation of the Revenue Conversion Factor for the Projected 12-Month Period Ending February 28, 2025
56	A-13	C-3	Aponte, Josnelly C.	JCA-39	Operating Revenues for the Projected 12-Month Period Ending February 28, 2025
57	A-13	C-4	Aponte, Josnelly C.	JCA-40	Cost of Fuel and Purchased Power for the Projected 12-Month Period Ending February 28, 2025
58	A-13	C-5	Aponte, Josnelly C.	JCA-41	Other Operating and Maintenance Expenses for the Projected 12-Month Period Ending February 28, 2025
59	A-13	C-5.a	Aponte, Josnelly C.	JCA-42	Summary of Inflation and Merit Increases Included in Projected Other Operation and Maintenance Expenses for the Projected 12-Month Period Ending February 28, 2025
60	A-13	C-6	Aponte, Josnelly C.	JCA-43	Depreciation and Amortization Expenses for the Projected 12-Month Period Ending February 28, 2025
61	A-13	C-7	Aponte, Josnelly C.	JCA-44	General Taxes for the Projected 12-Month Period Ending February 28, 2025
62	A-13	C-8	Aponte, Josnelly C.	JCA-45	Federal Income Taxes for the Projected 12-Month Period Ending February 28, 2025
63	A-13	C-9	Aponte, Josnelly C.	JCA-46	State Income Taxes for the Projected 12-Month Period Ending February 28, 2025

64	A-13	C-10	Aponte, Josnelly C.	JCA-47	Other (or Local) Taxes for the Projected 12-Month Period Ending February 28, 2025
65	A-13	C-11	Aponte, Josnelly C.	JCA-48	Allowance for Funds Used During Construction for the Projected 12-Month Period Ending February 28, 2025
66	A-13	C-12	Aponte, Josnelly C.	JCA-49	Tax Effect of Pro-Forma Interest Adjustment for the Projected 12-Month Period Ending February 28, 2025
67	A-13	C-13	Aponte, Josnelly C.	JCA-50	Tax Effect of Interest Synchronization Adjustment for the Projected 12-Month Period Ending February 28, 2025
68	A-13	C-14	Aponte, Josnelly C.	JCA-51	Development of Net Operating Income for the Projected 12-Month Period Ending February 28, 2025
69	A-14	D-1	Bleckman, Marc R.	MRB-1	Overall Rate of Return Summary
70	A-14	D-1a	Bleckman, Marc R.	MRB-2	Capital Structure Development
71	A-14	D-1b	Bleckman, Marc R.	MRB-3	Comparison of Development of Capital Structure
72	A-14	D-2	Bleckman, Marc R.	MRB-4	Cost of Long-Term Debt
73	A-14	D-3	Bleckman, Marc R.	MRB-5	Cost of Short-Term Debt
74	A-14	D-4	Bleckman, Marc R.	MRB-6	Cost of Preferred Stock
75	A-14	D-5	Wehner, Todd A.	TAW-1	Cost of Common Shareholders' Equity
76	A-14	D-6	Bleckman, Marc R.	MRB-7	Short-Term Debt Utilization
77	A-15	E-1	Breuring, Eugene M.	EMB-2	Annual Service Area Calendar Sales by Major Customer Classes and System Output – 5-Year Projected
78	A-15	E-2	Breuring, Eugene M.	EMB-3	Test Year Total Company Electric Revenues & Deliveries
79	A-15	E-3	Breuring, Eugene M.	EMB-4	Electric Deliveries & Customer Counts by Rate Category
80	A-15	E-4	Breuring, Eugene M.	EMB-5	Calculation of Annual System Load Factor
81	A-16	F-1	Davis, Emily A.	EAD-1	Electric Cost-of-Service Study – Projected 12-Month Period Ending Feb 28, 2025 Version 1
82	A-16	F-1.1	Davis, Emily A.	EAD-2	Electric Cost-of-Service Study – Projected 12-Month Period Ending Feb 28, 2025 Version 2
83	A-16	F-2	Gast, Alex M.	AMG-1	Summary of Present and Proposed Pro Forma Revenues by Rate Schedule
84	A-16	F-2.1	Gast, Alex M.	AMG-2	Calculation of Rate Design Targets
85	A-16	F-3	Gast, Alex M.	AMG-3	Present and Proposed Revenue Detail
86	A-16	F-4	Gast, Alex M.	AMG-4	Comparison of Present and Proposed Monthly Bills
87	A-16	F-5	Hurd, Shawn C.	SCH-2	Proposed Tariff Sheets (MPSC No. 3 Redlined Version)
88	A-17	-	Aponte, Josnelly C.	JCA-52	Regulatory Asset and Amortization of the PowerMIDrive Costs
89	A-18	-	Aponte, Josnelly C.	JCA-53	Regulatory Asset and Amortization of the PowerMIFleet Costs
90	A-19	-	Aponte, Josnelly C.	JCA-54	Regulatory Asset and Amortization of the Karn Plant Retention and Separation Costs
91	A-20	-	Aponte, Josnelly C.	JCA-55	Regulatory Asset and Amortization of the Campbell Retention and Separation Costs
92	A-21	-	Aponte, Josnelly C.	JCA-56	Regulatory Asset and Amortization of the Campbell 1-2 Plants Net Book Value
93	A-22	-	Aponte, Josnelly C.	JCA-57	Regulatory Asset and Amortization of the Campbell 3 Plant Net Book Value
94	A-23	-	Aponte, Josnelly C.	JCA-58	Regulatory Asset and Amortization of the Karn 1-2 Decommissioning Costs
95	A-24	-	Aponte, Josnelly C.	JCA-59	Regulatory Asset and Amortization of the Karn 1-2, Classic 7, and Campbell Ash Disposal
96	A-25	-	Aponte, Josnelly C.	JCA-60	Regulatory Asset and Amortization of the Campbell Decommissioning Costs
97	A-26	-	Aponte, Josnelly C.	JCA-61	Demand Response Revenue Requirement
98	A-27	-	Aponte, Josnelly C.	JCA-62	2022 Distribution Deferral
99	A-28	-	Aponte, Josnelly C.	JCA-63	2023 Distribution Deferral
100	A-29	-	Battaglia, Jeffrey E.	JEB-1	Karn Units 1&2 Decommissioning Plan Expenditures
101	A-30	-	Battaglia, Jeffrey E.	JEB-2	Actual Coal Ash Disposal Expenditures for 2022
102	A-31	-	Battaglia, Jeffrey E.	JEB-3	Projected JH Campbell Units 1, 2 & 3 Decommissioning Expense-March 2023
103	A-32	-	Bleckman, Marc R.	MRB-8	Current and Historical Credit Ratings

104	A-33	-	Bleckman, Marc R.	MRB-9	Recent Utility Corporate Bond Issuances
105	A-34	-	Bleckman, Marc R.	MRB-10	Peer Company Commission Authorized Equity Ratios
106	A-35	-	Bleckman, Marc R.	MRB-11	State Regulatory Evaluations
107	A-36	-	Bleckman, Marc R.	MRB-12	Moody's Investors Service May 10, 2021 Credit Opinion
108	A-37	-	Bleckman, Marc R.	MRB-13	S&P January 23, 2023 Report: Industry Top Trends – North America Regulated Utilities
109	A-38	-	Bleckman, Marc R.	MRB-14	S&P January 27, 2021 Credit Opinion
110	A-39	-	Blumenstock, Richard T.	RTB-1	Generating Unit Periodic Outages
111	A-40	-	Blumenstock, Richard T.	RTB-2	Generating Unit Availability Projections
112	A-41	-	Blumenstock, Richard T.	RTB-4	Summary of the Generation O&M Expense for the Years 2022 through February 2025
113	A-42	-	Bolden, Pamela L.	PLB-1	Line Clearing O&M Expense
114	A-43	-	Bolden, Pamela L.	PLB-2	Line Clearing Ramp-up Plan Estimated Potential Service Restoration Cost Avoidance
115	A-44	-	Bolden, Pamela L.	PLB-3	2023 HVD Line Clearing Work Plan
116	A-45	-	Bolden, Pamela L.	PLB-4	Line Clearing Reliability Results
117	A-46	-	Bolden, Pamela L.	PLB-5	Justification of a 7-Year Cycle v Other Cycles
118	A-47	-	Bolden, Pamela L.	PLB-6	Summary of O&M Expenses Projected Using Merit and Inflation
119	A-48	-	Bolden, Pamela L.	PLB-7	Forestry Management Process Review
120	A-49	-	Bolden, Pamela L.	PLB-8	2023 LVD Line Clearing Work Plan
121	A-50	-	Bolden, Pamela L.	PLB-9	Tree Outage Influence on CAIDI
122	A-51	-	Breuring, Eugene M.	EMB-6	Estimated Electric Rate Case PSCR Factor
123	A-52	-	Carveth, Adam S.	ASC-2	Fleet Responsibility Costs
124	A-53	-	Carveth, Adam S.	ASC-3	Detailed List of Projected Electric Capital Expenditures for the Years 2022, 2023, 2024 and test year 12 months ending February 28, 2025
125	A-54	-	Carveth, Adam S.	ASC-4	Summary of Fleet Tooling Actual & Projected Capital Expenditures
126	A-55	-	Clark, Thomas P.	TPC-1	Build Transfer Agreement by and between Mustang Mile Solar Energy LLC and Consumers Energy Company, dated January 21, 2021 (Confidential)
127	A-56	-	Clark, Thomas P.	TPC-2	Contract Change Order #2 to the Build Transfer Agreement by and between Mustang Mile Solar Energy LLC and Consumers Energy Company, dated October 21, 2022 (Confidential)
128	A-57	-	Clark, Thomas P.	TPC-3	Board Resolution Approving A Capital Expenditure For The Mustang Mile Solar Energy LLC Project, dated January 21, 2021
129	A-58	-	Clark, Thomas P.	TPC-4	2019 IRP Competitive Solicitation - Mustang Mile Solar Project - Milestone Schedule
130	A-59	-	Clark, Thomas P.	TPC-5	Projected Capital Expenditures for Company-owned IRP Solar Resources
131	A-60	-	Clark, Thomas P.	TPC-6	Build Transfer Agreement by and between Washtenaw Solar Energy, LLC and Consumers Energy Company, dated October 4, 2021 (Confidential)
132	A-61	-	Clark, Thomas P.	TPC-7	Board Resolution Approving A Capital Expenditure For The Washtenaw Solar Energy LLC Project, dated September 29, 2021
133	A-62	-	Clark, Thomas P.	TPC-8	2020 IRP Competitive Solicitation - Washtenaw Solar Project - Milestone Schedule
134	A-63	-	Clark, Thomas P.	TPC-9	Module Sale Agreement by and between First Solar Inc and Consumers Energy Company, dated June 10, 2022 (Confidential)
135	A-64	-	Clark, Thomas P.	TPC-10	Purchase Order by and between First Solar Inc and Consumers Energy Company, dated June 13, 2022 (Confidential)
136	A-65	-	Clark, Thomas P.	TPC-11	Substation Transformer Agreement by and between Virginia Transformer Company and Consumers Energy Company, dated September 4, 2022 (Confidential)
137	A-66	-	Clark, Thomas P.	TPC-12	Purchase Order by and between Virginia Transformer Company and Consumers Energy Company, dated September 23, 2022 (Confidential)
138	A-67	-	Clark, Thomas P.	TPC-13	Board Resolution Approving A Capital Expenditure For The Muskegon Solar Energy Center Project, dated January 25, 2023

139	A-68	-	Clark, Thomas P.	TPC-14	2021 IRP Competitive Solicitation – Muskegon Solar Energy Center Project - Milestone Schedule
140	A-69	-	Clark, Thomas P.	TPC-15	IRP Battery Energy Storage System Stakeholder Session
141	A-70	-	Clark, Thomas P.	TPC-16	IRP Battery Energy Storage System – Milestone Schedule
142	A-71	-	Clark, Thomas P.	TPC-17	Armstrong Battery Energy Storage System Project – Milestone Schedule
143	A-72	-	Coker, Jason R.	JRC-1	Summary of Business Case Costs and Benefits 2007 – 2032
144	A-73	-	Conrad, Amy M.	AMC-1	EICP Performance Measures
145	A-74	-	Conrad, Amy M.	AMC-2	Target Pay Level Market Analysis
146	A-75	-	Conrad, Amy M.	AMC-3	Summary of Actual and Projected – Annual Incentive O&M Expenses
147	A-76	-	Cumberworth, Audra L.	ALC-2	Synopses Containing Descriptions, Scope, Benefits, Implementation Dates, and Detailed Costs of Actual and Projected Electric & Common Capital Expenditures and O&M Expenses for the years 2022, 2023, 2024 and 2025
148	A-77	-	Cumberworth, Audra L.	ALC-3	Summary of Actual and Projected Security Operations O&M Expense for the Years 2022, 2023, 2024, and Test Year 12 Months Ending February 28, 2025
149	A-78	-	Cumberworth, Audra L.	ALC-4	Summary of Actual and Projected Security Investments O&M Expenses for the Years 2022, 2023, 2024, and Test Year 12 Months Ending February 28, 2025
150	A-79	-	Davis, Emily A.	EAD-3	Capacity Related Cost and Charge Calculation (SRM)
151	A-80	-	Davis, Emily A.	EAD-4	Substation Ownership Credit
152	A-81	-	Davis, Emily A.	EAD-5	Allocation of Surcharges
153	A-82	-	Foster, Matthew J.	MJF-1	Summary of Projected Electric & Common O&M Expense for the Years 2022, 2023, 2024 and the 12 Months Ending February 28, 2025
154	A-83	-	Foster, Matthew J.	MJF-2	Electric Projected Corporate Services O&M Expense for the Years 2022, 2023, 2024 and the 12 Months Ending February 28, 2025
155	A-84	-	Foster, Matthew J.	MJF-3	S&P Global Market Intelligence Ranking of Consumers Energy Electric A&G Costs for 2021
156	A-85	-	Foster, Matthew J.	MJF-4	Electric Uncollectible Accounts Expense for the Years 2022, 2023, 2024 and the 12 Months Ending February 28, 2025
157	A-86	-	Foster, Matthew J.	MJF-5	Electric Injuries & Damages Expense for the Years 2018 Through the 12 Months Ending February 28, 2025
158	A-87	-	Gast, Alex M.	AMG-5	Long-Term Industrial Load Retention Rate (LTILRR) Revenues (Confidential)
159	A-88	-	Gast, Alex M.	AMG-6	Demand Response (DR) Surcharge
160	A-89	-	Gast, Alex M.	AMG-7	Electric Rate Case (ERC) Deferral Surcharge
161	A-90	-	Gast, Alex M.	AMG-8	Investment Recovery Mechanism (IRM) Surcharge
162	A-91	-	Grob, Kendra K.	KKG-1	Summary of Actual and Projected Benefits O&M Expenses for the Years 2022, 2023, 2024, and Test Year Ending February 28, 2025
163	A-92	-	Grob, Kendra K.	KKG-2	CMS Energy – Pension Plans A and B - ASC 715 Pension Expense Estimates (\$ millions)
164	A-93	-	Grob, Kendra K.	KKG-3	CMS Energy - ASC 715 OPEB Expense Estimates (\$ millions)
165	A-94	-	Grob, Kendra K.	KKG-4	CMS Energy – Actuarial Letter of Support for 2023 Year Projections (Confidential)
166	A-95	-	Guinn, Quentin A.	QAG-2	Summary of Actual and Projected O&M Expenses
167	A-96	-	Guinn, Quentin A.	QAG-3	Detailed List of Projected Electric Capital Expenditures
168	A-97	-	Guinn, Quentin A.	QAG-4	Marshall Roadway Paving - Plan Summary
169	A-98	-	Guinn, Quentin A.	QAG-5	Lansing Service Center – Plan Summary
170	A-99	-	Guinn, Quentin A.	QAG-6	Hastings Service Center – Plan Summary
171	A-100	-	Guinn, Quentin A.	QAG-7	Greenville Storage Building – Plan Summary
172	A-101	-	Guinn, Quentin A.	QAG-8	Ludington Storage Building – Plan Summary
173	A-102	-	Guinn, Quentin A.	QAG-9	Hardy Headquarter Building – Plan Summary
174	A-103	-	Guinn, Quentin A.	QAG-10	Kalamazoo Service Center – Plan Summary
175	A-104	-	Guinn, Quentin A.	QAG-11	Jackson Dispatch – Plan Summary

176	A-105	-	Guinn, Quentin A.	QAG-12	Parnall 1-3 – Plan Summary
177	A-106	-	Guinn, Quentin A.	QAG-13	Grand Rapids LVD UG – Plan Summary
178	A-107	-	Guinn, Quentin A.	QAG-14	Control/Dispatch Consolidation – Plan Summary
179	A-108	-	Guinn, Quentin A.	QAG-15	South Monroe Renovation Conceptual Cost Estimate
180	A-109	-	Hale, Cullen M.	CMH-2	Summary of Actual & Projected O&M Expenses – Customer Experience and Operations
181	A-110	-	Henry, Matthew S.	MSH-1	Bridge Period Grid Modernization Projects
182	A-111	-	Henry, Matthew S.	MSH-2	Test Year Grid Modernization Projects
183	A-112	-	Henry, Matthew S.	MSH-3	EPRI Report: “Consumers Energy Grid Modernization Roadmap Assessment”
184	A-113	-	Henry, Matthew S.	MSH-4	Grid Modernization BCA
185	A-114		Henry, Matthew S.	MSH-5	EPRI Report: “DER Management Systems (DERMS) Adoption Pathways” (Confidential)
186	A-115	-	Henry, Matthew S.	MSH-6	2021 and 2022 Security Assessments
187	A-116		Herrygers, Steven J.	SJH-1	Consumers Energy Electric Asset Management High Voltage Distribution (HVD) Pole Inspection Specifications
188	A-117		Herrygers, Steven J.	SJH-2	Pole Inspection Plan by HVD Line
189	A-118		Herrygers, Steven J.	SJH-3	Summary Projected Electric Capital Expenditures for the Bridge Period 14 Month Ending February 29th, 2024
190	A-119		Herrygers, Steven J.	SJH-4	Summary Projected Electric Capital Expenditures for the Test Year 12 Months Ending February 28, 2025
191	A-120		Hurd, Shawn C.	SCH-1	Summary of Tariff Changes
192	A-121		Kelly, Michael P.	MPK-2	Electric Distribution Actual and Projected Capital Expenditures Master
193	A-122		Kelly, Michael P.	MPK-3	Electric Distribution Summary of 5yr Historical Electric Capital Expenditures
194	A-123		Kelly, Michael P.	MPK-4	New Business Program Summary of Actual and Projected Electric Capital Expenditures
195	A-124	-	Kelly, Michael P.	MPK-5	Reliability Program Summary of Actual and Projected Capital Expenditures
196	A-125	-	Kelly, Michael P.	MPK-6	Capacity Program Summary of Actual and Projected Capital Expenditures
197	A-126	-	Kelly, Michael P.	MPK-7	Demand Failures Program Summary of Actual and Projected Capital Expenditures
198	A-127	-	Kelly, Michael P.	MPK-8	Asset Relocation Program Summary of Actual and Projected Capital Expenditures
199	A-128	-	Kelly, Michael P.	MPK-9	Electric Other Program Summary of Actual and Projected Capital Expenditures
200	A-129	-	Kelly, Michael P.	MPK-10	Summary of Actual & Projected Electric & Common O&M Expenses Master
201	A-130	-	Kelly, Michael P.	MPK-11	Summary of Actual & Projected Electric & Common O&M Expenses – Operations and Engineering
202	A-131	-	Kelly, Michael P.	MPK-12	Summary of 5yr Historical O&M Expenses
203	A-132	-	Kelly, Michael P.	MPK-13	Summary of Actual & Projected Electric & Common O&M Expenses – O&M Programs, part 1
204	A-133	-	Kelly, Michael P.	MPK-14	Summary of Actual & Projected O&M Expenses Detail, part 2
205	A-134	-	Kelly, Michael P.	MPK-15	Summary of Actual & Projected O&M Expenses Cost Splits
206	A-135	-	Kelly, Michael P.	MPK-16	Summary of O&M Expenses Projected Using Inflation
207	A-136	-	Kelly, Michael P.	MPK-17	2021 Electric Distribution Infrastructure Investment Plan
208	A-137	-	Kelly, Michael P.	MPK-18	2021 Electric System Loss Study Report
209	A-138	-	Kelly, Michael P.	MPK-19	Electric System Reliability Maintenance Guidelines
210	A-139	-	Kelly, Michael P.	MPK-20	2018-2022 Distribution Unit Costs
211	A-140	-	Kelly, Michael P.	MPK-21	Environmental Justice Reliability and Hosting Capacity Maps
212	A-141	-	Lynd, Donald A.	DAL-1	2023-2024 Bridge Period LVD Project List
213	A-142	-	Lynd, Donald A.	DAL-2	2024-2025 Test Year LVD Project List
214	A-143	-	Lynd, Donald A.	DAL-3	Conservation Voltage Reduction Measurement and Verification Independent Assessment

215	A-144	-	McLean, Steven Q.	SQM-2	Summary of Actual & Projected O&M Expenses – Demand Response
216	A-145	-	McLean, Steven Q.	SQM-3	Electrification Pilot
217	A-146	-	Metz, Megan L.	MLM-1	Power Supply Cost Recovery Forecast for March 1, 2024 – February 28, 2025
218	A-147	-	Metz, Megan L.	MLM-2	Summary of PPA Changes from September 30, 2022 to February 1, 2023
219	A-148	-	Metz, Megan L.	MLM-3	Consumers Energy Forecast - Energy Market Sales Revenue and Total Variable Expense
220	A-149	-	Monroe, Adam J.	AJM-1	Overview of Consumers Energy River Hydro Facilities
221	A-150	-	Monroe, Adam J.	AJM-2	Consumers Energy River Hydro Brochure
222	A-151	-	Monroe, Adam J.	AJM-3	November 1, 2019 Letter from FERC to Consumers Energy Requiring Utilization of BOC
223	A-152	-	Monroe, Adam J.	AJM-4	April 28, 2017 Letter from FERC to Consumers Energy Requiring Spillway Assessment
224	A-153	-	Myers, Heidi J.	HJM-1	Distribution Investment Recovery Mechanism – Annual Revenue Requirement
225	A-154	-	Myrom, Jeffrey A.	JAM-2	Transportation Electrification Program Deferred Asset Costs
226	A-155	-	Slieff, James C.	JCS-1	Summary of Historical Year, Bridge Period, and Projected Test Year Operations Resources – Electric LVD Workplan; Summary of YE Actual and YE Projected Operations Resources – Electric LVD Workplan
227	A-156	-	Slieff, James C.	JCS-2	Summary of Historical Year, Bridge Period, and Projected Test Year Operations Resources – Electric HVD Workplan; Summary of YE Actual and YE Projected Operations Resources – Electric HVD Workplan
228	A-157	-	Stuart, R. Michael	RMS-1	2023 EICP Operational Goals
229	A-158	-	Stuart, R. Michael	RMS-2	2023 Customer Benefits: Employee Safety
230	A-159	-	Stuart, R. Michael	RMS-3	2023 Customer Benefits: Reliability
231	A-160	-	Stuart, R. Michael	RMS-4	2023 Customer Benefits: Culture Index
232	A-161	-	VanBlarcum, Brian J.	BJV-1	Development of the Property Tax Rate for the Test Year
233	A-162	-	VanBlarcum, Brian J.	BJV-2	Amortization of Excess Deferred Federal Income Taxes for the Test Year and Tax Reform Regulatory Liability & Amortization
234	A-163	-	VanBlarcum, Brian J.	BJV-3	Development of Production Tax Credits for the Test Year
235	A-164	-	Wehner, Todd A.	TAW-2	Wolfe Research: “The Fleishman Daily 11/21/22”
236	A-165	-	Wehner, Todd A.	TAW-3	Wells Fargo Equity Research: “DTE: Challenging Electric Rate Order – Bears Monitoring”
237	A-166	-	Wehner, Todd A.	TAW-4	UBS Equity Research: “DTE Energy Co: Electric Rate Case Revenue Light”
238	A-167	-	Wehner, Todd A.	TAW-5	Moody’s Investor Service November 10, 2022 Report
239	A-168	-	Wehner, Todd A.	TAW-6	Fama and French: “The CAPM is Wanted, Dead or Alive”
240	A-169	-	Wehner, Todd A.	TAW-7	Financial Times: “The time has come for the CAPM to RIP”
241	A-170	-	Wehner, Todd A.	TAW-8	Chartoff, Mayo, and Smith: “The Case Against the Use of the Capital Asset Pricing Model in Public Utility Ratemaking”
242	A-171	-	Wehner, Todd A.	TAW-9	Chretien and Coggins: “Cost of Equity for Energy Utilities: Beyond the CAPM”
243	A-172	-	Wehner, Todd A.	TAW-10	Federal Reserve Bank of New York Staff Reports: “The Equity Risk Premium: A Review of Models”
244	A-173	-	Wehner, Todd A.	TAW-11	Brattle Group: “Estimating the Cost of Equity for Regulated Companies”
245	A-174	-	Wehner, Todd A.	TAW-12	Value Line: “Using Beta”
246	A-175	-	Wehner, Todd A.	TAW-13	Gordon and Shapiro: “Capital Equipment Analysis”
247	A-176	-	Weller, Heather N.	HNW-1	Consumers Energy Digital Three-Year Plan for the Years 2023 – 2025 (Appendix B – Confidential)
248	A-177	-	Weller, Heather N.	HNW-2	Summary of Actual and Projected Information Technology Operations O&M Expense for the Years 2022, 2023, 2024, and Test Year 12 Months Ending February 28, 2025

249	A-178	-	Weller, Heather N.	HNW-3	Historical and Projected 13-Month Average of IT Cloud Computing Prepaid Balance for the historical years 2022 - 13-month balance ending December 31,2022, and for the projected years 2024 – 13-month balance ending February 28, 2025
250	A-179	-	Weller, Heather N.	HNW-4	Summary of Actual and Projected Information Technology Investments O&M Expense for the Years 2022, 2023, 2024 and Test Year 12 Months Ending February 28, 2025
251	A-180	-	Weller, Heather N.	HNW-6	Synopses Containing Descriptions, Scope, Benefits, Implementation Dates and Detailed Costs of Actual and Projected Electric & Common Capital Expenditures and O&M Expenses For the Years 2022, 2023, 2024 and 2025
252	A-181	-	Weller, Heather N.	HNW-7	Business Case Executive Summaries for Historical, Bridge Period, and Test Year Projects
253	A-182	-	Weller, Heather N.	HNW-8	Asset Refresh Programs Projected Electric and Common Capital Expenditures, For the Projected Year 2024 and Test Year Ending February 28, 2025, and For the Historical and Projected Years 2022 and 2023
254	A-183	-	Weller, Heather N.	HNW-9	Application Currency Programs Projected Electric and Common Capital and O&M Expenditures for the Years 2024, 2025, and Test Year 12 Months Ending February 28, 2025 (Confidential)
255	A-184	-	Weller, Heather N.	HNW-10	Projected Versus Actual Enhancement Capital Expenditures and O&M Expense Summary and Analysis
256	A-185	-	Weller, Heather N.	HNW-11	Enhancement Worklist Detail for Years 2016 through February 16, 2023

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

JOSNELLY C. APONTE

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

Schedule: A-1**MICHIGAN PUBLIC SERVICE COMMISSION**

Consumers Energy Company

Revenue Deficiency (Sufficiency)

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-1 (JCA-1)

Schedule: A-1

Page: 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)	(d)
Line No	Description	Source	Total (\$000)	Jurisdictional (\$000)
1	Rate base	Exhibit No.: A-2 (JCA-3)	12,647,343	12,599,436
2	Adjusted net operating income	Exhibit No.: A-3 (JCA-9)	645,347	657,238
3	Overall rate of return	Line 2 / Line 1	5.10%	5.22%
4	Required rate of return	Exhibit No.: A-4 (JCA-22)	5.80%	5.80%
5	Income requirements	Line 1 * Line 4	734,073	731,292
6	Income deficiency (sufficiency)	Line 5 - Line 2	88,726	74,054
7	Revenue conversion factor	Exhibit No.: A-3 (JCA-10)	1.3391	1.3391
8	Revenue deficiency (sufficiency)	Line 6 * Line 7	118,810	99,164

Schedule: A-2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Historical Financial Metrics - Financial Basis
 ELECTRIC RESULTS ONLY
 (\$000)

Case No.: U-21389
 Exhibit No.: A-1 (JCA-2)
 Schedule: A-2
 Page: 1 of 6
 Witness: JCAponte
 Date: May 2023

Line No	(a) Description	(b)	(c)	(d)	(e)	(f)
		Calendar Year				
		2018	2019	2020	2021	2022
1	Operating revenues	4,561,203	4,439,232	4,372,177	4,957,857	5,419,082
2	Operating expenses	3,745,412	3,632,901	3,531,392	4,156,834	4,630,615
3	Pre-tax operating income	815,791	806,331	840,785	801,024	788,467
4	Income taxes	108,918	134,677	115,416	117,193	108,800
5	Net operating income	706,874	671,655	725,370	683,831	679,667
6	Other income and deductions	35,098	47,194	41,819	83,027	102,863
7	AFUDC	5,095	5,918	7,211	8,207	5,520
8	Interest charges	(210,785)	(214,929)	(202,399)	(193,601)	(206,904)
9	Preferred stock dividends	(1,193)	(1,160)	(1,111)	(1,081)	(1,078)
10	Net income available to common stockholders	535,088	508,678	570,889	580,383	580,068
11	Average common equity	4,808,892	5,282,036	5,806,036	6,260,365	6,372,772
12	Earned rate of return on common equity	11.13%	9.63%	9.83%	9.27%	9.10%
13	Authorized rate of return on common equity	10.00% {1}	10.00% {2}	10.00% {3}	9.90% {3}	9.90% {4}

Schedule: A-2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Historical Financial Metrics - Financial Basis
 ELECTRIC RESULTS ONLY
 (\$000)

Case No.: U-21389
 Exhibit No.: A-1 (JCA-2)
 Schedule: A-2
 Page: 2 of 6
 Witness: JCAponte
 Date: May 2023

Line No	(a) Description	(b)	(c)	(d)	(e)	(f)
		Calendar Year				
		2018	2019	2020	2021	2022
EBIT Interest Coverage Ratio						
14	Pre-tax operating income	815,791	806,331	840,785	801,024	788,467
15	Other income and deductions	35,098	47,194	41,819	83,027	102,863
16	AFUDC	5,095	5,918	7,211	8,207	5,520
17	Total EBIT	855,985	859,443	889,815	892,257	896,850
18	Interest charges	210,785	214,929	202,399	193,601	206,904
19	EBIT interest coverage ratio	4.06	4.00	4.40	4.61	4.33
EBITDA Interest Coverage Ratio						
20	Total EBIT	855,985	859,443	889,815	892,257	896,850
21	Depreciation and amortization	682,307	713,007	739,148	771,634	757,258
22	Total EBITDA	1,538,292	1,572,450	1,628,963	1,663,891	1,654,108
23	Interest charges	210,785	214,929	202,399	193,601	206,904
24	EBITDA interest coverage ratio	7.30	7.32	8.05	8.59	7.99
Funds Flow from Operations (FFO) Interest Coverage Ratio						
25	Net operating income	706,874	671,655	725,370	683,831	679,667
26	Depreciation and amortization	682,307	713,007	739,148	771,634	757,258
27	Deferred income taxes	124,791	38,192	321,515	(85,580)	166,168
28	AFUDC	5,095	5,918	7,211	8,207	5,520
29	Other major recurring non-cash items	-	-	-	-	-
30	Interest paid	196,495	200,213	207,328	198,499	202,104
Less:						
31	Operating lease adjustment to depreciator	-	-	-	-	-
32	Subtotal	1,715,561	1,628,983	2,000,572	1,576,591	1,810,716
33	Interest charges	210,785	214,929	202,399	193,601	206,904
34	FFO interest coverage ratio	8.14	7.58	9.88	8.14	8.75

Schedule: A-2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Historical Financial Metrics - Financial Basis
ELECTRIC RESULTS ONLY
(\$000)

Case No.: U-21389
Exhibit No.: A-1 (JCA-2)
Schedule: A-2
Page: 3 of 6
Witness: JCAponte
Date: May 2023

Line No	(a) Description	(b)	(c)	(d)	(e)	(f)
		Calendar Year				
		2018	2019	2020	2021	2022
Overall Fixed Charge Coverage Ratio						
35	Net income available to common stockholders	535,088	508,678	570,889	580,383	580,068
36	Interest charges	210,785	214,929	202,399	193,601	206,904
37	Subtotal numerator	745,874	723,606	773,289	773,984	786,972
38	Interest charges	210,785	214,929	202,399	193,601	206,904
39	Preferred stock dividends	1,193	1,160	1,111	1,081	1,078
40	Subtotal denominator	211,978	216,089	203,510	194,682	207,982
41	Overall fixed charge coverage ratio	3.52	3.35	3.80	3.98	3.78
Cash Flow Coverage of Dividends Ratio						
42	Net income available to common stockholders	535,088	508,678	570,889	580,383	580,068
43	Depreciation and amortization	682,307	713,007	739,148	771,634	757,258
44	Deferred income taxes	124,791	38,192	321,515	(85,580)	166,168
45	Subtotal	1,342,186	1,259,876	1,631,553	1,266,437	1,503,494
46	Common dividends {5}	428,071	406,942	456,712	464,306	464,055
47	Cash flow coverage of dividends ratio	3.14	3.10	3.57	2.73	3.24
Common Dividend Payout Ratio						
48	Common dividends {5}	428,071	406,942	456,712	464,306	464,055
49	Net income available to common stockholders	535,088	508,678	570,889	580,383	580,068
50	Common dividend payout ratio	80%	80%	80%	80%	80%
Permanent Capitalization						
51	Long term debt	6,809,306	7,263,181	8,131,338	8,437,664	10,208,351
52	Preferred stock	37,315	37,315	37,315	37,315	37,315
53	Common equity	6,884,117	7,700,854	8,519,584	9,242,745	10,118,365
54	Total permanent capita	13,730,738	15,001,350	16,688,236	17,717,724	20,364,031
55	Long term debt	49.59%	48.42%	48.72%	47.62%	50.13%
56	Preferred stock	0.27%	0.25%	0.22%	0.21%	0.18%
57	Common equity	50.14%	51.33%	51.05%	52.17%	49.69%
58	Total permanent capita	100.00%	100.00%	100.00%	100.00%	100.00%

Schedule: A-2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Historical Financial Metrics - Ratemaking Basis
ELECTRIC RESULTS ONLY
(\$000)

Case No.: U-21389
Exhibit No.: A-1 (JCA-2)
Schedule: A-2
Page: 4 of 6
Witness: JCAponte
Date: May 2023

		(a)	(b)	(c)	(d)	(e)	(f)
Line No	Description	Calendar Year					
		2018	2019	2020	2021	2022	
59	Operating revenues	4,471,514	4,355,517	4,294,102	4,884,658	5,228,924	
60	Operating expenses	3,792,887	3,625,943	3,482,006	4,219,304	4,515,341	
61	Pre-tax operating income	678,627	729,574	812,096	665,354	713,583	
62	Income taxes	58,632	82,880	166,654	34,134	57,803	
63	Net operating income	619,995	646,693	645,441	631,220	655,781	
64	Tax impact of pro forma interest on NOI	(12,859)	(13,290)	(5,605)	(4,444)	(4,034)	
65	AFUDC	5,070	5,884	7,172	8,166	5,491	
66	Interest charges	(159,794)	(161,635)	(171,758)	(166,478)	(184,127)	
67	Preferred stock dividends	(1,070)	(1,026)	(1,010)	(973)	(931)	
68	Net income available to common stockholders and JDITC	451,342	476,626	474,240	467,492	472,180	
69	Return assignable to JDITC	(3,213)	(3,547)	(3,642)	(3,428)	(3,651)	
70	Net income available to common stockholders	448,130	473,079	470,598	464,064	468,529	
71	Average common equity	4,306,516	4,543,533	4,955,705	5,189,180	5,480,655	
72	Earned rate of return on common equity	10.41%	10.41%	9.50%	8.94%	8.55%	
73	Authorized return on equity	10.00% {1}	10.00% {2}	10.00% {3}	9.90% {3}	9.90% {4}	

Schedule: A-2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Historical Financial Metrics - Ratemaking Basis
ELECTRIC RESULTS ONLY
(\$000)

Case No.: U-21389
Exhibit No.: A-1 (JCA-2)
Schedule: A-2
Page: 5 of 6
Witness: JCAponte
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)
Line		Calendar Year				
No	Description	2018	2019	2020	2021	2022
	<u>EBIT Interest Coverage Ratio</u>					
74	Pre-tax operating income	678,627	729,574	812,096	665,354	713,583
75	AFUDC	5,070	5,884	7,172	8,166	5,491
76	Total EBIT	683,697	735,457	819,267	673,520	719,075
77	Interest charges, net of pro forma interest	146,935	148,345	166,153	162,034	180,093
78	EBIT interest coverage ratio	4.65	4.96	4.93	4.16	3.99
	<u>EBITDA Interest Coverage Ratio</u>					
79	Total EBIT	683,697	735,457	819,267	673,520	719,075
80	Depreciation and amortization	665,312	694,677	720,318	751,470	748,017
81	Total EBITDA	1,349,008	1,430,134	1,539,586	1,424,990	1,467,092
82	Interest charges, net of pro forma interest	146,935	148,345	166,153	162,034	180,093
83	EBITDA interest coverage ratio	9.18	9.64	9.27	8.79	8.15
	<u>Funds Flow from Operations (FFO) Interest Coverage Ratio</u>					
84	Net operating income	619,995	646,693	645,441	631,220	655,781
85	Depreciation and amortization	665,312	694,677	720,318	751,470	748,017
86	Deferred income tax	124,791	38,192	321,515	(85,580)	166,168
87	AFUDC	5,070	5,884	7,172	8,166	5,491
88	Other major recurring non-cash items	-	-	-	-	-
89	Interest paid	196,495	200,213	207,328	198,499	202,104
	Less:					
90	Operating lease adjustment to depreciation	-	-	-	-	-
91	Subtotal	1,611,662	1,585,658	1,901,774	1,503,775	1,777,561
92	Interest charges, net of pro forma interest	146,935	148,345	166,153	162,034	180,093
93	FFO interest coverage ratio	10.97	10.69	11.45	9.28	9.87
	<u>Overall Fixed Charge Coverage Ratio</u>					
94	Net income available to common stockholders	448,130	473,079	470,598	464,064	468,529
95	Interest charges, net of pro forma interest	146,935	148,345	166,153	162,034	180,093
96	Subtotal numerator	595,065	621,424	636,751	626,098	648,623
97	Interest charges, net of pro forma interest	146,935	148,345	166,153	162,034	180,093
98	Preferred stock dividends	1,070	1,026	1,010	973	931
99	Subtotal denominator	148,005	149,371	167,162	163,007	181,025
100	Overall fixed charge coverage ratio	4.02	4.16	3.81	3.84	3.58

Schedule: A-2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Historical Financial Metrics - Ratemaking Basis
ELECTRIC RESULTS ONLY
(\$000)

Case No.: U-21389
Exhibit No.: A-1 (JCA-2)
Schedule: A-2
Page: 6 of 6
Witness: JCAponte
Date: May 2023

		(a)	(b)	(c)	(d)	(e)	(f)
Line		Calendar Year					
No	Description	2018	2019	2020	2021	2022	
<u>Cash Flow Coverage of Dividends Ratio</u>							
101	Net income available to common stockholders	448,130	473,079	470,598	464,064	468,529	
102	Depreciation and amortization	665,312	694,677	720,318	751,470	748,017	
103	Deferred income tax	124,791	38,192	321,515	(85,580)	166,168	
104	Subtotal	1,238,232	1,205,947	1,512,431	1,129,954	1,382,714	
105	Common dividends {5}	358,504	378,463	376,478	371,251	374,824	
106	Cash flow coverage of dividends ratio	3.45	3.19	4.02	3.04	3.69	
<u>Common Dividend Payout Ratio</u>							
107	Common dividends {5}	358,504	378,463	376,478	371,251	374,824	
108	Net income available to common stockholders	448,130	473,079	470,598	464,064	468,529	
109	Common Dividend Payout Ratio	80%	80%	80%	80%	80%	
<u>Permanent Capitalization</u>							
110	Long term debt, excluding securitization	6,311,870	6,997,291	7,880,109	8,214,999	9,009,833	
111	Preferred stock	37,315	37,315	37,315	37,315	37,315	
112	Common equity	6,904,894	7,728,910	8,555,596	9,274,520	10,133,854	
113	Total permanent capital	13,254,079	14,763,516	16,473,020	17,526,834	19,181,001	
114	Long term debt, excluding securitization	47.62%	47.40%	47.84%	46.87%	46.97%	
115	Preferred stock	0.28%	0.25%	0.23%	0.21%	0.19%	
116	Common equity	52.10%	52.35%	51.94%	52.92%	52.83%	
117	Total permanent capital	100.00%	100.00%	100.00%	100.00%	100.00%	

Notes

- {1} Case No. U-18322 Final Order dated July 24, 2018
{2} Case No. U-20134 Settlement Agreement dated January 10, 2019
{3} Case No. U-20697 Final Order dated December 17, 2020
{4} Case No. U-20963 Final Order dated December 22, 2021
{5} CECO pays dividends to the parent as a total company. This number represents the company's 80% dividend policy applied to net income available to common stockholders

Schedule: B-1

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Rate Base
For the Historical Year Ended December 31, 2022

Case No.: U-21389
Exhibit No.: A-2 (JCA-3)
Schedule: B-1
Page 1 of 1
Witness: JCAponte
Date: May 2023

	(a)	(b)	(c)	(d)
Line No	Description	Source	Total (\$000)	Jurisdictional (\$000)
1	Total utility plant	Exhibit No.: A-2 (JCA-4)	18,230,527	18,158,506
2	Depreciation reserve	Exhibit No.: A-2 (JCA-5)	6,899,110	6,868,884
3	Net utility plant	Line 1 - Line 2	11,331,417	11,289,622
4	Customer advances for construction	Exhibit No.: A-2 (JCA-7), Column (d.1), Line 2	(57,395)	(57,361)
5	Utility plant rate base	Sum Lines 3 - 4	11,274,022	11,232,261
6	Working capital	Exhibit No.: A-2 (JCA-6)	1,373,322	1,367,175
7	Total rate base	Line 5 + Line 6	12,647,343	12,599,436

Schedule: B-2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Total Utility Plant
 For the Historical Year Ended December 31, 2022

Case No.: U-21389
 Exhibit No.: A-2 (JCA-4)
 Schedule: B-2
 Page 1 of 1
 Witness: JCAponte
 Date: May 2023

	(a)	(b)	(c)	(d)	(e)
Line		MPSC		Total	Jurisdictional
No	Description	Account Number	Source	(\$000)	(\$000)
1	Gross Plant, incl. Plant Held For Future Use	101, 105	Exhibit No.: A-2 (JCA-7), Column (d.1), Line 3	19,046,599	18,971,938
2	Exclude: Renewables plant-in-service			(1,286,401)	(1,281,362)
3	Deferred debits	182		-	-
4	Construction work in progress	107	Exhibit No.: A-2 (JCA-7), Column (d.1), Line 7	534,865	532,136
5	Exclude: Renewables CWIP			(64,536)	(64,207)
6	Total utility plant		Sum Lines 1 - 5	<u>18,230,527</u>	<u>18,158,506</u>

Schedule: B-3

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Depreciation Reserve and Other Deductions
For the Historical Year Ended December 31, 2022

Case No.: U-21389
Exhibit No.: A-2 (JCA-5)
Schedule: B-3
Page 1 of 1
Witness: JCAponte
Date: May 2023

	(a)	(b)	(c)	(d)
Line No	Description	Source	Total (\$000)	Jurisdictional (\$000)
1	Accumulated depreciation, amortization and depletion of plant-in-service	Exhibit No.: A-2 (JCA-7), Column (d.1), Line 6	5,902,935	5,877,073
2	Add: Cost of removal regulatory liability		1,194,276	1,189,043
3	Exclude: Renewables depreciation reserve		(198,100)	(197,233)
4	Total depreciation reserve and other deduction	Sum Lines 1 - 3	<u>6,899,110</u>	<u>6,868,884</u>

Schedule: B-4**MICHIGAN PUBLIC SERVICE COMMISSION**Consumers Energy Company

Working Capital

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-2 (JCA-6)

Schedule: B-4

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No.	(a) Description	(b) Source	(c) (\$000)
1	Cash, cash equivalents, and restricted cash	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 10	20,157
2	Accounts and notes receivable	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 11	284,586
3	Fuel stock	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 12	47,745
4	Materials and supplies	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 13	134,417
5	Allowances	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 14	14
6	Prepayments	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 16	769,539
7	Accrued revenue	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 17	302,353
8	Deferred taxes	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 18	228,375
9	Deferred debits	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 19	635,437
10	Total assets	Sum Lines 1 - 9	2,422,622
11	Accounts payable	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 23	558,140
12	Customer deposits	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 24	18,319
13	Accrued interest	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 25	49,157
14	Dividends payable	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 26	38,060
15	Accrued taxes	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 27	242,540
16	Other liabilities	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 28	65,860
17	Deferred credits	Exhibit No.: A-2 (JCA-7), Column (e.1), Line 31	77,226
18	Total liabilities	Sum Lines 11 - 17	1,049,301
19	Total working capital	Line 10 - Line 18	1,373,322
20	Jurisdictional factor		0.995524
21	Total working capital - jurisdictional	Line 19 * Line 20	1,367,175

Schedule: B-5

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 13-Month Average Balance Sheet Summary
 For the Historical Year Ended December 31, 2022

Case No.: U-21389
 Exhibit No.: A-2 (JCA-7)
 Schedule: B-5
 Page 1 of 1
 Witness: JCAponte
 Date: May 2023

Source: WP-JCA-7

	(a)	(b)	(c.1)	(c.2)	(d.1)	(d.2)	(d.3)	(d.4)	(e.1)	(e.2)	(e.3)
Line	EXCLUDED				PLANT				WORKING CAPITAL		
No.	Description	Total	Rate Base	Sub. Invest.	Electric	Gas	MGP	Non-Utility	Electric	Gas	Non-Utility
1	Utility plant	\$28,949,624,858	\$ 596,106,072	\$ -	\$ 18,142,653,137	\$ 10,210,865,649	\$ -	\$ -	\$ -	\$ -	\$ -
2	Common plant	1,461,563,329	3,651,731	-	903,946,098	553,965,500	-	-	-	-	-
3	Gross plant	30,411,188,187	599,757,803	-	19,046,599,235	10,764,831,149	-	-	-	-	-
4	Utility plant accumulated depreciator	(7,747,215,458)	(135,537,031)	-	(5,317,044,124)	(2,294,634,303)	-	-	-	-	-
5	Common plant accumulated depreciator	(932,998,950)	(5,734,881)	-	(585,890,504)	(341,373,565)	-	-	-	-	-
6	Total depreciation	(8,680,214,408)	(141,271,911)	-	(5,902,934,629)	(2,636,007,868)	-	-	-	-	-
7	Construction work in progress	909,086,737	3,011,447	-	534,864,958	371,210,332	-	(0)	-	-	-
8	Net plant	22,640,060,516	461,497,338	-	13,678,529,564	8,500,033,613	-	(0)	-	-	-
9	Other property	155,473,230	120,218,263	(7,497,295)	-	-	-	22,589,514	-	-	20,162,748
10	Cash, cash equivalents, and restricted cash	36,022,773	4,476,043	-	-	-	-	-	20,157,171	11,389,560	-
11	Accounts and notes receivable	453,917,097	22,918,715	-	-	-	-	-	284,585,938	139,012,746	7,399,697
12	Fuel stock	47,744,753	-	-	-	-	-	-	47,744,753	-	-
13	Materials and supplies	180,076,313	21	-	-	-	-	-	134,416,728	45,659,564	-
14	Allowances	11,861,028	11,846,740	-	-	-	-	-	14,288	-	-
15	Gas stored underground	636,586,676	-	-	-	-	-	-	-	636,586,676	-
16	Prepayments	1,273,770,365	-	-	-	-	-	-	769,538,868	504,231,606	(109)
17	Accrued revenue	690,330,884	212,102,088	-	-	-	-	-	302,352,753	175,876,043	-
18	Deferred taxes	2,302,871,509	1,915,770,507	-	-	-	-	-	228,374,946	158,449,181	276,875
19	Deferred debits	1,613,578,867	600,815,150	-	219,064	-	108,743,662	-	635,436,762	268,157,179	207,049
20	Total assets	30,042,294,011	3,349,644,866	(7,497,295)	13,678,748,628	8,500,033,613	108,743,662	22,589,514	2,422,622,208	1,939,362,555	28,046,260
21	Debt, leases, and financial obligations	(8,904,831,647)	(8,904,831,647)	-	-	-	-	-	-	-	-
22	Notes payable	(290,428,624)	(290,428,624)	-	-	-	-	-	-	-	-
23	Accounts payable	(794,708,290)	(3,187,918)	-	-	-	-	-	(558,139,988)	(230,984,761)	(2,395,623)
24	Customer deposits	(29,537,390)	-	-	-	-	-	-	(18,318,532)	(11,218,859)	-
25	Accrued interest	(76,605,873)	(37,548)	-	-	-	-	-	(49,156,865)	(27,258,324)	(153,137)
26	Dividends payable	(59,283,013)	-	-	-	-	-	-	(38,059,694)	(21,104,753)	(118,566)
27	Accrued taxes	(449,094,286)	-	-	-	-	-	-	(242,540,013)	(233,723,641)	27,169,367
28	Other liabilities	(252,485,808)	(151,727,906)	-	-	-	(3,788,988)	-	(65,860,033)	(31,108,880)	-
29	Customer advances	(94,823,136)	-	-	(57,395,361)	(37,427,776)	-	-	-	-	-
30	Deferred income and investment taxes	(5,814,109,156)	(5,814,109,156)	-	-	-	-	-	-	-	-
31	Deferred credits	(3,388,236,281)	(823,275,792)	-	(1,201,063,398)	(1,213,585,747)	(55,331,305)	-	(77,225,582)	(15,890,824)	(1,863,634)
32	Total liabilities	(20,154,143,506)	(15,987,598,592)	-	(1,258,458,758)	(1,251,013,523)	(59,120,294)	-	(1,049,300,706)	(571,290,042)	22,638,408
33	Common equity	(9,850,835,704)	(9,850,835,704)	-	-	-	-	-	-	-	-
34	Preferred stock	(37,314,800)	(37,314,800)	-	-	-	-	-	-	-	-
35	Total equity	(9,888,150,504)	(9,888,150,504)	-	-	-	-	-	-	-	-
36	Net assets (liabilities)	\$ -	\$ (22,526,104,230)	\$ (7,497,295)	\$ 12,420,289,870	\$ 7,249,020,090	\$ 49,623,369	\$ 22,589,514	\$ 1,373,321,502	\$ 1,368,072,514	\$ 50,684,668

Schedule: B-6

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Point-In-Time Working Capital Balance Sheet Summary
For the Historical Year Ended December 31, 2022

Source: WP-JCA-6

Case No.: U-21389
Exhibit No.: A-2 (JCA-8)
Schedule: B-6
Page 1 of 1
Witness: JCAponte
Date: May 2023

	(a)	(b)	(c.1)	(c.2)	(d.1)	(d.2)	(d.3)	(d.4)	(e.1)	(e.2)	(e.3)
Line	EXCLUDED				PLANT				WORKING CAPITAL		
No.	Description	Total	Rate Base	Sub. Invest.	Electric	Gas	MGP	Non-Utility	Electric	Gas	Non-Utility
1	Utility plant	\$29,999,811,989	\$ 686,839,993	\$ -	\$ 18,641,169,306	\$ 10,671,802,691	\$ -	\$ -	\$ -	\$ -	\$ -
2	Common plant	1,404,321,491	3,926,158	-	868,284,400	532,110,933	-	-	-	-	-
3	Gross plant	31,404,133,481	690,766,151	-	19,509,453,706	11,203,913,624	-	-	-	-	-
4	Utility plant accumulated depreciator	(8,036,587,692)	(157,944,290)	-	(5,547,576,293)	(2,331,067,110)	-	-	-	-	-
5	Common plant accumulated depreciator	(881,485,490)	(5,836,586)	-	(553,273,718)	(322,375,186)	-	-	-	-	-
6	Total depreciation	(8,918,073,183)	(163,780,875)	-	(6,100,850,011)	(2,653,442,297)	-	-	-	-	-
7	Construction work in progress	984,850,814	679,297	-	610,031,056	374,140,460	-	-	-	-	-
8	Net plant	23,470,911,112	527,664,573	-	14,018,634,751	8,924,611,788	-	-	-	-	-
9	Other property	160,378,431	118,900,551	(6,879,568)	-	-	-	27,117,777	-	-	21,239,671
10	Cash, cash equivalents, and restricted cash	40,928,508	2,319,973	-	-	-	-	-	24,669,397	13,939,137	-
11	Accounts and notes receivable	492,640,038	40,543,953	-	-	-	-	-	298,931,318	144,717,311	8,447,455
12	Fuel stock	59,159,159	-	-	-	-	-	-	59,159,159	-	-
13	Materials and supplies	206,028,524	21	-	-	-	-	-	155,227,413	50,801,090	-
14	Allowances	10,188,572	10,174,284	-	-	-	-	-	14,288	-	-
15	Gas stored underground	839,459,584	-	-	-	-	-	-	-	839,459,584	-
16	Prepayments	1,230,430,481	-	-	-	-	-	-	744,577,837	485,852,759	(115)
17	Accrued revenue	1,075,345,207	410,620,924	-	-	-	-	-	329,050,050	335,674,233	-
18	Deferred taxes	2,455,836,457	1,868,833,305	-	-	-	-	-	345,746,172	240,761,945	495,035
19	Deferred debits	1,800,871,873	654,486,926	-	1,522,508	-	108,282,211	-	724,697,210	311,679,467	203,551
20	Total assets	31,842,177,946	3,633,544,510	(6,879,568)	14,020,157,259	8,924,611,788	108,282,211	27,117,777	2,682,072,843	2,422,885,528	30,385,598
21	Debt, leases, and financial obligations	(10,160,744,281)	(10,160,744,281)	-	-	-	-	-	-	-	-
22	Notes payable	(253,080,460)	(253,080,460)	-	-	-	-	-	-	-	-
23	Accounts payable	(864,905,995)	-	-	-	-	-	-	(643,565,432)	(214,292,840)	(7,047,723)
24	Customer deposits	(33,226,380)	-	-	-	-	-	-	(20,781,559)	(12,444,821)	-
25	Accrued interest	(89,385,716)	(7,321)	-	-	-	-	-	(57,380,930)	(31,818,709)	(178,757)
26	Dividends payable	-	-	-	-	-	-	-	-	-	-
27	Accrued taxes	(573,638,149)	-	-	-	-	-	-	(342,072,132)	(231,246,686)	(319,332)
28	Other liabilities	(207,282,163)	(145,178,053)	-	-	-	(4,712,912)	-	(41,743,026)	(15,648,171)	-
29	Customer advances	(96,251,384)	-	-	(58,993,369)	(37,258,015)	-	-	-	-	-
30	Deferred income and investment taxes	(5,914,756,441)	(5,914,756,441)	-	-	-	-	-	-	-	-
31	Deferred credits	(3,493,226,917)	(922,578,338)	-	(1,190,155,456)	(1,242,520,674)	(58,174,149)	-	(70,184,879)	(6,962,285)	(2,651,135)
32	Total liabilities	(21,686,497,886)	(17,396,344,895)	-	(1,249,148,825)	(1,279,778,689)	(62,887,061)	-	(1,175,727,957)	(512,413,512)	(10,196,946)
33	Common equity	(10,118,365,260)	(10,118,365,260)	-	-	-	-	-	-	-	-
34	Preferred stock	(37,314,800)	(37,314,800)	-	-	-	-	-	-	-	-
35	Total equity	(10,155,680,060)	(10,155,680,060)	-	-	-	-	-	-	-	-
36	Net assets (liabilities)	\$ -	\$ (23,918,480,445)	\$ (6,879,568)	\$ 12,771,008,433	\$ 7,644,833,099	\$ 45,395,149	\$ 27,117,777	\$ 1,506,344,886	\$ 1,910,472,015	\$ 20,188,652

Schedule: C-1

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Adjusted Net Operating Income
For the Historical Year Ended December 31, 2022

Case No.: U-21389
Exhibit No.: A-3 (JCA-9)
Schedule: C-1
Page 1 of 1
Witness: JCAponte
Date: May 2023

(a)		(b)	(c)	(d)
Line No.	Description	Source	Total (\$000)	Jurisdictional (\$000)
1	Sales revenues	Exhibit No.: A-3 (JCA-11)	4,963,156	4,961,988
2	Sales for resale	Exhibit No.: A-3 (JCA-11)	240,958	240,958
3	Other electric revenues	Exhibit No.: A-3 (JCA-11)	160,123	136,953
4	Total operating revenues	Sum Lines 1 - 3	5,364,237	5,339,899
5	Fuel expense	Exhibit No.: A-3 (JCA-12)	663,350	656,232
6	Purchased & interchange power	Exhibit No.: A-3 (JCA-12)	1,959,097	1,938,073
7	Operations and maintenance	Exhibit No.: A-3 (JCA-13)	893,311	889,864
8	Depreciation and amortization	Exhibit No.: A-3 (JCA-14)	751,446	748,017
9	Property taxes	Exhibit No.: A-3 (JCA-15)	203,698	202,818
10	General taxes	Exhibit No.: A-3 (JCA-15)	35,840	35,618
11	Other (local) taxes	Exhibit No.: A-3 (JCA-18)	1,323	1,328
12	State income taxes	Exhibit No.: A-3 (JCA-17)	44,456	44,628
13	Federal income taxes	Exhibit No.: A-3 (JCA-16)	57,580	57,803
14	Total operating expenses	Sum Lines 5 - 13	4,610,100	4,574,381
15	Net operating income	Line 4 - Line 14	754,137	765,518
16	AFUDC	Exhibit No.: A-3 (JCA-19)	5,520	5,491
17	Net operating income, including AFUDC	Line 15 + Line 16	759,656	771,009
<u>Adjustments to Net Operating Income</u>				
18	Weather normalization	Exhibit No.: A-13 (JCA-51), Line 2	(41,107)	(40,897)
19	Purchased power administration fee revenue	Exhibit No.: A-13 (JCA-51), Line 3	(457)	(455)
20	Excess MDNR fees	Exhibit No.: A-13 (JCA-51), Line 4	334	332
21	MISO reliability schedule 10 charge	Exhibit No.: A-13 (JCA-51), Line 5	343	342
22	Surcharge revenues and expenses/amortization	Exhibit No.: A-13 (JCA-51), Line 6	(72,954)	(72,623)
23	Jobwork revenues	Exhibit No.: A-13 (JCA-51), Line 7	1,967	1,958
24	Interest income on cash operating accounts	Exhibit No.: A-13 (JCA-51), Line 8	371	370
25	Compensation expenses - Executive annual physicals	Exhibit No.: A-13 (JCA-51), Line 9	8	8
26	Compensation expenses - EICP	Exhibit No.: A-13 (JCA-51), Line 10	-	-
27	Compensation expenses - Corporate Giving - staff salaries	Exhibit No.: A-13 (JCA-51), Line 11	11	10
28	Compensation expenses - Corporate Communications - s	Exhibit No.: A-13 (JCA-51), Line 12	137	137
29	Advertising expenses	Exhibit No.: A-13 (JCA-51), Line 13	-	-
30	Dues & donations expenses	Exhibit No.: A-13 (JCA-51), Line 14	462	459
31	Jobwork expenses	Exhibit No.: A-13 (JCA-51), Line 15	(1,643)	(1,635)
32	Interest expense on security deposits	Exhibit No.: A-13 (JCA-51), Line 16	(193)	(192)
33	One-time Legal Consultant Expense	Exhibit No.: A-13 (JCA-51), Line 17	2,726	2,712
34	Voluntary Separation Plan expenses	Exhibit No.: A-13 (JCA-51), Line 18	4,045	4,024
35	Segregated Cell Insurance Expense	Exhibit No.: A-13 (JCA-51), Line 20	(4,309)	(4,287)
36	Pro Forma Interest Adjustment	Exhibit No.: A-13 (JCA-51), Line 21	(4,350)	(4,334)
37	Interest Synchronization Adjustment	Exhibit No.: A-13 (JCA-51), Line 22	301	300
38	Total net operating income adjustments	Sum Lines 18 - 37	(114,309)	(113,771)
39	Adjusted net operating income	Line 17 + Line 38	645,347	657,238

Schedule: C-2

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Calculation of the Revenue Conversion Factor

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-3 (JCA-10)

Schedule: C-2

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No.	(a)	(b)	(c)
	Description	Source	Amount
1	Income base - before taxes		100.0000
2	State income tax	Line 1 * 5.31% State Income Tax Rate	5.3100
3	Other (local) income tax	Line 1 * 0.16% Other (Local) Income Tax Rate	<u>0.1600</u>
4	Federal income tax base	Line 1 - Line 2 - Line 3	94.5300
5	Federal income tax	Line 4 * 21.00% Federal Income Tax Rate	<u>19.8513</u>
6	Income base - after taxes	Line 4 - Line 5	<u>74.6787</u>
7	Revenue conversion factor	Line 1 / Line 6	<u><u>1.3391</u></u>

Schedule: C-3

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Operating Revenues
For the Historical Year Ended December 31, 2022

Case No.: U-21389
Exhibit No.: A-3 (JCA-11)
Schedule: C-3
Page 1 of 1
Witness: JCAponte
Date: May 2023

(a)		(b)	(c)
Line No	Description	Source	Total (\$000)
1	Sales revenues	P-521 p. 300(M), Line 12 minus Line 17	4,963,156
2	Sales for resale	P-521 p. 300(M), Line 14	240,958
3	Other electric revenues	P-521 p. 300(M), Line 30	<u>160,123</u>
4	Total operating revenues	Sum Lines 1 - 3	<u><u>5,364,237</u></u>

Schedule: C-4**MICHIGAN PUBLIC SERVICE COMMISSION**Consumers Energy Company

Cost of Fuel and Purchased Power

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-3 (JCA-12)

Schedule: C-4

Page 1 of 1

Witness: JCAponte

Date: May 2023

(a)		(b)	(c)
Line No.	Description	Source	Total (\$000)
1	Fuel expense - Steam Power	P-521 p. 320, Line 5	336,055
2	Fuel expense - Other Power Generation	P-521 p. 321, Line 63	340,475
3	Fuel handling	WP-JCA-9	(13,180)
4	Total Fuel Expense	Sum Lines 1 - 3	663,350
5	Purchased power	P-521 p. 321, Line 76	1,484,172
6	Transmission expenses	P-521 p. 321, Line 112	468,500
7	Regional transmission and market operating expense	P-521 p. 322, Line 131	8,310
8	Load dispatch	WP-JCA-10	(1,885)
9	Total purchased and net interchange power	Sum Lines 5 - 8	1,959,097
10	Total cost of fuel and purchased power	Line 4 + Line 9	2,622,447

Schedule: C-5

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Operation and Maintenance Expenses
 For the Historical Year Ended December 31, 2022

Case No.: U-21389
 Exhibit No.: A-3 (JCA-13)
 Schedule: C-5
 Page 1 of 3
 Witness: JCAponte
 Date: May 2023

(a)		(b)	(c)
Line No.	Description	Source	Total (\$000)
1	Power production expenses	P-521 p. 321, Line 80 minus Exhibit No.: A-3 (JCA-12), Lines 4-	183,601
2	Transmission expenses	(-) WP-JCA-10	1,885
3	Distribution expenses	P-521 p. 322, Line 156	327,465
4	Customer accounts expenses	P-521 p. 322, Line 164	75,665
5	Customer service & information expenses	P-521 p. 323, Line 171	251,843
6	Sales expenses	P-521 p. 323, Line 178	-
7	Administrative & general expenses	P-521 p. 323, Line 197	53,750
8	Subtotal	Sum Lines 1 - 7	894,208
9	Gains from disposition of allowances	P-521 p. 115, Line 22 (e)	(898)
10	Total other operation and maintenance expenses	Line 8 + Line 9	893,311

Schedule C5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Historical Operation and Maintenance Expenses by Witness

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-3 (JCA-13)

Schedule: C-5

Page: 3 of 3

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(i)	(i)	(i)	(j)	(k)	(l)	(m)	(n)
Line No.	FERC Account	Description	Blumenstock	Bolden	Conrad	Cumberworth	Foster	Grob	Guinn	Hale	Kelly	McLean	Myrom	Weller	Total O&M	MPSC P-521 Pg 320-323	Variance
Items Included in CEC Co Other O&M Expense Not Included in the MPSC P-521																	
88	94040010	Limited Term Utility -Electric	-	-	-	-	-	-	-	-	-	-	-	10,240	10,240	10,240	-
89	94160000	Expense Jobbing & CTR	-	-	-	12,402	-	-	-	-	-	-	-	175,356	187,758	187,758	-
90	94265010	Other Deductions	(0)	-	980,944	-	0	-	-	-	45,619	-	-	-	1,026,562	1,026,562	-
91		Allocation Difference	-	-	-	(185,707)	-	-	-	-	-	-	-	(319,303)	(505,010)	(505,010)	-
92		Total O&M by Witness (Sum Lines 87-91)	\$ 150,030,678	\$ 102,003,474	\$ 2,675,767	\$ 9,704,976	\$ 63,716,010	\$ (42,199,611)	\$ 17,651,027	\$ 43,981,740	\$ 214,437,063	\$ 37,881,510	\$ -	\$ 53,071,180	\$ 652,953,814	\$ 3,517,374,421	\$ 2,864,420,607
Reconciliation of MPSC P-521 to Witness Other O&M Expense																	
93	95011010	FUEL														(336,055,164)	(336,055,164)
94	95012010	FUEL - NONFUEL CLAUS														13,179,851	13,179,851
95	95020010	STEAM EXPENSES														(368,139)	(368,139)
96	95460010	SUPV & ENGG-OTH PWR														(1,930,623)	(1,930,623)
97	95470010	FUEL-OTHER PWR GEN														(339,985,190)	(339,985,190)
98	95480010	GEN EXPENSES														(75,808)	(75,808)
99	95490010	MISC OTHER PWR GEN														(5,847,337)	(5,847,337)
100	95510010	MITC SUPV & ENGIN														(251,136)	(251,136)
101	95520010	MAINT OF STRUCTURES														(1,414,359)	(1,414,359)
102	95530010	MAINT OF ELECT PLT														(11,694,504)	(11,694,504)
103	95550010	PURCHASED POWER														(1,484,172,064)	(1,484,172,064)
104	95560010	SYS CONTROL AND DISP														113,707	113,707
105	95614010	SCHEDUL DISPATCH SVC														(12,071,709)	(12,071,709)
106	95618010	RELIAB PLAN STND SVC														(459,849)	(459,849)
107	95650010	TRANS OF ELEC OTHR														(454,543,216)	(454,543,216)
108	95757010	MKT ADMIN COMPL SVCS														(8,309,570)	(8,309,570)
109	95841010	OPS ENERG STOR EQUIP														(47,370)	(47,370)
110	95922010	MITC ENERGY STOR EQUI														(1,532)	(1,532)
111	99041010	UNCOLL ACCTS-ELEC														333,890	333,890
112	99070010	SUPRV CUST SVC-ELEC														3,012,064	3,012,064
113	99080010	CUST ASSIST-ELEC														(212,451,588)	(212,451,588)
114	99200010	A & G SAL-ELECT														(14,070)	(14,070)
115	99230010	OUTSIDE SVCS ELEC														(3,650,734)	(3,650,734)
116	99302010	MISC GEN EXP-ELECT														(7,859,689)	(7,859,689)
117		Rounding Adjustments														143,533	143,533
118		Total O&M by Witness (Sum Lines 92-117)														\$ 652,953,814	\$ -

Schedule: C-6

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Depreciation and Amortization Expenses
For the Historical Year Ended December 31, 2022

Case No.: U-21389
Exhibit No.: A-3 (JCA-14)
Schedule: C-6
Page 1 of 1
Witness: JCAponte
Date: May 2023

Line No.	(a) Description	(b) Source	(c) (\$000)
1	Steam production plant - depreciation expense	P-521 p. 321, Line 2(b) through 6(b)	314,817
2	Transmission plant - depreciation expense	P-521 p. 336M, Line 7(b)	-
3	Distribution plant - depreciation expense	P-521 p. 336M, Line 8(b)	298,444
4	General plant - depreciation expense	P-521 p. 336M, Line 9(b)	11,411
5	Common plant - electric - depreciation expense	P-521 p. 336M, Line 10(b)	20,523
6	Total amortization of limited-term electric plant (Acct 404)	P-521 p. 336M, Line 11(c)	871
7	Total amortization of other electric plant (Acct 405)	P-521 p. 336M, Line 11(d)	71,862
8	Amortization of utility plant acquisition adjustment	P-521 p. 115, Line 9 (e)	5,554
9	Regulatory debits	P-521 p. 115, Line 12 (e)	<u>27,964</u>
10	Subtotal	Sum Lines 1 - 9	751,446
11	Less: transmission expense	Line 2 *-1	<u>-</u>
12	Total depreciation and amortization expenses	Line 10 + Line 11	<u><u>751,446</u></u>

Schedule C-7**MICHIGAN PUBLIC SERVICE COMMISSION**Consumers Energy Company

General Taxes

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-3 (JCA-15)

Schedule: C-7

Page 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)
Line No.	Description	Source	Total (\$000)
1	Real and personal property taxes	P-521 p. 262.1 (i), Line 43	203,698
2	Federal unemployment tax	P-521 p. 262.1 (i), Line 4	125
3	Federal FICA tax	P-521 p. 262.1 (i), Line 6	25,257
4	State income tax	P-521 p. 262.1 (i), Line 17	735
5	Total payroll taxes	Sum Lines 2 - 4	26,117
6	State fuel tax	P-521 p. 262.1 (i), Line 19	-
7	State MPSC assessment	P-521 p. 262.1 (i), Line 35	9,723
8	State franchise tax	P-521 p. 262.1 (i), Line 22	-
9	State sales and use tax absorbed by company	P-521 p. 262.1 (i), Line 23-27	-
10	Other general taxes	Sum Lines 6 - 9	9,723
11	Total general taxes	Line 1 + Line 5 + Line 10	239,538

Schedule: C-8**MICHIGAN PUBLIC SERVICE COMMISSION**Consumers Energy Company

Federal Income Taxes

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-3 (JCA-16)

Schedule: C-8

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No.	(a) Description	(b) Source	(c) (\$000)
1	Federal income taxes	P-521 p. 115, Line 15 (e)	(68,683)
2	Provision for deferred income taxes	P-521 p. 115, Line 17 (e)	463,047
3	Provision for deferred income taxes - credit	(-) P-521 p. 115, Line 18 (e)	(315,111)
4	Investment tax credit adjustment, net	P-521 p. 115, Line 19 (e)	18,232
4	Deferred state income tax	Tax Department	(38,551)
5	Deferred other (local) income tax	Tax Department	<u>(1,354)</u>
6	Total federal income taxes	Sum Lines 1 - 5	<u><u>57,580</u></u>

Schedule: C-9

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
State Income Taxes
For the Historical Year Ended December 31, 2022

Case No.: U-21389
Exhibit No.: A-3 (JCA-17)
Schedule: C-9
Page 1 of 1
Witness: JCAponte
Date: May 2023

Line No.	(a)	(b)	(c)
	Description	Source	(\$000)
1	State income taxes	P-521 p. 262.1 (i), Line 13-15	5,905
2	Deferred state income taxes	Tax Department	<u>38,551</u>
3	Total state income taxes	Line 1 + Line 2	<u><u>44,456</u></u>

Schedule: C-10

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Other (or Local) Taxes

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-3 (JCA-18)

Schedule: C-10

Page 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)
Line No.	Description	Source	(\$000)
1	Other (local) income taxes	P-521 p. 262.1 (i), Line 44	(31)
2	Deferred other (local) income taxes	Tax Department	<u>1,354</u>
3	Total other (local) income taxes	Line 1 + Line 2	<u>1,323</u>

Schedule: C-11

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Allowance for Funds Used During Construction

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-3 (JCA-19)

Schedule: C-11

Page 1 of 1

Witness: JCAponte

Date: May 2023

(a)	(b)	(c)
Line		
No.	Description	Source
	(\$000)	
1	Total allowance for funds used during construction	WP-JCA-11
		<u>5,520</u>

Schedule: C-12**MICHIGAN PUBLIC SERVICE COMMISSION**Consumers Energy Company

Pro Forma Interest Adjustment

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-3 (JCA-20)

Schedule: C-12

Page 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)
Line No.	Description	Source	Total (\$000)
1	Rate base	Exhibit No.: A-2 (JCA-3)	12,647,343
2	Weighted cost of debt {1}	Exhibit No.: A-4 (JCA-22)	0.0145
3	Allowable interest expense	Line 1 * Line 2	183,636
4	Actual interest expense {2}	WP-JCA-11, WP-JCA-15	200,816
5	Increase/ (decrease) in allowable interest deduction	Line 3 - Line 4	(17,181)
6	Impact on taxable income	Line 5 * -1	17,181
7	Impact on local income taxes	Line 6 * 0.16% Other (Local) Income Tax Rate	27
8	Impact on state income taxes	Line 6 * 5.31% State Income Tax Rate	912
9	Impact on federal taxable income	Line 6 - Line 7 - Line 8	16,241
10	Impact on federal income tax	Line 9 * 21.00% Federal Income Tax Rate	3,411
11	Total impact on net operating income	(Line 7 + Line 8 + Line 10) * -1	(4,350)

Notes

{1} Excludes JDITC

{2} Excludes PPA lease interest expense

Schedule: C-13**MICHIGAN PUBLIC SERVICE COMMISSION**

Consumers Energy Company

Interest Synchronization Adjustment

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-3 (JCA-21)

Schedule: C-13

Page 1 of 1

Witness: JCAponte

Date: May 2023

(a)		(b)	(c)
Line No.	Description	Source	Total (\$000)
1	Rate base	Exhibit No.: A-2 (JCA-3)	12,647,343
2	JDITC debt-related portion of the capital structure	Exhibit No.: A-4 (JCA-22)	<u>0.0026</u>
3	Portion of rate base funded by JDITC	Line 1 * Line 2	32,351
4	Cost of JDITC - debt portion	Exhibit No.: A-4 (JCA-22)	<u>0.0368</u>
5	Allowable JDITC interest expense	Line 3 * Line 4	1,190
6	Impact on taxable income	Line 5 * -1	(1,190)
7	Impact on local income taxes	Line 6 * 0.16% Other (Local) Income Tax Rate	(2)
8	Impact on state income taxes	Line 6 * 5.31% State Income Tax Rate	(63)
9	Impact on federal taxable income	Line 6 - Line 7 - Line 8	<u>(1,125)</u>
10	Impact on federal income tax	Line 9 * 21.00% Federal Income Tax Rate	(236)
11	Total impact on net operating income	(Line 7 + Line 8 + Line 10) * -1	<u><u>301</u></u>

Schedule: D-1

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Overall Rate of Return Summary
For the Historical Year Ended December 31, 2022

Case No.: U-21389
Exhibit No.: A-4 (JCA-22)
Schedule: D-1
Page 1 of 1
Witness: JCAponte
Date: May 2023

(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
		Capital Structure				Weighted Cost			
Line		Amount	Percent	Percent	Cost Rate	Permanent	Total Cost	Conversion	Pre-Tax
No	Description	(\$000) {1}	Permanent Capital	of Total Capital	%	Capital	%	Factor	Return
1	Long term debt	8,650,491	46.59%	38.08%	3.68% {2}	1.71%	1.40%		1.40%
2	Preferred stock	37,315	0.20%	0.16%	4.50% {3}	0.01%	0.01%	1.3391	0.01%
3	Common equity	9,880,631	53.21%	43.50%	9.90% {4}	5.27%	4.31%	1.3391	5.77%
4	Permanent capital	18,568,436							
5	Notes payable	118,110		0.52%	7.92% {5}		0.04%		0.04%
6	Advanced renewable reg liability	28,988		0.13%	7.92% {5}		0.01%		0.01%
7	Short term debt	147,098							
8	Deferred federal income tax	3,874,254		17.06%	0.00%		0.00%		0.00%
9	Deferred JDITC - long term debt	58,102		0.26%	3.68%		0.01%		0.01%
10	Deferred JDITC - preferred stock	251		0.00%	4.50%		0.00%	1.3391	0.00%
11	Deferred JDITC - common equity	66,364		0.29%	9.90%		0.03%	1.3391	0.04%
12	Total deferred JDITC {6}	124,716							
13	Total	22,714,505					5.80%		7.28%

Notes

- {1} Amount reflects the 13-month average balance as of December 31, 2022
{2} Exhibit No.: A-4 (JCA-23)
{3} Exhibit No.: A-4 (JCA-25)
{4} Exhibit No.: A-4 (JCA-26)
{5} Exhibit No.: A-4 (JCA-24)
{6} The total deferred JDITC balance on line 12 is allocated to lines 9 through 11 based on the percent of permanent capital in column (c)

Schedule: D-2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Cost of Long Term Debt (Excluding Securitization)
For the Historical Year Ended December 31, 2022

Case No.: U-21389
Exhibit No.: A-4 (JCA-23)
Schedule: D-2
Page 1 of 1
Witness: JCAponte
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
					Amount		Underwriting	Net	Cost Based		
		Original	Stated	Interest	of	Price to	& Financing	Proceeds	on Net	Amount	Annual
Line		Issue	Maturity	Rate	Offering	Public	Expense	to the	Proceeds	Outstanding	Cost
No	Description	Date	Date	%	(\$000)	%	%	Company	%	(\$000) (1)	(\$000)
First Mortgage Bonds											
1	FMB 5.80% - \$175m - Due 09/15/2035	2005-08-11	2035-09-15	5.800%	175,000	99.81%	0.875%	98.76%	5.888%	175,000	10,304
2	FMB 6.17% - \$50m - Due 09/01/2040	2010-09-01	2040-09-01	6.170%	50,000	100.00%	0.906%	99.02%	6.243%	50,000	3,121
3	FMB 4.97% - \$50m - Due 10/15/2040	2010-10-15	2040-10-15	4.970%	50,000	100.00%	0.500%	99.46%	5.005%	50,000	2,502
4	FMB 3.19% - \$52m - Due 12/15/2024	2012-12-17	2024-12-15	3.190%	51,500	100.00%	0.501%	99.46%	3.244%	51,500	1,671
5	FMB 3.39% - \$36m - Due 12/15/2027	2012-12-17	2027-12-15	3.390%	35,500	100.00%	0.501%	99.46%	3.436%	35,500	1,220
6	FMB 4.31% - \$263m - Due 12/15/2042	2012-12-17	2042-12-15	4.310%	263,000	100.00%	0.501%	99.47%	4.342%	263,000	11,418
7	FMB 3.95% - \$425m - Due 05/15/2043	2013-05-17	2043-05-15	3.950%	425,000	99.84%	0.875%	98.81%	4.019%	425,000	17,079
8	FMB 3.375% - \$325m - Due 08/15/2023	2013-08-09	2023-08-15	3.375%	325,000	99.95%	0.650%	99.18%	3.472%	325,000	11,284
9	FMB 3.125% - \$250m - Due 08/31/2024	2014-08-18	2024-08-31	3.125%	250,000	99.90%	0.650%	99.11%	3.230%	250,000	8,074
10	FMB 4.35% - \$250m - Due 08/31/2064	2014-08-18	2064-08-31	4.350%	250,000	99.14%	0.875%	98.14%	4.443%	250,000	11,107
11	FMB 4.10% - \$250m - Due 11/15/2045	2015-11-02	2045-11-15	4.100%	250,000	99.91%	0.875%	98.86%	4.167%	250,000	10,417
12	FMB 3.25% - \$450m - Due 08/15/2046	2016-08-10	2046-08-15	3.250%	450,000	99.22%	0.875%	98.16%	3.347%	450,000	15,063
13	FMB 3.95% - \$350m - Due 07/15/2047	2017-02-15	2047-07-15	3.950%	350,000	99.58%	0.875%	98.52%	4.035%	350,000	14,122
14	FMB 3.18% - \$40m - Due 09/28/2032 (Private Placement)	2017-09-28	2032-09-28	3.180%	40,000	100.00%	0.301%	99.67%	3.208%	40,000	1,283
15	FMB 3.52% - \$125m - Due 09/28/2037 (Private Placement)	2017-09-28	2037-09-28	3.520%	125,000	100.00%	0.402%	99.56%	3.551%	125,000	4,439
16	FMB 3.86% - \$20m - Due 09/28/2052 (Private Placement)	2017-09-28	2052-09-28	3.860%	20,000	100.00%	0.703%	99.22%	3.901%	20,000	780
17	FMB 3.18% - \$60m - Due 11/15/2032 (Private Placement)	2017-11-15	2032-11-15	3.180%	60,000	100.00%	0.301%	99.67%	3.208%	60,000	1,925
18	FMB 3.6% - \$350m - Due 8/15/2032	2022-08-11	2032-08-15	3.600%	350,000	99.93%	0.650%	99.14%	3.703%	350,000	12,961
19	FMB 3.52% - \$210m - Due 11/15/2037 (Private Placement)	2017-11-15	2037-11-15	3.520%	210,000	100.00%	0.402%	99.56%	3.551%	210,000	7,457
20	FMB 3.86% - \$30m - Due 11/15/2052 (Private Placement)	2017-11-15	2052-11-15	3.860%	30,000	100.00%	0.703%	99.22%	3.901%	30,000	1,170
21	FMB 4.05% - \$550m - Due 05/15/2048	2018-05-14	2048-05-15	4.050%	550,000	98.97%	0.875%	97.89%	4.174%	550,000	22,955
22	FMB 3.68% - \$100m - Due 10/01/2027 (Private Placement)	2018-10-01	2027-10-01	3.680%	100,000	100.00%	0.145%	99.84%	3.701%	100,000	3,701
23	FMB 4.01% - \$215m - Due 10/01/2038 (Private Placement)	2018-10-01	2038-10-01	4.010%	215,000	100.00%	0.323%	99.65%	4.036%	215,000	8,677
24	FMB 4.28% - \$185m - Due 10/01/2057 (Private Placement)	2018-10-01	2057-10-01	4.280%	185,000	100.00%	0.629%	99.32%	4.316%	185,000	7,985
25	FMB 3.80% - \$300m - Due 11/15/2028	2018-11-13	2028-11-15	3.800%	300,000	99.75%	0.650%	99.00%	3.922%	300,000	11,765
26	FMB 4.35% - \$550m - Due 04/15/2049	2018-11-13	2049-04-15	4.350%	550,000	99.62%	0.875%	98.56%	4.436%	550,000	24,400
27	FMB 3.75% - \$300m - Due 02/15/2050	2019-05-28	2050-02-15	3.750%	300,000	98.83%	0.875%	97.75%	3.876%	300,000	11,627
28	FMB 3.10% - \$550m - Due 08/15/2050	2019-09-03	2050-08-15	3.100%	550,000	99.29%	0.875%	98.32%	3.186%	550,000	17,521
29	FMB variable (a) - \$76m - Due 09/15/2069	2019-09-19	2069-09-15	2.993%	75,649	100.00%	1.000%	98.01%	3.071%	75,649	2,323
30	FMB 3.50% - \$75m - Due 08/01/2051	2020-03-17	2051-08-01	3.500%	575,000	99.61%	0.875%	98.51%	3.579%	575,000	20,580
31	FMB 2.50% - \$525m - Due 05/01/2060	2020-05-13	2060-05-01	2.500%	525,000	99.15%	0.875%	98.10%	2.576%	525,000	13,526
32	FMB variable (b) - \$134m - Due 05/20/2070	2020-05-20	2070-05-20	2.684%	134,349	100.00%	1.000%	98.71%	2.731%	134,349	3,670
33	FMB variable (c) - \$126m - Due 10/07/2070	2020-10-07	2070-10-07	2.048%	126,497	100.00%	1.000%	98.71%	2.090%	126,497	2,644
34	FMB 0.35% - \$300m - Due 06/01/2023	2020-12-14	2023-06-01	0.350%	300,000	99.97%	0.300%	99.49%	0.559%	300,000	1,678
35	FMB 2.65% - \$300m - Due 8/15/2052	2021-08-12	2052-08-15	2.650%	300,000	99.92%	0.875%	98.85%	2.705%	300,000	8,116
36	FMB 4.2% - \$450m - Due 9/1/2052	2022-08-11	2052-09-01	4.200%	450,000	99.93%	0.875%	98.79%	4.272%	450,000	19,222
37	Total First Mortgage Bonds				8,996,495					8,996,495	327,789
Other Long-Term Debt											
38	MSF-VRBs	2021-10-07	2035-04-01	0.88%	35,000	100.00%	0.000%	98.62%	0.928%	35,000	325
39	2019 MSF VRBs	2019-10-01	2049-10-01	1.80%	75,000	100.00%	0.450%	98.45%	1.868%	75,000	1,401
40	Total other long term debt				110,000					110,000	1,726
41	Total long term debt, gross									9,106,495	329,515
42	Amortization of loss on reacquired debt										4,420
43	Total unamortized (premiums)/discounts									(27,194)	
44	Total long term debt before unamortized debt costs								3.68%	9,079,301	333,935
45	Unamortized debt costs									(69,468)	
46	Total long term debt									9,009,833	

Notes

{1} Amount reflects the outstanding balance for the 12-months ended December 31, 2022

Schedule: D-3**MICHIGAN PUBLIC SERVICE COMMISSION**Consumers Energy Company

Cost of Short Term Debt

For the Historical Year Ended December 31, 2022

Case No.: U-21389

Exhibit No.: A-4 (JCA-24)

Schedule: D-3

Page 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)	(d)	(e)
					Weighted Average Cost of Short Term Debt
Line No	Description	Amount (\$000)	Percent Total Capital {1}	Cost Rate {2}	(\$000)
1	Notes payable	118,110	0.52%		
2	Advanced renewable reg liability	<u>28,988</u>	0.13%		
3	Cost of short term debt	147,098		7.92%	<u><u>11,645</u></u>

Notes

{1} Exhibit No.: A-4 (JCA-22)

{2} Treasury Department

Schedule: D-4

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Cost of Preferred Stock
 For the Historical Year Ended December 31, 2022

Case No.: U-21389
 Exhibit No.: A-4 (JCA-25)
 Schedule: D-4
 Page 1 of 1
 Witness: JCAponte
 Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
								Total Value of Outstanding Proceeds	Cost Rate	Annual Dollar Amount
Line No	Description	Annual Dividend (\$000)	Par Value (\$000)	Discount or Premium (\$000)	Finance Expense (\$000)	Net Proceeds Received (\$000)	Number of Shares Outstanding	(\$000)	%	(\$000)
1	Preferred stock	4.50	100	-	-	100	373,148	37,315	4.50%	<u>1,679</u>

Schedule: D-5

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Cost of Common Equity
For the Historical Year Ended December 31, 2022

Case No.: U-21389
Exhibit No.: A-4 (JCA-26)
Schedule: D-5
Page 1 of 1
Witness: JCAponte
Date: May 2023

(a)

Line No	Description
1	The Company's authorized cost of common equity is 9.90%. This rate was authorized by the Michigan Public Service Commission in Electric Rate Case U-20963 issued on 12/22/2021.

Schedule: A-1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Revenue Deficiency (Sufficiency)

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-11 (JCA-27)

Schedule: A-1

Page 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)	(d)
Line No.	Description	Source	Total (\$000)	Jurisdictional (\$000)
1	Rate base	Exhibit No.: A-12 (JCA-30)	14,404,493	14,354,217
2	Adjusted net operating income	Exhibit No.: A-13 (JCA-37)	717,340	722,563
3	Overall rate of return	Line 2 / Line 1	4.98%	5.03%
4	Required rate of return	Exhibit No.: A-14 (MRB-1), Schedule D-1	6.11%	6.11%
5	Income requirement	Line 1 * Line 4	880,283	877,211
6	Income deficiency (sufficiency)	Line 5 - Line 2	162,943	154,648
7	Revenue conversion factor	Exhibit No.: A-13 (JCA-38)	1.3391	1.3391
8	Revenue deficiency (sufficiency)	Line 6 * Line 7	218,193	207,084

Schedule: A-2

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Financial Metrics - Ratemaking Basis

For the Projected 12-Month Period Ending February 28, 2025

ELECTRIC RESULTS ONLY

(\$000)

Case No.: U-21389

Exhibit No.: A-11 (JCA-28)

Schedule: A-2

Page 1 of 3

Witness: JCAponte

Date: May 2023

Line No.	(a) Description	(b)	(c)
		No Rate Relief	Full Rate Relief
1	Operating revenues	5,076,926	5,284,010
2	Operating expenses	4,283,833	4,283,833
3	Pre-tax operating income	793,093	1,000,177
4	Income taxes	96,173	148,609
5	Net operating income	696,920	851,568
6	Tax effect of pro forma interest on NOI	Included in Line 4	Included in Line 4
7	AFUDC	25,643	25,643
8	Interest charges, net of tax benefit	(246,358)	(246,358)
9	Preferred stock dividends	(889)	(889)
10	Net income available to common stockholders and JDITC	475,316	629,964
11	Return assignable to JDITC	(3,447)	(3,447)
12	Net income available to common stockholders	471,869	626,517
13	Average common equity	6,112,363	6,112,363
14	Earned rate of return on common equity	7.72%	10.25%
15	Authorized rate of return on common equity {1}	10.25%	10.25%

Schedule: A-2

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Financial Metrics - Ratemaking Basis

For the Projected 12-Month Period Ending February 28, 2025

ELECTRIC RESULTS ONLY

(\$000)

Case No.: U-21389

Exhibit No.: A-11 (JCA-28)

Schedule: A-2

Page 2 of 3

Witness: JCAponte

Date: May 2023

Line No	(a) Description	(b) No Rate Relief	(c) Full Rate Relief
	<u>EBIT Interest Coverage Ratio</u>		
16	Pre-tax operating income	793,093	1,000,177
17	AFUDC	25,643	25,643
18	Total EBIT	818,736	1,025,820
19	Interest charges, net of tax benefit	246,358	246,358
20	EBIT interest coverage ratio	3.32	4.16
	<u>EBITDA Interest Coverage Ratio</u>		
21	Total EBIT	818,736	1,025,820
22	Depreciation and amortization	730,079	730,079
23	EBITDA	1,548,814	1,755,898
24	Interest charges, net of tax benefit	246,358	246,358
25	EBITDA interest coverage ratio	6.29	7.13
	<u>FFO Interest Coverage Ratio {2}</u>		
26	Net operating income	696,920	851,568
27	Depreciation and amortization	730,079	730,079
28	Deferred income taxes	166,168	166,168
29	AFUDC	25,643	25,643
30	Other major recurring non-cash items	-	-
31	Interest paid	202,104	202,104
	Less:		
32	Operating lease adjustment to depreciation	-	-
33	Subtotal	1,820,913	1,975,561
34	Interest charges, net of tax benefit	246,358	246,358
35	FFO interest coverage ratio	7.39	8.02
	<u>Overall Fixed Charge Coverage Ratio</u>		
36	Net income available to common stockholders	471,869	626,517
37	Interest charges, net of tax benefit	246,358	246,358
38	Subtotal numerator	718,227	872,875
39	Interest charges, net of tax benefit	246,358	246,358
40	Preferred stock dividends	889	889
41	Subtotal denominator	247,247	247,247
42	Overall fixed charge coverage ratio	2.90	3.53

Schedule: A-2

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Financial Metrics - Ratemaking Basis

For the Projected 12-Month Period Ending February 28, 2025

ELECTRIC RESULTS ONLY

(\$000)

Case No.: U-21389

Exhibit No.: A-11 (JCA-28)

Schedule: A-2

Page 3 of 3

Witness: JCAponte

Date: May 2023

Line No	(a) Description	(b) No Rate Relief	(c) Full Rate Relief
<u>Cash Flow Coverage of Dividend Ratio</u>			
43	Net income available to common stockholders	471,869	626,517
44	Depreciation and amortization	730,079	730,079
45	Deferred taxes	166,168	166,168
46	Subtotal	1,368,116	1,522,763
47	Common dividends {3}	377,496	501,214
48	Cash flow coverage of dividends ratio	3.62	3.04
<u>Common Dividend Payout Ratio</u>			
49	Common dividends {3}	377,496	501,214
50	Net income available to common stockholders	471,869	626,517
51	Common dividend payout ratio	80%	80%
<u>Permanent Capitalization</u>			
52	Long term debt, excluding securitization	10,833	10,833
53	Preferred stock	37	37
54	Common equity	11,543	11,543
55	Permanent capitalization	22,413	22,413
56	Long term debt, excluding securitization	48.33%	48.33%
57	Preferred stock	0.17%	0.17%
58	Common equity	51.50%	51.50%
59	Permanent capitalization	100.00%	100.00%

Notes

- {1} The company is requesting an ROE of 10.25% in this case
- {2} FFO = Funds Flow from Operations
- {3} Consumers Energy pays dividends to the parent as a total company. This number represents the company's 80% dividend policy applied to Net Income Available to Common Stockholders

Schedule: A-3

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of the Historical and Projected Revenue Requirement
For the Projected 12-Month Period Ending February 28, 2025
(\$000)

Case No.: U-21389
Exhibit No.: A-11 (JCA-29)
Schedule: A-3
Page 1 of 1
Witness: JCAponte
Date: May 2023

(a)		(b)	(c)	(d)	(e)	(f)
Line No.	Description	Source		Jurisdictional		
		Historical Year	Projected Test Year	Historical Year	Change	Test Year
1	Net utility plant	Exhibit No.: A-2 (JCA-3)	Exhibit No.: A-12 (JCA-30)	11,232,261	96,095	11,328,356
2	Working capital	Exhibit No.: A-2 (JCA-3)	Exhibit No.: A-12 (JCA-30)	1,367,175	1,658,686	3,025,861
3	Rate base	Line 1 + Line 2	Line 1 + Line 2	12,599,436	1,754,781	14,354,217
4	Rate of return	Exhibit No.: A-4 (JCA-22)	Exhibit No.: A-14 (MRB-1), Schedule D-	5.80%		6.11%
5	Income requirement	Line 3 * Line 4	Line 3 * Line 4	731,292	145,918	877,211
6	Adjusted net operating income	Exhibit No.: A-3 (JCA-9)	Exhibit No.: A-13 (JCA-37)	657,238	65,325	722,563
7	Income deficiency (sufficiency)	Line 5 - Line 6	Line 5 - Line 6	74,054	80,594	154,648
8	Revenue conversion factor	Exhibit No.: A-3 (JCA-10)	Exhibit No.: A-13 (JCA-38)	1.3391		1.3391
9	Revenue deficiency (sufficiency)	Line 7 * Line 8	Line 7 * Line 8	99,164	107,920	207,084

Schedule: B-1**MICHIGAN PUBLIC SERVICE COMMISSION**Consumers Energy Company

Rate Base

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-12 (JCA-30)

Schedule: B-1

Page 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)	(d)
Line No.	Description	Source	Total (\$000)	Jurisdictional (\$000)
1	Total utility plant	Exhibit No.: A-12 (JCA-32)	17,265,169	17,217,600
2	Depreciation reserve	Exhibit No.: A-12 (JCA-33)	5,848,183	5,831,886
3	Net utility plant	Line 1 - Line 2	11,416,986	11,385,714
4	Customer advances for construction	Exhibit No.: A-2 (JCA-7)	(57,395)	(57,358)
5	Utility plant rate base	Sum Lines 3 - 4	11,359,591	11,328,356
6	Working capital	Exhibit No.: A-12 (JCA-34)	3,044,903	3,025,861
7	Total rate base	Line 5 + Line 6	14,404,493	14,354,217

Schedule: B-1a

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Development of Rate Base

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-12 (JCA-31)

Schedule: B-1a

Page 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)	(d)
Line No.	Description	Source	Total (\$000)	Jurisdictional (\$000)
1	Net utility plant	Exhibit No.: A-2 (JCA-3)	11,274,022	11,232,261
2	Working capital	Exhibit No.: A-2 (JCA-3)	1,373,322	1,367,175
3	Total rate base - historical year	Line 1 + Line 2	12,647,343	12,599,436
<u>Change in Rate Base</u>				
4	Change in net utility plant {1}	Line 10	85,569	96,095
5	Change in working capital	Line 13	1,671,581	1,658,686
6	Net change in rate base	Sum Lines 4 - 5	1,757,150	1,754,781
7	Total rate base - projected year	Line 3 + Line 6	14,404,493	14,354,217
<u>Change in Net Utility Plant</u>				
8	Net utility plant - projected year	Exhibit No.: A-12 (JCA-30)	11,359,591	11,328,356
9	Net utility plant - historical year	Exhibit No.: A-2 (JCA-3)	11,274,022	11,232,261
10	Change in net utility plant	Line 8 - Line 9	85,569	96,095
<u>Change in Working Capital</u>				
11	Working capital - projected year	Exhibit No.: A-12 (JCA-34)	3,044,903	3,025,861
12	Working capital - historical year	Exhibit No.: A-2 (JCA-6)	1,373,322	1,367,175
13	Change in working capital	Sum Lines 11 - 12	1,671,581	1,658,686
<u>Change in Jurisdictional Factors</u>				
14	Net utility plant - projected year - jurisdictional	Exhibit No.: A-12 (JCA-30)	11,328,356	
15	Net utility plant - projected year - total electric	Exhibit No.: A-12 (JCA-30)	11,359,591	
16	Jurisdictional factor	Line 14 / Line 15	0.9973	
17	Net utility plant - historical year - jurisdictional	Exhibit No.: A-2 (JCA-3)	11,232,261	
18	Net utility plant - historical year - jurisdictional using new factor	Exhibit No.: A-2 (JCA-3), Line 5 * Line 16	11,243,022	
19	Change in net utility plant due to change in factor	Line 18 - Line 17	10,761	
20	Working capital - projected year - jurisdictional	Exhibit No.: A-12 (JCA-30)	3,025,861	
21	Working capital - projected year - total electric	Exhibit No.: A-12 (JCA-30)	3,044,903	
22	Jurisdictional factor	Line 20 / Line 21	0.9937	
23	Working capital - historical year - jurisdictional	Exhibit No.: A-2 (JCA-3)	1,367,175	
24	Working capital - historical year - jurisdictional using new factor	Exhibit No.: A-2 (JCA-3), Line 6 * Line 22	1,364,733	
25	Change in working capital due to change in factor	Line 24 - Line 23	(2,441)	

Notes

{1} Includes impact of change in utility plant jurisdictional factors calculated on line 19

Schedule: B-2

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Total Utility Plant

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-12 (JCA-32)

Schedule: B-2

Page 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)	(d)	(e)
Line No.	Description	MPSC Account Number	Source	Total (\$000)	Jurisdictional (\$000)
1	Plant-in-service	101	WP-JCA-23	16,521,245	16,478,839
2	Plant held of future use	105		11,234	11,190
3	Construction work in progress	107	WP-JCA-24	732,689	727,572
4	Total utility plant		Sum Lines 1 - 3	17,265,169	17,217,600

Schedule: B-3**MICHIGAN PUBLIC SERVICE COMMISSION**

Consumers Energy Company

Depreciation Reserve

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-12 (JCA-33)

Schedule: B-3

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No.	(a) Description	(b) Source	(c) (\$000)
1	Distribution	WP-JCA-25	3,625,214
2	Production (Hydro)	WP-JCA-25	311,322
3	Production (Hydro Ludington)	WP-JCA-25	237,449
4	Production (Steam)	WP-JCA-25	272,063
5	Production (Solar)	WP-JCA-25	3,672
6	Production (Other)	WP-JCA-25	428,501
7	Intangibles	WP-JCA-25	528,057
8	E-GP Structures	WP-JCA-25	43,665
9	E-GP Computers	WP-JCA-25	7,227
10	E-GP Transportation	WP-JCA-25	65,651
11	E-GP Other	WP-JCA-25	26,060
12	C-GP Structures	WP-JCA-25	81,422
13	C-GP Computers	WP-JCA-25	25,553
14	C-GP Transportation	WP-JCA-25	64,845
15	C-GP Other	WP-JCA-25	38,301
16	Zeeland Acq Adjustment	WP-JCA-25	89,181
17	Total depreciation reserve and other deductions	Sum Lines 1 - 16	5,848,183
18	Jurisdictional factor	Cost of Service Study	0.997213
19	Total depreciation reserve and other deductions - jurisdictional	Line 17 * Line 18	5,831,886

Schedule: B-4

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Working Capital

For the Projected 12-Month Period Ending February 28, 2025

(\$000)

Case No.: U-21389

Exhibit No.: A-12 (JCA-34)

Schedule: B-4

Page 1 of 1

Witness: JCAponte

Date: May 2023

(a) (b) (c) (d) (e) (f)

Line No	Description	Historical Year Working Capital	Change	December 2022 Working Capital	Test Year Adjustments	Test Year Working Capital
1	Cash, cash equivalents, and restricted cash	20,157	-	20,157	33,522	53,679
2	Accounts and notes receivable	284,586	-	284,586	-	284,586
3	Fuel stock	47,745	-	47,745	-	47,745
4	Materials and supplies	134,417	-	134,417	-	134,417
5	Allowances	14	-	14	-	14
6	Prepayments	769,539	-	769,539	1,569	771,108
7	Accrued revenue	302,353	-	302,353	-	302,353
8	Deferred taxes	228,375	-	228,375	-	228,375
9	Deferred debits	635,437	-	635,437	1,659,650	2,295,087
10	Total assets	2,422,622	-	2,422,622	1,694,742	4,117,364
11	Accounts payable	558,140	-	558,140	-	558,140
12	Customer deposits	18,319	-	18,319	-	18,319
13	Accrued interest	49,157	-	49,157	-	49,157
14	Dividends payable	38,060	-	38,060	-	38,060
15	Accrued taxes	242,540	-	242,540	28,680	271,220
16	Other liabilities	65,860	-	65,860	-	65,860
17	Deferred credits	77,226	-	77,226	(5,519)	71,706
18	Total liabilities	1,049,301	-	1,049,301	23,161	1,072,461
19	Total working capital	1,373,322	-	1,373,322	1,671,581	3,044,903
20	Jurisdictional factor					0.993747
21	Total working capital - jurisdictional					3,025,861

Notes

Column (b): Exhibit No.: A-2 (JCA-6)
Column (c): Column (d) - Column (b)
Column (d): Exhibit No.: A-12 (JCA-35) Page 1 of 1, Column (b)
Column (e): Exhibit No.: A-12 (JCA-35) Page 1 of 1, Column (j)
Column (f): Column (d) + Column (e)

Schedule: B-4a

MICHIGAN PUBLIC SERVICE COMMISSION
 Consumers Energy Company
 Development of Working Capital
 For the Projected 12-Month Period Ending February 28, 2025
 (\$000)

Case No.: U-21389
 Exhibit No.: A-12 (JCA-35)
 Schedule: B-4a
 Page 1 of 1
 Witness: JCAponte
 Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
							PowerMIDrive and PowerMIFleet	Karn & Campbell Retention	Campbell NBV, Ash Disposal, Decommissioning	Year End Customer Reserve	Cash	Total	Test Year Working
Line No	Description	December 2022 Working Capital	Pension Adjustment	OPEB Adjustment	Accrued Tax Adjustment	Prepaid Cloud Computing Adjustment	Adjustment	Adjustment	g	Adjustment	Adjustment	Adjustments	Capital
1	Cash, cash equivalents, and restricted cash	20,157	-	-	-	-	-	-	-	-	33,522	33,522	53,679
2	Accounts and notes receivable	284,586	-	-	-	-	-	-	-	-	-	-	284,586
3	Fuel stock	47,745	-	-	-	-	-	-	-	-	-	-	47,745
4	Materials and supplies	134,417	-	-	-	-	-	-	-	-	-	-	134,417
5	Allowances	14	-	-	-	-	-	-	-	-	-	-	14
6	Prepayments	769,539	-	-	-	1,569	-	-	-	-	-	1,569	771,108
7	Accrued revenue	302,353	-	-	-	-	-	-	-	-	-	-	302,353
8	Deferred taxes	228,375	-	-	-	-	-	-	-	-	-	-	228,375
9	Deferred debits	635,437	97,425	169,426	-	-	(2,430)	18,181	1,377,048	-	-	1,659,650	2,295,087
10	Total assets	2,422,622	97,425	169,426	-	1,569	(2,430)	18,181	1,377,048	-	33,522	1,694,742	4,117,364
11	Accounts payable	558,140	-	-	-	-	-	-	-	-	-	-	558,140
12	Customer deposits	18,319	-	-	-	-	-	-	-	-	-	-	18,319
13	Accrued interest	49,157	-	-	-	-	-	-	-	-	-	-	49,157
14	Dividends payable	38,060	-	-	-	-	-	-	-	-	-	-	38,060
15	Accrued taxes	242,540	-	-	28,680	-	-	-	-	-	-	28,680	271,220
16	Other liabilities	65,860	-	-	-	-	-	-	-	-	-	-	65,860
17	Deferred credits	77,226	-	-	-	-	-	-	-	(5,519)	-	(5,519)	71,706
18	Total liabilities	1,049,301	-	-	28,680	-	-	-	-	(5,519)	-	23,161	1,072,461
19	Total working capital	1,373,322	97,425	169,426	(28,680)	1,569	(2,430)	18,181	1,377,048	5,519	33,522	1,671,581	3,044,903

Notes

Column (c): WP-JCA-28
 Column (d): WP-JCA-29
 Column (e): WP-JCA-30
 Column (f): WP-JCA-31
 Column (g): WP-JCA-32
 Column (h): WP-JCA-33
 Column (i): WP-JCA-34
 Column (j): SAP Account 2460600 average balance

Schedule: B-5

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Capital Spending
 For the Projected 12-Month Period Ending February 28, 2025
 (\$000)

Case No.: U-21389
 Exhibit No.: A-12 (JCA-36)
 Schedule: B-5
 Page 1 of 1
 Witness: JCAponte
 Date: May 2023

Line No.	(a) Description	(b) Source	(c)	(d)	(e)	(f)	(g)	(h)	(i)
			Historical	Projected				Approved Capital Expenditures	Projected Capital Expenditures in
								in U-21224 (1)	U-21224
			12 Months Ended 12/31/2022	12 Months Ended 12/31/2023	2 Months Ended 2/29/2024	14 Months Ended 2/29/2024	12 Months Ended 2/28/2025	12 Months Ended 12/31/2023	12 Months Ended 12/31/2023
1	Electric Distribution	Exhibit No.: A-12 (MPK-1), Sch 5.7	801,543	690,614	123,715	814,329	770,680	-	746,451
2	Streetlighting	Exhibit No.: A-12 (DAS-1), Sch 5.10	14,901	16,335	2,782	19,117	16,705	-	19,882
3	Generation	Exhibit No.: A-12 (RTB-3), Sch 5.1	141,372	1,293,935	78,679	1,372,614	375,970	-	1,258,404
4	Information Technology	Exhibit No.: A-12 (HNW-5), Sch 5.11	36,804	34,967	6,384	41,351	38,495	-	42,924
5	Security	Exhibit No.: A-12 (ALC-1), Sch 5.3	5,656	1,664	1,031	2,695	7,038	-	4,361
6	Electric Vehicles	Exhibit No.: A-12 (JAM-1), Sch 5.9	111	4,899	613	5,511	3,064	-	4,370
7	Operations Support	Exhibit No.: A-12 (QAG-1), Sch 5.5	28,181	25,349	3,734	29,082	39,798	-	25,178
8	Fleet Services	Exhibit No.: A-12 (ASC-1), Sch 5.3	20,079	19,633	2,209	21,842	20,863	-	19,545
9	Corporate	Exhibit No.: A-12 (MJF-6), Sch 5.4	537	558	95	653	572	-	614
10	Customer Experience & Operations	Exhibit No.: A-12 (CMH-1), Sch 5.6	52	310	-	310	-	-	350
11	Demand Response	Exhibit No.: A-12 (SQM-1), Sch 5.8	10,678	10,920	1,152	12,072	7,185	-	8,771
12	Total capital expenditures		1,059,913	2,099,183	220,394	2,319,577	1,280,369	-	2,130,850

Notes

(1) Case No. U-21224 resulted in a settlement agreement that did not state the approved capital spending for all spending categorie

Schedule: C-1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Adjusted Net Operating Income

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-13 (JCA-37)

Schedule: C-1

Page 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)	(d)
Line No.	Description	Source	Total (\$000)	Jurisdictional (\$000)
1	Sales revenues	Exhibit No.: A-13 (JCA-39)	4,943,202	4,943,202
2	Wholesale revenues	Exhibit No.: A-13 (JCA-39)	24,339	-
3	Other electric revenues	Exhibit No.: A-13 (JCA-39)	134,737	133,724
4	Total operating revenues	Sum Lines 1 - 3	5,102,278	5,076,926
5	Power supply costs	Exhibit No.: A-13 (JCA-40)	2,589,397	2,564,230
6	Operations and maintenance	Exhibit No.: A-13 (JCA-41)	705,078	702,708
7	Depreciation and amortization	Exhibit No.: A-13 (JCA-43)	733,075	730,079
8	Property taxes	Exhibit No.: A-13 (JCA-44)	250,200	249,436
9	General taxes	Exhibit No.: A-13 (JCA-44)	37,563	37,381
10	Other (local) taxes	Exhibit No.: A-13 (JCA-47)	1,244	1,232
11	State income taxes	Exhibit No.: A-13 (JCA-46)	36,604	36,890
12	Federal income taxes	Exhibit No.: A-13 (JCA-45)	57,601	58,051
13	Total operating expenses	Sum Lines 5 - 12	4,410,761	4,380,006
14	Net operating income	Line 4 - Line 13	691,517	696,920
15	AFUDC	Exhibit No.: A-13 (JCA-48)	25,823	25,643
16	Net operating income, including AFUDC	Line 14 + Line 15	717,340	722,563
	<u>Net Operating Income Adjustments</u>			
17	Income tax effect of interest {a}		Included in Lines 10, 11, and 12	
18	Interest synchronization adjustment {a}		Included in Lines 10, 11, and 12	
19	Total adjusted net operating income	Line 16	717,340	722,563

Notes

{a} Income tax effect of interest and the interest synchronization adjustment are included in the calculation of local, state, and federal income tax. The separate calculations can be found on Exhibit No.: A-13 (JCA-49) and Exhibit No.: A-13 (JCA-50)

Schedule: C-2

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Calculation of the Revenue Conversion Factor

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-13 (JCA-38)

Schedule: C-2

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No.	(a) Description	(b) Source	(c) Amount
1	Income base - before taxes		100.0000
2	State income tax	Line 1 * 5.31% State Income Tax Rate	5.3100
3	Other (local) income tax	Line 1 * 0.16% Other {Local} Income Tax Rate	<u>0.1600</u>
4	Federal income tax base	Line 1 - Line 2 - Line 3	94.5300
5	Federal income tax	Line 4 * 21.00% Federal Income Tax Rate	<u>19.8513</u>
6	Income base - after taxes	Line 4 - Line 5	<u>74.6787</u>
7	Revenue conversion factor	Line 1 / Line 6	<u><u>1.3391</u></u>

Schedule: C-3

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Operating Revenues

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-13 (JCA-39)

Schedule: C-3

Page 1 of 1

Witness: JCAponte

Date: May 2023

(a)		(b)	(c)
Line No.	Description	Source	Total (\$000)
1	Sales revenues	Exhibit No.: A-15 (EMB-3), Schedule E-2	4,943,202
2	Wholesale revenues	Exhibit No.: A-15 (EMB-3), Schedule E-2	24,339
3	Other electric revenue	Exhibit No.: A-15 (EMB-3), Schedule E-2	134,737
4	Total operating revenues	Sum Lines 1 - 3	<u>5,102,278</u>

Schedule: C-4

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Cost of Fuel and Purchased Power
For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389
Exhibit No.: A-13 (JCA-40)
Schedule: C-4
Page 1 of 1
Witness: JCAponte
Date: May 2023

Line No.	(a) Description	(b) Source	(c) (\$000)
1	Total fuel and purchased power	Exhibit No.: A-51 (EMB-6)	<u>2,589,397</u>
2	Jurisdictional factor	Cost of Service Study	<u>0.9903</u>
3	Total fuel and purchased power - jurisdictional	Line 1 * Line 2	<u>2,564,230</u>

Schedule: C-5

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Other Operation and Maintenance Expenses
For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389
Exhibit No.: A-13 (JCA-41)
Schedule: C-5
Page 1 of 1
Witness: JCAponte
Date: May 2023

(a)		(b)	(c)	(d)	(e)
Line No	Description	Source	Historical	Change	Projected
			Year (\$000)	(\$000)	Year (\$000)
1	Electric Division - Electric & Common	Exhibit No.: A-129 (MPK-10)	214,437	(6,890)	207,547
2	Forestry	Exhibit No.: A-47 (PLB-6)	102,003	16,853	118,856
3	Generation	Exhibit No.: A-41 (RTB-4)	150,031	14,221	164,252
4	Operations Support	Exhibit No.: A-95 (QAG-2)	17,651	(1,465)	16,186
5	Information Technology Operations	Exhibit No.: A-177 (HNW-2)	42,043	2,112	44,155
6	Information Technology Investments	Exhibit No.: A-179 (HNW-4)	11,028	636	11,664
7	Information Technology - Security Operations	Exhibit No.: A-77 (ALC-3)	8,784	221	9,006
8	Information Technology - Security Investments	Exhibit No.: A-78 (ALC-4)	921	(185)	735
9	Customer Interactions	Exhibit No.: A-109 (CMH-2)	18,212	88	18,301
10	Billing & Payment	Exhibit No.: A-109 (CMH-2)	25,769	775	26,544
11	Demand Response	Exhibit No.: A-144 (SQM-2)	37,882	3,368	41,250
12	Pension Plans A/B	Exhibit No.: A-91 (KKG-1)	(29,860)	(13,467)	(43,327)
13	Defined Company Contribution Plan	Exhibit No.: A-91 (KKG-1)	14,105	778	14,883
14	401(k) Employees' Savings Plan	Exhibit No.: A-91 (KKG-1)	12,974	710	13,684
15	Active Health Care/Life Insurance/LTD	Exhibit No.: A-91 (KKG-1)	26,332	1,549	27,881
16	Retiree Health Care and Life Insurance	Exhibit No.: A-91 (KKG-1)	(69,030)	29,422	(39,608)
17	Other Benefits	Exhibit No.: A-91 (KKG-1)	3,279	215	3,494
18	Corporate Services	Exhibit No.: A-82 (MJF-1)	38,247	10,370	48,617
19	Uncollectible Expense	Exhibit No.: A-82 (MJF-1)	18,817	(1,298)	17,519
20	Injuries & Damages	Exhibit No.: A-82 (MJF-1)	6,652	(2,669)	3,983
21	Incentive Compensation	Exhibit No.: A-75 (AMC-3)	2,218	185	2,403
22	Job Work Expense	Exhibit No.: A-15 (EMB-3), Schedule E-2	2,200	-	2,200
23	Interest expense on security deposits	WP-JCA-22	258	-	258
24	DR Reconciliation	S.McLean Testimony	-	(5,406)	(5,406)
25	Total operation and maintenance expenses:	Sum Lines 1 - 23	654,954	50,124	705,078
26	Jurisdictional factor	Cost of Service Study			0.9966
27	Total operation and maintenance expenses - jurisdictiona	Line 25 * Line 26			702,708

Schedule: C-5a

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Inflation and Merit Increases Included in Other Operation and Maintenance Expenses

For the Projected 12-Month Period Ending February 28, 2025

(\$000)

Case No.: U-21389

Exhibit No.: A-13 (JCA-42)

Schedule: C-5a

Page 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description	Source	Projected Adjustments					Total Projected Adjustments	Projected 12 Months Ended 2/28/2025
			Historical 12 Months Ended 12/31/2022	Inflation for the 12 Months Ended 12/31/2023	Inflation for the 12 Months Ended 12/31/2024	2 Months Ended 2/28/2025	Other Adjustments		
							Σ (d) thru (g)	(c) + (h)	
1	Electric Division - Electric & Common	Exhibit No.: A-50 (MPK-15)	214,437	6,753	3,958	644	(18,244)	(6,890)	207,547
2	Forestry	Exhibit No.: A-47 (PLB-6)	102,003	3,978	2,201	412	10,262	16,853	118,856
3	Generation	Exhibit No.: A-41 (RTB-4)	150,031	5,851	3,429	558	4,383	14,221	164,252
4	Operations Support	Exhibit No.: A-95 (QAG-2)	17,651	159	97	15	(1,736)	(1,465)	16,186
5	Information Technology Operations	Exhibit No.: A-177 (HNW-2)	42,043	399	234	38	1,441	2,112	44,155
6	Information Technology Investments	Exhibit No.: A-179 (HNW-4)	11,028	-	-	-	636	636	11,664
7	Information Technology - Security Operation	Exhibit No.: A-77 (ALC-3)	8,784	105	61	10	46	221	9,006
8	Information Technology - Security Investment	Exhibit No.: A-78 (ALC-4)	921	-	-	-	(185)	(185)	735
9	Customer Interactions	Exhibit No.: A-109 (CMH-2)	18,212	710	416	68	(1,106)	88	18,301
10	Billing & Payment	Exhibit No.: A-109 (CMH-2)	25,769	1,005	589	96	(915)	775	26,544
11	Demand Response	Exhibit No.: A-144 (SQM-2)	37,882	1,477	866	141	884	3,368	41,250
12	Pension Plans A/B	Exhibit No.: A-91 (KKG-1)	(29,860)	(1,165)	(683)	(111)	(11,509)	(13,467)	(43,327)
13	Defined Company Contribution Plan	Exhibit No.: A-91 (KKG-1)	14,105	550	322	52	(147)	778	14,883
14	401(k) Employees' Savings Plan	Exhibit No.: A-91 (KKG-1)	12,974	506	297	48	(141)	710	13,684
15	Active Health Care/Life Insurance/LTD	Exhibit No.: A-91 (KKG-1)	26,332	1,027	602	98	(178)	1,549	27,881
16	Retiree Health Care and Life Insurance	Exhibit No.: A-91 (KKG-1)	(69,030)	(2,692)	(1,578)	(257)	33,949	29,422	(39,608)
17	Other Benefits	Exhibit No.: A-91 (KKG-1)	3,279	128	75	12	-	215	3,494
18	Corporate Services	Exhibit No.: A-83 (MJF-2)	38,247	1,954	1,145	186	7,085	10,370	48,617
19	Uncollectible Expense	Exhibit No.: A-85 (MJF-4)	18,817	-	-	-	(1,298)	(1,298)	17,519
20	Injuries & Damages	Exhibit No.: A-86 (MJF-5)	6,652	-	-	-	(2,669)	(2,669)	3,983
21	Incentive Compensation	Exhibit No.: A-75 (AMC-3)	2,218	87	52	8	39	185	2,403
22	Job Work Expense	Exhibit No.: A-15 (EMB-3), Schedule E-2	2,200	-	-	-	-	-	2,200
23	Interest expense on security deposits	WP-JCA-22	258	-	-	-	-	-	258
24	DR Reconciliation	S.McLean Testimony	-	-	-	-	(5,406)	(5,406)	(5,406)
25	Projected inflation of other O&M expenses	Sum Lines 1 - 24	654,954	20,832	12,084	2,018	15,190	50,124	705,078

Schedule: C-6

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Depreciation and Amortization Expenses

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-13 (JCA-43)

Schedule: C-6

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No.	(a) Description	(b) Source	(c) (\$000)
1	Distribution	WP-JCA-26	353,060
2	Production (Hydro)	WP-JCA-26	47,273
3	Production (Hydro Ludington)	WP-JCA-26	17,604
4	Production (Steam)	WP-JCA-26	16,835
5	Production (Solar)	WP-JCA-26	5,767
6	Production (Other)	WP-JCA-26	48,631
7	Intangibles	WP-JCA-26	59,918
8	E-GP Structures	WP-JCA-26	4,433
9	E-GP Computers	WP-JCA-26	3,048
10	E-GP Transportation	WP-JCA-26	-
11	E-GP Other	WP-JCA-26	4,147
12	C-GP Structures	WP-JCA-26	2,824
13	C-GP Computers	WP-JCA-26	9,273
14	C-GP Transportation	WP-JCA-26	-
15	C-GP Other	WP-JCA-26	5,772
16	Zeeland Acq Adjustment	WP-JCA-26	5,455
17	PowerMiDrive	Exhibit No.: A-17 (JCA-52)	2,097
18	PowerMiFleet	Exhibit No.: A-18 (JCA-53)	559
19	Karn Plant Retention and Separation Costs	Exhibit No.: A-19 (JCA-54)	4,146
20	Campbell Plant Retention and Separation Costs	Exhibit No.: A-20 (JCA-55)	14,074
21	Campbell Net Book Value Amortization	Exhibits No.: A-21 (JCA-56) & A-22 (JCA-57)	125,469
22	Karn 1-2 Decommissioning Costs Amortization	Exhibit No.: A-23 (JCA-58)	1,344
23	Karn 1-2 and Classic 7 Ash Disposal Amortization	Exhibit No.: A-24 (JCA-59)	1,182
24	Campbell Decommissioning Costs Amortization	Exhibit No.: A-25 (JCA-60)	166
25	Total depreciation and amortization expenses	Sum Lines 1 - 24	<u>733,075</u>
26	Jurisdictional factor	Cost of Service Study	<u>0.9959</u>
27	Total depreciation and amortization expenses - jurisdictional	Line 25 * Line 26	<u>730,079</u>

Schedule: C-7

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

General Taxes

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-13 (JCA-44)

Schedule: C-7

Page 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)	(d)
Line No.	Description	Source	Total (\$000)	Jurisdictional (\$000)
1	Real and personal property taxes	Exhibit No.: A-161 (BJV-1)	250,200	249,436
	<u>Projected payroll taxes</u>			
2	Total payroll taxes - historical year	Exhibit No.: A-3 (JCA-15)	26,117	
3	Inflation factor 2023	WP-JCA-50	103.9%	
4	Inflation factor 2024	WP-JCA-50	102.2%	
5	Inflation factor 2025	WP-JCA-50	102.1%	
6	Total payroll taxes - projected year	Line 2 * Line 3 * Line 4 * Line 5	28,315	28,221
	<u>Projected other general taxes</u>			
7	State MPSC assessment - historical year	Exhibit No.: A-3 (JCA-15)	9,723	
8	Total operating revenues - historical year	Exhibit No.: A-3 (JCA-11)	5,364,237	
9	Average State MPSC assessment rate	Line 7 / Line 8	0.18%	
10	Total operating revenues - projected year	Exhibit No.: A-13 (JCA-39)	5,102,278	
11	Average State MPSC assessment rate	Line 9	0.18%	
12	Projected State MPSC assessment	Line 10 * Line 11	9,249	
13	Incremental change in State MPSC assessment	Line 12 - Line 7	(475)	
14	Other general taxes - historical year	Exhibit No.: A-3 (JCA-15)	9,723	
15	Total other general taxes - projected year	Line 13 + Line 14	9,249	9,160
16	Total general taxes	Line 1 + Line 6 + Line 15	287,763	286,817

Schedule: C-8

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Federal Income Taxes

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-13 (JCA-45)

Schedule: C-8

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No	(a) Description	(b) Source	(c) (\$000)
1	Operating revenues	Exhibit No.: A-13 (JCA-39)	5,102,278
2	Power supply costs	Exhibit No.: A-13 (JCA-40)	(2,589,397)
3	Other O&M expenses	Exhibit No.: A-13 (JCA-41)	(705,078)
4	Depreciation and amortization expenses	Exhibit No.: A-13 (JCA-43)	(733,075)
5	General taxes	Exhibit No.: A-13 (JCA-44)	(287,763)
6	State income taxes	Exhibit No.: A-13 (JCA-46), Line 22 * -1	(36,604)
7	Other (local) income taxes	Exhibit No.: A-13 (JCA-47), Line 20 * -1	(1,244)
8	Net operating income	Sum Lines 1 - 7	749,118
9	Allowable interest expense	Exhibit: Exhibit No.: A-13 (JCA-49)	(245,986)
10	Allowable JDITC interest expense	Exhibit: Exhibit No.: A-13 (JCA-50)	(1,235)
11	Pre-tax operating income	Sum Lines 8 - 10	501,897
<u>Permanent Differences</u>			
12	Meals and entertainment	Tax Department	448
13	Non-deductible parking	Tax Department	102
14	Total permanent differences	Sum Lines 12 - 13	550
<u>Temporary Differences</u>			
15	Book depreciation	Exhibit No.: A-13 (JCA-43)	733,075
16	Tax depreciation	Tax Department	(770,630)
17	263A - mixed service costs deduction	Tax Department	(25,213)
18	Tax repairs	Tax Department	(217,000)
19	Contributions in aid of construction	Tax Department	15,376
20	Cost of removal	Tax Department	(186,343)
21	Gain/loss on ACRS/MACRS dispositions	Tax Department	(68,129)
22	Software expense	Tax Department	(15,380)
23	Tax interest during construction	Tax Department	30,372
24	Bad debt allowance	Tax Department	-
25	OPEB - book expense	Exhibit No.: A-13 (JCA-41), Line 16	(39,608)
26	OPEB - payments	Tax Department	-
27	Other revenue reserves	Tax Department	1,990
28	Pension book expense	Exhibit No.: A-13 (JCA-41), Line 12	(43,327)
29	Pension - payments	Tax Department	-
30	Securitization Amortization	Tax Department	99,318
31	Premium, discount & debt expense amortization	Tax Department	2,554
32	Real and personal property taxes	Tax Department	(34,838)
33	Renewable energy reserve	Tax Department	36,888
34	Deferred state income taxes	Tax Department	37,907
35	Deferred city income taxes	Tax Department	1,147
36	Total temporary differences	Sum Lines 15 - 33	(441,842)
37	Federal taxable income	Line 11 + Line 14 + Line 36	60,605
<u>Calculation of Current Federal Income Tax Expenses</u>			
38	Current federal income taxes	Line 37 * 21.00% Federal Income Tax Rate	12,727
39	Production tax credits	Tax Department	(6,548)
40	Research & development credit	Tax Department	(2,018)
41	Total current federal income tax expenses	Line 38 + Line 40	4,161
<u>Calculation of Deferred Federal Income Tax Expenses</u>			
42	Deferred federal income taxes	(Line 36 * 21.00% Federal Income Tax Rate) * -1	92,787
43	FAS 109	Tax Department	(119)
44	Investment tax credit amortization	Tax Department	(3,699)
45	TCJA amortization - ARAM	Tax Department	(27,538)
46	TCJA amortization - non-ARAM	Tax Department	(4,148)
47	TCJA Tracker	Tax Department	(6,492)
48	Repealed Medicare subsidy benefit	Tax Department	2,649
49	Total deferred federal income tax expenses	Sum Lines 42 - 48	53,440
50	Total federal income taxes	Line 41 + Line 49	57,601
51	Jurisdictional factor	Cost of Service Study	1,0078
52	Total federal income taxes - jurisdictional	Line 50 * Line 51	58,051

Schedule: C-9

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

State Income Taxes

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-13 (JCA-46)

Schedule: C-9

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No	(a) Description	(b) Source	(c) (\$000)
1	Operating revenues	Exhibit No.: A-13 (JCA-39)	5,102,278
2	Power supply costs	Exhibit No.: A-13 (JCA-40)	(2,589,397)
3	Other O&M expenses	Exhibit No.: A-13 (JCA-41)	(705,078)
4	Depreciation and amortization expenses	Exhibit No.: A-13 (JCA-43)	(733,075)
5	General taxes	Exhibit No.: A-13 (JCA-44)	(287,763)
6	Net operating income	Sum Lines 1 - 5	786,965
7	Interest expense	Exhibit No.: A-13 (JCA-49)	(245,986)
8	Interest synchronization	Exhibit No.: A-13 (JCA-50)	(1,235)
9	Pre-tax operating income	Sum Lines 6 - 8	539,744
10	Total permanent differences	Exhibit No.: A-13 (JCA-45)	550
11	Schedule M temp diff, excluding DSIT and DCIT	Exhibit No.: A-13 (JCA-45)	(480,895)
12	Depreciation/gain-loss adjustment	Tax Department	(83,945)
13	State taxable income	Sum Lines 9 - 12	(24,546)
14	Current state income taxes	Line 13 * 5.31% State Income Tax Rate	(1,303)
<u>Calculation of Deferred State Income Tax Expense</u>			
15	Deferred state income tax expenses, tax effected	((Lines 11 + 12) * 5.31% State Income Tax Rate) * -1	29,993
16	FAS 109 - AFUDC-equity flow-through reversal	Tax Department	205
17	Deferred - amortization of Medicare Subsidy benefit - state	Tax Department	454
18	Deferred - amortization of state income tax transition reg asset	Tax Department	7,255
19	Deferred state income taxes	Sum Lines 15 - 18	37,907
20	Total state income taxes	Line 14 + Line 19	36,604
21	Jurisdictional factor	Cost of Service Study	1.0078
22	Total state income taxes - jurisdictional	Line 20 * Line 21	36,890

Schedule: C-10

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Other (or Local) Taxes

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-13 (JCA-47)

Schedule: C-10

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No.	(a) Description	(b) Source	(c) (\$000)
1	Operating revenues	Exhibit No.: A-13 (JCA-39)	5,102,278
2	Power supply costs	Exhibit No.: A-13 (JCA-40)	(2,589,397)
3	Other O&M expenses	Exhibit No.: A-13 (JCA-41)	(705,078)
4	Depreciation and amortization expenses	Exhibit No.: A-13 (JCA-43)	(733,075)
5	General taxes	Exhibit No.: A-13 (JCA-44)	(287,763)
6	State income taxes	Exhibit No.: A-13 (JCA-46)	(36,604)
7	Net operating income	Sum Lines 1 - 6	750,362
8	Interest expense	Exhibit: Exhibit No.: A-13 (JCA-49)	(245,986)
9	Interest synchronization	Exhibit: Exhibit No.: A-13 (JCA-50)	(1,235)
10	Pre-tax operating income	Sum Lines 7 - 9	503,141
11	Total permanent differences	Exhibit No.: A-13 (JCA-45)	550
12	Schedule M temp diff, excluding DCIT	Exhibit No.: A-13 (JCA-45)	(442,988)
13	Other (local) taxable income	Sum Lines 10 - 12	60,702
14	Current other (local) income taxes	Line 13 * 0.16% Other (Local) Income Tax Rate	97
<u>Calculation of Deferred Other (Local) Income Tax Expenses</u>			
15	Deferred other (local) income tax expenses, tax effected	(Line 12 * 0.16% Other (Local) Income Tax Rate) * -1	709
16	Deferred - amortization of reg asset	Exhibit No.: A-13 (JCA-45)	438
17	Total deferred other (local) income taxes	Line 15 + Line 16	1,147
18	Total other (local) income taxes	Line 14 + Line 17	1,244
19	Jurisdictional factor	Cost of Service Study	0.9905
20	Total other (local) income taxes - jurisdictional	Line 18 * Line 19	1,232

Schedule: C-11

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Allowance for Funds Used During Construction

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-13 (JCA-48)

Schedule: C-11

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No	(a)	(b)	(c)
	Description	Source	(\$000)
1	Total allowance for funds used during construction	WP-JCA-28	<u>25,823</u>
2	Jurisdictional factor	Cost of Service Study	<u>0.9930</u>
3	Total allowance for funds used during construction - jurisdictional	Line 1 * Line 2	<u>25,643</u>

Schedule: C-12**MICHIGAN PUBLIC SERVICE COMMISSION**Consumers Energy Company

Pro Forma Interest Adjustment

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-13 (JCA-49)

Schedule: C-12

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No.	(a) Description	(b) Source	(c) (\$000)
1	Rate base	Exhibit No.: A-12 (JCA-31)	14,404,493
2	Weighted cost of debt {1}	Exhibit No.: A-14 (MRB-1), Schedule D-1	<u>0.0171</u>
3	Allowable interest expense	Line 1 * Line 2	245,986
4	Pro forma interest expense - historical year	Exhibit No.: A-3 (JCA-20)	<u>183,636</u>
5	Increase/ (decrease) in allowable interest deduction	Line 3 - Line 4	62,350
6	Impact on taxable income	Line 5 * -1	(62,350)
7	Impact on state and local income taxes	Line 6 * 5.47% State and Local Income Tax Rate	<u>(3,411)</u>
8	Impact on federal taxable income	Line 6 - Line 7	(58,939)
9	Impact on federal income taxes	Line 8 * 21.00% Federal Income Tax Rate	<u>(12,377)</u>
10	Total impact on net operating income	(Line 7 + Line 9) * -1	<u><u>15,788</u></u>

Notes

{1} Excludes JDITC

Schedule C-13**MICHIGAN PUBLIC SERVICE COMMISSION**Consumers Energy Company

Interest Synchronization Adjustment

For the Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-13 (JCA-50)

Schedule: C-13

Page 1 of 1

Witness: JCAponte

Date: May 2023

Line No.	(a) Description	(b) Source	(c) (\$000)
1	Rate base	Exhibit No.: A-12 (JCA-31)	14,404,493
2	JDITC debt-related portion of the capital structure	Exhibit No.: A-14 (MRB-1), Schedule D-1	0.0021
3	Portion of rate base funded by JDITC	Line 1 * Line 2	29,812
4	Cost of JDITC - debt portion	Exhibit No.: A-14 (MRB-1), Schedule D-1	0.0414
5	Allowable JDITC interest expense	Line 3 * Line 4	1,235
6	Allowable JDITC interest expense - historical year	Exhibit No.: A-3 (JCA-21)	1,190
7	Increase/ (decrease) in allowable JDITC interest expense	Line 5 - Line 6	45
8	Impact on taxable income	Line 7 * -1	(45)
9	Impact on state and local income taxes	Line 8 * 5.47% State and Local Income Tax Rate	(2)
10	Impact on federal taxable income	Line 8 - Line 9	(43)
11	Impact on federal income taxes	Line 10 * 21.00% Federal Income Tax Rate	(9)
12	Total impact on net operating income	(Line 9 + Line 11) * -1	11

Schedule: C-14

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Development of Adjusted Net Operating Income
For the Projected 12-Month Period Ending February 28, 2025
(\$000)

Case No.: U-21389
Exhibit No.: A-13 (JCA-51)
Schedule: C-14
Page 1 of 1
Witness: JCAponte
Date: May 2023

Line No.	(a) Description	(b) Source	(c) Sales	(d) Wholesale	(e) Other Revenue	(f) Fuel P&I	(g) Other O&M	(h) Deprec.	(i) Property Tax	(j) Other Taxes	(k) CIT	(l) MCIT	(m) FIT	(n) NOI	(o) AFUDC	(p) Adjusted NOI
1	Net operating income - historical year		4,963,156	240,958	160,123	2,622,447	893,311	751,446	203,698	35,840	1,323	44,456	57,580	754,137	5,520	759,656
	Historical Year Adjustments															
2	Weather normalization	Exhibit No.: A-15 (EMB-3), Schedule E-2	(55,044)	-	-	-	-	-	-	-	(88)	(2,923)	(10,927)	(41,107)	-	(41,107)
3	Purchased power administration fee revenue	WP-JCA-12	(612)	-	-	-	-	-	-	-	(1)	(33)	(122)	(457)	-	(457)
4	Excess MDNR fees	WP-JCA-13	447	-	-	-	-	-	-	-	1	24	89	334	-	334
5	MISO reliability schedule 10 charge	WP-JCA-14	460	-	-	-	-	-	-	-	1	24	91	343	-	343
6	Surcharge revenues and expenses/amortization	WP-JCA-15	(361,034)	-	-	-	(237,073)	(70,541)	(9,733)	54,004	(156)	(5,187)	(19,393)	(72,954)	-	(72,954)
7	Jobwork revenues	Exhibit No.: A-15 (EMB-3), Schedule E-2	-	-	2,633	-	-	-	-	-	4	140	523	1,967	-	1,967
8	Interest income on cash operating accounts	WP-JCA-16	-	-	497	-	-	-	-	-	1	26	99	371	-	371
9	Compensation expenses - Executive annual physicals	WP-JCA-17	-	-	-	-	(10)	-	-	-	0	1	2	8	-	8
10	Compensation expenses - EICP	N/A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
11	Compensation expenses - Corporate Giving - staff salaries	WP-JCA-19	-	-	-	-	(14)	-	-	-	0	1	3	11	-	11
12	Compensation expenses - Corporate Communications - staff salaries	WP-JCA-19	-	-	-	-	(184)	-	-	-	0	10	36	137	-	137
13	Advertising expenses	WP-JCA-20	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14	Dues & donations expenses	WP-JCA-21	-	-	-	-	(618)	-	-	-	1	33	123	462	-	462
15	Jobwork expenses	Exhibit No.: A-15 (EMB-3), Schedule E-2	-	-	-	-	2,200	-	-	-	(4)	(117)	(437)	(1,643)	-	(1,643)
16	Interest expense on security deposits	WP-JCA-22	-	-	-	-	258	-	-	-	(0)	(14)	(51)	(193)	-	(193)
17	One-time Legal Consultant Expense	Exhibit No.: A-83 (MJF-2)	-	-	-	-	(3,651)	-	-	-	6	194	725	2,726	-	2,726
18	Voluntary Separation Plan expenses	WP-JCA-18	-	-	-	-	(5,417)	-	-	-	9	288	1,075	4,045	-	4,045
19	Segregated Cell Insurance Expense	WP-JCA-52	-	-	-	-	5,771	-	-	-	(9)	(306)	(1,146)	(4,309)	-	(4,309)
20	Pro Forma Interest Adjustment	Exhibit No.: A-3 (JCA-20)	-	-	-	-	-	-	-	-	27	912	3,411	(4,350)	-	(4,350)
21	Interest Synchronization Adjustment	Exhibit No.: A-3 (JCA-21)	-	-	-	-	-	-	-	-	(2)	(63)	(236)	301	-	301
22	Adjusted net operating income - historical year		<u>4,547,372</u>	<u>240,958</u>	<u>163,254</u>	<u>2,622,447</u>	<u>654,573</u>	<u>680,905</u>	<u>193,965</u>	<u>89,844</u>	<u>1,112</u>	<u>37,465</u>	<u>31,445</u>	<u>639,828</u>	<u>5,520</u>	<u>645,347</u>
	Test Year Adjustments															
23	Sales revenues		395,830	-	-	-	-	-	-	-	633	21,019	78,577	295,601	-	295,601
24	Wholesale revenues		-	(216,619)	-	-	-	-	-	-	(347)	(11,502)	(43,002)	(161,768)	-	(161,768)
25	Other electric revenues		-	-	(28,517)	-	-	-	-	-	(46)	(1,514)	(5,661)	(21,296)	-	(21,296)
26	Fuel and P&I expenses		-	-	-	(33,050)	-	-	-	-	53	1,755	6,561	24,681	-	24,681
27	Other operation and maintenance		-	-	-	-	50,505	-	-	-	(81)	(2,682)	(10,026)	(37,716)	-	(37,716)
28	Depreciation and amortization		-	-	-	-	-	52,169	-	-	(83)	(2,770)	(10,356)	(38,959)	-	(38,959)
29	Property taxes		-	-	-	-	-	-	56,235	-	(90)	(2,986)	(11,163)	(41,996)	-	(41,996)
30	Other general taxes		-	-	-	-	-	-	-	(52,281)	84	2,776	10,378	39,043	-	39,043
31	Other (local) income taxes		-	-	-	-	-	-	-	-	8	-	(2)	(6)	-	(6)
32	State income taxes		-	-	-	-	-	-	-	-	-	(4,956)	1,041	3,915	-	3,915
33	Federal income taxes		-	-	-	-	-	-	-	-	-	-	9,809	(9,809)	-	(9,809)
34	AFUDC		-	-	-	-	-	-	-	-	-	-	-	-	20,304	20,304
35	Adjusted net operating income - projected test year		<u>4,943,202</u>	<u>24,339</u>	<u>134,737</u>	<u>2,589,397</u>	<u>705,078</u>	<u>733,075</u>	<u>250,200</u>	<u>37,563</u>	<u>1,244</u>	<u>36,604</u>	<u>57,601</u>	<u>691,517</u>	<u>25,823</u>	<u>717,340</u>
36	Jurisdictional factor		<u>1.0000</u>	<u>-</u>	<u>0.9925</u>	<u>0.9903</u>	<u>0.9966</u>	<u>0.9959</u>	<u>0.9969</u>	<u>0.9952</u>	<u>0.9905</u>	<u>1.0078</u>	<u>1.0078</u>	<u>1.0078</u>	<u>0.9930</u>	<u>-</u>
37	Adjusted net operating income - projected test year - jurisdictional		<u>4,943,202</u>	<u>-</u>	<u>133,724</u>	<u>2,564,230</u>	<u>702,708</u>	<u>730,079</u>	<u>249,436</u>	<u>37,381</u>	<u>1,232</u>	<u>36,890</u>	<u>58,051</u>	<u>696,920</u>	<u>25,643</u>	<u>722,563</u>

Notes

Line 1, Col. (c) - Col. (p) Source: Exhibit No.: A-3 (JCA-9)
Line 37, Col. (c) - Col. (p) Source: Exhibit No.: A-13 (JCA-37)

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Item Description	Actual 2019	Actual 2020	Actual 2021	Actual 2022	Forecast 2023	Forecast 2024	Forecast 2025	Forecast 2026	Forecast 2027
1	Electric Vehicle Expenditures ⁴	\$ 917	\$ 2,896	\$ 2,947	\$ 3,876					
	Amortization Expense									
2	2019	-	183	183	183	183	183			
3	2020	-	-	579	579	579	579	579		
4	2021	-	-	-	589	589	589	589	589	
5	2022	-	-	-	-	775	775	775	775	775
6	Total amortization expense	<u>\$ -</u>	<u>\$ 183</u>	<u>\$ 763</u>	<u>\$ 1,352</u>	<u>\$ 2,127</u>	<u>\$ 2,127</u>	<u>\$ 1,944</u>	<u>\$ 1,365</u>	<u>\$ 775</u>
	Item Description	Avg Balance Feb 28, 2025 ⁽³⁾	12 Mos Ending Feb 29, 2024 ⁽¹⁾	12 Mos Ending Feb 28, 2025 ⁽²⁾						
7	Test Year Amortization Expense			<u>\$ 2,097</u>						
8	Electric Vehicle Deferred Balance	<u>\$ 4,808</u>	<u>\$ 5,857</u>	<u>\$ 3,760</u>						

⁽¹⁾ Sum of Line 1 columns (b) through (e) Less (sum of Line 6 columns (b) through (f) + Line 6 column (g) + 12 * 2)
⁽²⁾ Amortization Expense = [(Line 6 column (g)) + 12 * 10 (Line 6 column (h)) + 12 * 2)
Deferred Balance = Sum of Line 1 columns (b) through (e) Less (sum of Line 6 columns (b) through (g) + Line 6 column (h) + 12 * 2)
⁽³⁾ (Line 8 column (b) Plus Line 6 column (c)) + 2

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Item Description	Actual 2019	Actual 2020	Actual 2021	Actual 2022	Forecast 2023	Forecast 2024	Forecast 2025	Forecast 2026	Forecast 2026
1	Electric Vehicle Expenditures ⁴	\$ -	\$ -	\$ 1,009	\$ 1,785					
	Amortization Expense									
2	2019	-	-	-	-	-	-	-	-	-
3	2020	-	-	-	-	-	-	-	-	-
4	2021	-	-	-	202	202	202	202	202	-
5	2022	-	-	-	-	357	357	357	357	357
6	Total amortization expense	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ 202</u>	<u>\$ 559</u>	<u>\$ 559</u>	<u>\$ 559</u>	<u>\$ 559</u>	<u>\$ 357</u>
	Item Description	Avg Balance Feb 28, 2025 ⁽³⁾	12 Mos Ending Feb 29, 2024 ⁽¹⁾	12 Mos Ending Feb 28, 2025 ⁽²⁾						
7	Test Year Amortization Expense			<u>\$ 559</u>						
8	Electric Vehicle Deferred Balance	<u>\$ 1,660</u>	<u>\$ 1,940</u>	<u>\$ 1,381</u>						

⁽¹⁾ Sum of Line 1 columns (b) through (e) Less (sum of Line 6 columns (b) through (f) + Line 6 column (g) + 12 * 2)
⁽²⁾ Amortization Expense = [(Line 6 column (g)) + 12 * 10 (Line 6 column (h)) + 12 * 2)
Deferred Balance = Sum of Line 1 columns (b) through (e) Less (sum of Line 6 columns (b) through (g) + Line 6 column (h) + 12 * 2)
⁽³⁾ (Line 8 column (b) Plus Line 6 column (c)) + 2

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	(t)
Line No.	Item Description	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039
1	Karn Expenditures Deferred	\$ 7,016	\$ 4,828	\$ 1,762																
	Amortization Expense																			
2	2021		2,339	2,339	2,339	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3	2022			1,609	1,609	1,609	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	2023				587	587	587	-	-	-	-	-	-	-	-	-	-	-	-	-
5	2024					-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
6	Total amortization expense	<u>\$ -</u>	<u>\$ 2,339</u>	<u>\$ 3,948</u>	<u>\$ 4,535</u>	<u>\$ 2,197</u>	<u>\$ 587</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>
		Avg Balance Feb 28, 2025 ⁽³⁾	12 Mos Ending Feb 29, 2024 ⁽¹⁾	12 Mos Ending Feb 28, 2025 ⁽²⁾																
	Item Description																			
7	Test Year Amortization Expense			<u>\$ 4,146</u>																
8	Karn Expenditures Deferred Balance	<u>\$ 4,491</u>	<u>\$ 6,563</u>	<u>\$ 2,418</u>																

⁽¹⁾ Sum of Line 1 columns (b) through (d) Less (sum of Line 6 columns (b) through (d) + Line 6 column (e) ÷ 12 * 2)
⁽²⁾ Amortization Expense = [(Line 6 column (e)) ÷ 12 * 10 (Line 6 column (f)) ÷ 12 * 2)
Deferred Balance = Sum of Line 1 columns (b) through (d) Less (sum of Line 6 columns (b) through (e) + Line 6 column (f) ÷ 12 * 2)
⁽³⁾ (Line 8 column (b) Plus Line 6 column (c)) ÷ 2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Regulatory Asset and Amortization
Campbell Plant Retention and Separation Costs
(\$000)

Case No.: U-21389
Exhibit No.: A-20 (JCA-55)
Page: 1 of 1
Witness: JCAponte
Date: May 2023

Line No.	(a) Item Description	(b) 2021	(c) 2022	(d) 2023	(e) 2024	(f) 2025	(g) 2026	(h) 2027	(i) 2028	(j) 2029	(k) 2030	(l) 2031	(m) 2032	(n) 2033	(o) 2034	(p) 2035	(q) 2036	(r) 2037	(s) 2038	(t) 2039
1	Campbell Expenditures Deferred	\$ -	\$ 23,904	\$ 16,516	\$ 10,814	\$ 3,044														
2	Amortization Expense																			
3	2023			7,968	7,968	7,968	-	-	-	-	-	-	-	-	-	-	-	-	-	(23,904)
4	2024				5,505	5,505	5,505	-	-	-	-	-	-	-	-	-	-	-	-	7,387
5	2025					3,605	3,605	3,605	-	-	-	-	-	-	-	-	-	-	-	5,703
6	2026						1,015	1,015	1,015	-	-	-	-	-	-	-	-	-	-	(3,044)
6	Total amortization expense	\$ -	\$ -	\$ 7,968	\$ 13,473	\$ 17,078	\$ 10,125	\$ 4,619	\$ 1,015	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (13,858)

Item Description	Avg Balance Feb 28, 2025 ⁽³⁾	12 Mos Ending Feb 29, 2024 ⁽¹⁾	12 Mos Ending Feb 28, 2025 ⁽²⁾
7 Test Year Amortization Expense			\$ 14,074
8 Campbell Expenditures Deferred Balance	\$ 29,731	\$ 32,009	\$ 27,454

⁽¹⁾ Sum of Line 1 columns (b) through (d) + Line 1 column (e) + 12 * 2 Less (sum of Line 6 columns (b) through (d) + Line 6 column (e) + 12 * 2)

⁽²⁾ Amortization Expense = [(Line 6 column (e)) + 12 * 10 (Line 6 column (f)) + 12 * 2]

Deferred Balance = Sum of Line 1 columns (b) through (e) + Line 1 column (f) + 12 * 2 Less (sum of Line 6 columns (b) through (e) + Line 6 column (f) + 12 * 2)

⁽³⁾ (Line 8 column (b) Plus Line 6 column (c)) + 2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Regulatory Asset and Amortization
Campbell 1-2 Plants Net Book Value
(\$000)

Case No.: U-21389
Exhibit No.: A-21 (JCA-56)
Page: 1 of 1
Witness: JCAponte
Date: May 2023

	(a)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Line No.	Item Description	2022	2 Months Ending 02/28/23	12 Months Ending 02/29/24	12 Months Ending 02/28/25	12 Months Ending 02/28/26	12 Months Ending 02/28/27	12 Months Ending 02/29/28	12 Months Ending 02/28/29	12 Months Ending 02/28/30	12 Months Ending 02/28/31	3 Months Ending 5/31/2031
1	Beginning Plant in Service		\$ 1,119,994	\$ 1,120,350								
2	Additions ⁽¹⁾		356	7,215								
3	Ending Plant in Service	\$ 1,119,994	\$ 1,120,350	\$ 1,127,565								
4	Beginning Accumulated Depreciation		\$ 560,612	\$ 570,643								
5	Depreciation Expense ⁽²⁾		10,031	62,730								
6	Ending Accumulated Depreciation	\$ 560,612	\$ 570,643	\$ 633,373								
7	Unrecovered Net Book Value ⁽³⁾	\$ 559,382	\$ 549,707	\$ 494,192	\$ 494,192	\$ 426,028	\$ 357,863	\$ 289,699	\$ 221,534	\$ 153,370	\$ 85,206	\$ 17,041
8	Amortization Expense ⁽⁴⁾				\$ 68,164	\$ 68,164	\$ 68,164	\$ 68,164	\$ 68,164	\$ 68,164	\$ 68,164	\$ 17,041
9	Campbell 1&2 NBV Regulatory Asset Balance ⁽⁵⁾				426,028	357,863	289,699	221,534	153,370	85,206	17,041	(0)
10	Campbell 1&2 NBV Regulatory Asset Balance Test Year ⁽⁶⁾				\$ 460,110							

Notes:

⁽¹⁾ Additions as of 02/29/2024 include CWIP Balance as of 12/31/2022

⁽²⁾ Source: WP-JCA-55

⁽³⁾ Ending Plant in Service Less Ending Accumulated Depreciation

⁽⁴⁾ Unrecovered Net Book Value + 87 months x months in the period

⁽⁵⁾ Unrecovered Net Book Value Less Amortization Expense

⁽⁶⁾ (Line 9 column (f) Plus Line 11 column (f)) + 2

	(a)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	(t)	(u)	(v)
		2	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	3
Line No.	Item Description	2022	Months Ending 02/28/23	Months Ending 02/29/24	Months Ending 02/28/25	Months Ending 02/28/26	Months Ending 02/28/27	Months Ending 02/29/28	Months Ending 02/28/29	Months Ending 02/28/30	Months Ending 02/28/31	Months Ending 2/29/2032	Months Ending 2/28/2033	Months Ending 2/28/2034	Months Ending 2/28/2035	Months Ending 2/29/2036	Months Ending 2/28/2037	Months Ending 2/28/2038	Months Ending 2/28/2039	Months Ending 2/29/2040	Months Ending 6/31/2040
1	Beginning Plant in Service		\$ 1,764,862	\$ 1,766,031																	
2	Additions ⁽¹⁾		1,169	14,231																	
3	Ending Plant in Service	\$ 1,764,862	\$ 1,766,031	\$ 1,780,262																	
4	Beginning Accumulated Depreciation		\$ 774,552	\$ 784,822																	
5	Depreciation Expense ⁽²⁾		10,270	64,234																	
6	Ending Accumulated Depreciation	\$ 774,552	\$ 784,822	\$ 849,056																	
7	Unrecovered Net Book Value ⁽³⁾	\$ 990,310	\$ 981,209	\$ 931,206	\$ 931,206	\$ 873,901	\$ 816,596	\$ 759,291	\$ 701,986	\$ 644,681	\$ 587,376	\$ 530,071	\$ 472,766	\$ 415,461	\$ 358,156	\$ 300,851	\$ 243,546	\$ 186,241	\$ 128,936	\$ 71,631	\$ 14,326
8	Amortization Expense ⁽⁴⁾				\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 57,305	\$ 14,326
9	Campbell 3 NBV Regulatory Asset Balance ⁽⁵⁾				873,901	816,596	759,291	701,986	644,681	587,376	530,071	472,766	415,461	358,156	300,851	243,546	186,241	128,936	71,631	14,326	(0)
10	Campbell 3 NBV Regulatory Asset Balance Test Year ⁽⁶⁾				\$ 902,553																

Notes:

⁽¹⁾ Additions as of 02/29/2024 include CWIP Balance as of 12/31/2022

⁽²⁾ Source: WP-JCA-56

⁽³⁾ Ending Plant in Service Less Ending Accumulated Depreciation

⁽⁴⁾ Unrecovered Net Book Value ÷ 195 months x months in the period

⁽⁵⁾ Unrecovered Net Book Value Less Amortization Expense

⁽⁶⁾ (Line 9 column (f) Plus Line 11 column (f)) ÷ 2

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Regulatory Asset and Amortization

Karn 1-2 Decommissioning Costs

(\$000)

Case No.: U-21389

Exhibit No.: A-23 (JCA-58)

Page: 1 of 1

Witness: JCAponte

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Item Description	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
1	Karn 1-2 Decommissioning Costs Deferred (JEB-1)	\$ 10,755									
2	Amortization Expense 2022			1,344	1,344	1,344	1,344	1,344	1,344	1,344	1,344
3	Total amortization expense	<u>\$ -</u>	<u>\$ -</u>	<u>\$ 1,344</u>	<u>\$ 1,344</u>	<u>\$ 1,344</u>	<u>\$ 1,344</u>	<u>\$ 1,344</u>	<u>\$ 1,344</u>	<u>\$ 1,344</u>	<u>\$ 1,344</u>
		Avg Balance Feb 28, 2025 ⁽³⁾	12 Mos Ending Feb 29, 2024 ⁽¹⁾	12 Mos Ending Feb 28, 2025 ⁽²⁾							
	Item Description										
4	Test Year Amortization Expense			<u>\$ 1,344</u>							
5	Karn 1-2 Decommissioning Costs Deferred Balance	<u>\$ 9,859</u>	<u>\$ 10,531</u>	<u>\$ 9,187</u>							

⁽¹⁾ Line 1 column (b) Less Line 6 column (d) ÷ 12 * 2

⁽²⁾ Amortization Expense = [(Line 6 column (d)) ÷ 12 * 10 (Line 6 column (e)) ÷ 12 * 2]
Deferred Balance = Line 1 column (b) Less (sum of Line 6 column (d) + Line 6 column (e) ÷ 12 * 2)

⁽³⁾ (Line 8 column (b) Plus Line 6 column (c)) ÷ 2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Regulatory Asset and Amortization
Karn 1-2, Classic 7, and Campbell Ash Disposal
(\$000)

Case No.: U-21389
Exhibit No.: A-24 (JCA-59)
Page: 1 of 1
Witness: JCAponte
Date: May 2023

Line No.	(a) Item Description	(b) 2022	(c) 2023	(d) 2024	(e) 2025	(f) 2026	(g) 2027	(h) 2028	(i) 2029	(j) 2030	(k) 2031	(l) 2032	(m) 2033
1	Karn 1-2, Classic 7, and Campbell Ash Disposal Costs Deferred (JEB-2)	\$ 11,816											
2	Amortization Expense 2022			1,182	1,182	1,182	1,182	1,182	1,182	1,182	1,182	1,182	1,182
3	Total amortization expense	<u>\$ -</u>	<u>\$ -</u>	<u>\$ 1,182</u>	<u>\$ 1,182</u>	<u>\$ 1,182</u>	<u>\$ 1,182</u>	<u>\$ 1,182</u>	<u>\$ 1,182</u>	<u>\$ 1,182</u>	<u>\$ 1,182</u>	<u>\$ 1,182</u>	<u>\$ 1,182</u>
		Avg Balance Feb 28, 2025 ⁽³⁾	12 Mos Ending Feb 29, 2024 ⁽¹⁾	12 Mos Ending Feb 28, 2025 ⁽²⁾									
	Item Description												
4	Test Year Amortization Expense			<u>\$ 1,182</u>									
5	Karn 1-2, Classic 7, and Campbell Ash Disposal Costs Deferred Balance	<u>\$ 11,028</u>	<u>\$ 11,619</u>	<u>\$ 10,437</u>									

⁽¹⁾ Line 1 column (b) Less Line 6 column (d) ÷ 12 * 2

⁽²⁾ Amortization Expense = [(Line 6 column (d)) ÷ 12 * 10 (Line 6 column (e)) ÷ 12 * 2]
Deferred Balance = Line 1 column (b) Less (sum of Line 6 column (d) + Line 6 column (e) ÷ 12 * 2)

⁽³⁾ (Line 8 column (b) Plus Line 6 column (c)) ÷ 2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Regulatory Asset and Amortization
Campbell Decommissioning Costs
(\$000)

Case No.: U-21389
Exhibit No.: A-25 (JCA-60)
Page: 1 of 1
Witness: JCAponte
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Line No.	Item Description	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
1	Campbell Decommissioning Costs Deferred (JEB-3)	\$ 1,659											
2	Amortization Expense 2022			166	166	166	166	166	166	166	166	166	166
3	Total amortization expense	<u>\$ -</u>	<u>\$ -</u>	<u>\$ 166</u>	<u>\$ 166</u>	<u>\$ 166</u>	<u>\$ 166</u>	<u>\$ 166</u>	<u>\$ 166</u>	<u>\$ 166</u>	<u>\$ 166</u>	<u>\$ 166</u>	<u>\$ 166</u>
		Avg Balance Feb 28, 2025 ⁽³⁾	12 Mos Ending Feb 29, 2024 ⁽¹⁾	12 Mos Ending Feb 28, 2025 ⁽²⁾									
	Item Description												
4	Test Year Amortization Expense			<u>\$ 166</u>									
5	Campbell Decommissioning Costs Deferred Balance	<u>\$ 1,549</u>	<u>\$ 1,632</u>	<u>\$ 1,466</u>									

⁽¹⁾ Line 1 column (b) Less Line 6 column (d) ÷ 12 * 2

⁽²⁾ Amortization Expense = [(Line 6 column (d)) ÷ 12 * 10 (Line 6 column (e)) ÷ 12 * 2]
Deferred Balance = Line 1 column (b) Less (sum of Line 6 column (d) + Line 6 column (e) ÷ 12 * 2)

⁽³⁾ (Line 8 column (b) Plus Line 6 column (c)) ÷ 2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Demand Response Revenue Requirement
(000)

Case No.: U-21389
Exhibit No.: A-26 (JCA-61)
Page: 1 of 3
Witness: JCAponte
Date: May 2023

Total Demand Response

Line	Description	Total	Source
(a)	(b)	(c)	(d)
1	Residential Revenue Requirement	\$28,687	Exhibit A-26 (JCA-61), p 2
2	C&I Revenue Requirement	23,507	Exhibit A-26 (JCA-61), p 3
3	Case No. U-21233 Reconciliation - Incentive	4,640	Company Witness McLean
4	Case No. U-21233 Reconciliation - Reg Liability	(10,046)	Company Witness McLean
5	Total Demand Response Revenue Requirement	<u>\$46,789</u>	

Residential Demand Response

Line	Description	2 Mos Ending 2/28/2023	12 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025	Total	
(a)	(b)	(c)	(d)	(e)	(f)	
1	Total Capital Spending	<u>\$ 1,848</u>	<u>\$ 9,852</u>	<u>\$ 6,838</u>	<u>\$ 18,538</u>	Exhibit A-12 (SQM-1)

Revenue Requirement Calculation:

					2/28/2025 Average	
2	Beginning Plant	\$ 46,602	\$ 48,450	\$ 58,302		
3	Closings	<u>1,848</u>	<u>9,852</u>	<u>6,838</u>		
4	Ending Plant	<u>\$ 48,450</u>	<u>\$ 58,302</u>	<u>\$ 65,140</u>	<u>\$ 61,721</u>	
5	Beginning CWIP	\$ 119	\$ 119	\$ 119		
6	Capital Spending	<u>1,848</u>	<u>9,852</u>	<u>6,838</u>		
7	Closings	<u>1,848</u>	<u>9,852</u>	<u>6,838</u>		
8	Ending CWIP	<u>\$ 119</u>	<u>\$ 119</u>	<u>\$ 119</u>	<u>\$ 119</u>	
9	Beginning Depreciation Reserve	\$ 8,604	\$ 9,277	\$ 13,982		
10	Depreciation Expense ⁽¹⁾	<u>673</u>	<u>4,704</u>	<u>5,440</u>		
11	End Depreciation Reserve	<u>\$ 9,277</u>	<u>\$ 13,982</u>	<u>\$ 19,422</u>	<u>\$ 16,702</u>	
12	Total Rate Base				<u>\$ 45,138</u>	Line 4 + Line 8 - Line11
13	Pre-Tax Rate of Return				<u>7.60%</u>	Exhibit A-14 (MRB-1)
14	Return on Investment				<u>\$ 3,431</u>	Line 12 * Line 13
15	Depreciation Expense				5,440	Line 10
16	O&M				19,050	Exhibit A-144 (SQM-2)
17	R&PP Tax				<u>767</u>	Footnote ⁽²⁾
18	Total Residential Revenue Requirement				<u>\$28,687</u>	

Footnotes

- (1) 2022 Composite Depreciation Rate 8.81%
Jan-Feb 2023 Depreciation portion 16.07%
- (2) Average Plant Balance Test Year \$ 61,840
Property Tax Rate 0.012396326 Source: Exh A-161 (BJV-1)

Commercial and Industrial Demand Response

Line	Description	2 Mos Ending 2/28/2023	12 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025	Total	
(a)	(b)	(c)	(d)	(e)	(f)	
1	Total Capital Spending	<u>\$ 42</u>	<u>\$ 330</u>	<u>\$ 347</u>	<u>\$ 718</u>	Exhibit A-12 (SQM-1)

Revenue Requirement Calculation:

					<u>2/28/2025 Average</u>	
2	Beginning Plant	\$ 6,664	\$ 6,705	\$ 7,035		
3	Closings	<u>42</u>	<u>330</u>	<u>347</u>		
4	Ending Plant	\$ 6,705	\$ 7,035	\$ 7,382	<u>\$ 7,209</u>	
5	Beginning CWIP	\$ 17	\$ 17	\$ 17		
6	Capital Spending	<u>42</u>	<u>330</u>	<u>347</u>		
7	Closings	<u>42</u>	<u>330</u>	<u>347</u>		
8	Ending CWIP	<u>\$ 17</u>	<u>\$ 17</u>	<u>\$ 17</u>	<u>\$ 17</u>	
9	Beginning Depreciation Reserve	\$ 1,851	\$ 1,989	\$ 2,867		
10	Depreciation Expense ⁽¹⁾	<u>137</u>	<u>878</u>	<u>922</u>		
11	End Depreciation Reserve	\$ 1,989	\$ 2,867	\$ 3,789	<u>\$ 3,328</u>	
12	Total Rate Base				<u>\$ 3,898</u>	Line 4 + Line 8 - Line 11
13	Pre-Tax Rate of Return				<u>7.60%</u>	Exhibit A-14 (MRB-1)
14	Return on Investment				\$ 296	Line 12 * Line 13
15	Depreciation Expense				922	Line 10
16	O&M				22,200	Exhibit A-144 (SQM-2)
17	R&PP Tax				<u>90</u>	Footnote ⁽²⁾
18	Total C&I Revenue Requirement				<u>\$23,507</u>	

Footnotes

(1) 2022 Composite Depreciation Rate 12.79%
Jan-Feb 2023 Depreciation portion 16.07%

(2) Average Plant Balance Test Year \$ 7,226

MICHIGAN PUBLIC SERVICE COMMISSION**Consumers Energy Company**

2022 Distribution Deferral

(000)

Case No.: U-21389

Exhibit No.: A-27 (JCA-62)

Page: 1 of 2

Witness: JCAponte

Date: May 2023

Line	Description	Actual 2022	U-20963 Approved 2022	Difference 2022
(a)	(b)	(c)	(d)	(e)
	<u>Capital Spending</u>			
1	New Business	174,753	123,172	51,581
2	Demand Failures	214,758	121,365	93,393
3	Asset Relocations	62,267	53,078	9,189
4	Total Capital Spending	\$ 451,777	\$ 297,615	\$ 154,162

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

2022 Distribution Deferral

(000)

Case No.: U-21389

Exhibit No.: A-27 (JCA-62)

Page: 2 of 2

Witness: JCAponte

Date: May 2023

Line	Description	Amount	Source
(a)	(b)	(c)	(e)
<u>Revenue Requirement Calculation:</u>			
1	Beginning Plant in Service	\$ -	
2	Additional Capital Spending	154,162	Page 1, Line 4
3	Ending Plant	\$ 154,162	Line 1 + Line 2
4	Average Plant in Service	\$ 77,081	Average (Line 1 & Line 3)
5	Beginning Depreciation Reserve	\$ -	
6	Depreciation Expense ⁽¹⁾	2,305	Line 4 x 2.99%
7	End Depreciation Reserve	\$ 2,305	Line 5 + Line 6
8	Average Depreciation Reserve	\$ 1,152	Average (Line 5 & Line 7)
9	Total Average Rate Base	\$ 75,929	Line 3 Less Line 8
10	Return on Investment ⁽¹⁾⁽²⁾	\$5,341	Line 9 x 7.03%
11	R&PP Tax ⁽¹⁾	910	Line 4 x 1.181%
12	Total 2021 Deferred Revenue Requirement	\$ 8,556	Line 6 + Line 10 + Line 11

⁽¹⁾ Depreciation rate of 2.99%, pretax cost of capital of 7.03%, and the property tax rate of 1.181% are from case No. U-20963.

⁽²⁾ For accounting purposes there is a contra regulatory asset account that includes the return on equity portion of the return on investment. The return on equity authorized for deferral cannot be recorded until collected.

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

2023 Distribution Deferral
(000)

Case No.: U-21389
Exhibit No.: A-28 (JCA-63)
Page: 1 of 1
Witness: JCAponte
Date: May 2023

		U-21224 Approved 2023
Line	Description	
(a)	(b)	(c)
	<u>Capital Spending</u>	
1	New Business	167,217
2	Demand Failures	148,356
3	Asset Relocations	55,122
4	Total Capital Spending	\$ 370,695

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

JEFFREY E. BATTAGLIA

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Karn Units 1 & 2 Decommissioning Plan Expenditures

Case No.: U-21389

Exhibit No.: A-29 (JEB-1)

Page: 1 of 1

Witness: JEBattaglia

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
	Period	2022-2026		2022		2023	2024	2025	2026
Line No.	Activity	Projected (2022)	Actual and Projected (2023)	Forecast	Actual	Updated Projected	Updated Projected	Updated Projected	Updated Projected
1	Utility Separation	-	2,453,260	-	1,905,828	547,432	-	-	-
2	Decommissioning	5,935,991	7,628,332	234,995	299,590	7,328,743	-	-	-
3	Abatement	14,508,377	14,750,163	-	-	-	14,750,163	-	-
4	Demolition	28,373,080	26,826,467	-	-	-	11,447,658	15,378,809	-
5	Asset Recovery	(16,745,760)	(17,895,829)	-	-	-	(7,200,000)	(10,695,829)	-
6	Environmental	8,165,981	9,645,887	-	-	340,000	7,119,982	2,185,905	-
7	Restoration	3,306,472	4,502,339	-	-	-	-	4,502,339	-
8	Inventory Write Off	6,235,589	6,971,387	1,235,589	2,721,387	4,250,000	-	-	-
9	Contingency	13,661,085	17,042,850	300,000	-	1,300,000	10,000,000	5,742,850	-
10	Project Indirects	20,057,335	25,129,349	2,926,722	5,828,423	4,175,419	9,197,650	5,148,956	778,900
11	Total	83,498,149	97,054,205	4,697,306	10,755,228	17,941,594	45,315,453	22,263,030	778,900

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy CompanyActual Coal Ash Disposal Plan Expenditures
(\$000's)

Case No.: U-21389

Exhibit No.: A-30 (JEB-2)

Page: 1 of 1

Witness: JEBattaglia

Date: May 2023

	Site	Projected 2022	Actuals 2022		
(a)	DE Karn	\$ 549	\$ 121	\$	428
(b)	JC Weadock	\$ 9,491	\$ 5,166	\$	4,325
(c)	BC Cobb	\$ 4,158	\$ 6,102	\$	(1,944)
(d)	JR Whiting	\$ -	\$ -	\$	-
(e)	JH Campbell	\$ -	\$ 427	\$	(427)
(f)	Total	\$ 14,198	\$ 11,816		

Footnote:

1 All expenses from 2022-2029 will recorded as regulatory asset

2 All expenses in the regulatory asset will be amortized for 10 years (beginning in 2029) so that all amoritization will be complete by 2039

3 Ground monitor costs beginning in 2030 will be expensed in same year incurred

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Projected JHC Decommissioning Plan Expenditures

Case No.: U-21389
 Exhibit No.: A-31 (JEB-3)
 Page: 1 of 1
 Witness: JEBattaglia
 Date: May 2023

Projected JH Campbell Units 1, 2 & 3 Decommissioning Expense-March 2023

(a) (b) (c) (d) (e)

Actual Actual
Jan-June July- Dec Total Actual

Line No.	Activity	Forecasted	2022	2022	2022
1	Utility Separation	19,596,869			
2	Decommissioning	11,542,385			
3	Abatement	14,652,459			
4	Demolition	77,493,000			
5	Asset Recovery	(27,736,946)			
6	Environmental	31,632,778			
7	Restoration	7,363,983			
8	Inventory Write Off	10,400,000			
9	Contingency	31,185,000			
10	Project Indirects	45,539,383	497,314	1,191,923	1,659,205
11	Total	221,668,911	497,314	1,191,923	1,659,205

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS
OF
MARC R. BLECKMAN
ON BEHALF OF
CONSUMERS ENERGY COMPANY

May 2023

Schedule D-1

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Overall Rate of Return Summary
for the Projected Year Ending February 28, 2025

Case No.: U-21389
Exhibit No.: A-14 (MRB-1)
Schedule: D-1
Page: 1 of 2
Witness: MRBleckman
Date: May 2023

Line No.	(a) Description	(b) (c) (d) Capital Structure			(e) Cost Rate %	(f) (g) (h) (i) Weighted Cost			
		Amount (\$000,000) (1)	Percent Permanent Capital (2)	Percent of Total Capital		Permanent Capital (7)	Total Cost % (8)	Conversion Factor	Pre-Tax Return (9)
1									
2	Long-Term Debt	\$ 10,833	48.33%	39.96%	4.14% (3)	2.00%	1.66%	1.0000	1.66%
3									
4	Preferred Stock	37	0.17%	0.14%	4.50% (4)	0.01%	0.01%	1.3391	0.01%
5									
6	Common Shareholder's Equity	<u>11,543</u>	<u>51.50%</u>	42.58%	10.25% (5)	5.28%	4.36%	1.3391	5.84%
7									
8	Total Permanent Capital	\$ 22,413	<u>100.00%</u>						
9									
10	Short-Term Debt	294		1.08%	4.79% (6)		0.05%	1.0000	0.05%
11									
12	Deferred Income Taxes	4,280		15.79%	0.00%		0.00%	1.0000	0.00%
13									
14	<u>Investment Tax Credit</u>								
15	Long-Term Debt	56		0.21%	4.14%		0.01%	1.0000	0.01%
16	Preferred Stock	0		0.00%	4.50%		0.00%	1.3391	0.00%
17	Common Equity	63		0.23%	10.25%		0.02%	1.3391	0.03%
18									
19	Total	<u>\$ 27,106</u>		<u>100.00%</u>			<u>6.11%</u>		<u>7.60%</u>

(1) See Exhibit A-14 (MRB-2), Schedule D-1a, Page 1.

(2) Excludes Short-term Debt, Deferred Income Taxes, and Investment Tax Credit to calculate the rate of return for Investment Tax Credit purposes in accordance with Internal Revenue Service Income Tax Regulation Section 1.46-6.

(3) See Exhibit A-14 (MRB-4), Schedule D-2.

(4) See Exhibit A-14 (MRB-6), Schedule D-4.

(5) See Exhibit A-14 (TAW-1), Schedule D-5.

(6) See Exhibit A-14 (MRB-5), Schedule D-3, Page 1.

(7) Column (c) x column (e).

(8) Column (d) x column (e).

(9) Column (g) x column (h).

Schedule D-1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Overall Rate of Return Summary - Adjusted
for the Projected Year Ending February 28, 2025

Case No.: U-21389

Exhibit No.: A-14 (MRB-1)

Schedule: D-1

Page: 2 of 2

Witness: MRBleckman

Date: May 2023

Line No.	(a) Description	Unadjusted		Adjusted	
		(b) 13-Month Avg. For Yr. Ended Feb. 2025	(c) Percent Permanent Capital	(d) 13-Month Avg. For Yr. Ended Feb. 2025	(e) Percent Permanent Capital
		Amount (\$000,000)		Amount (\$000,000)	
1					
2	Long-Term Debt	\$ 10,833 (1)	48.33%	\$ 11,939	50.76%
3					
4	Preferred Stock	37 (1)	0.17%	37 (1)	0.16%
5					
6	Common Shareholder's Equity	11,543 (1)	51.50%	11,543 (1)	49.08%
7					
8	Total Permanent Capital	\$ 22,413	100.00%	\$ 23,519	100.00%
9					
10					
11					
12	Long-Term Debt			\$ 10,833 (1)	
13					
14	Add: Securitization Debt			731 (2)	
15	Add: Short-Term Debt			294 (1)	
16	Add: Leases			81 (3)	
17				\$ 1,106	
18					
19	Long-Term Debt - Adjusted			\$ 11,939	
20					

(1) See Exhibit A-14 (MRB-1), Schedule D-1, Page 1.

(2) Projected Securitization debt balance for the 13-months ending February 2025.

(3) Projected using the Lease balance at December 31, 2022.

Schedule D-1a

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Capital Structure Development
for the Projected Year Ending February 28, 2025

Case No.: U-21389
Exhibit No.: A-14 (MRB-2)
Schedule: D-1a
Page: 1 of 4
Witness: MRBleckman
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
		Historical Capital Structure					Recommended Capital Structure		
		13-Month Avg. For Yr. Ended December 31, 2022							
		Amount	% of		Balances as of	Test Year	Amount	% of	
Line	Description	Outstanding	Permanent	% of	Dec. 31, 2022	Adjustments	Outstanding	Permanent	% of
No.		(000,000)	Capital	Total Capital	(000,000)	(000,000)	(000,000)	Capital	Total Capital
1									
2	Long-Term Debt	\$ 8,650	46.59%	38.08%	\$ 9,010	\$ 1,823	\$ 10,833	48.33%	39.96%
3									
4	Preferred Stock	37	0.20%	0.16%	37	-	37	0.17%	0.14%
5									
6	Common Equity	<u>9,881</u>	<u>53.21%</u>	43.50%	<u>10,134</u>	<u>1,409</u>	<u>11,543</u>	<u>51.50%</u>	42.58%
7									
8	Total Permanent Capital	\$ 18,568	<u>100.00%</u>		\$ 19,181	\$ 3,232	\$ 22,413	<u>100.00%</u>	
9									
10	Short-Term Debt, Incl Renewable Liability	147		0.65%	1,095	(801)	294		1.08%
11									
12	Deferred Income Taxes	3,874		17.06%	3,772	508	4,280		15.79%
13									
14	<u>Investment Tax Credit</u>								
15	Long-Term Debt	58		0.26%	52	4	56		0.21%
16	Preferred Stock	0		0.00%	0	0	0		0.00%
17	Common Equity	<u>66</u>		<u>0.29%</u>	<u>59</u>	<u>4</u>	<u>63</u>		<u>0.23%</u>
18									
19	Total Investment Tax Credit	<u>125</u>		<u>0.55%</u>	<u>112</u>	<u>8</u>	<u>120</u>		<u>0.44%</u>
20									
21	Total Capitalization	<u>\$ 22,715</u>		<u>100.00%</u>	<u>\$ 24,159</u>		<u>\$ 27,106</u>		<u>100.00%</u>

Sources and Support, by Column:

- (b) Consumers Energy General Ledger 13-month average balances as of December 31, 2022.
- (c) Each line 2, 4, and 6 in column (b) is divided by line 8, column (b).
- (d) Each line 2, 4, 6, 10, 12, 15, 16, 17, and 19 in column (b) is divided by line 21, column (b).
- (e) Consumers Energy General Ledger balances as of December 31, 2022.
- (f) Line 2 Debt maturities and debt issues, line 6 Adjustment for retained earnings and equity contributions, line 10 Adjustment to project short-term debt and renewable liability balance, line 12 Adjustment to project Deferred Income Taxes balance, lines 15-17 Adjustment to project Investment Tax Credit balance.
- (g) Column (e) + column (f). Represents 13-month averages.
- (h) Each line 2, 4, and 6 in column (g) is divided by line 8, column (g).
- (i) Each line 2, 4, 6, 10, 12, 15, 16, 17, and 19 in column (g) is divided by line 21, column (g).

MICHIGAN PUBLIC SERVICE COMMISSION

Case No.: U-21389
Exhibit No.: A-14 (MRB-2)
Schedule: D-1a
Page: 2 of 4
Witness: MRBleckman
Date: May 2023

Exhibit No.: A-14 (MRB-2)

Schedule: D-1a

Page: 2 of 4

Date: May 2023

[illegible]

Schedule D-1a

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Capital Structure Development - Projected Common Equity Balance
for the Projected Year Ending February 28, 2025

(in millions)

Case No.: U-21389
Exhibit No.: A-14 (MRB-2)
Schedule: D-1a
Page: 3 of 4
Witness: MRBleckman
Date: May 2023

Line No.	(a) Common Equity	(b) Actual Jan-22	(c) Actual Feb-22	(d) Actual Mar-22	(e) Actual Apr-22	(f) Actual May-22	(g) Actual Jun-22	(h) Actual Jul-22	(i) Actual Aug-22	(j) Actual Sep-22	(k) Actual Oct-22	(l) Actual Nov-22	(m) Actual Dec-22
1													
2	Beginning Balance	\$ 9,275	\$ 9,614	\$ 9,732	\$ 9,832	\$ 9,739	\$ 9,798	\$ 10,082	\$ 10,017	\$ 10,088	\$ 10,111	\$ 9,983	\$ 10,045
3	Ret. Earnings	(111)	118	100	(93)	59	49	(65)	71	23	(128)	62	89
4	Equity Infusion	450	-	-	-	-	235	-	-	-	-	-	-
5	Ending Balance	<u>\$ 9,614</u>	<u>\$ 9,732</u>	<u>\$ 9,832</u>	<u>\$ 9,739</u>	<u>\$ 9,798</u>	<u>\$ 10,082</u>	<u>\$ 10,017</u>	<u>\$ 10,088</u>	<u>\$ 10,111</u>	<u>\$ 9,983</u>	<u>\$ 10,045</u>	<u>\$ 10,134</u>
6													
7													
8													
9	Common Equity	Jan-23	Feb-23	Mar-23	Apr-23	May-23	Jun-23	Jul-23	Aug-23	Sep-23	Oct-23	Nov-23	Dec-23
10													
11	Beginning Balance	\$ 10,134	\$ 10,150	\$ 10,240	\$ 10,256	\$ 10,272	\$ 10,688	\$ 10,703	\$ 10,719	\$ 10,735	\$ 10,750	\$ 10,766	\$ 10,782
12	Ret. Earnings	16	16	16	16	16	16	16	16	16	16	16	16
13	Equity Infusion	-	75	-	-	400	-	-	-	-	-	-	-
14	Ending Balance	<u>\$ 10,150</u>	<u>\$ 10,240</u>	<u>\$ 10,256</u>	<u>\$ 10,272</u>	<u>\$ 10,688</u>	<u>\$ 10,703</u>	<u>\$ 10,719</u>	<u>\$ 10,735</u>	<u>\$ 10,750</u>	<u>\$ 10,766</u>	<u>\$ 10,782</u>	<u>\$ 10,798</u>
15													
16													
17													
18	Common Equity	Jan-24	Feb-24	Mar-24	Apr-24	May-24	Jun-24	Jul-24	Aug-24	Sep-24	Oct-24	Nov-24	Dec-24
19													
20	Beginning Balance	\$ 10,798	\$ 10,813	\$ 11,179	\$ 11,195	\$ 11,210	\$ 11,226	\$ 11,592	\$ 11,608	\$ 11,623	\$ 11,639	\$ 11,655	\$ 11,670
21	Ret. Earnings	16	16	16	16	16	16	16	16	16	16	16	16
22	Equity Infusion	-	350	-	-	-	350	-	-	-	-	-	-
23	Ending Balance	<u>\$ 10,813</u>	<u>\$ 11,179</u>	<u>\$ 11,195</u>	<u>\$ 11,210</u>	<u>\$ 11,226</u>	<u>\$ 11,592</u>	<u>\$ 11,608</u>	<u>\$ 11,623</u>	<u>\$ 11,639</u>	<u>\$ 11,655</u>	<u>\$ 11,670</u>	<u>\$ 11,686</u>
24													
25													
26													
27	Common Equity	Jan-25	Feb-25										
28													
29	Beginning Balance	\$ 11,686	\$ 11,702										
30	Ret. Earnings	16	16										
31	Equity Infusion	-	350										
32	Ending Balance	<u>\$ 11,702</u>	<u>\$ 12,068</u>										
33													
34													
35	Test Year 13-Month Average		<u>\$ 11,543</u>										

	Test Year Impact of Retained Earnings			Test Year Impact of Equity Infusions		
	2023	2024	2025	2023	2024	2025
Jan	\$ 16	\$ 204	\$ 393	\$ -	\$ 475	\$ 1,175
Feb	\$ 31	\$ 220	<u>\$ 409</u>	\$ 75	\$ 825	<u>\$ 1,525</u>
Mar	\$ 47	\$ 236		\$ 75	\$ 825	
Apr	\$ 63	\$ 251		\$ 75	\$ 825	
May	\$ 79	\$ 267		\$ 475	\$ 825	
Jun	\$ 94	\$ 283		\$ 475	\$ 1,175	
Jul	\$ 110	\$ 299		\$ 475	\$ 1,175	
Aug	\$ 126	\$ 314		\$ 475	\$ 1,175	
Sep	\$ 141	\$ 330		\$ 475	\$ 1,175	
Oct	\$ 157	\$ 346		\$ 475	\$ 1,175	
Nov	\$ 173	\$ 361		\$ 475	\$ 1,175	
Dec	<u>\$ 189</u>	<u>\$ 377</u>		<u>\$ 475</u>	<u>\$ 1,175</u>	
13-Month Avg.		<u>\$ 315</u>				<u>\$ 1,094</u>

Schedule D-1b

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Development of Capital Structure
for the Projected Year Ending February 28, 2025

Case No.: U-21389
Exhibit No.: A-14 (MRB-3)
Schedule: D-1b
Page: 1 of 1
Witness: MRBleckman
Date: May 2023

Line No.	(a) Description	(b) Financial Basis	(c) MPSC Ratemaking Basis
I.	LONG-TERM DEBT:		
a.	First Mortgage Bonds	Include	Include
b.	Trust Preferred Securities	Include	Include
c.	Other Subordinated LTD	Include	Include
d.	Unamortized Debt Premium	Include	Include
e.	Unamortized Debt Discount	Include	Include
f.	Unamortized Debt Expense	Include	Include
g.	Current Maturities	Exclude	Include
h.	Capitalized Leases	Include	Exclude
II.	SHORT-TERM DEBT	Include	Include
III.	PREFERRED STOCK EXPENSE		
a.	Preferred Stock	Include	Include
b.	Trust Preferred Securities	Exclude	Exclude
c.	Preferred Stock Expense	Include	Include
IV.	COMMON EQUITY		
a.	Common Stock Issued	Include	Include
b.	Premium on Common Stock	Include	Include
c.	Donations Received From Stockholders	Include	Include
d.	Common Stock Expense	Include	Include
e.	Gain on Reacquired Stock	Include	Include
f.	Miscellaneous Paid In Capital	Include	Include
g.	Mark-to-Market Accounting	Include	Exclude
h.	Appropriated Retained Earnings	Include	Include
i.	Unappropriated Retained Earnings	Include	Include
j.	FAS 90 (Abandoned Plant)	Include	Include
V.	DEFERRED ITC	Exclude	Include
VI.	DEFERRED TAXES	Exclude	Include
VII.	DEFERRED JDITC	Exclude	Include

Schedule D-3

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Short-Term Debt

for the Projected Year Ending February 28, 2025

(in millions)

Case No.: U-21389

Exhibit No.: A-14 (MRB-5)

Schedule: D-3

Page: 1 of 2

Witness: MRBleckman

Date: May 2023

Line No.	Description	(a) Average Borrowings	(b) Cost of Borrowings	(c) Cost Rate
1	Short-Term Debt - Short-Term Liquidity Facilities	\$ 242.7	\$ 12.0	
2				
3	Short-Term Debt - Renewable Liability	51.3	2.1	
4				
5	Total Short-Term Debt	<u>\$ 294.0</u>	<u>\$ 14.1</u>	<u>4.79%</u>

Sources:

Column (a): Average borrowings per Exhibit A-14 (MRB-7), Schedule D-6.

Column (b): Short-Term Debt - Short-Term Liquidity Facilities cost per Exhibit A-14 (MRB-5), Schedule D-3, page 2.

Short-Term Debt - Renewable Liability cost equal to the average borrowings (column (a)) times the projected interest on borrowings rate of 4.06%, per Exhibit A-14 (MRB-5), Schedule D-3, page 2.

Column (c) = column (b)/column (a).

Schedule D-3

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Short-Term Debt - Short-Term Liquidity Facilities
for the Projected Year Ending February 28, 2025
(in millions)

Case No.: U-21389
Exhibit No.: A-14 (MRB-5)
Schedule: D-3
Page: 2 of 2
Witness: MRBleckman
Date: May 2023

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
	<u>Summary</u>		<u>Test Year Assumption</u>					
	Facility	Agreement Date	Expiration	Facility Amount	Less: Avg. Borrowings	Less: Letters of Credit	Amount Unused ^a	Upfront Fee Amort
					(A)	(B)	(C)	(D)
1	1. JPMorgan Revolver	Jun. 2018	Dec. 2027	\$ 1,100.0	\$ -	\$ 29.3	\$ 1,070.7	\$ 0.3
2	2. Commercial Paper Facility	Sep. 2014	N/A	\$ 500.0	\$ 242.7	\$ -	\$ -	\$ -
3	3. Scotiabank Revolver	Nov. 2018	Nov. 2024	\$ 250.0	\$ -	\$ 27.3	\$ 222.7	\$ 0.3
4								
5								
6								
7								
8								
9								
10	Cost of Short-Term Debt - Short-Term Liquidity Facilities							
11								
12	(A) Interest on Borrowings - Calculated on the projected drawn balance at SOFR plus the spread on borrowings.							
13								
14			Plus:	Projected	Avg.			
15	Facility	SOFR ^b	Spread	Rate	Borrowings	Cost		
16	1. JPMorgan Revolver	4.06%	0.875%	4.94%	\$ -	\$ -		
17	2. Commercial Paper Facility	4.06%	0.000%	4.06%	242.7	9.9		
18	3. Scotiabank Revolver	4.06%	0.750%	4.81%	-	-		
19						\$ 9.9		
20								
21	(B) Letter of Credit Fees - Calculated on the projected letters of credit outstanding at a rate equal to the spread.							
22								
23				Projected	Letters			
24	Facility	Letter of Credit Type		Rate	of Credit	Cost		
25	1. JPMorgan Revolver	Regular Operating		0.875%	\$ 29.3	\$ 0.3		
26	2. Commercial Paper Facility	N/A		N/A	-	-		
27	3. Scotiabank Revolver	MISO Margin & Other		0.750%	27.3	0.2		
28						\$ 0.5		
29								
30	(C) Unused (Commitment) Fees - Calculated on the unused portion of the revolver at a rate stated in the facility agreement.							
31								
32				Projected	Amount			
33				Rate	Unused ^a	Cost		
34	1. JPMorgan Revolver			0.075%	\$ 1,070.7	\$ 0.8		
35	2. Commercial Paper Facility			N/A	-	-		
36	3. Scotiabank Revolver			0.075%	\$ 222.7	\$ 0.2		
37						\$ 1.0		
38								
39	(D) Amortization / Expense of Facility Fees - Fees paid upfront at the inception or amendment to the facility, amortized over the life of the facility.							
40								
41						Annual		
42						Cost		
43	1. JPMorgan Revolver					\$ 0.3		
44	2. Commercial Paper Facility					-		
45	3. Scotiabank Revolver					0.3		
46						\$ 0.6		
47								
48								
49	Total Cost of Short-Term Debt - Revolver / Commercial Paper						\$ 12.0	
50								
51								

^a Commercial Paper drawn balances go against, or are "backstopped," by the Company's JPMorgan Revolver. To the extent amounts are borrowed under the Commercial Paper Facility, the availability of the JPMorgan Revolver are reduced. These borrowings do not, however, reduce the "unused" portion of the revolver in calculating the unused (commitment) fees.

^b Projected SOFR rate per Exhibit A-14 (MRB-4), Schedule D-2. Forecasted SOFR assumed to closely approximate commercial paper rate.

Schedule D-4

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Preferred Stock

for the Projected Year Ending February 28, 2025

Case No.: U-21389
Exhibit No.: A-14 (MRB-6)
Schedule: D-4
Page: 1 of 1
Witness: MRBleckman
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
						Net Proceeds Received By Company	Number Of Shares Outstanding	Total Value Of Net Proceeds (000)	Cost Rates	Annual Cost (000)
Line No.	Description	Annual Dividend	Par Value	Finance Expense	(Premium) Discount					
	PREFERRED STOCK:									
1	\$4.500 Series	\$4.500	\$100.00	\$0.00	\$0.00	\$100.00	373,148	<u>\$37,315</u>	4.50%	<u>\$1,679</u>

Calculations:

Column (i) = (column (h)*column (b))/column (f)/column (h).

Column (j) = column (h)*column (i).

Schedule D-6

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Short-Term Debt Utilization
for the Projected Year Ending February 28, 2025
(in millions)

Case No.: U-21389
Exhibit No.: A-14 (MRB-7)
Schedule: D-6
Page: 1 of 1
Witness: MRBleckman
Date: May 2023

Ending Short-Term Debt - Short-Term Liquidity Facilities

Line No.	(a) Actual Jan-22	(b) Actual Feb-22	(c) Actual Mar-22	(d) Actual Apr-22	(e) Actual May-22	(f) Actual Jun-22	(g) Actual Jul-22	(h) Actual Aug-22	(i) Actual Sep-22	(j) Actual Oct-22	(k) Actual Nov-22	(l) Actual Dec-22
1												
2	\$ -	\$ 238	\$ -	\$ -	\$ 122	\$ 77	\$ 402	\$ 34	\$ 550	\$ 682	\$ 1,044	\$ 1,095
3												
4												
5	Jan-23	Feb-23	Mar-23	Apr-23	May-23	Jun-23	Jul-23	Aug-23	Sep-23	Oct-23	Nov-23	Dec-23
6												
7	\$ 500	\$ 19	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 320	\$ 300	\$ 521	\$ 565
8												
9												
10	Jan-24	Feb-24	Mar-24	Apr-24	May-24	Jun-24	Jul-24	Aug-24	Sep-24	Oct-24	Nov-24	Dec-24
11												
12	\$ 375	\$ 354	\$ 255	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 290	\$ 300	\$ 485	\$ 500
13												
14												
15	Jan-25	Feb-25										
16												
17	\$ 290	\$ 325										
18												
19	Test Year 13-Month Avg \$ 243											

Ending Short-Term Debt - Renewable Liability Balance ^a

	Actual Jan-22	Actual Feb-22	Actual Mar-22	Actual Apr-22	Actual May-22	Actual Jun-22	Actual Jul-22	Actual Aug-22	Actual Sep-22	Actual Oct-22	Actual Nov-22	Actual Dec-22
20												
21												
22												
23	\$ 20	\$ 29	\$ 29	\$ 35	\$ 36	\$ 32	\$ 33	\$ 32	\$ 23	\$ 27	\$ 35	\$ 32
24												
25												
26	Jan-23	Feb-23	Mar-23	Apr-23	May-23	Jun-23	Jul-23	Aug-23	Sep-23	Oct-23	Nov-23	Dec-23
27												
28	\$ 32	\$ 33	\$ 33	\$ 34	\$ 34	\$ 35	\$ 35	\$ 36	\$ 36	\$ 37	\$ 37	\$ 38
29												
30												
31	Jan-24	Feb-24	Mar-24	Apr-24	May-24	Jun-24	Jul-24	Aug-24	Sep-24	Oct-24	Nov-24	Dec-24
32												
33	\$ 39	\$ 41	\$ 43	\$ 45	\$ 46	\$ 48	\$ 50	\$ 51	\$ 53	\$ 55	\$ 57	\$ 58
34												
35												
36	Jan-25	Feb-25										
37												
38	\$ 59	\$ 61										
39												
40	Test Year 13-Month Avg \$ 51											
41												
42												
43												

^aProjected year-end Renewable Liability balances are consistent with the Company's Renewable Energy proceeding reflected in Case No. U-21197.

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Current and Historical Credit Ratings
for the Projected Year Ending February 28, 2025

Case No.: U-21389
Exhibit No.: A-32 (MRB-8)
Page: 1 of 1
Witness: MRBleckman
Date: May 2023

Line No.	Credit Ratings					
	(a)	(b)	(c)	(d)	(e)	(f)
1		Standard & Poors Ratings at Year End				
2	Current	2022	2021	2020	2019	2018
3	Consumers Energy Company					
4	Senior Secured Debt	A	A	A	A	A
5	Commercial Paper	A-2	A-2	A-2	A-2	A-2
6	Senior Unsecured Debt	N/A	N/A	N/A	N/A	N/A
7	Hybrid Preferred Securities	N/A	N/A	N/A	N/A	N/A
8	Preferred Stock	N/A	N/A	N/A	N/A	N/A
9	Outlook	Stable	Stable	Stable	Stable	Stable
10						
11	CMS Energy Corporation					
12	Senior Secured Debt	N/A	N/A	N/A	N/A	N/A
13	Senior Unsecured Debt	BBB	BBB	BBB	BBB	BBB
14	Junior Subordinated Debt	BBB-	BBB-	BBB-	BBB-	BBB-
15	Hybrid Preferred Securities	N/A	N/A	N/A	N/A	N/A
16	Preferred Stock	BBB-	BBB-	N/A	N/A	N/A
17	Outlook	Stable	Stable	Stable	Stable	Stable
18						
19						
20		Moody's Ratings at Year End				
21	Current	2022	2021	2020	2019	2018
22	Consumers Energy Company					
23	Senior Secured Debt	A1	A1	Aa3	Aa3	Aa3
24	Commercial Paper	P-2	P-2	P-1	P-1	P-1
25	Senior Unsecured Debt	N/A	N/A	N/A	N/A	N/A
26	Hybrid Preferred Securities	N/A	N/A	N/A	N/A	N/A
27	Preferred Stock	Baa1	Baa1	A3	A3	A3
28	Outlook	Stable	Stable	Stable	Stable	Stable
29						
30	CMS Energy Corporation					
31	Senior Secured Debt	A3	A3	A3	A3	A3
32	Senior Unsecured Debt	Baa2	Baa2	Baa1	Baa1	Baa1
33	Junior Subordinated Debt	Baa3	Baa3	Baa2	Baa2	Baa2
34	Hybrid Preferred Securities	N/A	N/A	N/A	N/A	N/A
35	Preferred Stock	Ba1	Ba1	N/A	N/A	N/A
36	Outlook	Stable	Stable	Negative	Stable	Stable
37						
38						
39		Fitch Ratings at Year End				
40	Current	2022	2021	2020	2019	2018
41	Consumers Energy Company					
42	Senior Secured Debt	A+	A+	A+	A+	A+
43	Commercial Paper	F-2	F-2	F-2	F-2	F-2
44	Senior Unsecured Debt	A	A	A	A	A
45	Hybrid Preferred Securities	N/A	N/A	N/A	N/A	N/A
46	Preferred Stock	BBB+	BBB+	BBB+	BBB+	BBB+
47	Outlook	Stable	Stable	Stable	Stable	Stable
48						
49	CMS Energy Corporation					
50	Senior Secured Debt	BBB+	BBB+	BBB+	BBB+	BBB+
51	Senior Unsecured Debt	BBB	BBB	BBB	BBB	BBB
52	Junior Subordinated Debt	BB+	BB+	BB+	BB+	BB+
53	Hybrid Preferred Securities	N/A	N/A	N/A	N/A	N/A
54	Preferred Stock	BB+	BB+	N/A	N/A	N/A
55	Outlook	Stable	Stable	Stable	Stable	Stable

Line No.	(a) Issue Date	(b) Ticker	(c) Issuer	(d) Type	(e) Amt (\$mm)	(f) Coupon	(g) Ratings		(i) Maturity	(j) Issue Spread (bp)	(k) Category
							Moody's	S&P			
1	01/06/20	EIX	Southern California Edison Co (reopening)	Secured	100	2.850%	A3	A-	08/01/29	+90	
2	01/06/20	EIX	Southern California Edison Co	Secured	500	3.650%	A3	A-	02/01/50	+140	2
3	01/06/20	SRE	Southern California Gas	Secured	650	2.550%	Aa2	A+	02/01/30	+77	
4	01/06/20	DUK	Duke Energy Carolinas	Secured	500	2.450%	Aa2	A	02/01/30	+68	
5	01/06/20	DUK	Duke Energy Carolinas (reopening)	Secured	400	3.200%	Aa2	A	08/15/49	+88	1
6	01/07/20	ES	Eversource Energy	Unsecured	350	3.450%	Baa1	BBB+	01/15/50	+115	3
7	01/07/20	PEG	Public Service Electric & Gas	Secured	300	2.450%	Aa3	A	01/15/30	+65	
8	01/07/20	PEG	Public Service Electric & Gas	Secured	300	3.150%	Aa3	A	01/01/50	+85	1
9	01/08/20	SO	Georgia Power Company	Unsecured	700	2.100%	Baa1	A-	07/30/23	+50	
10	01/08/20	SO	Georgia Power Company	Unsecured	300	2.650%	Baa1	A-	09/15/29	+95	
11	01/08/20	SO	Georgia Power Company	Unsecured	500	3.700%	Baa1	A-	01/30/50	+135	2
12	01/22/20	NRUC	National Rural Utilities Cooperative Finance Corp	Unsecured	500	1.750%	A2	A	01/21/22	+27	
13	01/22/20	NRUC	National Rural Utilities Cooperative Finance Corp	Secured	500	2.400%	A1	A	03/15/30	+68	
14	01/28/20	BRKHEC	Nevada Power Company	Secured	425	2.400%	A2	A+	04/01/30	+78	
15	01/28/20	BRKHEC	Nevada Power Company	Secured	300	3.125%	A2	A+	08/01/50	+102	
16	02/11/20	DTE	DTE Electric	Secured	600	2.250%	Aa3	A	03/01/30	+68	
17	02/11/20	DTE	DTE Electric	Secured	500	2.950%	Aa3	A	03/01/50	+90	1
18	02/18/20	EXC	Commonwealth Edison	Secured	350	2.200%	A1	A	03/01/30	+68	
19	02/18/20	EXC	Commonwealth Edison	Secured	650	3.000%	A1	A	03/01/50	+100	1
20	02/18/20	FE	FirstEnergy Corporation	Unsecured	300	2.050%	Baa3	BBB-	03/01/25	+70	
21	02/18/20	FE	FirstEnergy Corporation	Unsecured	650	2.650%	Baa3	BBB-	03/01/30	+110	
22	02/18/20	FE	FirstEnergy Corporation	Unsecured	850	3.400%	Baa3	BBB-	03/01/50	+140	
23	03/03/20	ETR	Entergy Louisiana	Secured	350	2.900%	A2	A	03/15/51	+130	1
24	03/03/20	AEP	American Electric Power	Unsecured	400	2.300%	Baa1	BBB+	03/01/30	+130	
25	03/03/20	AEP	American Electric Power	Unsecured	400	3.250%	Baa1	BBB+	03/01/50	+165	3
26	03/04/20	EIX	Southern California Edison	Secured	400	2.250%	A3	A-	06/01/30	+125	
27	03/04/20	EIX	Southern California Edison (reopening)	Secured	700	3.650%	A3	A-	02/01/50	+170	2
28	03/05/20	ETR	Entergy Texas (reopening)	Secured	175	3.550%	Baa1	A	09/30/49	+138	1
29	03/10/20	DUK	Duke Energy Indiana	Secured	550	2.750%	Aa3	A	04/01/50	+165	1
30	03/13/20	ETR	Entergy Arkansas (reopening)	Secured	100	4.000%	A2	A	06/01/28	+175	
31	03/13/20	AEP	Ohio Power Company	Unsecured	350	2.600%	A2	A-	04/01/30	+170	
32	03/17/20	CMS	Consumers Energy	Secured	575	3.500%	Aa3	A	08/01/51	+200	1
33	03/17/20	D	Dominion Energy, Inc.	Unsecured	400	3.300%	Baa2	BBB	03/15/25	+265	
34	03/17/20	D	Dominion Energy, Inc.	Unsecured	350	3.600%	Baa2	BBB	03/15/27	+275	
35	03/17/20	ONCRTLX	Oncor Electric Delivery Company	Secured	400	2.750%	A2	A+	05/15/30	+175	
36	03/17/20	ONCRTLX	Oncor Electric Delivery Company	Secured	400	3.700%	A2	A+	05/15/50	+205	
37	03/17/20	AEE	Union Electric Corp	Secured	465	2.950%	A2	A	03/15/30	+200	
38	03/20/20	BRKHEC	Berkshire Hathaway Energy Company	Unsecured	1,250	4.050%	A3	A-	04/15/25	+355	
39	03/23/20	ES	NSTAR Electric	Unsecured	400	3.950%	A1	A	04/01/30	+325	
40	03/24/20	NEE	Florida Power & Light Company	Secured	1,100	2.850%	Aa2	A+	04/01/25	+238	
41	03/25/20	BRKHEC	Berkshire Hathaway Energy Company	Unsecured	1,100	3.700%	A3	A-	04/15/30	+285	
42	03/25/20	BRKHEC	Berkshire Hathaway Energy Company	Unsecured	900	4.250%	A3	A-	04/15/50	+285	2
43	03/26/20	ED	Consolidated Edison Company of New York	Unsecured	600	3.350%	Baa1	A-	04/01/30	+255	
44	03/26/20	ED	Consolidated Edison Company of New York	Unsecured	1,000	3.950%	Baa1	A-	04/01/50	+255	2
45	03/27/20	XEL	Xcel Energy	Unsecured	600	3.400%	Baa1	BBB+	06/01/30	+270	
46	03/30/20	EXC	Exelon Corporation	Unsecured	1,250	4.050%	Baa2	BBB	04/15/30	+338	
47	03/30/20	EXC	Exelon Corporation	Unsecured	750	4.700%	Baa2	BBB	04/15/50	+338	4
48	03/30/20	EIX	Southern California Edison (reopening)	Secured	600	3.700%	A3	A-	08/01/25	+275	
49	03/30/20	LNT	Wisconsin Electric Power & Light	Unsecured	350	3.650%	A3	A	04/01/50	+235	1
50	03/30/20	OGE	Oklahoma Gas & Electric	Unsecured	300	3.250%	A3	A-	04/01/30	+260	
51	03/30/20	AEP	AEP Transmission	Unsecured	525	3.650%	A2	A-	04/01/50	+235	2
52	03/31/20	DTE	DTE Electric Co.	Secured	600	2.625%	Aa3	A	03/01/31	+195	
53	03/31/20	D	Dominion Energy	Unsecured	1,500	3.375%	Baa2	BBB	04/01/30	+280	
54	03/31/20	SRE	San Diego Gas & Electric	Secured	400	3.320%	A2	A	04/15/50	+200	1
55	03/31/20	AEE	Ameren Corporation	Unsecured	800	3.500%	Baa1	BBB	01/15/31	+285	
56	03/31/20	EIX	Edison International	Unsecured	400	4.950%	Baa3	BBB-	04/15/25	+463	
57	03/31/20	IDA	Idaho Power (reopening)	Secured	220	4.200%	A1	A-	03/01/48	+210	
58	04/01/20	NEE	NextEra Energy Capital Holdings	Unsecured	1,250	2.750%	Baa1	BBB+	05/01/25	+240	
59	04/01/20	PPL	PPL Capital Funding	Unsecured	1,000	4.125%	Baa2	BBB+	04/15/30	+350	
60	04/01/20	SO	Southern Company	Unsecured	1,000	3.700%	Baa2	BBB+	04/30/30	+310	
61	04/02/20	EVRG	Evergy Kansas Central	Secured	500	3.450%	A2	A	04/15/50	+220	1
62	04/06/20	BRKHEC	PacifiCorp	Secured	400	2.700%	A1	A+	09/15/30	+205	
63	04/06/20	BRKHEC	PacifiCorp	Secured	600	3.300%	A1	A+	03/15/51	+205	
64	04/06/20	FTSCN	Tucson Electric Power	Unsecured	350	4.000%	A3	A-	04/15/50	+275	2
65	04/07/20	AGR	Avangrid	Unsecured	750	3.200%	Baa1	BBB	04/15/25	+275	
66	04/07/20	NI	NISource	Unsecured	1,000	3.600%	Baa2	BBB+	05/01/30	+290	
67	04/07/20	NGGLN	Narragansett Electric	Unsecured	600	3.395%	A3	A-	04/19/30	+265	
68	04/08/20	AES	IPALCO Enterprises Inc	Secured	475	4.250%	Baa3	BBB-	05/01/30	+350	
69	04/08/20	AWK	American Water	Unsecured	500	2.800%	Baa1	A	05/01/30	+210	
70	04/08/20	AWK	American Water	Unsecured	500	3.450%	Baa1	A	05/01/50	+210	1
71	04/13/20	WTRG	Essential Utilities	Unsecured	500	2.704%	Baa2	A-	04/15/30	+195	
72	04/13/20	WTRG	Essential Utilities	Unsecured	600	3.351%	Baa2	A-	04/15/50	+195	2
73	04/29/20	OGS	ONE Gas Inc	Unsecured	300	2.000%	A2	A	05/15/30	+145	
74	04/29/20	CMS	Consumers Energy	Secured	525	2.500%	Aa3	A	05/01/60	+130	
75	05/05/20	NEE	NextEra Energy Capital Holdings	Unsecured	2,000	2.250%	Baa1	BBB+	06/01/30	+160	
76	05/06/20	PEG	Public Service Electric & Gas	Secured	375	2.700%	Aa3	A	05/01/50	+130	1
77	05/06/20	TVA	Tennessee Valley Authority	Unsecured	1,000	0.750%	Aaa	AA+	05/15/25	+43	
78	05/08/20	XEL	Public Service Co of Colorado	Secured	375	1.900%	A1	A	01/15/31	+125	
79	05/08/20	XEL	Public Service Co of Colorado	Secured	375	2.700%	A1	A	01/15/51	+140	1
80	05/11/20	ITC	ITC Holdings	Unsecured	700	2.950%	Baa2	BBB+	05/14/30	+225	
81	05/11/20	XEL	Southwestern Public Service Co	Secured	350	3.150%	A3	A	05/01/50	+175	1
82	05/12/20	EXC	Exelon Generation	Unsecured	900	3.250%	Baa2	BBB+	05/14/25	+295	
83	05/12/20	AEP	Appalachian Power	Unsecured	500	3.700%	Baa1	A-	05/01/50	+235	2
84	05/13/20	DUK	Duke Energy Corporation	Unsecured	500	2.450%	Baa1	BBB+	06/01/30	+185	
85	05/13/20	ETR	Entergy Corporation	Unsecured	600	2.800%	Baa2	BBB	06/15/30	+225	
86	05/13/20	ETR	Entergy Corporation	Unsecured	600	3.750%	Baa2	BBB	06/15/50	+250	4
87	05/14/20	PSD	Puget Energy	Secured	650	4.100%	Baa3	BBB-	06/15/30	+350	
88	05/18/20	EVRG	Evergy Metro Inc	Secured	400	2.250%	A2	A+	06/01/30	+155	
89	05/18/20	DUK	Duke Energy Ohio	Secured	400	2.125%	A2	A	06/01/30	+140	
90	05/18/20	DUK	Piedmont Natural Gas	Unsecured	400	3.350%	A3	A-	06/01/50	+195	2
91	05/18/20	XEL	Northern States Power Co/WI	Secured	100	3.050%	Aa3	A	05/01/51	+170	1
92	05/19/20	PNW	Arizona Public Service Co	Unsecured	600	3.350%	A2	A-	05/15/50	+195	2
93	05/19/20	PPL	Kentucky Utilities	Secured	500	3.300%	A1	A	06/01/50	+190	1
94	05/19/20	ETR	Entergy Mississippi	Secured	170	3.500%	A2	A	06/01/51	+215	1
95	05/26/20	LNT	Interstate Power & Light Company	Unsecured	400	2.300%	Baa1	A-	06/01/30	+165	
96	06/01/20	EXC	PECO Energy	Secured	350	2.800%	Aa3	A	06/15/50	+135	1

97
98 FRN: Floating Rate Note
99 SOFR: Secured Overnight Financing Rate
100 3mL: Three-Month London Interbank Offered Rate (LIBOR)
101 Source: Barclays Bank

Line No.	(a) Issue Date	(b) Ticker	(c) Issuer	(d) Type	(e) Amt (\$mm)	(f) Coupon	(g) Ratings		(i) Maturity	(j) Issue Spread (bp)	(k) Category
							Moody's	S&P			
1	07/07/21	ENELIM	Enel Finance International NV	Unsecured	750	2.875%	Baa1	BBB+	07/12/41	+110	
2	07/07/21	ENELIM	Enel Finance International NV	Unsecured	1,000	1.875%	Baa1	BBB+	07/12/28	+85	
3	07/07/21	ENELIM	Enel Finance International NV	Unsecured	1,250	1.375%	Baa1	BBB+	07/12/26	+70	
4	07/07/21	BRKHEC	Pacificorp	Secured	1,000	2.900%	A1	A+	06/15/52	+98	
5	07/20/21	BRKHEC	MidAmerican Energy Company	Secured	500	2.700%	Aa2	A+	08/01/52	+85	
6	08/02/21	CMS	Consumers Energy Company	Secured	300	2.650%	A1	A	08/15/52	+78	1
7	08/02/21	AEP	AEP Transmission	Unsecured	450	2.750%	A2	A-	08/15/51	+90	2
8	08/02/21	EIX	Southern California Edison Co	Secured	400	0.700%	A3	A-	08/01/23	+50	
9	08/02/21	EIX	Southern California Edison Co	Secured	450	0.975%	A3	A-	08/01/24	+60	
10	08/05/21	EXC	Commonwealth Edison Co	Secured	450	2.750%	A1	A	09/01/51	+87	1
11	08/09/21	SRE	San Diego Gas and Electric	Secured	750	2.950%	A1	A	08/15/51	+100	1
12	08/09/21	DUK	Duke Energy Progress	Secured	450	2.900%	Aa3	A	08/15/51	+93	1
13	08/09/21	DUK	Duke Energy Progress	Secured	650	2.000%	Aa3	A	08/15/31	+70	
14	08/10/21	D	Dominion Energy	Unsecured	1,000	2.250%	Baa2	BBB	08/15/31	+92	
15	08/10/21	ES	Eversource Energy	Unsecured	350	FRN	Baa1	BBB+	08/15/23	SOFR+25	
16	08/10/21	ES	Eversource Energy	Unsecured	300	1.400%	Baa1	BBB+	08/15/26	+60	
17	08/10/21	ES	NSTAR Electric Company	Unsecured	300	1.950%	A1	A	08/15/31	+67	
18	08/17/21	PEG	Public Service Electric and Gas Co	Secured	425	1.900%	Aa3	A	08/15/31	+68	
19	08/18/21	SWX	Southwest Gas Corp	Unsecured	300	3.180%	Baa1	A-	08/15/51	+128	2
20	08/23/21	BKH	Black Hills Corporation	Unsecured	600	1.037%	Baa2	BBB+	08/23/24	+60	
21	09/07/21	EXC	PECO Energy	Secured	375	2.850%	Aa3	A	09/15/51	+88	1
22	09/07/21	SO	Southern Co Gas Capital Corp	Unsecured	450	3.150%	Baa1	A-	09/30/51	+123	2
23	09/08/21	PSD	Puget Sound Energy	Secured	450	2.893%	A2	A-	09/15/51	+95	2
24	09/08/21	CNL	Cleco Power	Unsecured	325	FRN	A3	BBB+	06/15/23	SOFR+50	
25	09/09/21	AES	Ohio Power	Unsecured	500	2.900%	A3	A-	10/01/51	+103	2
26	09/13/21	SO	The Southern Company	Pref/Hybrid	1,250	1.875%	A-	BBB	09/15/81	+211	
27	09/13/21	TVA	Tennessee Valley Authority	Unsecured	500	1.500%	Aaa	AA+	09/15/31	+18	
28	09/13/21	LNT	Wisconsin Power & Light	Unsecured	300	1.950%	A3	A	09/16/31	+63	
29	09/21/21	AGR	New York State Electric & Gas	Unsecured	350	2.150%	Baa1	A-	10/01/31	+85	
30	09/21/21	ATO	Alamos Energy Corp	Unsecured	600	2.850%	A1	A-	02/15/52	+103	2
31	09/23/21	DUK	Duke Energy Corp	Pref/Hybrid	500	3.250%	Baa3	BBB-	01/15/82	+232	
32	09/28/21	ETR	Entergy Louisiana	Secured	1,000	0.950%	A2	A	10/01/24	+45	
33	10/27/21	NEE	NextEra Energy Capital Holdings Inc	Unsecured	1,300	FRN	Baa1	BBB+	11/03/23	SOFR+40	
34	10/29/21	XEL	Xcel Energy Inc	Unsecured	500	1.750%	Baa1	BBB+	03/15/27	+62	
35	10/29/21	XEL	Xcel Energy Inc	Unsecured	300	2.350%	Baa1	BBB+	11/15/31	+82	
36	11/01/21	AEP	Southwestern Electric Power Co	Unsecured	650	3.250%	Baa2	A-	11/01/51	+130	2
37	11/04/21	PEG	Public Service Enterprise Group	Unsecured	750	0.841%	Baa2	BBB	11/08/23	+43	
38	11/04/21	PEG	Public Service Enterprise Group	Unsecured	750	2.450%	Baa2	BBB	11/15/31	+95	
39	11/05/21	SO	Southern Company	Unsecured	600	FRN	Baa2	BBB	05/10/23	SOFR+37	
40	11/10/21	AEP	American Electric Power Co Inc	Pref/Hybrid	750	3.875%	Baa3	BBB	02/15/62	+268	
41	11/10/21	ETR	Entergy Mississippi LLC	Secured	200	2.550%	A2	A	12/01/33	+105	
42	11/10/21	ONCRTX	Oncor Electric Delivery	Secured	300	2.750%	A2	A+	05/15/30	+58	
43	11/10/21	ONCRTX	Oncor Electric Delivery	Secured	500	2.700%	A2	A+	11/15/51	+88	
44	11/10/21	PCG	Pacific Gas & Electric Company	Secured	900	1.700%	Baa3	BBB-	11/15/23	+125	
45	11/10/21	PCG	Pacific Gas & Electric Company	Secured	300	FRN	Baa3	BBB-	11/14/22	SOFR+115	
46	11/10/21	PCG	Pacific Gas & Electric Company	Secured	550	3.250%	Baa3	BBB-	06/01/31	+158	
47	11/10/21	WEC	Wisconsin Public Service	Unsecured	450	2.850%	A2	A-	11/15/51	+95	2
48	11/15/21	SO	Alabama Power Company	Unsecured	700	3.000%	A1	A-	03/15/52	+100	2
49	11/15/21	AEE	Ameren Corp	Unsecured	500	1.950%	Baa1	BBB	03/15/27	+70	
50	11/15/21	SRE	Sempra Energy	Pref/Hybrid	1,000	4.125%	Baa3	BBB-	04/01/52	+287	
51	11/15/21	D	Virginia Electric and Power Co	Unsecured	500	2.300%	A2	BBB+	11/15/31	+70	
52	11/15/21	D	Virginia Electric and Power Co	Unsecured	500	2.950%	A2	BBB+	11/15/51	+95	3
53	11/16/21	NEE	Florida Power & Light Co	Secured	1,200	2.875%	Aa2	A+	12/04/51	+85	
54	11/16/21	LNT	Interstate Power and Light Co	Unsecured	300	3.100%	Baa1	A-	11/30/51	+110	2
55	11/18/21	D	Dominion Energy South Carolina	Secured	400	2.300%	A3	A	12/01/31	+72	
56	11/29/21	FE	American Transmission Systems	Unsecured	600	2.650%	A3	BBB	01/15/32	+115	
57	11/29/21	ED	Consolidated Edison Co of New York Inc	Unsecured	600	3.200%	Baa1	A-	12/01/51	+132	2
58	11/29/21	ED	Consolidated Edison Co of New York Inc (reopening)	Unsecured	150	2.400%	Baa1	A-	06/15/31	+87	
59	11/29/21	DUK	Duke Energy Florida	Secured	650	2.400%	A1	A	12/15/31	+90	
60	11/29/21	DUK	Duke Energy Florida	Secured	500	3.000%	A1	A	12/15/51	+115	1
61	12/01/21	D	Dominion Energy Inc	Pref/Hybrid	750	4.350%	Ba1	BBB-	04/15/27	+320	
62	12/01/21	EIX	Southern California Edison Co	Secured	550	FRN	A3	A-	12/02/22	SOFR+47	
63	12/01/21	SR	Spire Missouri Inc	Secured	300	FRN	A1	A	12/02/24	SOFR+50	
64	12/06/21	WEC	WEC Energy Group	Unsecured	500	2.200%	Baa1	BBB+	12/15/28	+85	
65	12/08/21	NEE	NextEra Energy Capital Holdings	Unsecured	1,000	1.875%	Baa1	BBB+	01/15/27	+62	
66	12/08/21	NEE	NextEra Energy Capital Holdings	Unsecured	1,000	2.440%	Baa1	BBB+	01/15/32	+92	
67	12/08/21	NEE	NextEra Energy Capital Holdings	Unsecured	500	3.000%	Baa1	BBB+	01/15/52	+112	3
68	12/09/21	NEE	NextEra Energy Capital Holdings	Pref/Hybrid	600	3.800%	Baa2	BBB	03/15/82	+255	
69	01/03/22	AEP	American Electric Power Co Inc	Unsecured	805	2.031%	Baa3	BBB+	03/15/24	+85	
70	01/04/22	NGGLN	Niagara Mohawk Power	Unsecured	400	2.759%	Baa1	BBB+	01/10/31	+105	
71	01/04/22	D	Virginia Electric and Power Company	Unsecured	600	2.400%	A2	BBB+	03/30/32	+70	
72	01/04/22	D	Virginia Electric and Power Company (reopening)	Unsecured	400	2.950%	A2	BBB+	11/15/51	+98	3
73	01/06/22	SRE	Sempra Infrastructure Partners LP	Unsecured	400	3.250%	Baa3	BBB-	01/15/32	+165	
74	01/10/22	EIX	Southern California Edison Company	Secured	500	2.750%	A3	A-	02/01/32	+98	
75	01/10/22	EIX	Southern California Edison Company	Secured	700	3.450%	A3	A-	02/01/52	+135	2
76	01/11/22	ATO	Alamos Energy Corp (reopening)	Unsecured	200	2.625%	A1	A-	09/15/29	+68	
77	01/12/22	AQNCN	Algonquin Power & Utilities	Pref/Hybrid	750	4.750%	NR	BB+	01/18/82	+325	
78	01/12/22	NEE	Florida Power & Light Company	Unsecured	1,000	FRN	A1	A	01/12/24	SOFR+38	
79	01/12/22	NEE	Florida Power & Light Company	Secured	1,500	2.450%	Aa2	A+	02/03/32	+73	
80	01/12/22	NGGLN	KeySpan Gas East Corp	Unsecured	400	3.586%	Baa1	BBB+	01/18/52	+150	3
81	02/15/22	FTSCN	Tucson Electric Power Co	Unsecured	325	3.250%	A3	A-	05/15/32	+125	
82	02/16/22	DTE	DTE Electric Company	Secured	500	3.000%	Aa3	A	03/01/32	+100	
83	02/16/22	DTE	DTE Electric Company	Secured	400	3.650%	Aa3	A	03/01/52	+130	1
84	02/16/22	PCG	Pacific Gas and Electric Co	Secured	1,000	3.250%	Baa3	BBB-	02/16/24	+175	
85	02/16/22	PCG	Pacific Gas and Electric Co	Secured	400	4.200%	Baa3	BBB-	03/01/29	+220	
86	02/16/22	PCG	Pacific Gas and Electric Co	Secured	450	4.400%	Baa3	BBB-	03/01/32	+240	
87	02/16/22	PCG	Pacific Gas and Electric Co	Secured	550	5.250%	Baa3	BBB-	03/01/52	+290	
88	02/22/22	ES	Eversource Energy	Unsecured	650	2.900%	Baa1	BBB+	03/01/27	+105	
89	02/22/22	ES	Eversource Energy	Unsecured	650	3.375%	Baa1	BBB+	03/01/32	+145	
90	02/23/22	LNT	Alliant Energy Finance LLC	Unsecured	350	3.600%	Baa2	BBB+	03/01/32	+165	
91	02/23/22	CNP	CenterPoint Energy Houston Electric	Secured	300	3.000%	A2	A	03/01/32	+107	
92	02/23/22	CNP	CenterPoint Energy Houston Electric	Secured	500	3.600%	A2	A	03/01/52	+137	1
93	03/01/22	DUK	Duke Energy Carolinas	Secured	500	2.850%	Aa3	A	03/15/32	+115	
94	03/01/22	DUK	Duke Energy Carolinas	Secured	650	3.550%	Aa3	A	03/15/52	+145	1
95	03/02/22	EXC	Exelon Corporation	Unsecured	650	2.750%	Baa2	BBB	03/15/27	+100	
96	03/02/22	EXC	Exelon Corporation	Unsecured	650	3.350%	Baa2	BBB	03/15/32	+145	
97	03/02/22	EXC	Exelon Corporation	Unsecured	700	4.100%	Baa2	BBB	03/15/52	+180	4
98	03/02/22	ETR	Entergy Arkansas (reopening)	Secured	200	4.200%	A2	A	04/01/49	+160	
99	03/02/22	SO	Alabama Power Company	Unsecured	700	3.050%	A1	A-	03/15/32	+120	
100	03/07/22	SRE	San Diego Gas & Electric Company	Secured	500	3.000%	A1	A	03/15/32	+128	

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Recent Utility Corporate Bond Issuances
for the Projected Year Ending February 28, 2025

Case No.: U-21389
Exhibit No.: A-33 (MRB-9)
Page: 6 of 6
Witness: MRBleckman
Date: May 2023

Line No.	(a) Issue Date	(b) Ticker	(c) Issuer	(d) Type	(e) Amt (\$mm)	(f) Coupon	(g) Moody's	(h) S&P	(i) Maturity	(j) Issue Spread (bp)	(k) Category
1	09/08/22	ITC	ITC Holdings Corporation	Unsecured	600	4.950%	Baa2	BBB+	09/22/27	+158	
2	09/08/22	TVA	Tennessee Valley Authority	Unsecured	500	4.250%	Aaa	AA+	09/15/52	+100	
3	09/12/22	CNP	CenterPoint Energy Houston Electric	Secured	500	4.450%	A2	A	10/01/32	+115	
4	09/12/22	CNP	CenterPoint Energy Houston Electric	Secured	300	4.850%	A2	A	10/01/52	+138	1
5	09/12/22	ES	NSTAR Electric Company	Unsecured	400	4.950%	A1	A	09/15/52	+147	1
6	09/12/22	NGGLN	Niagara Mohawk Power	Unsecured	500	5.783%	Baa1	BBB+	09/16/52	+225	3
7	09/14/22	WEC	Wisconsin Electric Power Company	Unsecured	500	4.750%	A2	A-	09/30/32	+135	
8	09/22/22	WEC	WEC Energy Group	Unsecured	500	5.000%	Baa1	BBB+	09/27/25	+90	
9	09/22/22	WEC	WEC Energy Group	Unsecured	400	5.150%	Baa1	BBB+	10/01/27	+128	
10	09/22/22	FE	Ohio Edison	Unsecured	300	5.500%	A3	BBB	01/15/33	+185	
11	09/28/22	ATO	Atmos Energy	Unsecured	300	5.450%	A1	A-	10/15/32	+152	
12	09/28/22	ATO	Atmos Energy	Unsecured	500	5.750%	A1	A-	10/15/52	+195	2
13	10/03/22	SO	The Southern Co	Unsecured	500	5.150%	Baa2	BBB	10/06/25	+110	
14	10/03/22	SO	The Southern Co	Unsecured	500	5.700%	Baa2	BBB	10/15/32	+205	
15	10/03/22	EDPPL	EDP Finance BV	Unsecured	500	6.300%	Baa3	BBB	10/11/27	+245	
16	10/06/22	ENELIM	Enel Finance International NV	Unsecured	750	6.800%	Baa1	BBB+	10/14/25	+275	
17	10/06/22	ENELIM	Enel Finance America LLC	Unsecured	1,000	7.100%	Baa1	BBB+	10/14/27	+325	
18	10/06/22	ENELIM	Enel Finance International NV	Unsecured	1,250	7.500%	Baa1	BBB+	10/14/32	+400	
19	10/06/22	ENELIM	Enel Finance International NV	Unsecured	1,000	7.750%	Baa1	BBB+	10/14/52	+440	3
20	10/17/22	BRKHEC	Nevada Power Company	Secured	400	5.900%	A2	A+	05/01/53	+190	
21	10/20/22	NRUC	National Rural Utilities Cooperative Finance	Unsecured	500	5.450%	A2	A-	10/30/25	+85	
22	10/20/22	NRUC	National Rural Utilities Cooperative Finance	Unsecured	350	5.800%	A2	A-	01/15/33	+158	
23	10/31/22	AEP	American Electric Power Co Inc	Unsecured	500	5.750%	Baa2	BBB+	11/1/2027	+155	
24	10/31/22	AEP	American Electric Power Co Inc	Unsecured	500	5.950%	Baa2	BBB+	11/1/2032	+195	
25	11/03/22	EIX	Southern California Edison Co	Secured	750	5.950%	A3	A-	11/1/2032	+190	
26	11/03/22	EIX	Southern California Edison Co	Secured	750	5.850%	A3	A-	11/1/2027	+150	
27	11/04/22	PEG	Public Service Enterprise Group	Unsecured	700	5.850%	Baa2	BBB	11/15/2027	+155	
28	11/04/22	PNW	Arizona Public Service Co	Unsecured	400	6.350%	A3	BBB+	12/15/2032	+220	
29	11/07/22	EIX	Edison International	Unsecured	550	6.950%	Baa3	BBB-	11/15/2029	+280	
30	11/07/22	WEC	Wisconsin Public Service	Unsecured	300	5.350%	A2	A-	11/10/2025	+72	
31	11/07/22	SRE	Southern California Gas Co	Secured	600	6.350%	Aa3	A+	11/15/2052	+205	
32	11/07/22	DUK	Duke Energy Florida	Secured	500	5.950%	A1	A	11/15/2052	+167	1
33	11/08/22	AES	Indianapolis Power & Light	Secured	350	5.650%	A2	A-	12/1/2032	+155	
34	11/08/22	AEE	Ameren Illinois Co	Secured	350	5.900%	A1	A	12/1/2052	+163	1
35	11/09/22	ED	Consolidated Edison Co of New York Inc	Unsecured	700	6.150%	Baa1	A-	11/15/2052	+185	2
36	11/16/22	D	Dominion Energy	Unsecured	850	5.375%	Baa2	BBB	11/15/32	+175	
37	11/18/22	PCG	PG&E Recovery Funding LLC	Secured	215	5.045%	Aaa	AAA	07/15/34	+105	
38	11/18/22	PCG	PG&E Recovery Funding LLC	Secured	200	5.256%	Aaa	AAA	01/15/40	+135	
39	11/18/22	PCG	PG&E Recovery Funding LLC	Secured	568	5.536%	Aaa	AAA	07/15/49	+140	
40	11/21/22	NGGLN	New England Power Company	Unsecured	300	5.936%	A3	BBB+	11/25/52	+200	3
41	11/28/22	EVRG	Evergy Missouri West	Secured	300	5.150%	A3	A	12/15/2027	+130	
42	11/29/22	SWX	Southwest Gas Corp	Unsecured	300	5.800%	Baa1	BBB	12/1/2027	+190	
43	11/29/22	BRKHEC	PacifiCorp	Secured	1,100	5.350%	A1	A+	12/1/2053	+153	
44	11/30/22	PEG	Public Service Electric and Gas Co	Secured	400	4.900%	A1	A	12/15/32	+113	
45	12/05/22	DUK	Duke Energy Corporation	Unsecured	500	5.000%	Baa2	BBB	12/08/25	+88	
46	12/05/22	DUK	Duke Energy Corporation	Unsecured	500	5.000%	Baa2	BBB	12/08/27	+123	
47	12/07/22	NRUC	National Rural Utilities Cooperative Finance Corp	Unsecured	400	4.800%	A2	A-	3/15/2028	+120	
48	01/03/23	ES	Connecticut Light & Power Company	Secured	500	5.250%	A1	A+	01/15/53	+135	
49	01/03/23	OGE	Oklahoma Gas & Electric	Unsecured	450	5.400%	A3	A-	01/15/33	+165	
50	01/03/23	EXC	Commonwealth Edison	Secured	400	4.900%	A1	A	02/01/33	+115	
51	01/03/23	EXC	Commonwealth Edison	Secured	575	5.300%	A1	A	02/01/53	+140	1
52	01/03/23	DUK	Duke Energy Carolinas	Secured	900	4.950%	Aa3	A	1/15/2033	+120	
53	01/03/23	DUK	Duke Energy Carolinas	Secured	900	5.350%	Aa3	A	1/15/2053	+145	1
54	01/03/23	AEP	Public Service Co. of Oklahoma	Unsecured	475	5.250%	Baa1	A-	1/15/2033	+150	
55	01/03/23	ETR	Entergy Arkansas	Secured	425	5.150%	A2	A	1/15/2033	+140	
56	01/03/23	CMS	Consumers Energy	Secured	425	4.650%	A1	A	3/1/2028	+75	
57	01/04/23	PCG	Pacific Gas and Electric	Secured	750	6.150%	Baa3	BBB-	1/15/2033	+250	
58	01/04/23	PCG	Pacific Gas and Electric	Secured	750	6.750%	Baa3	BBB-	1/15/2053	+295	
59	01/04/23	ES	Public Service New Hampshire	Secured	300	5.150%	A1	A+	1/15/2053	+133	
60	01/10/23	WEC	WEC Energy Group	Unsecured	650	4.750%	Baa1	BBB+	1/9/2026	+85	
61	01/10/23	WEC	WEC Energy Group	Unsecured	450	4.750%	Baa1	BBB+	1/15/2028	+113	

Average Spread 2020-Mid-Jan. 2023

(1) 30-year Debt Issuance A Rated Securities	+130
(2) 30-year Debt Issuance A- Rated Securities	+154
(3) 30-year Debt Issuance BBB+ Rated Securities	+159
(4) 30-year Debt Issuance BBB Rated Securities	+225

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Peer Company Commission Authorized Equity Ratios
for the Projected Year Ending February 28, 2025

Case No.: U-21389
Exhibit No.: A-34 (MRB-10)
Page: 1 of 1
Witness: MRBleckman
Date: May 2023

(a) Line No.	(b) State	(c) Company	(d) Docket #	(e) Case Type	(f) Date Authorized	(g) Common Equity %
1	Illinois	Ameren Illinois Company	D-22-0297	Electric	12/01/22	50.00%
2	Arizona	Arizona Public Service Co.	D-E-01345A-19-0236	Electric	11/02/21	54.67%
3	California	Pacific Gas and Electric Company	A-22-04-008	Electric	12/15/22	52.00%
4	Wisconsin	Wisconsin Electric Power Company	D-5-UR-110	Electric	12/29/22	59.01%
5	Wisconsin	Wisconsin Public Service Corporation	D-6690-UR-127	Electric	12/22/22	53.79%
6	Wisconsin	Northern States Power Co.	D-4220-UR-125	Electric	11/18/21	52.50%
7	North Dakota	Northern States Power Co.	C-PU-20-441	Electric	08/18/21	52.50%
8	Colorado	Public Service Company of Colorado	D-21AL-0317E	Electric	03/16/22	55.92%
9	Texas	Southwestern Public Service Co	D-49831	Electric	08/27/20	54.62%
10	New Mexico	Southwestern Public Service Co	C-20-00238-UT	Electric	02/16/22	54.72%
11	Average Equity Ratio					<u><u>53.97%</u></u>

Source: Individual state commission electronic docket records search.

Selection Criteria:

1. Regulated subsidiary of the Company's ROE proxy group in Case No. U-21224
2. Rate order issued from January 1, 2020 through December 31, 2022.
3. Rate order available through online docket search that includes an authorized equity ratio
4. For each regulated subsidiary, latest authorized equity ratio for each state / jurisdiction.

Note: Excludes final orders received by in-state proxy DTE Energy as well as the Company.

Line No.	(a) Tier	(b) State	(c) Company	(d) Equity Ratio	(e) ROE	(f) (d) x (e) Rate of Return
1	Tier 1A	Alabama	Alabama Power	55.00%	10.87%	5.98%
2	Tier 1B	Florida	FPL	59.60%	10.60%	6.32%
3		Georgia	Georgia Power	56.00%	10.50%	5.88%
4		Pennsylvania	UGI (PA)	54.02%	9.85%	5.32%
5	Tier 1C ¹	Wisconsin	WEC	52.33%	10.14%	5.31%
6		Mississippi	Mississippi Power	53.00%	10.00%	5.30%
7		Iowa	IPL (IA)	51.00%	10.00%	5.10%
8		Michigan	Consumers Energy	50.75%	9.90%	5.02%
9		North Carolina	Duke Energy	52.00%	9.60%	4.99%
10		Michigan	Pro Forma - Consumers Energy	51.00%	9.70%	4.95%

Source: February 24, 2023 S&P Global Market Intelligence Report - RRA Regulatory Focus - State Regulatory Evaluations - Energy.

Columns (d) & (e): Company materials, S&P Global IQ, Sell-side and Credit Rating reports.

¹Tennessee excluded as the Tennessee Valley Authority (TVA) is regulated at the federal level.

MOODY'S INVESTORS SERVICE

CREDIT OPINION

10 May 2021

Update

 Rate this Research

RATINGS

Consumers Energy Company

Domicile	Jackson, Michigan, United States
Long Term Rating	Baa1
Type	Pref. Stock - Dom Curr
Outlook	Stable

Please see the [ratings section](#) at the end of this report for more information. The ratings and outlook shown reflect information as of the publication date.

Contacts

Jairo Chung +1.212.553.5123
VP-Senior Analyst
jairo.chung@moodyys.com

Jillian Cardona +1.212.553.4351
Associate Analyst
jillian.cardona@moodyys.com

Michael G. Haggarty +1.212.553.7172
Associate Managing Director
michael.haggarty@moodyys.com

Jim Hempstead +1.212.553.4318
MD - Global Infrastructure & Cyber Risk
james.hempstead@moodyys.com

CLIENT SERVICES

Americas	1-212-553-1653
Asia Pacific	852-3551-3077
Japan	81-3-5408-4100
EMEA	44-20-7772-5454

Consumers Energy Company

Update following downgrade to A1 secured

Summary

Consumers Energy Company's (Consumers Energy) credit reflects its operating profile as a vertically integrated electric and gas utility in a credit supportive regulatory environment in Michigan. Historically, Consumers Energy had produced strong and consistent credit metrics, including a cash flow from operations before working capital (CFO pre-WC) to debt ratio in the mid-20% range. However, metrics began to weaken slightly starting in 2018 primarily due to the gradual erosion in its allowed return on equity (ROE) and equity capitalization through rate case outcomes. In addition, tax reform had a negative impact on its credit metrics. By 2019, CFO pre-WC to debt for Consumers Energy had fallen to about 20%.

Consumers Energy's stand-alone financial performance has historically been affected by the significant debt at its parent company CMS Energy Corporation (CMS, Baa2 stable). CMS has made progress in reducing consolidated leverage as well as the percentage of parent debt in its capital structure, although it remains around 30% of total consolidated debt. All of Consumers' outstanding debt obligations are secured.

Recent developments

On 3 May 2021, we downgraded the ratings of Consumers Energy due to its weakened credit metrics. Although the regulatory environment in Michigan remains relatively credit supportive, the outcome of recent rate cases has put pressure on its credit metric ratios and we do not expect the ratios to recover back to historical levels.

In the latest electric rate case, the Michigan Public Service Commission (MPSC) authorized a 9.9% ROE and 51.1% equity capitalization. In the previous electric rate case, which was settled and approved in January 2019, the stated ROE was 10% and the assumed capital structure included a 52.5% equity capitalization. Considering both the continued robust utility investment program totaling over \$13 billion over the next five years and the lower ROE and equity capitalization, we expect Consumers Energy to produce a CFO pre-WC to debt ratio of around 20% over the next 2-3 years.

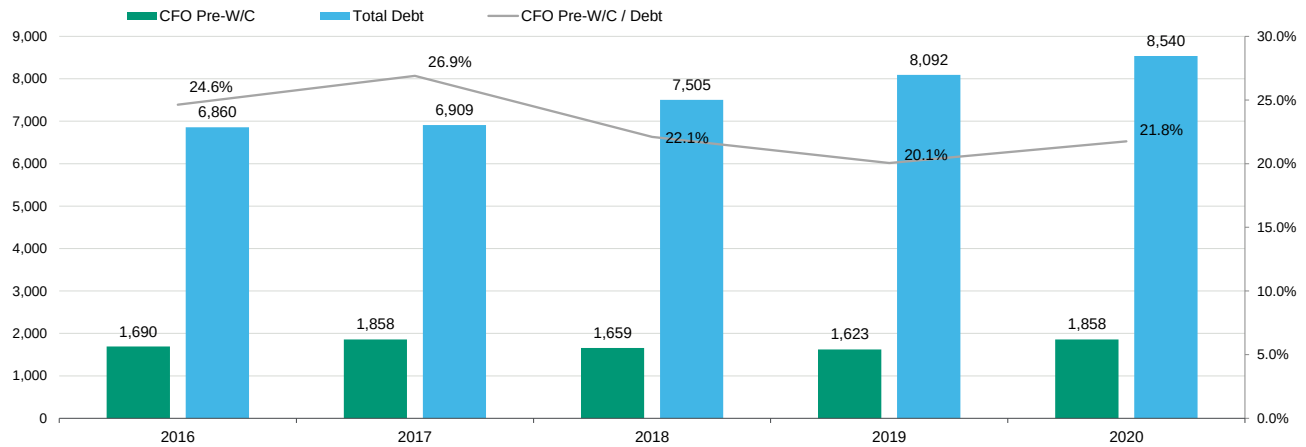
The rapid spread of the coronavirus outbreak, severe global economic shock, low oil prices, and asset price volatility are creating a severe and extensive credit shock across many sectors, regions and markets. The combined credit effects of these developments are unprecedented. We regard the coronavirus outbreak as a social risk under our ESG framework, given the substantial implications for public health and safety.

We expect Consumers Energy to be relatively resilient to recessionary pressures because of its rate regulated utility business. We continue to monitor changes in customers' usage, utility bill payment delinquency, and the regulatory response to counter these effects on

earnings and cash flow. We see these issues as temporary and not reflective of the core operations or long-term financial or credit profile of Consumers Energy.

Exhibit 1

Historical CFO Pre-WC, Total Debt and CFO Pre-WC to Debt



Source: Moody's Financial Metrics

Credit strengths

- » Credit supportive regulation despite financial pressure from recent rate case outcomes
- » Regulatory framework provides transparency and timely cost recovery

Credit challenges

- » Weakened financial metrics compared to historical levels
- » Maintaining continued regulatory support amid a robust capital investment program
- » High leverage at the parent company

Rating outlook

The stable outlook reflects our expectation that financial metrics will remain at current levels and that Consumers Energy will continue to benefit from a consistent and generally credit supportive regulatory environment. The stable outlook also incorporates our view that Consumers Energy will maintain prudent financial policies while managing through its robust investment cycle and that debt levels at either the parent or utility will not increase materially.

Factors that could lead to an upgrade

A rating upgrade could be considered if credit metrics improve such that CFO pre-WC to debt is above 21% on a sustained basis. In addition, if the Michigan regulatory framework becomes even more formulaic, transparent or timely with its suite of recovery mechanisms for Consumers Energy, a rating upgrade could be possible.

Factors that could lead to a downgrade

A rating downgrade could be considered if there is a material deterioration in the credit supportiveness of the Michigan regulatory environment; or if the CFO pre-WC to debt ratio declines below 18% on a sustained basis.

This publication does not announce a credit rating action. For any credit ratings referenced in this publication, please see the ratings tab on the issuer/entity page on www.moody's.com for the most updated credit rating action information and rating history.

Key indicators

Exhibit 2

Consumers Energy Company Key Indicators

	Dec-16	Dec-17	Dec-18	Dec-19	Dec-20
CFO Pre-W/C + Interest / Interest	6.9x	7.2x	6.3x	6.2x	6.6x
CFO Pre-W/C / Debt	24.6%	26.9%	22.1%	20.1%	21.8%
CFO Pre-W/C – Dividends / Debt	17.4%	19.3%	15.0%	12.7%	14.3%
Debt / Capitalization	43.4%	46.1%	46.4%	46.0%	44.5%

All ratios are based on 'Adjusted' financial data and incorporate Moody's Global Standard Adjustments for Non-Financial Corporations.

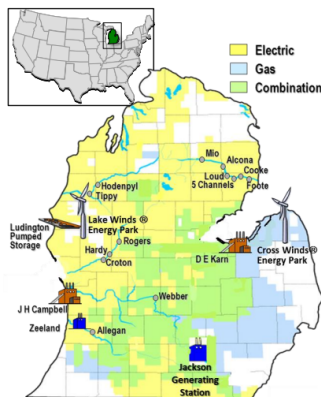
Source: Moody's Financial Metrics

Profile

Consumers Energy Company (Consumers Energy) is a vertically integrated electric and gas utility serving approximately 6.8 million customers in the state of Michigan with a rate base of approximately \$20 billion. Consumers Energy's electric operations account for approximately two thirds of its revenue, cash flow and asset base. Consumers Energy is the primary subsidiary of CMS Energy Corporation (CMS), representing over 90% of its consolidated operating revenue. In addition to Consumers Energy, CMS owns approximately 1,838 gross MW of unregulated, primarily natural gas-fired, generation located mostly within Michigan, and EnerBank, a FDIC-insured industrial bank providing unsecured consumer installment loans for financing home improvements. These businesses contribute modestly to consolidated results, and do not materially increase CMS's consolidated business risk profile.

Exhibit 3

Consumers Energy's Service Territory



Source: Company Presentations

Detailed credit considerations

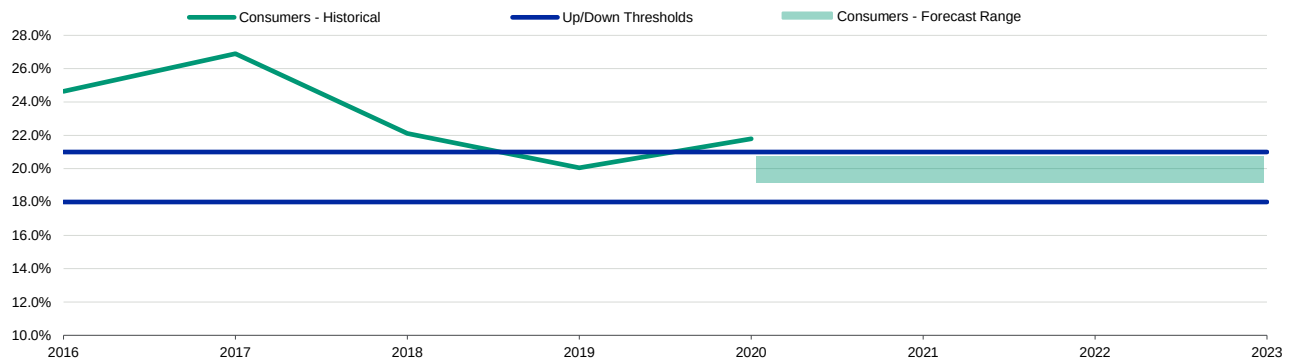
Weak credit metrics expected through high capital investment cycle

Consumers Energy had historically maintained strong financial metrics with its CFO pre-WC to debt in the mid-20% range. However, metrics began to weaken starting in 2018 primarily due a gradually declining ROE and a lower regulatory equity capital structure. Also, tax reform and an elevated capital expenditure program added downward pressure on the company's credit metric ratios. By 2019, the utility's CFO pre-WC to debt had fallen closer to 20%, reducing financial flexibility.

Consumers' capital program has increased from about \$1.2 billion in 2012 to about \$2.3 billion in 2020 and, as noted below, is expected to remain elevated. As of year-end 2020, CFO pre-WC to debt was weaker than its 3-year average CFO pre-WC to debt realized between 2017 and 2019. Considering both the continued robust utility investment program totaling over \$13 billion over the next five years and the lower ROE and equity capitalization, we expect Consumers Energy to maintain these lower CFO pre-WC to debt ratios of around 20% over the next 2-3 years.

Exhibit 4

Consumers Energy Historical and Projected CFO pre-WC to Debt



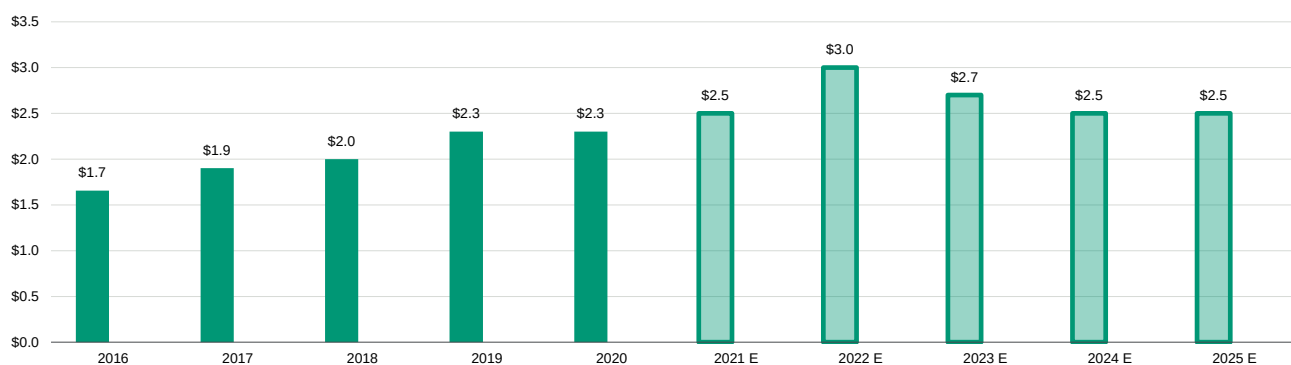
The up/down financial metric thresholds represent one of several factors that could result in an upgrade or downgrade of the rating if the metric is sustained above or below those levels.
Source: Moody's Financial Metrics and Estimates

Capital spending remains elevated

Consumers Energy is currently in the midst of a large capital investment plan and expects to spend approximately \$13 billion from 2021 to 2025. The magnitude of the plan, which is intended to improve reliability and efficiency while moving the company to a less carbon intensive future, will require continued regulatory support in order to maintain the company's current financial profile. In 2021, projected investments are approximately \$2.5 billion compared to around \$2.3 billion in 2020, \$2.3 billion in 2019, and \$2.0 billion in 2018.

Exhibit 5

Consumers Energy's elevated capital program will extend through 2025

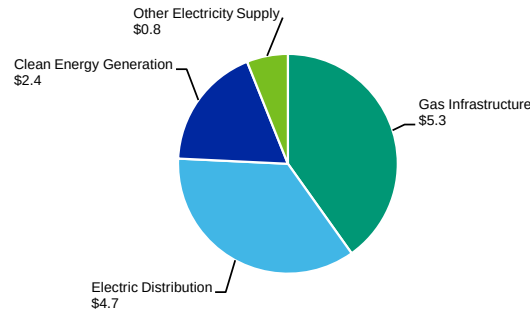


Source: Company Investor Presentation

Over the 2021-2025 period, projected capital investments will include maintenance capital of about \$10 billion (approximately \$4.7 billion for electric operations and \$5.3 billion for gas utility operations), and \$2.4 billion for clean energy generation and \$800 million for other electric supply needed to comply with state and federal laws and regulations. Consumers Energy's goal to keep rate increases modest should help to maintain regulatory support for the recovery of this spending. Funding for these forecasted investments will be provided by internally generated cash flow, the issuance of debt at Consumers and equity contributions from CMS.

Exhibit 6

Planned Capital Expenditures 2021 - 2025
\$ billions



Source: Company 10K

Credit supportive regulatory environment despite financial pressure from rate case outcomes

Consumers Energy is regulated in Michigan by the MPSC, which has a regulatory framework that we view to be more credit supportive than most other states. As a result of 2008 and 2016 energy legislation in Michigan, the regulatory framework was streamlined, improving both the rate case process and the timeliness of cost recovery. The 2016 legislation provided additional assurance of utility investment recovery by expanding the certificate of necessity (CON) process, which had already included pre-construction approval for large generating resources, into an integrated resource planning (IRP) process. The IRP process considers a wide range of factors including fuel cost, demand forecasts, resource adequacy, competitive pricing, environmental mandates and transmission options before constructing major projects. The legislation also lowered the threshold for major projects to \$100 million from \$500 million.

In June 2019, the MPSC approved Consumers Energy's latest IRP, which included a plan to reduce its carbon emissions by more than 90% from its 2005 levels by 2040. The plan included the retirement of the 503 MW Karn coal plant by 2023. Also, under Consumers Energy's renewable energy plan, the MPSC has approved the acquisition of up to 525 MW of new wind generation projects and authorized the utility to earn a 10.7% ROE on any projects approved by the MPSC. Also, the MPSC approved a 20-year Purchase Power Agreement (PPA) under which Consumers will purchase 100 MW of renewable capacity, energy and renewable energy credits from a 149 MW solar generating facility in Michigan, which is expected to be operational in 2022. The company plans to file its next IRP in June 2021, which will focus further on accelerating decarbonization, as well as maintaining grid reliability and resilience.

While Michigan's electricity restructuring had initially contemplated full competition in generation, the 2008 legislation limited the number of customers able to choose an alternative supplier at 10% of the prior year load in the utility's service territory. The December 2016 legislation maintained the 10% retail open access (ROA) limit but also provided the possibility for a periodic downward adjustments if ROA demand is below the 10% cap. The 2016 legislation also required, for the first time, that competitive retail suppliers demonstrate adequacy of electric supply for a multi-year period.

Timeliness of cost recovery based on a prescriptive suite of recovery mechanisms

Michigan utilities benefit from numerous formulaic rate adjustment mechanisms that provide a high degree of cash flow stability and assurance of recovery. For example, Consumers Energy has forward-looking Power Supply Cost Recovery (PSCR) and Gas Cost Recovery (GCR) mechanisms that are intended to ensure that it can recover prudently incurred power and gas supply costs. The PSCR covers fuel and purchased power costs as well as transmission and emission allowance costs. Differences between actual and forecast costs are deferred for recovery or refunded in the following year. The PSCR is a surcharge mechanism and provides a degree of base rate and cash flow stability, a credit positive. The GCR mechanism may be adjusted monthly within a capped range to minimize over/under recoveries, although interim gas inventory buildup could substantially increase the company's working capital financing when gas prices sharply increase.

Gas utilities in the state also benefit from revenue decoupling mechanisms (RDM) and programs designed to assure recovery of needed infrastructure improvements. Consumers Energy's RDM compares and adjusts for differences between weather normalized actual and authorized revenues. Consumers Energy's enhanced infrastructure replacement program (EIRP) is a MPSC authorized 25-year incremental investment program to upgrade natural gas infrastructure, including replacing approximately 540 miles of cast iron pipe and other high-risk components. Consumers currently projects that it will spend about \$100 million per year under the EIRP. These expenditures are recoverable through base rates.

Financial pressure from recent rate case outcomes

Consumers maintains an active regulatory schedule with its electric and gas general rate cases typically filed annually in an alternating pattern. The utility currently has an electric rate case pending.

Consumers' last electric rate case was completed in December 2020. The MPSC approved a \$90 million electric rate increase based on a 9.9% ROE and 51.1% equity layer. The order also approved the recovery of \$12.6 million related to investment in the distribution system. While we had expected the commission to maintain the 10% authorized ROE, we recognize the downward pressure on the ROE due to low Treasury rates.

In March 2021, Consumers filed an electric rate case application with the MPSC seeking a \$225 million annual increase, based on a 10.5% ROE and 52% equity layer. The company also requested recovery of \$121 million of new infrastructure investments related to the 2019 IRP, distribution system reliability and technology enhancements. We expect the rate case to conclude by the end of the year. If the rate case results in another decrease in both ROE and equity layer, we expect the utility's already weakened metrics to be further pressured.

Consumers' latest gas rate case was settled in September 2020. The MPSC authorized a \$144 million revenue increase, 9.9% ROE, 52.05% equity layer (both unchanged from the previous rate case), and continuation of a revenue decoupling mechanism. The company's rate base valuation was \$7.42 billion. As part of the settlement agreement, Consumers agreed not to file its next gas rate case until 1 December 2021. Between 1 October 2021 and 30 September 2022, the company will accelerate the amortization of tax liabilities in an amount forecasted to be approximately \$84.5 million.

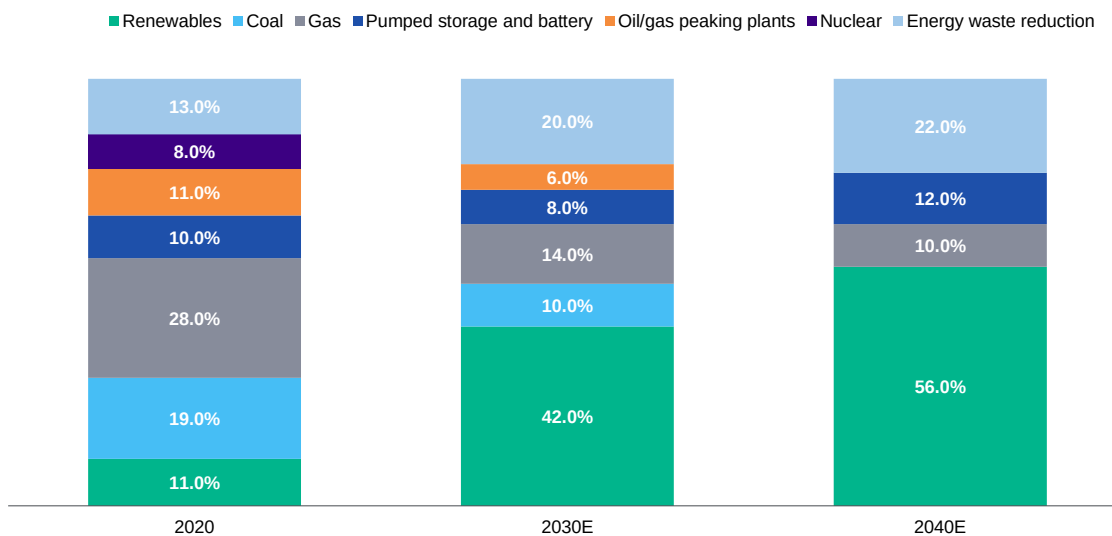
ESG considerations

Environmental

CMS is strongly positioned for the carbon transition with strategies and plans in place that substantially mitigate its carbon transition exposure. Consumers is currently in a transition to phase out its coal-fired generation and targets a reduction in its CO2 emissions by more than 90% from its 2005 levels by 2040. Michigan's regulatory framework supports the utility's transition plan by allowing certain investment cost recovery and renewable energy plan surcharges, for example.

Exhibit 7

Clean Energy Plan Electric Capacity by Fuel Source (MW)



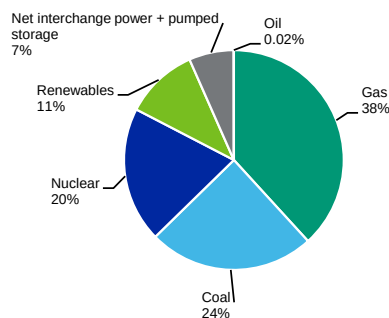
Source: Company filings

In June 2018, Consumers Energy filed its first ever IRP with the MPSC pursuant to the 2016 energy law, in accordance with requirements laid out in the legislation. The IRP was approved by the MPSC in June 2019. The IRP outlined a series of steps the company will take to reach its goal of a 90% carbon emissions reduction and to eliminate its use of coal generation by 2040. The combination and timeline of these programs are aimed at ensuring that the energy it provides remains reliable and affordable over the long-term.

Among the steps outlined in the plan is the termination of its PPA with the Palisades Nuclear Power Plant as well as the closing of units 1 and 2 (503 MW) of the Karn coal-fired plant and replacing it with 525 MW of wind and solar generation. The wind generation capacity addition has been approved by the commission through the Renewable Energy Plan. The IRP also included Consumers Energy's intention to fully exit coal generation by 2040 by closing all three units (1.3 GW) of the J.H. Campbell coal-fired plant. This capacity will be replaced over the next two decades with 5 GW of new solar generation by 2030 and an additional 1 GW by 2040. Even before the approval of the IRP, Consumers had been actively reducing its carbon footprint and moving toward more energy efficient, cleaner generating resources. Consumers' generating capacity has shifted from 41% coal in 2005 to 24% in 2020.

Exhibit 8

**2020 Owned and Purchased Generation Supply
GWh**



Source: Company filings

In 2020, over half of Consumers Energy's electric generation was provided through long-term power purchase agreements (PPAs) or market purchases. The majority of this production came from two PPAs, one with the 1,240 MW MCV gas-fired facility (approximately 16% of 2020 electric supply) that terminates in March 2025 and a second with the 813 MW Palisades nuclear facility (approximately 10% of 2020 electric supply) that is scheduled to terminate in 2022. The Palisades PPA is not expected to be renewed when it expires, while the company proposed to extend the MCV PPA for another five years in its IRP, albeit at a lower price. We expect that the utility will replace any ensuing energy and capacity shortfall with a mix of renewable generation and demand side management initiatives.

Michigan state statutes allow utilities to securitize related regulatory assets and stranded costs and the MPSC has a history of approving securitization bonds. Most recently, in 2020, the MPSC issued a securitization financing order to authorize Consumers to issue securitization bonds in order to finance the recovery of the remaining book value of Karn units 1 and 2, which are expected to retire in 2023. Prior to this in 2013, the MPSC authorized Consumers Energy's issuance of securitization bonds to finance the recovery of the remaining book value of seven smaller coal-fired electric generating plants that were retired in April 2016 and three smaller natural gas-fired units that were retired in June 2015. Approximately \$378 million of securitization bonds were issued through a securitization subsidiary in 2014. Principal and interest payments are made semi-annually through the maturity of the bonds in 2029. We expect similar regulatory treatment would be possible for other future plant retirements.

Social

Social risks are primarily related to Consumers Energy's customer and regulatory relations as well as demographic and societal trends. Consumers Energy's regulatory environment as well as its interaction with the MPSC are important in considering the utility's social risk. Also, the safety and reliability of its operations are extremely important social considerations. Given recent developments related to the COVID-19 pandemic, there is a possibility of increasing social risk, should unemployment remain high, making customers less able to absorb rate increases.

Governance

As a subsidiary of CMS, corporate governance considerations include the financial policy and risk management of its parent company. We note that a stable financial position is an important characteristic for managing environmental and social risks.

Liquidity analysis

We expect Consumers Energy's liquidity profile to be adequate over the next 12-18 months.

Consumers Energy has access to an \$850 million secured revolving credit facility expiring in June 2023 and as of 31 December 2020, its net availability was \$843 million after having \$7 million letters of credit outstanding. The facility includes a sustainability linked pricing metric, which permits an interest rate reduction for meeting targets related to environmental sustainability, specifically renewable generation, and highlights its commitment to shifting its exposure to a greater renewable content. Consumers Energy also maintains a \$250 million secured credit facility terminating in November 2022. These credit facilities provide support for working capital needs and act as a backstop to Consumer Energy's \$500 million commercial paper program. The credit facilities do not include a material adverse change representation for new borrowings, and have only one financial covenant, setting the maximum debt to capital at less than 65%. At the period ending 31 December 2020, debt to capital was 49%.

As of 31 December 2020, Consumers had no commercial paper outstanding under its CP program, no borrowings under its various credit facilities and \$38 million of letters of credit outstanding on a consolidated basis. In September and December 2020, Consumers redeemed its \$375 million first mortgage bond due in 2022 and its \$300 million term loan facility due in January 2021, respectively.

The utility's continuing capital expenditure program and dividend policy result in negative free cash flow for the foreseeable future. However, the company has a reasonable amount of external liquidity, demonstrated market access, and regularly receives capital contributions from its parent.

For the last twelve months ended 31 December 2020, Consumers generated approximately \$1.2 billion of cash from operations (CFO), invested \$2.2 billion in capital investments and distributed \$639 million in dividend payments to CMS, resulting in negative free cash flow (FCF) of approximately \$1.6 billion that was offset by parent contributions of \$650 million and incremental long-term debt. Consumers' policy is to grow its dividend with earnings, maintaining a payout ratio in the 80% range.

Structural considerations

Consumer Energy's strong stand-alone financial performance has been pressured by significant debt at CMS, the parent company. As of year-end 2020, CMS had approximately \$7 billion of consolidated debt outside of Consumers Energy, or approximately 46% of its consolidated total reported debt. Of this amount, approximately \$2.8 billion represented deposits of EnerBank, an insured industrial bank wholly owned by CMS, and is supported by approximately \$2.8 billion of notes receivable. Excluding self-funding EnerBank, we estimate that CMS's parent level debt to be approximately \$4.2 billion or about 34% of the total of Consumers Energy plus pure parent level debt. This still remains as a key driver of the relatively wide differential between the Consumers Energy and CMS credit profiles.

Rating methodology and scorecard factors

Exhibit 9

Methodology Scoring Factors Consumers Energy Company

Regulated Electric and Gas Utilities Industry [1][2]		Current FY 12/31/2020	Moody's 12-18 Month Forward View As of Date Published [3]
Factor 1 : Regulatory Framework (25%)	Measure	Score	Measure Score
a) Legislative and Judicial Underpinnings of the Regulatory Framework	A	A	A A
b) Consistency and Predictability of Regulation	Aa	Aa	Aa Aa
Factor 2 : Ability to Recover Costs and Earn Returns (25%)			
a) Timeliness of Recovery of Operating and Capital Costs	Aa	Aa	Aa Aa
b) Sufficiency of Rates and Returns	A	A	A A
Factor 3 : Diversification (10%)			
a) Market Position	Baa	Baa	Baa Baa
b) Generation and Fuel Diversity	Baa	Baa	Baa Baa
Factor 4 : Financial Strength (40%)			
a) CFO pre-WC + Interest / Interest (3 Year Avg)	6.4x	Aa	6.8x - 7x Aa
b) CFO pre-WC / Debt (3 Year Avg)	21.3%	Baa	19% - 21% Baa
c) CFO pre-WC – Dividends / Debt (3 Year Avg)	14.0%	Baa	12% - 14% Baa
d) Debt / Capitalization (3 Year Avg)	45.6%	Baa	44% - 46% Baa
Rating:			
Scorecard-Indicated Outcome Before Notching Adjustment		A2	A2
HoldCo Structural Subordination Notching		0	0
a) Scorecard-Indicated Outcome		A2	A2
b) Actual Rating Assigned		A1*	A1*

* Senior Secured Rating.

[1] All ratios are based on 'Adjusted' financial data and incorporate Moody's Global Standard Adjustments for Non-Financial Corporations.

[2] As of 12/31/2020.

[3] This represents Moody's forward view; not the view of the issuer; and unless noted in the text, does not incorporate significant acquisitions and divestitures.

Source: Moody's Financial Metrics

Appendix

Exhibit 10

Cash Flow and Credit Metrics

CF Metrics	Dec-16	Dec-17	Dec-18	Dec-19	Dec-20
As Adjusted					
FFO	1,739	1,825	1,760	1,752	1,994
+/- Other	-49	33	-101	-129	-136
CFO Pre-WC	1,690	1,858	1,659	1,623	1,858
+/- ΔWC	64	-65	1	-15	25
CFO	1,754	1,793	1,660	1,608	1,883
- Div	500	523	532	593	638
- Capex	1,673	1,649	1,834	2,100	2,184
FCF	-419	-379	-706	-1,085	-939
(CFO Pre-W/C) / Debt	24.6%	26.9%	22.1%	20.1%	21.8%
(CFO Pre-W/C - Dividends) / Debt	17.4%	19.3%	15.0%	12.7%	14.3%
FFO / Debt	25.4%	26.4%	23.5%	21.6%	23.3%
RCF / Debt	18.1%	18.9%	16.4%	14.3%	15.9%
Revenue	6,064	6,222	6,464	6,376	6,189
Interest Expense	288	298	315	312	332
Net Income	615	634	527	725	799
Total Assets	20,026	21,179	22,096	23,699	25,399
Total Liabilities	14,139	14,746	15,251	16,053	16,862
Total Equity	5,888	6,434	6,845	7,647	8,538

All figures and ratios are calculated using Moody's estimates and standard adjustments. Periods are Financial Year-End unless indicated. LTM = Last Twelve Months

Source: Moody's Financial Metrics

Exhibit 11

Peer Comparison Table

	Consumers Energy Company A1 (Stable)			DTE Electric Company (P)A2 (Stable)			DTE Gas Company A3 (Stable)			Northern States Power Company (Minnesota) A2 (Stable)			Wisconsin Power and Light Company A3 (Stable)		
(In US millions)	Dec-18	FYE Dec-19	FYE Dec-20	FYE Dec-18	FYE Dec-19	FYE Dec-20	FYE Dec-18	FYE Dec-19	FYE Dec-20	FYE Dec-19	FYE Dec-19	FYE Dec-20	FYE Dec-18	FYE Dec-19	FYE Dec-20
Revenue	6,464	6,376	6,189	5,298	5,224	5,506	1,415	1,462	1,396	5,122	5,112	5,101	1,453	1,476	1,395
CFO Pre-W/C	1,659	1,623	1,858	1,833	1,791	2,051	337	368	409	1,355	1,362	1,345	446	425	475
Total Debt	7,505	8,092	8,540	7,641	8,495	9,154	1,826	1,997	2,168	5,414	6,363	6,703	2,175	2,328	2,596
CFO Pre-W/C + Interest / Interest	6.3x	6.2x	6.6x	6.8x	6.3x	6.9x	5.5x	5.5x	6.0x	6.7x	6.5x	6.5x	6.5x	6.6x	6.5x
CFO Pre-W/C / Debt	22.1%	20.1%	21.8%	24.0%	21.1%	22.4%	18.5%	18.4%	18.9%	25.0%	21.4%	20.1%	20.5%	18.2%	18.3%
CFO Pre-W/C - Dividends / Debt	15.0%	12.7%	14.3%	18.0%	15.3%	16.5%	12.3%	12.3%	12.6%	16.6%	14.1%	14.0%	14.1%	12.1%	12.1%
Debt / Capitalization	46.4%	46.0%	44.5%	46.0%	47.2%	46.4%	43.9%	44.2%	43.9%	43.0%	45.0%	43.8%	44.6%	44.0%	45.0%

All figures & ratios calculated using Moody's estimates & standard adjustments. FYE = Financial Year-End.

Source: Moody's Financial Metrics

Ratings

Exhibit 12

Category	Moody's Rating
CONSUMERS ENERGY COMPANY	
Outlook	Stable
Sr Sec Bank Credit Facility	A1
First Mortgage Bonds	A1
Bkd Senior Secured	A1
Pref. Stock	Baa1
Commercial Paper	P-2
PARENT: CMS ENERGY CORPORATION	
Outlook	Stable
Sr Unsec Bank Credit Facility	Baa2
Senior Unsecured	Baa2
Jr Subordinate	Baa3

Source: Moody's Investors Service

© 2021 Moody's Corporation, Moody's Investors Service, Inc., Moody's Analytics, Inc. and/or their licensors and affiliates (collectively, "MOODY'S"). All rights reserved.

CREDIT RATINGS ISSUED BY MOODY'S CREDIT RATINGS AFFILIATES ARE THEIR CURRENT OPINIONS OF THE RELATIVE FUTURE CREDIT RISK OF ENTITIES, CREDIT COMMITMENTS, OR DEBT OR DEBT-LIKE SECURITIES, AND MATERIALS, PRODUCTS, SERVICES AND INFORMATION PUBLISHED BY MOODY'S (COLLECTIVELY, "PUBLICATIONS") MAY INCLUDE SUCH CURRENT OPINIONS. MOODY'S DEFINES CREDIT RISK AS THE RISK THAT AN ENTITY MAY NOT MEET ITS CONTRACTUAL FINANCIAL OBLIGATIONS AS THEY COME DUE AND ANY ESTIMATED FINANCIAL LOSS IN THE EVENT OF DEFAULT OR IMPAIRMENT. SEE APPLICABLE MOODY'S RATING SYMBOLS AND DEFINITIONS PUBLICATION FOR INFORMATION ON THE TYPES OF CONTRACTUAL FINANCIAL OBLIGATIONS ADDRESSED BY MOODY'S CREDIT RATINGS. CREDIT RATINGS DO NOT ADDRESS ANY OTHER RISK, INCLUDING BUT NOT LIMITED TO: LIQUIDITY RISK, MARKET VALUE RISK, OR PRICE VOLATILITY. CREDIT RATINGS, NON-CREDIT ASSESSMENTS ("ASSESSMENTS"), AND OTHER OPINIONS INCLUDED IN MOODY'S PUBLICATIONS ARE NOT STATEMENTS OF CURRENT OR HISTORICAL FACT. MOODY'S PUBLICATIONS MAY ALSO INCLUDE QUANTITATIVE MODEL-BASED ESTIMATES OF CREDIT RISK AND RELATED OPINIONS OR COMMENTARY PUBLISHED BY MOODY'S ANALYTICS, INC. AND/OR ITS AFFILIATES. MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND PUBLICATIONS DO NOT CONSTITUTE OR PROVIDE INVESTMENT OR FINANCIAL ADVICE, AND MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND PUBLICATIONS ARE NOT AND DO NOT PROVIDE RECOMMENDATIONS TO PURCHASE, SELL, OR HOLD PARTICULAR SECURITIES. MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND PUBLICATIONS DO NOT COMMENT ON THE SUITABILITY OF AN INVESTMENT FOR ANY PARTICULAR INVESTOR. MOODY'S ISSUES ITS CREDIT RATINGS, ASSESSMENTS AND OTHER OPINIONS AND PUBLISHES ITS PUBLICATIONS WITH THE EXPECTATION AND UNDERSTANDING THAT EACH INVESTOR WILL, WITH DUE CARE, MAKE ITS OWN STUDY AND EVALUATION OF EACH SECURITY THAT IS UNDER CONSIDERATION FOR PURCHASE, HOLDING, OR SALE.

MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS, AND PUBLICATIONS ARE NOT INTENDED FOR USE BY RETAIL INVESTORS AND IT WOULD BE RECKLESS AND INAPPROPRIATE FOR RETAIL INVESTORS TO USE MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS OR PUBLICATIONS WHEN MAKING AN INVESTMENT DECISION. IF IN DOUBT YOU SHOULD CONTACT YOUR FINANCIAL OR OTHER PROFESSIONAL ADVISER.

ALL INFORMATION CONTAINED HEREIN IS PROTECTED BY LAW, INCLUDING BUT NOT LIMITED TO, COPYRIGHT LAW, AND NONE OF SUCH INFORMATION MAY BE COPIED OR OTHERWISE REPRODUCED, REPACKAGED, FURTHER TRANSMITTED, TRANSFERRED, DISSEMINATED, REDISTRIBUTED OR RESOLD, OR STORED FOR SUBSEQUENT USE FOR ANY SUCH PURPOSE, IN WHOLE OR IN PART, IN ANY FORM OR MANNER OR BY ANY MEANS WHATSOEVER, BY ANY PERSON WITHOUT MOODY'S PRIOR WRITTEN CONSENT.

MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND PUBLICATIONS ARE NOT INTENDED FOR USE BY ANY PERSON AS A BENCHMARK AS THAT TERM IS DEFINED FOR REGULATORY PURPOSES AND MUST NOT BE USED IN ANY WAY THAT COULD RESULT IN THEM BEING CONSIDERED A BENCHMARK.

All information contained herein is obtained by MOODY'S from sources believed by it to be accurate and reliable. Because of the possibility of human or mechanical error as well as other factors, however, all information contained herein is provided "AS IS" without warranty of any kind. MOODY'S adopts all necessary measures so that the information it uses in assigning a credit rating is of sufficient quality and from sources MOODY'S considers to be reliable including, when appropriate, independent third-party sources. However, MOODY'S is not an auditor and cannot in every instance independently verify or validate information received in the rating process or in preparing its Publications.

To the extent permitted by law, MOODY'S and its directors, officers, employees, agents, representatives, licensors and suppliers disclaim liability to any person or entity for any indirect, special, consequential, or incidental losses or damages whatsoever arising from or in connection with the information contained herein or the use of or inability to use any such information, even if MOODY'S or any of its directors, officers, employees, agents, representatives, licensors or suppliers is advised in advance of the possibility of such losses or damages, including but not limited to: (a) any loss of present or prospective profits or (b) any loss or damage arising where the relevant financial instrument is not the subject of a particular credit rating assigned by MOODY'S.

To the extent permitted by law, MOODY'S and its directors, officers, employees, agents, representatives, licensors and suppliers disclaim liability for any direct or compensatory losses or damages caused to any person or entity, including but not limited to by any negligence (but excluding fraud, willful misconduct or any other type of liability that, for the avoidance of doubt, by law cannot be excluded) on the part of, or any contingency within or beyond the control of, MOODY'S or any of its directors, officers, employees, agents, representatives, licensors or suppliers, arising from or in connection with the information contained herein or the use of or inability to use any such information.

NO WARRANTY, EXPRESS OR IMPLIED, AS TO THE ACCURACY, TIMELINESS, COMPLETENESS, MERCHANTABILITY OR FITNESS FOR ANY PARTICULAR PURPOSE OF ANY CREDIT RATING, ASSESSMENT, OTHER OPINION OR INFORMATION IS GIVEN OR MADE BY MOODY'S IN ANY FORM OR MANNER WHATSOEVER.

Moody's Investors Service, Inc., a wholly-owned credit rating agency subsidiary of Moody's Corporation ("MCO"), hereby discloses that most issuers of debt securities (including corporate and municipal bonds, debentures, notes and commercial paper) and preferred stock rated by Moody's Investors Service, Inc. have, prior to assignment of any credit rating, agreed to pay to Moody's Investors Service, Inc. for credit ratings opinions and services rendered by it fees ranging from \$1,000 to approximately \$5,000,000. MCO and Moody's Investors Service also maintain policies and procedures to address the independence of Moody's Investors Service credit ratings and credit rating processes. Information regarding certain affiliations that may exist between directors of MCO and rated entities, and between entities who hold credit ratings from Moody's Investors Service and have also publicly reported to the SEC an ownership interest in MCO of more than 5%, is posted annually at www.moody.com under the heading "Investor Relations — Corporate Governance — Director and Shareholder Affiliation Policy."

Additional terms for Australia only: Any publication into Australia of this document is pursuant to the Australian Financial Services License of MOODY'S affiliate, Moody's Investors Service Pty Limited ABN 61 003 399 657 AFSL 336969 and/or Moody's Analytics Australia Pty Ltd ABN 94 105 136 972 AFSL 383569 (as applicable). This document is intended to be provided only to "wholesale clients" within the meaning of section 761G of the Corporations Act 2001. By continuing to access this document from within Australia, you represent to MOODY'S that you are, or are accessing the document as a representative of, a "wholesale client" and that neither you nor the entity you represent will directly or indirectly disseminate this document or its contents to "retail clients" within the meaning of section 761G of the Corporations Act 2001. MOODY'S credit rating is an opinion as to the creditworthiness of a debt obligation of the issuer, not on the equity securities of the issuer or any form of security that is available to retail investors.

Additional terms for Japan only: Moody's Japan K.K. ("MJKK") is a wholly-owned credit rating agency subsidiary of Moody's Group Japan G.K., which is wholly-owned by Moody's Overseas Holdings Inc., a wholly-owned subsidiary of MCO. Moody's SF Japan K.K. ("MSFJ") is a wholly-owned credit rating agency subsidiary of MJKK. MSFJ is not a Nationally Recognized Statistical Rating Organization ("NRSRO"). Therefore, credit ratings assigned by MSFJ are Non-NRSRO Credit Ratings. Non-NRSRO Credit Ratings are assigned by an entity that is not a NRSRO and, consequently, the rated obligation will not qualify for certain types of treatment under U.S. laws. MJKK and MSFJ are credit rating agencies registered with the Japan Financial Services Agency and their registration numbers are FSA Commissioner (Ratings) No. 2 and 3 respectively.

MJKK or MSFJ (as applicable) hereby disclose that most issuers of debt securities (including corporate and municipal bonds, debentures, notes and commercial paper) and preferred stock rated by MJKK or MSFJ (as applicable) have, prior to assignment of any credit rating, agreed to pay to MJKK or MSFJ (as applicable) for credit ratings opinions and services rendered by it fees ranging from JPY125,000 to approximately JPY550,000,000.

MJKK and MSFJ also maintain policies and procedures to address Japanese regulatory requirements.

CLIENT SERVICES

Americas	1-212-553-1653
Asia Pacific	852-3551-3077
Japan	81-3-5408-4100
EMEA	44-20-7772-5454

Industry Top Trends

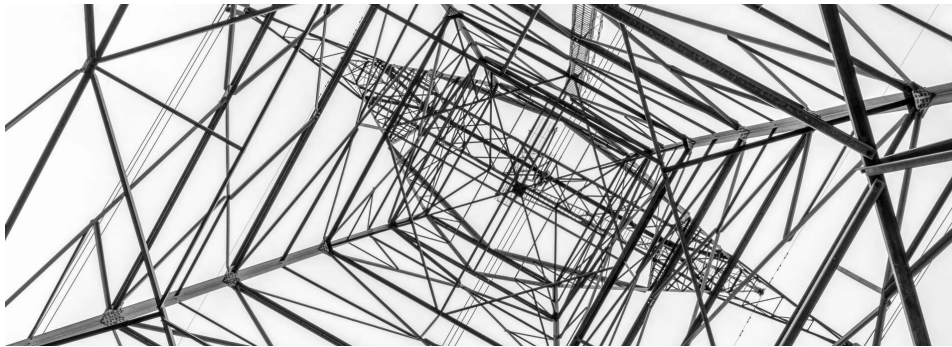
S&P Global
Ratings

North America Regulated Utilities

The industry's outlook remains negative

January 23, 2023

This report does not constitute a rating action



What's changed?

Credit Quality. For the third consecutive year, downgrades outpaced upgrades and the industry's median rating fell to 'BBB+' from 'A-'.

Interest rates. Interest rates increased by about 400 basis points during 2022, pressuring the industry's financial performance.

Securitization. In 2022, the industry benefited from the issuance of about \$15 billion in securitization bonds, recovering storm, hurricane, and wildfire costs.

What are the key assumptions for 2023?

Inflation decreases. S&P Global's economist expects consumer price index (CPI) to decrease to below 4% by year-end 2023.

Robust capital spending. The industry's 2022 capital spending was at an all-time high of more than \$190 billion and we expect it will be even higher in 2023.

Increasing asset sales. To support the funding of the industry's capital spending, we expect sales of minority interests in utilities and sales of non-utility assets to exceed \$20 billion in 2023.

What are the key risks around the baseline?

Commodity prices. Natural gas prices remain significantly above their pre-inflation price of about \$2.50/mmBTU, pressuring the customer bill.

Regulatory risk. The customer bill is increasing, which could pressure the industry's ability to effectively manage regulatory risk.

Physical risks. Recovery of severe winter storms, hurricanes, tornadoes, and wildfires may become increasingly challenging as the pace of such incidents appear to be intensifying.

Contacts

Gabe Grosberg

New York
+1 212 438 6043
gabe.grosberg@spglobal.com

Gerrit Jepsen

New York
+1 212 438 2529
gerrit.jepsen@spglobal.com

Obioma Ugboaja

New York
+1 212 438 7406
obioma.ugboaja@spglobal.com

Matthew O'Neill

New York
+1 212 438 4295
matthew.oneill@spglobal.com

Industry Top Trends 2023: North America Regulated Utilities

Ratings Trends: North America Regulated Utilities

Chart 1

Ratings distribution

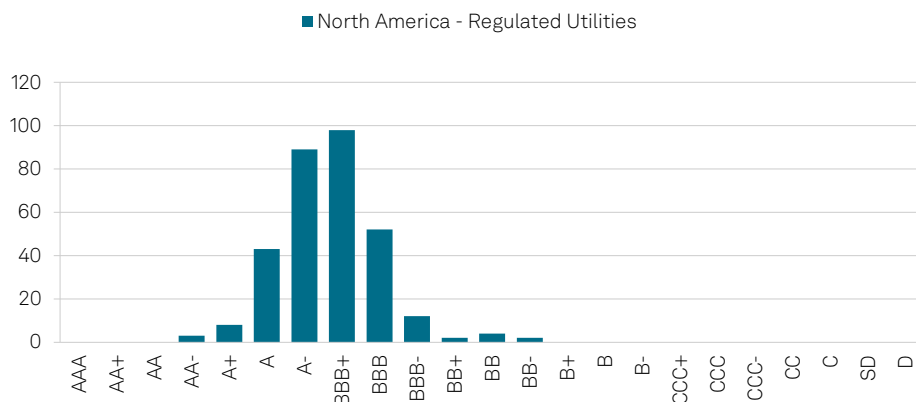


Chart 2

Ratings outlooks

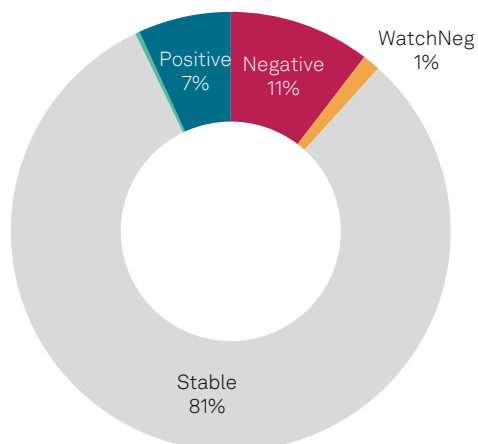
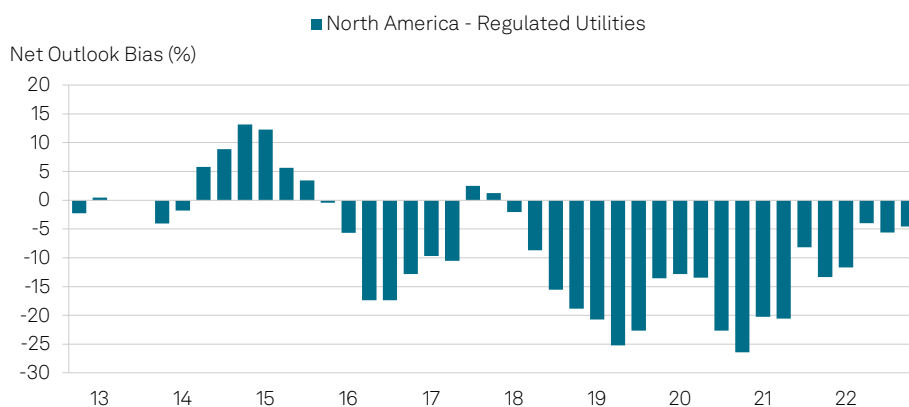


Chart 3

Ratings outlook net bias



Source: S&P Global Ratings. Ratings data measured at quarter-end.

Industry Top Trends 2023: North America Regulated Utilities

Industry Credit Metrics: North America Regulated Utilities

Chart 4

Debt / EBITDA (median, adjusted)

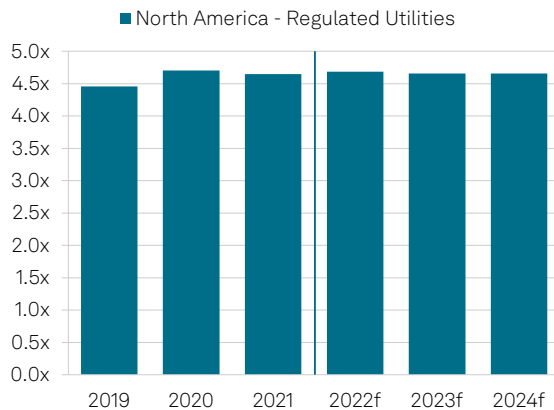


Chart 5

FFO / Debt (median, adjusted)

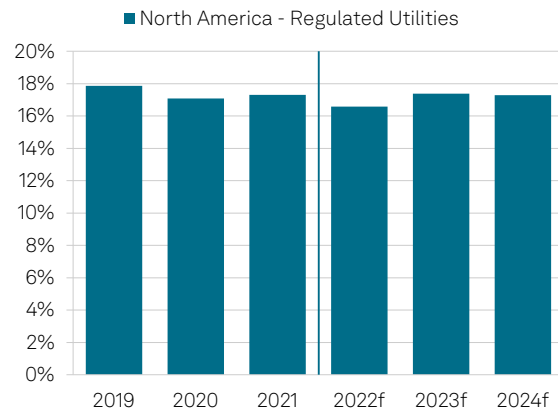


Chart 6

Cash flow and primary uses

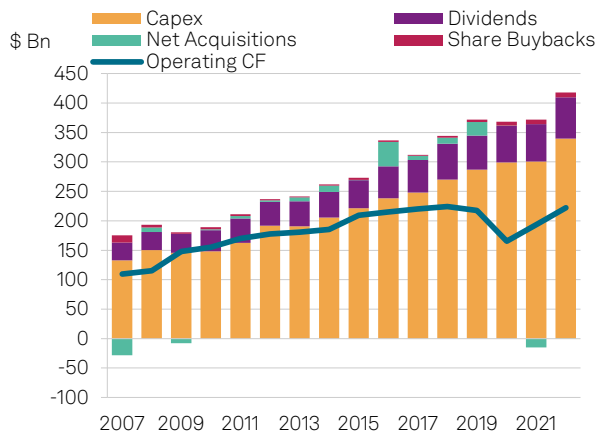
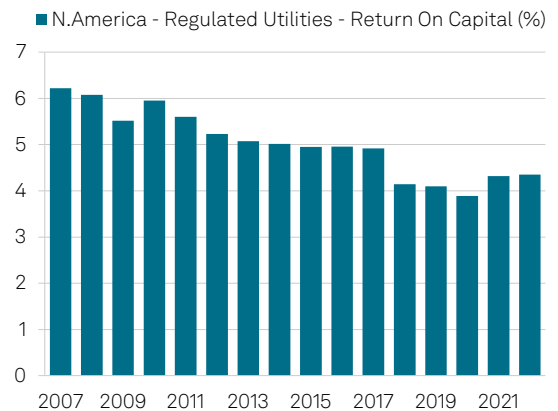


Chart 7

Return on capital employed



Source: S&P Global Ratings, S&P Capital IQ.

Revenue growth shows local currency growth weighted by prior-year common-currency revenue share. All other figures are converted into U.S. dollars using historic exchange rates. Forecasts are converted at the last financial year-end spot rate. FFO—Funds from operations. Most recent (2022) figures for cash flow and primary uses and return on capital employed use the last 12 months' data.

Industry Top Trends 2023: North America Regulated Utilities

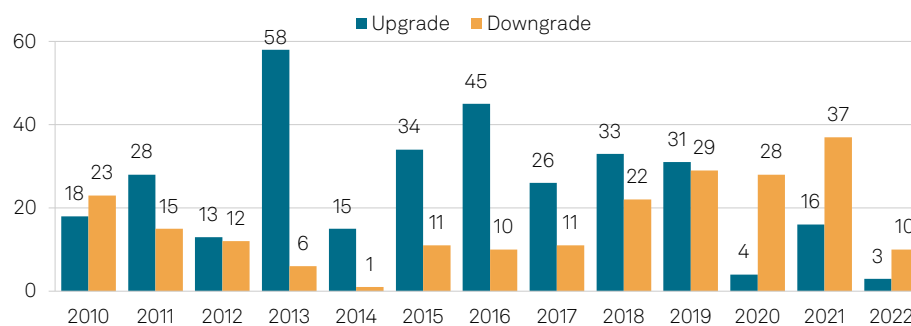
Industry Outlook

Ratings trends and outlook

The industry outlook remains negative and has been negative since early 2020. Over this timeframe downgrades have outpaced upgrades by more than 3:1 (see chart 8). While the industry's percentage of negative outlooks has decreased to about 15% from 35% at year-end 2020, prolonged inflationary risks or a deeper-than-expected recession could harm the industry's credit quality in 2023.

Chart 8

North America Regulated Utilities Upgrades And Downgrades



Data as of Jan. 11, 2023. Source: S&P Global Ratings.

Main assumptions about 2023 and beyond

1. Minimal financial cushion

More than 40% of the industry is strategically managing their financial performance with only minimal financial cushion, reflecting funds from operations (FFO) to debt that is less than 100 basis points above the downgrade threshold. Because utility cash flows are typically more stable than those of many other industries, this strategy of limiting excess credit capacity works well under ordinary conditions. However, when unexpected risks occur or base case assumptions deviate from expectations, the utility's credit quality can weaken, as we've seen over the past three years.

2. Consistent access to the capital markets

Because of the industry's high capital spending and consistent dividends, negative discretionary cashflow is regularly more than \$100 billion annually. To fund this large deficit, the industry requires consistent access to the capital markets. Rising interest rates, decreasing equity prices, and inflation could hamper consistent access the capital markets, potentially pressuring credit quality. Typically, most of the funding of negative discretionary cashflow is from new debt issuance with the balance from common equity, hybrid securities, and asset sales. For 2023, we expect the industry will rely on a lower percentage of common equity compared to prior years and instead rely on a higher percentage of asset sales.

3. Energy transition

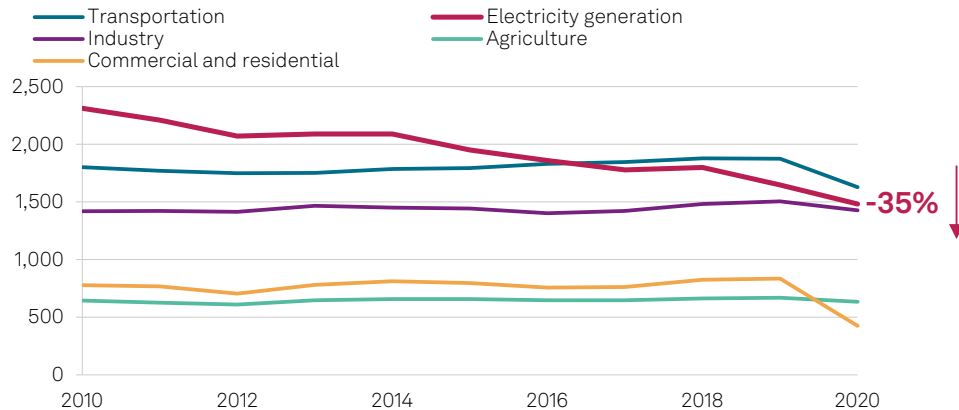
Over the past decade, the utility industry reduced its reliance on coal-fired generation by more than 50% and more than doubled its generation from renewable energy. We expect that by the end of the decade it will reduce coal-fired generation by about another 50% and will fully phase out coal by about 2040.

Industry Top Trends 2023: North America Regulated Utilities

The industry is aggressively reducing its greenhouse gas (GHG) emissions. Over the past decade, the industry reduced its GHG emissions by about 35%, mostly by replacing coal-fired generation with natural-gas-fired generation, which has about half the carbon emissions (see chart 9). However, we expect that during this decade GHG emissions will decrease by closer to 50% because the remaining coal closures will mostly be replaced with renewable generation and batteries.

Chart 9

Greenhouse Gas Emissions By Economic Sector (2010-2020)

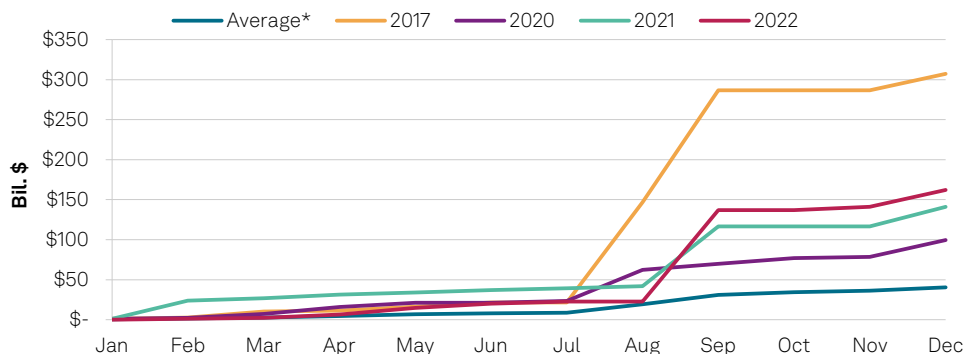


Source: U.S. Environmental Protection Agency.

Physical risks appear to be continuing at a record pace. During the past several years, the U.S. has experienced record levels of damages from storms and hurricanes, driving costs up for utilities (see chart 10). Accordingly, we downgraded several utilities when these rising costs led to higher leverage and weakening financial measures. Downgrades would have been substantially higher if not for regulators permitting the use of securitization, which we view as credit-supportive. For 2023, we again expect that the industry will significantly rely on securitization to recover more than \$5 billion of storm related costs.

Chart 10

U.S. Weather And Climate Billion Dollar Disaster Cost



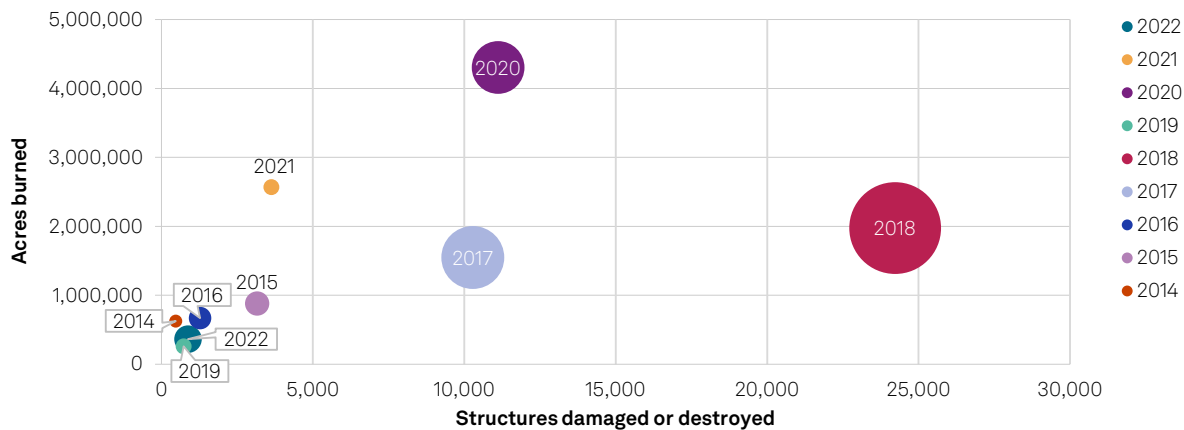
*1980-2022 average. Source: National Oceanic and Atmospheric Administration.

Industry Top Trends 2023: North America Regulated Utilities

California Wildfires. While other physical risks (such as severe storms) affected many utilities during 2022, California wildfires were less destructive for the second consecutive year (see chart 11). California utilities are most exposed to wildfire risks because of climate change and inverse condemnation. During the past several years California utilities have been proactively reducing their risks by investing in technology and system hardening. The technology helps the utility to predict the timing and location of a high-wind event, and utilities have found that proactively shutting down parts of their system is an effective way to avoid a catastrophic wildfire.

Chart 11

California Annual Wildfire Activity



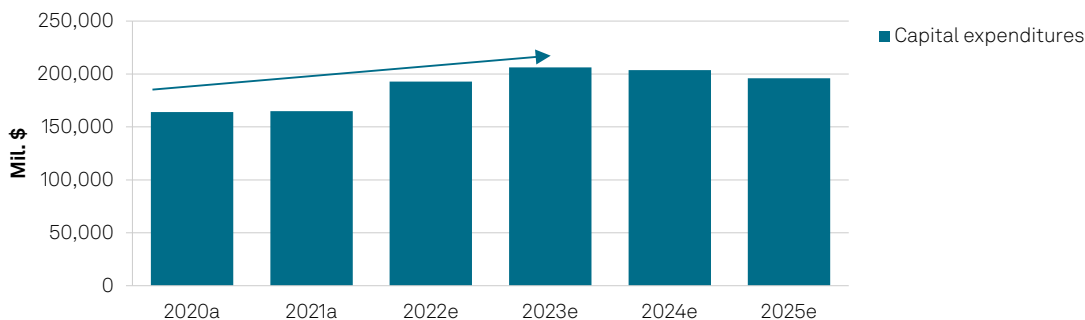
Bubble size represents number of fatalities. Source: The California Department of Forestry and Fire Protection.

Net-zero goals could be delayed. Most of the industry is committed to a net-zero emissions date between 2030 and 2050. As customer bill pressure increases, the industry may be forced to slow the pace of the energy transition, delaying the timeframe to reach net zero carbon emission.

Large projects could become more challenging. We expect the industry's 2023 capital spending to reach a record of more than \$200 billion, but given the macro risks large projects--including offshore wind--could become increasingly challenged (see chart 12). Supply chain delays and rising interest rates increase the probability that projects are not completed on time and on budget. While these risks affect all projects, larger and more complex projects are disproportionately affected, increasing the probability that some of these projects could be delayed or even canceled.

Chart 12

North America Regulated Utilities Rising Capital Expenditures



Data as of Jan. 11, 2023. Capital expenditures represent North American investor-owned electric, gas, and water utilities. a--Actual. e--Estimate. Source: S&P Global Ratings.

Industry Top Trends 2023: North America Regulated Utilities

Credit metrics and financial policy

Over the past decade the industry's financial measures have weakened from a combination of rising capital spending, regulatory lag, and lower authorized return on equity (ROE). The industry's return on capital was about 6% a decade ago and today is closer to 4%. More recently, we have seen instances where not only is the authorized ROE lowered but also the equity ratio is lowered. These results have weakened the industry's financial measures, pressuring credit quality. Under our base case of moderating inflationary risks during 2023, we expect the industry's credit measures to generally remain flat. However, if inflationary risks persist, it may further pressure the customer bill, potentially decreasing the level of regulatory credit support, weakening the industry's financial performance.

Key risks or opportunities around the baseline

1. Inflation reduction act (IRA)

While this legislation will benefit much of the industry, there are aspects of the law that will be detrimental to a few companies. The key benefits are the expansion of tax credits and the transferability of these credits. The legislation provides long-term tax credits for renewables, batteries, nuclear power, and hydrogen. We expect the use of these credits will significantly increase because of the relatively easy transferability of these tax credit. Conversely, the IRA also establishes an alternative minimal tax, which we expect will increase taxes for large transmission and distribution holding companies, pressuring their financial measures.

2. Recession

S&P Global's economists forecast the likelihood of a 2023 recession at greater than 50%. Should the recession be more severe than expected, unemployment could rise faster than expected, increasing the industry's uncollectibles and pressuring its financial measures.

3. Affordability of customer bill

Customer bills may become less affordable because of rising commodity prices, interest rates, inflation, and capital spending. During 2022, Henry Hub natural gas prices, the U.S. benchmark, peaked at about \$9 per mmBTU. Although prices have since retreated to about \$4/mmBTU and the forward curve reflects \$3.50-\$4.50/mmBTU, they remain substantially higher than pre-inflation levels, pressuring the customer bill. While we estimate the industry's average electric bill represents only about 2.5% of after-tax household income, sharp increases and bill volatility often results in increasing customer dissatisfaction that can ultimately heighten regulatory scrutiny and constrain the industry's ability to effectively manage regulatory risk.

Political interference could increase. As the customer bill continues to increase, the industry could experience increasing political interference, which we view as negative for credit quality. Most recently, the Canadian province of Nova Scotia enacted an amendment to the Public Utilities Act that overrode the regulatory process of the Nova Scotia Utility and Review Board, limiting rate increases. We viewed this as political interference that will undermine the regulatory construct, materially weakening the regulatory jurisdiction's predictability, and increase the uncertainty for its utilities and stakeholders. As such, we downgraded Nova Scotia Power Inc. by two notches.

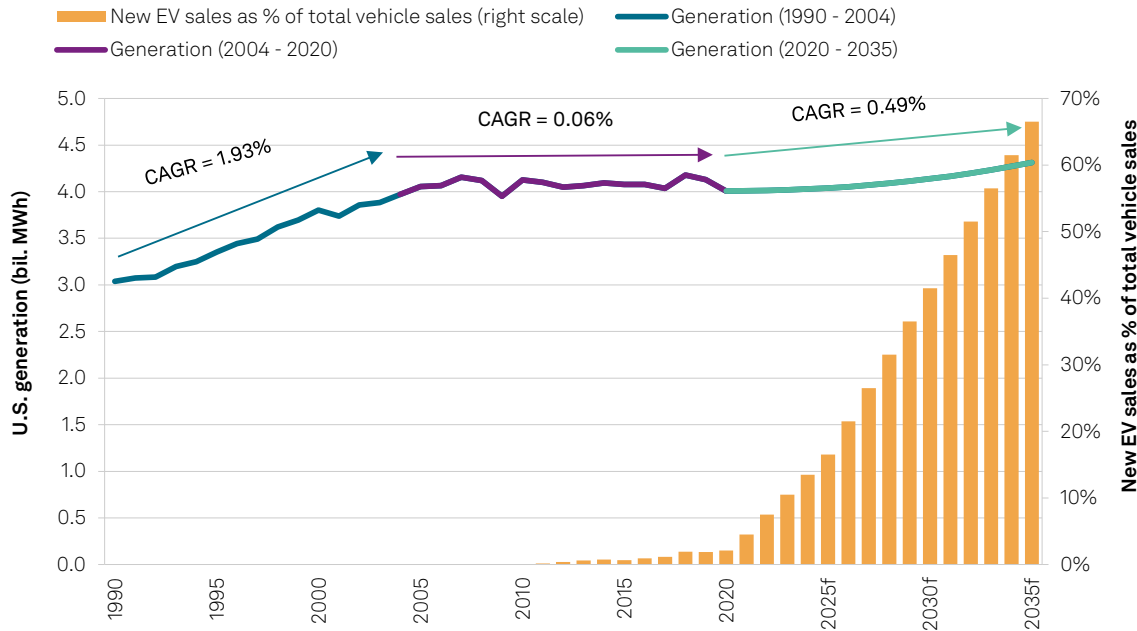
Growth of electric vehicles. While electric vehicles represented only about 6% of new U.S. car sales in 2022, we expect that by 2025 they will represent 15%-20%, leading to higher electricity sales (see chart 13). Should this growth trend persist, it is likely that by the next decade electric

Industry Top Trends 2023: North America Regulated Utilities

vehicles will materially contribute to the industry's sales growth, despite our expectations for continued conservation, which is the driver for flat sales growth over the last decade and a half.

Chart 13

U.S. Electric Vehicle Growth: Poised To Boost Demand



f--Forecast. CAGR--Compound annual growth rate. Sources: EIA, Alternative Fuels Data Center, S&P Global Ratings.

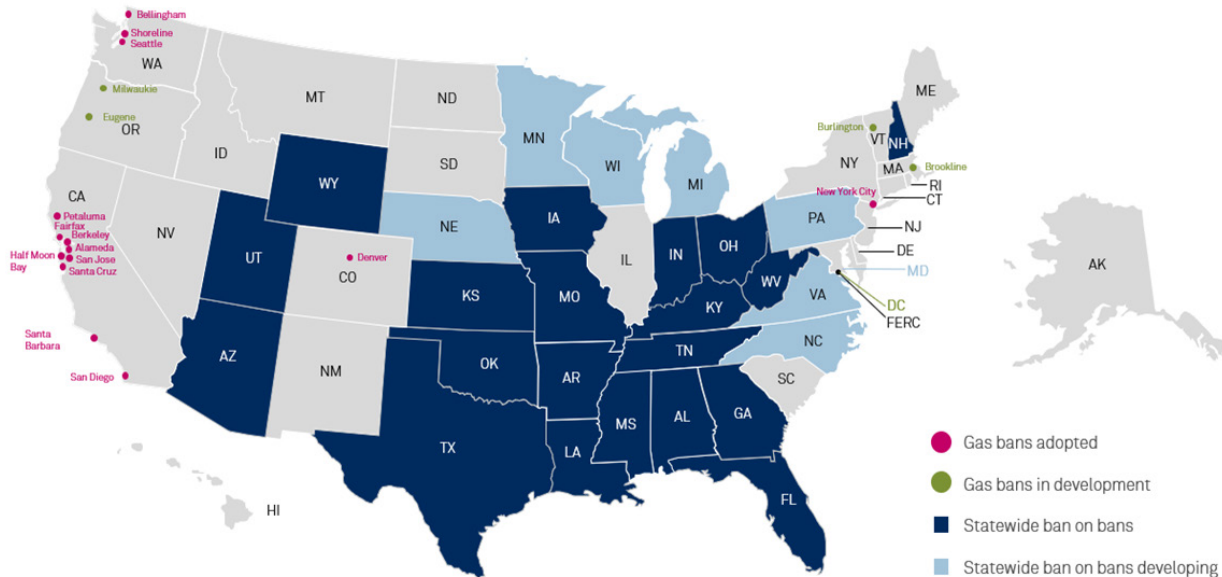
New technologies to reduce GHG emission. We believe an additional benefit of the IRA is it encourages further investments in carbon-free technologies. We expect the industry will continue to invest in hydrogen and renewable natural gas, and that these technologies will blend within the natural gas distribution system and account for up to 5% of the electric generation portfolio by 2040.

Rising risks for the natural gas local distribution industry. The industry's sales growth is becoming more challenging, with several cities banning new gas connections (see chart 14). While all industries typically require growth to increase valuations, credit quality could be preserved if this regulated industry can consistently reduce its regulatory lag and maintain regulatory support for its existing assets, despite its growth challenges.

Industry Top Trends 2023: North America Regulated Utilities

Chart 14

Ban On New Gas Connections



Data as of Nov. 1, 2022. Source: S&P Global Ratings.

Security. The 2021 breach of the Colonial Pipeline highlights the cybersecurity risks facing critical infrastructure assets. More recently, the shooting attack on Duke Energy's North Carolina substation that caused more than 45,000 customers to be without power underscores the industry's ongoing physical security risks. Overall, the industry's ongoing vigilance in these areas is critical to maintaining its credit quality.

Related Research

- [North American Utility Regulatory Jurisdictions Update: Nova Scotia Assessment Revised Downward Again After Passage Of Bill 212](#), Dec. 1, 2022
- [Inflation And High Yields: N.A. Power & Utilities: Inflation Could Create Some Pressure For Utilities](#), Nov. 29, 2022
- [Although Not Immune, North American Investor-Owned Utilities Are Still Well-Positioned To Handle Inflation](#), Aug. 18, 2022
- [Constructive Financial Policy Is Determining Credit Quality--Not Equity Pricing--For N.A. Regulated Utilities](#), June 8, 2022
- [Utilities' Early Retirement Of Coal Generation Increases Uncertainty Over Recouping Stranded Investments](#), April 11, 2022

Industry Top Trends 2023: North America Regulated Utilities

Copyright 2023 © by Standard & Poor's Financial Services LLC. All rights reserved.

No content (including ratings, credit-related analyses and data, valuations, model, software or other application or output therefrom) or any part thereof (Content) may be modified, reverse engineered, reproduced or distributed in any form by any means, or stored in a database or retrieval system, without the prior written permission of Standard & Poor's Financial Services LLC or its affiliates (collectively, S&P). The Content shall not be used for any unlawful or unauthorized purposes. S&P and any third-party providers, as well as their directors, officers, shareholders, employees or agents (collectively S&P Parties) do not guarantee the accuracy, completeness, timeliness or availability of the Content. S&P Parties are not responsible for any errors or omissions (negligent or otherwise), regardless of the cause, for the results obtained from the use of the Content, or for the security or maintenance of any data input by the user. The Content is provided on an "as is" basis. S&P PARTIES DISCLAIM ANY AND ALL EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, ANY WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE OR USE, FREEDOM FROM BUGS, SOFTWARE ERRORS OR DEFECTS, THAT THE CONTENT'S FUNCTIONING WILL BE UNINTERRUPTED OR THAT THE CONTENT WILL OPERATE WITH ANY SOFTWARE OR HARDWARE CONFIGURATION. In no event shall S&P Parties be liable to any party for any direct, indirect, incidental, exemplary, compensatory, punitive, special or consequential damages, costs, expenses, legal fees, or losses (including, without limitation, lost income or lost profits and opportunity costs or losses caused by negligence) in connection with any use of the Content even if advised of the possibility of such damages.

Credit-related and other analyses, including ratings, and statements in the Content are statements of opinion as of the date they are expressed and not statements of fact. S&P's opinions, analyses, and rating acknowledgment decisions (described below) are not recommendations to purchase, hold, or sell any securities or to make any investment decisions, and do not address the suitability of any security. S&P assumes no obligation to update the Content following publication in any form or format. The Content should not be relied on and is not a substitute for the skill, judgment and experience of the user, its management, employees, advisors and/or clients when making investment and other business decisions. S&P does not act as a fiduciary or an investment advisor except where registered as such. While S&P has obtained information from sources it believes to be reliable, S&P does not perform an audit and undertakes no duty of due diligence or independent verification of any information it receives. Rating-related publications may be published for a variety of reasons that are not necessarily dependent on action by rating committees, including, but not limited to, the publication of a periodic update on a credit rating and related analyses.

To the extent that regulatory authorities allow a rating agency to acknowledge in one jurisdiction a rating issued in another jurisdiction for certain regulatory purposes, S&P reserves the right to assign, withdraw, or suspend such acknowledgment at any time and in its sole discretion. S&P Parties disclaim any duty whatsoever arising out of the assignment, withdrawal, or suspension of an acknowledgment as well as any liability for any damage alleged to have been suffered on account thereof.

S&P keeps certain activities of its business units separate from each other in order to preserve the independence and objectivity of their respective activities. As a result, certain business units of S&P may have information that is not available to other S&P business units. S&P has established policies and procedures to maintain the confidentiality of certain nonpublic information received in connection with each analytical process.

S&P may receive compensation for its ratings and certain analyses, normally from issuers or underwriters of securities or from obligors. S&P reserves the right to disseminate its opinions and analyses. S&P's public ratings and analyses are made available on its Web sites, www.spglobal.com/ratings (free of charge) and www.ratingsdirect.com (subscription) and may be distributed through other means, including via S&P publications and third-party redistributors. Additional information about our ratings fees is available at www.spglobal.com/ratings/usratingsfees.

STANDARD & POOR'S, S&P and RATINGSDIRECT are registered trademarks of Standard & Poor's Financial Services LLC.



Consumers Energy Co.

Primary Credit Analyst:

Kevin M Sheridan, New York + 1 (212) 438 3022; kevin.sheridan@spglobal.com

Secondary Contact:

Gabe Grosberg, New York + 1 (212) 438 6043; gabe.grosberg@spglobal.com

Table Of Contents

Credit Highlights

Outlook

Our Base-Case Scenario

Company Description

Peer Comparison

Business Risk

Financial Risk

Liquidity

Environmental, Social, And Governance

Group Influence

Issue Ratings – Subordination Risk Analysis

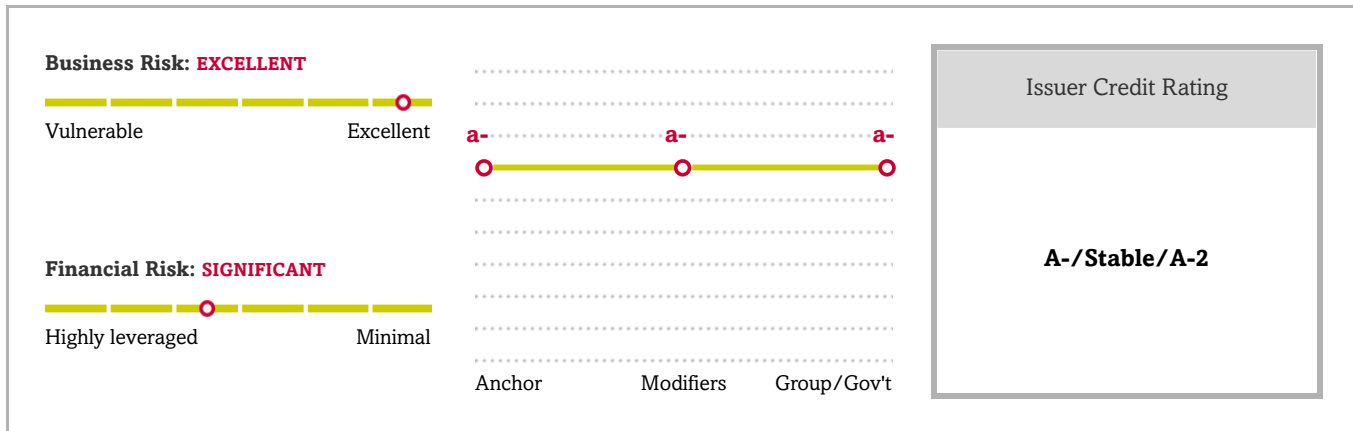
Issue Ratings - Recovery Analysis

Ratings Score Snapshot

Table Of Contents (cont.)

Related Criteria

Consumers Energy Co.



Credit Highlights

Overview

Key strengths	Key risks
Larger-than-average vertically integrated electric utility and gas distribution utility.	Limited geographic and regulatory diversity makes the company largely dependent on Michigan regulators to sustain credit quality.
Favorable regulatory construct in Michigan.	Significant exposure to carbon emission risk through high reliance on natural gas and coal-fired generation.
Effectively managed the associated risks of the COVID-19 pandemic.	Negative discretionary cash flow, reflecting robust capital spending, necessitating constant access to fairly priced capital markets.
Sufficient insulating measures for higher rating than group credit profile.	

The above-average regulatory environment in Michigan should continue to allow for consistent and constructive regulatory outcomes. Consumers Energy Co.'s most recent electric rate case order led to an increase in revenue of about \$90 million. This revenue increase incorporates the return of remaining deferred taxes from the Tax Cuts and Jobs Act (TCJA) to electric customers and reflects a lower return on equity (ROE; 9.9% compared to 10%) and a lower equity ratio (51.11% compared to 52.50%). Although we view resolving the effects of tax reform through this rate case as favorable, if lower ROEs and a lower equity ratio persist, credit quality could weaken.

The most recent gas rate case order led to an increase in revenue by \$144 million and maintained a consistent ROE of 9.9% and equity ratio of 52.05%. However, this outcome was slightly offset by a stay-out agreement that restricts Consumers Energy from filing another gas rate case before December 2021. We believe this outcome, while supportive of credit quality, may eventually lead to some regulatory lag because of the stay-out provision.

Because we assess the lower ROE and equity ratio in the electric rate case and the stay-out provision on the gas side as one-time decisions, we expect that the utility will continue to effectively manage regulatory risk through the use of various constructive rate mechanisms made available by the Michigan Public Service Commission (MPSC). In general, we view Michigan's regulatory construct as above average compared to peers because of the benefit of forward-looking test years, a stream-lined 10-month rate case process, and various constructive rate mechanisms--such as power supply and natural gas cost rider adjustments and partial decoupling for the gas

business--which help the company earn its allowed ROE and minimize regulatory lag.

The company effectively managed COVID-19-related risks. Given Consumers Energy's mostly residential customer base (about 60%), the company's exposure to the large commercial and industrial sales declines across the industry has been more limited than for many of its peers. Additionally, the company achieved cost savings of about \$100 million during the year by reducing operating and maintenance expenses, and it continues to pursue other cost efficiency strategies, offsetting incremental COVID-19-related expenses. Additionally, Consumers Energy received a constructive order from the MPSC in April 2020, which reduced the effect of bad debt expense on earnings by allowing the company to defer incremental uncollectible accounts related to the pandemic.

The company's elevated utility capital spending plan prioritizes infrastructure upgrades and renewable energy. Over the next five years, Consumers Energy plans to spend about \$9.4 billion to maintain and upgrade its gas infrastructure and electric distribution systems. The capital plan includes elevated investment of about \$5 billion in the gas segment, as well as about \$2.8 billion in electric supply projects, which primarily focus on renewable generation.

Consumers Energy has significant carbon emissions through its heavy reliance on natural gas and coal generation. While most of Consumers Energy's electricity is generated from natural gas (about 37%) and coal (about 29%), the company intends to reduce its carbon exposure through its Clean Energy Plan. The plan includes a 2040 long-term goal to significantly reduce carbon emissions of its owned generation by more than 90% from 2005 and eliminate coal-sourced electric generation. The plan also provides the foundation for the utility to achieve net-zero carbon emissions by 2040.

We expect Consumers Energy's consolidated credit measures to remain in the middle of the range for its financial risk profile category. Although we expect some modest weakening in financial metrics as a result of the recent electric rate case order, as well as some potential regulatory lag, we believe the company will continue to manage its funds from operations (FFO) to debt at about 20% over the next three years, consistent with the middle of the range for its financial risk profile category.

Outlook: Stable

The stable rating outlook on Consumers Energy reflects our expectation that management will focus on its core utility operations and reach constructive regulatory outcomes to avoid increasing business risk. We expect Consumers Energy will maintain stand-alone financial measures consistent with the middle of the range for its financial risk profile category, specifically FFO to debt at about 20%.

Downside scenario

We could lower our rating on Consumers Energy if its stand-alone financial measures weaken such that FFO to debt weakens to consistently below 15%. We could also lower our rating on Consumers Energy if we lower our rating on its parent, CMS Energy Corp.

Upside scenario

Although less likely, we could raise our rating on Consumers Energy if we raise our rating on CMS Energy and Consumers Energy's stand-alone financial measures improve, reflecting FFO to debt of consistently above 20%.

Our Base-Case Scenario

Assumptions

- Consistent rate case filings and use of existing regulatory mechanisms;
- Elevated capital spending over the forecast period averaging \$2.4 billion annually;
- Annual dividends of \$600 million-\$900 million annually;
- Equity distributions to parent CMS Energy, averaging \$530 million annually;
- All debt maturities are refinanced; and
- Continued negative discretionary cash flow will be financed in a balanced manner to support the regulated capital structure.

Key metrics

Table 1

Consumers Energy Co.--Key Metrics*

	2019a	2020e	2021e
FFO to debt (%)	22.1	20-22	19-21
Debt to EBITDA (x)	3.7	3.5-4.5	4-5
FFO interest coverage	7.2	6.5-7.5	6-7

*All figures adjusted by S&P Global Ratings. a--Actual. e--Estimate. FFO--Funds from operations.

Company Description

Consumers Energy is a subsidiary of CMS Energy and operates as an electric and gas utility serving about 3.6 million customers in Michigan. Consumers Energy's electric business operates as a vertically integrated utility that generates, distributes, and sells electricity. The electric utility sources about half of its generation from purchased power, rather than from its own plants. The company also sells, stores, and transports natural gas. It is based in Jackson, Mich.

Peer Comparison

Table 2

Consumers Energy Co.--Peer Comparison

Industry sector: Combo

	Consumers Energy Co.	DTE Electric Co.	Alliant Energy Corp.	Ameren Corp.	WEC Energy Group Inc.
Rating as of Jan. 19, 2021	A-/Stable/A-2	A-/Stable/A-2	A-/Stable/A-2	BBB+/Stable/A-2	A-/Stable/A-2
--Fiscal year ended Dec. 31, 2019--					

Table 2

Consumers Energy Co.--Peer Comparison (cont.)

Industry sector: Combo

	Consumers Energy Co.	DTE Electric Co.	Alliant Energy Corp.	Ameren Corp.	WEC Energy Group Inc.
(Mil. \$)					
Revenue	6,341.5	5,224.0	3,647.7	5,910.0	7,523.1
EBITDA	2,251.6	2,297.4	1,384.2	2,396.0	2,727.6
FFO	1,823.4	1,942.7	1,099.5	1,991.7	2,250.1
Interest expense	335.2	464.8	300.2	433.3	541.3
Cash interest paid	296.2	308.8	305.3	391.3	502.4
Cash flow from operations	1,688.4	1,739.7	629.5	2,132.7	2,401.9
Capital expenditure	2,190.9	2,192.3	1,612.9	2,422.0	2,302.9
FOCF	(502.5)	(452.6)	(983.4)	(289.3)	99.0
DCF	(1,094.5)	(946.6)	(1,316.0)	(793.3)	(796.2)
Cash and short-term investments	11.0	12.0	16.3	16.0	37.5
Debt	8,237.6	9,086.9	7,206.1	9,728.8	13,145.8
Equity	7,737.0	7,195.0	5,305.1	8,130.0	10,489.4
Adjusted ratios					
EBITDA margin (%)	35.5	44.0	37.9	40.5	36.3
Return on capital (%)	7.7	8.2	7.7	7.7	7.6
EBITDA interest coverage (x)	6.7	4.9	4.6	5.5	5.0
FFO cash interest coverage (x)	7.2	7.3	4.6	6.1	5.5
Debt/EBITDA (x)	3.7	4.0	5.2	4.1	4.8
FFO/debt (%)	22.1	21.4	15.3	20.5	17.1
Cash flow from operations/debt (%)	20.5	19.1	8.7	21.9	18.3
FOCF/debt (%)	(6.1)	(5.0)	(13.6)	(3.0)	0.8
DCF/debt (%)	(13.3)	(10.4)	(18.3)	(8.2)	(6.1)

FFO--Funds from operations. FOCF--Free operating cash flow. DCF--Discretionary cash flow.

Business Risk: Excellent

Our assessment of Consumers Energy's business risk profile reflects the company's lower-risk electric and natural gas utility operations. Consumers Energy is a larger-than-average utility that serves about 1.8 million electric customers and about 1.8 million natural gas customers throughout Michigan. In addition, about 80% of the company's electric customer revenue base is residential and commercial, providing stable cash flow and mitigating the company's exposure to industrial cyclicality. Consumers Energy is a wholly owned subsidiary of CMS Energy and contributes about 95% of CMS Energy's consolidated operations.

The MPSC regulates Consumers Energy. We view the regulatory environment in Michigan as above average

compared to peers as demonstrated through the company's benefit from forward-looking test years and a stream-lined 10-month rate case process. Consumers Energy also receives other constructive rate mechanisms, such as the Power Supply Cost Recovery and Gas Cost Recovery adjustment riders, as well as partial decoupling for the gas business, which annually reconciles actual weather-normalized non-fuel revenues with the revenues approved by the MPSC. These constructive rate mechanisms enable Consumers Energy to generally earn its allowed ROE and minimize regulatory lag.

The company's most recent electric rate case became effective in January 2021, and it increased revenue by about \$90 million. The order incorporated an equity ratio of 51.11%, an ROE of 9.9%, and the return of deferred tax liabilities from the TCJA to electric customers through rates. The order also approved the recovery of the balance of the financial compensation mechanism (FCM) approved in Consumers Energy's most recent integrated resource plan. We view the FCM as a constructive rate mechanism because it gives Consumers Energy another way to recover interim expenses by allowing the company to earn a financial incentive on certain power purchase agreements approved after Jan. 1, 2019, through a surcharge. Additionally, the company received approval of a new distributed generation tariff to replace the previous net metering tariff. We view this tariff as a useful mechanism to help Consumers Energy meet its demand response goals pursuant to Michigan's 2016 Energy Law. Although we do not view the lower equity ratio and ROE as supportive of credit quality, we believe this outcome is partially mitigated by the resolution of the effects of tax reform and the incorporation of new constructive rate mechanisms within the rate case. The most recent gas rate case order went into effect in October 2020, and it increased revenue by about \$144 million. The order included an equity ratio of 52.05%, an ROE of 9.9%, a stay-out agreement restricting the company from filing another gas rate case before 2021, and it accelerated amortization of certain tax liabilities beginning in October 2021. We view the outcome of this gas rate case as overall supportive of credit quality, despite some potential regulatory lag.

Financial Risk: Significant

We assess Consumers Energy's financial measures using our medial volatility table, reflecting the company's lower-risk regulated electric and gas utility operations and its effective management of regulatory risk. Under our base-case scenario, we expect elevated capital spending averaging \$2.4 billion annually over the forecast period, 2021 collective revenue increases of about \$234 million from the recent gas and electric rate cases, and a gas rate case stay-out until December 2021. We anticipate financial measures that are consistent with the middle of the company's financial risk category. Specifically, we forecast FFO to debt averaging about 20% over the outlook period.

Financial summary

Table 3

Consumers Energy Co.--Financial Summary

Industry sector: Combo					
--Fiscal year ended Dec. 31--					
	2019	2018	2017	2016	2015
(Mil. \$)					
Revenue	6,341.5	6,430.1	6,186.8	6,029.5	6,081.8
EBITDA	2,251.6	2,151.9	2,267.1	2,168.3	2,010.8

Table 3

Consumers Energy Co.--Financial Summary (cont.)

Industry sector: Combo

	--Fiscal year ended Dec. 31--				
	2019	2018	2017	2016	2015
FFO	1,823.4	1,665.1	1,953.7	1,820.4	1,779.5
Interest expense	335.2	354.9	347.4	332.8	340.3
Cash interest paid	296.2	330.9	314.4	297.8	315.3
Cash flow from operations	1,688.4	1,533.1	1,791.7	1,768.4	1,814.5
Capital expenditure	2,190.9	1,920.3	1,721.1	1,753.7	1,615.1
FOCF	(502.5)	(387.2)	70.6	14.7	199.4
DCF	(1,094.5)	(918.2)	(453.4)	(486.3)	(276.6)
Cash and short-term investments	11.0	39.0	44.0	131.0	50.0
Gross available cash	11.0	39.0	44.0	131.0	50.0
Debt	8,237.6	7,774.2	7,037.4	6,734.0	7,106.7
Equity	7,737.0	6,920.0	6,488.0	5,939.0	5,546.0
Adjusted ratios					
EBITDA margin (%)	35.5	33.5	36.6	36.0	33.1
Return on capital (%)	7.7	7.9	9.8	9.7	10.1
EBITDA interest coverage (x)	6.7	6.1	6.5	6.5	5.9
FFO cash interest coverage (x)	7.2	6.0	7.2	7.1	6.6
Debt/EBITDA (x)	3.7	3.6	3.1	3.1	3.5
FFO/debt (%)	22.1	21.4	27.8	27.0	25.0
Cash flow from operations/debt (%)	20.5	19.7	25.5	26.3	25.5
FOCF/debt (%)	(6.1)	(5.0)	1.0	0.2	2.8
DCF/debt (%)	(13.3)	(11.8)	(6.4)	(7.2)	(3.9)

FFO--Funds from operations. FOCF--Free operating cash flow. DCF--Discretionary cash flow.

Reconciliation

Table 4

Consumers Energy Co.--Reconciliation Of Reported Amounts With S&P Global Ratings' Adjusted Amounts

--Fiscal year ended Dec. 31, 2019--

Consumers Energy Co. reported amounts (mil. \$)

	Debt	Shareholders' equity	Revenue	EBITDA	Operating income	Interest expense	S&P Global Ratings' adjusted EBITDA	Cash flow from operations	Capital expenditure
	7,369.0	7,700.0	6,376.0	2,105.0	1,130.0	297.0	2,251.6	1,601.0	2,085.0
S&P Global Ratings' adjustments									
Cash taxes paid	--	--	--	--	--	--	(132.0)	--	--
Cash interest paid	--	--	--	--	--	--	(279.0)	--	--
Reported lease liabilities	106.0	--	--	--	--	--	--	--	--
Operating leases	--	--	--	9.0	1.5	1.5	(1.5)	7.5	--

Table 4

Consumers Energy Co.--Reconciliation Of Reported Amounts With S&P Global Ratings' Adjusted Amounts (cont.)									
Postretirement benefit obligations/deferred compensation	194.3	--	--	--	--	--	--	--	--
Accessible cash and liquid investments	(11.0)	--	--	--	--	--	--	--	--
Capitalized interest	--	--	--	--	--	4.0	(4.0)	(4.0)	(4.0)
Share-based compensation expense	--	--	--	21.0	--	--	--	--	--
Securitized stranded costs	(251.0)	--	(34.5)	(34.5)	(8.5)	(8.5)	8.5	(26.0)	--
Power purchase agreements	545.8	--	--	130.1	20.2	20.2	(20.2)	109.9	109.9
Asset-retirement obligations	374.5	--	--	21.0	21.0	21.0	--	--	--
Nonoperating income (expense)	--	--	--	--	10.0	--	--	--	--
Noncontrolling interest/minority interest	--	37.0	--	--	--	--	--	--	--
Debt: Other	(90.0)	--	--	--	--	--	--	--	--
Total adjustments	868.6	37.0	(34.5)	146.6	44.2	38.2	(428.2)	87.4	105.9
S&P Global Ratings' adjusted amounts									
	Debt	Equity	Revenue	EBITDA	EBIT	Interest expense	Funds from operations	Cash flow from operations	Capital expenditure
	8,237.6	7,737.0	6,341.5	2,251.6	1,174.2	335.2	1,823.4	1,688.4	2,190.9

Liquidity: Adequate

As of Sept. 30, 2020, we assess Consumers Energy's liquidity as adequate to cover its needs over the next 12 months, even if consolidated EBITDA declines 10%. We expect the company's liquidity sources will exceed uses by more than 1.1x during this period. Our assessment also reflects Consumers Energy's sound relationships with banks, satisfactory standing in the credit markets, and generally prudent risk management.

Principal liquidity sources	Principal liquidity uses
- Cash of about \$199 million; - Credit facility availability of about \$1.09 billion; and - Estimated cash FFO of about \$1.99 billion.	- Long-term debt maturities of about \$557 million; - Working capital outflows of about \$274 million; - Maintenance capital spending of about \$1.29 billion; and - Dividends of about \$690 million.

Environmental, Social, And Governance

We assess Consumers Energy's environmental risks as somewhat higher than peers' to reflect the relatively higher percentage of electricity generated from fossil fuels. We assess the company's social and governance risk factors as in line with those of its peers.

About 37% of Consumers Energy's owned and purchased generation use is from natural gas, 29% is from coal, 19% is from nuclear, and 10% is from renewables. The company's significant reliance on fossil fuels exposes it to the potential for changing environmental regulations that may result in higher operating costs, require significant capital investment, lead to premature plant closures, or increase operating risks. However, the company has been offsetting some of these risks through strategic reduction of its environmental liabilities through coal plant retirements and by increasing the proportion of natural gas and renewables in its generation portfolio. The utility's capital plan and its ongoing investment in MPSC-approved wind and solar projects demonstrate a commitment to invest in renewable generation. These planned utility investments are in line with the company's Clean Energy Plan, which includes a 2040 long-term goal to reduce carbon emissions of owned generation by more than 90% from 2005, eliminate the use of coal-sourced electric generation, and target net-zero carbon emissions.

On the gas side, the company's older assets are susceptible to natural gas leaks, which emit methane. Although aging infrastructure poses a clear operational risk, the company plans significant investment in gas infrastructure upgrades over the next few years and has announced a net-zero methane goal from its natural gas delivery system by 2030.

Group Influence

Under our group rating methodology, we consider CMS Energy as the parent of the group with a group credit profile (GCP) of 'bbb+'. We assess Consumers Energy as a core subsidiary of CMS Energy because we view the utility as integral to the group's identity, highly unlikely to be sold (95% of the consolidated company), and having a strong commitment from management, given the company's emphasis on maintaining the size of the regulated utility operations relative to the nonutility businesses.

Because Consumers Energy is operationally separate and sufficient insulating measures are in place, we rate the utility one notch above the GCP. Some of the key insulating measures in place include the following:

- Consumers Energy is a separate and stand-alone legal entity that functions independently (both financially and operationally), files its own rate cases, and is independently regulated by MPSC.
- Consumers Energy has its own records and books, including stand-alone audited financial statements.
- The utility has its own funding arrangements, including issuing its own long-term debt, and it has a separate committed credit facility to cover its short-term funding needs.
- Consumers Energy does not comingle funds, assets, or cash flow with parent CMS Energy or its other subsidiaries,

and it does not participate in a money pool.

- We believe there is a strong economic basis for CMS Energy to preserve Consumers Energy's credit strength, reflecting Consumers Energy's low-risk, profitable, and regulated utility business model. Consumers Energy is also a significant portion of CMS Energy, accounting for about 95% of the consolidated company.
- There are no cross-default provisions between parent CMS Energy and Consumers Energy that could directly lead to a default at the utility.

Issue Ratings – Subordination Risk Analysis

Capital structure

As of Sept. 30, 2020, Consumers Energy's capital structure consisted of about \$8.4 billion of long-term debt, including about \$8 billion in first mortgage bonds (FMBs).

Analytical conclusions

We base our 'A-2' short-term rating on Consumers Energy on our issuer credit rating.

Issue Ratings - Recovery Analysis

- We assign recovery ratings to FMBs issued by U.S. utilities, which can result in issue ratings being notched above an issuer credit rating on a utility, depending on the rating category and the extent of the collateral coverage. The FMBs issued by U.S. utilities are a form of secured utility bonds (SUBs) that qualify for a recovery rating as defined in our criteria.
- The recovery methodology is supported by our expectation of 100% recovery for secured bondholders in utility bankruptcies in the U.S. and our view that the factors that enhance recoveries (limited size of the creditor class and the durable value of utility rate-based assets during and after a reorganization, given the essential service provided and the high replacement cost) will persist.
- Under our SUB criteria, we calculate a ratio of our estimate of the value of the collateral pledged to bondholders relative to the amount of FMBs outstanding. FMB ratings can exceed an issuer credit rating on a utility by up to one notch in the 'A' category, two notches in the 'BBB' category, and three notches in speculative-grade categories, depending on the calculated ratio.
- Consumers Energy's FMBs benefit from a first-priority lien on substantially all of the utility's real property owned or subsequently acquired. Collateral coverage of more than 1.5x supports a recovery rating of '1+' and an issue rating one notch above the issuer credit rating.

Ratings Score Snapshot

Issuer Credit Rating

A-/Stable/A-2

Business risk: Excellent

- **Country risk:** Very low

- **Industry risk:** Very low
- **Competitive position:** Excellent

Financial risk: Significant

- **Cash flow/leverage:** Significant

Anchor: a-

Modifiers

- **Diversification/portfolio effect:** Neutral (no impact)
- **Capital structure:** Neutral (no impact)
- **Financial policy:** Neutral (no impact)
- **Liquidity:** Adequate (no impact)
- **Management and governance:** Satisfactory (no impact)
- **Comparable rating analysis:** Neutral (no impact)

Stand-alone credit profile : a-

- **Group credit profile:** bbb+
- **Entity status within group:** Insulated (no impact)

Related Criteria

- General Criteria: Hybrid Capital: Methodology And Assumptions, July 1, 2019
- General Criteria: Group Rating Methodology, July 1, 2019
- Criteria | Corporates | General: Corporate Methodology: Ratios And Adjustments, April 1, 2019
- Criteria | Corporates | General: Reflecting Subordination Risk In Corporate Issue Ratings, March 28, 2018
- General Criteria: Methodology For Linking Long-Term And Short-Term Ratings, April 7, 2017
- Criteria | Corporates | General: Methodology And Assumptions: Liquidity Descriptors For Global Corporate Issuers, Dec. 16, 2014
- General Criteria: Methodology: Industry Risk, Nov. 19, 2013
- Criteria | Corporates | Utilities: Key Credit Factors For The Regulated Utilities Industry, Nov. 19, 2013
- Criteria | Corporates | General: Corporate Methodology, Nov. 19, 2013
- General Criteria: Country Risk Assessment Methodology And Assumptions, Nov. 19, 2013
- Criteria | Corporates | Utilities: Collateral Coverage And Issue Notching Rules For '1+' And '1' Recovery Ratings On Senior Bonds Secured By Utility Real Property, Feb. 14, 2013
- General Criteria: Methodology: Management And Governance Credit Factors For Corporate Entities, Nov. 13, 2012
- General Criteria: Use Of CreditWatch And Outlooks, Sept. 14, 2009

Business And Financial Risk Matrix

Business Risk Profile	Financial Risk Profile					
	Minimal	Modest	Intermediate	Significant	Aggressive	Highly leveraged
Excellent	aaa/aa+	aa	a+/a	a-	bbb	bbb-/bb+
Strong	aa/aa-	a+/a	a-/bbb+	bbb	bb+	bb
Satisfactory	a/a-	bbb+	bbb/bbb-	bbb-/bb+	bb	b+
Fair	bbb/bbb-	bbb-	bb+	bb	bb-	b
Weak	bb+	bb+	bb	bb-	b+	b/b-
Vulnerable	bb-	bb-	bb-/b+	b+	b	b-

Ratings Detail (As Of January 27, 2021)*

Consumers Energy Co.

Issuer Credit Rating	A-/Stable/A-2
Commercial Paper	
Local Currency	A-2
Senior Secured	A

Issuer Credit Ratings History

30-Oct-2019	A-/Stable/A-2
03-Dec-2014	BBB+/Stable/A-2
11-Sep-2014	BBB/Positive/A-2

Related Entities

CMS Energy Corp.

Issuer Credit Rating	BBB+/Stable/A-2
Junior Subordinated	BBB-
Senior Unsecured	BBB

*Unless otherwise noted, all ratings in this report are global scale ratings. S&P Global Ratings' credit ratings on the global scale are comparable across countries. S&P Global Ratings' credit ratings on a national scale are relative to obligors or obligations within that specific country. Issue and debt ratings could include debt guaranteed by another entity, and rated debt that an entity guarantees.

Copyright © 2021 by Standard & Poor's Financial Services LLC. All rights reserved.

No content (including ratings, credit-related analyses and data, valuations, model, software or other application or output therefrom) or any part thereof (Content) may be modified, reverse engineered, reproduced or distributed in any form by any means, or stored in a database or retrieval system, without the prior written permission of Standard & Poor's Financial Services LLC or its affiliates (collectively, S&P). The Content shall not be used for any unlawful or unauthorized purposes. S&P and any third-party providers, as well as their directors, officers, shareholders, employees or agents (collectively S&P Parties) do not guarantee the accuracy, completeness, timeliness or availability of the Content. S&P Parties are not responsible for any errors or omissions (negligent or otherwise), regardless of the cause, for the results obtained from the use of the Content, or for the security or maintenance of any data input by the user. The Content is provided on an "as is" basis. S&P PARTIES DISCLAIM ANY AND ALL EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, ANY WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE OR USE, FREEDOM FROM BUGS, SOFTWARE ERRORS OR DEFECTS, THAT THE CONTENT'S FUNCTIONING WILL BE UNINTERRUPTED OR THAT THE CONTENT WILL OPERATE WITH ANY SOFTWARE OR HARDWARE CONFIGURATION. In no event shall S&P Parties be liable to any party for any direct, indirect, incidental, exemplary, compensatory, punitive, special or consequential damages, costs, expenses, legal fees, or losses (including, without limitation, lost income or lost profits and opportunity costs or losses caused by negligence) in connection with any use of the Content even if advised of the possibility of such damages.

Credit-related and other analyses, including ratings, and statements in the Content are statements of opinion as of the date they are expressed and not statements of fact. S&P's opinions, analyses and rating acknowledgment decisions (described below) are not recommendations to purchase, hold, or sell any securities or to make any investment decisions, and do not address the suitability of any security. S&P assumes no obligation to update the Content following publication in any form or format. The Content should not be relied on and is not a substitute for the skill, judgment and experience of the user, its management, employees, advisors and/or clients when making investment and other business decisions. S&P does not act as a fiduciary or an investment advisor except where registered as such. While S&P has obtained information from sources it believes to be reliable, S&P does not perform an audit and undertakes no duty of due diligence or independent verification of any information it receives. Rating-related publications may be published for a variety of reasons that are not necessarily dependent on action by rating committees, including, but not limited to, the publication of a periodic update on a credit rating and related analyses.

To the extent that regulatory authorities allow a rating agency to acknowledge in one jurisdiction a rating issued in another jurisdiction for certain regulatory purposes, S&P reserves the right to assign, withdraw or suspend such acknowledgment at any time and in its sole discretion. S&P Parties disclaim any duty whatsoever arising out of the assignment, withdrawal or suspension of an acknowledgment as well as any liability for any damage alleged to have been suffered on account thereof.

S&P keeps certain activities of its business units separate from each other in order to preserve the independence and objectivity of their respective activities. As a result, certain business units of S&P may have information that is not available to other S&P business units. S&P has established policies and procedures to maintain the confidentiality of certain non-public information received in connection with each analytical process.

S&P may receive compensation for its ratings and certain analyses, normally from issuers or underwriters of securities or from obligors. S&P reserves the right to disseminate its opinions and analyses. S&P's public ratings and analyses are made available on its Web sites, www.standardandpoors.com (free of charge), and www.ratingsdirect.com (subscription), and may be distributed through other means, including via S&P publications and third-party redistributors. Additional information about our ratings fees is available at www.standardandpoors.com/usratingsfees.

STANDARD & POOR'S, S&P and RATINGSDIRECT are registered trademarks of Standard & Poor's Financial Services LLC.

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

RICHARD T. BLUMENSTOCK

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Generating Unit Periodic Outages

March 1, 2024 Through February 28, 2025

Case No.: U-21389

Exhibit No.: A-39 (RTB-1)

Page: 1 of 1

Witness: RTBlumenstock

Date: May 2023

MAJOR OUTAGES: FOSSIL GENERATION AND LUDINGTON

<u>Line</u> <u>No.</u>	(a) <u>Unit</u>	(b) <u>Planned Days in test year</u>	(c) <u>Start</u>	(d) <u>Stop</u>
1	Karn 3	43	3/1/2024	4/13/2024
2	Karn 4	38	3/31/2024	5/8/2024
3	Campbell 2	30	4/1/2024	5/1/2024
4	Campbell 3	30	4/16/2024	5/16/2024
5	Karn 3	55	9/20/2024	11/14/2024
6	Ludington 4	121	2/10/2025	6/10/2025
7	Ludington 3	120	2/10/2025	6/10/2025

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Generating Unit Availability Projections
March 1, 2024 Through February 28, 2025

Case No.: U-21389
Exhibit No.: A-40 (RTB-2)
Page: 1 of 1
Witness: RTBlumenstock
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)
Line No.	Plant	Actual ROR 2018-2022	Projected ROR	Periodic Factor	Projected Availability	Actual NEV 2018-2022
1	Campbell 1	15.56%	16.00%	6.87%	78.23%	\$96,757,676
2	Campbell 2	17.33%	14.50%	17.99%	70.12%	\$89,647,979
3	Campbell 3	11.70%	8.00%	8.22%	84.44%	\$365,343,759
4	Karn 3	36.05%	16.50%	26.92%	61.02%	-\$8,099,064
5	Karn 4	29.51%	17.50%	10.43%	73.90%	-\$11,127,465
6	Ludington 1	3.52%	4.00%	9.55%	86.83%	\$11,804,455
7	Ludington 2	9.30%	4.00%	8.92%	87.44%	\$75,442
8	Ludington 3	1.70%	4.00%	15.45%	81.17%	-\$2,847,807
9	Ludington 4	7.82%	4.00%	17.17%	79.52%	\$2,532,288
10	Ludington 5	7.94%	4.00%	12.28%	84.21%	\$2,338,692
11	Ludington 6	3.93%	4.00%	11.09%	85.35%	\$18,927,307
12	Hydros	8.92%	5.60%	16.14%	79.16%	\$49,735,442
13	Zeeland CC	4.77%	4.00%	16.14%	80.51%	\$165,533,232
14	Zeeland 1A	3.11%	4.00%	16.14%	80.51%	\$14,565,093
15	Zeeland 1B	3.95%	4.00%	16.14%	80.51%	\$17,263,982
16	Jackson	6.38%	4.50%	4.45%	91.25%	\$152,007,851
17	Covert 1		1.14%	5.28%	93.64%	
18	Covert 2		1.74%	5.28%	93.07%	
19	Covert 3		0.96%	5.28%	93.81%	
20	Cross Winds EP ⁽¹⁾					\$105,083,258
21	Lake Winds EP ⁽¹⁾					\$36,022,420
22	Gratiot Farms Wind ⁽³⁾					\$24,658,099
23	Crescent Wind ⁽³⁾					\$23,355,382
24	Solar Gardens ⁽²⁾					\$914,786

(1) Reflects NEV for 2019-2022 only.
(2) Reflects NEV for 2020-2022 only.
(3) Reflects NEV for 2022 only.

Schedule: B-5.1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures
For the years 2022 through 2025
(\$000's)

Case No.: U-21389
Exhibit No.: A-12 (RTB-3)
Schedule: B-5.1
Page: 1 of 10
Witness: RTBlumenstock
Date: May 2023

Generation Capital Expenditures
(\$000)

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)
		Historical Year 12 Months Ended 12/31/2022	Projected Bridge Period		14 Mos Ending 2/29/2024	Projected Test Year 12 Mos Ending 2/28/2025
		12 Mos Ended 12/31/2023	2 Mos Ending 2/29/2024			
1	Steam Power Generation					
2	Environmental	\$ 1,108	\$ 3,024	\$ 6	\$ 3,031	\$ 48
3	Routine and Small CapEx	\$ 8,901	\$ 15,016	\$ 1,260	\$ 16,276	\$ 7,173
4	Total Steam Production	\$ 10,009	\$ 18,041	\$ 1,266	\$ 19,307	\$ 7,221
5	Hydraulic Power Generation					
6	Routine and Small CapEx	\$ 26,545	\$ 28,372	\$ 8,289	\$ 36,662	\$ 49,897
7	Total hydraulic production	\$ 26,545	\$ 28,372	\$ 8,289	\$ 36,662	\$ 49,897
8	Pumped Storage Generation					
9	Ludington Overhaul	\$ 8,796	\$ -	\$ -	\$ -	\$ -
10	Routine and Small CapEx	\$ 2,962	\$ 11,558	\$ 2,065	\$ 13,623	\$ 14,137
11	Total Pumped Storage Generation	\$ 11,757	\$ 11,558	\$ 2,065	\$ 13,623	\$ 14,137
12	Other Production Plant					
13	Routine and Small CapEx	\$ 93,060	\$ 1,231,991	\$ 69,392	\$ 1,301,383	\$ 317,717
14	Total Other Production Plant	\$ 93,060	\$ 1,231,991	\$ 69,392	\$ 1,301,383	\$ 317,717
15	SubTotal	\$ 141,372	\$ 1,289,962	\$ 81,012	\$ 1,370,974	\$ 388,972
16	Less Contingency	\$ -	\$ 590	\$ 217	\$ 807	\$ 1,083
17	Grand Total	\$ 141,372	\$ 1,289,372	\$ 80,795	\$ 1,370,168	\$ 387,888

	(a)	(b)	(c)	(d)	(e)
		Projected			
		2 Mos Ending 2/28/2023	12 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025	26 Mos Ending 2/28/2025
1	Steam Power Generation				
2	Environmental	\$ -	\$ 3,031	\$ 48	\$ 3,079
3	Routine and Small CapEx	\$ 4,677	\$ 428,780	\$ 276,630	\$ 710,087
4	Total Steam Production	\$ 4,677	\$ 431,811	\$ 276,679	\$ 713,166
5	Hydraulic Power Generation				
6	Routine and Small CapEx	\$ 2,005	\$ 34,657	\$ 49,897	\$ 86,558
7	Total hydraulic production	\$ 2,005	\$ 34,657	\$ 49,897	\$ 86,558
8	Pumped Storage Generation				
9	Ludington Overhaul			\$ -	\$ -
10	Routine and Small CapEx	\$ 1,151	\$ 12,472	\$ 14,137	\$ 27,759
11	Total Pumped Storage Generation	\$ 1,151	\$ 12,472	\$ 14,137	\$ 27,759
12	Other Production Plant				
13	Routine and Small CapEx	\$ 6,499	\$ 877,703	\$ 48,260	\$ 932,462
14	Total Other Production Plant	\$ 6,499	\$ 877,703	\$ 48,260	\$ 932,462
15	SubTotal	\$ 14,332	\$ 1,356,642	\$ 388,972	\$ 1,759,946
16	Less Contingency	\$ 77	\$ 730	\$ 1,083	\$ 1,890
17	Grand Total	\$ 14,256	\$ 1,355,912	\$ 387,888	\$ 1,758,056

Schedule: B-5.1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Summary of Actual and Projected Electric Capital Expenditures
For the years 2022 through 2025
(\$000's)

Case No.: U-21389
Exhibit No.: A-12 (RTB-3)
Schedule: B-5.1
Page: 2 of 10
Witness: RTBlumenstock
Date: May 2023

Generation Capital Expenditures											
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Description	Projected Bridge Year		Projected Bridge Period				Projected Test Year			
		12 Months Ended 12/31/2022		12 Months Ending 12/31/2023		2 Months Ending 2/29/2024		14 Months Ending 2/29/2024		12 Months Ending 2/28/2025	
1	JHCampbell 1&2	\$	4,067	\$	1,379	\$	14	\$	1,393	\$	272
2	Contractor		\$ 1,825	\$	902	\$	11	\$	913	\$	211
3	Labor		\$ 642	\$	-	\$	-	\$	-	\$	-
4	Materials		\$ 546	\$	311	\$	-	\$	311	\$	-
5	Business Expenses		\$ 2	\$	165	\$	3	\$	168	\$	6
6	Contingency		\$ -			\$	-	\$	-	\$	-
7	Other (Loadings, Chargebacks)		\$ 1,052			\$	-	\$	-	\$	-
8	JHCampbell 3	\$	7,935	\$	2,418	\$	19	\$	2,437	\$	360
9	Contractor		\$ 3,568	\$	1,655	\$	15	\$	1,671	\$	279
10	Labor		\$ 1,301	\$	60	\$	-	\$	60	\$	-
11	Materials		\$ 1,538	\$	523	\$	-	\$	523	\$	-
12	Business Expenses		\$ 7	\$	180	\$	4	\$	184	\$	81
13	Contingency			\$	-	\$	-	\$	-	\$	-
14	Other (Loadings, Chargebacks)		\$ 1,521	\$	-	\$	-	\$	-	\$	-
15	DEKarn 1&2	\$	(3,312)	\$	-	\$	-	\$	-	\$	-
16	Contractor		\$ (1,680)	\$	-	\$	-	\$	-	\$	-
17	Labor		\$ (460)	\$	-	\$	-	\$	-	\$	-
18	Materials		\$ (291)	\$	-	\$	-	\$	-	\$	-
19	Business Expenses		\$ (2)	\$	-	\$	-	\$	-	\$	-
20	Contingency			\$	-	\$	-	\$	-	\$	-
21	Other (Loadings, Chargebacks)		\$ (880)	\$	-	\$	-	\$	-	\$	-
22	DEKarn 3&4	\$	331	\$	11,220	\$	1,227	\$	12,446	\$	6,542
23	Contractor		\$ 292	\$	8,245	\$	1,105	\$	9,350	\$	5,882
24	Labor		\$ (115)	\$	157	\$	8	\$	165	\$	41
25	Materials		\$ 69	\$	826	\$	-	\$	826	\$	-
26	Business Expenses		\$ 0	\$	1,992	\$	88	\$	2,080	\$	494
27	Contingency			\$	-	\$	25	\$	25	\$	125
28	Other (Loadings, Chargebacks)		\$ 85	\$	-	\$	-	\$	-	\$	-
29	Zeeland	\$	21,590	\$	35,256	\$	2,622	\$	37,879	\$	15,286
30	Contractor		\$ 15,391	\$	31,551	\$	2,298	\$	33,848	\$	13,561
31	Labor		\$ 864	\$	344	\$	10	\$	353	\$	48
32	Materials		\$ 1,343	\$	175	\$	52	\$	227	\$	258
33	Business Expenses		\$ 15	\$	3,152	\$	214	\$	3,365	\$	1,169
34	Contingency			\$	35	\$	50	\$	85	\$	250
35	Other (Loadings, Chargebacks)		\$ 3,977			\$	-	\$	-	\$	-
36	Jackson Generating Station	\$	32,540	\$	14,062	\$	1,915	\$	15,977	\$	11,720
37	Contractor		\$ 22,830	\$	12,477	\$	1,750	\$	14,226	\$	10,721
38	Labor		\$ 1,122	\$	46	\$	5	\$	51	\$	25
39	Materials		\$ 1,686	\$	75	\$	49	\$	124	\$	263
40	Business Expenses		\$ 42	\$	1,404	\$	111	\$	1,515	\$	711
41	Contingency			\$	60	\$	-	\$	60	\$	-
42	Other (Loadings, Chargebacks)		\$ 6,860	\$	-	\$	-	\$	-	\$	-
43	Covert	\$	-	\$	825,714	\$	3,381	\$	829,095	\$	20,274
44	Contractor		\$ -	\$	9,765	\$	3,381	\$	13,146	\$	20,274
45	Labor		\$ -	\$	-	\$	-	\$	-	\$	-
46	Materials		\$ -	\$	-	\$	-	\$	-	\$	-
47	Business Expenses		\$ -	\$	815,949	\$	-	\$	815,949	\$	-
48	Contingency		\$ -	\$	-	\$	-	\$	-	\$	-
49	Other (Loadings, Chargebacks)		\$ -	\$	-	\$	-	\$	-	\$	-
50	Solar	\$	30,487	\$	355,715	\$	61,465	\$	417,180	\$	269,457
51	Contractor		\$ 24,534	\$	40,899	\$	15,801	\$	56,701	\$	75,238
52	Labor		\$ 1,225	\$	-	\$	-	\$	-	\$	-
53	Materials		\$ 3,172	\$	-	\$	-	\$	-	\$	-
54	Business Expenses		\$ 63	\$	41,403	\$	7,969	\$	49,372	\$	18,849
55	Contingency			\$	-	\$	-	\$	-	\$	-
56	Other (Loadings, Chargebacks)		\$ 1,493	\$	273,413	\$	37,694	\$	311,107	\$	175,371

MICHIGAN PUBLIC SERVICE COMMISSION

Schedule: B-5.1

Case No.: U-21389

Exhibit No.: A-12 (RTB-3)

Schedule: B-5.1

Page: 3 of 10

Witness: RTBlumenstock

Date: May 2023

Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures

For the years 2022 through 2025

(\$000's)

Generation Capital Expenditures

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
		Projected Bridge Year 12 Months Ended 12/31/2022		12 Months Ending 12/31/2023		2 Months Ending 2/29/2024		14 Months Ending 2/29/2024		Projected Test Year 12 Mos Ending 2/28/2025	
57	Classic 7	(120)		-		-					
58	Contractor		(68)		-		-		-		-
59	Labor		(17)		-		-		-		-
60	Materials				-		-		-		-
61	Business Expenses				-		-		-		-
62	Contingency				-		-		-		-
63	Other (Loadings, Chargebacks)		(34)		-		-		-		-
64	Hydros	26,545		28,372		8,289		36,662		49,897	
65	Contractor		16,457		22,594		6,569		29,163		39,612
66	Labor		3,043		37		15		52		92
67	Materials		1,972		-		-		-		-
68	Business Expenses		168		5,360		1,563		6,923		9,484
69	Contingency				382		142		524		708
70	Other (Loadings, Chargebacks)		4,905		-		-		-		-
71	Ludington	11,757		11,558		2,065		13,623		14,137	
72	Contractor		14,073		13,998		2,872		16,870		18,546
73	Labor		1,865		254		137		391		1,008
74	Materials		2,449		-		20		20		(5,628)
75	Business Expenses		48		(2,836)		(970)		(3,806)		176
76	Contingency				113		-		113		-
77	Other (Loadings, Chargebacks)		(6,677)		28		6		34		35
78	Admin and Other	8,443		1,243		9		1,252		979	
79	Contractor		2,345		740		-		740		645
80	Labor		599				-		-		-
81	Materials		953				-		-		-
82	Business Expenses		126		75		-		75		-
83	Contingency						-		-		-
84	Other (Loadings, Chargebacks)		4,420		428		9		437		334
85	Air Quality	1,695		-		-		-		-	
86	Contractor		315				-		-		-
87	Labor		129				-		-		-
88	Materials		867				-		-		-
89	Business Expenses		0				-		-		-
90	Contingency						-		-		-
91	Other (Loadings, Chargebacks)		383				-		-		-
92	RCRA	2,878		-		-		-		-	
93	Contractor		1,595				-		-		-
94	Labor		362				-		-		-
95	Materials		10				-		-		-
96	Business Expenses		5				-		-		-
97	Contingency						-		-		-
98	Other (Loadings, Chargebacks)		907				-		-		-
99	316b	(1,348)		-		-		-		-	
100	Contractor		(730)				-		-		-
101	Labor		(202)				-		-		-
102	Materials		(2)				-		-		-
103	Business Expenses		(9)				-		-		-
104	Contingency						-		-		-
105	Other (Loadings, Chargebacks)		(405)				-		-		-
106	SEEG	(2,865)		-		-		-		-	
107	Contractor		(1,595)				-		-		-
108	Labor		(352)				-		-		-
109	Materials		(10)				-		-		-
110	Business Expenses		(5)				-		-		-
111	Contingency						-		-		-
112	Other (Loadings, Chargebacks)		(903)				-		-		-
113	All Other Environmental	747		3,024		6		3,031		48	
114	Contractor		316		2,487		5		2,492		42
115	Labor		232		192		-		192		-
116	Materials		9		192		-		192		-
117	Business Expenses		2		155		1		156		6
118	Contingency		-				-		-		-
119	Other (Loadings, Chargebacks)		189				-		-		-
120	Total Capital	141,372	141,372	1,289,962	1,289,962	81,012	81,012	1,370,974	1,370,974	388,972	388,972

Schedule: B-5.1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Summary of Actual and Projected Electric Capital Expenditures
For the years 2022 through 2025
(\$000's)

Case No.: U-21389
Exhibit No.: A-12 (RTB-3)
Schedule: B-5.1
Page: 4 of 10
Witness: RTBlumenstock
Date: May 2023

Generation Capital Expenditures

Line No.	(a) Description	(b) (c) (d) (e) Projected			
		2 Mos Ending 2/28/2023	12 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025	26 Mos Ending 2/28/2025
1	JHCampbell 1&2	\$ 356	\$ 1,037	\$ 272	\$ 1,665
2	Contractor	\$ 516	\$ 397	\$ 211	\$ 1,124
3	Labor	\$ 3	\$ (3)	\$ -	\$ -
4	Materials	\$ (4)	\$ 315	\$ -	\$ 311
5	Business Expenses	\$ (160)	\$ 328	\$ 61	\$ 229
6	Contingency	\$ -	\$ -	\$ -	\$ -
7	Other (Loadings, Chargebacks)	\$ 1	\$ (1)	\$ -	\$ -
8	JHCampbell 3	\$ 1,169	\$ 1,268	\$ 360	\$ 2,797
9	Contractor	\$ 1,694	\$ (23)	\$ 279	\$ 1,950
10	Labor	\$ 8	\$ 52	\$ -	\$ 60
11	Materials	\$ (12)	\$ 535	\$ -	\$ 523
12	Business Expenses	\$ (523)	\$ 707	\$ 81	\$ 264
13	Contingency	\$ -	\$ -	\$ -	\$ -
14	Other (Loadings, Chargebacks)	\$ 2	\$ (2)	\$ -	\$ -
15	DEKarn 1&2	\$ -	\$ -	\$ -	\$ -
16	Contractor	\$ -	\$ -	\$ -	\$ -
17	Labor	\$ -	\$ -	\$ -	\$ -
18	Materials	\$ -	\$ -	\$ -	\$ -
19	Business Expenses	\$ -	\$ -	\$ -	\$ -
20	Contingency	\$ -	\$ -	\$ -	\$ -
21	Other (Loadings, Chargebacks)	\$ -	\$ -	\$ -	\$ -
22	DEKarn 3&4	\$ 1,420	\$ 11,027	\$ 6,542	\$ 18,988
23	Contractor	\$ 1,131	\$ 8,219	\$ 5,882	\$ 15,232
24	Labor	\$ 23	\$ 142	\$ 41	\$ 206
25	Materials	\$ -	\$ 826	\$ -	\$ 826
26	Business Expenses	\$ -	\$ 2,080	\$ 494	\$ 2,574
27	Contingency	\$ 3	\$ 22	\$ 125	\$ 150
28	Other (Loadings, Chargebacks)	\$ 262	\$ (262)	\$ -	\$ -
29	Zeeland	\$ 4,986	\$ 32,892	\$ 15,286	\$ 53,164
30	Contractor	\$ 4,320	\$ 29,528	\$ 13,561	\$ 47,409
31	Labor	\$ 47	\$ 307	\$ 48	\$ 401
32	Materials	\$ 32	\$ 195	\$ 258	\$ 485
33	Business Expenses	\$ 577	\$ 2,789	\$ 1,169	\$ 4,535
34	Contingency	\$ 11	\$ 74	\$ 250	\$ 335
35	Other (Loadings, Chargebacks)	\$ -	\$ -	\$ -	\$ -
36	Jackson Generating Station	\$ 1,475	\$ 14,502	\$ 11,720	\$ 27,697
37	Contractor	\$ 1,309	\$ 12,918	\$ 10,721	\$ 24,948
38	Labor	\$ 5	\$ 46	\$ 25	\$ 76
39	Materials	\$ 14	\$ 111	\$ 263	\$ 388
40	Business Expenses	\$ 142	\$ 1,373	\$ 711	\$ 2,226
41	Contingency	\$ 5	\$ 55	\$ -	\$ 60
42	Other (Loadings, Chargebacks)	\$ -	\$ -	\$ -	\$ -
43	Covert	\$ -	\$ 829,095	\$ 20,274	\$ 849,369
44	Contractor	\$ -	\$ 13,146	\$ 20,274	\$ 33,420
45	Labor	\$ -	\$ -	\$ -	\$ -
46	Materials	\$ -	\$ -	\$ -	\$ -
47	Business Expenses	\$ -	\$ 815,949	\$ -	\$ 815,949
48	Contingency	\$ -	\$ -	\$ -	\$ -
49	Other (Loadings, Chargebacks)	\$ -	\$ -	\$ -	\$ -
50	Solar	\$ 1,732	\$ 415,448	\$ 269,457	\$ 686,638
51	Contractor	\$ 281	\$ 56,420	\$ 75,238	\$ 131,939
52	Labor	\$ -	\$ -	\$ -	\$ -
53	Materials	\$ -	\$ -	\$ -	\$ -
54	Business Expenses	\$ 140	\$ 49,232	\$ 18,849	\$ 68,221
55	Contingency	\$ -	\$ -	\$ -	\$ -
56	Other (Loadings, Chargebacks)	\$ 1,312	\$ 309,796	\$ 175,371	\$ 486,478

MICHIGAN PUBLIC SERVICE COMMISSION

Schedule: B-5.1

Case No.: U-21389

Exhibit No.: A-12 (RTB-3)

Schedule: B-5.1

Page: 5 of 10

Witness: RTBlumenstock

Date: May 2023

Consumers Energy Company
Summary of Actual and Projected Electric Capital Expenditures
For the years 2022 through 2025
(\$000's)

Generation Capital Expenditures

	(a)	(b)	(c)	(d)	(e)
		Projected			
Line No.	Description	2 Mos Ending 2/28/2023	12 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025	26 Mos Ending 2/28/2025
57	Classic 7	\$ -	\$ -	\$ -	\$ -
58	Contractor	\$ -	\$ -	\$ -	\$ -
59	Labor	\$ -	\$ -	\$ -	\$ -
60	Materials	\$ -	\$ -	\$ -	\$ -
61	Business Expenses	\$ -	\$ -	\$ -	\$ -
62	Contingency	\$ -	\$ -	\$ -	\$ -
63	Other (Loadings, Chargebacks)	\$ -	\$ -	\$ -	\$ -
64	Hydros	\$ 2,005	\$ 34,657	\$ 49,897	\$ 86,558
65	Contractor	\$ 1,718	\$ 27,445	\$ 39,612	\$ 68,776
66	Labor	\$ 6	\$ 46	\$ 92	\$ 144
67	Materials	\$ -	\$ -	\$ -	\$ -
68	Business Expenses	\$ 224	\$ 6,699	\$ 9,484	\$ 16,407
69	Contingency	\$ 57	\$ 466	\$ 708	\$ 1,232
70	Other (Loadings, Chargebacks)	\$ -	\$ -	\$ -	\$ -
71	Ludington	\$ 1,151	\$ 12,472	\$ 14,137	\$ 27,759
72	Contractor	\$ 1,668	\$ 15,202	\$ 18,546	\$ 35,416
73	Labor	\$ 8	\$ 383	\$ 1,008	\$ 1,399
74	Materials	\$ (12)	\$ 33	\$ (5,628)	\$ (5,608)
75	Business Expenses	\$ (515)	\$ (3,291)	\$ 176	\$ (3,630)
76	Contingency	\$ -	\$ 113	\$ -	\$ 113
77	Other (Loadings, Chargebacks)	\$ 2	\$ 32	\$ 35	\$ 69
78	Admin and Other	\$ 38	\$ 1,214	\$ 979	\$ 2,231
79	Contractor	\$ 36	\$ 704	\$ 645	\$ 1,385
80	Labor	\$ -	\$ -	\$ -	\$ -
81	Materials	\$ -	\$ -	\$ -	\$ -
82	Business Expenses	\$ 2	\$ 73	\$ -	\$ 75
83	Contingency	\$ -	\$ -	\$ -	\$ -
84	Other (Loadings, Chargebacks)	\$ -	\$ 437	\$ 334	\$ 771
85	Air Quality	\$ -	\$ -	\$ -	\$ -
86	Contractor	\$ -	\$ -	\$ -	\$ -
87	Labor	\$ -	\$ -	\$ -	\$ -
88	Materials	\$ -	\$ -	\$ -	\$ -
89	Business Expenses	\$ -	\$ -	\$ -	\$ -
90	Contingency	\$ -	\$ -	\$ -	\$ -
91	Other (Loadings, Chargebacks)	\$ -	\$ -	\$ -	\$ -
92	RCRA	\$ -	\$ -	\$ -	\$ -
93	Contractor	\$ -	\$ -	\$ -	\$ -
94	Labor	\$ -	\$ -	\$ -	\$ -
95	Materials	\$ -	\$ -	\$ -	\$ -
96	Business Expenses	\$ -	\$ -	\$ -	\$ -
97	Contingency	\$ -	\$ -	\$ -	\$ -
98	Other (Loadings, Chargebacks)	\$ -	\$ -	\$ -	\$ -
99	316b	\$ -	\$ -	\$ -	\$ -
100	Contractor	\$ -	\$ -	\$ -	\$ -
101	Labor	\$ -	\$ -	\$ -	\$ -
102	Materials	\$ -	\$ -	\$ -	\$ -
103	Business Expenses	\$ -	\$ -	\$ -	\$ -
104	Contingency	\$ -	\$ -	\$ -	\$ -
105	Other (Loadings, Chargebacks)	\$ -	\$ -	\$ -	\$ -
106	SEEG	\$ -	\$ -	\$ -	\$ -
107	Contractor	\$ -	\$ -	\$ -	\$ -
108	Labor	\$ -	\$ -	\$ -	\$ -
109	Materials	\$ -	\$ -	\$ -	\$ -
110	Business Expenses	\$ -	\$ -	\$ -	\$ -
111	Contingency	\$ -	\$ -	\$ -	\$ -
112	Other (Loadings, Chargebacks)	\$ -	\$ -	\$ -	\$ -
113	All Other Environmental	\$ -	\$ 3,031	\$ 48	\$ 3,079
114	Contractor	\$ -	\$ 2,492	\$ 42	\$ 2,534
115	Labor	\$ -	\$ 192	\$ -	\$ 192
116	Materials	\$ -	\$ 192	\$ -	\$ 192
117	Business Expenses	\$ -	\$ 156	\$ 6	\$ 162
118	Contingency	\$ -	\$ -	\$ -	\$ -
119	Other (Loadings, Chargebacks)	\$ -	\$ -	\$ -	\$ -
120	SubTotal	\$ 14,332	\$ 1,356,642	\$ 388,972	\$ 1,759,946
121	Less Contingency	\$ 77	\$ 730	\$ 1,083	\$ 1,890
122	Grand Total	\$ 14,256	\$ 1,355,912	\$ 387,888	\$ 1,758,056

Schedule: B-5.1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures
For the years 2022 through 2025

(\$000's)

Case No.: U-21389

Exhibit No.: A-12 (RTB-3)

Schedule: B-5.1

Page: 6 of 10

Witness: RTBlumenstock

Date: May 2023

Generation Capital Expenditures

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)
		Historical Year	Projected Bridge Period			Projected Test Year
		12 Months Ended 12/31/2022	12 Months Ending 12/31/2023	2 Months Ending 2/29/2024	14 Months Ending 2/29/2024	12 Months Ending 2/28/2025
1	Contractor	\$ 99,467	\$ 145,313	\$ 33,808	\$ 179,120	\$ 185,012
2	Labor	\$ 10,236	\$ 1,090	\$ 174	\$ 1,264	\$ 1,213
3	Materials	\$ 14,311	\$ 2,102	\$ 121	\$ 2,223	\$ (5,107)
4	Business Expenses	\$ 464	\$ 866,998	\$ 8,983	\$ 875,981	\$ 31,030
5	Contingency	\$ -	\$ 590	\$ 217	\$ 807	\$ 1,083
6	Other (Loadings, Chargebacks)	\$ 16,894	\$ 273,869	\$ 37,709	\$ 311,579	\$ 175,740
	Total	\$ 141,372	\$ 1,289,962	\$ 81,012	\$ 1,370,974	\$ 388,972

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)
		12 Months Ending 12/31/2023	2 Months Ending 2/29/2024	Projected 14 Months Ending 2/29/2024	12 Months Ending 2/28/2025	26 Months Ending 2/28/2005
1	Contractor	\$ 145,313	\$ 33,808	\$ 179,120	\$ 185,012	\$ 364,133
2	Labor	\$ 1,090	\$ 174	\$ 1,264	\$ 1,213	\$ 2,477
3	Materials	\$ 2,102	\$ 121	\$ 2,223	\$ (5,107)	\$ (2,883)
4	Business Expenses	\$ 866,998	\$ 8,983	\$ 875,981	\$ 31,030	\$ 907,011
5	Contingency	\$ 590	\$ 217	\$ 807	\$ 1,083	\$ 1,890
6	Other (Loadings, Chargebacks)	\$ 273,869	\$ 37,709	\$ 311,579	\$ 175,740	\$ 487,318
	Total	\$ 1,289,962	\$ 81,012	\$ 1,370,974	\$ 388,972	\$ 1,759,946

Schedule: B-5.1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
 Summary of Projected Electric Capital Expenditures
 For the projected test year ending February 28, 2025
 (\$000's)

Case No.: U-21389
 Exhibit No.: A-12 (RTB-3)
 Schedule: B-5.1
 Page: 7 of 10
 Witness: RTBlumenstock
 Date: May 2023

Generation Capital Expenditures

(a)		(b)	(c)	(d)	(e)	(f)
Line		Projected Bridge Period		Projected Test Year		
No.	Description	14 Months Ending		12 Months Ending		Reference
		2/29/2024		2/28/2025		
1	Campbell 1&2 Non-Environmental	\$	1,393	\$	272	A-12 (RTB-3) Page 2 line 1 Columns (d) and (f)
2	Campbell 1&2 "All Other Environmental"		\$ 330		\$ 14	
3	Campbell 3 Non-Environmental	\$	2,437	\$	360	A-12 (RTB-3) Page 2 line 8 Columns (d) and (f)
4	Campbell 3 "All Other Environmental"		\$ 2,700		\$ 18	
5	Karn 1&2 Non-Environmental	\$	-	\$	-	A-12 (RTB-3) Page 2 line 15 Columns (d) and (f)
6	Karn 1&2 "All Other Environmental"		\$ -		\$ -	
7	Total Other Environmental		\$ 3,031		\$ 31	

MICHIGAN PUBLIC SERVICE COMMISSION

Schedule: B-5.1

Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures

For the years 2022 through 2025

Generation Capital Projects greater than \$1M

(\$000's)

Case No.: U-21389

Exhibit No.: A-12 (RTB-3)

Schedule: B-5.1

Page: 8 of 10

Witness: RTBlumenstock

Date: May 2023

Generation Capital Expenditures

Line No.	(a) Calendar Year	(b) Tier 1 Portfolio	(c) Tier 2 Portfolio	(d) Project Type	(e) Project Classification	(f) Work Item Description	(g) Projected ⁽¹⁾ Contingency	(h) Projected ⁽¹⁾ Amount	(i) Actual Amount
1	2022	Coal Generation	Campbell 1	Non-Routine	Environmental	Air Preheater Baskets and Seals	\$ -	\$ 1,470	\$ 1,819
2	2022	Coal Generation	Campbell 1	Routine	Environmental	Pulse Jet Fabric Filter Bag	\$ 200	\$ 1,019	\$ 1,040
3	2022	Coal Generation	Campbell 3	Non-Routine	Condition-based	House Service Air Compressor Replacement	\$ 267	\$ 1,053	\$ 1,207
4	2022	Coal Generation	Campbell 3	Routine	Environmental	Selective Catalytic Reduction Catalyst Management	\$ 500	\$ 1,960	\$ 1,196
5	2022	Coal Generation	Campbell 3	Non-Routine	Condition-based	Campbell Unit 3 Diesel Generator Controls	\$ 19	\$ 187	\$ 1,172
6	2023	Coal Generation	Campbell Site Commons	Non-Routine	Condition-based	North and South Pigeon Lake Jetties	\$ -	\$ 740	\$ 1,003
7	2022	Gas/Oil Generation	Karn 3&4 Commons	Non-Routine	Asset Separation	Unit Separation	\$ 500	\$ 7,207	\$ 1,605
8	2022	Gas Generation	Jackson Site Commons	Routine	Condition-based	GE Long Term Service Agreement FFH	\$ -	\$ 8,200	\$ 7,734
9	2022	Gas Generation	Jackson Site Commons	Non-Routine	Condition-based	Reverse Osmosis Pretreatment System	\$ 100	\$ 8,700	\$ 16,237
10	2022	Gas Generation	Jackson Site Commons	Routine	Condition-based	GE Long Term Service Agreement Historical Extra Work	\$ -	\$ 950	\$ 3,920
11	2022	Gas Generation	Zeeland Site Commons	Routine	Condition-based	Long Term Service Agreement - Running Capital Contract	\$ -	\$ 8,160	\$ 7,599
12	2022	Gas Generation	Zeeland Site Commons	Routine	Condition-based	Zeeland Unit 1A Generator Step Up Transformer Rewind	\$ -	\$ 5,500	\$ 8,878
13	2022	Hydro Generation	Croton	Routine	Condition-based	Croton 1 Wicket Gate	\$ 100	\$ 2,150	\$ 4,072
14	2022	Hydro Generation	Croton	Routine	Condition-based	Croton 2 Wicket Gate	\$ 100	\$ 2,150	
15	2022	Hydro Generation	Hardy	Routine	Safety	Replace powerhouse roof	\$ 100	\$ 1,575	\$ 1,711
16	2022	Hydro Generation	Hardy	Routine	Infrastructure	New Headquarters Building (previously was Croton HQ)	\$ 300	\$ 1,985	\$ 2,740
17	2022	Hydro Generation	Hardy	Non-Routine	Regulatory	Auxiliary Spillway Remediation	\$ 192	\$ 4,700	\$ 3,096
18	2022	Hydro Generation	Hodenpyl	Non-Routine	Condition-based	Spillway Hoist Replacement	\$ 100	\$ 1,900	\$ 1,712
19	2022	Hydro Generation	Hodenpyl	Non-Routine	Condition-based	Transformer Foundation	\$ 250	\$ 2,487	\$ 1,746
20	2022	Hydro Generation	Hodenpyl	Non-Routine	Safety	Electrical Safety Project	\$ -	\$ 2,220	\$ 3,170
21	2022	Hydro Generation	Mio	Non-Routine	Regulatory	Left Retaining Wall Replacement	\$ -	\$ 3,650	\$ 4,300
22	2022	Hydro Generation	Mio	Non-Routine	Safety	Electrical Safety Project	\$ 190	\$ 2,450	\$ 1,677
23	2022	Hydro Generation	Mio	Non-Routine	Regulatory	Mio Downstream reverse filter	\$ -	\$ 570	\$ 1,058
24	2022	Hydro Generation	Ludington Site Commons	Non-Routine	Safety	Powerhouse Roof Wearing Surface and Weather Proofing Replacement	\$ 220	\$ 3,090	\$ 2,325
25	2022	Hydro Generation	Ludington Site Commons	Non-Routine	Safety	Replace Lower Penstock Expansion Joint Chamber Waterstop	\$ 200	\$ 3,878	\$ 4,223
26	2022	Hydro Generation	Ludington Site Commons	Non-routine	Economic	Upgrade and Overhaul	\$ -	\$ 5,370	\$ 8,796
27	2022	Hydro Generation	Ludington Site Commons	Non-Routine	Condition-based	480V Motor Control Center for DLC	\$ -	\$ 845	\$ 1,538
28	2022	Renewables	Solar Commons	Non-Routine	New Generation	Solar - 2021 Bid Event (250 MW)	\$ -	\$ 20,333	\$ 22,789
29	Total 2022 Projects						\$ 3,337	\$ 104,498	\$ 118,365

Note:

(1) Projected amounts were taken from Case No. U-21224

MICHIGAN PUBLIC SERVICE COMMISSION

Schedule: B-5.1

Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures
For the years 2022 through 2025

Generation Capital Projects greater than \$1M
(\$000's)

Case No.: U-21389

Exhibit No.: A-12 (RTB-3)

Schedule: B-5.1

Page: 9 of 10

Witness: RTBlumenstock

Date: May 2023

Generation Capital Expenditures

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	
	Calendar	Tier 1	Tier 2	Project	Project	Class of Cost	Engineering	Project Bid	Full Internal		Planned	Project	Contingency	Projected	
Line No.	Year	Portfolio	Portfolio	Type	Classification	Estimate	Internal or External	Issued	Budget Approval	Work Item Description	Amount	Reduction	Amount	Amount	
1	Bridge Period	Coal Generation	Campbell 3	Routine	Environmental	Class 3	Internal	In Progress	Approved	SCR Catalyst Management	\$ 2,403	\$ 100	\$ -	\$ 2,303	
2	Bridge Period	Gas Generation	Covert Commons	Routine	Condition-based	Class 1	NA	Yes	Approved	Long Term Service Agreement - Running capital contract	\$ 9,761	\$ -	\$ -	\$ 9,761	
3	Bridge Period	Gas Generation	Covert Commons	Routine	Condition-based	Class 3	NA	No	Approved	Non LTSA Capital - Extras not included in contract	\$ 3,943	\$ -	\$ -	\$ 3,943	
4	Bridge Period	Gas Generation	Covert Commons	Non-Routine	Acquisition	Class 1	NA	Yes	Approved	Covert Generating Facility Acquisition	\$ 815,000	\$ -	\$ -	\$ 815,000	
5	Bridge Period	Gas/Oil Generation	Karn 3	Non-Routine	Condition-based	Class 3	Internal	No	Approved	Cooling Tower Rebuild	\$ 4,250	\$ 279	\$ -	\$ 3,971	
6	Bridge Period	Gas/Oil Generation	Karn 3&4	Non-Routine	Condition-based	Class 3	External	No	Approved	Tank Farm Storage Tank Heating Line Replacement	\$ 1,350	\$ 96	\$ -	\$ 1,254	
7	Bridge Period	Gas/Oil Generation	Karn 3&4	Non-Routine	Condition-based	Class 3	External	No	Approved	Sync Wire Replacement	\$ 1,450	\$ -	\$ -	\$ 1,450	
8	Bridge Period	Gas/Oil Generation	Karn 3&4 Commons	Non-Routine	Asset Separation	Class 3	External	No	Approved	Unit Separation	\$ 1,790	\$ -	\$ -	\$ 1,790	
9	Bridge Period	Gas Generation	Jackson Site Commons	Routine	Condition-based	Class 1	NA	Yes	Approved	GE Long Term Service Agreement FFH	\$ 9,571	\$ -	\$ -	\$ 9,571	
10	Bridge Period	Gas Generation	Jackson Site Commons	Non-Routine	Condition-based	Class 3	Internal	No	Approved	7EA Casing replacement & Hot section overhaul	\$ 2,288	\$ 200	\$ -	\$ 2,088	
11	Bridge Period	Gas Generation	Jackson Site Commons	Non-Routine	Condition-based	Class 3	External	No	Approved	HRSG Burner Element isolation Valves Addition	\$ 2,021	\$ -	\$ -	\$ 2,021	
12	Bridge Period	Gas Generation	Zeeland Site Commons	Routine	Condition-based	Class 1	NA	Yes	Approved	Long Term Service Agreement - Running Capital Contract	\$ 9,520	\$ -	\$ -	\$ 9,520	
13	Bridge Period	Gas Generation	Zeeland Site Commons	Routine	Condition-based	Class 1	NA	Yes	Approved	Milestone Outage Capital to GE - Part of LTSA Contract	\$ 7,870	\$ -	\$ -	\$ 7,870	
14	Bridge Period	Gas Generation	Zeeland Site Commons	Non-Routine	Condition-based	Class 1	Internal	Yes	Approved	Phase 2 599 699 345KV Breaker Replacement	\$ 1,302	\$ 79	\$ -	\$ 1,223	
15	Bridge Period	Gas Generation	Zeeland Site Commons	Non-Routine	Condition-based	Class 3	External	No	Approved	HRSG Casing Replacement	\$ 2,900	\$ 97	\$ -	\$ 2,803	
16	Bridge Period	Gas Generation	Zeeland Site Commons	Routine	Condition-based	Class 2	Internal	No	Approved	LTSA - Extras not included in contract (cranes, mobile equipment)	\$ 3,200	\$ 300	\$ -	\$ 2,900	
17	Bridge Period	Gas Generation	Zeeland Site Commons	Non-Routine	Condition-based	Class 2	External	Yes	Approved	Zeeland Unit 4 Field Rewind of Generator Rotor	\$ 1,250	\$ 45	\$ -	\$ 1,205	
18	Bridge Period	Gas Generation	Zeeland Site Commons	Non-Routine	Condition-based	Class 2	External	Yes	Approved	Zeeland Unit 3 Field Rewind	\$ 1,250	\$ 45	\$ -	\$ 1,205	
19	Bridge Period	Gas Generation	Zeeland Site Commons	Non-Routine	Condition-based	Class 3	Internal	Yes	Approved	Generator Step Up Transformer Replacement (Spare)	\$ 2,883	\$ -	\$ -	\$ 2,883	
20	Bridge Period	Gas Generation	Zeeland Site Commons	Non-Routine	Condition-based	Class 1	Internal	Yes	Approved	Zeeland Unit 1 Generator Step Up Transformer Rewind	\$ 4,700	\$ 96	\$ -	\$ 4,604	
21	Bridge Period	Hydro Generation	Alcona	Non-Routine	Infrastructure	Class 3	External	No	Approved	Core Wall Remediation Project	\$ 2,600	\$ 1,700	\$ -	\$ 1,910	
22	Bridge Period	Hydro Generation	Cooke	Non-Routine	Safety	Class 3	Internal	No	Approved	Spillway Hoist Replacement	\$ 2,100	\$ -	\$ -	\$ 2,100	
23	Bridge Period	Hydro Generation	Five Channels	Non-Routine	Regulatory	Class 3	External	No	Approved	Dead Bay and Log Chute Remediation	\$ 2,972	\$ 990	\$ -	\$ 1,982	
24	Bridge Period	Hydro Generation	Five Channels	Non-Routine	Regulatory	Class 3	External	No	Approved	Headgate Project	\$ 3,062	\$ 1,080	\$ -	\$ 1,982	
25	Bridge Period	Hydro Generation	Hardy	Non-Routine	Regulatory	Class 3	External	No	Approved	Auxiliary Spillway Remediation	\$ 3,459	\$ -	\$ -	\$ 3,459	
26	Bridge Period	Hydro Generation	Hodenpyl	Non-Routine	Condition-based	Class 3	Internal	No	Approved	Hodenpyl 1 Generator Rewind	\$ 3,310	\$ 210	\$ -	\$ 3,101	
27	Bridge Period	Hydro Generation	Hodenpyl	Non-Routine	Condition-based	Class 3	External	No	Approved	Spillway Hoist Replacement	\$ 2,958	\$ 94	\$ -	\$ 2,864	
28	Bridge Period	Hydro Generation	Webber	Non-Routine	Condition-based	Class 1	External	Yes	Approved	Unit 1 Generator Rewind	\$ 1,020	\$ -	\$ -	\$ 1,020	
29	Bridge Period	Hydro Generation	Rogers	Non-Routine	Regulatory	Class 4	External	No	Approved	Probable Maximum Flood Project	\$ 2,386	\$ 487	\$ -	\$ 1,899	
30	Bridge Period	Hydro Generation	Rogers	Non-Routine	Condition-based	Class 3	Internal	Yes	Approved	Unit 4 Generator Rewind	\$ 2,600	\$ -	\$ -	\$ 2,600	
31	Bridge Period	Hydro Generation	Mo	Non-Routine	Safety	Class 2	External	Yes	Approved	Electrical Safety Project	\$ 1,225	\$ 138	\$ -	\$ 1,086	
32	Bridge Period	Hydro Generation	Webber	Non-Routine	Infrastructure	Class 2	External	Yes	Approved	Webber Left Downstream Spillway Abutment Wall	\$ 3,200	\$ 225	\$ -	\$ 2,975	
33	Bridge Period	Hydro Generation	Ludington Site Commons	Non-Routine	Condition-based	Class 4	External	No	Approved	Oil Water Separator Replacement	\$ 1,263	\$ 100	\$ -	\$ 1,163	
34	Bridge Period	Hydro Generation	Ludington Site Commons	Non-Routine	Condition-based	Class 3	External	No	Approved	Unit 1-6 DCS Control Relay Replacement	\$ 1,348	\$ -	\$ -	\$ 1,348	
35	Bridge Period	Hydro Generation	Ludington Site Commons	Non-Routine	Safety	Class 3	External	No	Approved	Replace Lower Penstock Expansion Joint Chamber Waterstop	\$ 2,433	\$ 7	\$ -	\$ 2,425	
36	Bridge Period	Hydro Generation	Ludington Site Commons	Non-Routine	Safety	Class 1	External	Yes	Approved	Powerhouse Roof Wearing Surface and Weather Proofing Replacement	\$ 2,864	\$ 140	\$ -	\$ 2,724	
37	Bridge Period	Hydro Generation	Ludington Site Commons	Non-Routine	Regulatory	Class 1	Internal	Yes	Approved	Replace Barrier Net Panels	\$ 1,089	\$ -	\$ -	\$ 1,089	
38	Bridge Period	Renewables	Solar Commons	Non-Routine	New Generation	Class 1/BTA Executed 2021	CE Tech Specs/External Design	Yes	Approved	Solar - 2019 Bid Event (150 MW)	\$ 233,064	\$ -	\$ -	\$ 233,064	
39	Bridge Period	Renewables	Solar Commons	Non-Routine	New Generation	Class 1/BTA Executed 2021	CE Tech Specs/External Design	Yes	Approved	Solar - 2020 Bid Event (150 MW)	\$ 122,186	\$ -	\$ -	\$ 122,186	
40	Bridge Period	Renewables	Solar Commons	Non-Routine	New Generation	Class 1/Awarded Self-Build 2022	CE Tech Specs/External Design	Yes	Approved	Solar - 2021 Bid Event (Muskegon Solar) (250 MW)	\$ 61,930	\$ -	\$ -	\$ 61,930	
41	Total Bridge Period Projects											\$ 1,346,788	\$ 6,597	\$ -	\$ 1,340,171

Note:

(1) Planned amounts were reduced by project reductions and contingency to arrive at the projected amounts

Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures

For the years 2022 through 2025

Generation Capital Projects greater than \$1M

(\$000's)

Generation Capital Expenditures

Line No.	(a) Calendar Year	(b) Tier 1 Portfolio	(c) Tier 2 Portfolio	(d) Project Type	(e) Project Classification	(f) Class of Cost Estimate	(g) Engineering Internal or External	(h) Project Bid Issued	(i) Full Internal Budget Approval	(j) Work Item Description	(k)	(l)	(m)	(n)
											Planned Amount	Project Reduction	Contingency Amount	Projected Amount
1	Test Year	Gas Generation	Covert Commons	Routine	Condition-based	Class 1	NA	Yes	Approved	Long Term Service Agreement - Running capital contrak	\$ 13,848	-	\$ -	\$ 13,848
2	Test Year	Gas Generation	Covert Commons	Routine	Condition-based	Class 3	NA	No	Approved	Non Long Term Service Agreemen Capital - Extras not included in contra	\$ 5,510	-	\$ -	\$ 5,510
3	Test Year	Gas Generation	Jackson Site Commons	Non-Routine	Condition-based	Class 2	External	No	Approved	LM 1 - 6 SAC Extended Life Combusto	\$ 1,338	-	\$ -	\$ 1,338
4	Test Year	Gas Generation	Jackson Site Commons	Routine	Condition-based	Class 1	NA	Yes	Approved	GE Long Term Service Agreement FFF	\$ 8,245	-	\$ -	\$ 8,245
5	Test Year	Gas Generation	Zeeland Site Commons	Non-Routine	Condition-based	Class 2	Internal	Yes	Approved	Purchase of Site Spare GSL	\$ 3,917	-	\$ -	\$ 3,917
6	Test Year	Gas Generation	Zeeland Site Commons	Routine	Condition-based	Class 1	NA	Yes	Approved	Long Term Service Agreement - Running Capital Contra	\$ 8,160	-	\$ -	\$ 8,160
7	Test Year	Gas/Oil Generation	Karn 3	Non-Routine	Condition-based	Class 3	Internal	No	Approved	Karn 3 Cooling Tower Internal Structure Replacemer	\$ 5,000	-	\$ -	\$ 5,000
8	Test Year	Hydro Generation	Alcona	Non-Routine	Regulatory	Class 4	Internal	No	Approved	Risk Informed Decision Making Resolution	\$ 2,500	-	\$ -	\$ 2,500
9	Test Year	Hydro Generation	Alcona	Non-Routine	Infrastructure	Class 3	External	No	Approved	Core Wall Remediation Project	\$ 9,057	\$ -	\$ -	\$ 9,057
10	Test Year	Hydro Generation	Footle	Non-Routine	Condition-based	Class 3	External	No	Approved	Unit 2 Wicket Gates Replacement Project	\$ 1,253	-	\$ -	\$ 1,253
11	Test Year	Hydro Generation	Hardy	Routine	Safety	Class 3	External	No	Approved	Electrical Safety Project	\$ 1,500	-	\$ -	\$ 1,500
12	Test Year	Hydro Generation	Allegan	Non-Routine	Condition-based	Class 3	External	No	Approved	Unit 1 Wicket Gate Replacemen	\$ 1,333	-	\$ -	\$ 1,333
13	Test Year	Hydro Generation	Five Channels	Non-Routine	Safety	Class 2	External	Yes	Approved	Trash Rack Ergonomics Projec	\$ 1,728	-	\$ -	\$ 1,728
14	Test Year	Hydro Generation	Hardy	Non-Routine	Regulatory	Class 2	External	Yes	Approved	Hardy Splash Wall Replacemen	\$ 3,134	-	\$ -	\$ 3,134
15	Test Year	Hydro Generation	Webber	Non-Routine	Condition-based	Class 1	External	Yes	Approved	Unit 1 Generator Rewinx	\$ 4,750	-	\$ -	\$ 4,750
16	Test Year	Hydro Generation	Footle	Non-Routine	Safety	Class 3	External	No	Approved	ADA Ramp Investigation and replacemer	\$ 1,463	-	\$ -	\$ 1,463
17	Test Year	Hydro Generation	Hodenpyl	Non-Routine	Condition-based	Class 3	External	No	Approved	Downstream Wall	\$ 3,450	-	\$ -	\$ 3,450
18	Test Year	Hydro Generation	Rogers	Non-Routine	Regulatory	Class 4	External	No	Approved	Powerhouse Left Embankment Retaining Wa	\$ 1,083	-	\$ -	\$ 1,083
19	Test Year	Hydro Generation	Rogers	Non-Routine	Regulatory	Class 4	External	No	Approved	Probable Maximum Flood Projec	\$ 3,272	-	\$ -	\$ 3,272
20	Test Year	Hydro Generation	Cooke	Non-Routine	Condition-based	Class 3	External	No	Approved	Head Gate Replacement Projec	\$ 2,080	-	\$ -	\$ 2,080
21	Test Year	Hydro Generation	Ludington Site Commons	Non-routine	Condition-based	Class 3	External	No	Approved	Unit 1-6 DCS Control Relay Replacemer	\$ 2,501	-	\$ -	\$ 2,501
22	Test Year	Hydro Generation	Ludington Site Commons	Routine	Infrastructure	Class 3	External	No	Approved	Administrative Building Additior	\$ 2,714	-	\$ -	\$ 2,714
23	Test Year	Hydro Generation	Ludington Site Commons	Non-Routine	Regulatory	Class 1	Internal	Yes	Approved	Replace Barrier Net Panel	\$ 1,753	-	\$ -	\$ 1,753
24	Test Year	Hydro Generation	Ludington Site Commons	Non-Routine	Infrastructure	Class 4	External	No	Approved	Intake Gate and Gate House Mechanical Replaceme	\$ 1,118	-	\$ -	\$ 1,118
25	Test Year	Renewables	Solar Commons	Non-Routine	New Generation	Class 1/BTA Executed 2021	CE Tech Specs/External Design	Yes	Approved	Solar - 2019 Bid Event (Mustang Mile 150 MW	\$ 49,337	-	\$ -	\$ 49,337
26	Test Year	Renewables	Solar Commons	Non-Routine	New Generation	Class 1/BTA Executed 2021	CE Tech Specs/External Design	Yes	Approved	Solar - 2020 Bid Event (Washtenaw Solar) (150 MW	\$ 104,980	-	\$ -	\$ 104,980
27	Test Year	Renewables	Solar Commons	Non-Routine	New Generation	Class 1/Awarded Self-Build 2022	CE Tech Specs/External Design	Yes	Approved	Solar - 2021 Bid Event (Muskegon Solar) (250 MW	\$ 115,140	-	\$ -	\$ 115,140
28	Total Test Year Projects										\$ 360,164	\$ -	\$ -	\$ 360,164

Note:

(1) Planned amounts were reduced by project reductions and contingency to arrive at the projected amounts

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Summary of the Generation O&M Expense
For the Years 2022 through 2025
(\$000's)

Case No.: U-21389
Exhibit No.: A-41 (RTB-4)
Page: 1 of 3
Witness: RTBlumenstock
Date: May 2023

GENERATION OPERATION AND MAINTENANCE EXPENSES

Line No.	(a) Description	(b) Historical 12 Months Ended 12/31/2022	(c) Projected Bridge Period 14 Months Ending 02/29/2024	(d) Projected Test Year 12 Months Ending 02/28/2025
1	BASE O&M	\$ 106,819	\$ 136,474	\$ 126,416
2	ADJUSTED O&M			
3	Environmental Operations	\$ 10,802	\$ 8,714	\$ 5,860
4	Major Maintenance	\$ 32,411	\$ 27,278	\$ 31,976
5	Retention & Separation	\$ 2,339	\$ 14,917	\$ 18,220
6	TOTAL O&M	\$ 150,030.653	\$ 172,465.3	\$ 164,251.9

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
		2022 Actual	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2022	Merit & Inflation 12 Mos Ending Dec 31, 2023	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2023	Merit & Inflation 12 Mos Ending Dec 31, 2024	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2024	Merit & Inflation 2 Mos Ending 2/28/2025	Other Adjustments	Projected O&M 12 Mos Ending 2/28/2025
			(c) * Inflation Rate			(e) * Inflation Rate		(g) * Inflation Rate		(b) + (d) + (f) + (h) + (i)
1	Line Item 1	150,031	150,031	5,851	155,882	3,429	159,311	558	4,383	164,252
	Labor	91,790	91,790	3,580	95,370	2,098	97,468	341	2,699	100,509
	Material	6,931	6,931	270	7,201	158	7,359	26	188	7,573
	Contractor	16,999	16,999	663	17,662	389	18,051	63	474	18,588
	Non-Labor Overheads		0	0	0	0	0	0	0	0
	Non-Labor Other	34,311	34,311	1,338	35,649	784	36,433	128	1,022	37,582

Notes

	2023	2024	2025
4 Annual merit increase			
Annual merit increase	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated merit increase	3.9%	2.2%	0.4%
Annual inflation rates per WP-JCA-xx			
Annual inflation rates	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated inflation rate	3.9%	2.2%	0.4%

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of the Generation Major Maintenance O&M Expense

For the Years 2022 through 2025

(\$000's)

Case No.: U-21389

Exhibit No.: A-41 (RTB-4)

Page: 3 of 3

Witness: RTBlumenstock

Date: May 2023

GENERATION MAJOR MAINTENANCE EXPENSES

Line No.	Description	(a)	(b)	(c)	(d)
			Historical 12 Months Ended 12/31/2022	Projected Bridge Period 14 Months Ending 02/29/2024	Projected Test Year 12 Months Ending 02/28/2025
1	Major Maintenance				
2	Campbell 1&2	\$	3,307	\$ 1,875	\$ 1,782
3	Campbell 3	\$	3,196	\$ 2,623	\$ 2,229
4	Karn 1&2	\$	6,291	\$ 528	\$ 223
5	Karn 3&4	\$	607	\$ 1,579	\$ 2,017
6	Classic 7	\$	(636)	\$ 326	\$ 334
7	Zeeland Generating Station	\$	2,761	\$ 5,171	\$ 4,658
8	Jackson Generating Station	\$	7,546	\$ 3,237	\$ 3,061
9	Covert Generating Stations	\$	-	\$ 3,925	\$ 6,025
10	Ludington	\$	4,921	\$ 3,571	\$ 4,422
11	Hydros	\$	4,418	\$ 4,174	\$ 6,298
12	Solar	\$	-	\$ 144	\$ 778
13	Admin & Other	\$	-	\$ 125	\$ 150
14	TOTAL Major Maintenance	\$	32,411	\$ 27,278	\$ 31,976

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

PAMELA L. BOLDEN

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Line Clearing O&M Expense
2017-2022 Historical and 2023-2028 Projected

Case No.: U-21389
Exhibit No.: A-42 (PLB-1)
Page: 1 of 1
Witness: PLBorden
Date: May 2023

Line No.	(a) Description	(b) Expense Type	(c) Year	(d) 2017	(e) 2018	(f) 2019	(h) 2020	(i) 2021	(j) 2022	(k) 2023 Calendar Year	(l) 2024 Calendar Year	(m) 2025 Calendar Year	(n) 2026 Calendar Year	(o) 2027 Calendar Year	(p) 2028 Calendar Year	(q) Test Period 3/1/24-2/28/25
1	Electric System			Actual	Actual	Actual	Actual	Actual	Actual	Projected	Projected	Projected	Projected	Projected	Projected	Projected
2	Electric HVD			\$11,393,937	\$12,035,797	\$12,211,819	\$9,169,948	\$10,739,347	\$10,987,343	\$12,064,000	\$12,279,000	\$12,294,000	\$11,972,000	\$11,645,540	\$11,300,000	\$12,281,500
3	Contractor	Maintenance Clearing		\$7,658,022	\$8,438,081	\$7,571,492	\$4,511,969	\$4,488,246	\$5,157,938	\$5,108,885	\$5,315,945	\$5,313,370	\$5,286,488	\$5,179,735	\$4,811,782	\$5,315,516
4	Contractor	Brushing - Cut		\$2,388,351	\$2,682,986	\$3,823,127	\$2,365,927	\$2,797,068	\$3,063,144	\$3,244,000	\$3,324,000	\$3,324,000	\$3,000,000	\$2,752,100	\$2,750,000	\$3,324,000
5	Contractor	Brushing - Spray		N/A	N/A	N/A	\$1,395,683	\$2,407,477	\$1,565,658	\$2,575,000	\$2,490,000	\$2,490,000	\$2,500,000	\$2,500,000	\$2,500,000	\$2,490,000
6	Contractor	Demand Clearing		\$375,620	\$152,548	\$87,281	\$110,249	\$98,417	\$89,478	\$75,000	\$65,000	\$60,000	\$55,000	\$60,000	\$60,000	\$64,167
7	Contractor	Noxious Weed Control		\$20,546	\$70,093	\$52,416	\$91,873	\$95,891	\$152,737	\$95,000	\$95,000	\$95,000	\$95,000	\$95,000	\$95,000	\$95,000
8	Contractor	HVD Contract Planning		Note ¹	Note ¹	Note ¹	Note ¹	Note ¹	\$175,749	\$180,000	\$185,000	\$190,000	\$196,000	\$200,000	\$205,000	\$185,833
9	Labor	HVD Salaries		\$709,395	\$666,847	\$650,467	\$672,387	\$751,684	\$653,623	\$679,115	\$694,055	\$708,630	\$723,512	\$738,705	\$754,218	\$696,484
10	Other	HVD Expenses		\$242,003	\$25,242	\$27,036	\$21,859	\$100,564	\$129,016	\$107,000	\$110,000	\$113,000	\$116,000	\$120,000	\$124,000	\$110,500
11		Miles		1,019	1,081	1,102	1,116	1,178	1,106	1,164	1,166	1,168	1,171	1,173	1,173	1,166
12	Electric LVD			\$38,359,929	\$39,912,561	\$41,078,112	\$46,104,489	\$75,827,808	\$91,016,131	\$87,966,000	\$105,371,000	\$112,792,000	\$113,122,000	\$113,428,000	\$113,752,000	\$106,607,833
13	Contractor	Maintenance Clearing		\$29,372,207	\$31,038,484	\$32,173,215	\$36,613,687	\$62,121,855	\$76,623,355	\$72,639,394	\$87,887,025	\$94,966,260	\$95,141,216	\$95,413,285	\$95,664,366	\$89,066,897
14	Contractor	Repetitive Outage Zone		\$934,465	\$2,054,288	\$995,575	\$1,397,354	\$1,711,666	\$2,086,904	\$1,700,000	\$1,500,000	\$1,350,000	\$1,215,000	\$1,090,000	\$1,000,000	\$1,475,000
15	Contractor	CEMI Clearing		N/A	N/A	N/A	\$695,322	\$491,257	\$506,985	\$750,000	\$750,000	\$750,000	\$750,000	\$750,000	\$750,000	\$750,000
16	Contractor	First Zone Clearing		\$399,793	\$102,600	\$1,523,542	\$974,275	\$405,771	\$466,566	\$1,175,000	\$1,375,000	\$1,525,000	\$1,660,000	\$1,660,000	\$1,660,000	\$1,400,000
17	Contractor	LCP Zonal Clearing		N/A	N/A	N/A	N/A	N/A	N/A	N/A	\$750,000	\$750,000	\$750,000	\$750,000	\$750,000	\$750,000
18	Contractor	Demand Clearing		\$2,642,665	\$1,444,228	\$1,695,571	\$1,281,844	\$3,329,375	\$2,737,654	\$2,614,000	\$2,558,000	\$2,505,000	\$2,505,000	\$2,505,000	\$2,505,000	\$2,549,167
19	Contractor	Brushing - Spray		\$1,807,622	\$1,679,301	\$989,772	\$1,538,898	\$3,072,427	\$3,302,319	\$3,400,000	\$3,500,000	\$3,500,000	\$3,500,000	\$3,500,000	\$3,500,000	\$3,500,000
20	Contractor	Distribution Contract Planning		Note ²	Note ²	Note ²	Note ²	Note ²	\$1,300,689	\$1,530,221	\$2,029,266	\$2,340,961	\$2,411,190	\$2,483,526	\$2,558,032	\$2,081,215
21	Labor	Distribution Salaries		\$2,388,399	\$2,532,425	\$2,683,107	\$2,509,359	\$3,994,189	\$3,120,678	\$3,242,385	\$3,955,709	\$4,038,779	\$4,123,593	\$4,210,189	\$4,298,603	\$3,969,554
22	Other	Distribution Expenses		\$814,778	\$1,061,235	\$1,017,330	\$1,093,749	\$701,269	\$870,981	\$915,000	\$1,066,000	\$1,066,000	\$1,066,000	\$1,066,000	\$1,066,000	\$1,066,000
23		Miles		3,503	3,218	3,518	4,211	5,279	6,388	5,875	6,760	7,232	7,449	7,672	7,932	6,839
24	Electric Total	O&M Expense		\$49,753,866	\$51,948,358	\$53,289,931	\$55,274,437	\$86,567,155	\$102,003,474	\$100,030,000	\$117,650,000	\$125,086,000	\$125,094,000	\$125,073,540	\$125,052,000	\$118,889,334
25	Contractor	Clearing		\$45,599,291	\$47,662,609	\$48,911,991	\$50,977,083	\$81,019,448	\$95,752,737	\$93,376,279	\$109,609,970	\$116,628,630	\$116,457,704	\$116,255,120	\$116,046,148	\$110,779,746
26	Contractor	Contract Planning		Note ¹ , Note ²	Note ¹ , Note ²	Note ¹ , Note ²	Note ¹ , Note ²	Note ¹ , Note ²	\$1,476,438	\$1,710,221	\$2,214,266	\$2,530,961	\$2,607,190	\$2,683,526	\$2,763,032	\$2,267,049
27	Labor	Salaries		\$3,097,794	\$3,199,272	\$3,333,574	\$3,181,745	\$4,745,873	\$3,774,301	\$3,921,499	\$4,649,764	\$4,747,409	\$4,847,105	\$4,948,894	\$5,052,821	\$4,666,038
28	Other	Expenses		\$1,056,781	\$1,086,477	\$1,044,366	\$1,115,609	\$801,833	\$999,997	\$1,022,000	\$1,176,000	\$1,179,000	\$1,182,000	\$1,186,000	\$1,190,000	\$1,176,500
29		O&M Miles		4,522	4,299	4,620	5,326	6,457	7,494	7,039	7,926	8,400	8,620	8,845	9,105	8,005
30		HVD Cost/Mile		\$11,181	\$11,134	\$11,082	\$8,219	\$9,117	\$9,932	\$10,367	\$10,531	\$10,523	\$10,227	\$9,928	\$9,633	\$10,530
31		LVD Cost/Mile		\$10,951	\$12,403	\$11,677	\$10,949	\$14,364	\$14,248	\$14,973	\$15,587	\$15,596	\$15,186	\$14,785	\$14,341	\$15,589
32		LVD Target Cycle Miles		8,362	8,362	8,362	8,362	8,362	8,632	8,699	8,699	8,699	8,699	8,699	8,699	8,699
33		LVD O&M Program Miles		3,503	3,218	3,518	4,211	5,279	6,388	5,875	6,760	7,232	7,449	7,672	7,932	6,839
34		Capital Clearing Miles		442	489	301	258	252	221	250	325	325	325	325	326	327
35		Percent of 7 Year Cycle		47%	44%	46%	53%	66%	77%	70%	81%	87%	89%	92%	95%	82%
36		Effective Cycle Years		14.8	15.8	15.3	13.1	10.6	9.1	9.9	8.6	8.1	7.8	7.6	7.4	8.5

Note¹: HVD Contract Planners were not recorded separately in 2017 - 2021

Note²: LVD Contract Planners were not recorded separately in 2017 - 2021

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Line Clearing Ramp-up Plan Estimated Potential Service Restoration Cost Avoidance

Case No.: U-21389
Exhibit No.: A-43 (PLB-2)
Page: 1 of 1
Witness: PLBolden
Date: May 2023

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
1	<u>Year</u>	<u>2019</u> Actual	<u>2020</u> Actual	<u>2021</u> Actual	<u>2022</u> Actual	<u>2019-22</u> Average	<u>2023</u> Projected	<u>2024</u> Projected	<u>2025</u> Projected	<u>2026</u> Projected	<u>2027</u> Projected	<u>2028</u> Projected
2	Line Clearing Program Funding Level (\$M)	\$53.3	\$54.8	\$86.6	\$102.0	\$74.2	\$100.0	\$117.6	\$125.1	\$125.1	\$125.1	\$125.1
3	Actual Tree-Related Primary Outage Incidents with MED	11,247	9,337	12,665	8,256	10,376						
4	Projected Tree-Related Primary Outage Incidents with MED						10,080	9,885	9,690	9,494	9,299	9,103
5	Primary Voltage Outage Incidents Reduced with MED						296	491	687	882	1,078	1,273
6	Outage Incident Reduction Percent from 4-Year Average						2.85%	4.73%	6.62%	8.50%	10.38%	12.27%
7	Potential Service Restoration Cost Avoidance (\$M)						\$0.330	\$0.563	\$0.812	\$1.073	\$1.351	\$1.645

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
2023 HVD Line Clearing Work Plan

Case No.: U-21389
Exhibit No.: A-44 (PLB-3)
Page: 1 of 1
Witness: PLBorden
Date: May 2023

Line No.	(a) Description	(b) Miles	(c) Voltage	(d) Year Last Cleared	(e) Line Name	(f) Miles	(g) Voltage	(h) Year Last Cleared
1	ADRIAN #2	0.66	138	2019	KALAMAZOO TREATMENT	1.25	46	2019
2	AGNEW	11.78	46	2020	KARN-GARFIELD	0.02	46	2007
3	ALABASTER	27.99	46	2019	KAWKAWLIN	31.19	46	2019
4	ALLEDALE	25.98	46	2018	KELLOGGSVILLE	10.48	46	2006
5	AMASTEEL	0.05	138	2019	KENDRICK	6.57	138	2020
6	AMBER-WHITE LAKE	0.02	138	2004	LAKE SHORE	12.42	46	2015
7	AMPERSEE	5.26	46	2019	LAMOREAUX	14.00	46	2019
8	ARGENTA-VERONA	0.96	46	2000	LARKIN	11.12	46	2019
9	ATHENS	0.01	46	2015	LAWNDALE-MALLEABLE	0.03	138	2007
10	AUBURN	12.43	46	2019	LESLIE	28.50	46	2010
11	BARAGA-FOUR MILE	0.20	138	2019	LINCOLN	28.21	46	2019
12	BARAGA-RACE STREET	0.19	138	2019	LINDEN	0.01	46	2020
13	BARRY	2.01	46	2019	LIVINGSTON-GAYLORD	0.11	46	2020
14	BASS CREEK-STERNBERG	0.01	138	2020	LOOKING GLASS	5.33	138	2019
15	BATTLE CREEK-ISLAND RD	0.05	138	2019	MARKEY	20.58	46	2018
16	BEALS RD-HAZELWOOD	2.83	138	2020	MARQUETTE-BINGHAM	2.61	138	2019
17	BEARLS RD-HAZELWOOD	0.05	138	2020	MASON	0.06	46	2022
18	BIG RAPIDS	22.09	46	2014	MECOSTA	4.82	46	2019
19	BINGHAM-CORNELL	0.06	138	2019	MEDUSA-STOVER	14.74	138	2020
20	BINGHAM-AVERY SOLAR	0.60	46	2020	MENASHA	4.09	46	2018
21	BOYNE CITY	15.13	46	2019	METRO	16.40	46	2019
22	BRICKYARD-HOLTON	0.75	46	2019	MIDDLEVILLE	34.18	46	2019
23	BRONSON	23.15	46	2019	MILHAM-UPJOHN	0.59	138	2019
24	BRYE RD	0.02	46	2019	MILLERS PT-LINDBERGH	1.09	138	2021
25	BURNIPS-WAYLAND	0.02	69	2019	MORRICE	40.53	46	2019
26	BURR OAK	6.88	46	2019	MORROW-SIMPSON	3.36	138	2019
27	CAMPBELL-BLACK RIVER	2.19	138	2019	NASHVILLE	37.21	46	2010
28	CAMPBELL-BLENDON	0.06	138	2019	NEWAYGO	0.02	46	2019
29	CAMPBELL-BUCK CREEK	1.58	138	2019	NORTH ADAMS	23.01	46	2019
30	CAMPBELL-RANSOM	3.56	138	2004	NORTH KENT	11.68	46	2019
31	CAMPBELL-WEALTHY	0.02	138	2019	OAK ST	0.08	46	2013
32	CAMPBELL-BEALS RD	3.38	138	2019	OAKWOOD	6.39	46	2018
33	CEMENT CITY	15.09	46	2019	OKEMOS	12.13	46	2019
34	CHARLEVOIX	17.22	46	2005	ORIOLE	5.80	46	2019
35	CHARLOTTE	8.73	46	2015	OSCODA	12.09	46	2019
36	COLDWATER-MOORE RD	0.09	138	2017	OID	34.26	46	2019
37	COOKE	13.73	46	2015	PARAMOUNT	2.40	46	2019
38	COOLEY	2.26	46	2019	PARKWAY	3.04	46	2014
39	CORK ST-UPJOHN	1.70	138	2019	PARMA	30.32	46	2019
40	CORNELL-DELHI	0.01	138	2020	PETOSKEY	5.56	46	2019
41	CORNELL-TIHART	0.08	138	2019	PLASTER CREEK-GAINES	4.50	138	2019
42	DAVID	2.59	46	2019	PORTLAND-LOWELL	0.87	69	2019
43	DAVISON	13.05	46	2020	RACE STREET-SPAULDING	0.05	138	2019
44	DELHI-OAKLAND	2.36	138	2019	RAMONA	7.99	46	2019
45	DEXTER TRAIL	0.09	46	2022	RANSOM-BUCK CREEK	10.13	138	2019
46	DIETZ RD-GAYLORD	17.63	46	2019	RED CEDAR	7.77	46	2019
47	DUPONT-COBB	0.20	138	2019	SCOTTVILLE	24.69	46	2019
48	EDMORE	24.67	46	2013	SHELBY	25.53	46	2013
49	EDWARDS #1	1.33	138	2021	SILICON	0.02	46	1999
50	EDWARDS #2	1.21	138	2021	SILVER LAKE	8.63	46	2020
51	FENTON	5.59	46	2022	SOLVAY	3.85	138	2019
52	FILER CITY	0.30	138	2018	SOUTH BOARDMAN-RICHLAND	2.06	69	2019
53	FINE LAKE	21.57	46	2015	SPAULDING-BROADMOOR	0.03	138	2018
54	FOUR MILE-WEALTHY	0.05	138	2019	SPAULDING-BEALS RD	0.08	138	2018
55	FREMONT	9.04	46	2019	SPAULDING-VERGENNES	0.01	138	2018
56	GERRISH	17.63	46	2000	SPRING LAKE	3.04	46	2020
57	GOLDEN	19.71	46	2019	STAMPING PLANT-GAINES	0.12	138	2019
58	GOODALE	15.83	46	2019	THOMPSON-GAINES	1.62	138	2019
59	GRAND BLANC	27.28	46	2015	THOMPSON RD	0.76	138	2019
60	GRAYLING	5.58	46	2014	TITTABAWASSEE-WEADOCK	2.39	138	2019
61	HALSEY-OAKLAND	3.33	138	2019	UNION ST	23.06	46	2020
62	HAMMOND RD	1.80	46	2014	VAN WERT	12.34	138	2019
63	HANSEN	5.71	46	2019	VENICE	0.15	46	2020
64	HASTINGS	2.96	46	2019	W-23	0.07	138	2020
65	HERSEY-WEIDMAN	0.04	69	2019	WALDO	7.47	46	2014
66	HESPERIA SPUR	0.63	69	2019	WAYLAND	21.91	46	2019
67	HOLTON	28.02	46	2019	WENTWORTH	13.86	46	2020
68	IONIA #1	6.93	46	2019	WHITE LAKE-COBB	0.04	138	2018
69	IONIA #2	19.98	46	2015	WHITESTONE POINT	18.07	46	2019
70	ISLAND RD-DELHI	0.17	138	2019				
71	Total 2023 Work Plan Miles	1161.55						

Consumers Energy Company

Reliability Pre and Post Clearing for 2018, 2019, 2020, 2021, and 2022 Cleared IVD Circuits

	(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I)	(J)
1	ID	Description	Unit	Quantity	Unit Price	Amount	Discount	Net Amount	Tax	Total
2	Subtotal									
3	Grand Total									
4	Total									
5	Total									
6	Total									
7	Total									
8	Total									
9	Total									
10	Total									
11	Total									
12	Total									
13	Total									
14	Total									
15	Total									
16	Total									
17	Total									
18	Total									
19	Total									
20	Total									
21	Total									
22	Total									
23	Total									
24	Total									
25	Total									
26	Total									
27	Total									
28	Total									
29	Total									
30	Total									
31	Total									
32	Total									
33	Total									
34	Total									
35	Total									
36	Total									
37	Total									
38	Total									
39	Total									
40	Total									
41	Total									
42	Total									
43	Total									
44	Total									
45	Total									
46	Total									
47	Total									
48	Total									
49	Total									
50	Total									
51	Total									
52	Total									
53	Total									
54	Total									
55	Total									
56	Total									
57	Total									
58	Total									
59	Total									
60	Total									
61	Total									
62	Total									
63	Total									
64	Total									
65	Total									
66	Total									
67	Total									
68	Total									
69	Total									
70	Total									
71	Total									
72	Total									
73	Total									
74	Total									
75	Total									
76	Total									
77	Total									
78	Total									
79	Total									
80	Total									
81	Total									
82	Total									
83	Total									
84	Total									
85	Total									
86	Total									
87	Total									
88	Total									
89	Total									
90	Total									
91	Total									
92	Total									
93	Total									
94	Total									
95	Total									
96	Total									
97	Total									
98	Total									
99	Total									
100	Total									
101	Total									
102	Total									
103	Total									
104	Total									
105	Total									
106	Total									
107	Total									
108	Total									
109	Total									
110	Total									
111	Total									
112	Total									
113	Total									
114	Total									
115	Total									
116	Total									
117	Total									
118	Total									
119	Total									
120	Total									
121	Total									
122	Total									
123	Total									
124	Total									
125	Total									
126	Total									
127	Total									
128	Total									
129	Total									
130	Total									
131	Total									
132	Total									
133	Total									
134	Total									
135	Total									
136	Total									
137	Total									
138	Total									
139	Total									
140	Total									
141	Total									
142	Total									
143	Total									
144	Total									
145	Total									
146	Total									
147	Total									
148	Total									
149	Total									
150	Total									
151	Total									
152	Total									
153	Total									
154	Total									
155	Total									
156	Total									
157	Total									
158	Total									
159	Total									
160	Total									
161	Total									
162	Total									
163	Total									
164	Total									
165	Total									
166	Total									
167	Total									
168	Total									
169	Total									
170	Total									
171	Total									
172	Total									
173	Total									
174	Total									
175	Total									
176	Total									
177	Total									
178	Total									
179	Total									
180	Total									
181	Total									
182	Total									
183	Total									
184	Total									
185	Total									
186										

Case No.: 11-21389

Case No.: U-21389
Exhibit No.: A-45 (PLB-4)
Page: 2 of 10
Witness: PLBorden
Date: May 2023

Line Clearing Reliability Results
Reliability Pre and Post Clearing for 2018, 2019, 2020 and 2021 Cleared LVD Circuits
(minimum 50% of circuit miles cleared for year listed, primary voltage outage incidents)

Case No.: U-21389
Exhibit No.: A-45 (PLB-4)
Page: 5 of 10
Witness: PLBorden
Date: May 2023

Line	No.	Description	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	(t)	(u)	(v)	(w)	(x)	(y)	(z)	(aa)	(ab)	(ac)	(ad)	(ae)	(af)	(ag)	(ah)	(ai)	(aj)	(ak)	(al)	(am)	(an)	(ao)	(ap)	(aq)	(ar)	(as)	(at)	(au)	(av)	(aw)	(ax)	(ay)	(az)	(ba)	(bb)	(bc)	(bd)	(be)	(bf)	(bg)	(bh)	(bi)	(bj)	(bk)	(bl)	(bm)	(bn)	(bo)	(bp)	(bq)	(br)	(bs)	(bt)	(bu)	(bv)	(bw)	(bx)	(by)	(bz)	(ca)	(cb)	(cc)	(cd)	(ce)	(cf)	(cg)	(ch)	(ci)	(cj)	(ck)	(cl)	(cm)	(cn)	(co)	(cp)	(cq)	(cr)	(cs)	(ct)	(cu)	(cv)	(cw)	(cx)	(cy)	(cz)	(da)	(db)	(dc)	(dd)	(de)	(df)	(dg)	(dh)	(di)	(dj)	(dk)	(dl)	(dm)	(dn)	(do)	(dp)	(dq)	(dr)	(ds)	(dt)	(du)	(dv)	(dw)	(dx)	(dy)	(dz)	(ea)	(eb)	(ec)	(ed)	(ee)	(ef)	(eg)	(eh)	(ei)	(ej)	(ek)	(el)	(em)	(en)	(eo)	(ep)	(eq)	(er)	(es)	(et)	(eu)	(ev)	(ew)	(ex)	(ey)	(ez)	(fa)	(fb)	(fc)	(fd)	(fe)	(ff)	(fg)	(fh)	(fi)	(fj)	(fk)	(fl)	(fm)	(fn)	(fo)	(fp)	(fq)	(fr)	(fs)	(ft)	(fu)	(fv)	(fw)	(fx)	(fy)	(fz)	(ga)	(gb)	(gc)	(gd)	(ge)	(gf)	(gg)	(gh)	(gi)	(gj)	(gk)	(gl)	(gm)	(gn)	(go)	(gp)	(gq)	(gr)	(gs)	(gt)	(gu)	(gv)	(gw)	(gx)	(gy)	(gz)	(ha)	(hb)	(hc)	(hd)	(he)	(hf)	(hg)	(hh)	(hi)	(hj)	(hk)	(hl)	(hm)	(hn)	(ho)	(hp)	(hq)	(hr)	(hs)	(ht)	(hu)	(hv)	(hw)	(hx)	(hy)	(hz)	(ia)	(ib)	(ic)	(id)	(ie)	(if)	(ig)	(ih)	(ii)	(ij)	(ik)	(il)	(im)	(in)	(io)	(ip)	(iq)	(ir)	(is)	(it)	(iu)	(iv)	(iw)	(ix)	(iy)	(iz)	(ja)	(jb)	(jc)	(jd)	(je)	(jf)	(jg)	(jh)	(ji)	(jj)	(jk)	(jl)	(jm)	(jn)	(jo)	(jp)	(jq)	(jr)	(js)	(jt)	(ju)	(jv)	(jw)	(jx)	(jy)	(jz)	(ka)	(kb)	(kc)	(kd)	(ke)	(kf)	(kg)	(kh)	(ki)	(kj)	(kl)	(km)	(kn)	(ko)	(kp)	(kq)	(kr)	(ks)	(kt)	(ku)	(kv)	(kw)	(kx)	(ky)	(kz)	(la)	(lb)	(lc)	(ld)	(le)	(lf)	(lg)	(lh)	(li)	(lj)	(lk)	(ll)	(lm)	(ln)	(lo)	(lp)	(lq)	(lr)	(ls)	(lt)	(lu)	(lv)	(lw)	(lx)	(ly)	(lz)	(ma)	(mb)	(mc)	(md)	(me)	(mf)	(mg)	(mh)	(mi)	(mj)	(mk)	(ml)	(mn)	(mo)	(mp)	(mq)	(mr)	(ms)	(mt)	(mu)	(mv)	(mw)	(mx)	(my)	(mz)	(na)	(nb)	(nc)	(nd)	(ne)	(nf)	(ng)	(nh)	(ni)	(nj)	(nk)	(nl)	(nm)	(nn)	(no)	(np)	(nq)	(nr)	(ns)	(nt)	(nu)	(nv)	(nw)	(nx)	(ny)	(nz)	(oa)	(ob)	(oc)	(od)	(oe)	(of)	(og)	(oh)	(oi)	(oj)	(ok)	(ol)	(om)	(on)	(oo)	(op)	(oq)	(or)	(os)	(ot)	(ou)	(ov)	(ow)	(ox)	(oy)	(oz)	(pa)	(pb)	(pc)	(pd)	(pe)	(pf)	(pg)	(ph)	(pi)	(pj)	(pk)	(pl)	(pm)	(pn)	(po)	(pp)	(pq)	(pr)	(ps)	(pt)	(pu)	(pv)	(pw)	(px)	(py)	(pz)	(qa)	(qb)	(qc)	(qd)	(qe)	(qf)	(qg)	(qh)	(qi)
------	-----	-------------	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------

Note 1 - The percent cleared is the percent of miles cleared for the year listed versus the circuit configuration miles in 2020. Circuits cleared less than 50% not shown.

Case No.: U-21389

Exhibit No.: A-45 (PLB-4)
Page: 6 of 10

Witness: PLBolden
Date: May 2023

Note 1 - The percent cleared is the percent of miles cleared for the year listed versus the circuit configuration miles in 2020. Circuits cleared less than 50% not shown.

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Line Clearing Reliability Results

Reliability Pre and Post Clearing for 2018, 2019, 2020 and 2021 Cleared LVD Circuits
(minimum 50% of circuit miles cleared for year listed, primary voltage outage incidents)

Case No.: U-21389

Exhibit No.: A-45 (PLB-4)

Page: 7 of 10

Witness: PLBorden

Date: May 2023

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Description												
1	Excluding MEDs											
2	Circuits Cleared in 2019						Tree Related Outage Incidents					
3	FEEDER_ID	WHQ	Substation	Circuit	% Cleared	Circuit Mile	Comp Mile	Voltage (kV)	2018	2019	2020	2021
4	21302	JAC	SUMMIT	FRANCIS STREET	100.0%	10.72	10.72	4.8 W	2	2	2	2
5	95201	GVL	PECK ROAD	ORE-IDA	100.0%	11.42	11.42	14.4 W	1	0	0	0
6	40502	SMN	LASALLE	OTTER CREEK	100.0%	38.39	38.39	7.2 W	0	1	1	2
7	44801	LDG	PENTWATER	PENTWATER WIRE PROD	100.0%	15.45	15.45	7.2 W	11	5	2	4
8	1901	JAC	ROBERTS STREET	LEROY STREET	100.0%	16.16	16.16	4.8 W	4	1	0	2
9	32504	GRA	BOSTON SQUARE	MULICK PARK	100.0%	5.02	5.02	7.2 W	1	0	0	0
10	114502	ADR	CADMIUS	BAKER	100.0%	23.39	23.39	7.2 W	0	1	0	1
11	50803	GVL	BELDING	COOKS CORNERS	100.0%	81.07	81.07	4.8 W	11	6	9	12
12	154202	KAL	SPICEBUSH	LACOTA	100.0%	37.88	37.88	14.4 D	4	7	2	6
13	18302	ADR	MANITOU BEACH	ADDISON	100.0%	60.76	60.76	4.8 W	5	2	4	4
14	103801	SMN	JEFFS ROAD	U.S. 23	100.0%	11.14	11.14	7.2 W	0	1	0	0
15	31702	GRA	ABERDEEN	ABERDEEN	100.0%	14.74	14.74	7.2 W	2	3	0	1
16	93602	MUS	HAYES STREET	172ND AVENUE	100.0%	4.85	4.85	4.8 W	0	1	0	0
17	127404	WBR	DUQUITE	PINE RIVER	100.0%	65.93	65.93	14.4 W	22	12	4	3
18	40401	BCK	ALBER	TERRITORIAL	100.0%	11.30	11.30	4.8 W	2	4	0	0
19	163703	JAC	WILLIS ROAD	LIST ROAD	100.0%	49.98	49.98	14.4 W	26	12	12	20
20	138301	HML	MILL GROVE	BABYLON	100.0%	25.15	25.15	14.4 W	5	4	2	5
21	97201	MDL	EASTLAWN	EASTLAWN	100.0%	8.95	8.95	4.8 W	1	0	0	1
22	160202	BIG	BARRYTON	CHIPPEWA LAKE	100.0%	42.48	42.48	14.4 W	3	5	5	3
23	50403	OWS	OLIVER	KING STREET	100.0%	6.29	6.29	4.8 W	1	0	0	1
24	6707	OWS	OWOSSO	GOULD	100.0%	17.54	17.54	4.8 W	2	0	1	2
25	11701	FLT	LINDEN	LOBDELL	100.0%	24.43	24.43	4.8 W	4	2	1	0
26	6706	OWS	OWOSSO	STEWART	99.7%	30.25	30.16	4.8 W	0	0	1	0
27	130301	LAN	WEST ROAD	WOOD ROAD	99.5%	26.42	26.30	4.8 W	1	1	0	0
28	42902	ADR	BLISSFIELD	CITY	99.4%	20.99	20.86	7.2 W	1	0	2	1
29	155402	SMN	DUNBAR	HULL	99.4%	14.26	14.17	7.2 W	0	0	0	0
30	128403	KAL	MILLERS POINT	CONCORD	99.3%	20.45	20.31	4.8 W	0	0	0	0
31	149902	KAL	ZYLMAN	ZYLMAN	99.3%	20.51	20.37	4.8 W	1	1	0	0
32	40402	BCK	ALBER	ALBERS	99.3%	5.39	5.35	4.8 W	3	2	2	1
33	80902	BCK	FIFTEEN MILE RD	15 MILE ROAD	99.2%	55.48	55.01	14.4 W	5	3	7	8
34	87702	BNC	BAGLEY	OTSEGO LAKE	99.1%	42.27	41.88	14.4 W	5	5	2	8
35	21902	BCK	GOGUAC	LAKEVIEW	98.9%	8.97	8.87	4.8 W	0	1	0	0
36	128402	KAL	MILLERS POINT	HOLIDAY	98.9%	22.98	22.72	4.8 W	0	1	1	0
37	127502	TWS	EAST TAWAS	LINCOLN STREET	98.7%	61.43	60.62	14.4 W	6	6	8	3
38	149803	FLT	WEST FENTON	OVERPASS	98.6%	13.47	13.28	4.8 W	0	0	0	1
39	13401	ADR	TECUMSEH	PATTERSON	98.5%	9.80	9.66	7.2 W	0	0	0	0
40	29602	JAC	VANDERCOOK LK	ACKERSON LAKE	98.1%	43.25	42.45	4.8 W	2	4	5	4
41	6201	FLT	GRAND BLANC	GRAND BLANC	97.9%	13.64	13.35	4.8 W	0	0	0	0
42	19603	BCK	CONVIS	MAR CREEK	97.8%	39.07	38.22	14.4 W	10	6	5	5
43	122401	GRN	PETTIS ROAD	HONEY CREEK	97.0%	60.60	58.76	14.4 W	12	10	6	14
44	52601	BNC	BOYNE CITY	BOYNE CITY	96.4%	21.99	21.21	7.2 W	1	2	1	1
45	37602	JAC	BATTESEE	PLEASANT LAKE	96.2%	63.85	61.42	14.4 W	6	7	7	7
46	71701	JAC	SPRING ARBOR	CHAPEL	95.4%	25.35	24.20	4.8 W	2	2	1	6
47	81602	LAN	HAGADORN	BEAUMONT	95.4%	12.93	12.33	4.8 W	0	2	0	0
48	96201	BNC	BEAUGRAND	HOSPITAL	95.4%	10.38	9.90	7.2 W	1	0	0	0
49	79002	SAG	SEIDEL	BROCKWAY	95.3%	18.54	17.67	4.8 W	2	0	0	0
50	54403	LDG	ORIOLE	HAMLIN	94.9%	41.65	39.53	14.4 W	6	12	8	11
51	21301	JAC	SUMMIT	SOUTH STREET	94.3%	8.15	7.68	4.8 W	2	0	0	0
52	42401	CAD	MARION	GASCOM	92.0%	60.83	55.99	14.4 D	7	5	8	7
53	78902	GRE	ALTO	MCCORDS	91.6%	55.92	51.20	4.8 W	4	5	7	9
54	78202	TWS	GREENBUSH	GREENBUSH	90.9%	39.93	36.30	12.0 D	10	4	3	7
55	127902	JAC	CAMBRIDGE	IRISH HILLS	90.9%	31.80	28.91	4.8 W	2	1	2	3
56	10402	CLR	GLADWIN	SPRING	89.9%	11.07	9.95	4.8 W	5	0	1	0
57	77201	JAC	SCIPION	MOSHERVILLE	88.9%	48.16	42.79	4.8 W	7	6	1	8
58	125202	LAN	UPTON	MARKET PLACE	88.6%	11.03	9.78	4.8 W	3	0	0	0
59	57302	BIG	OHMAN ROAD	HERSEY	87.3%	46.54	40.62	14.4 W	12	5	4	10
60	138303	HML	MILL GROVE	DUMONT	87.1%	85.71	74.69	14.4 W	19	25	27	23
61	93601	MUS	HAYES STREET	HAYES STREET	84.2%	14.22	11.97	4.8 W	4	2	1	3
62	49701	BCK	OLIVET	AINGER	83.1%	28.08	23.34	4.8 W	3	2	2	5
63	71202	ADR	COLLEGE PARK	MADISON	77.1%	16.08	12.40	7.2 W	0	0	0	1
64	134701	SAG	BELL ROAD	RATHBUN	77.0%	28.07	21.60	14.4 W	4	6	1	4
65	44202	MUS	MONTAGUE	NORTH SHORE	76.6%	51.00	39.08	7.2 W	4	13	1	4
66	78201	TWS	GREENBUSH	HARRISVILLE	76.2%	29.58	22.54	12.0 D	2	0	2	5
67	104102	ALM	GILSON	ROCK LAKE	74.1%	18.31	13.57	14.4 W	3	1	0	0
68	1101	GRN	BELLA VISTA	BELLA VISTA	73.6%	9.99	7.35	4.8 W	3	0	1	3
69	75704	MUS	MAPLE GROVE	SHAW BOX	72.0%	9.63	6.94	7.2 W	0	3	0	0
70	30202	FRE	HOLTON	MAPLE ISLAND	71.8%	152.11	109.16	14.4 W	20	14	17	11
71	137002	BCY	HOTCHKISS	BAY VALLEY	71.4%	6.95	4.96	14.4 W	0	0	0	0
72	100901	CLR	SURREY	SURREY	69.2%	57.81	39.98	14.4 W	10	5	5	8
73	60102	CAD	TUSTIN	LUTHER	67.2%	92.48	62.15	14.4 W	7	8	8	9
74	42502	CLD	KINDERHOOK	GILEAD	66.4%	112.91	75.02	4.8 W	13	27	8	7
75	33302	TWS	AU GRES	AU GRES	65.7%	58.77	38.63	11.0 D	23	5	6	7
76	123102	BIG	BIG PRAIRIE	OXBOW	64.7%	33.71	21.80	4.8 W	6	13	5	8
77	3506	CAD	CADILLAC	BERRY LAKE	64.3%	21.09	13.56	14.4 D	9	8	0	0
78	137301	KAL	TWILIGHT	EAST TOWNE	63.2%	20.24	12.79	4.8 W	1	2	0	0
79	127702	MDL	LEVELY	STURGEON	60.9%	110.77	67.47	14.4 W	18	24	15	27
80	60101	CAD	TUSTIN	WVTV	57.9%	54.36	31.50	14.4 W	5	9	4	7
81	57301	BIG	OHMAN ROAD	EVART	56.2%	39.95	22.46	14.4 W	7	3	4	10
82	7101	SMN	ERIE	STRASBURG ROAD	55.7%	23.24	12.94	7.2 W	0	3	0	2
83	34401	GRN	HULL STREET	LIME LAKE	55.5%	59.53	33.03	14.4 W	20	20	8	10
84	54602	TRA	ALDEN	CLAM	54.8%	38.70	21.19	7.2 W	5	13	6	3
85	21202	KAL	PHILLIPS	FACTORY	54.5%	6.95	3.79	4.8 W	2	0	0	0
86	63502	BCK	LEVEL PARK	LEVEL PARK	54.3%	19.66	10.67	4.8 W	0	2	1	5
87	151101	TRA	LELAND	NARROWS	52.2%	15.00	7.83	7.2 W	2	0	3	4
88												
89				Totals		2815.29	2377.26		413	368	252	349
90				Percent Improvement					10.9%	39.0%	15.5%	42.4%
91								1 Yr Prior	Yr Trim	1 Yr Post	2 Yr Post	3 Yr Post
92				Totals		Only 100% cleared Circuits		107	69	45	69	46
93				Percent Improvement		Percent Improvement		35.5%	57.9%	35.5%	57.0%	
94								1 Yr Prior	Yr Trim	1 Yr Post	2 Yr Post	3 Yr Post

Note 1 - The percent cleared is the percent of miles cleared for the year listed versus the circuit configuration miles in 2020. Circuits cleared less than 50% not shown.

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Line Clearing Reliability Results
Reliability Pre and Post Clearing for 2018, 2019, 2020 and 2021 Cleared LVD Circuits
(minimum 50% of circuit miles cleared for year listed, primary voltage outage incidents)

Case No.: U-21389
Exhibit No.: A-45 (PLB-4)
Page: 9 of 10
Witness: PLBalden
Date: May 2023

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Description															
1	Excluding MEDs														
2	Circuits Cleared in 2018														
3	FEEDER ID	WHQ	Substation	Circuit	% Cleared	Circuit Miles	Comp Miles	Voltage (kV)	2017	2018	2019	2020	2021	2022	
4	24203	CAD	MCBAIN	LUCAS	100.0%	11.88	11.88	14.4 D	16	6	2	0	0	0	
5	39202	JAC	MORRELL	WEST AVENUE	100.0%	11.02	11.02	4.8 W	3	2	0	1	1	3	
6	155801	SMN	SCHOOL ROAD	MOROCCO	100.0%	48.56	48.56	7.2 W	5	7	1	3	11	3	
7	50302	GRN	NORTH PARK	COIT	100.0%	5.43	5.43	7.2 W	0	0	0	0	0	0	
8	702	MUS	MUSKEGON HEIGHTS	LEAHY	100.0%	2.88	2.88	7.2 W	1	0	1	0	0	0	
9	101801	KAL	COLONY FARM	BTR PARK	100.0%	3.94	3.94	14.4 W	1	0	0	0	0	0	
10	148302	CLR	MANN SIDING	MANN SIDING	100.0%	60.62	60.62	14.4 W	4	1	2	4	4	7	
11	19203	ADR	RIGA	BIERMAN	100.0%	39.56	39.56	7.2 W	1	0	0	0	0	0	
12	98203	GRA	LEFFINGWELL	RAPIDSTAN	100.0%	5.66	5.66	7.2 W	1	1	1	0	0	2	
13	93901	WBR	LYON MANOR	TREASURE	100.0%	17.40	17.40	4.8 W	7	5	0	3	4	6	
14	32501	GRA	BOSTON SQUARE	NELAND	100.0%	8.92	8.92	7.2 W	0	1	0	0	0	1	
15	704	MUS	MUSKEGON HEIGHTS	HEIGHTS	100.0%	11.07	11.07	7.2 W	3	0	1	0	2	1	
16	81503	LAN	HARPER ROAD	ONONDAGA	100.0%	51.21	51.21	14.4 W	9	4	2	3	7	11	
17	158201	WBR	RYNO	MORENCI	100.0%	70.77	70.77	14.4 W	30	5	1	13	7	5	
18	67504	MUS	MCCRACKEN	SHERMAN	100.0%	7.82	7.82	7.2 W	1	0	1	1	0	1	
19	42601	ADR	PITTSFORD	CHURCH ROAD	100.0%	109.83	109.83	14.4 W	11	10	8	5	4	15	
20	53001	FRE	WHITE CLOUD	WILLIAM STREET	100.0%	24.07	24.07	4.8 W	2	1	1	0	0	4	
21	157004	FLT	MILBOURNE	DARTMOUTH	100.0%	10.14	10.14	4.8 W	1	0	0	0	1	2	
22	50502	SAG	BRISTOL	MILLER	100.0%	5.76	5.76	4.8 W	0	0	1	0	0	1	
23	53801	FLT	COURT	WOODLAWN	100.0%	8.40	8.40	4.8 W	0	0	0	0	0	0	
24	128401	KAL	MILLERS POINT	MEADOW VIEW	99.3%	10.93	10.85	4.8 W	0	0	6	0	2	2	
25	10202	LAN	GRAND LEDGE	HARTEL ROAD	98.9%	13.27	13.12	4.8 W	0	2	0	0	0	0	
26	107601	ALM	ISABELLA	PICKARD	98.7%	14.99	14.79	4.8 W	0	0	0	0	0	0	
27	50501	SAG	BRISTOL	BRISTOL	98.2%	6.27	6.15	4.8 W	0	1	0	0	0	0	
28	29303	WBR	MARKEY	FOREST ESTATES	98.1%	21.04	20.63	4.8 W	2	7	2	1	3	6	
29	21303	JAC	SUMMIT	FOURTH STREET	97.8%	11.59	11.33	4.8 W	1	0	0	0	2	0	
30	39303	SAG	NIAGARA	NIAGARA	97.2%	9.09	8.83	4.8 W	2	0	0	1	1	0	
31	63403	KAL	KILGORE	MOUNT EVEREST	97.1%	8.36	8.12	4.8 W	2	2	0	0	1	0	
32	155401	SMN	DUNBAR	DUNBAR	97.0%	17.16	16.65	7.2 W	0	1	0	2	0	0	
33	37601	JAC	BATTEESE	COON HILL	97.0%	46.84	45.43	14.4 W	3	12	1	6	2	26	
34	57002	CLD	COLON	PALMER	96.2%	56.69	54.55	4.8 W	3	3	3	5	14	6	
35	26901	HML	OTSEGO	OTSEGO	96.2%	7.52	7.23	4.8 W	1	0	0	0	0	3	
36	140202	WBR	SIMMONS	CABIN LAKE	96.0%	14.16	13.59	14.4 W	0	1	1	1	1	1	
37	127701	MDL	LEVELY	ALLBRIGHT	95.6%	87.79	83.95	14.4 W	3	11	6	4	12	15	
38	149002	LAN	PACKARD	PACKARD	94.2%	18.66	17.58	14.4 W	2	0	0	0	1	1	
39	24301	KAL	AUSTIN	WEST LAKE	94.1%	29.89	28.13	4.8 W	4	3	5	2	1	6	
40	33802	CLR	HARRISON	HARRISON	94.0%	45.46	42.74	4.8 W	3	1	4	8	2	6	
41	48601	BCK	MORGAN	ORCHARD	93.6%	27.33	25.59	4.8 W	2	2	3	3	11	3	
42	132304	MDL	ORCHARD ROAD	ST ANDREWS	92.7%	16.05	14.89	4.8 W	0	0	1	2	2	0	
43	60701	MUS	GETTY	MARQUETTE	92.3%	26.56	24.53	7.2 W	2	1	3	2	2	0	
44	67404	LAN	DEXTER TRAIL	DANSVILLE	89.0%	46.24	41.16	14.4 W	6	4	2	3	4	5	
45	50101	KAL	EASTWOOD	TEXEL	88.7%	12.21	10.83	4.8 W	1	3	1	1	2	6	
46	104201	GRN	ROGUE RIVER	CANNON FARMS	88.7%	58.08	51.51	14.4 W	19	10	2	4	4	8	
47	71702	JAC	SPRING ARBOR	ARBOR HILLS	88.5%	52.74	46.69	4.8 W	5	5	7	3	10	6	
48	27502	KAL	OAKWOOD	PARKVIEW	88.3%	10.14	8.95	4.8 W	3	2	3	5	3	5	
49	22202	GRN	COWAN LAKE	RAMSDALL	87.0%	64.19	55.82	14.4 W	15	17	10	12	6	18	
50	156202	BCK	CRANBROOK	11 MILE ROAD	85.5%	23.18	19.82	14.4 W	2	0	2	2	2	3	
51	32002	KAL	STADIUM	COLLEGE	84.3%	6.14	5.18	4.8 W	0	0	1	2	3	0	
52	32503	GRA	BOSTON SQUARE	HALL	83.3%	10.45	8.71	7.2 W	1	2	1	0	2	1	
53	57001	CLD	COLON	COLON	83.1%	61.58	51.19	4.8 W	10	1	10	1	10	9	
54	6802	BEN	HOMESTEAD	BEULAH	82.7%	83.69	69.19	7.2 W	14	17	12	10	12	21	
55	147602	LAN	VAN ATTA	VAN ATTA	81.3%	54.62	44.42	4.8 W	4	0	0	2	3	2	
56	70801	GRA	MICHIGAN	LYDIA	81.2%	8.15	6.62	7.2 W	1	0	1	0	0	1	
57	161902	GRE	EMERSON	KILMER	81.0%	53.87	43.66	7.2 W	5	4	7	8	7	3	
58	129601	JAC	BLACKMAN	SANDSTONE	78.7%	53.20	41.89	14.4 W	13	14	8	4	11	7	
59	42002	BCK	BEADLE	SPAULDING	75.6%	47.05	35.56	4.8 W	11	7	6	15	12	7	
60	48001	BNC	PELLSTON	BURT LAKE	74.9%	73.81	55.25	13.2 D	24	14	8	20	23	16	
61	150001	WBR	WITHEY LAKE	PETTIT	74.6%	100.48	74.91	14.4 W	23	9	17	19	15	13	
62	82904	GRA	MULLINS	ROYAL VISTA	69.5%	34.94	24.27	7.2 W	2	8	2	1	0	2	
63	80901	BCK	FIFTEEN MILE RD	A DRIVE	67.7%	35.40	23.96	14.4 W	4	1	1	4	0	8	
64	106503	FLT	FOURTEENTH ST	LIBERTY ST	67.3%	7.95	5.35	4.8 W	2	0	0	0	0	2	
65	129401	JAC	BROUGHWELL	MINARD	62.3%	179.04	111.51	14.4 W	27	23	16	20	29	39	
66	154302	GVL	HARVARD LAKE	COURTLAND	61.2%	47.27	28.91	14.4 W	6	4	1	5	3	6	
67	35002	MUS	FRUITPORT	JUDSON ROAD	60.9%	38.71	23.56	7.2 W	5	4	14	6	4	10	
68	39702	KAL	GULL LAKE	WILLOW BEACH	60.0%	18.05	10.83	4.8 W	2	9	4	9	3	7	
69	151601	TWS	HUBBARD LAKE	HUBBARD LAKE	56.0%	15.41	8.63	14.4 W	4	0	4	2	2	4	
70	100502	OWS	CHAPIN	MARION	55.6%	108.47	60.34	4.8 W	21	10	6	9	6	2	
71	33803	CLR	HARRISON	STOCKWELL	54.5%	34.82	18.97	4.8 W	3	2	5	6	8	4	
72	75901	JAC	REYNOLDS	REYNOLDS	52.5%	47.74	25.06	14.4 W	10	8	9	10	4	13	
73	57301	BIG	OHMAN ROAD	EVART	51.0%	39.95	20.38	14.4 W	4	7	3	4	10	11	
74	7402	ADR	HUDSON	CITY	51.0%	48.38	24.67	4.8 W	5	9	3	7	5	3	
75	Totals					2,480.54	2,041.42		378	284	223	264	301	379	
76	Percent Improvement									24.9%	41.0%	30.2%	20.4%	-0.3%	
77									1 Yr Prior	Yr Trim	1 Yr Post	2 Yr Post	3 Yr Post	4 Yr Post	
78	Totals					Only 100% cleared Circuits				96	43	22	33	41	62
79	Percent Improvement									55.2%	77.1%	65.6%	57.3%	35.4%	
80									1 Yr Prior	Yr Trim	1 Yr Post	2 Yr Post	3 Yr Post	4 Yr Post	

Note 1 - The percent cleared is the percent of miles cleared for the year listed versus the circuit configuration miles in 2020. Circuits cleared less than 50% not shown.

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Line Clearing Reliability Results

Reliability Pre and Post Clearing for 2018, 2019, 2020 and 2021 Cleared LVD Circuits

(minimum 50% of circuit miles cleared for year listed, primary voltage outage incidents)

Case No.: U-21389

Exhibit No.: A-45 (PLB-4)

Page: 10 of 10

Witness: PLBorden

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Line No.	Description													
1	Including MEDs													
2	Circuits Cleared in 2018													
3	FEEDER ID	WHQ	Substation	Circuit	% Cleared	Circuit Miles	Comp Miles	Voltage (kV)	2017	2018	2019	2020	2021	2022
4	24203	CAD	MCBAIN	LUCAS	100.0%	11.88	11.88	14.4 D	17	10	2	0	1	0
5	39202	JAC	MORRELL	WEST AVENUE	100.0%	11.02	11.02	4.8 W	4	3	0	2	1	3
6	155801	SMN	SCHOOL ROAD	MOROCCO	100.0%	48.56	48.56	7.2 W	8	7	1	3	17	3
7	50302	GRN	NORTH PARK	COIT	100.0%	5.43	5.43	7.2 W	1	0	0	0	0	0
8	702	MUS	MUSKEGON HEIGHTS	LEAHY	100.0%	2.88	2.88	7.2 W	1	0	1	0	0	0
9	101801	KAL	COLONY FARM	BTR PARK	100.0%	3.94	3.94	14.4 W	2	0	0	0	0	0
10	148302	CLR	MANN SIDING	MANN SIDING	100.0%	60.62	60.62	14.4 W	5	8	6	14	9	7
11	19203	ADR	RIGA	BIERMAN	100.0%	39.56	39.56	7.2 W	1	0	0	0	0	0
12	98203	GRA	LEFFINGWELL	RAPIDSTAN	100.0%	5.66	5.66	7.2 W	1	1	1	0	0	2
13	93901	WBR	LYON MANOR	TREASURE	100.0%	17.40	17.40	4.8 W	13	6	0	4	6	6
14	32501	GRA	BOSTON SQUARE	NELAND	100.0%	8.92	8.92	7.2 W	1	1	0	0	0	1
15	704	MUS	MUSKEGON HEIGHTS	HEIGHTS	100.0%	11.07	11.07	7.2 W	3	1	1	0	2	1
16	81503	LAN	HARPER ROAD	ONONDAGA	100.0%	51.21	51.21	14.4 W	14	9	6	4	7	11
17	158201	WBR	RYNO	MORENCI	100.0%	70.77	70.77	14.4 W	40	8	2	17	8	5
18	67504	MUS	MCCRACKEN	SHERMAN	100.0%	7.82	7.82	7.2 W	1	0	1	1	1	1
19	42601	ADR	PITTSFORD	CHURCH ROAD	100.0%	109.83	109.83	14.4 W	18	12	9	6	8	15
20	53001	FRE	WHITE CLOUD	WILLIAM STREET	100.0%	24.07	24.07	4.8 W	4	4	4	1	1	4
21	157004	FLT	MILBOURNE	DARTMOUTH	100.0%	10.14	10.14	4.8 W	1	1	0	1	1	2
22	50502	SAG	BRISTOL	MILLER	100.0%	5.76	5.76	4.8 W	0	0	1	2	0	1
23	53801	FLT	COURT	WOODLAWN	100.0%	8.40	8.40	4.8 W	0	0	0	1	0	0
24	128401	KAL	MILLERS POINT	MEADOW VIEW	99.3%	10.93	10.85	4.8 W	0	0	6	0	2	2
25	10202	LAN	GRAND LEDGE	HARTEL ROAD	98.9%	13.27	13.12	4.8 W	0	2	0	0	1	0
26	107601	ALM	ISABELLA	PICKARD	98.7%	14.99	14.79	4.8 W	0	0	0	0	0	0
27	50501	SAG	BRISTOL	BRISTOL	98.2%	6.27	6.15	4.8 W	0	1	0	1	1	0
28	29303	WBR	MARKEY	FOREST ESTATES	98.1%	21.04	20.63	4.8 W	5	22	2	3	4	6
29	21303	JAC	SUMMIT	FOURTH STREET	97.8%	11.59	11.33	4.8 W	1	0	1	1	2	0
30	39303	SAG	NIAGARA	NIAGARA	97.2%	9.09	8.83	4.8 W	2	0	0	3	1	0
31	63403	KAL	KILGORE	MOUNT EVEREST	97.1%	8.36	8.12	4.8 W	2	2	0	0	1	0
32	155401	SMN	DUNBAR	DUNBAR	97.0%	17.16	16.65	7.2 W	0	1	0	2	0	0
33	37601	JAC	BATTESEE	COON HILL	97.0%	46.84	45.43	14.4 W	10	13	4	10	5	26
34	57002	CLD	COLON	PALMER	96.2%	56.69	54.55	4.8 W	3	4	4	7	24	6
35	26901	HML	OTSEGO	OTSEGO	96.2%	7.52	7.23	4.8 W	1	0	0	1	0	3
36	140202	WBR	SIMMONS	CABIN LAKE	96.0%	14.16	13.59	14.4 W	0	1	2	2	3	1
37	127701	MDL	LEVELY	ALLBRIGHT	95.6%	87.79	83.95	14.4 W	7	18	8	6	28	15
38	149002	LAN	PACKARD	PACKARD	94.2%	18.66	17.58	14.4 W	6	0	2	1	2	1
39	24301	KAL	AUSTIN	WEST LAKE	94.1%	29.89	28.13	4.8 W	7	4	9	5	1	6
40	33802	CLR	HARRISON	HARRISON	94.0%	45.46	42.74	4.8 W	4	2	6	13	6	6
41	48601	BCK	MORGAN	ORCHARD	93.6%	27.33	25.59	4.8 W	3	3	5	6	16	3
42	132304	MDL	ORCHARD ROAD	ST ANDREWS	92.7%	16.05	14.89	4.8 W	0	2	2	5	2	0
43	60701	MUS	GETTY	MARQUETTE	92.3%	26.56	24.53	7.2 W	2	1	3	2	5	0
44	67404	LAN	DEXTER TRAIL	DANSVILLE	89.0%	46.24	41.16	14.4 W	7	6	3	3	4	5
45	50101	KAL	EASTWOOD	TEXEL	88.7%	12.21	10.83	4.8 W	1	4	3	1	3	6
46	104201	GRN	ROGUE RIVER	CANNON FARMS	88.7%	58.08	51.51	14.4 W	20	11	11	5	7	8
47	71702	JAC	SPRING ARBOR	ARBOR HILLS	88.5%	52.74	46.69	4.8 W	8	6	9	3	15	6
48	27502	KAL	OAKWOOD	PARKVIEW	88.3%	10.14	8.95	4.8 W	5	3	4	6	3	5
49	22202	GRN	COWAN LAKE	RAMSDALL	87.0%	64.19	55.82	14.4 W	17	25	22	19	20	18
50	156202	BCK	CRANBROOK	11 MILE ROAD	85.5%	23.18	19.82	14.4 W	3	0	2	2	5	3
51	32002	KAL	STADIUM	COLLEGE	84.3%	6.14	5.18	4.8 W	0	0	1	3	5	0
52	32503	GRA	BOSTON SQUARE	HALL	83.3%	10.45	8.71	7.2 W	2	3	2	0	2	1
53	57001	CLD	COLON	COLON	83.1%	61.58	51.19	4.8 W	11	2	12	2	33	9
54	6802	BEN	HOMESTEAD	BEULAH	82.7%	83.69	69.19	7.2 W	19	22	15	18	32	21
55	147602	LAN	VAN ATTA	VAN ATTA	81.3%	54.62	44.42	4.8 W	5	1	0	3	6	2
56	70801	GRA	MICHIGAN	LYDIA	81.2%	8.15	6.62	7.2 W	1	0	1	0	0	1
57	161902	GRE	EMERSON	KILMER	81.0%	53.87	43.66	7.2 W	6	7	12	9	12	3
58	129601	JAC	BLACKMAN	SANDSTONE	78.7%	53.20	41.89	14.4 W	20	15	10	5	14	7
59	42002	BCK	BEADLE	SPAULDING	75.6%	47.05	35.56	4.8 W	12	7	10	22	19	7
60	48001	BNC	PELLSTON	BURT LAKE	74.9%	73.81	55.25	13.2 D	37	21	9	27	32	16
61	150001	WBR	WITHEY LAKE	PETTIT	74.6%	100.48	74.91	14.4 W	38	17	20	27	26	13
62	82904	GRA	MULLINS	ROYAL VISTA	69.5%	34.94	24.27	7.2 W	5	9	4	3	3	2
63	80901	BCK	FIFTEEN MILE RD	A DRIVE	67.7%	35.40	23.96	14.4 W	6	1	2	5	1	8
64	106503	FLT	FOURTEENTH ST	LIBERTY ST	67.3%	7.95	5.35	4.8 W	2	0	0	0	0	2
65	129401	JAC	BROUGHWELL	MINARD	62.3%	179.04	111.51	14.4 W	48	30	25	26	47	39
66	154302	GVL	HARVARD LAKE	COURTLAND	61.2%	47.27	28.91	14.4 W	8	7	6	9	10	6
67	35002	MUS	FRUITPORT	JUDSON ROAD	60.9%	38.71	23.56	7.2 W	5	6	22	7	8	10
68	39702	KAL	GULL LAKE	WILLOW BEACH	60.0%	18.05	10.83	4.8 W	5	12	5	12	6	7
69	151601	TWS	HUBBARD LAKE	HUBBARD LAKE	56.0%	15.41	8.63	14.4 W	5	1	4	3	3	4
70	100502	OWS	CHAPIN	MARION	55.6%	108.47	60.34	4.8 W	27	13	9	15	7	2
71	33803	CLR	HARRISON	STOCKWELL	54.5%	34.82	18.97	4.8 W	7	3	6	10	22	4
72	75901	JAC	REYNOLDS	REYNOLDS	52.5%	47.74	25.06	14.4 W	16	9	9	12	10	13
73	57301	BIG	OHMAN ROAD	EVART	51.0%	39.95	20.38	14.4 W	7	10	9	7	20	11
74	7402	ADR	HUDSON	CITY	51.0%	48.38	24.67	4.8 W	6	9	4	7	10	3
75	Totals					2480.54	2041.42		547	407	340	395	551	379
76	Percent Improvement									25.6%	37.8%	27.8%	-0.7%	30.7%
77									1 Yr Prior	Yr Trim	1 Yr Post	2 Yr Post	3 Yr Post	4 Yr Post
78	Totals				Only 100% cleared Circuits				135	71	35	56	62	62
79	Percent Improvement									47.4%	74.1%	58.5%	54.1%	54.1%
80									1 Yr Prior	Yr Trim	1 Yr Post	2 Yr Post	3 Yr Post	4 Yr Post

Note 1 - The percent cleared is the percent of miles cleared for the year listed versus the circuit configuration miles in 2020. Circuits cleared less than 50% not shown.

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Justification of 7-Year Cycle v Other Cycles

Case No.: U-21389
Exhibit No.: A-46 (PLB-5)
Page: 1 of 1
Witness: PLBorden
Date: May 2023

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
1		4.8/8.32 kV System				7.2/12.47 kV System				14.4/24.9 kV System			
	Years Since Last Cleared / Cycle Years	Incidents per Mile per Year	Incidents Per Year for Cycle Years	Contractor Cycle \$ Per Incident	Contractor \$ Annual LVD Cycle Cost	7.2 kV Incidents / Mile / Year	Incidents Per Year	Contractor Cycle \$ Per Incident	Contractor \$ Annual LVD Cycle Cost	14.4 kV Incidents / Mile / Year	Incidents Per Year	Contractor Cycle \$ Per Incident	Contractor \$ Annual LVD Cycle Cost
2	1	0.1951				0.2450				0.2445			
3	2	0.1999	6327	\$15,881	\$100,471,860	0.2611	2627	\$12,394	\$32,555,969	0.2593	4509	\$12,451	\$56,141,796
4	3	0.2025	6380	\$10,498	\$66,981,240	0.2720	2692	\$8,062	\$21,703,979	0.2693	4613	\$8,114	\$37,427,864
5	4	0.2035	6415	\$7,831	\$50,235,930	0.2795	2744	\$5,931	\$16,277,985	0.2757	4693	\$5,981	\$28,070,898
6	5	0.2035	6436	\$7,879	\$50,707,740	0.2854	2788	\$5,893	\$16,430,865	0.2797	4756	\$5,958	\$28,334,537
7	6	0.2031	6447	\$7,652	\$49,332,689	0.2915	2828	\$5,653	\$15,985,307	0.2825	4806	\$5,736	\$27,566,184
8	7	0.2029	6455	\$7,290	\$47,058,046	0.2996	2868	\$5,317	\$15,248,252	0.2853	4849	\$5,423	\$26,295,156
9	8	0.2035	6463	\$6,863	\$44,357,445	0.3115	2914	\$4,933	\$14,373,175	0.2893	4890	\$5,068	\$24,786,110
10	9	0.2055	6476	\$6,407	\$41,494,015	0.3290	2969	\$4,528	\$13,445,336	0.2957	4935	\$4,698	\$23,186,080
11	10	0.2095	6500	\$5,943	\$38,625,636	0.3539	3040	\$4,117	\$12,515,893	0.3057	4989	\$4,326	\$21,583,283
12	11	0.2161	6538	\$5,484	\$35,852,775	0.3880	3130	\$3,712	\$11,617,400	0.3205	5057	\$3,962	\$20,033,860
13	12	0.2259	6596	\$5,040	\$33,242,410	0.4331	3243	\$3,321	\$10,771,562	0.3413	5144	\$3,611	\$18,575,237
14	13	0.2395	6679	\$4,589	\$30,840,911	0.4910	3386	\$2,934	\$9,993,403	0.3693	5257	\$3,258	\$17,233,324
15	14	0.2575	6791	\$4,176	\$28,681,400	0.5635	3562	\$2,581	\$9,293,655	0.4057	5400	\$2,934	\$16,026,630
16	15	0.2805	6937	\$3,801	\$26,788,166	0.6524	3776	\$2,265	\$8,680,189	0.4517	5579	\$2,641	\$14,968,726

	Cost per Incident Reduced	Expected Incidents	Incident Reduction from 2019 Total	Contractor Full Circuit Dollars	Cycle 14.4/7.2/4.8	Years Since Last Cleared	2021 Contractor \$/Mile
18							
19	\$14,420.36	14100	5900	\$85,076,804	5/7/9	4	\$6,273
20	\$15,308.04	14079	5921	\$90,640,835	5/7/7	5	\$7,915
21	\$15,858.61	13980	6020	\$95,473,142	5/5/5	6	\$9,240
22	\$15,301.23	13999	6001	\$91,823,448	5/5/7	7	\$10,283
23	\$14,425.39	14020	5980	\$86,259,417	5/5/9	8	\$11,078
24	\$14,446.59	14060	5940	\$85,813,858	5/6/9	9	\$11,658
25	\$15,328.17	14039	5961	\$91,377,889	5/6/7	10	\$12,058
26						11	\$12,311
27	14.4 kV needs 5 year cycle - too many outages/complaints when longer					12	\$12,453
28	7.2 kV vary between 5-7 years for analysis					13	\$12,516
29	4.8 kV vary between 7 and 9 years for analysis					14	\$12,535
30						15	\$12,544
31						16	\$12,576
32						17	\$12,667
33						18	\$12,850

kV System Miles		
4.8	32034	3559
7.2	10380	1483
14.4	17900	3580
Total	60314	8622



2021 Cost per Mile for Years Since Last Cleared					
Years Since Last Cleared	Maintenance	Repetitive Outage	First Zone	CEMI	
3	\$4,280	\$14,845	\$5,278	\$16,559	
4	\$6,273	\$15,080	\$4,922	\$16,559	
5	\$7,915	\$15,354	\$4,669	\$16,559	
6	\$9,240	\$15,666	\$4,520	\$16,559	
7	\$10,283	\$16,017	\$4,475	\$16,559	
8	\$11,078	\$16,407	\$4,534	\$16,559	
9	\$11,658	\$16,835	\$4,697	\$16,559	
10	\$12,058	\$17,302	\$4,963	\$16,559	
11	\$12,311	\$17,808	\$5,333	\$16,559	
12	\$12,453	\$18,353	\$5,807	\$16,559	
13	\$12,516	\$18,936	\$6,385	\$16,559	
14	\$12,535	\$19,557	\$7,066	\$16,559	
15	\$12,544	\$20,218	\$7,851	\$16,559	
16	\$12,576	\$20,917	\$8,740	\$16,559	
17	\$12,667	\$21,654	\$9,733	\$16,559	
18	\$12,850	\$22,431	\$10,830	\$16,559	
19	\$13,158	\$23,246	\$12,030	\$16,559	
20	\$13,627	\$24,099	\$13,334	\$16,559	
21	\$14,290	\$24,992	\$14,742	\$16,559	
22	\$15,180	\$25,923	\$16,254	\$16,559	
23	\$16,333	\$26,892	\$17,869	\$16,559	
24	\$17,783	\$27,900	\$19,589	\$16,559	
25	\$19,562	\$28,947	\$21,412	\$16,559	
26	\$21,706	\$30,033	\$23,339	\$16,559	
27	\$24,248	\$31,157	\$25,369	\$16,559	
28	\$27,222	\$32,320	\$27,504	\$16,559	
29	\$30,662	\$33,521	\$29,742	\$16,559	
30+	\$34,603	\$34,762	\$32,084	\$16,559	

CEMI work not correlated to YSLT, \$16,559 is 2021 CEMI average

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Summary of Actual & Projected O&M Expenses
 Forestry Operations
 (\$000)

Case No.: U-21389
 Exhibit No.: A-47 (PLB-6)
 Page: 1 of 2
 Witness: PLBolden
 Date: May 2023

	(a)	(b)	(c)
Line No.	Description	2022 Actual	Projected 12-Mos Ending 2/28/2025
	Forestry Operations Administration and Work		
1	Planning	6,251	8,110
	Labor	3,774	4,666
	Material	0	0
	Contractor	1,476	2,267
	Non-Labor Overheads	1,000	1,177
	Non-Labor Other	0	
2	Line Clearing Contractor Costs	95,753	110,779
	Labor	0	0
	Material	0	0
	Contractor	95,753	110,779
	Non-Labor Overheads	0	0
	Non-Labor Other	0	0
3	Total "Description of Area" O&M Expenses	\$ 102,003	\$ 118,889
	Labor	3,774	4,666
	Material	0	0
	Contractor	97,229	113,046
	Non-Labor Overheads	1,000	1,177
	Non-Labor Other	0	0

Executive Summary

Report Name: Forestry Management Process Review
Report Number: 2018 - 14
Assessed Risk: Risk Based
Business Owner: Chris Niemi, Manager of Forestry Operations

Area of Focus

The scope was designed to perform an assessment of the Forestry Management Program. More specifically, the audit focused on evaluating the following:

- Forestry governance, processes, procedures, policies and controls;
- Roles, responsibilities, resources, and maturity of the organization; and
- Forestry planning, work management and other management monitoring activities.

2018 Catastrophic California Wildfires and Risk Considerations for CMS

In 2018, California experienced its most deadly and destructive wildfire season, burning nearly 2 million acres of land, causing billions of dollars in damages. A common risk that power and utility companies face is the potential to be held liable for damages and claims caused by wildfires if both of the following factors are present: (a) the utility's equipment caused the fire and (b) the utility is found to have acted negligently based on inappropriate behavior and/or inadequate forestry management. California utilities face a unique and significant risk as they can be held liable under the state law of "reverse condemnation" for the damages caused by wildfires started by their equipment even if they are not found to have acted negligently. The catastrophic impact of this risk recently presented itself with the news that PG&E, California's largest utility, filed for bankruptcy resulting from the billions of dollars in estimated potential liability resulting from the 2017 and 2018 wildfires. Multiple public sources (e.g. NY Times, Fortune, others) have cited that PG&E's bankruptcy filing includes over \$30 billion in estimated liabilities related to the wildfires.

More specifically, PGE is being investigated to determine first if their equipment caused the ignition of the fire in their service territory and second to further determine if they were negligent in the operation and maintenance of the associated equipment. Based on publicly available information, it is expected that if their equipment is determined to have caused the fire, they may be liable for the entire cost of the damage caused regardless if they were negligent. If they are also found to have been negligent in the operation of their equipment, additional civil and criminal penalties may apply.

What does this mean for CMS? IA identified the following key factors that reduce this type of significant risk for the Company:

- Per discussions with CE management, they believe California law exposes the California utilities to significantly more exposure from wildfire liability than Michigan law since Michigan law requires negligence to be determined in the operation and maintenance of utility equipment for liability to be assigned;
- Management is currently monitoring and helping address this risk through effective forestry management and line clearing practices as well as proactive monitoring of potential risks that could present themselves from climate changes;
- The fire risk season in Michigan is considered to be significantly shorter and less risky than the western states due to Michigan's climate. Based on statistics tracked and published by government agencies (Congressional Research Service, Michigan DNR), on average, wildfires in Michigan over the past 10 years represent less than 0.01% of wildfires across the US and are typically contained to only a few acres or less at a time; and
- Wildfires are more widely and rapidly accessible for first responders to prevent from spreading as compared to states with large forests in mountainous regions.

Positive Highlights

IA identified multiple positive highlights, including:

- A strong Forestry leadership team with substantial knowledge and experience in technical and field operations and a demonstrated desire to deliver solid forestry performance;
- A Forestry Management planning process that includes refined modeling and data analytics, which represents better industry practice and planning maturity ahead of industry peers;
- Detailed and timely measuring and reporting of third-party Forestry Management contractor performance, including strong quarterly contractor quality performance scoring process, which includes areas such as safety performance and on-call/storm and customer relations responsiveness; and
- Strong use of data visualization and strong participation in the Reliability Rally Room.

Executive Summary

Process Overview and Key Gaps Identified

The forestry process is complex, requiring assessment and monitoring of multiple factors that impact the identification, planning, scheduling and execution of program activities, that include, but are not limited to:

- Operational safety;
- Compliance to state and federal requirements and guidelines;
- Availability and timing of Operations and Maintenance (O&M) budget;
- Other company programs and investment activities requiring support from Forestry resources; and
- Monitoring contractor commercial terms and conditions; and Contractor resource constraints due to nation-wide seasonal demand and unpredictable events, such as regulatory changes and natural disasters.

Management is aware of the risks that each of the aforementioned areas can present related to forestry management. However, IA believes that management has an opportunity to increase the effectiveness and efficiency of the program through the following:

- Evaluating and outlining what risks and impacts to forestry management are controllable, semi-controllable, and non-controllable;
- Performing additional qualitative and/or quantitative risk assessments to understand the magnitude and direction of program risks; and
- Moving to a more proactive approach that is risk-informed, condition based and flexible in order to carry out forestry management activities at opportune times based on cost, risk and/or program objectives.

The Company plans to spend approximately \$285 million from 2018 through 2022 for the Forestry Management Program. Development of the risk management program capabilities will likely enhance managements' ability to deliver on program objectives and operate in a safe manner that reduces additional cost and time that could have been avoided.

Summary of Findings

IA identified one finding that does not represent a significant risk to ongoing operations, where enhancements can be made to improve program maturity and improve resource efficiency and effectiveness.

Summary of Findings			Mgmt. Response
#	Description	Priority Rating*	Target Completion Date
1.	Enhancements to Forestry Management Planning and Monitoring Managements' statistical model to provide the basis for planning, selecting, and prioritizing the annual forestry management plan is better industry practice. IA identified opportunities for management to enhance program planning and monitoring by incorporating additional risk considerations, which may allow management to take a more proactive approach to forestry management.	Green	4/30/19

*See Appendix B for definition of Priority Rating and Sarbanes Key Control.

Note 1: None of the findings are Sarbanes Key Controls.

Note 2: Management agrees with all of the findings identified.

Internal Audit Report CC Listing:

PKPoppee	BFRich	DMBaughman
RPHayes	GJRochow	MAMacdonald
JFBrossoit	GPBarba	CAShellberg
CAHendrian	GCPackard	CDuVall
BJHofmeister	MMGleespen	JMOConnor
DVRao	SMJohnson	Audit Committee
CMReynolds	LEYoungdahl	PwC

Executive Summary

Table of Contents

<i>Detailed Findings, Recommendations and Management Action Plans.....</i>	<i>4</i>
<i>Appendix A – Background and Scope Details</i>	<i>6</i>
<i>Appendix B – Definition of Red, Yellow, Green and Sarbanes Key Control.....</i>	<i>8</i>

Detailed Findings, Recommendations, and Management Action Plans

Finding 1: Enhancements to Forestry Management Planning and Monitoring	Sarbanes Portal Key Control or Risk Area: No
Action Plan Priority: Green	If Sarbanes Yes, Provide Reference: N/A
Description of Issue The Forestry Management team currently uses a statistical model to provide the basis for planning, selecting, and prioritizing circuits to be included in the annual plan. Management prioritizes the selected zones (i.e. segments of each circuit) for clearing based on current reliability performance with a goal to maximize the reliability benefit at the lowest cost. However, IA identified opportunities for management to enhance program planning and monitoring by incorporating risk considerations. For example, CMS Forestry management is cognizant of risk factors that could impact the achievability of program objectives, but does not perform and document a systematic risk assessment of the likelihood and magnitude of expected impacts on Forestry Management Program activities. Forestry Management Program risk management is complex, requiring assessment and developing mitigating responses across multiple factors. Better industry practice includes use of: • A risk framework to guide assessing the controllable, semi-controllable, and uncontrollable risks across functional activities such as Work Identification, Planning, Scheduling, Execution, and performance monitoring and reporting. The framework model should include appropriate figures-of-merit to estimate the impact of the risk on areas including but not limited to cost, productivity, and customer satisfaction; and • In-depth qualitative and quantitative assessment of risk across safety, regulatory, cost, customer and other key operational considerations and constraints. Risk topics may include, but not be limited to: 1. Operational safety; 2. Compliance to state and federal requirements and guidelines; 3. Available and timing of Operations and Maintenance (O&M) budget; 4. Other company programs and investment activities requiring support from Forestry resources, such as: a. HVD and LVD capital investment activities; and b. Storm restoration efforts; 5. Monitoring contractor commercial terms and conditions; and 6. Contractor resource constraints due to nation-wide seasonal demand and unpredictable events such as regulatory changes and natural disasters. Risk to CMS Energy Application of a robust and systematic assessment of risk may allow management better visibility and preparedness to anticipate potential regulatory, customer, financial and operational impacts to the efficiency and effectiveness of the forestry management program. Recommendation Management should consider the following: 1. Create a forestry risk management multi-dimensional framework where topics are assessed against: a. Risk Category: Risk categories might include inherently <u>Controllable Risk</u> (e.g., capable of being directly engaged or influenced); <u>Semi-Controllable Risk</u> (e.g., capable of being indirectly influenced); and <u>Uncontrollable Risk</u> (e.g., the risk response must be reactive and not directly or indirectly manageable). b. An appropriate functional range of risk topics such as Work Identification, Planning, Scheduling, Execution and Performance Monitoring and Reporting.	

Detailed Findings, Recommendations, and Management Action Plans

2. Develop one or more figures-of-merit to define appropriate dependent variables to assess risk across each topic area.
3. For each framework element, qualitatively or quantitatively assess the magnitude and direction of the uncertainty in the risk assessment. Elements may include, but not be limited to:
 - a. Operational safety;
 - b. Compliance to state and federal requirements and guidelines;
 - c. Available and timing of Operations and Maintenance (O&M) budget;
 - d. Other company programs and investment activities requiring support from Forestry resources, such as:
 - i. HVD and LVD capital investment activities; and
 - ii. Storm restoration efforts;
 - e. Monitoring contractor commercial terms and conditions; and
 - f. Contractor resource constraints due to nation-wide seasonal demand and unpredictable events such as regulatory changes and natural disasters;
4. To help perform the data analytics, Forestry may find it useful to utilize risk management assessment tools such as but not limited to the Microsoft Excel Add-in Risk Detective (see <http://www.riskdetective.com/>). Some minor training in the selected risk management assessment tool may be necessary;
5. Forestry should perform and update the risk assessment annually, or as needed based on changing field conditions, regulations, budget, or other areas determined by management that may impact the risk environment.

Management Action Plan:

Responsible Department Personnel:

MAP Owner: Christopher J. Niemi, Manager of Forestry Operations

MAP Support Role: John M. O'Connor, Senior Technical Analyst Lead

Management Action Plan:

Management agrees with IA's findings and will perform the following:

1. Create a forestry risk management multi-dimensional framework where topics are assessed against: Risk Category: Risk categories might include inherently Controllable Risk (e.g., capable of being directly engaged or influenced); Semi-Controllable Risk (e.g., capable of being indirectly influenced); and Uncontrollable Risk (e.g., the risk response must be reactive and not directly or indirectly manageable) that possibly affect Work Identification, Planning, Scheduling, Execution and Performance Monitoring and Reporting;
2. Define appropriate dependent variables to assess risk across each topic area;
3. For each framework element, qualitatively or quantitatively assess the magnitude and direction of the uncertainty in the risk assessment framework. Elements may include, but not be limited to:
 - a. Operational safety;
 - b. Compliance to state and federal requirements;
 - c. Operations and Maintenance (O&M) budget;
 - d. HVD and LVD capital work and storm restoration activities requiring support from Forestry resources;
 - e. Contractor resource constraints;
4. Utilize Microsoft Excel Add-in Risk Detective risk management assessment tool or similar tool with IT approval;
5. Perform and update the risk assessment annually by January 31st, or as needed based on changing field conditions, regulations, budget, or other areas determined by management that may impact the risk environment throughout the year.

Target Date: 4/30/19

Appendix A – Background and Scope Details

Background:

Consumers Energy's (CE) distributes electricity to approximately 1.8 million electric customers in Michigan. CE owns and operates 60,931 miles of transmission and distribution overhead electric lines as detailed in the table below.

Electric Overhead Lines	Miles
138 kilovolts transmission	214
138 kilovolts high-voltage distribution	188
46 kilovolts and 69 kilovolts high-voltage distribution	4,431
Low-voltage electric distribution	56,098
Total	60,931

CE's Vegetation Management Program helps ensure that trees and other vegetation do not interfere with the safe and reliable delivery of electricity. CE Foresters have either a bachelor or Master of Science degree in Forestry (or related field of study) or are an arborist certification through the International Society of Arboriculture. Additionally, CE utilizes contractors who employ qualified line-clearance arborists to maintain minimum clearances following industry-accepted standards necessary to minimize power outages.

CE utilizes a statistical model to provide the basis for planning, selecting, and prioritizing circuits to be included in the annual Vegetation Management plan. The statistical model output is based on circuit voltage, historic reliability performance and other factors, such as tree density and years since the circuit was last cleared of trees and brush. Management prioritizes the selected zones (i.e. segments of each circuit) for clearing based on current reliability performance, with a goal to maximize the reliability benefit at the lowest cost. For emergent events, demand trim work is conducted. While the annual trimming is focused within the legal right-of-way, trees located outside the easement area may be marked for removal if they are dead or dying and pose a significant risk to the reliable operation of the system.

Risk Overview:

Vegetation management program, policies, procedures, standards and training may be incomplete or not be in place, resulting in inconsistent or incomplete line clearance and other vegetation management operating practices, process weaknesses and performance deficiencies, which could result in operational, financial, regulatory and reputational impacts.

Audit Objectives:

The objectives of this audit were to evaluate the design and operating effectiveness of the overall Vegetation Management Program governance, processes, business practices, standards and controls to help ensure that the program is in line with the Company's expectations, regulatory standards and general business practices.

Audit Scope:

The audit scope focused on the following areas associated with Vegetation Management:

- Program governance, controls, policies, procedures and standards;
- Roles, responsibilities, resources, and maturity of the organization to support the Vegetation Management Program and provide tactical support to the organization in meeting the program requirements;
- Process related to monitoring of key contractual terms and conditions with third-party vendors, including timeliness of tree trimming activities, vendor performance, and the accuracy and completeness of invoices paid by CE for forestry / vegetation management activities;
- Key performance indicators, visual management and other tools used to monitor progress and the overall Program; and
- Processes to manage any special record keeping and data management requirements.

The audit scope also included field observations of vegetation management activities.

Out of Scope:

Storm related vegetation management activities for events of greater than 3% of CE customers (or which do not include local and out-of-state foreign /mutual assistance) resources were considered out of scope.

Appendix B – Definition of Red, Yellow, Green and Sarbanes Key Control

Period covered:

The audit focused on January 2014 through August 2018 to obtain enough data and information to assess and establish observations, trends and findings.

Procedures:

Internal Audit (IA) performed the following procedures:

1. Conducted an opening meeting with key process owners;
2. Performed a walkthrough and evaluated the processes and controls in place for the aforementioned areas in the audit scope section;
3. Obtained the documents, electronic files, and other records required to perform the review and evaluation;
4. Identified potential areas of risk and evaluated the mitigation activities (if available) for reasonableness and verified that the mitigations were being performed timely;
5. Prepared and validated observations and identified gaps;
6. Identified additional areas for more detailed analysis and testing;
7. Conducted follow-up interviews and obtained supplemental documentation, as needed;
8. Conducted weekly engagement status updates with key stakeholders and shared preliminary observations, report progress and communicated any additional needs;
9. Conducted a closing meeting with key stakeholders and shared observations;
10. Provided a draft audit report with findings, priority color coding and recommendations to management for review;
11. Obtained Management Action Plans (MAPs) from the process owner; and
12. Issued the final report.

Red

Represents a significant area of risk that requires immediate action and escalation to senior management and the Audit Committee. The condition(s) requires improvement with more than usual management involvement and monitoring until the risk is mitigated. The management action plan to remediate the risks identified would be considered a “high” priority item.

Yellow

Represents a moderate area of risk that requires management action. Management is required to provide countermeasures to remediate the risk identified and improve the overall control environment. The management action plan to remediate the risks identified would be considered a “medium” priority item.

Green

Represents a low area of risk that is unlikely to result in moderate or significant impacts to the Company. Management is required to provide countermeasures to remediate the risk identified OR accept the risk with the approval of senior management. The management action plan to remediate the risks identified would be considered a “low” priority item.

Sarbanes

Findings that relate to a SOX key control or risk area as determined by the Executive Director Internal Control and Compliance, will be identified by “yes” in the report.

Page: 1 of 5

Witness: PLBolden

Date: May 2023

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
	Description																
												</					

MICHIGAN PUBLIC SERVICE COMMISSION

Case No.: U-21389

Exhibit No.: A-49 (PLB-8)

Consumers Energy Company

Page: 4 of 5

2023 LVD Line Clearing Work Plan

Witness: PLBolden

Date: May 2023

191	74403 PINE RIVER	WRIGHT	ALM	4.8 W	1.39	1.39	158	0.00	35	17.71	0.000	0	\$0	113.7	0.00	1.39
192	74901 WURTSMITH	MISSION	TWS	14.4 W	30.10	30.10	970	143.00	12	61.57	0.498	2145	\$189	32.2	0.00	30.10
193	75902 REYNOLDS	SEARS	JAC	14.4 W	44.13	44.13	795	63.74	12	66.36	0.612	1721	\$438	18.0	0.00	44.13
194	76002 CHEYENNE	MCDONALDS	SAG	4.8 W	12.29	12.29	1320	190.00	11	49.50	0.081	190	\$873	107.4	0.00	12.29
195	76201 BEHNKE	ANGOLA ROAD	CLD	4.8 W	27.09	27.09	387	41.63	11	73.56	0.591	666	\$549	14.3	0.00	27.09
196	76602 JUDD ROAD	OTTAWA	FLT	4.8 W	9.20	9.20	400	66.88	35	36.22	0.870	535	\$232	43.5	0.00	9.20
197	76605 JUDD ROAD	CARMANWOOD	FLT	4.8 W	12.55	12.55	1636	328.75	30	50.99	0.319	1315	\$129	130.4	0.00	12.55
198	76901 SPRINKLE	BISHOP	KAL	4.8 W	0.31	0.31	2	0.00	35	9.87	0.000	0	\$0	6.5	0.00	0.31
199	77403 DEWEY	CONVENTION CENTER	GRA	7.2 W	2.95	2.95	791	0.00	24	21.60	0.000	0	\$0	268.1	0.00	2.95
200	77404 DEWEY	BRIDGEWATER	GRA	7.2 W	3.73	3.73	604	0.00	35	18.74	0.000	0	\$0	161.9	0.00	3.73
201	77604 KEATING	EAST BARNEY	MUS	7.2 W	9.49	9.49	889	81.67	7	34.48	0.316	245	\$523	93.7	9.49	0.00
202	78402 FOUNTAIN	DOWNTOWN	BCK	4.8 W	1.02	1.02	50	0.00	35	9.87	0.000	0	\$0	49.0	0.00	1.02
203	78405 FOUNTAIN	KELLOGG RESEARCH	BCK	4.8 W	0.02	0.02	2	0.00	35	9.87	0.000	0	\$0	100.0	0.00	0.02
204	79201 MIDDLEVILLE	BUSINESS	HST	4.8 W	51.94	51.94	1410	102.15	11	65.93	0.635	3371	\$208	27.1	0.00	51.94
205	79301 RAVINE	PATTERSON	KAL	4.8 W	14.71	14.71	1585	276.00	10	43.38	0.272	1104	\$180	107.7	0.00	14.71
206	79302 RAVINE	RAVINE	KAL	4.8 W	29.73	29.73	1655	94.60	13	63.89	0.841	2365	\$170	55.7	0.00	29.73
207	79602 LANDWER	VANWAGONER	MUS	7.2 W	4.82	4.82	155	10.00	33	19.19	0.207	10	\$6,736	32.2	0.00	4.82
208	81504 HARPER ROAD	AURELIUS	LAN	14.4 W	32.68	32.68	2349	352.00	19	37.45	0.245	2816	\$157	71.9	0.00	32.68
209	81802 ALPINE	WESTGATE	GRN	7.2 W	10.25	10.25	270	82.33	17	50.23	0.293	247	\$560	26.3	0.00	10.25
210	82402 DALE ROAD	BEAR CREEK	CLR	4.8 W	13.45	13.45	150	36.67	35	32.31	0.669	330	\$724	11.2	0.00	13.45
211	82902 MULLINS	ELMRIDGE	GRA	7.2 W	4.70	4.70	119	0.00	12	21.01	0.000	0	\$0	25.3	0.00	4.70
212	85702 COCHRAN	SNOW	LAN	14.4 W	28.04	28.04	1087	142.64	22	78.87	0.392	1569	\$241	38.8	0.00	28.04
213	86801 KOLASSA	KOSMERICK	CLD	14.4 W	107.85	61.18	1566	53.04		56.57				14.5	0.00	107.85 2022 Carryover
214	90001 DUPONT	E.I.DUPONT	MUS	11.0 W	0.01	0.01	0	0.00	35	13.16	0.000	0	\$0	0.0	0.00	0.01
215	90101 WAGER	FLINT PARK	FLT	4.8 W	11.48	11.48	1332	50.00	17	57.00	0.174	100	\$1,550	116.0	0.00	11.48
216	90301 LEE STREET	BLACK HILLS	GRA	7.2 W	2.89	2.89	85	0.00	34	18.35	0.000	0	\$0	29.4	0.00	2.89
217	90304 LEE STREET	KIRTLAND	GRA	7.2 W	12.91	12.91	3203	122.20	8	37.58	0.387	611	\$285	248.1	0.00	12.91
218	90402 KIESEL	EUCLED	BCY	4.8 W	14.21	14.21	1456	45.00	9	29.71	0.070	45	\$4,263	102.5	14.21	0.00
219	90403 KIESEL	HANDY	BCY	4.8 W	11.74	11.74	996	395.00	9	33.03	0.511	2370	\$67	84.8	11.74	0.00
220	90503 KENT AIRPORT	PATTERSON	GRE	7.2 W	3.07	3.07	51	0.00	34	18.35	0.000	0	\$0	16.6	0.00	3.07
221	90602 HARRIET	HARRIET	FLT	4.8 W	9.22	9.22	1151	16.00	35	54.55	0.325	48	\$2,593	124.8	0.00	9.22
222	90802 OAK STREET	WEST GANSON	JAC	4.8 W	8.51	8.51	1608	818.00	9	26.15	0.235	1636	\$70	189.0	8.51	0.00
223	91502 GRAND VALLEY	GRAND VALLEY	GRA	7.2 W	0.01	0.01	1	0.00	35	9.87	0.000	0	\$0	100.0	0.00	0.01
224	91603 AMPERSEE	NORTH COMMERCIAL	KAL	4.8 W	6.77	6.77	535	96.00	9	26.14	0.591	384	\$243	79.0	6.77	0.00
225	91702 CLAY	FREEZER	GRA	7.2 W	0.79	0.79	38	0.00	35	14.21	0.000	0	\$0	48.1	0.00	0.79
226	92301 BIG RAPIDS	FERRIS	BIG	7.2 D	0.84	0.84	1	0.00	11	11.12	0.000	0	\$0	1.2	0.00	0.84
227	92302 BIG RAPIDS	M-20	BIG	7.2 D	1.07	1.07	97	0.00	35	23.86	0.000	0	\$0	90.7	0.00	1.07
228	93802 RIX ROAD	UNDERPASS	KAL	4.8 W	53.64	53.64	2093	184.32	11	70.00	1.044	10322	\$75	39.0	0.00	53.64
229	93902 LYON MANOR	TOWN HALL	WBR	4.8 W	29.28	29.28	1510	266.31	15	61.06	0.888	6924	\$104	51.6	0.00	29.28
230	98204 LEFFINGWELL	NOTTINGHAM	GRA	7.2 W	11.72	4.60	691	108.13		37.27				59.0	0.00	11.72 2022 Carryover
231	99602 CENTER ROAD	EASTLAND	FLT	4.8 W	9.94	9.94	586	85.00	29	41.26	0.101	85	\$1,579	59.0	0.00	9.94
232	100201 SHAFFER	WOODLAND	GRE	7.2 W	0.86	0.86	6	0.00	35	15.55	0.000	0	\$0	7.0	0.00	0.86
233	100204 SHAFFER	EASTBROOK	GRE	7.2 W	2.57	2.57	35	0.00	21	14.57	0.000	0	\$0	13.6	0.00	2.57
234	100801 HARRIETTA	CABERFAE	CAD	14.4 D	10.40	10.40	199	80.80	3	16.83	0.962	808	\$173	19.1	10.40	0.00
235	100902 SURREY	MAIN STREET	CLR	14.4 W	37.75	37.75	1191	80.44	13	77.92	0.954	2896	\$215	31.5	0.00	37.75
236	101204 BARNUM CREEK	MALL	BCK	4.8 W	1.78	1.78	18	0.00	35	18.64	0.000	0	\$0	10.1	0.00	1.78
237	102102 LOCH ERIN	KINGSLEY	ADR	4.8 W	7.74	7.74	326	71.50	14	42.87	0.517	286	\$365	42.1	0.00	7.74
238	102201 FILLMORE	NORTH BLENDON	GRA	7.2 W	37.90	37.90	771	65.74	8	79.19	0.818	2038	\$264	20.3	0.00	37.90
239	102501 ROUND LAKE	ASPHALT	ADR	4.8 W	15.79	13.50	620	49.75		27.23				39.3	15.79	0.00 2022 Carryover
240	103302 SQUIRE HILL	HILLSIDE	FLT	4.8 W	16.45	16.45	1103	37.92	35	44.47	0.790	493	\$450	67.1	0.00	16.45
241	103802 JEFFS ROAD	ADLER ROAD	SMN	7.2 W	34.91	34.91	1933	41.96	7	59.01	0.659	965	\$488	55.4	34.91	0.00
242	106501 FOURTEENTH STREET	TOBIAS STREET	FLT	4.8 W	7.93	7.93	904	0.00	29	39.22	0.000	0	\$0	114.0	0.00	7.93
243	107503 OBERLIN	PRATT LAKE	CLR	14.4 W	74.99	74.99	1401	96.44	35	79.59	0.480	3472	\$355	18.7	0.00	74.99
244	108801 GOLDEN	SMITH'S CROSSING	MDL	4.8 W	3.29	3.29	70	65.33	21	24.64	0.912	196	\$227	21.3	0.00	3.29
245	109201 DAVENPORT	CONGRESS	SAG	4.8 W	17.94	17.94	2423	177.40	9	38.67	0.557	1774	\$137	135.1	17.94	0.00
246	109202 DAVENPORT	SCHAEFFER	SAG	4.8 W	1.57	1.57	31	0.00	35	21.46	0.000	0	\$0	19.7	0.00	1.57
247	111206 DRAKE ROAD	MAPLE HILL	KAL	4.8 W	3.68	3.68	395	68.00	34	22.98	0.543	136	\$365	107.3	0.00	3.68
248	111901 CHAUNCEY	AUSTIN	BCK	4.8 W	38.27	38.27	1457	45.40	24	74.22	1.124	1952	\$265	38.1	0.00	38.27
249	111902 CHAUNCEY	CASS	BCK	4.8 W	38.74	38.74	1531	64.72	11	49.45	0.645	1618	\$323	39.5	0.00	38.74
250	113401 HARING	SIXTH STREET	CAD	4.8 W	3.79	3.79	124	58.00	9	18.67	0.264	58	\$882	32.7	3.79	0.00
251	113601 EMERALD	BANBURY	KAL	4.8 W	9.15	9.15	2024	1487.20	9	45.58	0.546	7436	\$17	221.2	9.15	0.00
252	113604 EMERALD	EMERALD	KAL	4.8 W	0.68	0.68	4	4.00	35	9.87	2.941	8	\$2,760	5.9	0.00	0.68
253	114501 CADMUS	WINTER	ADR	7.2 W	8.31	8.31	1006	421.00	9	22.14	0.120	421	\$266	121.1	0.00	8.31
254	115902 BALLENGER	SALISBURY	FLT	4.8 W	6.11	6.11	1617	99.40	9	42.80	0.818	497	\$166	264.6	6.11	0.00
255	115903 BALLENGER	BRADLEY	FLT	4.8 W	13.37	13.37	2859	19.86	9	50.94	0.524	139	\$1,299	213.8	13.37	0.00
256	115904 BALLENGER	HOSPITAL	FLT	4.8 W	1.58	1.58	57	7.00	9	14.38	0.633	7	\$3,047	36.1	1.58	0.00

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

2023 LVD Line Clearing Work Plan

Case No.: U-21389

Exhibit No.: A-49 (PLB-8)

Page: 5 of 5

Witness: PLBolden

Date: May 2023

257	117901 SKYLARK	KING'S POINTE	FLT	4.8 W	22.79	22.80	1614	19.50		27.19					70.8	0.00	22.79 2022 Carryover
258	118302 BILLWOOD	LANSING ROAD	LAN	4.8 W	13.24	13.24	732	35.25	23	32.96	0.302	141	\$1,268		55.3	0.00	13.24
259	119601 PEWAMO	PEWAMO	LAN	12.0 D	0.81	0.81	5	0.00	13	11.25	0.000	0	\$0		6.2	0.00	0.81
260	123701 WILMOTT	WILMOTT	HML	4.8 W	40.11	40.11	2313	99.11	12	67.27	0.474	1883	\$288		57.7	0.00	40.11
261	124402 WESTWOOD	DURAND	JAC	4.8 W	7.05	7.05	1710	0	9	22.97	0.000	0	\$0		242.6	7.05	0.00
262	124502 CEDAR LAKE	VAN ET TEN	TWS	14.4 W	39.95	39.95	2193	244	15	63.43	0.626	6092	\$89		54.9	0.00	39.95
263	127403 DUQUITE	SAGANING	WBR	14.4 W	69.99	69.40	863	89		85.77					12.3	0.00	69.99 2022 Carryover
264	127601 COTTAGE GROVE	PREVO	BCY	14.4 W	126.49	39.50	1587	103		54.66					12.5	126.49	0.00 2022 Carryover
265	129902 ROSCOMMON	PIONEER	WBR	14.4 W	115.85	44.25	2164	67		60.07					18.7	0.00	115.85 2022 Carryover
266	131401 BELKNAP	BELKNAP	HML	4.8 W	13.94	13.94	167	11	26	34.18	1.793	274	\$730		12.0	0.00	13.94
267	134702 BELL ROAD	ALBEE	SAG	14.4 W	132.21	132.21	2429	123	17	69.39	0.514	8348	\$214		18.4	0.00	132.21
268	135202 FOREMAN	CUMBERLAND	GRE	4.8 W	43.03	43.03	956	75	20	68.71	0.488	1567	\$453		22.2	0.00	43.03
269	136602 M.A.E.	ALBAIN ROAD	SMN	7.2 W	13.25	13.25	892	37	7	32.17	0.075	37	\$4,834		67.3	13.25	0.00
270	136802 BROADMOOR	AIRWEST	GRE	7.2 W	3.12	3.12	127	0	34	15.26	0.000	0	\$0		40.7	0.00	3.12
271	136804 BROADMOOR	36TH STREET	GRE	7.2 W	3.84	3.84	72	0	34	15.26	0.000	0	\$0		18.8	0.00	3.84
272	138301 MILL GROVE	BABYLON	HML	14.4 W	25.18	25.18	611	109	5	53.14	0.635	1737	\$196		24.3	25.18	0.00
273	138302 MILL GROVE	ALLEGAN HYDRO	HML	14.4 W	48.08	48.08	1409	322	8	70.71	0.582	9014	\$95		29.3	0.00	48.08
274	147202 DORR CORNERS	100TH STREET	GRA	4.8 W	39.86	39.86	677	81	35	80.15	0.878	2852	\$189		17.0	0.00	39.86
275	148201 TITUS LAKE	CASINO	HML	4.8 W	2.66	2.66	24	0	20	14.29	0.000	0	\$0		9.0	0.00	2.66
276	149301 CHEESMAN	MONROE	ALM	4.8 W	10.12	10.12	374	0	35	16.58	0.000	0	\$0		37.0	0.00	10.12
277	150601 MISSION	UNIVERSITY	ALM	4.8 W	8.50	8.50	1203	1	35	13.70	0.118	1	\$114,750		141.5	0.00	8.50
278	150602 MISSION	THREE LEAVES	ALM	4.8 W	5.67	5.67	1243	0	35	18.64	0.000	0	\$0		219.2	0.00	5.67
279	151002 BALCOM	BANKERS	CLD	4.8 W	70.55	70.55	1394	78	9	61.52	0.936	5154	\$185		19.8	70.55	0.00
280	151202 PARAMOUNT	BIRD	GRN	7.2 W	37.14	37.14	2644	149	21	56.64	0.862	4764	\$105		71.2	0.00	37.14
281	151601 HUBBARD LAKE	HUBBARD LAKE	TWS	14.4 W	15.41	15.41	510	284	5	52.34	0.649	2841	\$81		33.1	15.41	0.00
282	154102 BLUE STAR	GANGES	HML	4.8 W	28.13	13.20	657	82		79.03					23.4	0.00	28.13 2022 Carryover
283	154601 PRICE ROAD	PRICE	MDL	14.4 W	39.42	39.42	1033	92	35	86.01	0.989	3575	\$205		26.2	0.00	39.42
284	156302 SCENIC LAKE	BEARD ROAD	OWS	4.8 W	25.51	10.40	918	62		59.39					36.0	0.00	25.51 2022 Carryover
285	157401 FOREST GROVE	FUTURE	GRA	7.2 W	0.03	0.03	0	0	35	9.87	0.000	0	\$0		0.0	0.00	0.03
286	157601 WEST CLARK LAKE	EAGLE POINT	JAC	4.8 W	13.35	13.24	496	59		49.81					37.2	0.00	13.35 2022 Carryover
287	160401 RATIGAN	GREELEY	GRN	14.4 W	42.65	12.60	711	51		83.17					16.7	0.00	42.65 2022 Carryover
288	162802 FISH LAKE	ROSE	FLT	4.8 W	22.12	22.12	1283	226	23	70.15	0.497	2489	\$120		58.0	0.00	22.12
289	163503 BENSTON	LEWIS	MUS	7.2 W	16.48	16.48	652	115	11	50.42	0.485	916	\$243		39.6	0.00	16.48
290	167403 SEVEN MILE	QUARTERLINE	BCY	14.4 W	12.94	12.94	146	53	13	31.05	0.386	265	\$659		11.3	0.00	12.94
291	168701 CLOVERLEAF	BUCHANAN	ALM	4.8 W	4.45	4.45	22	0	13	3.48	0.000	0	\$0		4.9	0.00	4.45

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

EUGÈNE M.J.A. BREURING

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

Schedule E-1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Annual Service Area Calendar Sales by Major Customer Classes and System Output

5-Year Calendar Historical

(GWh)

Case No.: U-21389
Exhibit No.: A-5 (EMB-1)
Schedule: E-1
Page: 1 of 2
Witness: EMBreuring
Date: May 2023

Line No.			(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
			<u>Total Company Electric Deliveries</u>					<u>Losses & Company Use</u>		
	Year		Residential	Commercial	Industrial	Other	Total	GWh	% of Output	System Output
1	2018	Hist	13,051	13,046	11,599	535	38,231	3,101	7.5%	41,331
2	2019	Hist	12,485	12,619	11,209	503	36,815	2,917	7.3%	39,732
3	2020	Hist	13,331	11,918	9,642	512	35,403	2,792	7.3%	38,195
4	2021	Hist	13,229	12,436	10,086	493	36,244	2,777	7.1%	39,021
5	2022	Hist	12,977	12,602	11,292	463	37,333	2,775	6.9%	40,108
			<u>Bundled Electric Deliveries</u>					<u>Losses & Company Use</u>		System Output
	Year		Residential	Commercial	Industrial	Other	Total	GWh	% of Output	
6	2018	Hist	13,051	12,034	8,833	535	34,453	2,997	8.0%	37,450
7	2019	Hist	12,485	11,638	8,424	503	33,050	2,862	8.0%	35,911
8	2020	Hist	13,331	11,014	6,952	512	31,810	2,742	7.9%	34,552
9	2021	Hist	13,229	11,504	7,375	493	32,600	2,660	7.5%	35,260
10	2022	Hist	12,977	11,676	8,455	463	33,570	2,756	7.6%	36,326
			<u>Choice Electric Deliveries</u>					<u>Losses & Company Use</u>		System Output
	Year		Residential	Commercial	Industrial	Other	Total	GWh	% of Output	
11	2018	Hist	-	1,012	2,766	-	3,778	104	2.7%	3,881
12	2019	Hist	-	980	2,785	-	3,765	56	1.5%	3,821
13	2020	Hist	-	904	2,689	-	3,593	49	1.4%	3,643
14	2021	Hist	-	932	2,712	-	3,644	116	3.1%	3,760
15	2022	Hist	-	926	2,837	-	3,763	20	0.5%	3,783

Schedule E-1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Annual Service Area Calendar Sales by Major Customer Classes

5-Year Calendar Historical - Weather-normalized

(GWh)

Case No.: U-21389

Exhibit No.: A-5 (EMB-1)

Schedule: E-1

Page: 2 of 2

Witness: EMBreuring

Date: May 2023

			(a)	(b)	(c)	(d)	(e)
<u>Total Company Electric Deliveries</u>							
Line No.	Year		Residential	Commercial	Industrial	Other	Total
1	2018	Hist	12,450	12,786	11,568	534	37,339
2	2019	Hist	12,452	12,653	11,215	503	36,824
3	2020	Hist	13,105	11,870	9,641	512	35,127
4	2021	Hist	12,747	12,258	10,059	492	35,557
5	2022	Hist	12,619	12,450	11,283	462	36,815

Bundled Electric Deliveries

	Year		Residential	Commercial	Industrial	Other	Total
6	2018	Hist	12,450	11,798	8,805	534	33,587
7	2019	Hist	12,452	11,672	8,431	503	33,058
8	2020	Hist	13,105	10,973	6,950	512	31,539
9	2021	Hist	12,747	11,338	7,350	492	31,927
10	2022	Hist	12,619	11,528	8,448	462	33,058

Choice Electric Deliveries

	Year		Residential	Commercial	Industrial	Other	Total
11	2018	Hist	-	988	2,764	-	3,752
12	2019	Hist	-	981	2,784	-	3,766
13	2020	Hist	-	897	2,691	-	3,587
14	2021	Hist	-	921	2,709	-	3,630
15	2022	Hist	-	922	2,835	-	3,757

Schedule E-1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Annual Service Area Calendar Sales by Major Customer Classes and System Output

5-Year Projected

(GWh)

Case No.: U-21389
Exhibit No.: A-15 (EMB-2)
Schedule: E-1
Page: 1 of 1
Witness: EMBreuring
Date: May 2023

Line No.			(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
			<u>Total Company Electric Deliveries</u>					<u>Losses & Company Use</u>		System Output
	Year		Residential	Commercial	Industrial	Other	Total	GWh	% of Output	
1	2023	Fcst	12,536	12,502	11,459	461	36,959	2,247	5.7%	39,206
2	2024	Fcst	12,572	12,221	11,438	494	36,725	2,170	5.6%	38,895
3	Test Year	Fcst	12,539	12,144	11,292	440	36,415	2,138	5.5%	38,553
4	2025	Fcst	12,585	11,979	11,731	152	36,448	2,096	5.4%	38,544
5	2026	Fcst	12,572	11,826	11,672	151	36,221	2,078	5.4%	38,300
6	2027	Fcst	12,562	11,806	11,818	150	36,335	2,009	5.2%	38,344
<u>Bundled Electric Deliveries</u>								<u>Losses & Company Use</u>		System Output
	Year		Residential	Commercial	Industrial	Other	Total	GWh	% of Output	
7	2023	Fcst	12,536	11,557	8,633	461	33,187	2,158	6.1%	35,345
8	2024	Fcst	12,572	11,295	8,646	494	33,007	2,083	5.9%	35,091
9	Test Year	Fcst	12,539	11,222	8,505	440	32,705	2,055	5.9%	34,760
10	2025	Fcst	12,585	11,070	8,984	152	32,791	2,014	5.8%	34,805
11	2026	Fcst	12,572	10,928	8,971	151	32,622	1,998	5.8%	34,620
12	2027	Fcst	12,562	10,908	9,121	150	32,741	1,931	5.6%	34,672
<u>Choice Electric Deliveries</u>								<u>Losses & Company Use</u>		System Output
	Year		Residential	Commercial	Industrial	Other	Total	GWh	% of Output	
13	2023	Fcst	-	945	2,826	-	3,771	90	2.3%	3,861
14	2024	Fcst	-	926	2,792	-	3,718	86	2.3%	3,804
15	Test Year	Fcst	-	923	2,786	-	3,709	83	2.2%	3,793
16	2025	Fcst	-	909	2,748	-	3,657	82	2.2%	3,739
17	2026	Fcst	-	898	2,701	-	3,599	81	2.2%	3,680
18	2027	Fcst	-	898	2,696	-	3,594	78	2.1%	3,672

Schedule E-2

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Test Year Total Company Electric Revenues & Deliveries

March 2024 - February 2025

Case No.: U-21389
 Exhibit No.: A-15 (EMB-3)
 Schedule: E-2
 Page: 1 of 1
 Witness: EMBreuring
 Date: May 2023

Line No.	Description	(a) 2022 Actual	(b) PSCR Factor	(c) Less Non-Base Tariff Items Surcharges	(d) Other Adjustments	(e) LTILRR	(f) Weather Normalization	(g) Change In Determinants	(h) Pro Forma	(i) Test Year PSCR Factor	(j) Total Company	(k) Test Year Non- Jurisdictional	(l) Jurisdictional
Adjusted Operating Revenues (\$000)													
<u>Cycle Billed Tariff Revenue</u>													
1	Base Tariff	\$ 2,482,287	\$ -	\$ -	\$ -	\$ -	\$ (28,999)	\$ (323,155)	\$ 318,173	\$ -	\$ 2,448,306	\$ 445	\$ 2,447,861
2	GSG Power Supply (1)	11,277	(11,277)	-	-	-	-	-	-	10,321	10,321	-	10,321
3	Base PSCR	1,879,841	-	-	-	-	(26,046)	(270,244)	251,224	-	1,834,775	23,665	1,811,110
4	PSCR Factor	54,854	(54,854)	-	-	-	-	-	-	180,558	180,558	107	180,451
5	Total Cycle Billed Tariff Revenue	4,428,258	(66,130)	-	-	-	(55,044)	(593,400)	569,397	190,879	4,473,960	24,217	4,449,743
<u>Cycle Billed Surcharge Revenue</u>													
6	Energy Optimization	194,120	-	(194,105)	-	(16)	-	-	-	-	(0)	-	(0)
7	Deferral	-	-	1	-	(1)	-	-	-	-	-	-	-
8	Renewable Energy	(104,490)	-	104,490	-	-	-	-	-	-	-	-	-
9	Low-Income Assistance Fund	19,659	-	(19,659)	-	(0)	-	-	-	-	(0)	-	(0)
10	Securitization (Classic7)	35,869	-	(34,093)	-	(1,776)	-	-	-	-	0	-	0
11	Other Provisions for Refund	(37,157)	-	37,157	-	-	-	-	-	-	-	-	-
12	Total Cycle Billed Surcharge Revenue	108,001	-	(106,208)	-	(1,793)	-	-	-	-	0	-	0
<u>Unbilled Revenue</u>													
14	Base Tariff	46,858	-	-	(46,931)	73	-	-	-	-	0	-	0
15	Base PSCR	2,246	-	-	(2,246)	-	-	-	-	-	-	-	-
16	PSCR Factor	1,203	(1,203)	-	-	-	-	-	-	-	-	-	-
17	Total Unbilled Revenue	50,307	(1,203)	-	(49,177)	73	-	-	-	-	0	-	0
18	PSCR Over/Under Recovery	408,808	(408,808)	-	-	-	-	-	-	-	-	-	-
19	Total Calendar Revenue	4,995,375	(476,142)	(106,208)	(49,177)	(1,720)	(55,044)	(593,400)	569,397	190,879	4,473,960	24,217	4,449,743
20	Miscellaneous Revenue (2) (3) (4)	151,075	(612)	-	(115,171)	96,812	-	-	-	-	132,104	1,013	131,091
21	Intersystem Sales Revenue	217,787	-	-	(217,787)	-	-	-	493,581	-	493,581	122	493,459
22	Total Operating Revenue	\$ 5,364,237	\$ (476,754)	\$ (106,208)	\$ (382,135)	\$ 95,092	\$ (55,044)	\$ (593,400)	\$ 1,062,978	\$ 190,879	\$ 5,099,644	\$ 25,352	\$ 5,074,293
<u>Revenue Adjustments</u>													
25	Jobwork Revenue	2,633	-	-	-	-	-	-	-	-	2,633	-	2,633
26	Jobwork Expense	(2,200)	-	-	-	-	-	-	-	-	(2,200)	-	(2,200)
27	Total Revenue Adjustments	433	-	-	-	-	-	-	-	-	433	-	433
28													
29	Adjusted Total Operating Revenue	\$ 5,364,670	\$ (476,754)	\$ (106,208)	\$ (382,135)	\$ 95,092	\$ (55,044)	\$ (593,400)	\$ 1,062,978	\$ 190,879	\$ 5,100,077	\$ 25,352	\$ 5,074,726
Electric Deliveries (MWh)													
32	Cycle Billed Bundled Service	33,406,059	-	-	-	2,067,241	(116,300)	(4,802,422)	-	-	30,554,578	328,226	30,226,352
33	Cycle Billed Self-generation	111,345	-	-	-	-	-	(9,651)	-	-	101,694	-	101,694
34	Cycle Billed Electric Choice	3,759,083	-	-	-	-	(4,417)	(40,751)	-	-	3,713,915	-	3,713,915
35	Intersystem Sales	2,996,465	-	-	(2,996,465)	-	-	-	-	-	-	-	-
36	Unbilled Bundled Service	52,649	-	-	(52,649)	185,753	-	(185,753)	-	-	-	-	-
37	Unbilled Electric Choice	3,984	-	-	(3,984)	-	-	-	-	-	-	-	-
38	Total Electric Deliveries	40,329,585	-	-	(3,053,098)	2,252,993	(120,718)	(5,038,576)	-	-	34,370,187	328,226	34,041,961

Notes:

(1) GSG Power Supply (ln2, col. i), (EMB-6)

(2) Miscellaneous Revenues (ln20, col. b): 2022 P-521, Page 301(M) footnote

(3) Miscellaneous Revenues (ln20, col.d): Authorized Return on Renewable Energy Assets (2022 P-521, pg 301.1) removed; and LTILRR customer base tariff revenues (WP-2, ln 19 col. d)

(4) Miscellaneous Revenues (ln20, col. e): Confidential Exhibit A-87 (AMG-5), page 2 of 2, ln20, col.p

Schedule E-3

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Electric Deliveries & Customer Counts by Rate Category

(Annual Deliveries in MWh)

Case No.: U-21389

Exhibit No.: A-15 (EMB-4)

Schedule: E-3

Page: 1 of 1

Witness: EMBreuring

Date: May 2023

Line No.	Description	(a)	(b)	(c)	(d)	(e)	(f)
		2022		Total Company		Jurisdictional	
		Avg. No. of Customers	Annual Deliveries	Avg. No. of Customers	Annual Deliveries	No. of Customers	Annual Deliveries
	Bundled Residential Service						
1	Standard Service RS	15,581	147,420	14,474	142,308	14,474	142,308
2	Peak Pwr Savers / Dynamic Pricing	5	(2)	-	-	-	-
3	Time-of-Day RT	1	58	-	-	-	-
4	Electric Vehicles REV-1	-	-	-	-	-	-
5	Electric Vehicles REV-2	-	8	-	-	-	-
6	Nighttime Savers (RPM)	3,646	36,588	13,773	105,411	13,773	105,411
7	Summer On-Peak (RSP)	1,610,624	12,697,623	1,618,090	12,185,915	1,618,090	12,185,915
8	Smart Hours (RSH)	15,723	136,746	16,512	125,514	16,512	125,514
9	Total Bundled Residential	1,645,580	13,018,441	1,662,850	12,559,148	1,662,850	12,559,148
	Bundled Secondary Service						
10	Secondary Energy-only GSD	198,297	4,114,927	196,122	3,996,698	196,122	3,996,698
11	Secondary Demand GSD	20,873	3,031,852	20,782	2,949,510	20,782	2,949,510
12	Secondary GSTU	845	38,445	918	27,489	918	27,489
13	Total Bundled Secondary	220,015	7,185,224	217,821	6,973,697	217,821	6,973,697
	Bundled Primary Service						
14	Primary Energy-only GP	1,513	805,117	1,479	805,656	1,479	805,656
15	Primary Demand GPD	1,117	7,604,517	796	4,580,914	796	4,580,914
16	Primary Time-of-Use GPTU	1,212	3,860,717	1,365	4,681,858	1,365	4,681,858
17	General Service Primary (EIP)	18	405,136	17	447,618	17	447,618
18	Special Contract (Rate_LED)	-	-	-	-	-	-
19	Total Bundled Primary	3,860	12,675,487	3,656	10,516,047	3,656	10,516,047
	Bundled Street Lighting Service						
20	Unmetered Lighting GUL	3,909	69,149	-	-	-	-
21	Metered Lighting GML	386	4,720	404	12,195	404	12,195
22	Unmetered GU	463	93,710	461	93,426	461	93,426
23	Unmetered Lighting GU-LED	789	16,845	-	-	-	-
24	Universal Unmetered Lighting UUL	-	-	4,686	71,839	4,686	71,839
25	Total Bundled Street Lighting	5,547	184,424	5,551	177,460	5,551	177,460
	Bundled Self-generation Service						
26	Self-generation GSG-1	-	-	-	-	-	-
27	Self-generation GSG-2	16	111,345	15	101,694	15	101,694
28	Total Bundled Self-generation	16	111,345	15	101,694	15	101,694
	Bundled Other Service						
29	Wholesale	1	325,294	1	310,379	-	-
30	Grand Rapids	1	17,189	1	17,847	-	-
31	Total Bundled Other	2	342,483	2	328,226	-	-
32	Cycle Billed Bundled Service	1,875,020	33,517,404	1,889,896	30,656,272	1,889,894	30,328,046
33	Unbilled	-	52,649	-	-	-	-
34	Calendar Bundled Service	1,875,020	33,570,053	1,889,896	30,656,272	1,889,894	30,328,046
	ROA Secondary Service						
35	Secondary Energy-only GSD	95	22,917	96	23,026	96	23,026
36	Secondary Demand GSD	461	173,831	462	172,462	462	172,462
37	Total ROA Secondary	556	196,748	558	195,488	558	195,488
	ROA Primary Service						
38	Primary Energy-only GP	59	75,408	59	75,604	59	75,604
39	Primary Demand GPD	341	3,486,927	338	3,442,823	338	3,442,823
40	Total ROA Primary	400	3,562,335	396	3,518,427	396	3,518,427
41	Cycle Billed ROA Service	956	3,759,083	954	3,713,915	954	3,713,915
42	Unbilled ROA	-	3,984	-	-	-	-
43	Calendar ROA Service	956	3,763,067	954	3,713,915	954	3,713,915
44	Cycle Billed Total Deliveries	1,875,976	37,276,487	1,890,850	34,370,187	1,890,848	34,041,961
45	Unbilled	-	56,633	-	-	-	-
46	Calendar Total Deliveries	1,875,976	37,333,120	1,890,850	34,370,187	1,890,848	34,041,961

Schedule E-4

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Calculation of Annual System Load Factor

2018 - 2022 Historical / 2023 - 2027 Forecast

Case No.: U-21389
Exhibit No.: A-15 (EMB-5)
Schedule: E-4
Page: 1 of 1
Witness: EMBreuring
Date: May 2023

Line No.			(a)	(b)	(c)
	Year		System Output GWh	System Peak Demand MW	Annual Load Factor
1	2018	Hist	41,331	8,084	58.4%
2	2019	Hist	39,732	8,039	56.4%
3	2020	Hist	38,195	8,215	52.9%
4	2021	Hist	39,021	7,951	56.0%
5	2022	Hist	40,108	8,061	56.8%
6	2023	Fcst	39,206	7,625	58.7%
7	2024	Fcst	38,895	7,585	58.4%
8	Test Year	Fcst	38,553	7,585	58.0%
9	2025	Fcst	38,544	7,536	58.4%
10	2026	Fcst	38,300	7,520	58.1%
11	2027	Fcst	38,344	7,541	58.0%

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Estimated Electric Rate Case PSCR Factor
 March 2024 - February 2025

Case No.: U-21389
 Exhibit No.: A-51 (EMB-6)
 Page: 1 of 1
 Witness: EMBreuring
 Date: May 2023

Line No.	Description	(a) Total Company	(b) Jurisdictional Factors	(c) Jurisdictional	(d) Non-Jurisdictional Factors	(e) Non-Jurisdictional
<u>Expenses (\$000)</u>						
1	System Power Supply Costs	(1) \$ 1,918,557	99.0%	\$ 1,899,391	1.00%	\$ 19,166
2	2022 Under-Recovery PSCR	(2) \$ 127,110	99.0%	\$ 125,840	1.00%	\$ 1,270
3	Transmission & Market Administrative Expense	(3) 543,730	99.1%	538,999	0.87%	4,731
4	PSCR Expenses	<u>\$ 2,589,397</u>		<u>\$ 2,564,230</u>		<u>\$ 25,167</u>
<u>PSCR Revenue Contributions (\$000)</u>						
5	Intersystem Sales Revenue	(4) \$ 493,581	100.0%	\$ 493,459	0.02%	\$ 122
6	Self-Generation	(6) 10,321	100.0%	10,321	0.00%	-
7	GI-2	-	100.0%	-	0.00%	-
8	Base Jurisdictional PSCR x \$0.05992	1,811,110	100.0%	1,811,110	0.00%	-
9	Base Non-Jurisdictional PSCR x \$0.05992	1,069	0.0%	-	100.00%	1,069
10	Long-Term Industrial Load Retention Rate	(5) 68,759	100.0%	68,759	0.00%	-
11	Wholesale Fuel Revenue	(7) 22,596	0.0%	-	100.00%	22,596
12	PSCR Revenue Contributions	<u>\$ 2,407,436</u>		<u>\$ 2,383,649</u>		<u>\$ 23,787</u>
13	PSCR (Over)/Under Recovery	<u>\$ 181,961</u>		<u>\$ 180,581</u>		<u>\$ 1,380</u>
<u>Electric Deliveries (MWh)</u>						
14	Jurisdictional Bundled Deliveries	(8)		32,472,424		
15	Less: Self-Generation	(9)		(101,694)		
16	Less: GI-2			-		
17	Less: Long-Term Industrial Load Retention Rate			(2,144,378)		
18	Total Bundled PSCR Deliveries			<u>30,226,352</u>		
<u>Estimated Jurisdictional PSCR Factor</u>						
19	PSCR (Over)/Under Recovery (\$000)			\$ 180,581		
20	Total Bundled PSCR Deliveries (MWh)			30,226,352		
21	Total Jurisdictional PSCR Factor (\$ per kWh)			<u>\$ 0.00597</u>		
<u>Estimated PSCR Recovery</u>						
22	PSCR Revenue Contributions	\$ 2,407,436		\$ 2,383,649		\$ 23,787
23	Plus: PSCR (Over)/Under Recovery	180,451		180,451		-
24	Non-jurisdictional PSCR Factor Contribution	107		-		107
25	Total PSCR Revenues	<u>\$ 2,587,994</u>		<u>\$ 2,564,100</u>		<u>\$ 23,893</u>
26	Total PSCR (Over)/Under Recovery	<u>\$ 1,403</u>		<u>\$ 130</u>		<u>\$ 1,273</u>
				Total Losses	0.0704	
				Less: Transmission	0.0000	
				System Losses	0.0704	
				System Efficiency	1.07573	
				PSCR Base @ Gen	0.05570	
				System Efficiency	1.07573	
				PSCR Base @ Del	<u>0.05992</u>	

Notes:

- (1) Exhibit A-146 (MLM-1), page 1, line 41; plus page 3, line 60; less page 1, line 35.
 (2) WP-JCA-57
 (3) Exhibit A-146 (MLM-1), page 1, line 35.
 (4) Exhibit A-146 (MLM-1), page 3, line 60.
 (5) Exhibit A-146 (MLM-1), page 1, line 42.
 (6) Self-Generation PSCR Revenue from WP-AMG-15
 (7) \$0.0752/kWh from Case No. U-21255, Exhibit A-1, Schedule 4, page 1 (Supplemental Power: Energy Charge rate)
 (8) Exhibit A-15 (EMB-4), Schedule E-3, line 34, column (f).
 (9) Exhibit A-15 (EMB-4), Schedule E-3, line 28, column (f).

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
Electric and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

ADAM S. CARVETH

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

Schedule: B-5.2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Summary of Actual and Projected Capital Expenditures
 Fleet Services
 (\$000)

Case No.: U-21389
 Exhibit No.: A-12 (ASC-1)
 Schedule: B-5.2
 Page: 1 of 2
 Witness: ASCarveth
 Date: May 2023

Line No	(a) Description	(b)	(c)	(d)	(e)	(f)
		Historical Year 12 Mos Ended 12/31/2022	Projected Bridge Period			Projected Test Year 12 Mos Ending 2/28/2025
		12 Mos Ended 12/31/2023	2 Mos Ending 2/29/2024	14 Mos Ending 2/29/2024		
1	Fleet Vehicle Capital Replacement Plan	17,083	19,035	2,168	21,203	19,494
2	Fleet Expansion	2,054	-	-	-	-
3	Fleet Vehicle Electrification	-	-	-	-	1,123
4	Fleet Business Partner Funded	686	33	-	33	-
5	Fleet Tools - Garage	256	240	41	281	247
6	Total Capital	20,079	19,308	2,209	21,517	20,863

		Projected			
		2 Mos Ending 2/28/2023	12 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025	26 Mos Ending 2/28/2025
7	Fleet Vehicle Capital Replacement Plan	7,386	13,817	19,494	40,697
8	Fleet Expansion	-	-	-	-
9	Fleet Vehicle Electrification	-	-	1,123	1,123
9	Fleet Business Partner Funded	-	33	-	33
10	Fleet Tools - Garage	40	241	247	528
11	Total Capital	7,426	14,091	20,863	42,380

Schedule: B-5.2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Actual and Projected Capital Expenditures
Fleet Services
(\$000)

Case No.: U-21389
Exhibit No.: A-12 (ASC-1)
Schedule: B-5.2
Page: 2 of 2
Witness: ASCarveth
Date: May 2023

Line No	(a) Description	(b)	(c)	(d)	(e)	(f)
		Historical Year	Projected Bridge Period		Projected Test Year	
		12 Mos Ended 12/31/2022	12 Mos Ended 12/31/2023	2 Mos Ending 2/29/2024	14 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025
1	Fleet Vehicle Capital Replacer	17,083	19,035	2,168	21,203	19,494
	Contractor	3,620	4,047	461	4,508	4,144
	Labor	-	-	-	-	-
	Materials	13,359	14,784	1,684	16,468	15,141
	Business Expenses	104	204	23	227	209
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-	-
2	Fleet Expansion	2,054	-	-	-	-
	Contractor	435	-	-	-	-
	Labor	-	-	-	-	-
	Materials	1,606	-	-	-	-
	Business Expenses	13	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-	-
3	Fleet Vehicle Electrification	-	-	-	-	1,123
	Contractor	-	-	-	-	239
	Labor	-	-	-	-	-
	Materials	-	-	-	-	872
	Business Expenses	-	-	-	-	12
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-	-
4	Fleet Business Partner Funde	686	33	-	33	-
	Contractor	145	7	-	7	-
	Labor	-	-	-	-	-
	Materials	536	26	-	26	-
	Business Expenses	4	0	-	0	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-	-
5	Fleet Tools - Garage	256	240	41	281	247
	Contractor	-	-	-	-	-
	Labor	-	-	-	-	-
	Materials	256	240	41	281	247
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-	-
6	Total Capital	20,079	19,308	2,209	21,517	20,863

		Projected			
		2 Mos Ending	12 Mos Ending	12 Mos Ending	26 Mos Ending
		2/28/2023	2/29/2024	2/28/2025	2/28/2025
7	Fleet Vehicle Capital Replacer	7,386	13,817	19,494	40,697
	Contractor	1,570	2,937	4,144	8,652
	Labor	-	-	-	-
	Materials	5,737	10,732	15,141	31,609
	Business Expenses	79	148	209	435
	Contingency	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-
8	Fleet Expansion	-	-	-	-
	Contractor	-	-	-	-
	Labor	-	-	-	-
	Materials	-	-	-	-
	Business Expenses	-	-	-	-
	Contingency	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-
9	Fleet Vehicle Electrification	-	-	1,123	1,123
	Contractor	-	-	239	239
	Labor	-	-	-	-
	Materials	-	-	872	872
	Business Expenses	-	-	12	12
	Contingency	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-
10	Fleet Business Partner Funde	-	33	-	33
	Contractor	-	7	-	7
	Labor	-	-	-	-
	Materials	-	26	-	26
	Business Expenses	-	0	-	0
	Contingency	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-
11	Fleet Tools - Garage	40	241	247	528
	Contractor	-	-	-	-
	Labor	-	-	-	-
	Materials	40	241	247	528
	Business Expenses	-	-	-	-
	Contingency	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-
12	Total Capital	7,426	14,091	20,863	42,380

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Fleet Responsibility Costs

Case No.: U-21389

Exhibit No.: A-52 (ASC-2)

Page: 1 of 1

Witness: ASCarveth

Date: May 2023

Categories	2016	2017	2018	2019	2020	2021	2022
Licensing	3,750,828	359,176	2,366,665	2,378,900	2,738,753	2,840,343	3,145,786
Technician	13,517,549	14,945,880	15,741,694	17,083,824	16,878,522	18,902,790	19,788,278
Parts	6,915,581	7,797,781	9,616,863	11,150,363	10,656,308	11,649,547	14,557,200
Outside Vendor	7,969,417	8,707,404	10,312,544	12,338,445	12,809,528	12,801,491	14,875,779
Fuel	10,986,225	14,273,104	16,894,736	15,398,764	12,808,774	17,865,478	25,105,355
Support Labor	4,179,501	4,310,356	4,639,242	5,203,200	5,096,942	5,034,713	5,031,440
Support Other	1,653,792	2,248,636	3,071,977	3,125,588	3,141,164	3,480,793	4,205,239
Total Cost	48,972,893	52,642,337	62,643,722	66,679,085	64,129,991	72,575,154	86,709,077

Line No.	(a)	(b)	Test Year							
			(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
			12 months ending 12/31/2022 Projected Spend	12 months ending 12/31/2022 Projected Unit Count	12 months ending 12/31/2023 Projected Spend	12 months ending 12/31/2023 Projected Unit Count	12 months ending 12/31/2024 Projected Spend	12 months ending 12/31/2024 Projected Unit Count	12 months ending 2/28/2025 Projected Spend	12 months ending 2/28/2025 Projected Unit Count
1	Fleet Vehicle Capital Replacement Plan		(\$000)	(units)	(\$000)	(units)	(\$000)	(units)	(\$000)	(units)
	<u>Vehicle Spec</u>	<u>Description</u>	17,083	122	19,035	157	19,454	143	19,494	151
07		SEDAN MID SIZE 4DR	-	-	188	4	227	5	227	5
22		1/2 TON PICKUP	27	4	620	11	726	19	1,083	19
23		VAN CARGO	1,396	24	930	12	323	4	323	4
26		1 TON PICKUP WITH SERVICE BODY SHORT CAB	672	18	192	2	-	-	-	-
28		3/4 TON PICKUP	180	8	555	8	585	3	228	3
36		1 TON PICKUP WITH SERVICE BODY	-	-	307	3	849	8	849	8
39		VAN CARGO HI ROOF	62	1	212	2	-	-	-	-
45		MECHANIC SERVICE TRUCK	329	2	-	-	183	1	183	1
54		5 YARD DUMP TRUCK SINGLE AXLE	57	1	157	1	-	-	-	-
55		10 YARD DUMP TRUCK TANDEM AXLE	34	1	-	-	-	-	-	-
57		MEDIUM DUTY CRANE TRUCK	272	1	-	-	-	-	-	-
59		STAKE RACK TRUCK	78	2	-	-	103	1	103	1
60		ELECTRIC SERVICE TRUCK	316	6	-	-	706	6	706	6
61		SUBSTATION OPERATIONS BUCKET TRUCK	225	1	-	-	-	-	-	-
64		SERVICE TRUCK WITH AERIAL DEVICE	-	-	4,148	16	803	3	803	3
65		STREETLIGHT TRUCK	-	-	-	-	1,298	3	1,298	3
67		SUBSTATION OPERATIONS SERVICE TRUCK	62	1	-	-	-	-	-	-
77		DIGGER DERRICK TRUCK	4,202	11	767	2	5,871	15	5,871	15
85		TANDUM AXLE TRACTOR WITH FIFTH WHEEL HITCH	190	2	172	1	183	1	183	1
89		BUCKET TRUCK	4,118	10	-	-	927	2	927	2
90		FLATBED TRUCK	153	2	153	2	476	3	476	3
96		BUCKET TRUCK	4,461	13	2,217	6	391	1	391	1
96 - Electric		BUCKET TRUCK - ELECTRIC	-	-	600	1	-	-	-	-
99		MISC. TRUCK	-	-	-	-	129	1	129	1
EQUIPMENT		EQUIPMENT	629	5	1,796	15	1,623	13	2,053	19
TRAILERS		TRAILERS	264	9	5,115	71	3,170	54	3,662	56
Lease Buyout - Electric/Common			702		906		881		-	
Rental Buyouts			11		-	-	-	-	-	-
Historical Disallowance - Toolkit Removal			(1,354)		-	-	-	-	-	-
2	Fleet Expansion		(\$000)	(units)	(\$000)	(units)	(\$000)	(units)	(\$000)	(units)
	<u>Vehicle Spec</u>	<u>Description</u>	2,054	34	-	-	-	-	-	-
22		1/2 TON PICKUP	19	4	-	-	-	-	-	-
23		VAN CARGO	-	-	-	-	-	-	-	-
90		FLATBED TRUCK	773	10	-	-	-	-	-	-
Equipment		EQUIPMENT	519	3	-	-	-	-	-	-
Trailers		TRAILERS	743	17	-	-	-	-	-	-
3	Fleet Vehicle Electrification		(\$000)	(units)	(\$000)	(units)	(\$000)	(units)	(\$000)	(units)
	<u>Vehicle Spec</u>	<u>Description</u>	-	-	-	-	1,123	-	1,123	-
22/28		Electric Chevy Silverado, (Count 19, Line 1, Spec 22)	-	-	-	-	823	-	823	-
		Altec JEMS, (Count 6, Line 1, Spec 64, 89, 96)	-	-	-	-	300	-	300	-
3	Fleet Business Partner Funded		(\$000)	(units)	(\$000)	(units)	(\$000)	(units)	(\$000)	(units)
	<u>Vehicle Spec</u>	<u>Description</u>	686	1	33	1	-	-	-	-
Generation Rebuild			686	1						
22		1/2 TON PICKUP	-	-	33	1				
4	Totals:		19,823	157	19,068	158	20,577	143	20,617	151

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Corporation

Summary of Fleet Tooling Actual & Projected Capital Expenditures

Fleet Services

For the Years 2022, 2023, 2024 and test year 12 months ending 02/28/2025

(\$000)

Case No.: U-21389

Exhibit No.: A-54 (ASC-4)

Page: 1 of 1

Witness: ASCarveth

Date: May 2023

Line No.	(a)	(b)		(c)		(d)		<i>Test Year</i> (e)	
		12 months ending 12/31/2022 Historical		12 months ending 12/31/2023 Projected Spend		12 months ending 12/31/2024 Projected Spend		12 months ending 2/28/2025 Projected Spend	
1	Fleet Tools - Garage								
	<u>Tool Description</u>								
	Shop Equipment	\$	92	\$	95	\$	110	\$	91
	Shop Tools	\$	120	\$	90	\$	68	\$	78
	Diagnostic Tools	\$	44	\$	55	\$	67	\$	78
2	Fleet Tools Totals:	\$	256	\$	240	\$	246	\$	247

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

THOMAS P. CLARK

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

EXHIBIT A-55 (TPC-1)
IS CONFIDENTIAL AND BEING FILED
UNDER SEAL WITH THE MPSC

EXHIBIT A-56 (TPC-2)
IS CONFIDENTIAL AND BEING FILED
UNDER SEAL WITH THE MPSC

Extract from the minutes of the meeting of the Board of Directors of Consumers Energy Company (the “Company”), duly called and held on January 21, 2021.

* * *

**RESOLUTIONS APPROVING A CAPITAL EXPENDITURE FOR THE
MUSTANG MILE SOLAR ENERGY LLC PROJECT**

RESOLVED: That, upon the recommendation of the Finance Committee of the Board of Directors (the “Board”) of Consumers Energy Company (the “Company”), the Board approves a capital expenditure in an amount of up to \$262 million for the Mustang Mile Solar Energy LLC project, a solar project located in Lenawee County, Michigan and expected to be up to 150 megawatts (the “Project”), as discussed at the meeting; and

RESOLVED FURTHER: That the officers of the Company, and each of them, are authorized and directed, for and on behalf of the Company, to execute any and all documents and take all actions and do all things they may deem necessary or appropriate, and as counsel may advise, to fully and timely initiate and complete the Project referred to in the foregoing resolution, without further action of the Board, unless the total Project cost exceeds the authorized amount by 10% or more as provided for under the Financial Authorities Policy.

Mustang Mile Solar Project – Milestone Schedule

Project	2020				2021				2022				2023				2024				
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	
Mustang Mile (BTA)	Development Phase												★	Execution Phase				★	★	★	★

Legend

Development Phase

Execution Phase

Operations Phase



Notice to Proceed



Substantial Completion



Capacity Test Completion



Final Completion



Reclamation & Vegetation Management

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Actual and Projected Capital Expenditures

Company Owned IRP Solar Resources

(\$000's)

Case No.: U-21389

Exhibit No.: A-59 (TPC-5)

Page: 1 of 1

Witness: TPClark

Date: May 2023

Line No	Description	Historical Year	Bridge Year	Projected Test Year
		12 Mos Ended 12/31/2022	14 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025
	(a)	(b)	(c)	(d)
1	Solar - 2019 IRP Solicitation - Mustang Mile Solar Project	\$ 162	\$ 233,064	\$ 49,337
2	Contractor	14	222,328	41,394
3	Labor	65	456	508
4	Materials	-	-	-
5	Business Expenses	28	792	93
6	Contingency	-	-	-
7	Other (Loadings, Chargebacks)	55	9,488	7,342
8	Solar - 2020 IRP Solicitation - Washtenaw Solar Project	\$ 147	\$ 122,186	\$ 104,980
9	Contractor	-	116,445	94,643
10	Labor	49	314	279
11	Materials	-	-	-
12	Business Expenses	43	808	350
13	Contingency	-	-	-
14	Other (Loadings, Chargebacks)	55	4,619	9,708
15	Solar - 2021 IRP Solicitation - Muskegon Solar Project	\$ 22,789	\$ 61,930	\$ 115,140
16	Contractor	22,132	56,643	105,397
17	Labor	68	314	553
18	Materials	-	-	-
19	Business Expenses	176	2,258	1,200
20	Contingency	-	-	-
21	Other (Loadings, Chargebacks)	413	2,715	7,990
15	Total Capital	<u>\$ 23,098</u>	<u>\$ 417,180</u>	<u>\$ 269,457</u>

EXHIBIT A-60 (TPC-6)
IS CONFIDENTIAL AND BEING FILED
UNDER SEAL WITH THE MPSC

Extract from the minutes of the meeting of the Special Board of Directors of Consumers Energy Company (the “Company”), duly called and held on September 29, 2021.

* * *

BOARD RESOLUTIONS APPROVING A CAPITAL EXPENDITURE FOR THE WASHTENAW SOLAR ENERGY LLC PROJECT

RESOLVED: That, upon the recommendation of Management of Consumers Energy Company (the “Company”), the Board of Directors (the “Board”) of the Company approves a capital expenditure in an amount of up to \$232 million for the Washtenaw Solar Energy LLC project, a solar project located in Washtenaw County, Michigan and expected to be up to 150 megawatts (the “Project”), as discussed at the meeting; and

RESOLVED FURTHER: That the officers of the Company, and each of them, are authorized and directed, for and on behalf of the Company, to execute any and all documents and take all actions and do all things they may deem necessary or appropriate, and as counsel may advise, to fully and timely initiate and complete the Project referred to in the foregoing resolution, without further action of the Board, unless the total Project cost exceeds the authorized amount by 10% or more as provided for under the Financial Authorities Policy.

Washtenaw Solar Project – Milestone Schedule

Project	2021				2022				2023				2024				2025			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Washtenaw (BTA)	<																			

Legend

Development Phase

Execution Phase

Operations Phase

★ Notice to Proceed

★ Substantial Completion

★ Capacity Test

★ Final Completion

★ Reclamation & Vegetation Management

EXHIBIT A-63 (TPC-9)
IS CONFIDENTIAL AND BEING FILED
UNDER SEAL WITH THE MPSC

EXHIBIT A-64 (TPC-10)
IS CONFIDENTIAL AND BEING FILED
UNDER SEAL WITH THE MPSC

EXHIBIT A-65 (TPC-11)
IS CONFIDENTIAL AND BEING FILED
UNDER SEAL WITH THE MPSC

EXHIBIT A-66 (TPC-12)
IS CONFIDENTIAL AND BEING FILED
UNDER SEAL WITH THE MPSC

Extract from the minutes of the meeting of the Board of Directors of Consumers Energy Company (the “Company”), duly called and held on January 25, 2023.

* * *

**PROPOSED FINANCE RESOLUTIONS APPROVING A CAPITAL
EXPENDITURE FOR THE MUSKEGON SOLAR ENERGY CENTER PROJECT**

RESOLVED: That the Finance Committee (the “Committee”) of Consumers Energy Company (the “Company”) approves a capital expenditure in an amount of \$354 million for the Muskegon Solar Energy Center Project, a solar project located in Muskegon County, Michigan and expected to be up to 250 megawatts (the “Project”), as discussed at the meeting; and

RESOLVED FURTHER: That the officers of the Company, and each of them, are authorized and directed, for and on behalf of the Company, to execute any and all documents and take all actions and do all things they may deem necessary or appropriate, and as counsel may advise, to fully and timely complete the Project without further action of the Committee unless the total Project cost exceeds the authorized amount by 10% or more as provided for under the Financial Authorities Policy.

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

2021 IRP Solicitation – Muskegon Solar Energy Center Project – Milestone
Schedule

Case No.: U-21389

Exhibit No.: A-68 (TPC-14)

Page: 1 of 1

Witness: TPClark

Date: May 2023

Muskegon Solar Energy Center Project – Milestone Schedule

Project	2022				2023				2024				2025				2026			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Muskegon Solar (BTA)																				

Legend

Development Phase

Execution Phase

Operations Phase



Notice to Proceed



Substantial Completion



Capacity Test Completion



Final Completion



Reclamation & Vegetation Management

Stakeholder Outreach Meeting

2021 IRP - Settlement Agreement Battery Deployment Program

August 17th, 2022



Agenda

10:00 – 10:05	Welcome and Introductions
10:05 – 10:20	Battery Deployment Program Overview
10:20 – 10:35	Battery Deployment Program Competitive Solicitation Overview
10:35 – 10:50	Stakeholder Feedback – Clarifying Questions
10:50 – 11:00	Written Feedback Instructions and Next Steps

Introductions

Consumers Energy Presenters

- Sara Walz | Director of Electric Supply Planning
- Chris McNalley | Senior Engineer Lead - Generation Planning Renewable Assets
- Demeury (Dex) Naranjo-Rodriguez | Planning Engineer - LVD Substation Planning
- Frank Gerzetich | Portfolio Manager - Supply Chain

Battery Deployment Program Overview

2021 Integrated Resource Plan Settlement Agreement

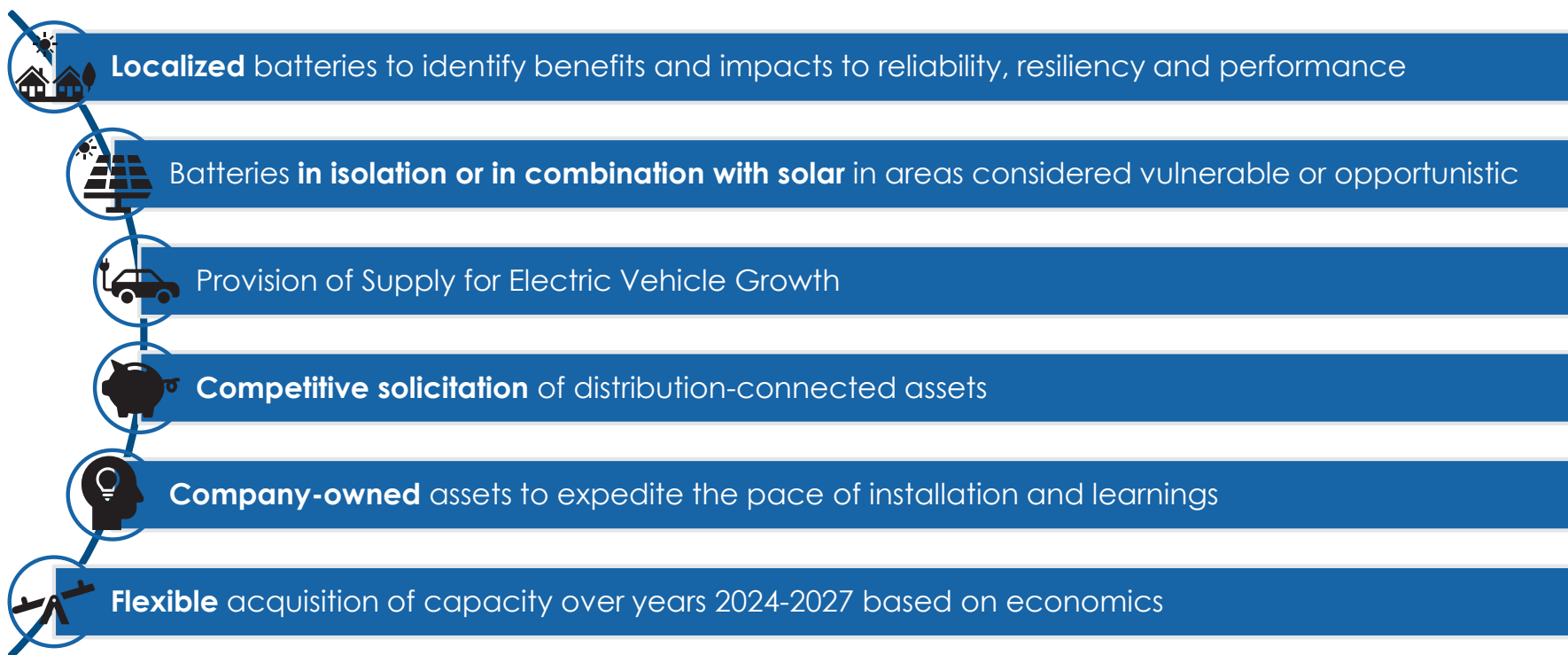
Battery Deployment Program:

The parties agree to the approval of the battery deployment program as proposed by Company witness Richard T. Blumenstock. The parties agree that **the Company will conduct stakeholder outreach to solicit feedback regarding the battery deployment program prior to the issuance of the first battery deployment program competitive solicitation.** The approval to recover the costs associated with the batteries acquired in the battery deployment program will be sought in future electric rate cases.

Stakeholder feedback is being requested prior to RFP for consideration. Incorporation of feedback is not guaranteed. The Company will provide written responses to all comments, suggestions, and questions.

Battery Deployment Program Key Elements

The battery deployment program is designed to expand the Company's learnings of potential benefits to the distribution system from storage. The program targets...



Proposed Program Deployment Schedule

The battery deployment program was initially designed to add distribution supply assets to conservatively follow potential electric vehicle (EV), other electrification or load growth.

Commercial Operation Date	Incremental Battery Capacity (MW)	Cumulative Battery Capacity (MW)
Dec 31, 2024	5	5
Dec 31, 2025	10	15
Dec 31, 2026	20	35
Dec 31, 2027	40	75

Capacity will be acquired flexibly for years 2024-2027, as economically viable through competitive solicitation.

Battery Deployment Program Competitive Solicitation Overview

Battery Storage Solicitation Structure

Product

- Single solicitation event
 - Initial 5 MW / 20 MWh
 - Options for up to 75 MW / 300 MWh
- Capacity on-line between 2024-2027
- See technical specification for additional information

Proposed Bid Option 1

Equipment Manufacturer / EPC Contract

- The Company identifies sites based on parameters of the program and then bids out for an Integrator contract.
- All engineering, procurement, and construction to be provided by single entity through construction.

Proposed Bid Option 2

Equipment Only Contract

- Separating award for major equipment and "balance of plant".
- The Company solicits bids for major equipment and "balance of plant" separately

Solicitation Timing

Exact date and timing subject to change

- RFP issued Q3 2022
- Bids due Q4 2022
- Bid award Q4 2022

EPC =
Engineering,
Procurement,
Construction

RFP Technical Specification and Siting

Battery System Criteria

Interconnection	Distribution Connected (<69 kV)
	Interconnection to Consumers Energy System
Size and Technology	0.1 MW or Larger (maximum 20MW)
	Battery Technologies – Flexible (Li-Ion, Iron Redox, Vanadium Flow, etc.)
Location	Vulnerable Communities (defined later) and Public Safety Requirements
Operating Parameters	Minimum Duration of 4 Hours
	Meets Qualification for MISO Market Participation
	Lifespan of Asset Minimum 15 Years
Configuration	Standalone or Renewable Hybrid Systems
Use Cases	Distribution Capacity Deferral, Voltage Support, Resiliency, Power Quality, Energy Arbitrage, Capacity Credit, etc.

RFP Evaluation Criteria



Siting Criteria

Located in Vulnerable Communities

Opportunistic Locations

Company-owned Land, Brownfields, Pollinator Habitat etc.

Value-Added Criteria

Minority and Women Business Development

Michigan / Union Labor and Content

Vulnerable communities are defined as communities that have an environmental or demographic "index", as examined by the Environmental Protection Agency (EPA) EJSCREEN Tool or MIEJScreen Tool, as scoring at the 75th percentile or above

Opportunistic locations are circuits or locations that would provide electric system benefits such as capacity deferral, resiliency improvements, or operational improvements

Clarifying Questions

Stakeholder Feedback Areas

1. Capacity Acquisition Schedule

- Single solicitation for up to 75 MW over the period of 2024-2027

2. Contract Options

- EPC Contract vs Equipment-only Contract

3. Evaluation Criteria

4. RFP Schedule Timing

Written Feedback Instructions and Next Steps

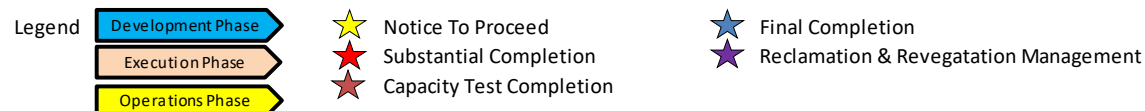
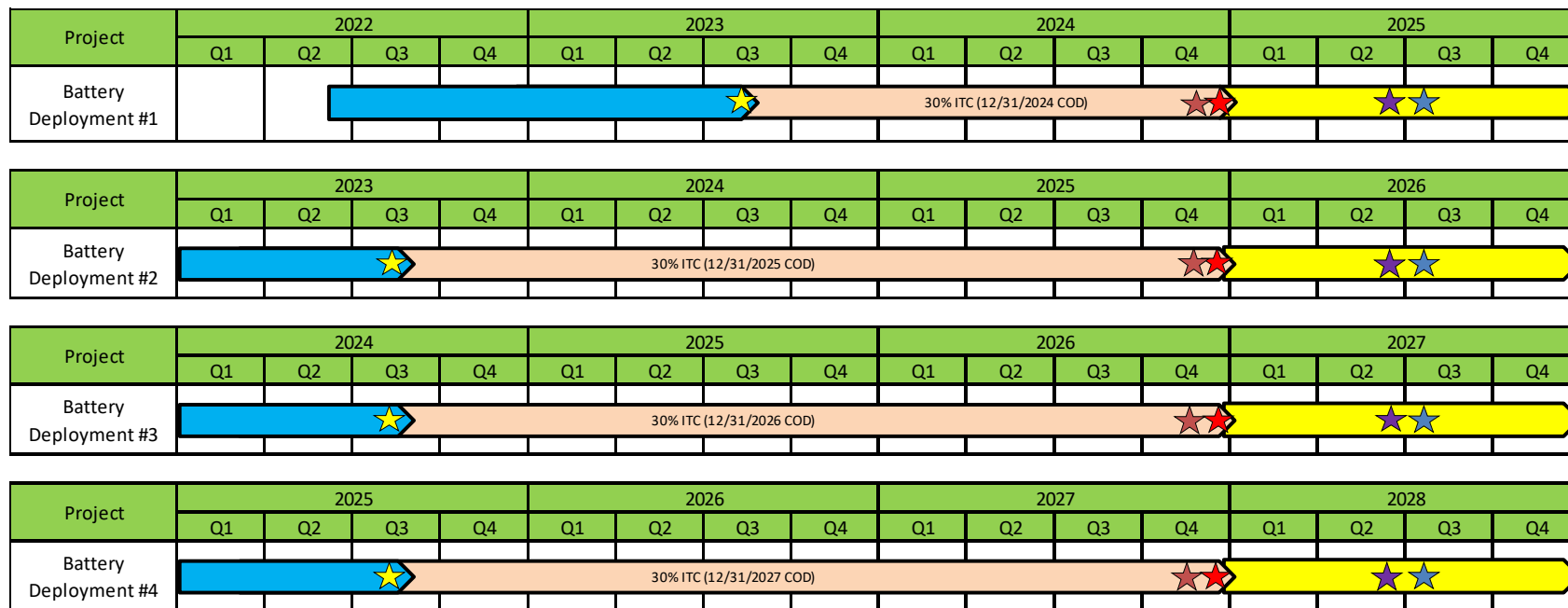
Please submit comments, suggestions, and follow up questions via email to Kenneth.Johnston@cmsenergy.com

Deadline to submit comments and recommendations is Wednesday, August 24th.

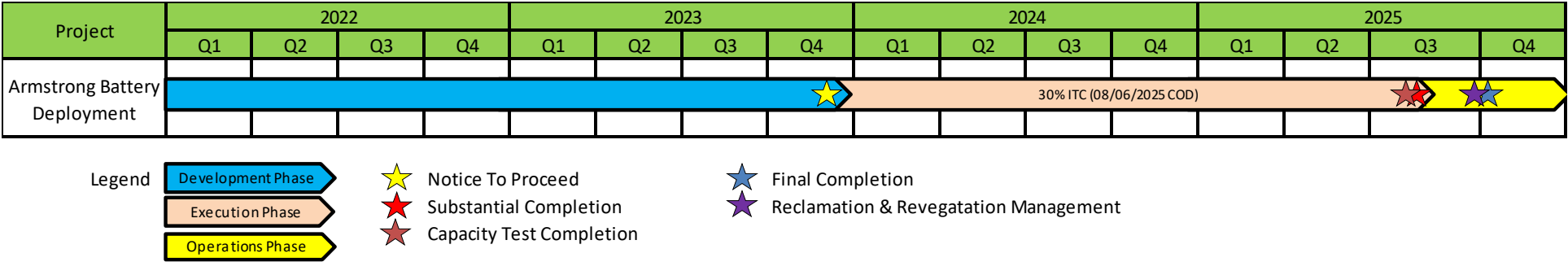
Materials presented today will be distributed to all participants.

Thank you for your participation!

Battery Energy System Storage – Milestone Schedule



Armstrong Battery Deployment – Milestone Schedule



STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBIT
OF
JASON R. COKER
ON BEHALF OF
CONSUMERS ENERGY COMPANY

May 2023

Smart Grid Program - Annual Capital Expenditures

Line No.	(a) Utility	(b) Description	(c) 2007	(d) 2008	(e) 2009	(f) 2010	(g) 2011	(h) 2012	(i) 2013	(j) 2014	(k) 2015	(l) 2016	(m) 2017	(n) 2018	(o) 2019	(p) 2020
1	Electric	Meters	\$ -	\$ 1,424	\$ 1,231	\$ 930	\$ 128	\$ 9,602	\$ 21,432	\$ 29,976	\$ 62,152	\$ 109,569	\$ 53,679	\$ -	\$ -	\$ -
2		Field Equipment/Facilities	\$ -	\$ 113	\$ 1,305	\$ 1,510	\$ 341	\$ 259	\$ 171	\$ 154	\$ 270	\$ 1,200	\$ 5,965	\$ 6,331	\$ 8,440	\$ 8,580
3		Software/Systems Development	\$ 7,889	\$ 8,214	\$ 12,259	\$ 15,909	\$ 19,654	\$ 20,770	\$ 32,100	\$ 31,990	\$ 45,038	\$ 24,918	\$ (1,734)	\$ 1,162	\$ 16,390	\$ 952
4		SG Infrastructure	\$ -	\$ -	\$ -	\$ 2,789	\$ 7,834	\$ 5,912	\$ 1,109	\$ 976	\$ 1,463	\$ 3,391	\$ 750	\$ 1,943	\$ 2,827	\$ 3,375
5		Pilot Prep & Project Management	\$ -	\$ 7,883	\$ 8,932	\$ 13,283	\$ 9,558	\$ 7,919	\$ 6,194	\$ 6,063	\$ 6,657	\$ 4,790	\$ 2,493	\$ -	\$ -	\$ -
6		Total Capital Costs before Avoidance - Electric	\$ 7,889	\$ 17,634	\$ 23,726	\$ 34,421	\$ 37,515	\$ 44,463	\$ 61,005	\$ 69,159	\$ 115,581	\$ 143,868	\$ 61,153	\$ 9,436	\$ 27,657	\$ 12,907
7		Avoided Capital Costs - Electric	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (306)	\$ (240)	\$ 410	\$ 352	\$ 263	\$ 274
8		Total Capital Costs Net - Electric	\$ 7,889	\$ 17,634	\$ 23,726	\$ 34,421	\$ 37,515	\$ 44,463	\$ 61,005	\$ 69,159	\$ 115,274	\$ 143,628	\$ 61,563	\$ 9,787	\$ 27,920	\$ 13,181
9																
10	Gas	Modules	\$ -	\$ -	\$ -	\$ 5	\$ -	\$ -	\$ -	\$ -	\$ 6,663	\$ 25,404	\$ 15,845	\$ -	\$ -	\$ -
11		Field Equipment/Facilities	\$ -	\$ 15	\$ 162	\$ 95	\$ 41	\$ 37	\$ 23	\$ 21	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
12		Software/Systems Development	\$ 1,076	\$ 1,120	\$ 1,524	\$ 2,143	\$ 2,664	\$ 2,832	\$ 4,377	\$ 4,362	\$ 5,995	\$ 3,359	\$ (54)	\$ 158	\$ 165	\$ 130
13		SG Infrastructure	\$ -	\$ -	\$ -	\$ 380	\$ 1,068	\$ 806	\$ 151	\$ 133	\$ 200	\$ 470	\$ 102	\$ 265	\$ 385	\$ 460
14		Pilot Prep & Project Management	\$ -	\$ 1,573	\$ 2,348	\$ 3,016	\$ 1,737	\$ 1,183	\$ 827	\$ 727	\$ 872	\$ 667	\$ 354	\$ -	\$ -	\$ -
15		Total Capital Costs before Avoidance - Gas	\$ 1,076	\$ 2,709	\$ 4,034	\$ 5,639	\$ 5,510	\$ 4,858	\$ 5,379	\$ 5,243	\$ 13,730	\$ 29,900	\$ 16,248	\$ 423	\$ 550	\$ 590
16		Avoided Capital Costs - Gas	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (10)	\$ (20)	\$ (34)	\$ (42)	\$ (43)	\$ (43)
17		Total Capital Costs Net - Gas	\$ 1,076	\$ 2,709	\$ 4,034	\$ 5,639	\$ 5,510	\$ 4,858	\$ 5,379	\$ 5,243	\$ 13,720	\$ 29,881	\$ 16,213	\$ 381	\$ 507	\$ 547
18																
19	Total	Meter/Module Purchases, including Set & Test	\$ -	\$ 1,424	\$ 1,231	\$ 935	\$ 128	\$ 9,602	\$ 21,432	\$ 29,976	\$ 68,816	\$ 134,973	\$ 69,524	\$ -	\$ -	\$ -
20		Field Equipment/Facilities	\$ -	\$ 128	\$ 1,466	\$ 1,605	\$ 382	\$ 296	\$ 194	\$ 174	\$ 270	\$ 1,200	\$ 5,965	\$ 6,331	\$ 8,440	\$ 8,580
21		Software/Systems Development	\$ 8,965	\$ 9,334	\$ 13,782	\$ 18,052	\$ 22,318	\$ 23,602	\$ 36,477	\$ 36,353	\$ 51,033	\$ 28,277	\$ (1,788)	\$ 1,320	\$ 16,555	\$ 1,082
22		SG Infrastructure	\$ -	\$ -	\$ -	\$ 3,170	\$ 8,903	\$ 6,719	\$ 1,260	\$ 1,109	\$ 1,663	\$ 3,861	\$ 853	\$ 2,208	\$ 3,212	\$ 3,835
23		Pilot Prep & Project Management	\$ -	\$ 9,456	\$ 11,280	\$ 16,299	\$ 11,294	\$ 9,102	\$ 7,021	\$ 6,790	\$ 7,529	\$ 5,457	\$ 2,848	\$ -	\$ -	\$ -
24		Total Capital Costs before Avoidance	\$ 8,965	\$ 20,342	\$ 27,760	\$ 40,060	\$ 43,025	\$ 49,320	\$ 66,384	\$ 74,402	\$ 129,311	\$ 173,768	\$ 77,401	\$ 9,859	\$ 28,207	\$ 13,498
25		Avoided Capital Costs	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (316)	\$ (260)	\$ 375	\$ 309	\$ 220	\$ 231
26		Total Capital Costs Net	\$ 8,965	\$ 20,342	\$ 27,760	\$ 40,060	\$ 43,025	\$ 49,320	\$ 66,384	\$ 74,402	\$ 128,994	\$ 173,508	\$ 77,776	\$ 10,168	\$ 28,428	\$ 13,729
27																
28																
29		Note:														
30		Cost of Removal	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 129	\$ 120	\$ 1	\$ -	\$ 59	\$ 23	\$ -	\$ -	\$ -
31		Total Capital Costs Incl COR before Avoidance	\$ 8,965	\$ 20,342	\$ 27,760	\$ 40,060	\$ 43,025	\$ 49,450	\$ 66,504	\$ 74,403	\$ 129,311	\$ 173,827	\$ 77,424	\$ 9,859	\$ 28,207	\$ 13,498
32		Cml Total Capital Costs Incl COR before Avoidance	\$ 8,965	\$ 29,307	\$ 57,067	\$ 97,127	\$ 140,152	\$ 189,602	\$ 256,106	\$ 330,509	\$ 459,820	\$ 633,647	\$ 711,071	\$ 720,930	\$ 749,137	\$ 762,634

Total NPV Benefit of -\$71.6 million based on nominal values 2007 - 2012 and present values 2013 - 2032.

Smart Grid Program - Annual Capital Expenditures

Line No.	(a) Utility	(b) Description	(c) 2021	(d) 2022	(e) 2023	(f) 2024	(g) 2025	(h) 2026	(i) 2027	(j) 2028	(k) 2029	(l) 2030	(m) 2031	(n) 2032	(o) Total 2007-2032
1	Electric	Meters	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 290,123
2		Field Equipment/Facilities	\$ 7,651	\$ 6,941	\$ 4,125	\$ 4,176	\$ 4,224	\$ 4,248	\$ 4,310	\$ 4,331	\$ 4,406	\$ 4,425	\$ 4,437	\$ 4,445	\$ 92,357
3		Software/Systems Development	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 235,511
4		SG Infrastructure	\$ 2,916	\$ 2,342	\$ 2,569	\$ 3,064	\$ 2,985	\$ 2,617	\$ 2,577	\$ 2,850	\$ 2,943	\$ 2,756	\$ 2,648	\$ 2,760	\$ 65,395
5		Pilot Prep & Project Management	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 73,771
6		Total Capital Costs before Avoidance - Electric	\$ 10,566	\$ 9,282	\$ 6,695	\$ 7,239	\$ 7,209	\$ 6,865	\$ 6,887	\$ 7,182	\$ 7,348	\$ 7,181	\$ 7,085	\$ 7,204	\$ 757,156
7		Avoided Capital Costs - Electric	\$ (882)	\$ 423	\$ 445	\$ 299	\$ 245	\$ 226	\$ 229	\$ 251	\$ 450	\$ 451	\$ 454	\$ 457	\$ 3,801
8		Total Capital Costs Net - Electric	\$ 9,685	\$ 9,705	\$ 7,140	\$ 7,539	\$ 7,454	\$ 7,091	\$ 7,116	\$ 7,432	\$ 7,798	\$ 7,632	\$ 7,539	\$ 7,661	\$ 760,957
9															
10	Gas	Modules	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 47,918
11		Field Equipment/Facilities	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 394
12		Software/Systems Development	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 29,851
13		SG Infrastructure	\$ 398	\$ 319	\$ 350	\$ 418	\$ 407	\$ 357	\$ 351	\$ 389	\$ 401	\$ 376	\$ 361	\$ 376	\$ 8,925
14		Pilot Prep & Project Management	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 13,305
15		Total Capital Costs before Avoidance - Gas	\$ 398	\$ 319	\$ 350	\$ 418	\$ 407	\$ 357	\$ 351	\$ 389	\$ 401	\$ 376	\$ 361	\$ 376	\$ 100,392
16		Avoided Capital Costs - Gas	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (707)
17		Total Capital Costs Net - Gas	\$ 355	\$ 276	\$ 307	\$ 375	\$ 364	\$ 314	\$ 309	\$ 346	\$ 358	\$ 333	\$ 318	\$ 333	\$ 99,685
18															
19	Total	Meter/Module Purchases, including Set & Test	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 338,040
20		Field Equipment/Facilities	\$ 7,651	\$ 6,941	\$ 4,125	\$ 4,176	\$ 4,224	\$ 4,248	\$ 4,310	\$ 4,331	\$ 4,406	\$ 4,425	\$ 4,437	\$ 4,445	\$ 92,750
21		Software/Systems Development	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 265,362
22		SG Infrastructure	\$ 3,313	\$ 2,661	\$ 2,920	\$ 3,482	\$ 3,392	\$ 2,974	\$ 2,929	\$ 3,239	\$ 3,344	\$ 3,132	\$ 3,009	\$ 3,136	\$ 74,320
23		Pilot Prep & Project Management	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 87,076
24		Total Capital Costs before Avoidance	\$ 10,964	\$ 9,602	\$ 7,045	\$ 7,657	\$ 7,616	\$ 7,221	\$ 7,239	\$ 7,570	\$ 7,750	\$ 7,557	\$ 7,446	\$ 7,581	\$ 857,549
25		Avoided Capital Costs	\$ (925)	\$ 380	\$ 402	\$ 257	\$ 202	\$ 184	\$ 186	\$ 208	\$ 407	\$ 408	\$ 411	\$ 414	\$ 3,094
26		Total Capital Costs Net	\$ 10,039	\$ 9,981	\$ 7,448	\$ 7,914	\$ 7,818	\$ 7,405	\$ 7,425	\$ 7,778	\$ 8,156	\$ 7,965	\$ 7,857	\$ 7,994	\$ 860,642
27															
28															
29		Note:													
30		Cost of Removal	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 332
31		Total Capital Costs Incl COR before Avoidance	\$ 10,964	\$ 9,602	\$ 7,045	\$ 7,657	\$ 7,616	\$ 7,221	\$ 7,239	\$ 7,570	\$ 7,750	\$ 7,557	\$ 7,446	\$ 7,581	\$ 857,881
32		Cml Total Capital Costs Incl COR before Avoidance	\$ 773,598	\$ 783,200	\$ 790,245	\$ 797,902	\$ 805,518	\$ 812,740	\$ 819,978	\$ 827,548	\$ 835,298	\$ 842,855	\$ 850,301	\$ 857,881	

Smart Grid Program - Benefits and O&M costs

Line No.	(a) Utility	(b) Description	(c) 2007	(d) 2008	(e) 2009	(f) 2010	(g) 2011	(h) 2012	(i) 2013	(j) 2014	(k) 2015	(l) 2016	(m) 2017	(n) 2018	(o) 2019	(p) 2020
33	Electric	Meter Reading	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (116)	\$ (1,878)	\$ (5,151)	\$ (9,767)	\$ (16,577)	\$ (19,196)	\$ (19,926)	\$ (20,325)
34		Uncollectible Expense	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (128)	\$ (3,685)	\$ (6,227)	\$ (8,181)	\$ (8,344)	\$ (8,511)
35		Other O&M	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (1,271)	\$ (1,115)	\$ (2,917)	\$ (4,690)	\$ (7,375)	\$ (7,942)	\$ (8,183)	\$ (8,508)
36		AC Load Control Avoided Generation, Transmission	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (135)	\$ (960)	\$ (3,502)	\$ (4,674)
37		Demand Response Avoided Generation, Transmission	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (0)	\$ (47)	\$ (490)	\$ (954)	\$ (1,809)
38		Theft Reduction	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (93)	\$ (919)	\$ (1,837)	\$ (4,787)	\$ (6,943)	\$ (8,342)	\$ (9,810)	\$ (15,126)	\$ (22,561)
39		AMI Induced Conservation & Efficiency Energy	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (2,906)	\$ (5,571)	\$ (7,991)	\$ (10,912)	\$ (8,513)
40		Load Management Conserved Energy	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
41		Demand Response Conserved Energy	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (1,417)
42		Property Tax (Savings) - Legacy Meters	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (33)	\$ (97)	\$ (235)	\$ (500)	\$ (835)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)
43		Terminal Value	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
44		Total Non Capital Benefits before Costs - Electric	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (126)	\$ (2,403)	\$ (5,064)	\$ (13,482)	\$ (28,827)	\$ (45,340)	\$ (55,636)	\$ (68,014)	\$ (77,384)
45		Common	\$ 382	\$ 67	\$ 545	\$ 1,165	\$ 1,094	\$ 2,586	\$ 5,977	\$ 4,853	\$ 6,466	\$ 6,821	\$ 8,606	\$ 9,360	\$ 9,502	\$ 9,724
46		Meters, Modules & Communications	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 97	\$ 963	\$ 1,310	\$ 1,671	\$ 3,226	\$ 4,898	\$ 5,227	\$ 5,228	\$ 5,228
47		Customer Engagement	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 1,538	\$ 2,444	\$ 1,956	\$ 1,833	\$ 4,179	\$ 5,275	\$ 7,938	\$ 21,493	\$ 9,299
48		Load Control Program	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 195	\$ 665	\$ 2,567	\$ 3,017	\$ 2,726	\$ 2,625
49		Demand Response Program	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 16	\$ 194	\$ 2,219	\$ 986	\$ 606	\$ 618
50		Property Taxes - AMI	\$ -	\$ 24	\$ 44	\$ 70	\$ 52	\$ 335	\$ 862	\$ 1,572	\$ 2,977	\$ 5,402	\$ 6,644	\$ 6,515	\$ 6,407	\$ 6,270
51		Total Non Capital Benefits Net - Electric	\$ 382	\$ 91	\$ 589	\$ 1,235	\$ 1,146	\$ 4,430	\$ 7,843	\$ 4,627	\$ (325)	\$ (8,340)	\$ (15,131)	\$ (22,594)	\$ (22,052)	\$ (43,620)
52																
53	Gas	Meter Reading	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 0	\$ (591)	\$ (1,965)	\$ (2,507)	\$ (2,602)	\$ (2,654)
54		Uncollectible Expense	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (168)	\$ (1,335)	\$ (2,144)	\$ (2,187)	\$ (2,231)
55		Other O&M	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (68)	\$ 10	\$ (1)	\$ (214)	\$ (411)	\$ (444)	\$ (454)	\$ (468)
56		Theft Reduction	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (235)	\$ (462)	\$ (531)	\$ (667)	\$ (890)	\$ (3,460)
57		AMI Induced Conservation & Efficiency Energy	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (199)	\$ (1,433)	\$ (2,247)	\$ (2,585)	\$ (2,705)
58		LAUF Gas Reduction Enabled	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (22)	\$ (173)	\$ (181)	\$ (190)
59		Terminal Value	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
60		Total Non Capital Benefits before Costs - Gas	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (68)	\$ 10	\$ (236)	\$ (1,634)	\$ (5,697)	\$ (8,182)	\$ (8,900)	\$ (11,709)
61		Common	\$ 367	\$ 64	\$ 87	\$ 143	\$ 73	\$ 127	\$ 36	\$ 11	\$ 216	\$ 746	\$ 1,173	\$ 1,276	\$ 1,296	\$ 1,326
62		Meters, Modules & Communications	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 5	\$ 6	\$ -	\$ 42	\$ 466	\$ 1,237	\$ 1,371	\$ 1,371	\$ 1,371
63		Property Taxes - AMI	\$ -	\$ -	\$ -	\$ 5	\$ 5	\$ 8	\$ 12	\$ 11	\$ 141	\$ 692	\$ 891	\$ 831	\$ 778	\$ 728
64		Total Non Capital Benefits Net - Gas	\$ 367	\$ 64	\$ 87	\$ 148	\$ 78	\$ 140	\$ (14)	\$ 32	\$ 164	\$ 271	\$ (2,396)	\$ (4,704)	\$ (5,455)	\$ (8,283)
65																
66	Total	Meter Reading	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (116)	\$ (1,878)	\$ (5,151)	\$ (10,358)	\$ (18,542)	\$ (21,704)	\$ (22,529)	\$ (22,979)
67		Uncollectible Expense	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (128)	\$ (3,685)	\$ (7,562)	\$ (10,325)	\$ (10,531)	\$ (10,742)
68		Other O&M	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (1,339)	\$ (1,105)	\$ (2,917)	\$ (4,905)	\$ (7,786)	\$ (8,386)	\$ (8,637)	\$ (8,976)
69		AC Load Control Avoided Generation, Transmission	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (135)	\$ (960)	\$ (3,502)	\$ (4,674)
70		Demand Response Avoided Generation, Transmission	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (0)	\$ (47)	\$ (490)	\$ (954)	\$ (1,809)
71		Theft Reduction	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (93)	\$ (919)	\$ (1,837)	\$ (5,023)	\$ (7,405)	\$ (8,873)	\$ (10,477)	\$ (16,016)	\$ (26,021)
72		AMI Induced Conservation & Efficiency Energy	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (3,105)	\$ (7,004)	\$ (10,238)	\$ (13,497)	\$ (11,218)
73		Load Management Conserved Energy	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
74		Demand Response Conserved Energy	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (1,417)
75		Property Tax (Savings) - Legacy Meters	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (33)	\$ (97)	\$ (235)	\$ (500)	\$ (835)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)
76		LAUF Gas Reduction Enabled	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (22)	\$ (173)	\$ (181)	\$ (190)
77		Terminal Value	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
78		Total Non Capital Benefits before Costs	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (126)	\$ (2,471)	\$ (5,055)	\$ (13,718)	\$ (30,461)	\$ (51,037)	\$ (63,819)	\$ (76,914)	\$ (89,093)
79		Common	\$ 749	\$ 131	\$ 632	\$ 1,308	\$ 1,167	\$ 2,713	\$ 6,013	\$ 4,864	\$ 6,682	\$ 7,567	\$ 9,779	\$ 10,636	\$ 10,798	\$ 11,050
80		Meters, Modules & Communications	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 102	\$ 969	\$ 1,310	\$ 1,713	\$ 3,692	\$ 6,135	\$ 6,598	\$ 6,599	\$ 6,599
81		Other O&M	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 1,538	\$ 2,444	\$ 1,956	\$ 1,833	\$ 4,179	\$ 5,275	\$ 7,938	\$ 21,493	\$ 9,299
82		Load Control Program	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 195	\$ 665	\$ 2,567	\$ 3,017	\$ 2,726	\$ 2,625
83		Demand Response Program	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 16	\$ 194	\$ 2,219	\$ 986	\$ 606	\$ 618
84		Property Taxes - AMI	\$ -	\$ 24	\$ 44	\$ 75	\$ 57	\$ 343	\$ 874	\$ 1,583	\$ 3,118	\$ 6,094	\$ 7,535	\$ 7,346	\$ 7,185	\$ 6,999
85		Total Non Capital Benefits Net	\$ 749	\$ 155	\$ 676	\$ 1,383	\$ 1,224	\$ 4,569	\$ 7,829	\$ 4,659	\$ (161)	\$ (8,069)	\$ (17,526)	\$ (27,298)	\$ (27,507)	\$ (51,903)

Smart Grid Program - Benefits and O&M Costs

Line No.	(a) Utility	(b) Description	(c) 2021	(d) 2022	(e) 2023	(f) 2024	(g) 2025	(h) 2026	(i) 2027	(j) 2028	(k) 2029	(l) 2030	(m) 2031	(n) 2032	(o) Total 2007-2032
33	Electric	Meter Reading	\$ (20,935)	\$ (21,563)	\$ (22,210)	\$ (22,876)	\$ (23,562)	\$ (24,269)	\$ (24,997)	\$ (25,747)	\$ (26,519)	\$ (27,315)	\$ (28,134)	\$ (28,978)	\$ (390,041)
34		Uncollectible Expense	\$ (8,681)	\$ (8,855)	\$ (9,032)	\$ (9,213)	\$ (9,397)	\$ (9,585)	\$ (9,776)	\$ (9,972)	\$ (10,171)	\$ (10,375)	\$ (10,582)	\$ (10,794)	\$ (151,509)
35		Other O&M	\$ (8,826)	\$ (9,104)	\$ (9,363)	\$ (9,613)	\$ (9,865)	\$ (10,124)	\$ (10,392)	\$ (10,669)	\$ (10,953)	\$ (11,247)	\$ (11,550)	\$ (11,863)	\$ (165,570)
36		AC Load Control Avoided Generation, Transmission	\$ (6,080)	\$ (7,333)	\$ (8,458)	\$ (9,395)	\$ (10,384)	\$ (11,391)	\$ (12,424)	\$ (13,498)	\$ (14,596)	\$ (15,718)	\$ (16,856)	\$ (18,009)	\$ (153,414)
37		Demand Response Avoided Generation, Transmission	\$ (10,982)	\$ (10,976)	\$ (10,968)	\$ (12,320)	\$ (14,135)	\$ (16,186)	\$ (18,299)	\$ (20,308)	\$ (20,695)	\$ (21,089)	\$ (21,489)	\$ (21,898)	\$ (202,645)
38		Theft Reduction	\$ (27,997)	\$ (35,010)	\$ (39,052)	\$ (39,833)	\$ (40,629)	\$ (41,442)	\$ (42,271)	\$ (43,116)	\$ (43,978)	\$ (44,858)	\$ (45,755)	\$ (46,670)	\$ (561,029)
39		AMI Induced Conservation & Efficiency Energy	\$ (8,683)	\$ (8,857)	\$ (9,034)	\$ (9,215)	\$ (9,399)	\$ (9,587)	\$ (9,779)	\$ (9,974)	\$ (10,174)	\$ (10,377)	\$ (10,585)	\$ (10,797)	\$ (152,354)
40		Load Management Conserved Energy	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
41		Demand Response Conserved Energy	\$ (1,577)	\$ (1,681)	\$ (1,744)	\$ (1,810)	\$ (1,865)	\$ (1,920)	\$ (1,978)	\$ (2,037)	\$ (2,097)	\$ (2,156)	\$ (2,215)	\$ (2,270)	\$ (24,767)
42		Property Tax (Savings) - Legacy Meters	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (18,767)
43		Terminal Value	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (244,979)
44		Total Non Capital Benefits before Costs - Electric	\$ (94,829)	\$ (104,445)	\$ (110,927)	\$ (115,340)	\$ (120,304)	\$ (125,571)	\$ (130,983)	\$ (136,388)	\$ (140,251)	\$ (144,202)	\$ (148,234)	\$ (152,324)	\$ (2,065,075)
45		Common	\$ 11,294	\$ 11,525	\$ 10,781	\$ 11,012	\$ 11,258	\$ 11,510	\$ 11,774	\$ 12,032	\$ 12,297	\$ 12,579	\$ 12,868	\$ 13,171	\$ 209,249
46		Meters, Modules & Communications	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 90,579
47		Customer Engagement	\$ 3,598	\$ 3,614	\$ 3,631	\$ 3,647	\$ 3,664	\$ 3,682	\$ 3,700	\$ 3,718	\$ 3,736	\$ 3,755	\$ 3,775	\$ 3,794	\$ 100,270
48		Load Control Program	\$ 2,530	\$ 1,690	\$ 587	\$ 1,573	\$ 1,590	\$ 1,608	\$ 1,627	\$ 1,647	\$ 1,668	\$ 1,689	\$ 1,712	\$ 1,736	\$ 31,450
49		Demand Response Program	\$ 874	\$ 416	\$ 460	\$ 669	\$ 683	\$ 696	\$ 710	\$ 724	\$ 739	\$ 754	\$ 769	\$ 784	\$ 12,918
50		Property Taxes - AMI	\$ 6,138	\$ 5,987	\$ 5,873	\$ 5,712	\$ 5,556	\$ 5,399	\$ 5,253	\$ 5,124	\$ 5,025	\$ 4,981	\$ 5,002	\$ 5,051	\$ 102,276
51		Total Non Capital Benefits Net - Electric	\$ (65,167)	\$ (75,985)	\$ (84,367)	\$ (87,499)	\$ (92,325)	\$ (97,448)	\$ (102,691)	\$ (107,915)	\$ (111,558)	\$ (115,217)	\$ (118,881)	\$ (122,560)	\$ (1,518,333)
52															
53	Gas	Meter Reading	\$ (2,734)	\$ (2,816)	\$ (2,901)	\$ (2,988)	\$ (3,077)	\$ (3,170)	\$ (3,265)	\$ (3,363)	\$ (3,464)	\$ (3,567)	\$ (3,674)	\$ (3,785)	\$ (49,122)
54		Uncollectible Expense	\$ (2,276)	\$ (2,321)	\$ (2,367)	\$ (2,415)	\$ (2,463)	\$ (2,512)	\$ (2,563)	\$ (2,614)	\$ (2,666)	\$ (2,719)	\$ (2,774)	\$ (2,829)	\$ (38,585)
55		Other O&M	\$ (484)	\$ (501)	\$ (519)	\$ (537)	\$ (556)	\$ (575)	\$ (595)	\$ (615)	\$ (636)	\$ (658)	\$ (680)	\$ (703)	\$ (9,111)
56		Theft Reduction	\$ (4,501)	\$ (5,570)	\$ (7,363)	\$ (7,601)	\$ (7,753)	\$ (7,908)	\$ (8,066)	\$ (8,227)	\$ (8,392)	\$ (8,560)	\$ (8,731)	\$ (8,906)	\$ (97,824)
57		AMI Induced Conservation & Efficiency Energy	\$ (2,846)	\$ (3,002)	\$ (3,174)	\$ (3,352)	\$ (3,419)	\$ (3,487)	\$ (3,557)	\$ (3,628)	\$ (3,701)	\$ (3,775)	\$ (3,850)	\$ (3,927)	\$ (50,888)
58		LAUF Gas Reduction Enabled	\$ (201)	\$ (213)	\$ (225)	\$ (238)	\$ (243)	\$ (248)	\$ (252)	\$ (256)	\$ (260)	\$ (263)	\$ (267)	\$ (270)	\$ (3,502)
59		Terminal Value	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (33,406)
60		Total Non Capital Benefits before Costs - Gas	\$ (13,042)	\$ (14,424)	\$ (16,550)	\$ (17,131)	\$ (17,511)	\$ (17,900)	\$ (18,297)	\$ (18,703)	\$ (19,119)	\$ (19,543)	\$ (19,977)	\$ (20,419)	\$ (282,439)
61		Common	\$ 1,540	\$ 1,572	\$ 1,470	\$ 1,502	\$ 1,535	\$ 1,570	\$ 1,606	\$ 1,641	\$ 1,677	\$ 1,715	\$ 1,755	\$ 1,796	\$ 26,319
62		Meters, Modules & Communications	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 22,324
63		Property Taxes - AMI	\$ 685	\$ 644	\$ 603	\$ 563	\$ 522	\$ 482	\$ 441	\$ 401	\$ 362	\$ 334	\$ 326	\$ 326	\$ 9,792
64		Total Non Capital Benefits Net - Gas	\$ (9,446)	\$ (10,837)	\$ (13,105)	\$ (13,695)	\$ (14,083)	\$ (14,477)	\$ (14,879)	\$ (15,290)	\$ (15,708)	\$ (16,123)	\$ (16,525)	\$ (16,925)	\$ (224,003)
65															
66	Total	Meter Reading	\$ (23,669)	\$ (24,379)	\$ (25,110)	\$ (25,864)	\$ (26,639)	\$ (27,439)	\$ (28,262)	\$ (29,110)	\$ (29,983)	\$ (30,882)	\$ (31,809)	\$ (32,763)	\$ (439,164)
67		Uncollectible Expense	\$ (10,957)	\$ (11,176)	\$ (11,399)	\$ (11,627)	\$ (11,860)	\$ (12,097)	\$ (12,339)	\$ (12,586)	\$ (12,838)	\$ (13,094)	\$ (13,356)	\$ (13,623)	\$ (190,094)
68		Other O&M	\$ (9,310)	\$ (9,605)	\$ (9,882)	\$ (10,150)	\$ (10,420)	\$ (10,699)	\$ (10,987)	\$ (11,284)	\$ (11,590)	\$ (11,905)	\$ (12,231)	\$ (12,566)	\$ (174,681)
69		AC Load Control Avoided Generation, Transmission	\$ (6,080)	\$ (7,333)	\$ (8,458)	\$ (9,395)	\$ (10,384)	\$ (11,391)	\$ (12,424)	\$ (13,498)	\$ (14,596)	\$ (15,718)	\$ (16,856)	\$ (18,009)	\$ (153,414)
70		Demand Response Avoided Generation, Transmission	\$ (10,982)	\$ (10,976)	\$ (10,968)	\$ (12,320)	\$ (14,135)	\$ (16,186)	\$ (18,299)	\$ (20,308)	\$ (20,695)	\$ (21,089)	\$ (21,489)	\$ (21,898)	\$ (202,645)
71		Theft Reduction	\$ (32,498)	\$ (40,580)	\$ (46,415)	\$ (47,434)	\$ (48,382)	\$ (49,350)	\$ (50,337)	\$ (51,344)	\$ (52,370)	\$ (53,418)	\$ (54,486)	\$ (55,576)	\$ (658,852)
72		AMI Induced Conservation & Efficiency Energy	\$ (11,529)	\$ (11,860)	\$ (12,209)	\$ (12,567)	\$ (12,818)	\$ (13,074)	\$ (13,336)	\$ (13,602)	\$ (13,875)	\$ (14,152)	\$ (14,435)	\$ (14,724)	\$ (203,245)
73		Load Management Conserved Energy	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
74		Demand Response Conserved Energy	\$ (1,577)	\$ (1,681)	\$ (1,744)	\$ (1,810)	\$ (1,865)	\$ (1,920)	\$ (1,978)	\$ (2,037)	\$ (2,097)	\$ (2,156)	\$ (2,215)	\$ (2,270)	\$ (24,767)
75		Property Tax (Savings) - Legacy Meters	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (18,767)
76		LAUF Gas Reduction Enabled	\$ (201)	\$ (213)	\$ (225)	\$ (238)	\$ (243)	\$ (248)	\$ (252)	\$ (256)	\$ (260)	\$ (263)	\$ (267)	\$ (270)	\$ (3,502)
77		Terminal Value	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (278,385)
78		Total Non Capital Benefits before Costs	\$ (107,871)	\$ (118,869)	\$ (127,477)	\$ (132,471)	\$ (137,815)	\$ (143,471)	\$ (149,280)	\$ (155,091)	\$ (159,370)	\$ (163,745)	\$ (168,211)	\$ (172,771)	\$ (2,347,514)
79		Common	\$ 12,835	\$ 13,097	\$ 12,251	\$ 12,513	\$ 12,793	\$ 13,080	\$ 13,380	\$ 13,672	\$ 13,974	\$ 14,294	\$ 14,623	\$ 14,967	\$ 235,568
80		Meters, Modules & Communications	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 112,903
81		Other O&M	\$ 3,598	\$ 3,614	\$ 3,631	\$ 3,647	\$ 3,664	\$ 3,682	\$ 3,700	\$ 3,718	\$ 3,736	\$ 3,755	\$ 3,775	\$ 3,794	\$ 100,270
82		Load Control Program	\$ 2,530	\$ 1,690	\$ 587	\$ 1,573	\$ 1,590	\$ 1,608	\$ 1,627	\$ 1,647	\$ 1,668	\$ 1,689	\$ 1,712	\$ 1,736	\$ 31,450
83		Demand Response Program	\$ 874	\$ 416	\$ 460	\$ 669	\$ 683	\$ 696	\$ 710	\$ 724	\$ 739	\$ 754	\$ 769	\$ 784	\$ 12,918
84		Property Taxes - AMI	\$ 6,823	\$ 6,631	\$ 6,477	\$ 6,275	\$ 6,078	\$ 5,881	\$ 5,695	\$ 5,526	\$ 5,388	\$ 5,315	\$ 5,328	\$ 5,377	\$ 112,068
85		Total Non Capital Benefits Net	\$ (74,613)	\$ (86,822)	\$ (97,472)	\$ (101,194)	\$ (106,408)	\$ (111,926)	\$ (117,570)	\$ (123,205)	\$ (127,266)	\$ (131,339)	\$ (135,406)	\$ (141,893)	\$ (1,742,336)

Smart Grid Program - Annual Calculation of Pretax Return on AMI Investment

Line No.	(a) Utility	(b) Description	(c) 2007	(d) 2008	(e) 2009	(f) 2010	(g) 2011	(h) 2012	(i) 2013	(j) 2014	(k) 2015	(l) 2016	(m) 2017	(n) 2018	(o) 2019	(p) 2020
86	Electric	Beginning Plant in Service Value	\$ -	\$ 7,889	\$ 25,487	\$ 49,111	\$ 83,376	\$ 116,739	\$ 151,423	\$ 199,494	\$ 251,849	\$ 345,377	\$ 460,025	\$ 486,608	\$ 459,004	\$ 457,704
87		Ending Plant in Service Value	\$ 7,889	\$ 25,487	\$ 49,111	\$ 83,376	\$ 116,739	\$ 151,423	\$ 199,494	\$ 251,849	\$ 345,377	\$ 460,025	\$ 486,608	\$ 459,004	\$ 457,704	\$ 429,447
88																
89		Average Plant in Service Value [(Line 86 + Line 87)/2]	\$ 3,945	\$ 16,688	\$ 37,299	\$ 66,244	\$ 100,058	\$ 134,081	\$ 175,459	\$ 225,672	\$ 298,613	\$ 402,701	\$ 473,317	\$ 472,806	\$ 458,354	\$ 443,575
90		Pre-tax Cost of Capital (Actual approved amounts through 2022, then most recently approved after 2022)			9.76%	9.76%	9.82%	9.13%	9.13%	9.13%	8.90%	8.90%	8.58%	7.32%	7.40%	7.40%
91																
92																
93		Pretax Return on AMI Investment	\$ -	\$ -	\$ 3,640	\$ 6,465	\$ 9,826	\$ 12,242	\$ 16,019	\$ 20,604	\$ 26,577	\$ 35,840	\$ 40,611	\$ 34,609	\$ 33,918	\$ 32,825
94																
95	Gas	Beginning Plant in Service Value	\$ -	\$ 1,076	\$ 3,769	\$ 7,641	\$ 13,186	\$ 18,032	\$ 21,449	\$ 25,061	\$ 28,177	\$ 39,235	\$ 65,277	\$ 76,472	\$ 71,418	\$ 66,459
96		Ending Plant in Service Value	\$ 1,076	\$ 3,769	\$ 7,641	\$ 13,186	\$ 18,032	\$ 21,449	\$ 25,061	\$ 28,177	\$ 39,235	\$ 65,277	\$ 76,472	\$ 71,418	\$ 66,459	\$ 61,502
97																
98		Average Plant in Service Value	\$ 538	\$ 2,422	\$ 5,705	\$ 10,413	\$ 15,609	\$ 19,741	\$ 23,255	\$ 26,619	\$ 33,706	\$ 52,256	\$ 70,874	\$ 73,945	\$ 68,939	\$ 63,981
99		Pre-tax Cost of Capital (Actual approved amounts through 2021, then most recently approved after 2021)			9.76%	9.76%	9.82%	9.13%	9.13%	9.13%	8.90%	8.90%	8.58%	7.32%	7.40%	7.40%
100																
101																
102		Pretax Return on AMI Investment	\$ -	\$ -	\$ 557	\$ 1,016	\$ 1,533	\$ 1,802	\$ 2,123	\$ 2,430	\$ 3,000	\$ 4,651	\$ 6,081	\$ 5,413	\$ 5,101	\$ 4,735
103																
104	Total	Beginning Plant in Service Value	\$ -	\$ 8,965	\$ 29,256	\$ 56,752	\$ 96,562	\$ 134,771	\$ 172,872	\$ 224,556	\$ 280,026	\$ 384,613	\$ 525,303	\$ 563,080	\$ 530,423	\$ 524,163
105		Ending Plant in Service Value	\$ 8,965	\$ 29,256	\$ 56,752	\$ 96,562	\$ 134,771	\$ 172,872	\$ 224,556	\$ 280,026	\$ 384,613	\$ 525,303	\$ 563,080	\$ 530,423	\$ 524,163	\$ 490,949
106																
107		Average Plant in Service Value	\$ 4,483	\$ 19,111	\$ 43,004	\$ 76,657	\$ 115,666	\$ 153,821	\$ 198,714	\$ 252,291	\$ 332,319	\$ 454,958	\$ 544,191	\$ 546,751	\$ 527,293	\$ 507,556
108		Pre-tax Cost of Capital (Actual approved amounts through 2022, then most recently approved after 2022)			9.76%	9.76%	9.82%	9.13%	9.13%	9.13%	8.90%	8.90%	8.58%	7.32%	7.40%	7.40%
109																
110		Pretax Return on AMI Investment	\$ -	\$ -	\$ 4,197	\$ 7,482	\$ 11,358	\$ 14,044	\$ 18,143	\$ 23,034	\$ 29,576	\$ 40,491	\$ 46,692	\$ 40,022	\$ 39,020	\$ 37,559

Smart Grid Program - Annual Calculation of Pretax Return on AMI Investment

Line No.	(a) Utility	(b) Description	(c) 2021	(d) 2022	(e) 2023	(f) 2024	(g) 2025	(h) 2026	(i) 2027	(j) 2028	(k) 2029	(l) 2030	(m) 2031	(n) 2032
86	Electric	Beginning Plant in Service Value	\$ 429,447	\$ 397,959	\$ 364,445	\$ 327,628	\$ 290,752	\$ 253,237	\$ 218,652	\$ 188,828	\$ 161,820	\$ 138,271	\$ 118,406	\$ 101,844
87		Ending Plant in Service Value	\$ 397,959	\$ 364,445	\$ 327,628	\$ 290,752	\$ 253,237	\$ 218,652	\$ 188,828	\$ 161,820	\$ 138,271	\$ 118,406	\$ 101,844	\$ 87,528
88														
89		Average Plant in Service Value [(Line 86 + Line 87)/2]	\$ 413,703	\$ 381,202	\$ 346,037	\$ 309,190	\$ 271,994	\$ 235,945	\$ 203,740	\$ 175,324	\$ 150,045	\$ 128,338	\$ 110,125	\$ 94,686
90		Pre-tax Cost of Capital (Actual approved amounts through 2022, then most recently approved after 2022)	7.07%	7.03%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%
91														
92														
93		Pretax Return on AMI Investment	\$ 29,249	\$ 26,814	\$ 24,445	\$ 21,842	\$ 19,214	\$ 16,668	\$ 14,393	\$ 12,385	\$ 10,600	\$ 9,066	\$ 7,779	\$ 6,689
94														
95	Gas	Beginning Plant in Service Value	\$ 61,502	\$ 56,319	\$ 51,035	\$ 45,758	\$ 40,524	\$ 35,252	\$ 30,525	\$ 26,552	\$ 22,930	\$ 19,648	\$ 16,723	\$ 14,156
96		Ending Plant in Service Value	\$ 56,319	\$ 51,035	\$ 45,758	\$ 40,524	\$ 35,252	\$ 30,525	\$ 26,552	\$ 22,930	\$ 19,648	\$ 16,723	\$ 14,156	\$ 11,768
97														
98		Average Plant in Service Value	\$ 58,911	\$ 53,677	\$ 48,397	\$ 43,141	\$ 37,888	\$ 32,888	\$ 28,538	\$ 24,741	\$ 21,289	\$ 18,185	\$ 15,439	\$ 12,962
99		Pre-tax Cost of Capital (Actual approved amounts through 2022, then most recently approved after 2022)	7.07%	7.03%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%
100														
101														
102		Pretax Return on AMI Investment	\$ 4,165	\$ 3,776	\$ 3,419	\$ 3,048	\$ 2,676	\$ 2,323	\$ 2,016	\$ 1,748	\$ 1,504	\$ 1,285	\$ 1,091	\$ 916
103														
104	Total	Beginning Plant in Service Value	\$ 490,949	\$ 454,278	\$ 415,480	\$ 373,386	\$ 331,276	\$ 288,489	\$ 249,177	\$ 215,380	\$ 184,750	\$ 157,919	\$ 135,128	\$ 116,000
105		Ending Plant in Service Value	\$ 454,278	\$ 415,480	\$ 373,386	\$ 331,276	\$ 288,489	\$ 249,177	\$ 215,380	\$ 184,750	\$ 157,919	\$ 135,128	\$ 116,000	\$ 99,296
106														
107		Average Plant in Service Value	\$ 472,614	\$ 434,879	\$ 394,433	\$ 352,331	\$ 309,882	\$ 268,833	\$ 232,279	\$ 200,065	\$ 171,334	\$ 146,524	\$ 125,564	\$ 107,648
108		Pre-tax Cost of Capital (Actual approved amounts through 2022, then most recently approved after 2022)	7.07%	7.03%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%	7.06%
109														
110		Pretax Return on AMI Investment	\$ 33,414	\$ 30,589	\$ 27,864	\$ 24,890	\$ 21,891	\$ 18,991	\$ 16,409	\$ 14,133	\$ 12,103	\$ 10,351	\$ 8,870	\$ 7,605

1000

Smart Grid Program - Calculation of Net Present Value of Net Revenue Requirements - Total

Line No.	(a) Year	(b) Description	(c) Pretax Return on AMI Investment	(d) AMI Depreciation	(e) Revenue Requirements - Avoided Capital Costs	(f) Total Non-Capital Benefits Net	(g) Net Revenue Requirements	(h) NPV Factor	(i) NPV of Net Revenue Requirements
111	2009		\$4.2	+	\$0.1	+	\$0.7	=	\$5.0
112	2010		\$7.5	+	0.2	+	1.4	=	9.0
113	2011		\$11.4	+	4.8	+	1.2	=	17.4
114	2012	First Year of Electric AMI Meter Installations	\$14.0	+	10.9	+	4.6	=	29.5
115	2013		\$18.1	+	14.5	+	7.8	=	40.4
116	2014		\$23.0	+	18.8	+	4.7	=	46.2
117	2015	First Year of Gas Module Installations	\$29.6	+	24.7	+	(0.2)	=	53.8
118	2016		\$40.5	+	33.2	+	(8.1)	=	65.2
119	2017	Electric AMI Meter and Gas Module Installations complete	\$46.7	+	40.0	+	(17.5)	=	68.7
120	2018		\$40.0	+	42.5	+	(27.3)	=	50.9
121	2019		\$39.0	+	44.5	+	(27.5)	=	51.9
122	2020		\$37.6	+	46.7	+	(51.9)	=	28.4
123	2021		\$33.4	+	47.6	+	(74.6)	=	2.6
124	2022		\$30.6	+	48.4	+	(86.8)	=	(11.5)
125	2023		\$27.9	+	49.1	+	(97.5)	=	(23.9)
126	2024		\$24.9	+	49.8	+	(101.2)	=	(29.7)
127	2025		\$21.9	+	50.4	+	(106.4)	=	(37.0)
128	2026		\$19.0	+	46.5	+	(111.9)	=	(49.1)
129	2027		\$16.4	+	41.0	+	(117.6)	=	(62.6)
130	2028		\$14.1	+	38.2	+	(123.2)	=	(73.1)
131	2029		\$12.1	+	34.6	+	(127.3)	=	(82.6)
132	2030		\$10.4	+	30.3	+	(131.3)	=	(92.4)
133	2031		\$8.9	+	26.6	+	(135.4)	=	(101.5)
134	2032		\$7.6	+	24.3	+	(139.5)	=	(108.9)
135	2032	Model Terminal Value					(286.1)	=	(286.1)
136		Total 2009-2032	\$538.7		\$767.7		(\$45.1)		(\$1,750.9)
							(\$489.6)		(\$59.6)

Smart Grid Program - Calculation of Net Present Value of Net Revenue Requirements - Electric Portion

(a)		(b)	(c)		(d)	(e)		(f)		(g)	(h)		(i)
Line No.	Year	Description	Pretax Return on AMI Investment	AMI Depreciation		Revenue Requirements - Avoided Capital Costs		Total Non-Capital Benefits Net	Net Revenue Requirements	NPV Factor	NPV of Net Revenue Requirements		
137	2009		\$3.640	+	\$0.1	+	\$0.0	+	\$0.6	=	\$4.3	x 1.000000	= \$4.3
138	2010		\$6.465	+	0.2	+	0.0	+	1.2	=	7.9	x 1.000000	= 7.9
139	2011		\$9.826	+	4.2	+	0.0	+	1.1	=	15.1	x 1.000000	= 15.1
140	2012	First Year of Electric AMI Meter Installations	\$12.242	+	9.5	+	0.0	+	4.4	=	26.2	x 1.000000	= 26.2
141	2013		\$16.019	+	12.8	+	(0.1)	+	7.8	=	36.5	x 0.916338	= 33.5
142	2014		\$20.604	+	16.7	+	(0.3)	+	4.6	=	41.6	x 0.839676	= 34.9
143	2015	First Year of Gas Module Installations	\$26.577	+	22.1	+	(0.4)	+	(0.3)	=	47.9	x 0.774313	= 37.1
144	2016		\$35.840	+	29.3	+	(0.4)	+	(8.3)	=	56.4	x 0.711031	= 40.1
145	2017	Electric AMI Meter and Gas Module Installations complete	\$40.611	+	34.9	+	(0.4)	+	(15.1)	=	60.0	x 0.662599	= 39.8
146	2018		\$34.609	+	37.0	+	(4.4)	+	(22.6)	=	44.7	x 0.654510	= 29.2
147	2019		\$33.918	+	39.0	+	(4.2)	+	(22.1)	=	46.7	x 0.606694	= 28.3
148	2020		\$32.825	+	41.2	+	(4.0)	+	(43.6)	=	26.4	x 0.564892	= 14.9
149	2021		\$29.249	+	42.1	+	(3.8)	+	(65.2)	=	2.3	x 0.540742	= 1.3
150	2022		\$26.814	+	42.8	+	(3.6)	+	(76.0)	=	(10.0)	x 0.506738	= (5.1)
151	2023		\$24.445	+	43.5	+	(3.4)	+	(84.4)	=	(19.8)	x 0.471966	= (9.3)
152	2024		\$21.842	+	44.1	+	(3.2)	+	(87.5)	=	(24.7)	x 0.440825	= (10.9)
153	2025		\$19.214	+	44.7	+	(2.9)	+	(92.3)	=	(31.3)	x 0.411739	= (12.9)
154	2026		\$16.668	+	41.4	+	(2.7)	+	(97.4)	=	(42.0)	x 0.384572	= (16.2)
155	2027		\$14.393	+	36.7	+	(2.5)	+	(102.7)	=	(54.1)	x 0.359197	= (19.4)
156	2028		\$12.385	+	34.2	+	(2.3)	+	(107.9)	=	(63.6)	x 0.335497	= (21.3)
157	2029		\$10.600	+	30.9	+	(2.0)	+	(111.6)	=	(72.1)	x 0.313360	= (22.6)
158	2030		\$9.066	+	27.0	+	(1.8)	+	(115.2)	=	(80.9)	x 0.292684	= (23.7)
159	2031		\$7.779	+	23.6	+	(1.5)	+	(118.9)	=	(89.0)	x 0.273373	= (24.3)
160	2032		\$6.689	+	21.5	+	(1.3)	+	(122.6)	=	(95.7)	x 0.255335	= (24.4)
161	2032	Model Terminal Value							(251.8)	=	(251.8)	x 0.255061	= (64.2)
162		Total 2009-2032	\$472.3		\$679.5		(\$45.1)		(\$1,525.6)		(\$418.8)		\$58.3

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS
OF
AMY M. CONRAD
ON BEHALF OF
CONSUMERS ENERGY COMPANY

May 2023

OFFICER PLAN		
Category	2022 MEASURE	Weighting
Safety	Employee Safety (Recordable Incident Rate & High-Risk Injuries (equally weighted) and Zero fatalities)	5%
People	Culture Index (Employee Empowerment, Employee Engagement and DEI)	5%
Customer	Customer Experience - CXi (Survey measuring Customer Experience)	5%
Reliability	Electric Reliability - SAIDI (System Average Interruption Duration Index)	5%
Price	Affordability (O&M Savings through Waste Elimination)	5%
Environmental	Methane Emission Reduction (Reduction of Methane Emissions through replacements of Mains and Services, and through implantation of gas venting protocols)	5%
Total Operational Goals		30%
Financial	Earnings (Earnings Per Share)	70%
Total Financial Goals		70%

NON-OFFICER PLAN		
Category	2022 MEASURE	Weighting
Safety	Employee Safety (Recordable Incident Rate & High-Risk Injuries (equally weighted) and Zero fatalities)	8%
People	Culture Index (Employee Empowerment, Employee Engagement and DEI)	8%
Customer	Customer Experience - CXi (Survey measuring Customer Experience)	8%
Reliability	Electric Reliability - SAIDI (System Average Interruption Duration Index)	8%
Price	Affordability (O&M Savings through Waste Elimination)	8%
Environmental	Methane Emission Reduction (Reduction of Methane Emissions through replacements of Mains and Services, and through implantation of gas venting protocols)	8%
Total Operational Goals		50%
Financial	Earnings (Earnings Per Share)	50%
Total Financial Goals		50%

2022 Plan Year						
	50% Threshold	75%	100% Target	125%	150%	175% Max
Employee Safety (Recordable Incidents Rate, High-Risk Injuries ¹ & No fatalities ²)	a.1.74 b. 34	a.1.31 b.32	a.1.20 b.30	a.1.15 b.28	a.1.10 b.26	a.1.03 b.24
Culture Index (Empowerment, Employee Engagement & DEI)	1 index at Target ³	2 index at Target ³	All 3 indexes at Target ³	2 indexes at Target & 1 Index above Target	1 index at Target & 2 Indexes above Target	3 Indexes above Target
Customer Experience (Survey Measuring CXi)	69	71	73	75	77	79
Electric Reliability (SAIDI)	190	185	180	175	165	152
Affordability⁴ (O&M Savings through Waste Elimination)	\$31M	\$35M	\$39M	\$43M	\$47M	\$55M
Methane Emission (Reduction through replacement of mains and services)	301MT	318MT	335MT	351MT	378MT	400MT

- 1 High-Risk = Recordable or non-recordable injury caused by a release of high energy
- 2 A fatality = zero payout for employee safety
- 3 Target = +2 from 2021 actual. Baseline will be reaffirmed in 2022 with move to new survey platform.
- 4 The Company is not seeking recovery of this measure as it is financial in nature.

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Target Pay Level Market Analysis
Non-officer

Case No.: U-21389
Exhibit No.: A-74 (AMC-2)
Page: 1 of 1
Witness: AMConrad
Date: May 2023

Exempt Jobs:
2022 Market study

Job Family	# of Employees	Avg. Annual Salary	Market Data	Avg. Salary vs. Market
<u>ACCOUNTING ANALYST</u>	40	\$ 112,717	\$ 118,091	-4.6%
<u>IT BUSINESS</u>	15	\$ 98,208	\$ 99,224	-1.0%
<u>IT PROJECT MGR</u>	21	\$ 153,028	\$ 143,538	6.6%
<u>BUSINESS SUPPORT</u>	150	\$ 89,748	\$ 87,101	3.0%
<u>COMMUNICATIONS</u>	23	\$ 94,892	\$ 93,826	1.1%
<u>ELEC./GAS FIELD LEADER</u>	227	\$ 127,986	\$ 128,370	-0.3%
<u>LEARNING & DEVELOPMENT</u>	67	\$ 95,050	Benchmark Pending	
<u>ENGINEER</u>	270	\$ 113,220	\$ 114,367	-1.0%
<u>ENGINEER TECH</u>	137	\$ 95,805	\$ 105,951	-9.6%
<u>ENVIRONMENTAL PLANNER</u>	15	\$ 113,700	\$ 115,502	-1.6%
<u>FORESTERY</u>	22	\$ 82,712	\$ 84,761	-2.4%
<u>GENERAL TECHICAL ANALYST</u>	58	\$ 91,074	\$ 90,521	0.6%
<u>HUMAN RESOURCES</u>	42	\$ 100,261	Benchmark Pending	
<u>SAFETY</u>	10	\$ 103,968	\$ 101,803	2.1%
<u>LABORATORY TECH</u>	11	\$ 108,078	\$ 101,135	6.9%
<u>RATE ANALYST</u>	13	\$ 108,382	\$ 107,418	0.9%
<u>EXCUTIVE ASSISTANT</u>	22	\$ 84,423	\$ 84,031	0.5%
<u>MAINTENANCE/PRODUCTION SUPV</u>	41	\$ 120,753	\$ 138,965	-13.1%
<u>SYSTEM CONTROLLER</u>	25	\$ 118,297	\$ 154,708	-23.5%
<u>PLANNER/SCHEDULER</u>	24	\$ 87,950	Benchmark Pending	
<u>CORPORATE ACCOUNT MANAGER</u>	20	\$ 120,443	\$ 123,512	-2.5%
<u>EPC PROJECT MANAGER</u>	62	\$ 129,055	Benchmark Pending	
<u>GENERATION ASSET MANAGER</u>	16	\$ 119,878	Benchmark Pending	
<u>FINANCIAL ANALYST</u>	29	\$ 112,930	\$ 117,922	-4.2%
<u>TAX</u>	7	\$ 118,257	\$ 112,485	5.1%
<u>IT TECHNICAL</u>	156	\$ 109,704	\$ 108,214	1.4%
<u>IT SECURITY</u>	49	\$ 120,196	\$ 111,427	7.9%
<u>IT ARCHITECTURE</u>	7	\$ 169,715	\$ 156,375	8.5%
Average				-0.8%

Total exempt employees as a
% of market survey matches 40%

% of total base salaries Exempt & Non-Exempt 48%

Non-Exempt:
2022 Market study

Job Family	# of Employees	Avg. Annual Salary	Market Data	Avg. Salary vs. Market
Administrative Support Job Family	172	\$57,541	\$50,307	14.4%
Customer Service Revenue Recovery Job Family	33	\$53,813	Benchmark Pending	
Operations Support Job Family	10	\$70,240	\$71,244	-1.4%
Technical Support Job Family	79	\$92,860	\$100,337	-7.5%
Distribution Project Delivery Job Family	147	\$82,499	Benchmark Pending	
Technician Job Family	391	\$74,358	\$85,247	-12.8%
Paralegal Job Family	2	\$95,090	\$108,353	-12.2%
Dispatcher Job Family	71	\$65,516	Benchmark Pending	
Average				-1.3%

Total non-exempt employees as a
% of market survey matches 71%

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Summary of Actual & Projected O&M Expenses
Annual Incentive O&M Expenses
(\$000)

Case No.: U-21389
Exhibit No.: A-75 (AMC-3)
Page: 1 of 2
Witness: AMConrad
Date: May 2023

	(a)	(b)	(c)
Line No.	Description	2022 Actual	Projected 12-Mos Ending 2/28/2025
1	Annual Incentive - Officer	779	795
	Labor (1)	779	795
	Material		
	Contractor		
	Non-Labor Overheads		
	Non-Labor Other		
2	Annual Incentive - Non Officer (EICP)	1,439	1,608
	Labor (1)	1,439	1,608
	Material		
	Contractor		
	Non-Labor Overheads		
	Non-Labor Other		
3	Total Annual Incentive O&M Expenses	\$ 2,218	\$ 2,403
	Labor	2,218	2,403
	Material	0	0
	Contractor	0	0
	Non-Labor Overheads	0	0
	Non-Labor Other	0	0

Footnotes

- (1) Includes only O&M for operational incentive goals
(2) Amounts represent 2022 EICP assuming payout at 100% for operational incentive goals
Amount of actual payout based on 2022 incentive operational goal results were:

Officer	1,122
Non-officer	1,554
Total	<u>\$2,676</u>

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Summary of O&M Expenses Projected Using Inflation
Annual Incentive O&M Expenses
(\$000)

Case No.: U-21389
Exhibit No.: A-75 (AMC-3)
Page: 2 of 2
Witness: AMConrad
Date: May 2023

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
		2022 Actual	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2022	Merit & Inflation 12 Mos Ending Dec 31, 2023	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2023	Merit & Inflation 12 Mos Ending Dec 31, 2024	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2024	Merit & Inflation 2 Mos Ending 2/28/2025	Other Adjustments	Projected O&M 12 Mos Ending 2/28/2025
				(c) * Inflation Rate		(e) * Inflation Rate		(g) * Inflation Rate		(b) + (d) + (f) + (h) + (i)
1	Annual Incentive - Officer	779	779	30	775	17	792	3	-34	795
	Labor	779	779	30	775	17	792	3	-34	795
	Material			0		0		0		0
	Contractor			0		0		0		0
	Non-Labor Overheads			0		0		0		0
	Non-Labor Other			0		0		0		0
2	Annual Incentive - Non Officer (EICP)	1,439	1,439	56	1,568	34	1,602	6	73	1,608
	Labor	1,439	1,439	56	1,568	34	1,602	6	73	1,608
	Material			0		0		0		0
	Contractor			0		0		0		0
	Non-Labor Overheads			0		0		0		0
	Non-Labor Other			0		0		0		0
3	Total Annual Incentive O&M Expenses	\$ 2,218	\$ 2,218	\$ 87	\$ 2,343	\$ 52	\$ 2,394	\$ 8	\$ 39	\$ 2,403
	Labor	2,218	2,218	87	2,343	52	2,394	8	39	2,403
	Material	0	0	0	0	0	0	0	0	0
	Contractor	0	0	0	0	0	0	0	0	0
	Non-Labor Overheads	0	0	0	0	0	0	0	0	0
	Non-Labor Other	0	0	0	0	0	0	0	0	0

Notes

	2023	2024	2025
4 Annual merit increase			
Annual merit increase	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated merit increase	3.9%	2.2%	0.4%
Annual inflation rates per WP-JCA-xx			
Annual inflation rates	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated inflation rate	3.9%	2.2%	0.4%

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

AUDRA L. CUMBERWORTH

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

Schedule B-5.3

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Actual and Project Capital Expenditures
 Security
 (\$000)

Case No.: U-21389
 Exhibit No.: A-12 (ALC-1)
 Schedule: B-5.3
 Page: 1 of 1
 Witness: ALCumberworth
 Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)
Line No.	Description	Historical Year 12 Mos Ended 12/31/2022	Projected Bridge Period 12 Mos Ending 12/31/2023	2 Mos Ending 2/29/2024	14 Mos Ending 2/29/2024	Projected Test Year 12 Mos Ending 2/28/2025
1	Enhancements	\$ 2,849	\$ 509	\$ 91	\$ 600	\$ 543
	Software	820	-	-	-	-
	Materials	7	-	-	-	-
	Labor	278	401	72	473	435
	Contractor Costs	1,626	-	-	-	-
	Overhead & Others	118	109	18	127	109
	Contingency	-	-	-	-	-
2	Security	\$ 2,807	\$ 1,155	\$ 941	\$ 2,095	\$ 6,494
	Software	475	-	-	-	-
	Materials	1,835	985	442	1,427	2,567
	Labor	115	136	108	244	854
	Contractor Costs	268	-	365	365	2,771
	Overhead & Others	115	34	26	60	302
	Contingency	-	-	-	-	-
	Total Capital	\$ 5,656	\$ 1,664	\$ 1,031	\$ 2,695	\$ 7,038
Projected						
		2 Mos Ending 2/28/2023	12 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025	26 Mos Ending 2/28/2025	
3	Enhancements	\$ 85	\$ 515	\$ 543	\$ 1,143	
	Software	-	-	-	-	
	Materials	-	-	-	-	
	Labor	67	406	435	908	
	Contractor Costs	-	-	-	-	
	Overhead & Others	18	109	109	235	
	Contingency	-	-	-	-	
4	Security	\$ 192	\$ 1,903	\$ 6,494	\$ 8,590	
	Software	-	-	-	-	
	Materials	164	1,263	2,567	3,994	
	Labor	23	221	854	1,098	
	Contractor Costs	-	365	2,771	3,136	
	Overhead & Others	6	54	302	362	
	Contingency	-	-	-	-	
	Total Capital	\$ 277	\$ 2,418	\$ 7,038	\$ 9,733	

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)
	SPEND YEAR	PROJECT NAME	PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/ BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric
1	2022	Enhancements-Security-Capital	Enhancements	Software	This initiative will use Capital funding to make enhancements to existing technology and to address requests generated by changing business requirements. O&M is included within this project to complete expense activities associated with Capital enhancements in accordance with accounting rules. As business conditions improve and change, compliance requirements evolve, and new capabilities are needed, new requirements surface that call for smaller effort software application changes (enhancements). Enhancing applications requires a short timeframe between inception and implementation and cannot and should not wait for rate case approval at an individual line-item level. Failure to make these changes to applications can have a direct negative impact on key customer and business processes, increase support costs, limit the Company's ability to consistently meet objectives, and increase security risk. The value of technology enhancements in the Security Portfolio lies in: (1) enhancing security protections by funding emerging or unplanned cyber security activities resulting from audits, incidents, or a changing threat landscape; (2) reducing the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage additional functionality available in the technology. The scope of technology applications encompasses: (1) making necessary system changes, and (2) updating documentation related to the changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as cyber security related platforms, cyber security incident response, physical security, security awareness, risk management, privacy, and compliance. Prior to implementing an enhancement, a review is completed to identify the best solution. During that review, requests for this funding are governed by a cross-functional board that routinely evaluates and prioritizes the work. Security enhancements fortify the Company's ability to make software technology changes as part of process improvements and regulatory changes, and to meet legally required system changes in a fast and nimble manner. Without funding for enhancements, the Company will be limited in its ability to quickly provide needed capabilities and improvements.	2022-12-31	(1.00)	2,848,804	819,685	7,001	278,446	1,625,659	117,813	141,707	0	0	55,135	86,214	358
2	2022	Enhancements-Security-O&M	Enhancements	Software	This initiative will use O&M funding to make enhancements to existing technology and to address requests generated by changing business requirements. As business conditions improve and change, compliance requirements evolve, and new capabilities are needed, new requirements surface that call for smaller effort software application changes (enhancements). Enhancing applications requires a short timeframe between inception and implementation and cannot and should not wait for rate case approval at an individual line-item level. Failure to make these changes to applications can have a direct negative impact on key customer and business processes, increase support costs, limit the Company's ability to consistently meet objectives, and increase security risk. The value of technology enhancements in the Security Portfolio lies in: (1) enhancing security protections by funding emerging or unplanned cyber security activities resulting from audits, incidents, or a changing threat landscape; (2) reducing the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage additional functionality available in the technology. While these small-work technology efforts are neither projects nor operational work, funding for resources is still required to maintain business agility in the digital environment. Included in the implementation are small changes and functionality improvement to existing technology investments for Security. The scope of technology applications encompasses: (1) making necessary system changes, and (2) updating documentation related to the changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as cyber security related platforms, cyber security incident response, physical security, security awareness, risk management, privacy and compliance. Prior to implementing an enhancement, a review is completed to identify the best solution. During that review, requests for this funding are governed by a cross-functional board comprised of representatives from each area that routinely evaluates and prioritizes the work and assesses requests for value using established cost-to-benefit ratio acceptance criteria. In addition, the overall enhancements budget is reviewed annually, and the alternative of a zero-budget allocation for enhancements is considered. This project fortifies the Company's ability to make software technology changes as part of process improvements and regulatory changes, and to meet legally required system changes. Without funding for enhancements, the Company will be limited in its ability to quickly provide needed capabilities and improvements.	2022-12-31	(1.00)	0	0	0	0	0	263,371	0	0	186,504	75,051	1,816	
3	SUBTOTAL 2022 Enhancements							2,848,804	819,685	7,001	278,446	1,625,659	117,813	405,078	0	0	241,639	161,265	2,174
4	2022	Badge Reader/Lock and Key Management System	Security	Hardware	This project will implement a physical smart lock and key management system throughout the Company's Electric service territories. Lack of key control makes facilities, specifically in Electric operations, vulnerable to accidental or intentional disruption to power supply which could cause large scale outages. Install badge readers, smart locks, and credentials on smart devices to grant access to electric assets. Current estimates show there are approximately 12,000 locks throughout the state, and the Company does not have a system to properly manage ownership of the associated physical keys or control over who uses the keys. Locks are not unique in nature and can be easily duplicated. Today's lock and key system allows for 24-hour site access without having the ability to limit outside contractor access. Lack of key control makes facilities, specifically in Electric operations, vulnerable to accidental or intentional disruption which could cause large scale outages. Completion of this project will provide value to the Company by: (1) providing an extra layer of protection which is our first defense against criminal acts; (2) determining core functionalities needed to ensure proper lock and key management state-wide for electric operations; (3) implementing a smart lock and key solution that will provide the physical security team remote deactivation capabilities and is easily audited; (4) installing a smart lock system; and (5) Leveraging credentials on mobile device and badge readers that can read both physical cards and digital. The scope of this project includes: (1) a determination of the different levels and functions of different smart lock systems available for electric facilities; and (2) purchase and implementation against a site plan of a lock and key management system specific for needs of Electric Operations.	2027-11-01	(1.00)	0	0	0	0	0	6,482	0	0	6,482	0	0	
5	2022	Passive Vulnerability Assessment	Security	Hardware	The Passive Vulnerability Assessment project will evaluate, procure, and deploy a tool to passively provide vulnerability assessment services to Operational technology (OT) Control Networks. Currently insufficient information exists to make security judgements within critical control networks. This project will obtain asset and vulnerability data from networks that may suffer negative impacts to reliability from active vulnerability scanning. A pilot will identify capabilities and determine approach between on-site deployment vs. off-site deployment. The project creates value for the Company by: (1) obtaining a more accurate and timely lists of assets; (2) improving the ability to make data-driven operational and security decisions; (3) completing an assessment of existing vulnerabilities in the environment without the risk of negatively impacting the reliability of the Company's electric generation and energy delivery systems; and (4) increasing incident response capability, speed, and effectiveness. The scope of this project encompasses evaluating a zero-cost proof-of-concept to determine capabilities and drive the decision to implement an on-site or off-site deployment strategy. As part of the review process the alternative considered was to implement a traditional vulnerability scanner to interrogate all systems on the network. However, this was determined not to be the best solution to fit the Company's requirements.	2020-12-01	(1.00)	(8,749)	(17,350)	10,083	(2,336)	447	406	8,838	0	128	2,189	6,206	335
6	2022	Physical Security Asset Refresh	Security	Hardware	This project will enhance or replace physical security assets to provide improved visibility and incident resolution related to security concerns. The Company has several thousand physical security asset devices currently in use including security cameras, motion detectors, intrusion detection systems, and card access systems. Current limitations include the lack of integrated solutions for centralized management, situational awareness, real time monitoring, compliance with regulations and guidelines, and faster responses to emergencies and incidents. This could result in the increase of potential security vulnerabilities, associated penalties, and reputational damage. The value provided by completing the project is to: (1) maintain compliance with State and Federal Regulations; (2) reduce redundancies by decreasing multiple camera dependencies and reducing gaps in functionality; and (3) optimize overall system performance. Included in the project is the enhancement or replacement of assets including (1) advanced door systems at Company buildings; (2) security cameras for monitoring capabilities; and (3) gate and lock systems, which include security cameras, motion detectors, intrusion detection systems, and card access systems. Alternatives considered include: (1) not refreshing physical security assets, and (2) defer a portion of the refresh of physical security assets per an asset refresh cycle industry standard. Alternatives 1 and 2 were not selected due to the risk of security concerns, incident resolution, and the inability to meet Federal Energy Regulatory commission (FERC) requirements. Alternative 2 was selected as it maintains compliance, reduces redundancies and gaps in functionality, and optimizes overall performance of physical security systems.	2022-12-31	(1.00)	869,992	0	1,717,521	(99,118)	(722,560)	(25,851)	110,095	0	2,574	104,252	1,253	2,015

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)			
	SPEND YEAR	PROJECT NAME	PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/ BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-Electric	MATERIAL COSTS-Electric	LABOR COSTS-Electric	CONTRACTOR COSTS-Electric	OVERHEAD & OTHER COSTS-Electric	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-Electric	MATERIAL COSTS-Electric	LABOR COSTS-Electric	CONTRACTOR COSTS-Electric	OVERHEAD & OTHER COSTS-Electric			
7	2022	Radar Intrusion Detection	Security	Hardware	This project will deploy new radar detection technology to enable the detection and tracking of individuals at Company sites. The radar will alert in real time and automatically dispatch Company security or law enforcement to investigate. The Company currently lacks the ability to detect potential intruders at Company sites across the state, including critical infrastructure sites such as city gates, compressor stations, sub stations, generation plants, and hydro sites. Currently, three sites have flawed fence intrusion detection technology that result in the recurrence of multiple false alarms that require an employee to investigate. Additionally, the current detection technology is unable to identify the difference between animals and humans, cannot set different boundaries, and prohibits the expansion of intrusion detection at many other critical sites. Any investigation that has to be done is hampered by the limitations of the current system, requiring manual frame by frame review of video playback is manually intensive and time consuming. Completion of this project will provide value to the Company through evaluating and selecting a more reliable intrusion detection solution that reduces the number of false alarms and prevents resources from being sent to a site for unnecessary investigations. Additionally, the project will improve the efficiency of investigations by reducing the amount of time it takes to review playback to develop a suspect or investigative leads. In most cases this will be eliminated completely. This added layer of security benefits both the Company through waste elimination, as well as the Company's customers through the safety and security of assets necessary to continue to serve. The scope of this project encompasses: (1) implementing radar intrusion detection options; (2) establishing a priority list of sites where the solution will be rolled out; and (3) installing updated radar intrusion at physical sites in the Company's service territory with a focus on critical infrastructure sites, including but not limited to gas compressor stations, electric substations, and hydro-electric sites. Alternative solutions considered included: (1) keeping the current fence intrusion detection solution; or (2) implementing a new intrusion detection solution. The preferred option is to implement a new solution, given the above limitations with the current solution.	2022-12-22	(1.00)	1,083,552	0	171,613	56,422	837,533	17,983	2,832	0	0	2,832	0	0	0	0	0
8	2022	Asset Refresh Program - Cyber Security	Security	Network	This project will replace cyber security infrastructure to support increasing system and application demands and to prevent system failures and service interruptions. When enterprise software or cyber security infrastructure used to support and enhance customer interactions is obsolete, these assets are more expensive to support and can be more difficult to keep current with security updates. This project will create value by maintaining the currency of the cyber security infrastructure for core enterprise software. These components are used to ensure the stability of technology for business operations. The scope of this project consists of: (1) annually replacing a subset of cyber security firewalls and servers in keeping with a 3-5 year hardware lifecycle; and (2) performing application upgrades. As part of the review process, the alternative solutions considered were to: (1) upgrade or replace only those assets identified in the plan; (2) upgrade or replace a portion of the assets identified in the plan. Option 1 was not chosen based on a continued refresh cycle for cyber security assets to avoid security risks, system vulnerabilities, and out-of-warranty repair costs. Option 2 was not chosen due to the security risk inherent with not replacing assets as per established standard refresh cycles enabling increased system vulnerabilities, and out-of-warranty repair costs.	2022-12-31	(1.00)	31,070	2,041	24,531	2,229	1,555	713	55,420	1,782	0	6,717	46,921	0	0	0	
9	2022	Security Island Protocol	Security	Network	This project will implement a mutual distrust between all Operational Technology (OT) and Information Technology (IT) networks. Recent cyber-attacks have shown the criticality of ensuring that OT systems are fully separate and can operate even while the corporate network is under attack. The Company currently does not have standard procedures for separating the OT systems from the corporate systems if they were under a cyber-attack, and has never tested the capability. This project provides value by ensuring network traffic destined for the corporate core routers, must first pass through a corporate, security managed firewall and come from an OT managed firewall. Additionally, any network traffic destined towards OT networks must first pass through an OT managed firewall and come from a corporate, security managed firewall. This project scope will include: (1) Create a playbook to perform severing procedure at mutual distrust locations and document all impacts (business or otherwise); (2) Evaluate additional sever points to best protect energy delivery operations; (3) Creation of mutual distrust architecture standard and documentation of all mutual distrust points in the environment; (4) Evaluate and implement an internet proxy solution within the operational technology network with specific allow lists; (5) Perform an exercise to test severing procedure and document impacts; (6) All OT systems and connectivity in the Utility regardless of electric or gas are in scope. CMS Enterprises is out of scope. As part of the review process alternative solutions having this capability was considered a must have and due to recent cyber-attacks, the effort was prioritized against other planned investments.	2021-10-13	(1.00)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	2022	AccessNOW	Security	Software	The AccessNOW project will implement configurable identity, Access Management functionality and best practices, and will enable compliance enforcement. This project will add value by: (1) reducing waste and failure points through automation; (2) improving and standardizing the business partner experience; (3) centralizing access management; (4) ensuring regulatory compliance; and (5) ensuring system stability and continuous improvement. The project scope includes implementing additional integrations to Active Directory domains that are not currently connected to the AccessNOW application and integration to the Company's SAP system(s) to allow for automation of SAP role access provisioning. In addition, the project scope includes: (1) design, configuration and testing of technical connections; (2) completion of application support pack upgrades to the next version(s) to maintain system stability and to stay current with vendor releases; and (3) review and implement enhancements to improve and standardize the business partner experience. As part of the review process the alternative solution considered was the use of manual processes. This option was not chosen because it was deemed too costly and inefficient.	2021-12-31	(0.95)	255,742	141,718	0	5,046	97,880	11,097	186,628	0	0	83,834	101,534	1,259	0	0	
11	2022	Application Currency-Security-Capital	Security	Software	This project will utilize both Capital and O&M funding to keep applications current for security and reliability. O&M is included in this project to complete expense activities associated with Capital upgrades. The Company manages a large number of applications in the technology landscape that require regular version upgrades to maintain vendor-supported software versions. Without vendor supported versions, the Company loses the ability to receive version updates and upgrades to address defects, patch security vulnerabilities, protect against cyber threats, protect data, and add new features. Failure to upgrade these applications can have a direct negative impact on key customer and business processes, increase support costs, increase unplanned outages, and increase cyber security vulnerabilities. Maintaining the appropriate versions of applications through application currency upgrades in the Security portfolio adds value by: (1) enabling the Company to maintain vendor support; (2) remediating vendor security vulnerabilities and enhancing security protections; (3) addressing vendor defects that impair stability and functionality, leading to fewer incidents due to outdated software; (4) addressing version interdependencies and compatibility between systems; and (5) allowing the Company to leverage new functionality available in the upgrades; (6) provides customer value through closing the gaps on vulnerabilities that can impact reliability; (7) provides a mechanism for proactive actions to stay ahead of evolving cyber threats. This is essential to delivering safe, reliable, and affordable service to the Company's customers. The application upgrades in scope are regularly prioritized based on considerations that include application criticality; number of versions behind the current available version; security and operational risk; operational impacts of performing the upgrade; ability to defer; and cost. The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes; (4) testing the upgraded software; and (5) updating documentation related to the integration changes. Applications within the Security portfolio are routinely evaluated to determine if and what upgrade efforts are necessary to maintain an appropriate level of currency, as well as the priority of those efforts. During that review, the alternative of delaying the timing of the individual upgrades is considered based on: (1) maintaining an optimal balance between keeping the application current and risking failure; (2) an increased number of incidents; (3) paying increased support costs; and (4) preventing employees from performing their daily tasks. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to keep the large number of applications in the technology landscape secure and operational through upgrades. Without these upgrades, the Company will fall further behind in maintaining vendor-supported software versions, increasing the cost and complexity of the upgrade in the future.	2022-12-31	(1.00)	258,399	223,153	0	23,818	0	11,428	1,505	1,122	0	383	0	0	0	0	0

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)				
	SPEND YEAR	PROJECT NAME	PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/ BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric				
12	2022	Application Currency-Security-O&M	Security	Software	This project will utilize both Capital and O&M funding to keep applications current for security and reliability. O&M is included in this project to complete expense activities associated with Capital upgrade. The Company manages a large number of applications in the technology landscape that require regular version upgrades to maintain vendor-supported software versions. Without vendor supported versions, the Company loses the ability to receive version updates and upgrades to address defects, patch security vulnerabilities, protect against cyber threats, protect data, and add new features. Failure to upgrade these applications can have a direct negative impact on key customer and business processes, increase support costs, increase unplanned outages, and increase cyber security vulnerabilities. Maintaining the appropriate versions of applications through application currency upgrades in the Security portfolio adds value by: (1) enabling the Company to maintain vendor support; (2) remediating vendor security vulnerabilities and enhancing security protections; (3) addressing vendor defects that impair stability and functionality; (4) addressing version interdependencies and compatibility between systems; and (5) allowing the Company to leverage additional functionality available in the upgrades. The application upgrades in scope are regularly prioritized based on considerations that include application criticality, number of versions behind the current available version, security and operational risk, operational impacts of performing the upgrade, ability to defer, and cost. The application upgrades in scope are regularly prioritized based on considerations that include application criticality, number of versions behind the current available version, security and operational risk, operational impacts of performing the upgrade, ability to defer, and cost. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, (4) testing the upgraded software, and (5) updating documentation related to the integration changes. Applications within the Security portfolio are routinely evaluated to determine if upgrade efforts are necessary to maintain an appropriate level of currency and then prioritized accordingly. During that evaluation process, the alternative of delaying the individual upgrades is considered based on: (1) maintaining an optimal balance between keeping the application current and risking failure; (2) an increase in the number of incidents; (3) paying increased support costs; and (4) prevention of employees from performing their daily tasks. This project makes ongoing upgrades and support for these applications possible and fortifies the Company's ability to keep the large number of security applications in the technology landscape secure and operational through upgrades. Without these upgrades, the Company will fall further behind in maintaining vendor-supported software versions, increasing the cost and complexity of the upgrade in the future.	2022-12-31	(1.00)								107,671	32,093							
13	2022	Business Continuity - Disaster Recovery Planning Tool	Security	Software	Prior period adjustment.	N/A	N/A	(8,599)	0	0	(5,463)	(1,159)	(1,976)	6,603	0	0	5,074	1,076	453				
14	2022	Mass Notification	Security	Software	The Mass Notification project will interact with the current Siemens fire panels to take advantage of communications protocols in existence at Consumers Energy physical sites. Similar to current capabilities used to announce fire events, the mass notification tool will allow for other event notifications to building participants. In addition, mass notification has capabilities to involve personal device notification via texting, email alerts, and social media updates. Included in the project is implementing communications software to tap into current fire panels in physical locations, allowing for remote connectivity to sites for access to software, allowing for personal device communications, tapping into current building security software, allowing for emailing of key threats/events, and allowing for social media interfacing. The value of completing the project is directly tied to employee safety, which is increased by allowing better communication to employees during emergency situations. As part of our review process the alternative solution considered was not to implement the capability and to continue notifying individual Consumers Energy site managers of life safety situations through emails, phone calls, and/or text messages.	2020-10-01	(1.06)	0	0	(901)	0	901	0	0	0	0	0	0	0	0	0		
15	2022	Operational Security Assessment	Security	Software	The Operational Security Assessment project is for the implementation of an automate tracking tool to plan audits and inspections, monitor progress in real-time and track corrective actions in one central and secure location. The implementation of the identified product will help to decrease vulnerabilities that could threaten Company assets. This project will establish: (1) a centralized location to securely manage and store assessments; (2) implement an automated tracking system; (3) prioritize deficiencies; and (4) track completion with an interface to the SAP work order system. This will decrease the manual effort needed by the physical assessors to complete their day to day work. With the automation of tracking audits and inspections the physical security team can: (1) increase their audits by 20%; (2) reduce the likelihood of impactful security events to customers; and (3) increase efficiency, which in turn reduces potential operating costs for the Company. An additional feature of the identified solution is the ability for executive leadership to track the assessment scores in real-time and identify long trends enabling proactive prevention. As part of the review process the alternative solution considered was to continue using the manual process of tracking in an Excel spreadsheet and entering tickets in the Request Management Center (RMC) SharePoint site but this strategy was deemed too costly and inefficient.	2021-06-30	(1.00)	(24,629)	0	0	0	(24,629)	0	0	0	0	0	0	0	0	0	0	
16	2022	Privacy Management Platform	Security	Software	The project will implement a solution for the management of a range of privacy functions including: data inventory, risk assessments, consent management, individual rights management, and request tracking. Additionally, the platform will support compliance monitoring and reporting as well as industry changes impacting the Privacy Program. The Company lacks a platform to manage an ever-changing regulatory environment and legislative changes that impact the Privacy Management Program. At current, the Privacy Management Program is managed using disparate systems, spreadsheets, and manual analysis. New requirements are expected in all phases of the privacy program management, including: privacy laws and regulation compliance, facilitated compliance program decision making, intake and fulfillment of data subject rights and requests, increased risk discovery and mitigation, and data subject access request. Without a common platform to manage program requirements, the Company risks timely compliance with regulatory commitment and successful overall program management. The project will add value by: (1) Implementing a solution for privacy management to manage sensitive data, data locations, data management processes, and data sharing controls; (2) providing new functionality for the collection documentation of data subject consent for data usage and management of internal and external data subject access requests; (3) provisioning a dashboard for privacy incidents; and (4) enabling compliance management for regulatory and legislative requirements for customer privacy and information security. This project encompasses the implementation of a new privacy management platform that includes: (1) data management, (2) process assessment, (3) customer consent management, (4) Artificial Intelligence, (5) data subject access and request management, (6) regulatory insights, and (7) dashboards and reporting capabilities. Alternative solutions considered include: (1) continue existing practices of manual oversight, (2) build a solution to meet our needs, or (3) buy a platform to perform these services. Alternative 1 was not selected because it does not mitigate the risk associated with manual management of increasing levels of regulatory and legislative privacy management oversight. Alternative 2 is not currently preferred as the team foresees a significant amount of customization, resource expertise and future system maintenance being needed. Alternative 3 is preferred because buying platform will allow us to keep current with the changing environment and meet requirements quicker.	2022-06-30	(1.00)	350,297	125,106	(87,941)	134,380	77,978	100,774	(31,830)	(34,515)	0	2,685	0	0	0	0	0	0
17	2022	Security Analytics	Security	Software	Prior period adjustment.	N/A	N/A	0	0	0	0	0	0	53,158	0	0	961	48,450	3,747				
18	SUBTOTAL 2022 Security								2,807,074	474,666	1,834,906	114,978	267,949	114,575	513,258	482	2,701	284,223	217,399	8,158			
19	TOTAL 2022								5,655,879	1,294,553	1,841,907	393,424	1,895,607	232,387	918,336	482	2,702	526,162	378,664	10,327			
20	2023	Enhancements-Security-Capital	Enhancements	Software	This initiative will use Capital funding to make enhancements to existing technology and to address requests generated by changing business requirements. O&M is included within this project to complete expense activities associated with Capital enhancements in accordance with accounting rules. As business conditions improve and change, compliance requirements evolve, and new capabilities are needed, new requirements surface that call for smaller effort software application changes (enhancements). Enhancing applications requires a short timeframe between inception and implementation and cannot and should not wait for rate case approval at an individual line-item level. Failure to make these changes to applications can have a direct negative impact on key customer and business processes, increase support costs, limit the Company's ability to consistently meet objectives, and increase security risk. The value of technology enhancements in the Security Portfolio lies in: (1) enhancing security protections by funding emerging or unplanned cyber security activities resulting from audits, incidents, or a changing threat landscape; (2) reducing the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage additional functionality available in the technology. The scope of technology applications encompasses: (1) making necessary system changes, and (2) updating documentation related to the changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as cyber security related platforms, cyber security incident response, physical security, security awareness, risk management, privacy, and compliance. Prior to implementing an enhancement, a review is completed to identify the best solution. During that review, requests for this funding are governed by a cross-functional board that routinely evaluates and prioritizes the work. Security enhancements fortify the Company's ability to make software technology changes as part of process improvements and regulatory changes, and to meet legally required system changes in a fast and nimble manner. Without funding for enhancements, the Company will be limited in its ability to quickly provide needed capabilities and improvements.	2023-12-31	(1.00)	509,400	0	0	400,728	0	108,672	48,750	0	0	39,000	0	0	0	0	0	5,750

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)
	SPEND YEAR	PROJECT NAME	PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/ BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-Electric	MATERIAL COSTS-Electric	LABOR COSTS-Electric	CONTRACTOR COSTS-Electric	OVERHEAD & OTHER COSTS-Electric	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-Electric	MATERIAL COSTS-Electric	LABOR COSTS-Electric	CONTRACTOR COSTS-Electric	OVERHEAD & OTHER COSTS-Electric
	2023	Enhancements-Security-O&M	Enhancements	Software	This initiative will use O&M funding to make enhancements to existing technology and to address requests generated by changing business requirements. As business conditions improve and change, compliance requirements evolve, and new capabilities are needed, new requirements surface that call for smaller effort software application changes (enhancements). Enhancing applications requires a short timeframe between inception and implementation and cannot and should not wait for rate case approval at an individual line-item level. Failure to make these changes to applications can have a direct negative impact on key customer and business processes, increase support costs, limit the Company's ability to consistently meet objectives, and increase security risk. The value of technology enhancements in the Security Portfolio lies in: (1) enhancing security protections by funding emerging or unplanned cyber security activities resulting from audits, incidents, or a changing threat landscape; (2) reducing the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage additional functionality available in the technology. While these small-work technology efforts are neither projects nor operational work, funding for resources is still required to maintain business agility in the digital environment. Included in the implementation are small changes and functionality improvement to existing technology investments for security. The scope of technology applications encompasses: (1) making necessary system changes, and (2) updating documentation related to the changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as cyber security related platforms, cyber security incident response, physical security, security awareness, risk management, privacy and compliance. Prior to implementing an enhancement, a review is completed to identify the best solution. During that review, requests for this funding are governed by a cross-functional board comprised of representatives from each area that routinely evaluates and prioritizes the work and assesses requests for value using established cost-to-benefit ratio acceptance criteria. In addition, the overall enhancements budget is reviewed annually, and the alternative of a zero-budget allocation for enhancements is considered. This project fortifies the Company's ability to make software technology changes as part of process improvements and regulatory changes, and to meet legally required system changes. Without funding for enhancements, the Company will be limited in its ability to quickly provide needed capabilities and improvements.	2023-12-31	(1.00)	0	0	0	0	0	0	349,692	0	0	279,753	0	69,938
21																			
22	SUBTOTAL	2021 Enhancements						509,400	0	0	400,728	0	106,672	396,442	0	0	318,753	0	79,688
	2023	Physical Security Asset Refresh	Security	Hardware	This project will enhance or replace physical security assets to provide improved visibility and incident resolution related to security concerns. The Company has several thousand physical security asset devices currently in use including security cameras, motion detectors, intrusion detection systems, and card access systems. Current limitations include the lack of integrated solutions for centralized management, situational awareness, real time monitoring, compliance with regulations and guidelines, and faster responses to emergencies and incidents. This could result in the increase of potential security vulnerabilities, associated penalties, and reputational damage. The value provided by completing the project is to: (1) maintain compliance with State and Federal Regulations; (2) reduce redundancies by decreasing multiple camera dependencies and reducing gaps in functionality; and (3) optimize overall system performance. Included in the project is the enhancement or replacement of assets including: (1) advanced door systems at Company buildings; (2) security cameras for monitoring capabilities; and (3) gate and lock systems, which include security cameras, motion detectors, intrusion detection systems, and card access systems. Alternatives considered include: (1) not refreshing physical security assets, and (2) defer a portion of the refresh of physical security assets per an asset refresh cycle industry standard. Alternatives 1 and 2 were not selected due to the risk of security concerns, incident resolution, and the inability to meet Federal Energy Regulatory commission (FERC) requirements. Alternative 2 was selected as it maintains compliance, reduces redundancies and gaps in functionality, and optimizes overall performance of physical security systems.	2023-12-31	(1.00)	1,086,720	0	984,840	81,504	0	20,376	3,900	0	3,120	0	0	780
23																			
	2023	Application Currency-Security-Capital	Security	Software	This project will utilize both Capital and O&M funding to keep applications current for security and reliability. O&M is included in this project to complete expense activities associated with Capital upgrades. The Company manages a large number of applications in the technology landscape that require regular version upgrades to maintain vendor-supported software versions. Without vendor supported versions, the Company loses the ability to receive version updates and upgrades to address defects, patch security vulnerabilities, protect against cyber threats, protect data, and add new features. Failure to upgrade these applications can have a direct negative impact on key customer and business processes, increase support costs, increase unplanned outages, and increase cyber security vulnerabilities. Maintaining the appropriate versions of applications through application currency upgrades in the Security portfolio adds value by: (1) enabling the Company to maintain vendor support; (2) remediating vendor security vulnerabilities and enhancing security protections; (3) addressing vendor defects that impair stability and functionality, leading to fewer incidents due to outdated software; (4) addressing version interdependencies and compatibility between systems; and (5) allowing the Company to leverage new functionality available in the upgrades; (6) provides customer value through closing the gaps on vulnerabilities that can impact reliability; (7) provides a mechanism for proactive actions to stay ahead of evolving cyber threats. This is essential to delivering safe, reliable, and affordable service to the Company's customers. The application upgrades in scope are regularly prioritized based on considerations that include application criticality; number of versions behind the current available version; security and operational risk; operational impacts of performing the upgrade; ability to defer; and cost. The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes; (4) testing the upgraded software; and (5) updating documentation related to the integration changes. Applications within the Security portfolio are routinely evaluated to determine if and what upgrade efforts are necessary to maintain an appropriate level of currency, as well as the priority of those efforts. During that review, the alternative of delaying the timing of the individual upgrades is considered based on: (1) maintaining an optimal balance between keeping the application current and risking failure; (2) an increased number of incidents; (3) paying increased support costs, and (4) preventing employees from performing their daily tasks. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to keep the large number of applications in the technology landscape secure and operational through upgrades. Without these upgrades, the Company will fall further behind in maintaining vendor-supported software versions, increasing the cost and complexity of the upgrade in the future.	2023-12-31	(1.00)	67,920	0	0	54,336	0	13,584	6,500	0	0	5,200	0	1,300
24																			
	2023	Application Currency-Security-O&M	Security	Software	This project will utilize both Capital and O&M funding to keep applications current for security and reliability. O&M is included in this project to complete expense activities associated with Capital upgrade. The Company manages a large number of applications in the technology landscape that require regular version upgrades to maintain vendor-supported software versions. Without vendor supported versions, the Company loses the ability to receive version updates and upgrades to address defects, patch security vulnerabilities, protect against cyber threats, protect data, and add new features. Failure to upgrade these applications can have a direct negative impact on key customer and business processes, increase support costs, increase unplanned outages, and increase cyber security vulnerabilities. Maintaining the appropriate versions of applications through application currency upgrades in the Security portfolio adds value by: (1) enabling the Company to maintain vendor support; (2) remediating vendor security vulnerabilities and enhancing security protections; (3) addressing vendor defects that impair stability and functionality; (4) addressing version interdependencies and compatibility between systems; and (5) allowing the Company to leverage additional functionality available in the upgrades. The application upgrades in scope are regularly prioritized based on considerations that include application criticality; number of versions behind the current available version; security and operational risk; operational impacts of performing the upgrade; ability to defer; and cost. The application upgrades in scope are regularly prioritized based on considerations that include application criticality; number of versions behind the current available version; security and operational risk; operational impacts of performing the upgrade; ability to defer; and cost. The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes; (4) testing the upgraded software; and (5) updating documentation related to the integration changes. Applications within the Security portfolio are routinely evaluated to determine if upgrade efforts are necessary to maintain an appropriate level of currency and then prioritized accordingly. During that evaluation process, the alternative of delaying the individual upgrades is considered based on: (1) maintaining an optimal balance between keeping the application current and risking failure; (2) an increase in the number of incidents; (3) paying increased support costs; and (4) prevention of employees from performing their daily tasks. This project makes ongoing upgrades and support for these applications possible and fortifies the Company's ability to keep the large number of security applications in the technology landscape secure and operational through upgrades. Without these upgrades, the Company will fall further behind in maintaining vendor-supported software versions, increasing the cost and complexity of the upgrade in the future.	2023-12-31	(1.00)	0	0	0	0	0	37,805	0	0	30,244	0	7,561	
25																			
26	SUBTOTAL	2023 Security						1,124,640	0	984,840	131,840	0	31,960	46,205	0	0	38,544	0	9,641
27	TOTAL	2023						1,664,040	0	984,840	536,568	0	142,632	446,646	0	0	357,317	0	89,329

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	
	SPEND YEAR	PROJECT NAME	PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/ BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric	
28	2024	Enhancements-Security-Capital	Enhancements	Software	This initiative will use Capital funding to make enhancements to existing technology and to address requests generated by changing business requirements. O&M is included in this project to complete expense activities associated with Capital enhancements in accordance with accounting rules. As business conditions improve and change, compliance requirements evolve, and new capabilities are needed, new requirements surface that call for smaller effort software application changes (enhancements). Enhancing applications requires a short timeframe between inception and implementation and cannot and should not wait for rate case approval at an individual line-item level. Failure to make these changes to applications can have a direct negative impact on key customer and business processes, increase support costs, limit the Company's ability to consistently meet objectives, and increase security risk. The value of technology enhancements in the Security Portfolio lies in: (1) enhancing security protections by funding emerging or unplanned cyber security activities resulting from audits, incidents, or a changing threat landscape; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage additional functionality available in the technology. The scope of technology applications encompasses: (1) making necessary system changes, and (2) updating documentation related to the changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as cyber security related platforms, cyber security incident response, physical security, security awareness, risk management, privacy, and compliance. Prior to implementing an enhancement, a review is completed to identify the best solution. During that review, requests for this funding are governed by a cross-functional board that routinely evaluates and prioritizes the work. Security enhancements fortify the Company's ability to make software technology changes as part of process improvements and regulatory changes, and to meet legally required system changes in a fast and nimble manner. Without funding for enhancements, the Company will be limited in its ability to quickly provide needed capabilities and improvements.	2024-12-31	(1.00)	543,360	0	0	434,688	0	108,672	26,000	0	0	20,800	0	0	5,200
29	2024	Enhancements-Security-O&M	Enhancements	Software	This initiative will use O&M funding to make enhancements to existing technology and to address requests generated by changing business requirements. As business conditions improve and change, compliance requirements evolve, and new capabilities are needed, new requirements surface that call for smaller effort software application changes (enhancements). Enhancing applications requires a short timeframe between inception and implementation and cannot and should not wait for rate case approval at an individual line-item level. Failure to make these changes to applications can have a direct negative impact on key customer and business processes, increase support costs, limit the Company's ability to consistently meet objectives, and increase security risk. The value of technology enhancements in the Security Portfolio lies in: (1) enhancing security protections by funding emerging or unplanned cyber security activities resulting from audits, incidents, or a changing threat landscape; (2) reducing the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage additional functionality available in the technology. While these small-work technology efforts are neither projects nor operational work, funding for resources is still required to maintain business agility in the digital environment. Included in the implementation are small changes and functionality improvement to existing technology investments for Security. The scope of technology applications encompasses: (1) making necessary system changes, and (2) updating documentation related to the changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as cyber security related platforms, cyber security incident response, physical security, security awareness, risk management, privacy and compliance. Prior to implementing an enhancement, a review is completed to identify the best solution. During that review, requests for this funding are governed by a cross-functional board comprised of representatives from each area that routinely evaluates and prioritizes the work and assesses requests for value using established cost-to-benefit ratio acceptance criteria. In addition, the overall enhancements budget is reviewed annually, and the alternative of a zero-budget allocation for enhancements is considered. This project fortifies the Company's ability to make software technology changes as part of process improvements and regulatory changes, and to meet legally required system changes. Without funding for enhancements, the Company will be limited in its ability to quickly provide needed capabilities and improvements.	2024-12-31	(1.00)	0	0	0	0	0	378,045	0	0	302,436	0	0	75,609	
30	SUBTOTAL 2024 Enhancements							543,360	0	0	434,688	0	108,672	404,045	0	0	323,236	0	80,809	
31	2024	Badge Reader/Lock and Key Management System	Security	Hardware	This project will implement a physical smart lock and key management system throughout the Company's Electric service territories. Lack of key control makes facilities, specifically in Electric operations, vulnerable to accidental or intentional disruption to power supply which could cause large scale outages. Install badge readers, smart locks, and credentials on smart devices to grant access to electric assets. Current estimates show there are approximately 12,000 locks throughout the state, and the Company does not have a system to properly manage ownership of the associated physical keys or control over who uses the keys. Locks are not unique in nature and can be easily duplicated. Today's lock and key system allows for 24-hour site access without having the ability to limit outside contractor access. Lack of key control makes facilities, specifically in Electric operations, vulnerable to accidental or intentional disruption which could cause large scale outages. Completion of this project will provide value to the Company by: (1) providing an extra layer of protection which is our first defense against criminal acts; (2) determining core functionalities needed to ensure proper lock and key management state-wide for electric operations; (3) implementing a smart lock and key solution that will provide the physical security team remote deactivation capabilities and is easily audited; (4) installing a smart lock system; and (5) Leveraging credentials on mobile device and badge readers that can read both physical cards and digital. The scope of this project includes: (1) a determination of the different levels and functions of different smart lock systems available for electric facilities; and (2) purchase and implementation against a site plan of a lock and key management system specific for needs of Electric Operations.	2027-11-01	(1.00)	1,698,000	0	169,800	339,600	1,103,700	84,900	92,950	0	28,600	52,000	0	12,350	
32	2024	Physical Security Asset Refresh	Security	Hardware	This project will enhance or replace physical security assets to provide improved visibility and incident resolution related to security concerns. The Company has several thousand physical security asset devices currently in use including security cameras, motion detectors, intrusion detection systems, and card access systems. Current limitations include the lack of integrated solutions for centralized management, situational awareness, real time monitoring, compliance with regulations and guidelines, and faster responses to emergencies and incidents. This could result in the increase of potential security vulnerabilities, associated penalties, and reputational damage. The value provided by completing the project is to: (1) maintain compliance with State and Federal Regulations; (2) reduce redundancies with less of a need for multiple cameras and reducing gaps in functionality; and (3) optimize overall system performance. Included in the project is the enhancement or replacement of assets including: (1) advanced door systems at Company buildings; (2) security cameras for monitoring capabilities; and (3) gate and lock systems, which include security cameras, motion detectors, intrusion detection systems, and card access systems. Alternatives considered include: (1) not refreshing physical security assets, and (2) defer a portion of the refresh of physical security assets per an asset refresh cycle industry standard. Alternatives 1 and 2 were not selected due to the risk of security concerns, incident resolution, and the inability to meet FERC requirements. The Alternative selected maintains compliance, reduces redundancies and gaps in functionality, and optimizes overall performance of physical security systems.	2024-12-31	(1.00)	2,105,520	0	1,494,240	81,504	509,400	20,376	6,500	0	3,120	2,600	780		
33	2024	Asset Refresh Program -Cyber Security	Security	Network	This project will replace cyber security infrastructure to support the security system and applications and to prevent system failures and service interruptions. When enterprise software or cyber security infrastructure used to support and enhance customer interactions is obsolete, these assets are more expensive to support and can be more difficult to keep current with security updates. This project will create value by maintaining the currency of the cyber security infrastructure for core enterprise software. These are used to ensure the stability of technology for business operations. The scope of this project consists of: (1) annually replacing a subset of cyber security firewalls and servers in keeping with a 3-5 year hardware lifecycle; and (2) performing application upgrades. As part of the review process, the alternatives considered were to: (1) upgrade or replace only those assets identified in the plan; (2) upgrade or replace a portion of the assets identified in the plan. Option 1 was not chosen based on continued refresh cycle for cyber security assets to avoid security risks, system vulnerabilities, and out-of-warranty repair costs. Options 2 was not chosen due to the security risk inherent with not replacing assets as per standard refresh cycles increased system vulnerabilities, and out-of-warranty repair costs.	2024-12-31	(1.00)	252,360	0	168,240	67,296	0	16,824	16,250	0	16,250	0	0	16,250	

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	
	SPEND YEAR	PROJECT NAME	PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/ BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric	
34	2024	Network Packet Capture and Storage	Security	Network	Boundary network traffic and holding this data for a period of time, allows this data to be utilized for identifying security threats, identification of network behavior, and forensic cyber analysis. This data can be utilized in both real time and investigative analysis. Consumers Energy does not have the capability to capture, store and analyze full content network traffic at its internet connection. This limits the organization's ability to detect, respond and forensically analyze attacks from a network perspective. The company will gain value from this project through: (1) Decrypting and recording all internet communication to and from the company; (2) storing this data for historic use; (3) Allow a cyber analyst to leverage this information when investigating questionable activity or an incident; (4) Enable us to further enhance our security posture and program (following the Center for Internet Security Controls Framework). This project will: (1) develop evaluation criteria to assess solutions that provide the ability to collect encrypted traffic from Internet Boundary Devices (i.e. Firewall), securely store this data for the retention period determined, and provide tools to utilize this data for search capabilities and analysis. This criteria will be based on industry best practices, such as NIST 800-86 Guide to Integrating Forensic Techniques into Incident Response and SANS research papers on Full Packet Capture, (2) Evaluate technical solutions and complete a design validation based on established criteria, (3) select and implement the best fit solution, (4) design new processes, standards, audits and metrics for consistent implementation and maintenance, (5) Training key stakeholders, systems administrators, and operations on how to properly use the system. Alternatives considered include: (1) continue utilization of existing network firewall real time information and storage, (2) capture less effective data, such as network NetFlow information. These alternatives provide limited benefit in the ability to decrypt all internet network traffic and the ability to retain this information for an extended period of time.	2024-12-31	(1.00)	841,200	0	785,120	44,864	0	11,216	32,500	0	0	26,000	0	0	6,500
35	2024	Access Now Upgrade 2024	Security	Software	The project will upgrade the AccessNow Application and Web Portal to modern Identity Manager Version. Upgrade AccessNow Application to modern version to be compliant with One Identity support agreement, upgrade Web Portal to standardized architecture, implement new functionality and features. Current AccessNow version will be end of life by December 2022 and will lose support from One Identity. Additionally, the application contains numerous customized workflows and integrations to support CE business and IT processes. This customization requires One Identity professional services to support ongoing use of the application and when changes are required. Current version of AccessNow is lacking proper data analytics and reporting for end users in order to identify waste elimination opportunities. Since AccessNow is an enterprise wide platform that supports access provisioning, access controls, and processes that support regulatory compliance including SOX, NERC CIP, TSA, and MPSC requirements, the risk of operating without vendor support includes the following: inability to perform critical defect fixes due to lack of vendor support, impacting ability to perform critical business functions; Software incompatibility with integrated systems such as Active Roles Server, upgrade other dependent software or hardware assets; Compliance issues, including non-compliance with regulatory standards, and the risk of fines that may result from non-compliance; increased operations and maintenance costs; Degraded or poor performance, or application reliability risk (downtime risk). Upgrade AccessNow application and web portal to current versioning to obtain at least 3 years of in support compliance as well as introduce new functionality including: 1) Simplifying processes through standardization of the web portal from customized to standard Angular code. Will reduce the dependency of professional services through reduction in complexity by transforming customized portal design into standardized Angular code 2) New and improved portal design for more intuitive end user experience and reduce human struggle in access request processes. 3) More responsive portal and better performance reducing user wait time in requesting access. 4) Implement analytic capabilities for real time analysis of processes and workflows to identify waste elimination opportunities. The project scope includes upgrading AccessNow Application and Web Portal to current Identity Manager Version. Remove customized code in web portal and transform into Out of The Box (OOTB) functionality in new version to standardize support of the web portal. Upgrade AccessNow Application to current version to implement new functionality including transforming customized code to standard OOTB functionality, new and improved front end user interface design, implement analytic capabilities for real time analysis of workflows and processes. Alternatives considered include: (1) Implement hot fixes to version 8.1.5 to ensure support through summer of 2023 -150-250k (3 months of professional services); (2) Upgrade only the application to current supported version. Will not include the new web portal, but will include new application functionality. 400k; (3) Migrate AccessNow functionality to existing Sailpoint platform to manage access requests, including Active Directory, SAP, and disconnected systems as well as existing API integrations. Estimated cost: \$1.25 million; (4) Remain on current version that will be unsupported as of Jan 2023. Any break / fix will require One Identity professional services. Patches and enhancements may introduce additional risk of version instability. The option selected was to proceed with the Access Now 2024 upgrade which would provide stream lined One Identity architecture and operations, onboarding processes, and a better end user experience.	2024-11-15	(1.00)	679,200	0	31,960	61,128	577,320	6,792	65,000	0	0	58,500	0	0	6,500
36	2024	Application Currency-Security-Capital	Security	Software	This project will utilize both Capital and O&M funding to keep applications current for security and reliability. O&M is included in this project to complete expense activities associated with Capital upgrades. The Company manages a large number of applications in the technology landscape that require regular version upgrades to maintain vendor-supported software versions. Without vendor supported versions, the Company loses the ability to receive version updates and upgrades to address defects, patch security vulnerabilities, protect against cyber threats, protect data, and add new features. Failure to upgrade these applications can have a direct negative impact on key customer and business processes, increase support costs, increase unplanned outages, and increase cyber security vulnerabilities. Maintaining the appropriate versions of applications through application currency upgrades in the Security portfolio adds value by: (1) enabling the Company to maintain vendor support; (2) remediating vendor security vulnerabilities and enhancing security protections; (3) addressing vendor defects that impair stability and functionality, leading to fewer incidents due to outdated software; (4) addressing version interdependencies and compatibility between systems; and (5) allowing the Company to leverage new functionality available in the upgrades; (6) provides customer value through closing the gaps on vulnerabilities that can impact reliability; (7) provides a mechanism for proactive actions to stay ahead of evolving cyber threats. This is essential to delivering safe, reliable, and affordable service to the Company's customers. The application upgrades in scope are regularly prioritized based on considerations that include application criticality, number of versions behind the current available version; security and operational risk; operational impacts of performing the upgrade; ability to defer; and cost. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, (4) testing the upgraded software, and (5) updating documentation related to the integration changes. Applications within the Security portfolio are routinely evaluated to determine if and what upgrade efforts are necessary to maintain an appropriate level of currency, as well as the priority of those efforts. During that review, the alternative of delaying the timing of the individual upgrades is considered based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company	2024-12-31	(1.00)	67,920	0	0	54,336	0	13,584	6,500	0	0	5,200	0	0	1,300

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)			
	SPEND YEAR	PROJECT NAME	PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/ BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric			
37	2024	Application Currency-Security-O&M	Security	Software	This project will utilize both Capital and O&M funding to keep applications current for security and reliability. O&M is included in this project to complete expense activities associated with Capital upgrade. The Company manages a large number of applications in the technology landscape that require regular version upgrades to maintain vendor-supported software versions. Without vendor-supported versions, the Company loses the ability to receive version updates and upgrades to address defects, patch security vulnerabilities, protect against cyber threats, protect data, and add new features. Failure to upgrade these applications can have a direct negative impact on key customer and business processes, increase support costs, increase unplanned outages, and increase cyber security vulnerabilities. Maintaining the appropriate versions of applications through application currency upgrades in the Security portfolio adds value by: (1) enabling the Company to maintain vendor support; (2) remediating vendor security vulnerabilities and enhancing security protections; (3) addressing vendor defects that impair stability and functionality; (4) addressing version interdependencies and compatibility between systems; and (5) allowing the Company to leverage additional functionality available in the upgrades. The application upgrades in scope are regularly prioritized based on considerations that include application criticality, number of versions behind the current available version, security and operational risk, operational impacts of performing the upgrade, ability to defer, and cost. The application upgrades in scope are regularly prioritized based on considerations that include application criticality, number of versions behind the current available version, security and operational risk, operational impacts of performing the upgrade, ability to defer, and cost. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, (4) testing the upgraded software, and (5) updating documentation related to the integration changes. Applications within the Security portfolio are routinely evaluated to determine if upgrade efforts are necessary to maintain an appropriate level of currency and then prioritized accordingly. During that evaluation process, the alternative of delaying the individual upgrades is considered based on: (1) maintaining an optimal balance between keeping the application current and risking failure; (2) an increase in the number of incidents; (3) paying increased support costs; and (4) prevention of employees from performing their daily tasks. This project makes ongoing upgrades and support for these applications possible and fortifies the Company's ability to keep the large number of security applications in the technology landscape secure and operational through upgrades. Without these upgrades, the Company will fall further behind in maintaining vendor-supported software versions, increasing the cost and complexity of the upgrade in the future.	2024-12-31	(1.00)	0	0	0	0	0	0	0	37,805	0	0	30,244	0	7,561		
38	SUBTOTAL 2024 Security							6,644,200	0	2,651,360	648,738	2,190,420	113,692	317,990	0	28,600	191,114	2,600	14,990			
39	TOTAL	2024						6,644,200	0	2,651,360	1,083,416	2,190,420	252,364	661,550	0	28,600	514,550	2,600	115,800			
40	2025	Enhancements-Security-Capital	Enhancements	Software	This initiative will use Capital funding to make enhancements to existing technology and to address requests generated by changing business requirements. O&M is included in this project to complete expense activities associated with Capital enhancements in accordance with accounting rules. As business conditions improve and change, compliance requirements evolve, and new capabilities are needed, new requirements surface that call for smaller effort software application changes (enhancements). Enhancing applications requires a short timeframe between inception and implementation and cannot and should not wait for rate case approval at an individual line-item level. Failure to make these changes to applications can have a direct negative impact on key customer and business processes, increase support costs, limit the Company's ability to consistently meet objectives, and increase security risk. The value of technology enhancements in the Security Portfolio lies in: (1) enhancing security protections by funding emerging or unplanned cyber security activities resulting from audits, incidents, or a changing threat landscape; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage additional functionality available in the technology. The scope of technology applications encompasses: (1) making necessary system changes, and (2) updating documentation related to the changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as cyber security related platforms, cyber security incident response, physical security, security awareness, risk management, privacy, and compliance. Prior to implementing an enhancement, a review is completed to identify the best solution. During that review, requests for this funding are governed by a cross-functional board that routinely evaluates and prioritizes the work. Security enhancements fortify the Company's ability to make software technology changes as part of process improvements and regulatory changes, and to meet legally required system changes in a fast and nimble manner. Without funding for enhancements, the Company will be limited in its ability to quickly provide needed capabilities and improvements.	2025-12-31	(1.00)	543,360	0	0	434,688	0	108,672	26,000	0	0	0	0	0	20,800	0	5,200
41	2025	Enhancements-Security-O&M	Enhancements	Software	This initiative will use O&M funding to make enhancements to existing technology and to address requests generated by changing business requirements. As business conditions improve and change, compliance requirements evolve, and new capabilities are needed, new requirements surface that call for smaller effort software application changes (enhancements). Enhancing applications requires a short timeframe between inception and implementation and cannot and should not wait for rate case approval at an individual line-item level. Failure to make these changes to applications can have a direct negative impact on key customer and business processes, increase support costs, limit the Company's ability to consistently meet objectives, and increase security risk. The value of technology enhancements in the Security Portfolio lies in: (1) enhancing security protections by funding emerging or unplanned cyber security activities resulting from audits, incidents, or a changing threat landscape; (2) reducing the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage additional functionality available in the technology. While these small-work technology efforts are neither projects nor operational work, funding for resources is still required to maintain business agility in the digital environment. Included in the implementation are small changes and functionality improvement to existing technology investments for Security. The scope of technology applications encompasses: (1) making necessary system changes, and (2) updating documentation related to the changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as cyber security related platforms, cyber security incident response, physical security, security awareness, risk management, privacy and compliance. Prior to implementing an enhancement, a review is completed to identify the best solution. During that review, requests for this funding are governed by a cross-functional board comprised of representatives from each area that routinely evaluates and prioritizes the work and assesses requests for value using established cost-to-benefit ratio acceptance criteria. In addition, the overall enhancements budget is reviewed annually, and the alternative of a zero-budget allocation for enhancements is considered. This project fortifies the Company's ability to make software technology changes as part of process improvements and regulatory changes, and to meet legally required system changes. Without funding for enhancements, the Company will be limited in its ability to quickly provide needed capabilities and improvements.	2025-12-31	(1.00)	0	0	0	0	0	0	0	378,045	0	0	302,436	0	75,609		
42	SUBTOTAL 2025 Enhancements							543,360	0	0	434,688	0	108,672	40,040	0	0	121,236	0	80,809			
43	2025	Badge Reader/Lock and Key Management System	Security	Hardware	This project will implement a physical smart lock and key management system throughout the Company's Electric service territories. Lack of key control makes facilities, specifically in Electric operations, vulnerable to accidental or intentional disruption to power supply which could cause large scale outages. Install badge readers, smart locks, and credentials on smart devices to grant access to electric assets. Current estimates show there are approximately 12,000 locks throughout the state, and the Company does not have a system to properly manage ownership of the associated physical keys or control over who uses the keys. Locks are not unique in nature and can be easily duplicated. Today's lock and key system allows for 24-hour site access without having the ability to limit outside contractor access. Lack of key control makes facilities, specifically in Electric operations, vulnerable to accidental or intentional disruption which could cause large scale outages. Completion of this project will provide value to the Company by: (1) providing an extra layer of protection which is our first defense against criminal acts; (2) determining core functionalities needed to ensure proper lock and key management state-wide for electric operations; (3) implementing a smart lock and key solution that will provide the physical security team remote deactivation capabilities and is easily audited; (4) installing a smart lock system; and (5) leveraging credentials on mobile device and badge readers that can read both physical cards and digital. The scope of this project includes: (1) a determination of the different levels and functions of different smart lock systems available for electric facilities; and (2) purchase and implementation against a site plan of a lock and key management system specific for needs of Electric Operations. Alternatives considered include: (1) Continue to use current Corbin locks with no key control.	2027-11-01	(1.00)	1,698,000	0	169,800	339,600	1,103,700	84,900	27,950	0	6,750	5,850	0	12,350			

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)
	SPEND YEAR	PROJECT NAME	PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/ BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric
44	2025	Physical Security Asset Refresh	Security	Hardware	This project will enhance or replace physical security assets to provide improved visibility and incident resolution related to security concerns. The Company has several thousand physical security assets currently in use including security cameras, motion detectors, intrusion detection systems, and card access systems. Current limitations include the lack of integrated solutions for centralized management, situational awareness, real time monitoring, compliance with regulations and guidelines, and faster responses to emergencies and incidents. This could result in the increase of potential security vulnerabilities, associated penalties, and reputational damage. The value provided by completing the project is to: (1) maintain compliance with State and Federal Regulations; (2) reduce redundancies with less of a need for multiple cameras and reducing gaps in functionality; and (3) optimize overall system performance. Included in the project is the enhancement or replacement of assets including: (1) advanced door systems at Company buildings; (2) security cameras for monitoring capabilities; and (3) gate and lock systems, which include security cameras, motion detectors, intrusion detection systems, and card access systems. Alternatives considered include: (1) not refreshing physical security assets, and (2) defer a portion of the refresh of physical security assets per an asset refresh cycle industry standard. Alternatives 1 and 2 were not selected due to the risk of security concerns, incident resolution, and the inability to meet FERC requirements. The alternative selected maintains compliance, reduces redundancies and gaps in functionality, and optimizes overall performance of physical security systems.	2025-12-31	(1.00)	2,105,520	0	1,494,240	81,504	509,400	20,376	6,500	0	0	3,120	2,600	780
45	2025	Asset Refresh Program - Cyber Security	Security	Network	This project will replace cyber security infrastructure to support increasing system and application demands and to prevent system failures and service interruptions. When enterprise software or cyber security infrastructure used to support and enhance customer interactions is obsolete, these assets are more expensive to support and can be more difficult to keep current with security updates. This project will create value by maintaining the currency of the cyber security infrastructure for core enterprise software. These are used to ensure the stability of technology for business operations. The scope of this project consists of: (1) annually replacing a subset of cyber security firewalls and servers in keeping with a 3-5 year hardware lifecycle; and (2) performing application upgrades. As part of the review process, the alternatives considered were to: (1) upgrade or replace only those assets identified in the plan; (2) upgrade or replace a portion of the assets identified in the plan. Option 1 was not chosen based on continued refresh cycle for cyber security assets to avoid security risks, system vulnerabilities, and out-of-warranty repair costs. Options 2 was not chosen due to the security risk inherent with not replacing assets as per standard refresh cycles increased system vulnerabilities, and out-of-warranty repair costs.	2025-12-31	(1.00)	252,360	0	168,240	67,296	0	16,824	16,250	0	0	16,250	0	0
46	2025	Critical Substation Upgrade	Security	Network	This project will implement enhanced security measures at our critical substations in order to better understand and deter criminal activity, and respond to threats in real-time. With the enhanced security posture, Consumers Energy will be better positioned to mitigate malicious activity and increase the reliability of electric delivery to our customers. Today, Consumers Energy currently deems 16 substations as business critical to our electric system while an additional 34 substations fall under NERC-CIP compliance. Recent attacks to critical infrastructure have increased the likelihood of an adversary exploiting vulnerabilities at our critical substation increasing the risk for large scale outages and customer impact. Our current security measures do not allow us to detect, deter or monitor potential threats on site. Without the ability to monitor these assets in real-time, we are vulnerable to an adversary causing destruction without knowledge of when or how the event took place. Consumers Energy would take a phased implementation approach to further mitigate risk to assets deemed most vulnerable. The objective of this project is to implement enhanced security measures to better detect, deter and respond to threats to our electric substations in real-time. This enhanced security posture will align with other utilities that own and operate substations to the electric system while improving reliability. Enhanced security measures that will need to be implemented include but are not limited to: (1) access controls; (2) access readers; (3) cameras; (4) video and audio programming; (5) fence intrusion (which will be accomplished through camera systems); (6) subcontractor work of trenching poles, installing equipment etc.; (7) Additional facility lighting; (8) security equipment maintenance.	2027-11-01	0.00	6,250,000	0	312,500	1,000,000	4,062,500	875,000	500,000	0	0	375,000	0	125,000
47	2025	Application Currency-Security-Capital	Security	Software	These activities will be performed at the following company locations: Acme (Acme), Algoma (Sparta), Alma (Alma), Amber (Ludington), Barry (Hastings), Beak Road (Wyoming), Beecher (Adrian), Beveridge (Flint), Blackstone (Jackson), Boardman (Traverse City), Brickyard (Holtom), Broadmoor (Grand Rapids), Campbell (West Olive), Cannon (Rockford), Cement City (Cement City), Churchill (Leslie), Clearwater (Kalkaska), Cleveland (Spring Lake), Cole Creek (Flushing), Crescent (Hillsdale), Croton Dam (Newaygo), David (Pewamo), Deja (Edmore), Dort (Mt. Morris), Dowling (Hudson), Edenville (Dum (Hope), Edwards (Kalamazoo), Ellsworth (Grand Rapids), Elm Street (Battle Creek), Eureka (Greenville), Farr Road (Okeema), Glesner (Hazeland), Gosport Farms (Middletown), Groat Road (Erie), Groat (Gladwin), Hazelwood (Plainville), Hemphill (Flint), Hile Road (Muskegon), Hodorsky Dam (Wexford), Holland Road (Saginaw), HSC Hemlock Semiconductor (Hemlock), Hughes Road (Marshall), Island Road (Charlotte), Jackson Plymouth (Jackson), Lafayette (Battle Creek), Layton (Chesaning), Lindbergh (Kalamazoo), Looking Glass (Lansing), Manlius (New Richmond), McNally (Charlevoix), Miles Road (East Jordan), Monitor (Bay City), Moore Road (Hillsdale), Morrow (Comstock), Muskegon Heights (Muskegon), Oceana (Hart), Ogemaw (West Branch), Page Avenue (Jackson), Pasadena (Flint), Pingree (Alden), Raisin (Tecumseh), Rice Creek (Albion), Riverview (Kalamazoo), Silicon Hemlock Semiconductor (Hemlock), Simpson (Mendon), Sonoma (Battle Creek), Spartan (East Lansing), Thompson Road (Clarksville), Trowbridge (Allegan), Twining (Twining), Vernon (Clare), VeVay (Mason), Vrooman (Jackson), Washtenaw (Chelsea), Wealthy (Grand Rapids), and Willard (Birch Run). Maintain our current security perimeter fencing and conduct routine inspections of the site. Without funding for monitoring equipment, we are unable to prevent, detect or respond to threats in real-time. This project will utilize both Capital and O&M funding to keep applications current for security and reliability. O&M is included in this project to complete expense activities associated with Capital upgrades. The Company manages a large number of applications in the technology landscape that require regular version upgrades to maintain vendor-supported software versions. Without vendor supported versions, the Company loses the ability to receive version updates and upgrades to address defects, patch security vulnerabilities, protect against cyber threats, protect data, and add new features. Failure to upgrade these applications can have a direct negative impact on key customer and business processes, increase support costs, increase unplanned outages, and increase cyber security vulnerabilities. Maintaining the appropriate versions of applications through application currency upgrades in the Security portfolio adds value by: (1) enabling the Company to maintain vendor support; (2) remediating vendor security vulnerabilities and enhancing security protections; (3) addressing vendor defects that impair stability and functionality, leading to fewer incidents due to outdated software; (4) addressing version interdependencies and compatibility between systems; and (5) allowing the Company to leverage new functionality available in the upgrades; (6) provides customer value through closing the gaps on vulnerabilities that can impact reliability; (7) provides a mechanism for proactive actions to stay ahead of evolving cyber threats. This is essential to delivering safe, reliable, and affordable service to the Company's customers. The application upgrades in scope are regularly prioritized based on considerations that include application criticality; number of versions behind the current available version; security and operational risk; operational impacts of performing the upgrade; ability to defer; and cost. The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes, (4) testing the upgraded software, and (5) updating documentation related to the integration changes. Applications within the Security portfolio are routinely evaluated to determine if and what upgrade efforts are necessary to maintain an appropriate level of currency, as well as the priority of those efforts. During that review, the alternative of delaying the timing of the individual upgrades is considered based on: (1) maintaining an optimal balance between keeping the application current and rising failures; (2) an increased number of incidents; (3) paying increased support costs, and (4) preventing employees from performing their daily tasks. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company	2025-12-31	(1.00)	67,920	0	0	54,336	0	13,584	6,500	0	0	5,200	0	1,300

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)
	SPEND YEAR	PROJECT NAME	PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/ BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS- Electric	MATERIAL COSTS- Electric	LABOR COSTS- Electric	CONTRACTOR COSTS- Electric	OVERHEAD & OTHER COSTS- Electric
48	2025	Application Currency-Security-O&M	Security	Software	This project will utilize both Capital and O&M funding to keep applications current for security and reliability. O&M is included in this project to complete expense activities associated with Capital upgrade. The Company manages a large number of applications in the technology landscape that require regular version upgrades to maintain vendor-supported software versions. Without vendor-supported versions, the Company loses the ability to receive version updates and upgrades to address defects, patch security vulnerabilities, protect against cyber threats, protect data, and add new features. Failure to upgrade these applications can have a direct negative impact on key customer and business processes, increase support costs, increase unplanned outages, and increase cyber security vulnerabilities. Maintaining the appropriate versions of applications through application currency upgrades in the Security portfolio adds value by: (1) enabling the Company to maintain vendor support; (2) remediating vendor security vulnerabilities and enhancing security protections; (3) addressing vendor defects that impair stability and functionality; (4) addressing version interdependencies and compatibility between systems; and (5) allowing the Company to leverage additional functionality available in the upgrades. The application upgrades in scope are regularly prioritized based on considerations that include application criticality, number of versions behind the current available version, security and operational risk, operational impacts of performing the upgrade, ability to defer, and cost. The application upgrades in scope are regularly prioritized based on considerations that include application criticality, number of versions behind the current available version, security and operational risk, operational impacts of performing the upgrade, ability to defer, and cost. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, (4) testing the upgraded software, and (5) updating documentation related to the integration changes. Applications within the Security portfolio are routinely evaluated to determine if upgrade efforts are necessary to maintain an appropriate level of currency and then prioritized accordingly. During that evaluation process, the alternative of delaying the individual upgrades is considered based on: (1) maintaining an optimal balance between keeping the application current and risking failure; (2) an increase in the number of incidents; (3) paying increased support costs; and (4) prevention of employees from performing their daily tasks. This project makes ongoing upgrades and support for these applications possible and fortifies the Company's ability to keep the large number of security applications in the technology landscape secure and operational through upgrades. Without these upgrades, the Company will fall further behind in maintaining vendor-supported software versions, increasing the cost and complexity of the upgrade in the future.	2025-12-31	(1.00)	0	0	0	0	0	0	0	37,805	0	30,244	0	7,561
49	2025	HR Process Integration	Security	Software	HR (Human Resources) Process Integration is a project to Automate HR joiner, mover, leaver actions from the SAP HR system to the Identity and Access Management tool (IAM). Modernize, consolidate, and eliminate redundant integrations and systems that require numerous hand offs and are highly customized requiring numerous teams to support. Automation from SAP HR to the IAM tool's SAP HR module would greatly improve the real time transmission of employee HR actions without multiple hand offs of processing which include highly customized scripts prone to failure due to the number of required time sensitive steps to process. Additionally, this automation project would eliminate the need for the IT Security Database (ITSOB) by consolidating functionality. SAP HR batch file processes introduce failure points, delays, and data integrity issues because of time based batching processes from SAP (Employee Central, Redwood (Batch Job Orchestration System), Bitalk (File Mover System), and custom scripts (logic processing scripts) over a decade old. When and if processes do not execute or complete on a specified timeline, failures occur in the end-to-end process. When these failures occur, employee action data such as new or transferred employees do not get updated in Active Roles Server, Active Directory, and subsequently CE's IAM tool. An example of a failure outcome is not allowing new employees to log in to their computer or request access since their user ID would not be available or populate in AccessNow. These failures apply to both the HR Daily processes and the resulting IT Security Database that consumes the same data. Currently in the IT Security Database, an SAP Access transfer control does not exist to ensure SAP access is appropriate for employees who transfer to a different job position (for example from a Regulated role to a Non-Regulated role). This introduces SOX and MPSC Code of Conduct Compliance risk as employees potentially retain access not commensurate with their job responsibilities for extended periods of time. 1) Automate current HR Daily chronological batch file jobs and customized scripts into API direct processing of employee joiner, mover, leaver actions from SAP HR to CE's Identity and Access Management tool. 2) Allow for real time processing of employee actions and associated access requirements. 3) Eliminate failures in the process that have resulted in major IT incidents and compliance deficiencies due to access issues related to employee job changes. 4) Retire the custom built ITSOB by modernizing the functionality within the CE's IAM tool allowing for the elimination of labor support and infrastructure of the ITSOB. 5) Reduce SOX risk through creation of a SOX Control for SAP Access attestations upon transfer of an employee to a new job function. Automate employee joiner, mover, leaver actions from SAP HR to CE's IAM tool via API. Remove aged batch processing system, file mover system, and custom scripts managing the existing HR Daily Processes. Retire ITSOB and implement a new attestation control for SAP access upon employee job transfer. 1) As part of the review process of current HR Daily processing, it's possible to continue to utilize and maintain existing HR Daily processing systems and infrastructure. However, it would require the continuation of support of multiple connected systems and custom scripting. These systems are managed by sensitive chronologically based jobs that are required to execute in a very specific order to not produce failures. Multiple teams support a different tool in the process. It is more risk to manage numerous hand offs in the process and support multiple teams and infrastructure to manage the employee HR actions. By automating these actions via the IAM HR module instead, CE is able to eliminate numerous hand offs and reduce risk and ongoing expense as a result.	2025-11-01	0.00	370,164	0	0	336,204	0	33,960	25,250	0	33,400	0	5,850	
50	SUBTOTAL 2025 Security							10,743,964	0	2,144,780	1,878,940	5,675,600	1,044,644	624,315	0	9,750	499,064	2,600	102,841
51	TOTAL 2025							11,287,324	0	2,144,780	2,313,628	5,675,600	1,153,318	1,028,900	0	9,750	782,300	2,600	233,650

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of Actual and Projected Security Operations O&M Expense

For the Years 2022, 2023, 2024 and Test Year 12 Months Ending February 28, 2025

(\$000)

Case No.: U-21389

Exhibit No.: A-77 (ALC-3)

Page: 1 of 2

Witness: ALCumberworth

Date: May 2023

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)
		Historical	Projected		Test Year	
		12 Mos Ended 12/31/2022	12 Mos Ending 12/31/2023	12 Mos Ending 12/31/2024	2 Mos Ending 2/28/2025	12 Mos Ending 2/28/2025
1	Labor	2,682	3,274	3,274	535	3,274
2	Contracts	5,867	5,484	5,484	922	5,484
3	Business Expense	209	248	248	41	248
4	Material	26	-	-	-	-
5	Total Operations Expense	<u>\$ 8,784</u>	<u>\$ 9,006</u>	<u>\$ 9,006</u>	<u>\$ 1,498</u>	<u>\$ 9,006</u>

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of O&M Expenses Projected Using Merit and Inflation

For the Years 2022, 2023, 2024 and Test Year 12 Months Ending February 28, 2025
(\$000)

Case No.: U-21389

Exhibit No.: A-77 (ALC-3)

Page: 2 of 2

Witness: ALCumberworth

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	2022 Actual	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2022	Merit & Inflation 12 Mos Ending Dec 31, 2023	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2023	Merit & Inflation 12 Mos Ending Dec 31, 2024	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2024	Merit & Inflation 2 Mos Ending 2/28/2025	Other Adjustments	Projected O&M 12 Mos Ending Dec 31, 2023
				(c) * Inflation Rate		(e) * Inflation Rate		(g) * Inflation Rate		(b) + (d) + (f) + (h) + (i)
1	Operations O&M	8,784	2,682	105	2,786	61	2,848	10	46	9,006
	Labor		2,682	105	2,786	61	2,848	10	416	3,274
	Contracts	5,867		0		0		0	-384	5,484
	Business Expense	209		0		0		0	39	248
	Material	26		0		0		0	-26	0
3	Total Operations O&M Expenses	\$ 8,784	\$ 2,682	\$ 105	\$ 2,786	\$ 61	\$ 2,848	\$ 10	\$ 46	\$ 9,006
	Labor	2,682	2,682	105	2,786	61	2,848	10	416	3,274
	Contracts	5,867	0	0	0	0	0	0	-384	5,484
	Business Expense	209	0	0	0	0	0	0	39	248
	Material	26	0	0	0	0	0	0	-26	0

Notes

	2023	2024	2025
4 Annual merit increase			
Annual merit increase	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated merit increase	3.9%	2.2%	0.4%
5 Annual inflation rates per WP-JCA-51			
Annual inflation rates	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated inflation rate	3.9%	2.2%	0.4%

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of Actual and Projected Security Investments O&M Expenses

For the Years 2022, 2023, 2024 and Test Year 12 Months Ending February 28, 2025

(\$000)

Case No.: U-21389

Exhibit No.: A-78 (ALC-4)

Page: 1 of 2

Witness: ALCumberworth

Date: May 2023

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)
		Historical 12 Mos Ended 12/31/2022	12 Mos Ending 12/31/2023	Projected 12 Mos Ending 12/31/2024	2 Mos Ending 2/28/2025	Test Year 12 Mos Ending 2/28/2025
1	Investments Planning	\$ 2	\$ 13	\$ 13	\$ 2	13
2	Labor	2	0	0	0	0
3	Contracts	0	13	13	2	13
4	Business Expense	0	0	0	0	0
5	Material	0	0	0	0	0
6	Investments O&M	\$ 918	\$ 447	\$ 662	\$ 171	723
7	Labor	526	357	515	130	559
8	Software	0	0	0	0	0
9	Material	3	0	29	2	25
10	Contractor Costs	379	0	3	0	3
11	Overhead & Others	10	89	116	39	135
12	Total Investments Expense	\$ 921	\$ 459	\$ 674	\$ 173	\$ 735

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of O&M Expenses Projected Using Merit and Inflation

For the Years 2022, 2023, 2024 and Test Year 12 Months Ending February 28, 2025

(\$000)

Case No.: U-21389

Exhibit No.: A-78 (ALC-4)

Page: 2 of 2

Witness: ALCumberworth

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	2022 Actual	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2022	Merit & Inflation 12 Mos Ending Dec 31, 2023	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2023	Merit & Inflation 12 Mos Ending Dec 31, 2024	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2024	Merit & Inflation 2 Mos Ending 2/28/2025	Other Adjustments	Projected O&M 12 Mos Ending 2/28/2025
			(c) * Inflation Rate			(e) * Inflation Rate		(g) * Inflation Rate		(b) + (d) + (f) + (h) + (i)
1	Investments Planning	2	0	0	0	0	0	0	10	13
	Labor	2		0		0		0	-2	0
	Contracts	0		0		0		0	13	13
	Business Expense	0		0		0		0	0	0
	Material	0		0		0		0	0	0
2	Investments O&M	918	0	0	0	0	0	0	-196	723
	Labor	526		0		0		0	33	559
	Software	0		0		0		0	0	0
	Material	3		0		0		0	23	25
	Contractor Costs	379		0		0		0	-376	3
	Overhead & Others	10		0		0		0	125	135
3	Total Investments O&M Expenses	\$ 921	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (185)	\$ 735

Notes

	2023	2024	2025
4 Annual merit increase			
Annual merit increase	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated merit increase	3.9%	2.2%	0.4%
5 Annual inflation rates per WP-JCA-51			
Annual inflation rates	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated inflation rate	3.9%	2.2%	0.4%

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

EMILY A. DAVIS

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Summary RETURN (SUMMARY)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional	
1	Rate Base	14,404,493	14,353,301	8,064,222	3,609,948	2,432,603	203,087	43,441	51,192	
2	Revenue									
3	Total Rate Revenue	4,283,187	4,258,970	2,212,572	998,917	1,006,592	38,997	1,893	24,217	
4	Total Revenue Credits	819,090	817,953	373,681	191,920	237,636	3,161	11,555	1,137	
5	Total Revenue	5,102,277	5,076,923	2,586,253	1,190,837	1,244,228	42,158	13,448	25,354	
6	Expenses:									
7	Fuel and P&I Expense	2,045,668	2,025,198	871,305	478,151	659,442	10,441	5,859	20,470	
8	Transmission Expense	543,730	538,999	248,150	126,951	160,721	1,997	1,179	4,731	
9	Other O & M Expense	705,078	702,679	421,199	164,015	108,490	7,454	1,523	2,399	
10	Depreciation & Amortization Expense	733,075	730,035	410,534	181,959	124,853	11,117	1,573	3,039	
11	Other Taxes	325,611	324,927	185,718	78,928	54,485	4,799	997	684	
12	Federal Income Taxes	57,601	58,060	34,551	12,367	10,476	488	178	(459)	
13	Total Expenses	4,410,762	4,379,898	2,171,458	1,042,370	1,118,466	36,296	11,309	30,864	
14	Net Operating Income	691,515	697,025	414,796	148,467	125,762	5,862	2,139	(5,510)	
15	Other Income Adjustments	25,823	25,641	13,035	6,216	6,164	170	55	182	
16	Adjusted Net Operating Income	717,339	722,666	427,831	154,683	131,926	6,032	2,194	(5,327)	
17	Rate of Return on Rate Base	4.98%	5.03%	5.31%	4.28%	5.42%	2.97%	5.05%	-10.41%	
18	Index of Return (Jurisdictional)		100	105	85	108	59	100		
17	Return on Rate Base @ 6.11%	880,283	877,155	492,818	220,610	148,661	12,411	2,655	3,128	
18	Income Deficiency (Sufficiency)	162,945	154,489	64,987	65,927	16,735	6,379	461	8,456	
19	Revenue Deficiency (Sufficiency)	218,194	206,871	87,023	88,281	22,409	8,542	617	11,323	
	Additional Rev Requirement	0	0	0	0	0	0	0	0	
20	Total Revenue Deficiency (Sufficiency)	218,194	206,871	87,023	88,281	22,409	8,542	617	11,323	
21	Revenue Requirement/Total Cost of Service	5,320,472	5,283,795	2,673,276	1,279,118	1,266,637	50,699	14,065	36,677	
22	Less: Revenue Credits	819,090	817,953	373,681	191,920	237,636	3,161	11,555	1,137	
23	Proposed Rate Design Revenue	4,501,381	4,465,842	2,299,594	1,087,198	1,029,001	47,539	2,510	35,540	
24	Production: Net Capacity Cost	203,104	201,304	98,945	48,968	52,746	376	269	1,800	
25	Production: Capacity Related Cost Offset	1,092,719	1,081,287	501,065	262,946	312,315	3,423	1,538	11,432	
26	Production: Non-Capacity Related Cost	1,446,700	1,425,605	593,946	335,404	492,640	8,612	(4,997)	21,094	
27	Distribution: Demand Related Cost	1,519,102	1,517,930	925,628	393,246	159,920	33,570	5,567	1,173	
28	Distribution: Customer Related Cost	239,757	239,716	180,011	46,634	11,379	1,558	134	41	
29	Full Service MWH Sales	30,656,265	30,328,039	12,559,141	6,973,697	10,516,047	177,460	101,694	328,226	
30	ROA MWH Sales	3,713,915	3,713,915	-	195,488	3,518,427	-	-	-	
31	MWH Sales	34,370,180	34,041,954	12,559,141	7,169,185	14,034,473	177,460	101,694	328,226	
32	Customers	1,890,850	1,890,848	1,662,850	218,379	4,052	5,551	15	2	

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Residential/Secondary RETURN (SUMMARY)						
Line No.	(a) Description	(b) Rate RS	(c) Total Residential	(d) Rate GS	(e) Rate GSD	(f) Total Commercial Secondary
1	Rate Base	8,064,222	8,064,222	2,274,904	1,335,044	3,609,948
	Revenue					
3	Total Rate Revenue	2,212,572	2,212,572	614,759	384,158	998,917
4	Total Revenue Credits	373,681	373,681	114,543	77,377	191,920
5	Total Revenue	2,586,253	2,586,253	729,302	461,535	1,190,837
6	Expenses:					
7	Fuel and P&I Expense	871,305	871,305	279,217	198,934	478,151
8	Transmission Expense	248,150	248,150	76,014	50,938	126,951
9	Other O & M Expense	421,199	421,199	104,283	59,732	164,015
10	Depreciation & Amortization Expense	410,534	410,534	114,872	67,086	181,959
11	Other Taxes	185,718	185,718	49,765	29,163	78,928
12	Federal Income Taxes	34,551	34,551	8,085	4,281	12,367
13	Total Expenses	2,171,458	2,171,458	632,235	410,135	1,042,370
14	Net Operating Income	414,796	414,796	97,067	51,400	148,467
15	Other Income Adjustments	13,035	13,035	3,809	2,407	6,216
16	Adjusted Net Operating Income	427,831	427,831	100,876	53,807	154,683
17	Rate of Return on Rate Base	5.31%	5.31%	4.43%	4.03%	4.28%
18	Index of Return (Jurisdictional)	105	105	88	80	85
17	Return on Rate Base @ 6.11%	492,818	492,818	139,023	81,587	220,610
18	Income Deficiency (Sufficiency)	64,987	64,987	38,147	27,780	65,927
19	Revenue Deficiency (Sufficiency)	87,023	87,023	51,082	37,199	88,281
	Additional Rev Requirement	0	0	0	0	0
20	Total Revenue Deficiency (Sufficiency)	87,023	87,023	51,082	37,199	88,281
21	Revenue Requirement/Total Cost of Service	2,673,276	2,673,276	780,384	498,734	1,279,118
22	Less: Revenue Credits	373,681	373,681	114,543	77,377	191,920
23	Proposed Rate Design Revenue	2,299,594	2,299,594	665,841	421,357	1,087,198
24	Production: Net Capacity Cost	98,945	98,945	30,018	18,950	48,968
25	Production: Capacity Related Cost Offset	501,065	501,065	157,380	105,566	262,946
26	Production: Non-Capacity Related Cost	593,946	593,946	193,424	141,980	335,404
27	Distribution: Demand Related Cost	925,628	925,628	247,696	145,550	393,246
28	Distribution: Customer Related Cost	180,011	180,011	37,322	9,311	46,634
29	Full Service MWH Sales	12,559,141	12,559,141	4,024,187	2,949,510	6,973,697
30	ROA MWH Sales	-	-	23,026	172,462	195,488
31	MWH Sales	12,559,141	12,559,141	4,047,213	3,121,972	7,169,185
32	Customers	1,662,850	1,662,850	197,136	21,244	218,379

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Primary & Lighting RETURN (SUMMARY)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Line No.	Description	Rate GP	Rate GPTU Vlt 1	Rate GPTU Vlt 2	Rate GPTU Vlt 3	Rate GPD Vlt 1	Rate GPD Vlt 2	Rate GPD Vlt 3	Rate EIP Vlt 1	Rate EIP Vlt 2	Rate EIP Vlt 3	Total Primary	Rate GML	Rate UUL	Rate GU	Total Lighting & Unmetered
1	Rate Base	250,769	55,851	102,146	948,047	133,346	311,725	595,595	22,657	9,850	2,618	2,432,603	6,318	171,707	25,062	203,087
	Revenue															
3	Total Rate Revenue	91,875	33,312	51,637	375,888	78,990	131,069	214,317	23,615	5,006	884	1,006,592	1,472	28,187	9,338	38,997
4	Total Revenue Credits	20,535	8,483	12,564	89,729	19,545	31,665	49,512	4,670	815	119	237,636	170	1,032	1,959	3,161
5	Total Revenue	112,409	41,796	64,202	465,616	98,535	162,734	263,828	28,285	5,821	1,002	1,244,228	1,642	29,219	11,296	42,158
6	Expenses:															
7	Fuel and P&I Expense	53,035	24,225	35,594	239,709	60,344	89,051	133,217	20,272	3,469	526	659,442	661	3,894	5,885	10,441
8	Transmission Expense	14,154	5,590	8,393	60,614	12,948	20,674	31,975	5,432	799	142	160,721	118	653	1,226	1,997
9	Other O & M Expense	10,615	2,735	4,753	42,413	6,225	13,559	26,185	1,448	442	115	108,490	284	5,957	1,212	7,454
10	Depreciation & Amortization Expense	12,847	3,326	5,513	49,370	7,336	15,279	29,583	1,130	367	102	124,853	302	9,533	1,282	11,117
11	Other Taxes	5,549	1,287	2,324	20,690	3,373	7,281	13,224	472	229	56	54,485	141	4,103	556	4,799
12	Federal Income Taxes	1,246	356	586	4,061	639	1,299	2,279	(36)	40	5	10,476	10	390	87	488
13	Total Expenses	97,446	37,518	57,163	416,857	90,866	147,144	236,463	28,718	5,346	945	1,118,466	1,516	24,531	10,248	36,296
14	Net Operating Income	14,963	4,277	7,039	48,759	7,669	15,590	27,365	(433)	476	57	125,762	126	4,688	1,048	5,862
15	Other Income Adjustments	581	201	310	2,426	422	797	1,347	63	14	3	6,164	5	113	52	170
16	Adjusted Net Operating Income	15,544	4,478	7,349	51,185	8,091	16,387	28,712	(370)	490	60	131,926	131	4,801	1,100	6,032
17	Rate of Return on Rate Base	6.20%	8.02%	7.19%	5.40%	6.07%	5.26%	4.82%	-1.63%	4.97%	2.28%	5.42%	2.08%	2.80%	4.39%	2.97%
18	Index of Return (Jurisdictional)	123	159	143	107	121	104	96	(32)	99	45	108	41	56	87	59
17	Return on Rate Base @ 6.11%	15,325	3,413	6,242	57,937	8,149	19,050	36,398	1,385	602	160	148,661	386	10,493	1,532	12,411
18	Income Deficiency (Sufficiency)	(219)	(1,065)	(1,107)	6,752	58	2,663	7,686	1,755	112	100	16,735	255	5,693	431	6,379
19	Revenue Deficiency (Sufficiency)	(294)	(1,426)	(1,482)	9,041	77	3,567	10,292	2,350	150	134	22,409	341	7,623	578	8,542
	Additional Rev Requirement	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
20	Total Revenue Deficiency (Sufficiency)	(294)	(1,426)	(1,482)	9,041	77	3,567	10,292	2,350	150	134	22,409	341	7,623	578	8,542
21	Revenue Requirement/Total Cost of Service	112,115	40,370	62,720	474,658	98,612	166,300	274,120	30,634	5,972	1,137	1,266,637	1,984	36,842	11,874	50,699
22	Less: Revenue Credits	20,535	8,483	12,564	89,729	19,545	31,665	49,512	4,670	815	119	237,636	170	1,032	1,959	3,161
23	Proposed Rate Design Revenue	91,581	31,886	50,155	384,929	79,067	134,636	224,608	25,964	5,157	1,018	1,029,001	1,813	35,810	9,915	47,539
24	Production: Net Capacity Cost	5,165	1,805	2,720	21,518	3,491	6,753	11,295	-	-	-	52,746	1	-	375	376
25	Production: Capacity Related Cost Offset	28,970	10,796	16,181	122,244	22,870	40,273	65,867	4,289	714	111	312,315	151	784	2,487	3,423
26	Production: Non-Capacity Related Cost	38,080	18,126	26,641	174,201	47,641	66,618	96,801	20,604	3,387	542	492,640	602	3,515	4,495	8,612
27	Distribution: Demand Related Cost	16,710	1,045	4,362	63,062	4,615	20,067	47,705	987	1,014	351	159,920	1,005	30,077	2,487	33,570
28	Distribution: Customer Related Cost	2,656	114	251	3,904	450	925	2,940	84	42	14	11,379	54	1,433	71	1,558
29	Full Service MWH Sales	805,656	397,758	574,708	3,709,392	1,033,644	1,448,667	2,098,603	374,401	64,003	9,214	10,516,047	12,195	71,839	93,426	177,460
30	ROA MWH Sales	75,604	-	-	-	1,130,638	1,351,423	960,761	-	-	-	3,518,427	-	-	-	-
31	MWH Sales	881,260	397,758	574,708	3,709,392	2,164,283	2,800,090	3,059,364	374,401	64,003	9,214	14,034,473	12,195	71,839	93,426	177,460
32	Customers	1,537	19	41	1,305	36	96	1,002	6	7	4	4,052	404	4,686	461	5,551

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Summary RATE BASE (SUMMARY)										
Line No.	(a) Description	(b)	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Net Plant									
2	Production		2,617,318	2,594,124	1,230,699	625,616	725,603	7,506	4,700	23,194
3	Transmission		-	-	-	-	-	-	-	-
4	Distribution		8,113,300	8,106,810	4,847,430	2,124,473	929,416	173,164	32,327	6,490
5	General/Common/Intangible		686,368	684,060	408,835	161,413	105,208	7,166	1,437	2,309
6	Plant Purchased/Sold		0	0	0	0	0	0	0	0
7	Total Net Plant		11,416,986	11,384,994	6,486,964	2,911,503	1,760,227	187,836	38,465	31,992
8	Working Capital									
9	Total Current Assets		4,117,364	4,095,109	2,237,437	979,531	835,150	34,881	8,111	22,255
10	Total Current Liabilities		1,072,461	1,069,447	624,897	266,382	156,963	18,261	2,943	3,015
11	Total Working Capital		3,044,903	3,025,663	1,612,540	713,149	678,187	16,620	5,167	19,240
	Adjustments to Rate Base									
13	Additions to Rate Base		0	0	0	0	0	0	0	0
14	Deductions from Rate Base		57,395	57,356	35,281	14,704	5,811	1,368	191	40
15	Total Adjustments to Rate Base		(57,395)	(57,356)	(35,281)	(14,704)	(5,811)	(1,368)	(191)	(40)
16	Total Rate Base		14,404,493	14,353,301	8,064,222	3,609,948	2,432,603	203,087	43,441	51,192

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Residential/Secondary RATE BASE (SUMMARY)						
	(a)	(b)	(c)	(d)	(e)	(h)
Line No.	Description	Rate RS	Total Residential	Rate GS	Rate GSD	Commercial Secondary
1	Net Plant					
2	Production	1,230,699	1,230,699	378,039	247,577	625,616
3	Transmission	-	-	-	-	-
4	Distribution	4,847,430	4,847,430	1,363,738	760,735	2,124,473
5	General/Common/Intangible	408,835	408,835	102,247	59,166	161,413
6	Plant Purchased/Sold	0	0	0	0	0
7	Total Net Plant	6,486,964	6,486,964	1,844,025	1,067,478	2,911,503
8	Working Capital					
9	Total Current Assets	2,237,437	2,237,437	608,512	371,019	979,531
10	Total Current Liabilities	624,897	624,897	168,233	98,148	266,382
11	Total Working Capital	1,612,540	1,612,540	440,278	272,871	713,149
	Adjustments to Rate Base					
13	Additions to Rate Base	0	0	0	0	0
14	Deductions from Rate Base	35,281	35,281	9,399	5,305	14,704
15	Total Adjustments to Rate Base	(35,281)	(35,281)	(9,399)	(5,305)	(14,704)
16	Total Rate Base	8,064,222	8,064,222	2,274,904	1,335,044	3,609,948

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Primary & Lighting RATE BASE (SUMMARY)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(j)	(k)	(l)	(q)	(r)	(t)	(u)	(v)
Line No.	Description	Rate GP	Rate GPTU Vit 1	Rate GPTU Vit 2	Rate GPTU Vit 3	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate EIP Vit 1	Rate EIP Vit 2	Rate EIP Vit 3	Total Primary	Rate GML	Rate UUL	Rate GU	Total Lighting & Unmetered
1	Net Plant															
2	Production	66,848	25,610	38,238	285,989	52,640	93,998	153,076	7,682	1,329	194	725,603	272	1,568	5,666	7,506
3	Transmission	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Distribution	110,530	5,784	25,502	356,150	28,876	118,522	269,588	6,024	6,455	1,984	929,416	5,136	155,694	12,334	173,164
5	General/Common/Intangible	10,563	2,569	4,529	41,780	5,686	13,063	25,648	919	350	101	105,208	259	5,759	1,147	7,166
6	Plant Purchased/Sold	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	Total Net Plant	187,941	33,963	68,269	683,919	87,202	225,583	448,312	14,626	8,134	2,279	1,760,227	5,667	163,021	19,148	187,836
8	Working Capital															
9	Total Current Assets	79,934	25,198	40,228	327,956	54,395	106,388	188,645	9,453	2,410	543	835,150	1,215	25,864	7,802	34,881
10	Total Current Liabilities	16,419	3,275	6,201	61,565	8,073	19,548	39,649	1,385	657	192	156,963	527	15,933	1,801	18,261
11	Total Working Capital	63,515	21,923	34,027	266,391	46,322	86,839	148,996	8,068	1,754	351	678,187	688	9,930	6,001	16,620
	Adjustments to Rate Base															
13	Additions to Rate Base	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14	Deductions from Rate Base	687	35	150	2,263	177	698	1,713	37	38	13	5,811	37	1,244	87	1,368
15	Total Adjustments to Rate Base	(687)	(35)	(150)	(2,263)	(177)	(698)	(1,713)	(37)	(38)	(13)	(5,811)	(37)	(1,244)	(87)	(1,368)
16	Total Rate Base	250,769	55,851	102,146	948,047	133,346	311,725	595,595	22,657	9,850	2,618	2,432,603	6,318	171,707	25,062	203,087

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Summary O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional		
1	Production Expense										
2	Fuel Expense	1,039,289	1,028,589	431,039	241,533	346,869	5,891	3,257	10,700		
3	Purchased & Interchange Power Expense	1,006,379	996,609	440,267	236,618	312,573	4,550	2,602	9,770		
4	Total Fuel and P&I Expense	2,045,668	2,025,198	871,305	478,151	659,442	10,441	5,859	20,470		
5	Fossil O&M Exp	99,427	98,471	44,079	23,426	30,271	443	252	955		
6	Nuclear O&M Exp	0	-	-	-	-	-	-	-		
7	Hydro O&M Exp	19,168	18,987	8,612	4,531	5,720	79	45	181		
8	Other Power Gen O&M Exp	72,922	72,276	34,289	17,431	20,216	209	131	646		
9	Other Power Supply O&M Exp	8,579	8,503	4,034	2,051	2,378	25	15	76		
10	Total Production O&M Expense	200,095	198,236	91,014	47,438	58,585	756	444	1,859		
11	Total Production (Inc. Fuel and P&I) O&M Expense	2,245,763	2,223,435	962,320	525,589	718,027	11,196	6,303	22,329		
12	Transmission & Distribution Expense										
13	Trans O&M Exp	543,730	538,999	248,150	126,951	160,721	1,997	1,179	4,731		
14	Other O&M Adjustments	0	0	0	0	0	0	0	0		
15	Distr Oper Exp	113,332	113,309	73,359	27,391	8,123	4,225	212	22		
16	Distr Maint Exp	215,272	215,217	138,773	53,891	20,816	1,089	647	55		
17	Total Transmission & Distribution O&M Expense	872,333	867,525	460,282	208,233	189,660	7,311	2,039	4,808		
18	Customer Related Expense										
19	Customer Accounts Exp	57,278	57,173	43,340	8,884	4,677	264	8	104		
20	Customer Service Exp	5,151	5,113	2,468	968	1,641	24	12	38		
21	Other Customer Exp	18,882	18,882	15,616	3,085	76	104	0	0		
22	Total Customer O&M Expense	81,310	81,168	61,425	12,937	6,393	392	21	143		
23	Admin & General Expense	95,069	94,749	56,628	22,357	14,572	993	199	320		
24	Total O&M Expense	3,294,476	3,266,876	1,540,654	769,116	928,652	19,891	8,561	27,600		

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Residential/Secondary O&M (SUMMARY)						
	(a)	(b)	(c)	(d)	(e)	(h)
Line No.	Description	Rate RS	Total Residential	Rate GS	Rate GSD	Commercial Secondary
1	Production Expense					
2	Fuel Expense	431,039	431,039	139,769	101,764	241,533
3	Purchased & Interchange Power Expense	440,267	440,267	139,448	97,170	236,618
4	Total Fuel and P&I Expense	871,305	871,305	279,217	198,934	478,151
5	Fossil O&M Exp	44,079	44,079	13,828	9,599	23,426
6	Nuclear O&M Exp	-	-	-	-	-
7	Hydro O&M Exp	8,612	8,612	2,689	1,842	4,531
8	Other Power Gen O&M Exp	34,289	34,289	10,533	6,898	17,431
9	Other Power Supply O&M Exp	4,034	4,034	1,239	811	2,051
10	Total Production O&M Expense	91,014	91,014	28,288	19,150	47,438
11	Total Production (Inc. Fuel and P&I) O&M Expense	962,320	962,320	307,505	218,084	525,589
12	Transmission & Distribution Expense					
13	Trans O&M Exp	248,150	248,150	76,014	50,938	126,951
14	Other O&M Adjustments	0	0	0	0	0
15	Distr Oper Exp	73,359	73,359	17,739	9,652	27,391
16	Distr Maint Exp	138,773	138,773	34,081	19,810	53,891
17	Total Transmission & Distribution O&M Expense	460,282	460,282	127,834	80,399	208,233
18	Customer Related Expense					
19	Customer Accounts Exp	43,340	43,340	6,723	2,160	8,884
20	Customer Service Exp	2,468	2,468	591	377	968
21	Other Customer Exp	15,616	15,616	2,697	388	3,085
22	Total Customer O&M Expense	61,425	61,425	10,012	2,925	12,937
23	Admin & General Expense	56,628	56,628	14,162	8,195	22,357
24	Total O&M Expense	1,540,654	1,540,654	459,513	309,604	769,116

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Primary & Lighting O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	(t)	(u)	(v)
Line No.	Description	Rate GP	Rate GPTU Vit 1	Rate GPTU Vit 2	Rate GPTU Vit 3	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate EIP Vit 1	Rate EIP Vit 2	Rate EIP Vit 3	Total Primary	Rate GML	Rate UUL	Rate GU	Lighting & Unmetered							
1	Production Expense																						
2	Fuel Expense	27,045	12,848	18,815	123,859	33,121	47,241	69,421	12,130	2,074	315	346,869	394	2,321	3,177	5,891							
3	Purchased & Interchange Power Expense	25,990	11,378	16,778	115,850	27,223	41,810	63,795	8,143	1,395	211	312,573	268	1,573	2,708	4,550							
4	Total Fuel and P&I Expense	53,035	24,225	35,594	239,709	60,344	89,051	133,217	20,272	3,469	526	659,442	661	3,894	5,885	10,441							
5	Fossil O&M Exp	2,555	1,112	1,641	11,373	2,453	3,983	6,258	748	129	19	30,271	26	153	264	443							
6	Nuclear O&M Exp	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-							
7	Hydro O&M Exp	492	208	308	2,171	454	750	1,187	125	22	3	5,720	4	25	49	79							
8	Other Power Gen O&M Exp	1,862	714	1,065	7,968	1,467	2,619	4,265	214	37	5	20,216	8	44	158	209							
9	Other Power Supply O&M Exp	219	84	125	937	173	308	502	25	4	1	2,378	1	5	19	25							
10	Total Production O&M Expense	5,128	2,118	3,139	22,449	4,546	7,660	12,212	1,112	192	28	58,585	39	227	490	756							
11	Total Production (Inc. Fuel and P&I) O&M Expense	58,163	26,343	38,733	262,158	64,890	96,711	145,429	21,384	3,662	554	718,027	700	4,121	6,375	11,196							
12	Transmission & Distribution Expense																						
13	Trans O&M Exp	14,154	5,590	8,393	60,614	12,948	20,674	31,975	5,432	799	142	160,721	118	653	1,226	1,997							
14	Other O&M Adjustments	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0							
15	Distr Oper Exp	1,029	15	169	3,410	68	778	2,579	14	42	19	8,123	91	3,974	160	4,225							
16	Distr Maint Exp	2,421	45	512	8,571	202	2,379	6,471	39	128	48	20,816	95	661	334	1,089							
17	Total Transmission & Distribution O&M Expense	17,603	5,649	9,073	72,595	13,219	23,832	41,025	5,485	969	209	189,660	304	5,288	1,719	7,311							
18	Customer Related Expense																						
19	Customer Accounts Exp	442	154	238	1,738	368	604	994	112	23	4	4,677	15	199	50	264							
20	Customer Service Exp	104	46	67	434	253	327	358	44	7	1	1,641	2	11	11	24							
21	Other Customer Exp	29	0	1	24	1	2	19	0	0	0	76	8	88	9	104							
22	Total Customer O&M Expense	574	201	306	2,196	621	933	1,371	155	31	5	6,393	24	298	70	392							
23	Admin & General Expense	1,463	356	627	5,787	788	1,809	3,552	127	49	14	14,572	36	798	159	993							
24	Total O&M Expense	77,804	32,550	48,740	342,735	79,518	123,285	191,377	27,152	4,710	782	928,652	1,064	10,504	8,323	19,891							

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Summary ALLOCATORS 1										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
<u>Input Allocation Schedules</u>										
	Energy @ Generation	100	100.000	98.970	41.868	23.248	32.942	0.592	0.321	1.030
	Energy On-Peak @ Generation	101	100.000	98.970	39.804	24.342	34.120	0.422	0.282	1.030
	Energy Off-Peak @ Generation	102	100.000	98.970	43.604	21.754	32.483	0.770	0.360	1.030
	Energy On-Peak @ Generation Summer	103	100.000	98.970	40.944	24.619	32.842	0.331	0.234	1.030
	Energy Off-Peak @ Generation Summer	104	100.000	98.970	43.786	21.985	32.216	0.672	0.312	1.030
	Energy On-Peak @ Generation Non-Summer	105	100.000	98.970	39.117	24.175	34.891	0.477	0.310	1.030
	Energy Off-Peak @ Generation Non-Summer	106	100.000	98.970	43.500	21.622	32.635	0.826	0.387	1.030
	Energy Critical On-Peak @ Gen	107	100.000	98.970	44.276	23.898	30.355	0.237	0.204	1.030
	Energy Summer Mid-Peak @ Gen	108	100.000	98.970	39.708	24.886	33.765	0.366	0.246	1.030
	12CP Dmd @ Generation	120	100.000	99.130	45.639	23.348	29.559	0.367	0.217	0.870
	4CP Dmd @ Generation	121	100.000	99.114	48.716	24.110	25.970	0.185	0.132	0.886
	Classpeak @ Transmission	127	100.000	99.152	45.551	21.276	30.497	0.384	1.444	0.848
	Production Revenue	141	100.000	99.208	45.172	23.040	30.567	0.422	0.007	0.792
	Distribution Revenue	142	100.000	99.963	63.537	23.861	10.591	1.859	0.114	0.037
	Total Rate Revenue	143	100.000	99.457	51.249	23.312	23.957	0.897	0.042	0.543
	Billed Sales	150	100.000	99.045	36.541	20.859	40.833	0.516	0.296	0.955
	Number Of Customers	160	100.000	100.000	87.942	11.549	0.214	0.294	0.001	0.000
	Weighted Customer	170	100.000	99.994	60.629	31.913	7.403	0.022	0.027	0.006
<u>*Calculated Allocation Schedules</u>										
	4CP 75/0/25	220	100.000	99.114	47.021	23.903	27.723	0.287	0.180	0.886
	4CP 75/0/25 Exc WFR	222	100.000	99.975	47.429	24.111	27.965	0.289	0.181	0.025
	4CP Dmd @ Gen Jurisdictional	224	100.000	100.000	49.152	24.325	26.202	0.187	0.134	-
	12CP Demand @ Subtrans	226	100.000	100.000	48.256	24.687	26.544	0.388	0.124	-
	Class Peak @ Subtransmission	122	100.000	99.952	48.878	22.830	26.283	0.412	1.549	0.048
	Classpeak @ Primary	230	100.000	100.000	53.663	25.065	20.809	0.453	0.010	-
	Classpeak @ Secondary	231	100.000	100.000	67.772	31.656	-	0.572	-	-
	Classpeak for Streetlighting	233	100.000	99.021	-	-	-	99.021	-	0.979
	Classpeak @ Single Phase	235	100.000	100.000	67.772	31.656	-	0.572	-	-
	Billed Sales - Primary	253	100.000	100.000	45.644	26.055	27.654	0.645	0.002	-
	Customers - Residential	260	100.000	100.000	100.000	-	-	-	-	-
	Customers - Drops	261	100.000	100.000	-	100.000	-	-	-	-
	Customers - NonPID	263	100.000	100.000	88.373	11.606	-	0.021	-	-
	Customers - NonMunicipal	264	100.000	100.000	88.201	11.583	0.215	-	0.001	0.000
	Customers - Commercial & Primary	267	100.000	100.000	-	98.172	1.822	-	0.007	-
	Customer Count (CCC)	161	100.000	100.000	93.927	6.073	-	-	-	-
	Customer Count (BCC)	162	100.000	100.000	-	92.017	3.363	4.608	0.012	-
	PIS - HVD (Voltage 1)	301	100.000	99.152	45.551	21.276	30.497	0.384	1.444	0.848
	PIS - HVD (Voltage 2)	302	100.000	99.952	48.878	22.830	26.283	0.412	1.549	0.048
	PIS - HVD (Voltage 1) Subs (FERC 361/362)	303	100.000	99.152	45.551	21.276	30.497	0.384	1.444	0.848
	PIS - HVD (Voltage 2) Subs (FERC 361/362)	304	100.000	99.952	48.878	22.830	26.283	0.412	1.549	0.048

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Residential/Secondary ALLOCATORS 1							
	(a)	(b)	(c)	(d)	(e)	(h)	
Line No.	Description	Alloc	Rate RS	Total Residential	Rate GS	Rate GSD	Commercial Secondary
<u>Input Allocation Schedules</u>							
	Energy @ Generation	100	41.868	41.868	13.415	9.833	23.248
	Energy On-Peak @ Generation	101	39.804	39.804	14.389	9.954	24.342
	Energy Off-Peak @ Generation	102	43.604	43.604	12.178	9.576	21.754
	Energy On-Peak @ Generation Summer	103	40.944	40.944	14.636	9.982	24.619
	Energy Off-Peak @ Generation Summer	104	43.786	43.786	12.353	9.632	21.985
	Energy On-Peak @ Generation Non-Summer	105	39.117	39.117	14.239	9.936	24.175
	Energy Off-Peak @ Generation Non-Summer	106	43.500	43.500	12.078	9.544	21.622
	Energy Critical On-Peak @ Gen	107	44.276	44.276	14.141	9.757	23.898
	Energy Summer Mid-Peak @ Gen	108	39.708	39.708	14.820	10.066	24.886
	12CP Dmd @ Generation	120	45.639	45.639	13.980	9.368	23.348
	4CP Dmd @ Generation	121	48.716	48.716	14.780	9.330	24.110
	Classpeak @ Transmission	127	45.551	45.551	13.185	8.092	21.276
	Production Revenue	141	45.172	45.172	13.750	9.290	23.040
	Distribution Revenue	142	63.537	63.537	15.445	8.416	23.861
	Total Rate Revenue	143	51.249	51.249	14.311	9.001	23.312
	Billed Sales	150	36.541	36.541	11.775	9.083	20.859
	Number Of Customers	160	87.942	87.942	10.426	1.123	11.549
	Weighted Customer	170	60.629	60.629	25.838	6.075	31.913
<u>*Calculated Allocation Schedules</u>							
	4CP 75/0/25	220	47.021	47.021	14.444	9.459	23.903
	4CP 75/0/25 Exc WFR	222	47.429	47.429	14.569	9.541	24.111
	4CP Dmd @ Gen Jurisdictional	224	49.152	49.152	14.912	9.414	24.325
	12CP Demand @ Subtrans	226	48.256	48.256	14.782	9.905	24.687
	Class Peak @ Subtransmission	122	48.878	48.878	14.148	8.682	22.830
	Classpeak @ Primary	230	53.663	53.663	15.533	9.532	25.065
	Classpeak @ Secondary	231	67.772	67.772	19.617	12.039	31.656
	Classpeak for Streetlighting	233	-	-	-	-	-
	Classpeak @ Single Phase	235	67.772	67.772	19.617	12.039	31.656
	Billed Sales - Primary	253	45.644	45.644	14.709	11.346	26.055
	Customers - Residential	260	100.000	100.000	-	-	-
	Customers - Drops	261	-	-	90.272	9.728	100.000
	Customers - NonPID	263	88.373	88.373	10.477	1.129	11.606
	Customers - NonMunicipal	264	88.201	88.201	10.456	1.127	11.583
	Customers - Commercial & Primary	267	-	-	88.622	9.550	98.172
	Customer Count (CCC)	161	93.927	93.927	6.013	0.060	6.073
	Customer Count (BCC)	162	-	-	75.266	16.750	92.017
	PIS - HVD (Voltage 1)	301	45.551	45.551	13.185	8.092	21.276
	PIS - HVD (Voltage 2)	302	48.878	48.878	14.148	8.682	22.830
	PIS - HVD (Voltage 1) Subs (FERC 361/362)	303	45.551	45.551	13.185	8.092	21.276
	PIS - HVD (Voltage 2) Subs (FERC 361/362)	304	48.878	48.878	14.148	8.682	22.830

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Primary & Lighting ALLOCATORS 1																	
Line	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(j)	(k)	(l)	(q)	(r)	(t)	(u)	(v)	
No.	Description	Alloc	Rate GP	Rate GPTU Vit 1	Rate GPTU Vit 2	Rate GPTU Vit 3	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate EIP Vit 1	Rate EIP Vit 2	Rate EIP Vit 3	Total Primary	Rate GML	Rate UUL	Rate GU	Lighting & Unmetered
<u>Input Allocation Schedules</u>																	
100	Energy @ Generation	2.584	1.247	1.823	11.908	2.886	4.386	6.702	1.174	0.203	0.030	32.942	0.041	0.239	0.311	0.592	
101	Energy On-Peak @ Generation	2.925	1.233	1.834	12.711	2.998	4.402	6.589	1.198	0.189	0.040	34.120	0.020	0.120	0.282	0.422	
102	Energy Off-Peak @ Generation	2.177	1.246	1.788	10.876	3.465	4.751	6.814	1.132	0.216	0.018	32.483	0.063	0.368	0.339	0.770	
103	Energy On-Peak @ Generation Summer	2.822	1.171	1.703	12.388	2.739	4.282	6.437	1.100	0.165	0.035	32.842	0.011	0.068	0.252	0.331	
104	Energy Off-Peak @ Generation Summer	2.126	1.204	1.714	10.941	3.327	4.752	6.824	1.111	0.199	0.017	32.216	0.052	0.310	0.309	0.672	
105	Energy On-Peak @ Generation Non-Summer	2.988	1.271	1.914	12.906	3.155	4.475	6.681	1.257	0.203	0.043	34.891	0.026	0.150	0.301	0.477	
106	Energy Off-Peak @ Generation Non-Summer	2.205	1.269	1.831	10.839	3.543	4.750	6.809	1.145	0.226	0.018	32.635	0.069	0.401	0.356	0.826	
107	Energy Critical On-Peak @ Gen	2.604	1.057	1.557	11.604	2.285	3.945	6.152	0.996	0.136	0.019	30.355	0.001	0.004	0.233	0.237	
108	Energy Summer Mid-Peak @ Gen	2.903	1.214	1.757	12.679	2.907	4.407	6.542	1.139	0.176	0.041	33.765	0.015	0.092	0.259	0.366	
120	12CP Dmd @ Generation	2.603	1.028	1.544	11.148	2.381	3.802	5.881	0.999	0.147	0.026	29.559	0.022	0.120	0.225	0.367	
121	4CP Dmd @ Generation	2.543	0.889	1.339	10.595	1.719	3.325	5.561	-	-	-	25.970	0.000	-	0.185	0.185	
127	Classpeak @ Transmission	2.507	0.826	1.121	8.702	4.242	5.283	6.589	0.890	0.287	0.050	30.497	0.038	0.207	0.139	0.384	
141	Production Revenue	2.420	1.168	1.705	11.256	2.737	4.144	6.138	0.836	0.143	0.021	30.567	0.024	0.138	0.260	0.422	
142	Distribution Revenue	1.652	0.055	0.281	4.188	0.232	1.080	2.947	0.059	0.076	0.021	10.591	0.056	1.659	0.144	1.859	
143	Total Rate Revenue	2.166	0.800	1.234	8.917	1.908	3.130	5.082	0.579	0.121	0.021	23.957	0.035	0.641	0.222	0.897	
150	Billed Sales	2.564	1.157	1.672	10.792	6.297	8.147	8.901	1.089	0.186	0.027	40.833	0.035	0.209	0.272	0.516	
160	Number Of Customers	0.081	0.001	0.002	0.069	0.002	0.005	0.053	0.000	0.000	0.000	0.214	0.021	0.248	0.024	0.294	
170	Weighted Customer	2.482	0.040	0.096	2.546	0.084	0.208	1.912	0.013	0.015	0.007	7.403	0.022	-	-	0.022	
<u>*Calculated Allocation Schedules</u>																	
220	4CP 75/0/25	2.554	0.978	1.461	10.927	2.011	3.591	5.849	0.294	0.051	0.007	27.723	0.010	0.060	0.216	0.287	
222	4CP 75/0/25 Exc WFR	2.576	0.987	1.474	11.022	2.029	3.623	5.900	0.296	0.051	0.007	27.965	0.010	0.060	0.218	0.289	
224	4CP Dmd @ Gen Jurisdictional	2.566	0.896	1.351	10.689	1.734	3.355	5.611	-	-	-	26.202	0.000	-	0.186	0.187	
226	12CP Demand @ Subtrans	2.703	-	1.632	11.787	-	4.020	6.218	-	0.155	0.028	26.544	0.023	0.127	0.238	0.388	
122	Class Peak @ Subtransmission	2.642	-	1.203	9.337	-	5.669	7.071	-	0.308	0.053	26.283	0.041	0.222	0.149	0.412	
230	Classpeak @ Primary	2.736	-	-	10.251	-	-	7.763	-	-	0.059	20.809	0.045	0.244	0.164	0.453	
231	Classpeak @ Secondary	-	-	-	-	-	-	-	-	-	-	-	0.057	0.308	0.207	0.572	
233	Classpeak for Streetlighting	-	-	-	-	-	-	-	-	-	-	-	15.504	83.517	-	99.021	
235	Classpeak @ Single Phase	-	-	-	-	-	-	-	-	-	-	-	0.057	0.308	0.207	0.572	
253	Billed Sales - Primary	3.021	-	-	13.481	-	-	11.119	-	-	0.033	27.654	0.044	0.261	0.340	0.645	
260	Customers - Residential	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
261	Customers - Drops	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
263	Customers - NonPID	-	-	-	-	-	-	-	-	-	-	-	0.021	-	-	0.021	
264	Customers - NonMunicipal	0.082	0.001	0.002	0.069	0.002	0.005	0.053	0.000	0.000	0.000	0.215	-	-	-	-	
267	Customers - Commercial & Primary	0.691	0.008	0.018	0.586	0.016	0.043	0.450	0.002	0.003	0.002	1.822	-	-	-	-	
161	Customer Count (CCC)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
162	Customer Count (BCC)	1.276	0.016	0.034	1.083	0.030	0.080	0.831	0.005	0.006	0.004	3.363	0.335	3.889	0.383	4.608	
301	PIS - HVD (Voltage 1)	2.507	0.826	1.121	8.702	4.242	5.283	6.589	0.890	0.287	0.050	30.497	0.038	0.207	0.139	0.384	
302	PIS - HVD (Voltage 2)	2.642	-	1.203	9.337	-	5.669	7.071	-	0.308	0.053	26.283	0.041	0.222	0.149	0.412	
303	PIS - HVD (Voltage 1) Subs (FERC 361/362)	2.507	0.826	1.121	8.702	4.242	5.283	6.589	0.890	0.287	0.050	30.497	0.038	0.207	0.139	0.384	
304	PIS - HVD (Voltage 2) Subs (FERC 361/362)	2.642	-	1.203	9.337	-	5.669	7.071	-	0.308	0.053	26.283	0.041	0.222	0.149	0.412	

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Summary ALLOCATORS 2										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
	<u>*Calculated Allocation Schedules</u>									
	PIS - OH LVD System	305	100.000	100.000	63.810	29.805	5.843	0.538	0.003	-
	PIS - LVD Distribution	306	100.000	99.998	65.270	26.535	5.255	2.934	0.004	0.002
	PIS- OH (LVD & HVD) & Services	307	100.000	99.987	66.525	25.068	7.746	0.437	0.211	0.013
	PIS- UG LVD Distribution	308	100.000	100.000	64.901	30.314	4.235	0.548	0.002	(0.000)
	PIS- Total Distribution	309	100.000	99.931	61.471	25.619	10.125	2.384	0.332	0.069
	PIS- Distribution Services	310	100.000	100.000	94.495	5.505	-	-	-	(0.000)
	PIS- Streetlighting Equipment	311	100.000	99.925	-	-	-	99.925	-	0.075
	PIS- Line Equipment	312	100.000	100.000	66.239	30.939	2.262	0.559	0.001	(0.000)
	PIS- Meters	313	100.000	99.994	60.629	31.913	7.403	0.022	0.027	0.006
	PIS - General	315	100.000	99.664	59.565	23.517	15.328	1.044	0.209	0.336
	Total PIS	316	100.000	99.738	58.308	25.076	14.233	1.830	0.290	0.262
	Distribution Depreciation	317	100.000	99.954	65.304	24.343	7.180	2.939	0.187	0.046
	CWIP	330	100.000	99.294	50.480	24.072	23.869	0.659	0.214	0.706
	Rate Base- Total	390	100.000	99.645	55.984	25.061	16.888	1.410	0.302	0.355
	Dist Op Expense (LVD) excl Sup & Eng	400	100.000	99.985	65.103	24.205	6.717	3.803	0.158	0.015
	Dist Maint Expense (LVD) excl Sup & Eng	401	100.000	99.987	65.880	25.244	8.150	0.515	0.198	0.013
	Dist Op Expense (HVD Voltage 1) excl Sup & Eng	402	100.000	99.145	46.131	21.795	29.668	0.348	1.204	0.855
	Dist Maint Expense (HVD Voltage 1) excl Sup & Eng	403	100.000	99.336	46.329	21.935	29.569	0.342	1.161	0.664
	Dist Op Expense (HVD Voltage 2) excl Sup & Eng	404	100.000	99.952	48.878	22.830	26.283	0.412	1.549	0.048
	Dist Maint Expense (HVD Voltage 2) excl Sup & Eng	405	100.000	99.952	48.878	22.830	26.283	0.412	1.549	0.048
	Total HVD Distribution O&M Expense	406	100.000	99.817	48.338	22.639	26.973	0.398	1.469	0.183
	Total Distribution O&M Expense (excl. HVD)	407	100.000	99.987	65.633	24.865	7.617	1.690	0.183	0.013
	Total Customer Accounts Expense (excl. Supv)	408	100.000	99.818	75.667	15.510	8.166	0.460	0.015	0.182
	Total Customer Accounts & Service Expense	409	100.000	99.771	73.377	15.782	10.120	0.461	0.032	0.229
	Jurisdictional Distribution O&M	414	100.000	100.000	64.571	24.741	8.809	1.617	0.262	-
	Pre Tax NOI	439	100.000	100.797	59.984	21.470	18.186	0.848	0.309	(0.797)
	Depreciation & Amortization Expense	442	100.000	99.585	56.002	24.821	17.031	1.516	0.215	0.415
	Non PSCR O&M Expense	443	100.000	99.660	59.738	23.262	15.387	1.057	0.216	0.340
	Distribution Depreciation Expense	444	100.000	99.950	62.431	25.900	8.736	2.639	0.244	0.050
	Gen/Comm/Int Depreciation Expense	445	100.000	99.664	59.565	23.517	15.328	1.044	0.209	0.336
	Production Labor	500	100.000	99.114	47.021	23.903	27.723	0.287	0.180	0.886
	Total Labor	502	100.000	99.664	59.565	23.517	15.328	1.044	0.209	0.336
	50% O&M, 50% Net Plant	600	100.000	99.595	54.567	25.019	18.277	1.412	0.320	0.405
	50/50 PIS & Labor	601	100.000	99.701	58.937	24.297	14.781	1.437	0.250	0.299

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Residential/Secondary ALLOCATORS 2							
	(a)		(b)	(c)	(d)	(e)	(h)
Line No.	Description	Alloc	Rate RS	Total Residential	Rate GS	Rate GSD	Commercial Secondary
	<u>*Calculated Allocation Schedules</u>						
	PIS - OH LVD System	305	63.810	63.810	18.470	11.335	29.805
	PIS - LVD Distribution	306	65.270	65.270	17.082	9.453	26.535
	PIS- OH (LVD & HVD) & Services	307	66.525	66.525	15.776	9.291	25.068
	PIS- UG LVD Distribution	308	64.901	64.901	18.786	11.529	30.314
	PIS- Total Distribution	309	61.471	61.471	16.376	9.243	25.619
	PIS- Distribution Services	310	94.495	94.495	4.969	0.536	5.505
	PIS- Streetlighting Equipment	311	-	-	-	-	-
	PIS- Line Equipment	312	66.239	66.239	19.173	11.766	30.939
	PIS- Meters	313	60.629	60.629	25.838	6.075	31.913
	PIS - General	315	59.565	59.565	14.897	8.620	23.517
	Total PIS	316	58.308	58.308	15.844	9.232	25.076
	Distribution Depreciation	317	65.304	65.304	15.401	8.942	24.343
	CWIP	330	50.480	50.480	14.751	9.321	24.072
	Rate Base- Total	390	55.984	55.984	15.793	9.268	25.061
	Dist Op Expense (LVD) excl Sup & Eng	400	65.103	65.103	15.690	8.514	24.205
	Dist Maint Expense (LVD) excl Sup & Eng	401	65.880	65.880	15.990	9.253	25.244
	Dist Op Expense (HVD Voltage 1) excl Sup & Eng	402	46.131	46.131	13.477	8.318	21.795
	Dist Maint Expense (HVD Voltage 1) excl Sup & Eng	403	46.329	46.329	13.558	8.377	21.935
	Dist Op Expense (HVD Voltage 2) excl Sup & Eng	404	48.878	48.878	14.148	8.682	22.830
	Dist Maint Expense (HVD Voltage 2) excl Sup & Eng	405	48.878	48.878	14.148	8.682	22.830
	Total HVD Distribution O&M Expense	406	48.338	48.338	14.022	8.617	22.639
	Total Distribution O&M Expense (excl. HVD)	407	65.633	65.633	15.879	8.986	24.865
	Total Customer Accounts Expense (excl. Supv)	408	75.667	75.667	11.738	3.772	15.510
	Total Customer Accounts & Service Expense	409	73.377	73.377	11.717	4.065	15.782
	Jurisdictional Distribution O&M	414	64.571	64.571	15.774	8.968	24.741
	Pre Tax NOI	439	59.984	59.984	14.037	7.433	21.470
	Depreciation & Amortization Expense	442	56.002	56.002	15.670	9.151	24.821
	Non PSCR O&M Expense	443	59.738	59.738	14.790	8.472	23.262
	Distribution Depreciation Expense	444	62.431	62.431	16.866	9.034	25.900
	Gen/Comm/Int Depreciation Expense	445	59.565	59.565	14.897	8.620	23.517
	Production Labor	500	47.021	47.021	14.444	9.459	23.903
	Total Labor	502	59.565	59.565	14.897	8.620	23.517
	50% O&M, 50% Net Plant	600	54.567	54.567	15.658	9.361	25.019
	50/50 PIS & Labor	601	58.937	58.937	15.370	8.926	24.297

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

Projected 12-Month Period Ending February 28, 2025
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Primary & Lighting ALLOCATORS 2																	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(j)	(k)	(l)	(q)	(r)	(t)	(u)	(v)	
Line No.	Description	Alloc	Rate GP	Rate GPTU Vit 1	Rate GPTU Vit 2	Rate GPTU Vit 3	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate EIP Vit 1	Rate EIP Vit 2	Rate EIP Vit 3	Total Primary	Rate GML	Rate UUL	Rate GU	Lighting & Unmetered
<u>*Calculated Allocation Schedules</u>																	
	PIS - OH LVD System	305	0.768	-	-	2.879	-	-	2.180	-	-	0.016	5.843	0.054	0.290	0.195	0.538
	PIS - LVD Distribution	306	0.807	0.003	0.007	2.504	0.006	0.016	1.895	0.001	0.001	0.014	5.255	0.072	2.708	0.153	2.934
	PIS- OH (LVD & HVD) & Services	307	0.902	0.007	0.163	3.302	0.037	0.766	2.501	0.008	0.042	0.019	7.746	0.044	0.235	0.158	0.437
	PIS- UG LVD Distribution	308	0.557	-	-	2.086	-	-	1.580	-	-	0.012	4.235	0.055	0.295	0.198	0.548
	PIS- Total Distribution	309	1.196	0.062	0.261	3.943	0.309	1.216	2.985	0.064	0.066	0.022	10.125	0.065	2.167	0.152	2.384
	PIS- Distribution Services	310	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	PIS- Streetlighting Equipment	311	-	-	-	-	-	-	-	-	-	-	-	1.183	98.742	-	99.925
	PIS- Line Equipment	312	0.297	-	-	1.114	-	-	0.844	-	-	0.006	2.262	0.056	0.301	0.202	0.559
	PIS- Meters	313	2.482	0.040	0.096	2.546	0.084	0.208	1.912	0.013	0.015	0.007	7.403	0.022	-	-	0.022
	PIS - General	315	1.539	0.374	0.660	6.087	0.828	1.903	3.737	0.134	0.051	0.015	15.328	0.038	0.839	0.167	1.044
	Total PIS	316	1.508	0.279	0.545	5.580	0.708	1.769	3.645	0.118	0.062	0.018	14.233	0.051	1.612	0.167	1.830
	Distribution Depreciation	317	0.827	0.040	0.144	2.952	0.207	0.677	2.235	0.043	0.037	0.017	7.180	0.070	2.719	0.151	2.939
	CWIP	330	2.250	0.777	1.201	9.393	1.636	3.085	5.218	0.244	0.055	0.011	23.869	0.021	0.437	0.201	0.659
	Rate Base- Total	390	1.741	0.388	0.709	6.582	0.926	2.164	4.135	0.157	0.068	0.018	16.888	0.044	1.192	0.174	1.410
	Dist Op Expense (LVD) excl Sup & Eng	400	0.869	0.009	0.125	2.868	0.041	0.578	2.171	0.008	0.032	0.016	6.717	0.081	3.581	0.141	3.803
	Dist Maint Expense (LVD) excl Sup & Eng	401	0.993	0.007	0.154	3.514	0.036	0.719	2.661	0.007	0.039	0.020	8.150	0.044	0.315	0.156	0.515
	Dist Op Expense (HVD Voltage 1) excl Sup & I	402	2.514	0.837	1.161	9.048	3.780	4.925	6.401	0.727	0.234	0.041	29.668	0.031	0.169	0.147	0.348
	Dist Maint Expense (HVD Voltage 1) excl Sup	403	2.520	0.841	1.171	9.131	3.700	4.866	6.378	0.697	0.225	0.039	29.569	0.030	0.162	0.149	0.342
	Dist Op Expense (HVD Voltage 2) excl Sup & I	404	2.642	-	1.203	9.337	-	5.669	7.071	-	0.308	0.053	26.283	0.041	0.222	0.149	0.412
	Dist Maint Expense (HVD Voltage 2) excl Sup	405	2.642	-	1.203	9.337	-	5.669	7.071	-	0.308	0.053	26.283	0.041	0.222	0.149	0.412
	Total HVD Distribution O&M Expense	406	2.616	0.176	1.196	9.291	0.776	5.503	6.926	0.147	0.291	0.050	26.973	0.039	0.210	0.149	0.398
	Total Distribution O&M Expense (excl. HVD)	407	0.947	0.008	0.142	3.277	0.035	0.664	2.481	0.007	0.036	0.019	7.617	0.058	1.483	0.150	1.690
	Total Customer Accounts Expense (excl. Supv	408	0.771	0.269	0.416	3.034	0.642	1.055	1.736	0.195	0.041	0.007	8.166	0.026	0.347	0.087	0.460
	Total Customer Accounts & Service Expense	409	0.874	0.321	0.489	3.478	0.994	1.491	2.166	0.249	0.049	0.008	10.120	0.026	0.336	0.098	0.461
	Jurisdictional Distribution O&M	414	1.050	0.018	0.207	3.647	0.082	0.961	2.755	0.016	0.052	0.020	8.809	0.056	1.411	0.150	1.617
	Pre Tax NOI	439	2.164	0.619	1.018	7.051	1.109	2.254	3.957	(0.063)	0.069	0.008	18.186	0.018	0.678	0.152	0.848
	Depreciation & Amortization Expense	442	1.752	0.454	0.752	6.735	1.001	2.084	4.035	0.154	0.050	0.014	17.031	0.041	1.300	0.175	1.516
	Non PSCR O&M Expense	443	1.506	0.388	0.674	6.015	0.883	1.923	3.714	0.205	0.063	0.016	15.387	0.040	0.845	0.172	1.057
	Distribution Depreciation Expense	444	1.152	0.045	0.197	3.477	0.220	0.900	2.631	0.046	0.049	0.019	8.736	0.067	2.429	0.143	2.639
	Gen/Comm/Int Depreciation Expense	445	1.539	0.374	0.660	6.087	0.828	1.903	3.737	0.134	0.051	0.015	15.328	0.038	0.839	0.167	1.044
	Production Labor	500	2.554	0.978	1.461	10.927	2.011	3.591	5.849	0.294	0.051	0.007	27.723	0.010	0.060	0.216	0.287
	Total Labor	502	1.539	0.374	0.660	6.087	0.828	1.903	3.737	0.134	0.051	0.015	15.328	0.038	0.839	0.167	1.044
	50% O&M, 50% Net Plant	600	1.806	0.452	0.795	6.979	1.133	2.371	4.348	0.284	0.087	0.021	18.277	0.046	1.180	0.187	1.412
	50/50 PIS & Labor	601	1.523	0.327	0.603	5.834	0.768	1.836	3.691	0.126	0.056	0.017	14.781	0.045	1.226	0.167	1.437

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Summary RETURN (SUMMARY)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional	
1	Rate Base	14,404,493	14,354,217	8,063,705	3,595,591	2,448,185	203,126	43,609	50,276	
2	Revenue									
3	Total Rate Revenue	4,283,187	4,258,970	2,212,572	998,917	1,006,592	38,997	1,893	24,217	
4	Total Revenue Credits	819,090	817,956	371,907	191,014	240,331	3,154	11,551	1,134	
5	Total Revenue	5,102,277	5,076,926	2,584,478	1,189,931	1,246,923	42,151	13,443	25,351	
6	Expenses:									
7	Fuel and P&I Expense	2,045,668	2,025,232	869,901	477,456	661,585	10,436	5,855	20,436	
8	Transmission Expense	543,730	538,999	248,150	126,951	160,721	1,997	1,179	4,731	
9	Other O & M Expense	705,078	702,708	421,139	163,414	109,171	7,457	1,527	2,370	
10	Depreciation & Amortization Expense	733,075	730,079	410,405	180,973	126,082	11,050	1,569	2,996	
11	Other Taxes	325,611	324,939	185,739	78,732	54,662	4,805	1,001	672	
12	Federal Income Taxes	57,601	58,051	34,536	12,488	10,357	493	178	(450)	
13	Total Expenses	4,410,762	4,380,007	2,169,869	1,040,013	1,122,578	36,237	11,310	30,755	
14	Net Operating Income	691,515	696,919	414,609	149,918	124,345	5,914	2,134	(5,404)	
15	Other Income Adjustments	25,823	25,643	12,976	6,178	6,264	170	55	180	
16	Adjusted Net Operating Income	717,339	722,562	427,585	156,096	130,608	6,084	2,189	(5,224)	
17	Rate of Return on Rate Base	4.98%	5.03%	5.30%	4.34%	5.33%	3.00%	5.02%	-10.39%	
18	Index of Return (Jurisdictional)		100	105	86	106	60	100		
17	Return on Rate Base @ 6.11%	880,283	877,211	492,787	219,733	149,613	12,413	2,665	3,072	
18	Income Deficiency (Sufficiency)	162,945	154,649	65,202	63,636	19,005	6,329	476	8,296	
19	Revenue Deficiency (Sufficiency)	218,194	207,085	87,310	85,214	25,448	8,475	638	11,109	
	Additional Rev Requirement	0	0	0	0	0	0	0	0	
20	Total Revenue Deficiency (Sufficiency)	218,194	207,085	87,310	85,214	25,448	8,475	638	11,109	
21	Revenue Requirement/Total Cost of Service	5,320,472	5,284,012	2,671,789	1,275,145	1,272,371	50,626	14,081	36,460	
22	Less: Revenue Credits	819,090	817,956	371,907	191,014	240,331	3,154	11,551	1,134	
23	Proposed Rate Design Revenue	4,501,381	4,466,056	2,299,882	1,084,131	1,032,040	47,472	2,530	35,326	
24	Production: Net Capacity Cost	202,916	201,135	97,930	48,465	54,101	372	266	1,781	
25	Production: Capacity Related Cost Offset	1,092,719	1,081,396	497,109	260,998	318,353	3,408	1,527	11,323	
26	Production: Non-Capacity Related Cost	1,446,700	1,425,603	595,745	336,295	489,936	8,619	(4,992)	21,097	
27	Distribution: Demand Related Cost	1,519,429	1,518,346	929,150	391,784	158,300	33,515	5,596	1,083	
28	Distribution: Customer Related Cost	239,617	239,577	179,947	46,588	11,349	1,558	134	41	
29	Full Service MWH Sales	30,656,265	30,328,039	12,559,141	6,973,697	10,516,047	177,460	101,694	328,226	
30	ROA MWH Sales	3,713,915	3,713,915	-	195,488	3,518,427	-	-	-	
31	MWH Sales	34,370,180	34,041,954	12,559,141	7,169,185	14,034,473	177,460	101,694	328,226	
32	Customers	1,890,850	1,890,848	1,662,850	218,379	4,052	5,551	15	2	



Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Residential/Secondary RETURN (SUMMARY)						
	(a)	(b)	(c)	(d)	(e)	(f)
Line No.	Description	Rate RS	Total Residential	Rate GS	Rate GSD	Total Commercial Secondary
1	Rate Base	8,063,705	8,063,705	2,267,110	1,328,481	3,595,591
	Revenue					
3	Total Rate Revenue	2,212,572	2,212,572	614,759	384,158	998,917
4	Total Revenue Credits	371,907	371,907	113,989	77,025	191,014
5	Total Revenue	2,584,478	2,584,478	728,748	461,183	1,189,931
6	Expenses:					
7	Fuel and P&I Expense	869,901	869,901	278,790	198,665	477,456
8	Transmission Expense	248,150	248,150	76,014	50,938	126,951
9	Other O & M Expense	421,139	421,139	103,939	59,475	163,414
10	Depreciation & Amortization Expense	410,405	410,405	114,422	66,551	180,973
11	Other Taxes	185,739	185,739	49,647	29,084	78,732
12	Federal Income Taxes	34,536	34,536	8,146	4,342	12,488
13	Total Expenses	2,169,869	2,169,869	630,958	409,055	1,040,013
14	Net Operating Income	414,609	414,609	97,790	52,128	149,918
15	Other Income Adjustments	12,976	12,976	3,786	2,392	6,178
16	Adjusted Net Operating Income	427,585	427,585	101,577	54,520	156,096
17	Rate of Return on Rate Base	5.30%	5.30%	4.48%	4.10%	4.34%
18	Index of Return (Jurisdictional)	105	105	89	82	86
17	Return on Rate Base @ 6.11%	492,787	492,787	138,547	81,186	219,733
18	Income Deficiency (Sufficiency)	65,202	65,202	36,970	26,666	63,636
19	Revenue Deficiency (Sufficiency)	87,310	87,310	49,506	35,708	85,214
	Additional Rev Requirement	0	0	0	0	0
20	Total Revenue Deficiency (Sufficiency)	87,310	87,310	49,506	35,708	85,214
21	Revenue Requirement/Total Cost of Service	2,671,789	2,671,789	778,254	496,891	1,275,145
22	Less: Revenue Credits	371,907	371,907	113,989	77,025	191,014
23	Proposed Rate Design Revenue	2,299,882	2,299,882	664,265	419,866	1,084,131
24	Production: Net Capacity Cost	97,930	97,930	29,710	18,755	48,465
25	Production: Capacity Related Cost Offset	497,109	497,109	156,186	104,813	260,998
26	Production: Non-Capacity Related Cost	595,745	595,745	193,970	142,325	336,295
27	Distribution: Demand Related Cost	929,150	929,150	247,109	144,676	391,784
28	Distribution: Customer Related Cost	179,947	179,947	37,291	9,297	46,588
29	Full Service MWH Sales	12,559,141	12,559,141	4,024,187	2,949,510	6,973,697
30	ROA MWH Sales	-	-	23,026	172,462	195,488
31	MWH Sales	12,559,141	12,559,141	4,047,213	3,121,972	7,169,185
32	Customers	1,662,850	1,662,850	197,136	21,244	218,379

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Primary & Lighting RETURN (SUMMARY)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Line No.	Description	Rate GP	Rate GPTU Vlt 1	Rate GPTU Vlt 2	Rate GPTU Vlt 3	Rate GPD Vlt 1	Rate GPD Vlt 2	Rate GPD Vlt 3	Rate EIP Vlt 1	Rate EIP Vlt 2	Rate EIP Vlt 3	Total Primary	Rate GML	Rate UUL	Rate GU	Total Lighting & Unmetered
1	Rate Base	251,674	54,909	101,542	945,875	131,084	308,773	591,222	50,214	10,330	2,562	2,448,185	6,336	171,754	25,036	203,126
	Revenue															
3	Total Rate Revenue	91,875	33,312	51,637	375,888	78,990	131,069	214,317	23,615	5,006	884	1,006,592	1,472	28,187	9,338	38,997
4	Total Revenue Credits	20,443	8,450	12,515	89,339	19,481	31,540	49,302	7,779	1,289	194	240,331	170	1,032	1,952	3,154
5	Total Revenue	112,318	41,763	64,152	465,227	98,471	162,609	263,618	31,393	6,295	1,077	1,246,923	1,642	29,219	11,290	42,151
6	Expenses:															
7	Fuel and P&I Expense	52,961	24,200	35,555	239,404	60,295	88,955	133,057	22,726	3,847	586	661,585	661	3,894	5,880	10,436
8	Transmission Expense	14,154	5,590	8,393	60,614	12,948	20,674	31,975	5,432	799	142	160,721	118	653	1,226	1,997
9	Other O & M Expense	10,645	2,717	4,733	42,325	6,188	13,468	26,017	2,475	489	113	109,171	285	5,960	1,212	7,457
10	Depreciation & Amortization Expense	12,829	3,287	5,474	49,116	7,246	15,131	29,319	3,009	548	122	126,082	301	9,473	1,276	11,050
11	Other Taxes	5,569	1,272	2,316	20,677	3,335	7,239	13,166	816	219	53	54,662	141	4,108	556	4,805
12	Federal Income Taxes	1,243	361	591	4,082	650	1,318	2,313	(236)	30	5	10,357	10	394	88	493
13	Total Expenses	97,401	37,427	57,062	416,217	90,663	146,785	235,848	34,222	5,932	1,020	1,122,578	1,517	24,483	10,237	36,237
14	Net Operating Income	14,917	4,336	7,090	49,010	7,808	15,824	27,770	(2,829)	363	57	124,345	125	4,736	1,053	5,914
15	Other Income Adjustments	578	199	308	2,412	419	791	1,338	182	30	5	6,264	6	113	52	170
16	Adjusted Net Operating Income	15,495	4,535	7,398	51,421	8,227	16,615	29,109	(2,647)	393	62	130,608	131	4,849	1,105	6,084
17	Rate of Return on Rate Base	6.16%	8.26%	7.29%	5.44%	6.28%	5.38%	4.92%	-5.27%	3.81%	2.44%	5.33%	2.07%	2.82%	4.41%	3.00%
18	Index of Return (Jurisdictional)	122	164	145	108	125	107	98	(105)	76	48	106	41	56	88	60
17	Return on Rate Base @ 6.11%	15,380	3,356	6,205	57,804	8,011	18,870	36,131	3,069	631	157	149,613	387	10,496	1,530	12,413
18	Income Deficiency (Sufficiency)	(115)	(1,179)	(1,192)	6,383	(216)	2,255	7,022	5,716	238	94	19,005	256	5,648	425	6,329
19	Revenue Deficiency (Sufficiency)	(154)	(1,579)	(1,597)	8,547	(289)	3,020	9,403	7,654	319	126	25,448	343	7,562	569	8,475
	Additional Rev Requirement	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
20	Total Revenue Deficiency (Sufficiency)	(154)	(1,579)	(1,597)	8,547	(289)	3,020	9,403	7,654	319	126	25,448	343	7,562	569	8,475
21	Revenue Requirement/Total Cost of Service	112,164	40,184	62,555	473,774	98,181	165,629	273,021	39,047	6,614	1,203	1,272,371	1,985	36,782	11,859	50,626
22	Less: Revenue Credits	20,443	8,450	12,515	89,339	19,481	31,540	49,302	7,779	1,289	194	240,331	170	1,032	1,952	3,154
23	Proposed Rate Design Revenue	91,720	31,733	50,040	384,435	78,701	134,089	223,719	31,268	5,325	1,010	1,032,040	1,815	35,750	9,907	47,472
24	Production: Net Capacity Cost	5,112	1,786	2,693	21,297	3,455	6,684	11,179	1,609	248	39	54,101	1	-	371	372
25	Production: Capacity Related Cost Offset	28,763	10,724	16,072	121,384	22,732	40,005	65,418	11,196	1,780	281	318,353	151	784	2,472	3,408
26	Production: Non-Capacity Related Cost	38,174	18,159	26,690	174,593	47,704	66,741	97,006	17,495	2,909	466	489,936	602	3,515	4,502	8,619
27	Distribution: Demand Related Cost	17,013	952	4,335	63,255	4,365	19,740	47,184	887	358	212	158,300	1,007	30,017	2,491	33,515
28	Distribution: Customer Related Cost	2,659	112	250	3,906	445	920	2,932	82	31	12	11,349	54	1,433	71	1,558
29	Full Service MWH Sales	805,656	397,758	574,708	3,709,392	1,033,644	1,448,667	2,098,603	374,401	64,003	9,214	10,516,047	12,195	71,839	93,426	177,460
30	ROA MWH Sales	75,604	-	-	-	1,130,638	1,351,423	960,761	-	-	-	3,518,427	-	-	-	-
31	MWH Sales	881,260	397,758	574,708	3,709,392	2,164,283	2,800,090	3,059,364	374,401	64,003	9,214	14,034,473	12,195	71,839	93,426	177,460
32	Customers	1,537	19	41	1,305	36	96	1,002	6	7	4	4,052	404	4,686	461	5,551

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Summary RATE BASE (SUMMARY)										
Line No.	(a) Description	(b)	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Net Plant									
2	Production		2,617,301	2,594,325	1,221,778	621,202	739,197	7,472	4,676	22,977
3	Transmission		-	-	-	-	-	-	-	-
4	Distribution		8,113,317	8,107,296	4,860,915	2,118,837	921,685	173,319	32,540	6,020
5	General/Common/Intangible		686,368	684,093	408,574	160,712	106,198	7,168	1,441	2,275
6	Plant Purchased/Sold		0	0	0	0	0	0	0	0
7	Total Net Plant		11,416,986	11,385,714	6,491,266	2,900,750	1,767,081	187,960	38,657	31,272
8	Working Capital									
9	Total Current Assets		4,117,364	4,095,367	2,232,956	974,859	844,644	34,807	8,101	21,997
10	Total Current Liabilities		1,072,461	1,069,506	625,146	265,356	157,775	18,271	2,957	2,956
11	Total Working Capital		3,044,903	3,025,861	1,607,810	709,503	686,869	16,535	5,144	19,041
	Adjustments to Rate Base									
13	Additions to Rate Base		0	0	0	0	0	0	0	0
14	Deductions from Rate Base		57,395	57,358	35,371	14,662	5,764	1,369	192	37
15	Total Adjustments to Rate Base		(57,395)	(57,358)	(35,371)	(14,662)	(5,764)	(1,369)	(192)	(37)
16	Total Rate Base		14,404,493	14,354,217	8,063,705	3,595,591	2,448,185	203,126	43,609	50,276

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Residential/Secondary RATE BASE (SUMMARY)						
	(a)	(b)	(c)	(d)	(e)	(h)
Line No.	Description	Rate RS	Total Residential	Rate GS	Rate GSD	Commercial Secondary
1	Net Plant					
2	Production	1,221,778	1,221,778	375,333	245,868	621,202
3	Transmission	-	-	-	-	-
4	Distribution	4,860,915	4,860,915	1,360,788	758,049	2,118,837
5	General/Common/Intangible	408,574	408,574	101,841	58,871	160,712
6	Plant Purchased/Sold	0	0	0	0	0
7	Total Net Plant	6,491,266	6,491,266	1,837,962	1,062,788	2,900,750
8	Working Capital					
9	Total Current Assets	2,232,956	2,232,956	606,178	368,682	974,859
10	Total Current Liabilities	625,146	625,146	167,653	97,704	265,356
11	Total Working Capital	1,607,810	1,607,810	438,525	270,978	709,503
	Adjustments to Rate Base					
13	Additions to Rate Base	0	0	0	0	0
14	Deductions from Rate Base	35,371	35,371	9,377	5,285	14,662
15	Total Adjustments to Rate Base	(35,371)	(35,371)	(9,377)	(5,285)	(14,662)
16	Total Rate Base	8,063,705	8,063,705	2,267,110	1,328,481	3,595,591

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Primary & Lighting RATE BASE (SUMMARY)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(j)	(k)	(l)	(q)	(r)	(t)	(u)	(v)
Line No.	Description	Rate GP	Rate GPTU Vit 1	Rate GPTU Vit 2	Rate GPTU Vit 3	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate EIP Vit 1	Rate EIP Vit 2	Rate EIP Vit 3	Total Primary	Rate GML	Rate UUL	Rate GU	Total Lighting & Unmetered
1	Net Plant															
2	Production	66,382	25,447	37,993	284,049	52,325	93,389	152,058	23,254	3,726	575	739,197	272	1,568	5,633	7,472
3	Transmission	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Distribution	112,230	5,158	25,395	357,816	27,226	116,916	267,564	5,254	2,825	1,302	921,685	5,155	155,787	12,377	173,319
5	General/Common/Intangible	10,582	2,548	4,504	41,648	5,641	12,958	25,457	2,300	452	108	106,198	260	5,763	1,146	7,168
6	Plant Purchased/Sold	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	Total Net Plant	189,194	33,153	67,891	683,513	85,192	223,263	445,079	30,808	7,003	1,985	1,767,081	5,687	163,118	19,155	187,960
8	Working Capital															
9	Total Current Assets	79,687	25,001	39,968	326,127	53,982	105,564	187,207	22,377	3,974	759	844,644	1,215	25,823	7,769	34,807
10	Total Current Liabilities	16,510	3,213	6,168	61,492	7,923	19,365	39,363	2,939	630	174	157,775	529	15,942	1,801	18,271
11	Total Working Capital	63,177	21,788	33,800	264,635	46,059	86,199	147,843	19,439	3,344	585	686,869	686	9,881	5,968	16,535
Adjustments to Rate Base																
13	Additions to Rate Base	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14	Deductions from Rate Base	697	31	149	2,273	167	689	1,700	32	17	8	5,764	38	1,244	87	1,369
15	Total Adjustments to Rate Base	(697)	(31)	(149)	(2,273)	(167)	(689)	(1,700)	(32)	(17)	(8)	(5,764)	(38)	(1,244)	(87)	(1,369)
16	Total Rate Base	251,674	54,909	101,542	945,875	131,084	308,773	591,222	50,214	10,330	2,562	2,448,185	6,336	171,754	25,036	203,126

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Summary O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional		
1	Production Expense										
2	Fuel Expense	1,039,289	1,028,589	431,039	241,533	346,869	5,891	3,257	10,700		
3	Purchased & Interchange Power Expense	1,006,379	996,644	438,862	235,923	314,716	4,544	2,598	9,736		
4	Total Fuel and P&I Expense	2,045,668	2,025,232	869,901	477,456	661,585	10,436	5,855	20,436		
5	Fossil O&M Exp	99,427	98,475	43,917	23,346	30,518	443	251	952		
6	Nuclear O&M Exp	0	-	-	-	-	-	-	-		
7	Hydro O&M Exp	19,168	18,988	8,573	4,512	5,779	79	45	180		
8	Other Power Gen O&M Exp	72,922	72,282	34,041	17,308	20,595	208	130	640		
9	Other Power Supply O&M Exp	8,579	8,503	4,005	2,036	2,423	24	15	75		
10	Total Production O&M Expense	200,095	198,248	90,536	47,201	59,315	754	442	1,847		
11	Total Production (Inc. Fuel and P&I) O&M Expense	2,245,763	2,223,480	960,437	524,657	720,899	11,189	6,298	22,283		
12	Transmission & Distribution Expense										
13	Trans O&M Exp	543,730	538,999	248,150	126,951	160,721	1,997	1,179	4,731		
14	Other O&M Adjustments	0	0	0	0	0	0	0	0		
15	Distr Oper Exp	113,332	113,313	73,497	27,302	8,075	4,227	213	18		
16	Distr Maint Exp	215,272	215,225	139,090	53,714	20,679	1,092	651	47		
17	Total Transmission & Distribution O&M Expense	872,333	867,537	460,737	207,966	189,474	7,316	2,044	4,796		
18	Customer Related Expense										
19	Customer Accounts Exp	57,278	57,173	43,340	8,884	4,677	264	8	104		
20	Customer Service Exp	5,151	5,113	2,468	968	1,641	24	12	38		
21	Other Customer Exp	18,882	18,882	15,616	3,085	76	104	0	0		
22	Total Customer O&M Expense	81,310	81,168	61,425	12,937	6,393	392	21	143		
23	Admin & General Expense	95,069	94,753	56,591	22,260	14,709	993	200	315		
24	Total O&M Expense	3,294,476	3,266,939	1,539,190	767,821	931,476	19,890	8,562	27,537		

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Residential/Secondary O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(h)
Line No.	Description		Rate RS	Total Residential	Rate GS	Rate GSD	Commercial Secondary
1	Production Expense						
2	Fuel Expense		431,039	431,039	139,769	101,764	241,533
3	Purchased & Interchange Power Expense		438,862	438,862	139,022	96,901	235,923
4	Total Fuel and P&I Expense		869,901	869,901	278,790	198,665	477,456
5	Fossil O&M Exp		43,917	43,917	13,778	9,568	23,346
6	Nuclear O&M Exp		-	-	-	-	-
7	Hydro O&M Exp		8,573	8,573	2,677	1,835	4,512
8	Other Power Gen O&M Exp		34,041	34,041	10,457	6,850	17,308
9	Other Power Supply O&M Exp		4,005	4,005	1,230	806	2,036
10	Total Production O&M Expense		90,536	90,536	28,143	19,059	47,201
11	Total Production (Inc. Fuel and P&I) O&M Expense		960,437	960,437	306,933	217,724	524,657
12	Transmission & Distribution Expense						
13	Trans O&M Exp		248,150	248,150	76,014	50,938	126,951
14	Other O&M Adjustments		0	0	0	0	0
15	Distr Oper Exp		73,497	73,497	17,691	9,610	27,302
16	Distr Maint Exp		139,090	139,090	33,987	19,727	53,714
17	Total Transmission & Distribution O&M Expense		460,737	460,737	127,691	80,275	207,966
18	Customer Related Expense						
19	Customer Accounts Exp		43,340	43,340	6,723	2,160	8,884
20	Customer Service Exp		2,468	2,468	591	377	968
21	Other Customer Exp		15,616	15,616	2,697	388	3,085
22	Total Customer O&M Expense		61,425	61,425	10,012	2,925	12,937
23	Admin & General Expense		56,591	56,591	14,106	8,154	22,260
24	Total O&M Expense		1,539,190	1,539,190	458,743	309,078	767,821

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Primary & Lighting O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(q)	(r)	(t)	(u)	(v)
Line No.	Description	Rate GP	Rate GPTU Vit 1	Rate GPTU Vit 2	Rate GPTU Vit 3	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate EIP Vit 1	Rate EIP Vit 2	Rate EIP Vit 3	Total Primary	Rate GML	Rate UUL	Rate GU	Lighting & Unmetered		
1	Production Expense																	
2	Fuel Expense	27,045	12,848	18,815	123,859	33,121	47,241	69,421	12,130	2,074	315	346,869	394	2,321	3,177	5,891		
3	Purchased & Interchange Power Expense	25,916	11,352	16,740	115,545	27,174	41,714	63,635	10,596	1,773	271	314,716	268	1,573	2,703	4,544		
4	Total Fuel and P&I Expense	52,961	24,200	35,555	239,404	60,295	88,955	133,057	22,726	3,847	586	661,585	661	3,894	5,880	10,436		
5	Fossil O&M Exp	2,546	1,109	1,636	11,338	2,448	3,972	6,239	1,031	173	26	30,518	26	153	264	443		
6	Nuclear O&M Exp	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
7	Hydro O&M Exp	490	208	307	2,162	452	748	1,183	192	32	5	5,779	4	25	49	79		
8	Other Power Gen O&M Exp	1,850	709	1,059	7,914	1,458	2,602	4,237	648	104	16	20,595	8	44	157	208		
9	Other Power Supply O&M Exp	218	83	125	931	172	306	498	76	12	2	2,423	1	5	18	24		
10	Total Production O&M Expense	5,103	2,109	3,126	22,345	4,529	7,627	12,157	1,948	321	48	59,315	39	227	488	754		
11	Total Production (Inc. Fuel and P&I) O&M Expense	58,065	26,309	38,681	261,748	64,824	96,582	145,214	24,674	4,168	634	720,899	700	4,121	6,368	11,189		
12	Transmission & Distribution Expense																	
13	Trans O&M Exp	14,154	5,590	8,393	60,614	12,948	20,674	31,975	5,432	799	142	160,721	118	653	1,226	1,997		
14	Other O&M Adjustments	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
15	Distr Oper Exp	1,043	14	168	3,420	65	767	2,554	13	19	12	8,075	91	3,976	160	4,227		
16	Distr Maint Exp	2,458	40	509	8,596	191	2,346	6,409	41	56	31	20,679	95	662	335	1,092		
17	Total Transmission & Distribution O&M Expense	17,655	5,644	9,070	72,629	13,204	23,788	40,939	5,485	874	186	189,474	304	5,291	1,721	7,316		
18	Customer Related Expense																	
19	Customer Accounts Exp	442	154	238	1,738	368	604	994	112	23	4	4,677	15	199	50	264		
20	Customer Service Exp	104	46	67	434	253	327	358	44	7	1	1,641	2	11	11	24		
21	Other Customer Exp	29	0	1	24	1	2	19	0	0	0	76	8	88	9	104		
22	Total Customer O&M Expense	574	201	306	2,196	621	933	1,371	155	31	5	6,393	24	298	70	392		
23	Admin & General Expense	1,466	353	624	5,769	781	1,795	3,526	319	63	15	14,709	36	798	159	993		
24	Total O&M Expense	77,760	32,507	48,681	342,342	79,431	123,098	191,049	30,633	5,135	840	931,476	1,065	10,508	8,317	19,890		

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Summary ALLOCATORS 1										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
<u>Input Allocation Schedules</u>										
	Energy @ Generation	100	100.000	98.970	41.868	23.248	32.942	0.592	0.321	1.030
	Energy On-Peak @ Generation	101	100.000	98.970	39.804	24.342	34.120	0.422	0.282	1.030
	Energy Off-Peak @ Generation	102	100.000	98.970	43.604	21.754	32.483	0.770	0.360	1.030
	Energy On-Peak @ Generation Summer	103	100.000	98.970	40.944	24.619	32.842	0.331	0.234	1.030
	Energy Off-Peak @ Generation Summer	104	100.000	98.970	43.786	21.985	32.216	0.672	0.312	1.030
	Energy On-Peak @ Generation Non-Summer	105	100.000	98.970	39.117	24.175	34.891	0.477	0.310	1.030
	Energy Off-Peak @ Generation Non-Summer	106	100.000	98.970	43.500	21.622	32.635	0.826	0.387	1.030
	Energy Critical On-Peak @ Gen	107	100.000	98.970	44.276	23.898	30.355	0.237	0.204	1.030
	Energy Summer Mid-Peak @ Gen	108	100.000	98.970	39.708	24.886	33.765	0.366	0.246	1.030
	12CP Dmd @ Generation	120	100.000	99.130	45.639	23.348	29.559	0.367	0.217	0.870
	4CP Dmd @ Generation	121	100.000	99.122	48.261	23.884	26.662	0.183	0.131	0.878
	Classpeak @ Transmission	127	100.000	99.173	46.035	21.365	29.926	0.388	1.459	0.827
	Production Revenue	141	100.000	99.208	45.172	23.040	30.567	0.422	0.007	0.792
	Distribution Revenue	142	100.000	99.963	63.537	23.861	10.591	1.859	0.114	0.037
	Total Rate Revenue	143	100.000	99.457	51.249	23.312	23.957	0.897	0.042	0.543
	Billed Sales	150	100.000	99.045	36.541	20.859	40.833	0.516	0.296	0.955
	Number Of Customers	160	100.000	100.000	87.942	11.549	0.214	0.294	0.001	0.000
	Weighted Customer	170	100.000	99.994	60.629	31.913	7.403	0.022	0.027	0.006
<u>*Calculated Allocation Schedules</u>										
	4CP 75/0/25	220	100.000	99.122	46.681	23.734	28.243	0.285	0.179	0.878
	4CP 75/0/25 Exc WFR	222	100.000	99.975	47.081	23.939	28.487	0.288	0.180	0.025
	4CP Dmd @ Gen Jurisdictional	224	100.000	100.000	48.689	24.096	26.898	0.185	0.132	-
	12CP Demand @ Subtrans	226	100.000	100.000	48.256	24.687	26.544	0.388	0.124	-
	Class Peak @ Subtransmission	122	100.000	99.972	49.156	22.814	26.029	0.415	1.558	0.028
	Classpeak @ Primary	230	100.000	100.000	53.808	24.973	20.756	0.454	0.010	-
	Classpeak @ Secondary	231	100.000	100.000	67.909	31.518	-	0.573	-	-
	Classpeak for Streetlighting	233	100.000	99.429	-	-	-	99.429	-	0.571
	Classpeak @ Single Phase	235	100.000	100.000	67.909	31.518	-	0.573	-	-
	Billed Sales - Primary	253	100.000	100.000	45.644	26.055	27.654	0.645	0.002	-
	Customers - Residential	260	100.000	100.000	100.000	-	-	-	-	-
	Customers - Drops	261	100.000	100.000	-	100.000	-	-	-	-
	Customers - NonPID	263	100.000	100.000	88.373	11.606	-	0.021	-	-
	Customers - NonMunicipal	264	100.000	100.000	88.201	11.583	0.215	-	0.001	0.000
	Customers - Commercial & Primary	267	100.000	100.000	-	98.172	1.822	-	0.007	-
	Customer Count (CCC)	161	100.000	100.000	93.927	6.073	-	-	-	-
	Customer Count (BCC)	162	100.000	100.000	-	92.017	3.363	4.608	0.012	-
	PIS - HVD (Voltage 1)	301	100.000	99.173	46.035	21.365	29.926	0.388	1.459	0.827
	PIS - HVD (Voltage 2)	302	100.000	99.972	49.156	22.814	26.029	0.415	1.558	0.028
	PIS - HVD (Voltage 1) Subs (FERC 361/362)	303	100.000	99.173	46.035	21.365	29.926	0.388	1.459	0.827
	PIS - HVD (Voltage 2) Subs (FERC 361/362)	304	100.000	99.972	49.156	22.814	26.029	0.415	1.558	0.028

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Residential/Secondary ALLOCATORS 1						
Line No.	(a) Description	(b) Alloc	(c) Rate RS Total Residential	(d) Rate GS	(e) Rate GSD	(h) Commercial Secondary
<u>Input Allocation Schedules</u>						
	Energy @ Generation	100	41.868	13.415	9.833	23.248
	Energy On-Peak @ Generation	101	39.804	14.389	9.954	24.342
	Energy Off-Peak @ Generation	102	43.604	12.178	9.576	21.754
	Energy On-Peak @ Generation Summer	103	40.944	14.636	9.982	24.619
	Energy Off-Peak @ Generation Summer	104	43.786	12.353	9.632	21.985
	Energy On-Peak @ Generation Non-Summer	105	39.117	14.239	9.936	24.175
	Energy Off-Peak @ Generation Non-Summer	106	43.500	12.078	9.544	21.622
	Energy Critical On-Peak @ Gen	107	44.276	14.141	9.757	23.898
	Energy Summer Mid-Peak @ Gen	108	39.708	14.820	10.066	24.886
	12CP Dmd @ Generation	120	45.639	13.980	9.368	23.348
	4CP Dmd @ Generation	121	48.261	14.642	9.243	23.884
	Classpeak @ Transmission	127	46.035	13.246	8.119	21.365
	Production Revenue	141	45.172	13.750	9.290	23.040
	Distribution Revenue	142	63.537	15.445	8.416	23.861
	Total Rate Revenue	143	51.249	14.311	9.001	23.312
	Billed Sales	150	36.541	11.775	9.083	20.859
	Number Of Customers	160	87.942	10.426	1.123	11.549
	Weighted Customer	170	60.629	25.838	6.075	31.913
<u>*Calculated Allocation Schedules</u>						
	4CP 75/0/25	220	46.681	14.340	9.394	23.734
	4CP 75/0/25 Exc WFR	222	47.081	14.464	9.475	23.939
	4CP Dmd @ Gen Jurisdictional	224	48.689	14.771	9.325	24.096
	12CP Demand @ Subtrans	226	48.256	14.782	9.905	24.687
	Class Peak @ Subtransmission	122	49.156	14.145	8.669	22.814
	Classpeak @ Primary	230	53.808	15.483	9.490	24.973
	Classpeak @ Secondary	231	67.909	19.541	11.977	31.518
	Classpeak for Streetlighting	233	-	-	-	-
	Classpeak @ Single Phase	235	67.909	19.541	11.977	31.518
	Billed Sales - Primary	253	45.644	14.709	11.346	26.055
	Customers - Residential	260	100.000	-	-	-
	Customers - Drops	261	-	90.272	9.728	100.000
	Customers - NonPID	263	88.373	10.477	1.129	11.606
	Customers - NonMunicipal	264	88.201	10.456	1.127	11.583
	Customers - Commercial & Primary	267	-	88.622	9.550	98.172
	Customer Count (CCC)	161	93.927	6.013	0.060	6.073
	Customer Count (BCC)	162	-	75.266	16.750	92.017
	PIS - HVD (Voltage 1)	301	46.035	13.246	8.119	21.365
	PIS - HVD (Voltage 2)	302	49.156	14.145	8.669	22.814
	PIS - HVD (Voltage 1) Subs (FERC 361/362)	303	46.035	13.246	8.119	21.365
	PIS - HVD (Voltage 2) Subs (FERC 361/362)	304	49.156	14.145	8.669	22.814

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Primary & Lighting ALLOCATORS 1																	
Line	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(j)	(k)	(l)	(q)	(r)	(t)	(u)	(v)	
No.	Description	Alloc	Rate GP	Rate GPTU Vit 1	Rate GPTU Vit 2	Rate GPTU Vit 3	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate EIP Vit 1	Rate EIP Vit 2	Rate EIP Vit 3	Total Primary	Rate GML	Rate UUL	Rate GU	Lighting & Unmetered
<u>Input Allocation Schedules</u>																	
100	Energy @ Generation	2.584	1.247	1.823	11.908	2.886	4.386	6.702	1.174	0.203	0.030	32.942	0.041	0.239	0.311	0.592	
101	Energy On-Peak @ Generation	2.925	1.233	1.834	12.711	2.998	4.402	6.589	1.198	0.189	0.040	34.120	0.020	0.120	0.282	0.422	
102	Energy Off-Peak @ Generation	2.177	1.246	1.788	10.876	3.465	4.751	6.814	1.132	0.216	0.018	32.483	0.063	0.368	0.339	0.770	
103	Energy On-Peak @ Generation Summer	2.822	1.171	1.703	12.388	2.739	4.282	6.437	1.100	0.165	0.035	32.842	0.011	0.068	0.252	0.331	
104	Energy Off-Peak @ Generation Summer	2.126	1.204	1.714	10.941	3.327	4.752	6.824	1.111	0.199	0.017	32.216	0.052	0.310	0.309	0.672	
105	Energy On-Peak @ Generation Non-Summer	2.988	1.271	1.914	12.906	3.155	4.475	6.681	1.257	0.203	0.043	34.891	0.026	0.150	0.301	0.477	
106	Energy Off-Peak @ Generation Non-Summer	2.205	1.269	1.831	10.839	3.543	4.750	6.809	1.145	0.226	0.018	32.635	0.069	0.401	0.356	0.826	
107	Energy Critical On-Peak @ Gen	2.604	1.057	1.557	11.604	2.285	3.945	6.152	0.996	0.136	0.019	30.355	0.001	0.004	0.233	0.237	
108	Energy Summer Mid-Peak @ Gen	2.903	1.214	1.757	12.679	2.907	4.407	6.542	1.139	0.176	0.041	33.765	0.015	0.092	0.259	0.366	
120	12CP Dmd @ Generation	2.603	1.028	1.544	11.148	2.381	3.802	5.881	0.999	0.147	0.026	29.559	0.022	0.120	0.225	0.367	
121	4CP Dmd @ Generation	2.519	0.880	1.327	10.496	1.703	3.294	5.509	0.793	0.122	0.019	26.662	0.000	-	0.183	0.183	
127	Classpeak @ Transmission	2.564	0.733	1.121	8.783	3.997	5.231	6.567	0.775	0.123	0.032	29.926	0.039	0.209	0.141	0.388	
141	Production Revenue	2.420	1.168	1.705	11.256	2.737	4.144	6.138	0.836	0.143	0.021	30.567	0.024	0.138	0.260	0.422	
142	Distribution Revenue	1.652	0.055	0.281	4.188	0.232	1.080	2.947	0.059	0.076	0.021	10.591	0.056	1.659	0.144	1.859	
143	Total Rate Revenue	2.166	0.800	1.234	8.917	1.908	3.130	5.082	0.579	0.121	0.021	23.957	0.035	0.641	0.222	0.897	
150	Billed Sales	2.564	1.157	1.672	10.792	6.297	8.147	8.901	1.089	0.186	0.027	40.833	0.035	0.209	0.272	0.516	
160	Number Of Customers	0.081	0.001	0.002	0.069	0.002	0.005	0.053	0.000	0.000	0.000	0.214	0.021	0.248	0.024	0.294	
170	Weighted Customer	2.482	0.040	0.096	2.546	0.084	0.208	1.912	0.013	0.015	0.007	7.403	0.022	-	-	0.022	
<u>*Calculated Allocation Schedules</u>																	
220	4CP 75/0/25	2.536	0.972	1.452	10.853	1.999	3.568	5.810	0.888	0.142	0.022	28.243	0.010	0.060	0.215	0.285	
222	4CP 75/0/25 Exc WFR	2.558	0.981	1.464	10.946	2.017	3.599	5.860	0.896	0.144	0.022	28.487	0.010	0.060	0.217	0.288	
224	4CP Dmd @ Gen Jurisdictional	2.541	0.888	1.339	10.588	1.718	3.323	5.558	0.800	0.123	0.020	26.898	0.000	-	0.185	0.185	
226	12CP Demand @ Subtrans	2.703	-	1.632	11.787	-	4.020	6.218	-	0.155	0.028	26.544	0.023	0.127	0.238	0.388	
122	Class Peak @ Subtransmission	2.689	-	1.197	9.379	-	5.586	7.012	-	0.132	0.035	26.029	0.041	0.223	0.150	0.415	
230	Classpeak @ Primary	2.776	-	-	10.266	-	-	7.675	-	-	0.038	20.756	0.045	0.244	0.164	0.454	
231	Classpeak @ Secondary	-	-	-	-	-	-	-	-	-	-	-	0.057	0.308	0.207	0.573	
233	Classpeak for Streetlighting	-	-	-	-	-	-	-	-	-	-	-	15.572	83.857	-	99.429	
235	Classpeak @ Single Phase	-	-	-	-	-	-	-	-	-	-	-	0.057	0.308	0.207	0.573	
253	Billed Sales - Primary	3.021	-	-	13.481	-	-	11.119	-	-	0.033	27.654	0.044	0.261	0.340	0.645	
260	Customers - Residential	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
261	Customers - Drops	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
263	Customers - NonPID	-	-	-	-	-	-	-	-	-	-	-	0.021	-	-	0.021	
264	Customers - NonMunicipal	0.082	0.001	0.002	0.069	0.002	0.005	0.053	0.000	0.000	0.000	0.215	-	-	-	-	
267	Customers - Commercial & Primary	0.691	0.008	0.018	0.586	0.016	0.043	0.450	0.002	0.003	0.002	1.822	-	-	-	-	
161	Customer Count (CCC)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
162	Customer Count (BCC)	1.276	0.016	0.034	1.083	0.030	0.080	0.831	0.005	0.006	0.004	3.363	0.335	3.889	0.383	4.608	
301	PIS - HVD (Voltage 1)	2.564	0.733	1.121	8.783	3.997	5.231	6.567	0.775	0.123	0.032	29.926	0.039	0.209	0.141	0.388	
302	PIS - HVD (Voltage 2)	2.689	-	1.197	9.379	-	5.586	7.012	-	0.132	0.035	26.029	0.041	0.223	0.150	0.415	
303	PIS - HVD (Voltage 1) Subs (FERC 361/362)	2.564	0.733	1.121	8.783	3.997	5.231	6.567	0.775	0.123	0.032	29.926	0.039	0.209	0.141	0.388	
304	PIS - HVD (Voltage 2) Subs (FERC 361/362)	2.689	-	1.197	9.379	-	5.586	7.012	-	0.132	0.035	26.029	0.041	0.223	0.150	0.415	

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Summary ALLOCATORS 2										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
	<u>*Calculated Allocation Schedules</u>									
	PIS - OH LVD System	305	100.000	100.000	63.949	29.680	5.828	0.540	0.003	-
	PIS - LVD Distribution	306	100.000	99.998	65.371	26.435	5.254	2.935	0.004	0.002
	PIS- OH (LVD & HVD) & Services	307	100.000	99.989	66.663	24.978	7.698	0.438	0.213	0.011
	PIS- UG LVD Distribution	308	100.000	100.000	65.039	30.186	4.224	0.549	0.002	(0.000)
	PIS- Total Distribution	309	100.000	99.936	61.627	25.545	10.043	2.386	0.335	0.064
	PIS- Distribution Services	310	100.000	100.000	94.495	5.505	-	-	-	(0.000)
	PIS- Streetlighting Equipment	311	100.000	99.956	-	-	-	99.956	-	0.044
	PIS- Line Equipment	312	100.000	100.000	66.376	30.806	2.256	0.560	0.001	(0.000)
	PIS- Meters	313	100.000	99.994	60.629	31.913	7.403	0.022	0.027	0.006
	PIS - General	315	100.000	99.669	59.527	23.415	15.472	1.044	0.210	0.331
	Total PIS	316	100.000	99.743	58.344	24.981	14.296	1.832	0.291	0.257
	Distribution Depreciation	317	100.000	99.957	65.439	24.260	7.129	2.941	0.188	0.043
	CWIP	330	100.000	99.302	50.248	23.926	24.256	0.659	0.214	0.698
	Rate Base- Total	390	100.000	99.651	55.980	24.962	16.996	1.410	0.303	0.349
	Dist Op Expense (LVD) excl Sup & Eng	400	100.000	99.989	65.221	24.124	6.679	3.805	0.159	0.011
	Dist Maint Expense (LVD) excl Sup & Eng	401	100.000	99.990	66.015	25.155	8.104	0.517	0.199	0.010
	Dist Op Expense (HVD Voltage 1) excl Sup & Eng	402	100.000	99.164	46.442	21.826	29.328	0.351	1.216	0.836
	Dist Maint Expense (HVD Voltage 1) excl Sup & Eng	403	100.000	99.352	46.608	21.955	29.272	0.344	1.172	0.648
	Dist Op Expense (HVD Voltage 2) excl Sup & Eng	404	100.000	99.972	49.156	22.814	26.029	0.415	1.558	0.028
	Dist Maint Expense (HVD Voltage 2) excl Sup & Eng	405	100.000	99.972	49.156	22.814	26.029	0.415	1.558	0.028
	Total HVD Distribution O&M Expense	406	100.000	99.836	48.618	22.630	26.709	0.400	1.479	0.164
	Total Distribution O&M Expense (excl. HVD)	407	100.000	99.990	65.762	24.779	7.574	1.692	0.184	0.010
	Total Customer Accounts Expense (excl. Supv)	408	100.000	99.818	75.667	15.510	8.166	0.460	0.015	0.182
	Total Customer Accounts & Service Expense	409	100.000	99.771	73.377	15.782	10.120	0.461	0.032	0.229
	Jurisdictional Distribution O&M	414	100.000	100.000	64.707	24.659	8.752	1.619	0.263	-
	Pre Tax NOI	439	100.000	100.781	59.957	21.680	17.981	0.855	0.309	(0.781)
	Depreciation & Amortization Expense	442	100.000	99.591	55.984	24.687	17.199	1.507	0.214	0.409
	Non PSCR O&M Expense	443	100.000	99.664	59.729	23.177	15.483	1.058	0.217	0.336
	Distribution Depreciation Expense	444	100.000	99.954	62.559	25.824	8.685	2.641	0.246	0.046
	Gen/Comm/Int Depreciation Expense	445	100.000	99.669	59.527	23.415	15.472	1.044	0.210	0.331
	Production Labor	500	100.000	99.122	46.681	23.734	28.243	0.285	0.179	0.878
	Total Labor	502	100.000	99.669	59.527	23.415	15.472	1.044	0.210	0.331
	50% O&M, 50% Net Plant	600	100.000	99.600	54.586	24.937	18.343	1.413	0.321	0.400
	50/50 PIS & Labor	601	100.000	99.706	58.936	24.198	14.884	1.438	0.251	0.294

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Residential/Secondary ALLOCATORS 2							
	(a)		(b)	(c)	(d)	(e)	(h)
Line No.	Description	Alloc	Rate RS	Total Residential	Rate GS	Rate GSD	Commercial Secondary
	<u>*Calculated Allocation Schedules</u>						
	PIS - OH LVD System	305	63.949	63.949	18.401	11.278	29.680
	PIS - LVD Distribution	306	65.371	65.371	17.027	9.408	26.435
	PIS- OH (LVD & HVD) & Services	307	66.663	66.663	15.728	9.250	24.978
	PIS- UG LVD Distribution	308	65.039	65.039	18.715	11.471	30.186
	PIS- Total Distribution	309	61.627	61.627	16.337	9.208	25.545
	PIS- Distribution Services	310	94.495	94.495	4.969	0.536	5.505
	PIS- Streetlighting Equipment	311	-	-	-	-	-
	PIS- Line Equipment	312	66.376	66.376	19.100	11.706	30.806
	PIS- Meters	313	60.629	60.629	25.838	6.075	31.913
	PIS - General	315	59.527	59.527	14.838	8.577	23.415
	Total PIS	316	58.344	58.344	15.790	9.191	24.981
	Distribution Depreciation	317	65.439	65.439	15.356	8.904	24.260
	CWIP	330	50.248	50.248	14.663	9.263	23.926
	Rate Base- Total	390	55.980	55.980	15.739	9.223	24.962
	Dist Op Expense (LVD) excl Sup & Eng	400	65.221	65.221	15.647	8.477	24.124
	Dist Maint Expense (LVD) excl Sup & Eng	401	66.015	66.015	15.942	9.212	25.155
	Dist Op Expense (HVD Voltage 1) excl Sup & Eng	402	46.442	46.442	13.502	8.325	21.826
	Dist Maint Expense (HVD Voltage 1) excl Sup & Eng	403	46.608	46.608	13.576	8.379	21.955
	Dist Op Expense (HVD Voltage 2) excl Sup & Eng	404	49.156	49.156	14.145	8.669	22.814
	Dist Maint Expense (HVD Voltage 2) excl Sup & Eng	405	49.156	49.156	14.145	8.669	22.814
	Total HVD Distribution O&M Expense	406	48.618	48.618	14.023	8.607	22.630
	Total Distribution O&M Expense (excl. HVD)	407	65.762	65.762	15.832	8.947	24.779
	Total Customer Accounts Expense (excl. Supv)	408	75.667	75.667	11.738	3.772	15.510
	Total Customer Accounts & Service Expense	409	73.377	73.377	11.717	4.065	15.782
	Jurisdictional Distribution O&M	414	64.707	64.707	15.730	8.930	24.659
	Pre Tax NOI	439	59.957	59.957	14.141	7.538	21.680
	Depreciation & Amortization Expense	442	55.984	55.984	15.609	9.078	24.687
	Non PSCR O&M Expense	443	59.729	59.729	14.741	8.435	23.177
	Distribution Depreciation Expense	444	62.559	62.559	16.824	9.000	25.824
	Gen/Comm/Int Depreciation Expense	445	59.527	59.527	14.838	8.577	23.415
	Production Labor	500	46.681	46.681	14.340	9.394	23.734
	Total Labor	502	59.527	59.527	14.838	8.577	23.415
	50% O&M, 50% Net Plant	600	54.586	54.586	15.612	9.325	24.937
	50/50 PIS & Labor	601	58.936	58.936	15.314	8.884	24.198

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

Projected 12-Month Period Ending February 28, 2025
Version 2
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Primary & Lighting ALLOCATORS 2																	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(k)	(l)	(q)	(r)	(t)	(u)	(v)	
Line No.	Description	Alloc	Rate GP	Rate GPTU Vit 1	Rate GPTU Vit 2	Rate GPTU Vit 3	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate EIP Vit 1	Rate EIP Vit 2	Rate EIP Vit 3	Total Primary	Rate GML	Rate UUL	Rate GU	Lighting & Unmetered
<u>*Calculated Allocation Schedules</u>																	
	PIS - OH LVD System	305	0.780	-	-	2.883	-	-	2.155	-	-	0.011	5.828	0.054	0.290	0.195	0.540
	PIS - LVD Distribution	306	0.818	0.003	0.007	2.513	0.006	0.016	1.879	0.001	0.001	0.009	5.254	0.072	2.709	0.154	2.935
	PIS- OH (LVD & HVD) & Services	307	0.916	0.006	0.162	3.311	0.035	0.755	2.476	0.007	0.018	0.012	7.698	0.044	0.236	0.159	0.438
	PIS- UG LVD Distribution	308	0.565	-	-	2.089	-	-	1.562	-	-	0.008	4.224	0.055	0.295	0.199	0.549
	PIS- Total Distribution	309	1.215	0.055	0.260	3.961	0.291	1.200	2.962	0.056	0.029	0.014	10.043	0.066	2.168	0.152	2.386
	PIS- Distribution Services	310	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	PIS- Streetlighting Equipment	311	-	-	-	-	-	-	-	-	-	-	-	1.189	98.768	-	99.956
	PIS- Line Equipment	312	0.302	-	-	1.116	-	-	0.834	-	-	0.004	2.256	0.056	0.301	0.203	0.560
	PIS- Meters	313	2.482	0.040	0.096	2.546	0.084	0.208	1.912	0.013	0.015	0.007	7.403	0.022	-	-	0.022
	PIS - General	315	1.542	0.371	0.656	6.068	0.822	1.888	3.709	0.335	0.066	0.016	15.472	0.038	0.840	0.167	1.044
	Total PIS	316	1.518	0.273	0.542	5.576	0.692	1.752	3.618	0.253	0.056	0.016	14.296	0.052	1.613	0.167	1.832
	Distribution Depreciation	317	0.841	0.036	0.144	2.964	0.195	0.669	2.216	0.038	0.016	0.011	7.129	0.070	2.720	0.151	2.941
	CWIP	330	2.240	0.771	1.194	9.339	1.623	3.063	5.183	0.705	0.118	0.020	24.256	0.021	0.437	0.200	0.659
	Rate Base- Total	390	1.747	0.381	0.705	6.567	0.910	2.144	4.104	0.349	0.072	0.018	16.996	0.044	1.192	0.174	1.410
	Dist Op Expense (LVD) excl Sup & Eng	400	0.881	0.008	0.125	2.875	0.038	0.570	2.150	0.007	0.014	0.010	6.679	0.081	3.582	0.141	3.805
	Dist Maint Expense (LVD) excl Sup & Eng	401	1.008	0.007	0.153	3.523	0.034	0.708	2.634	0.007	0.017	0.013	8.104	0.045	0.316	0.156	0.517
	Dist Op Expense (HVD Voltage 1) excl Sup & I	402	2.556	0.760	1.159	9.097	3.577	4.877	6.373	0.778	0.123	0.030	29.328	0.032	0.171	0.148	0.351
	Dist Maint Expense (HVD Voltage 1) excl Sup	403	2.559	0.766	1.168	9.173	3.504	4.819	6.349	0.780	0.123	0.030	29.272	0.030	0.164	0.150	0.344
	Dist Op Expense (HVD Voltage 2) excl Sup & I	404	2.689	-	1.197	9.379	-	5.586	7.012	-	0.132	0.035	26.029	0.041	0.223	0.150	0.415
	Dist Maint Expense (HVD Voltage 2) excl Sup	405	2.689	-	1.197	9.379	-	5.586	7.012	-	0.132	0.035	26.029	0.041	0.223	0.150	0.415
	Total HVD Distribution O&M Expense	406	2.662	0.160	1.191	9.334	0.735	5.427	6.874	0.163	0.130	0.034	26.709	0.039	0.211	0.150	0.400
	Total Distribution O&M Expense (excl. HVD)	407	0.961	0.007	0.142	3.286	0.034	0.654	2.457	0.006	0.016	0.012	7.574	0.058	1.483	0.151	1.692
	Total Customer Accounts Expense (excl. Supv	408	0.771	0.269	0.416	3.034	0.642	1.055	1.736	0.195	0.041	0.007	8.166	0.026	0.347	0.087	0.460
	Total Customer Accounts & Service Expense	409	0.874	0.321	0.489	3.478	0.994	1.491	2.166	0.249	0.049	0.008	10.120	0.026	0.336	0.098	0.461
	Jurisdictional Distribution O&M	414	1.066	0.017	0.206	3.657	0.078	0.948	2.728	0.016	0.023	0.013	8.752	0.057	1.412	0.151	1.619
	Pre Tax NOI	439	2.157	0.627	1.025	7.087	1.129	2.288	4.016	(0.409)	0.052	0.008	17.981	0.018	0.685	0.152	0.855
	Depreciation & Amortization Expense	442	1.750	0.448	0.747	6.700	0.989	2.064	3.999	0.411	0.075	0.017	17.199	0.041	1.292	0.174	1.507
	Non PSCR O&M Expense	443	1.510	0.385	0.671	6.003	0.878	1.910	3.690	0.351	0.069	0.016	15.483	0.040	0.845	0.172	1.058
	Distribution Depreciation Expense	444	1.169	0.041	0.196	3.496	0.207	0.888	2.614	0.040	0.022	0.013	8.685	0.067	2.430	0.143	2.641
	Gen/Comm/Int Depreciation Expense	445	1.542	0.371	0.656	6.068	0.822	1.888	3.709	0.335	0.066	0.016	15.472	0.038	0.840	0.167	1.044
	Production Labor	500	2.536	0.972	1.452	10.853	1.999	3.568	5.810	0.888	0.142	0.022	28.243	0.010	0.060	0.215	0.285
	Total Labor	502	1.542	0.371	0.656	6.068	0.822	1.888	3.709	0.335	0.066	0.016	15.472	0.038	0.840	0.167	1.044
	50% O&M, 50% Net Plant	600	1.815	0.446	0.792	6.973	1.119	2.354	4.324	0.418	0.083	0.019	18.343	0.046	1.180	0.187	1.413
	50/50 PIS & Labor	601	1.530	0.322	0.599	5.822	0.757	1.820	3.663	0.294	0.061	0.016	14.884	0.045	1.226	0.167	1.438

*Calculated allocation schedules may vary between production and distribution functions. Combined production and distribution shown.

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Electric Cost-of-Service Study
Capacity Related Cost and Charge Calculation
Projected 12-Month Period Ending February 28, 2025

Case No.: U-21389
Exhibit No.: A-79 (EAD-3)
Page: 1 of 1
Witness: EADavis
Date: May 2023

Line No.	(a) Description	(b) <u>Total</u> <u>Electric</u> (\$000)	(c) <u>Capacity</u> <u>Charge</u>	(d) <u>Formula</u>
1	Total Production Related Cost	\$ 2,742,335		
2	<u>Non-Capacity Related Cost:</u>			
3	Fuel Expense	\$ 1,039,289		
4	Purchased & Interchanged	593,953		
5	Energy Related Other O&M Expense	59,634		
6	PSCR Revenue Credits	(684,353)		
7	Non-PSCR Revenue Credits	(105,552)		
8	Transmission Expense	543,730		
9	Total Non-Capacity Related Cost	\$ 1,446,700		Σ Lines 3:8
10	Total Capacity Related Cost	\$ 1,295,635		Line 1 - Line 9
11	<u>Offsets:</u>			
12	Energy Market Sales	\$ 1,967,929		
13	Off-System Energy Sales	12,000		
14	Ancillary Service Sales	11,874		
15	Bilateral Energy Sales	-		
16	Total Revenue	\$ 1,991,803		Σ Lines 12:15
17	Related Fuel Cost	899,084		
18	Total Revenue Less Fuel Cost	\$ 1,092,719		Line 16 - Line 17
19	Net Capacity Cost	<u>\$ 202,916</u>		Line 10 - Line 18
20	Capacity Charge Demand (MW)		7,526	
21	Capacity Charge (\$/MW-Day)		\$73.87	[(Line 19 x 1,000) ÷ Line 20] ÷ 365

Source:
Lines 1-9: Exhibit A-16 (EAD-2) Schedule F-1.1
Lines 12-15, 17, 20 Company witness Megan L. Metz

Line	(a) Description	(b) Rate GPD Voltage 1		(c) Rate GPD Voltage 2		(d) Rate GPTU Voltage 1		(e) Rate GPTU Voltage 2		(f) Rate EIP Voltage 1		(g) Rate EIP Voltage 2	
1	Plant in Service	\$	25,771	\$	44,127	\$	4,723	\$	9,454	\$	4,995	\$	1,042
2	Depreciation Reserve		(5,959)		(9,318)		(1,092)		(1,996)		(1,155)		(220)
3	Net Plant		19,811		34,809		3,631		7,458		3,840		822
4	Working Capital		1,347		2,786		450		707		360		87
5	CWIP (HV Distribution)		269		146		48		31		52		3
6	Substation Rate Base	\$	21,428	\$	37,741	\$	4,129	\$	8,196	\$	4,252	\$	912
7	Pre-Tax Return		7.60%		7.60%		7.60%		7.60%		7.60%		7.60%
8	Pre-Tax Return (\$)	\$	1,629	\$	2,869	\$	314	\$	623	\$	323	\$	69
9	Depreciation Expense		616		1,039		113		223		119		25
10	O&M Expense		108		186		20		40		21		4
11	Other Taxes		835		1,018		155		225		162		33
12	Revenue Credits		(282)		(179)		(45)		(37)		(48)		(6)
13	Total Revenue Requirement	\$	2,906	\$	4,932	\$	557	\$	1,073	\$	577	\$	126
14	Max Demand (MW)		4,690		6,163		1,164		1,610		1,163		467
15	Substation Ownership Credit (kW)	\$	0.6196	\$	0.8001	\$	0.4783	\$	0.6666	\$	0.4961	\$	0.2695

Source: Exhibit A-16 (EAD-2) Schedule F-1.1
WP-AMG-3-4
WP-EAD-192-193

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Allocation of Surcharges (\$000)

Case No.: U-21389
Exhibit No.: A-81 (EAD-5)
Page: 1 of 3
Witness: EADavis
Date: May 2023

Line No	Description	Total Electric	Source
1	Electric Rate Case ("ERC") Deferral	\$ 8,556	Exhibit JCA-62
2	Demand Response ("DR") Revenue Req	\$ 52,195	Exhibit JCA-61
3	Reconciliation - Incentive	\$ 4,640	Exhibit JCA-61
4	Reconciliation - Reg Asset	\$ (10,046)	Exhibit JCA-61
5	Total DR Revenue Requirement	\$ 46,789	
6	Investment Recovery Mechanism ("IRM")		
7	IRM 12 Mo. Ending 2/28/25	\$ 6,263	Exhibit HJM-1
8	IRM 12 Mo. Ending 2/28/26	\$ 17,748	
9	IRM 12 Mo. Ending 2/28/27	\$ 29,208	

Line No	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
			(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
10	<u>Demand Response Surcharge</u>									
11	1/ 4CP 75/0/25 (Allocator)	220	100.0000	99.1221	46.6808	23.7344	28.2427	0.2855	0.1787	0.8779
12	Allocation of DR Revenue Req		\$ 52,195	\$ 51,737	\$ 24,365	\$ 12,388	\$ 14,741	\$ 149	\$ 93	\$ 458
13	Allocation of DR Incentive		\$ 4,640	\$ 4,599	\$ 2,166	\$ 1,101	\$ 1,310	\$ 13	\$ 8	\$ 41
14	Allocation of DR Reg Asset		\$ (10,046)	\$ (9,958)	\$ (4,689)	\$ (2,384)	\$ (2,837)	\$ (29)	\$ (18)	\$ (88)
15	Total Allocation of DR Surcharge		\$ 46,789	\$ 46,378	\$ 21,841	\$ 11,105	\$ 13,214	\$ 134	\$ 84	\$ 411
16	<u>ERC Deferral Surcharge</u>									
17	1/ Distribution: Demand Related Cost		1,519,429	1,518,346	929,150	391,784	158,300	33,515	\$ 5,596	\$ 1,083
18	1/ Distribution: Customer Related Cost		239,617	239,577	179,947	46,588	11,349	1,558	\$ 134	\$ 41
19	Total Distribution Costs		\$ 1,759,046	\$ 1,757,922	\$ 1,109,097	\$ 438,372	\$ 169,649	\$ 35,073	\$ 5,730	\$ 1,124
20	Total Distribution Costs (Allocator)		1.0000	0.9994	0.63051	0.2492	0.0964	0.0199	0.0033	0.0006
21	Allocation of ERC Surcharge		\$ 8,556	\$ 8,550	\$ 5,394	\$ 2,132	\$ 825	\$ 171	\$ 28	\$ 5
22	Allocation of IRM Surcharge (Ending 2/28/25)		\$ 6,263	\$ 6,259	\$ 3,949	\$ 1,561	\$ 604	\$ 125	\$ 20	\$ 4
23	Allocation of IRM Surcharge (Ending 2/28/26)		\$ 17,748	\$ 17,737	\$ 11,190	\$ 4,423	\$ 1,712	\$ 354	\$ 58	\$ 11
24	Allocation of IRM Surcharge (Ending 2/28/27)		\$ 29,208	\$ 29,189	\$ 18,416	\$ 7,279	\$ 2,817	\$ 582	\$ 95	\$ 19

Source:

1/ Exhibit A-16 (EAD-2) Schedule F-1.1

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Allocation of Surcharges (\$000)

Case No.: U-21389
Exhibit No.: A-81 (EAD-5)
Page: 2 of 3
Witness: EADavis
Date: May 2023

Line No	Description	Alloc	Rate Residential (a)	Total Residential (b)	Rate GS (c)	Rate GSD (d)	Total Commercial Secondary (e)
1	<u>Demand Response Surcharge</u>						
2	1/ 4CP 75/0/25 (Allocator)	220	46.6808	46.6808	14.3405	9.3940	23.7344
3	Allocation of DR Revenue Req	\$	24,365	\$ 24,365	\$ 7,485	\$ 4,903	\$ 12,388
4	Allocation of DR Incentive	\$	2,166	\$ 2,166	\$ 665	\$ 436	\$ 1,101
5	Allocation of DR Reconciliation	\$	(4,689)	\$ (4,689)	\$ (1,441)	\$ (944)	\$ (2,384)
6	Total Allocation of DR Surcharge	\$	21,841	\$ 21,841	\$ 6,710	\$ 4,395	\$ 11,105
	<u>ERC Deferral Surcharge</u>						
7	1/ Distribution: Demand Related Cost		929,150	929,150	247,109	144,676	391,784
8	1/ Distribution: Customer Related Cost		179,947	179,947	37,291	9,297	46,588
9	Total Distribution Costs	\$	1,109,097	\$ 1,109,097	\$ 284,399	\$ 153,973	\$ 438,372
10	Total Distribution Costs (Allocator)		0.6305	0.6305	0.1617	0.0875	0.2492
11	Allocation of ERC Surcharge	\$	5,394	\$ 5,394	\$ 1,383	\$ 749	\$ 2,132
12	Allocation of IRM Surcharge (Ending 2/28/25)	\$	3,949	\$ 3,949	\$ 1,013	\$ 548	\$ 1,561
13	Allocation of IRM Surcharge (Ending 2/28/26)	\$	11,190	\$ 11,190	\$ 2,869	\$ 1,554	\$ 4,423
14	Allocation of IRM Surcharge (Ending 2/28/27)	\$	18,416	\$ 18,416	\$ 4,722	\$ 2,557	\$ 7,279

Source:
1/ Exhibit A-16 (EAD-2) Schedule F-1.1

Line No	Description	Alloc	Rate GP (a)	Rate GPTU Vit 1 (b)	Rate GPTU Vit 2 (c)	Rate GPTU Vit 3 (d)	Rate GPD Vit 1 (e)	Rate GPD Vit 2 (f)	Rate GPD Vit 3 (g)	Rate EIP Vit 1 (h)	Rate EIP Vit 2 (i)	Rate EIP Vit 3 (j)	Total Primary (k)	Rate GML (l)	Rate UUL (m)	Rate GU (n)	Total Lighting & Unmetered (o)
1	Demand Response Surcharge																
2	1/ 4CP 75/0/25 (Allocator)	220	2.5363	0.9723	1.4516	10.8527	1.9992	3.5681	5.8097	0.8885	0.1424	0.0220	28.2427	0.0104	0.0599	0.2152	0.2855
3	Allocation of DR Revenue Req		\$ 1,324	\$ 507	\$ 758	\$ 5,665	\$ 1,043	\$ 1,862	\$ 3,032	\$ 464	\$ 74	\$ 11	\$ 14,741	\$ 5	\$ 31	\$ 112	\$ 149
4	Allocation of DR Incentive		\$ 118	\$ 45	\$ 67	\$ 504	\$ 93	\$ 166	\$ 270	\$ 41	\$ 7	\$ 1	\$ 1,310	\$ 0	\$ 3	\$ 10	\$ 13
5	Allocation of DR Reconciliation		\$ (255)	\$ (98)	\$ (146)	\$ (1,090)	\$ (201)	\$ (358)	\$ (584)	\$ (89)	\$ (14)	\$ (2)	\$ (2,837)	\$ (1)	\$ (6)	\$ (22)	\$ (29)
6	Total Allocation of DR Surcharge		\$ 1,187	\$ 455	\$ 679	\$ 5,078	\$ 935	\$ 1,669	\$ 2,718	\$ 416	\$ 67	\$ 10	\$ 13,214	\$ 5	\$ 28	\$ 101	\$ 134
	ERC Deferral Surcharge																
7	1/ Distribution: Demand Related Cost		17,013	952	4,335	63,255	4,365	19,740	47,184	887	358	212	158,300	1,007	30,017	2,491	33,515
8	1/ Distribution: Customer Related Cost		2,659	112	250	3,906	445	920	2,932	82	31	12	11,349	54	1,433	71	1,558
9	Total Distribution Costs		\$ 19,672	\$ 1,064	\$ 4,585	\$ 67,161	\$ 4,810	\$ 20,660	\$ 50,116	\$ 968	\$ 389	\$ 224	\$ 169,649	\$ 1,061	\$ 31,450	\$ 2,562	\$ 35,073
10	Total Distribution Costs (Allocator)		0.01118	0.00060	0.00261	0.03818	0.00273	0.01174	0.02849	0.00055	0.00022	0.00013	0.09644	0.00060	0.01788	0.00146	0.01994
11	Allocation of ERC Surcharge		\$ 96	\$ 5	\$ 22	\$ 327	\$ 23	\$ 100	\$ 244	\$ 5	\$ 2	\$ 1	\$ 825	\$ 5	\$ 153	\$ 12	\$ 171
12	Allocation of IRM Surcharge (Ending 2/28/25)		\$ 70	\$ 4	\$ 16	\$ 239	\$ 17	\$ 74	\$ 178	\$ 3	\$ 1	\$ 1	\$ 604	\$ 4	\$ 112	\$ 9	\$ 125
13	Allocation of IRM Surcharge (Ending 2/28/26)		\$ 198	\$ 11	\$ 46	\$ 678	\$ 49	\$ 208	\$ 506	\$ 10	\$ 4	\$ 2	\$ 1,712	\$ 11	\$ 317	\$ 26	\$ 354
14	Allocation of IRM Surcharge (Ending 2/28/27)		\$ 327	\$ 18	\$ 76	\$ 1,115	\$ 80	\$ 343	\$ 832	\$ 16	\$ 6	\$ 4	\$ 2,817	\$ 18	\$ 522	\$ 43	\$ 582

Source:
1/ Exhibit A-16 (EAD-2) Schedule F-1.1

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

MATTHEW J. FOSTER

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Summary of Projected Electric O&M Expense
(\$000)

Case No.: U-21389
Exhibit No.: A-82 (MJF-1)
Page: 1 of 1
Witness: MJFoster
Date: May 2023

Corporate O&M

Line No.	(a) Program Description	(b) Source	(c)	(d)	(e)	(f)
			2022 Actual	2023 Projected	2024 Projected	12 Months Ending 2/28/2025 Projected
1	Adjusted Corporate Services Expense	Exhibit A-83 (MJF-2)	\$ 38,247	\$ 40,201	\$ 41,346	\$ 48,617
2	Uncollectible Expense	Exhibit A-85 (MJF-4)	\$ 18,817	23,796	24,731	17,519
3	Injuries & Damages Expense	Exhibit A-86 (MJF-5)	\$ 6,652	3,884	3,969	3,983
4	TOTAL O&M EXPENSES		<u><u>\$ 63,716</u></u>	<u><u>\$ 67,881</u></u>	<u><u>\$ 70,046</u></u>	<u><u>\$ 70,119</u></u>

Case No.: U-21389
Exhibit No.: A-83 (MJF-2)
Page: 1 of 1
Witness: MJFoster
Date: May 2023

[illegible]

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

S&P Global Market Intelligence Ranking of Consumers Energy Electric A&G Costs for 2021

Ranked by A&G per Customer (less Pension and Benefits)

(Companies over 500K Customers)

Case No.: U-21389

Exhibit No.: A-84 (MJF-3)

Page: 1 of 1

Witness: MJFoster

Date: May 2023

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Rank		Total	Total Admin & Gen:	Total Admin & Gen:	Total Admin & Gen:	Total Admin & Gen:	
No.	Company Name	Customers	O&M Expense	Pension & Benefits	Excluding Pension & Benefits	Excluding Pension & Benefits	\$/Customer
		(actual)	(\$000)	(\$000)	(d) - (e)	(f)/(c) x \$1,000	Ranking
1	PacifiCorp	2,002,863	173,646	124,791	48,855	\$24.39	1
2	MidAmerican Energy Company	804,312	42,906	13,175	29,731	\$36.96	2
3	Consumers Energy Company	1,871,096	63,193	-22,908	86,101	\$46.02	3
4	Ohio Edison Company	1,062,269	-11,546	-60,882	49,336	\$46.44	4
5	Ohio Power Company	1,511,448	81,850	3,559	78,291	\$51.80	5
6	Cleveland Electric Illuminating Company	755,212	-256	-43,435	43,179	\$57.17	6
7	Florida Power & Light Company	5,214,219	361,970	43,213	318,757	\$61.13	7
8	Public Service Electric and Gas Company	2,323,293	42,600	-100,917	143,517	\$61.77	8
9	Metropolitan Edison Company	581,453	-18,374	-55,145	36,771	\$63.24	9
10	Pennsylvania Electric Company	588,261	-42,295	-81,119	38,824	\$66.00	10
11	West Penn Power Company	733,761	5,937	-42,998	48,935	\$66.69	11
12	Duke Energy Ohio, Inc.	735,921	58,902	9,415	49,487	\$67.24	12
13	Nevada Power Company	984,770	87,099	16,611	70,488	\$71.58	13
14	Jersey Central Power & Light Company	1,150,247	-59,142	-143,682	84,540	\$73.50	14
15	Commonwealth Edison Company	4,095,262	414,223	111,942	302,281	\$73.81	15
16	Public Service Company of Colorado	1,535,755	161,790	46,272	115,518	\$75.22	16
17	Wisconsin Electric Power Company	1,144,822	109,624	19,671	89,953	\$78.57	17
18	Central Maine Power Company	628,246	60,486	7,323	53,163	\$84.62	18
19	Public Service Company of Oklahoma	568,226	53,075	4,486	48,589	\$85.51	19
20	Virginia Electric and Power Company	2,698,553	315,512	84,094	231,418	\$85.76	20
21	PECO Energy Co.	1,682,172	173,123	26,550	146,573	\$87.13	21
22	Puget Sound Energy, Inc.	1,196,851	150,921	35,236	115,685	\$96.66	22
23	Oklahoma Gas and Electric Company	874,591	122,311	33,619	88,692	\$101.41	23
24	Eversource Energy	570,014	134,302	75,245	59,057	\$103.61	24
25	Consolidated Edison Company of New York, Inc.	3,530,574	556,760	190,843	365,917	\$103.64	25
26	Appalachian Power Company	964,443	105,166	5,145	100,021	\$103.71	26
27	PPL Electric Utilities Corporation	1,466,284	165,133	9,135	155,998	\$106.39	27
28	Arizona Public Service Company	1,317,266	124,483	-16,279	140,762	\$106.86	28
29	Tampa Electric Company	802,049	135,015	49,077	85,938	\$107.15	29
30	Dayton Power and Light Company	532,418	67,333	9,714	57,619	\$108.22	30
31	Ameren Illinois Company	1,228,566	139,515	6,031	133,484	\$108.65	31
32	DTE Electric Company	2,249,459	390,790	142,722	248,068	\$110.28	32
33	Baltimore Gas and Electric Company	1,320,805	197,122	45,367	151,755	\$114.90	33
34	Southwestern Electric Power Company	546,238	76,841	11,624	65,217	\$119.39	34
35	Northern States Power Company - MN	1,522,745	263,008	80,397	182,611	\$119.92	35
36	NSTAR Electric Company	1,475,929	177,106	-1,969	179,075	\$121.33	36
37	Duke Energy Carolinas, LLC	2,764,923	426,290	79,752	346,538	\$125.33	37
38	New York State Electric & Gas Corporation	913,633	132,294	12,759	119,535	\$130.83	38
39	Kentucky Utilities Company	565,153	103,770	25,435	78,335	\$138.61	39
40	Atlantic City Electric Company	564,929	91,434	9,767	81,667	\$144.56	40
41	Duke Energy Indiana, LLC	860,972	157,563	33,009	124,554	\$144.67	41
42	Public Service Company of New Hampshire	529,986	81,563	4,282	77,281	\$145.82	42
43	Union Electric Company	1,244,264	192,583	7,861	184,722	\$148.46	43
44	Indiana Michigan Power Company	604,531	112,140	18,529	93,611	\$154.85	44
45	Connecticut Light and Power Company	1,272,008	218,117	18,184	199,933	\$157.18	45
46	Potomac Electric Power Company	914,279	158,351	14,464	143,887	\$157.38	46
47	Niagara Mohawk Power Corporation	1,706,025	327,873	50,445	277,428	\$162.62	47
48	Delmarva Power & Light Company	539,709	96,157	6,511	89,646	\$166.10	48
49	Duke Energy Progress, LLC	1,644,311	359,421	78,734	280,687	\$170.70	49
50	Idaho Power Company	596,394	162,042	56,595	105,447	\$176.81	50
51	Entergy Louisiana, LLC	1,106,510	330,272	133,223	197,049	\$178.08	51
52	Georgia Power Company	2,657,949	468,321	-7,159	475,480	\$178.89	52
53	Entergy Arkansas, LLC	727,743	211,904	81,099	130,805	\$179.74	53
54	Alabama Power Company	1,510,098	294,909	14,314	280,595	\$185.81	54
55	Duquesne Light Company	607,350	128,748	13,810	114,938	\$189.25	55
56	Dominion Energy South Carolina, Inc.	771,620	188,917	38,984	149,933	\$194.31	56
57	Massachusetts Electric Company	1,344,807	315,907	52,766	263,141	\$195.67	57
58	Portland General Electric Company	912,209	242,115	62,572	179,543	\$196.82	58
59	Duke Energy Florida, LLC	1,899,991	409,717	13,583	396,134	\$208.49	59
60	Indianapolis Power & Light Company	514,140	134,376	14,142	120,234	\$233.85	60
61	Narragansett Electric Company	501,117	141,580	22,182	119,398	\$238.26	61
62	Public Service Company of New Mexico	540,035	173,891	22,600	151,291	\$280.15	62
63	San Diego Gas & Electric Company	1,478,307	656,792	56,506	600,286	\$406.06	63
64	Pacific Gas and Electric Company	5,623,301	2,972,623	390,655	2,581,968	\$459.16	64
65	Southern California Edison Company	5,192,855	2,730,166	52,119	2,678,047	\$515.72	65

S&P Global Market Intelligence, 55 Water Street, New York, NY 10041

Columns c-e from S&P Global for regulated electric companies with more than 500,000 customers.

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Electric Uncollectible Accounts Expense

(\$000)

Case No.: U-21389

Exhibit No.: A-85 (MJF-4)

Page: 1 of 2

Witness: MJFoster

Date: May 2023

	(a)	(b)	(c)	(d)	(e)
		2022	2023	2024	12 Months Ending
Line	Description	Actual	Projected	Projected	2/28/2025
No.					Projected
1	Uncollectible Accounts Expense	\$ 18,817	\$ 23,796 ¹	\$ 24,731 ²	\$ 17,519 ³

¹ 2023 projected amount based on forecasted O&M² 2024 projected amount based on forecasted O&M³ Exhibit A-85 (MJF-4) Page 2, line 9, column (e)

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Electric Uncollectible Accounts Expense

for 12 Mos. Ending 2/28/2025

(\$000)

Case No.: U-21389

Exhibit No.: A-85 (MJF-4)

Page: 2 of 2

Witness: MJFoster

Date: May 2023

Line No.	(a) Year	(b) Gross Charge-Offs	(c) Less Recoveries	(d) Net Write-Offs	(e) Total Electric Service Revenue MPSC P-521 P. 304.1 col (c) + P. 305 col (c)	(f) BDLR col (d) / col (e)
1	2018	28,943	12,282	16,661	4,382,878	0.380%
2	2019	27,032	11,100	15,932	4,249,553	0.375%
3	2020	22,393	11,965	10,428	4,147,450	0.251%
4	2021	26,335	10,844	15,491	4,681,931	0.331%
5	2022	27,543	8,726	18,817	4,737,236	0.397%
6	3-Year Average 2018, 19, 22	\$ 27,839	\$ 10,703	\$ 17,137	\$ 4,456,556	0.385%
7	Test Year Total Company Electric Revenues and Deliveries Exhibit A-15 (EMB-3), Schedule E-2, Page 1 of 1 Row 19, Column (I) - Row 19, Column (c)				\$ 4,555,951	
8	3-Year Average 2018, 19, 22					0.385%
9	Test Year Total Uncollectible Accounts Expense				\$ 17,518.782	

MICHIGAN PUBLIC SERVICE COMMISSION**Consumers Energy Company**

Electric Injuries & Damages Expense

for the Years 2017 Through the 12 Months Ending December 31, 2023

(\$000)

Case No.: U-21389

Exhibit No.: A-86 (MJF-5)

Page: 1 of 1

Witness: MJFoster

Date: May 2023

Line No.	(a) Program Description	(b) 2018 Actual	(c) 2019 Actual	(d) 2020 Actual	(e) 2021 Actual	(f) 2022 Actual	(g) 12 Mos Ending 12/31/2023 Projected	(h) 12 Mos Ending 12/31/2024 Projected	(i) 12 Mos Ending 2/28/2025 Projected
1	Electric Injuries & Damages	¹ \$ 2,958	\$ 2,396	\$ 2,021	\$ 2,694	\$ 5,567	\$ 3,127	\$ 3,196	\$ 3,207
2	Internal Legal Costs	² 559	379	452	412	424	445	\$ 455	\$ 457
3	Workers' Compensation	³ 458	176	(139)	401	661	311	\$ 318	\$ 319
4	Total Electric Injuries & Damages	<u>\$ 3,975</u>	<u>\$ 2,951</u>	<u>\$ 2,334</u>	<u>\$ 3,507</u>	<u>\$ 6,652</u>	<u>\$ 3,884</u>	<u>\$ 3,969</u>	<u>\$ 3,983</u>
		<u>2023</u>	<u>2024</u>	<u>2025</u>					
Annual merit increase									
Annual merit increase		3.9%	2.2%	2.1%					
Number of months in the period		<u>12</u>	<u>12</u>	<u>2</u>					
Pro-rated merit increase		3.9%	2.2%	0.4%					
Annual inflation rates per WP-JCA-51									
Annual inflation rates		3.9%	2.2%	2.1%					
Number of months in the period		<u>12</u>	<u>12</u>	<u>2</u>					
Pro-rated inflation rate		3.9%	2.2%	0.4%					

¹ Electric Injuries & Damages costs are 2018 - 2022 (actual expense). Projected 2023 expense based on five-year average of actual expenses for years 2018 - 2022. Projected 2024 and test year based on a five-year average plus inflation.

² Legal costs are 2018 - 2022 (actual expense). Projected 2023 expense based on five-year average of actual expenses for years 2018 - 2022. Projected 2024 and test year based on a five-year average plus inflation.

³ Electric Workers' Compensation costs are 2018 - 2022 (actual expense). Projected 2023 expense based on five-year average of actual expenses for years 2018 - 2022. Projected 2024 and test year based on a five-year average plus inflation.

Schedule: B-5.4

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Actual and Projected Capital Expenditures

Electric Corporate Services Capital Expenditures

(\$000)

Case No.: U-21389

Exhibit No.: A-12 (MJF-6)

Schedule: B-5.4

Page: 1 of 1

Witness: MJFoster

Date: May 2023

Line No	(a) Description	(b)	(c)	(d)	(e)	(f)
		Historical Year 12 Mos Ended 12/31/2022	Projected Bridge Period		Projected Test Year	
		12 Mos Ended 12/31/2022	12 Mos Ended 12/31/2023	2 Mos Ending 2/29/2024	14 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025
1	Corporate Capital	537	558	95	653	572
2	Contractor	139	144	25	169	148
3	Labor	37	38	7	45	39
4	Materials	352	366	62	428	375
5	Business Expenses	-	-	-	-	-
6	Contingency	-	-	-	-	-
7	Other (Loadings, Chargebacks)	9	9	2	11	10
8	Total Corporate Capital	537	558	95	653	572

		Projected			
		2 Mos Ending 2/28/2023	12 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025	26 Mos Ending 2/28/2025
9	Corporate Capital	93	560	572	1,225
10	Contractor	24	145	148	317
11	Labor	6	39	39	84
12	Materials	61	367	375	803
13	Business Expenses	-	-	-	-
14	Contingency	-	-	-	-
15	Other (Loadings, Chargebacks)	2	9	10	21
16	Total Corporate Capital	93	560	572	1,225

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS
OF
ALEX M. GAST
ON BEHALF OF
CONSUMERS ENERGY COMPANY

May 2023

Schedule F-2.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Present and Proposed Pro Forma Revenues by Rate Schedule

Total Revenues

Case No.: U-21389
Exhibit No.: A-16 (AMG-1)
Schedule: F-2.0
Page: 1 of 3
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)
Line				Revenue		Net Increase / (Decrease)	
No.	Description	Customers	Sales	Present	Proposed	Revenue	Percent
		Mthly	MWh	\$000	\$000	\$000	%
FULL SERVICE							
Residential Class							
1	Summer On-peak RSP	1,618,090	12,185,915	\$ 2,220,094	\$ 2,292,963	\$ 72,869	3.3
2	Smart Hours RSH	16,512	125,514	22,787	23,568	781	3.4
3	Night Time Savers RPM	13,773	105,411	18,638	19,458	819	4.4
4	Non-Transmitting Meters RSM	14,474	142,301	26,030	26,936	906	3.5
5	Total Residential Class	1,662,850	12,559,141	\$ 2,287,550	\$ 2,362,925	\$ 75,375	3.3
Secondary Class							
6	Energy-only GS	196,122	3,996,698	\$ 633,550	\$ 687,940	\$ 54,390	8.6
7	Time-of-Use GSTU	918	27,489	4,193	4,601	408	9.7
8	Demand GSD	20,782	2,949,510	395,241	432,209	36,968	9.4
9	Total Secondary	217,821	6,973,697	\$ 1,032,984	\$ 1,124,750	\$ 91,766	8.9
Primary Class							
10	Energy-only GP	1,479	805,656	\$ 95,337	\$ 95,677	\$ 340	0.4
	Voltage Level 1	6	15,684	1,568	1,523	(45)	(2.9)
	Voltage Level 2	38	31,696	3,414	3,354	(60)	(1.8)
	Voltage Level 3	1,435	758,276	90,355	90,800	445	0.5
11	Demand GPD	796	4,580,914	428,713	435,150	6,437	1.5
12	Time-of-Use GPTU	1,365	4,681,858	488,788	501,703	12,916	2.6
13	Energy Intensive EIP	17	447,618	32,177	35,547	3,370	10.5
14	Total Primary	3,656	10,516,047	\$ 1,045,014	\$ 1,068,077	\$ 23,062	2.2
Lighting & Unmetered Class							
15	Metered Lighting GML	404	12,195	\$ 1,545	\$ 1,900	\$ 355	23.0
16	Universal Unmetered Lighting UUL	4,686	71,839	28,616	36,492	7,876	27.5
17	Unmetered GU	461	93,426	9,896	10,484	588	5.9
18	Total Lighting & Unmetered	5,551	177,460	\$ 40,056	\$ 48,876	\$ 8,819	22.0
Self-generation Class							
19	Small Self-generation GSG-1	-	-	\$ -	\$ -	\$ -	NA
20	Large Self-generation GSG-2	15	101,694	12,214	12,585	371	3.0
21	Total Self-generation	15	101,694	\$ 12,214	\$ 12,585	\$ 371	3.0
22	Total Full Service	1,889,894	30,328,039	\$ 4,417,818	\$ 4,617,212	\$ 199,394	4.5
ROA SERVICE							
Secondary Class							
23	Energy-only GS	96	23,026	\$ 1,041	\$ 1,383	\$ 342	32.9
24	Demand GSD	462	172,462	6,526	8,305	1,780	27.3
25	Total Secondary	558	195,488	\$ 7,567	\$ 9,688	\$ 2,122	28.0
Primary Class							
26	Energy-only GP	59	75,604	\$ 1,347	\$ 1,645	\$ 298	22.1
27	Demand GPD	338	3,442,823	23,011	27,674	4,663	20.3
28	Total Primary	396	3,518,427	\$ 24,358	\$ 29,319	\$ 4,961	20.4
29	Total ROA Service	954	3,713,915	\$ 31,925	\$ 39,008	\$ 7,083	22.2
30	Total Jurisdictional Service	1,890,848	34,041,954	\$ 4,449,743	\$ 4,656,220	\$ 206,477	4.6
31	Plus: Rounding			-	608	608	
32	Total Jurisdictional Revenues			\$ 4,449,743	\$ 4,656,828	\$ 207,085	4.7
33	Less: PSCR Factor Revenues			190,772	190,772	-	
34	Less: DR Surcharge			-	59,182	59,182	
35	Less: IRM Surcharge			-	6,263	6,263	
36	Total Jurisdictional Base Revenues			\$ 4,258,971	\$ 4,400,610	\$ 141,640	3.3
37	Total Jurisdictional Revenues			4,449,743	4,656,828	207,085	4.7
38	Plus: ERC Deferral Surcharge			-	8,558	8,558	
39	Total Jurisdictional Revenues Requested			\$ 4,449,743	\$ 4,665,386	\$ 215,643	4.8

Schedule F-2.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Present and Proposed Pro Forma Revenues by Rate Schedule

Production & Transmission Revenues

Case No.: U-21389

Exhibit No.: A-16 (AMG-1)

Schedule: F-2.0

Page: 2 of 3

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)
Line No.	Description	Sales	Revenue		Net Increase / (Decrease)	
			Present	Proposed	Revenue	Percent
		MWh	\$000	\$000	\$000	%
FULL SERVICE						
Residential Class						
1	Summer On-peak RSP	12,185,915	\$ 1,309,577	\$ 1,228,684	\$ (80,893)	(6.2)
2	Smart Hours RSH	125,514	13,356	12,553	(803)	(6.0)
3	Night Time Savers RPM	105,411	10,706	10,195	(511)	(4.8)
4	Non-Transmitting Meters RSM	142,301	15,556	14,666	(890)	(5.7)
5	Total Residential Class	12,559,141	\$ 1,349,195	\$ 1,266,099	\$ (83,096)	(6.2)
Secondary Class						
6	Energy-only GS	3,996,698	\$ 407,937	\$ 402,473	\$ (5,464)	(1.3)
7	Time-of-Use GSTU	27,489	2,744	2,741	(4)	(0.1)
8	Demand GSD	2,949,510	277,469	283,827	6,357	2.3
9	Total Secondary	6,973,697	\$ 688,151	\$ 689,040	\$ 889	0.1
Primary Class						
10	Energy-only GP	805,656	\$ 79,920	\$ 77,084	\$ (2,836)	(3.5)
	Voltage Level 1	15,684	1,527	1,472	(55)	(3.6)
	Voltage Level 2	31,696	3,131	3,017	(114)	(3.6)
	Voltage Level 3	758,276	75,261	72,594	(2,667)	(3.5)
11	Demand GPD	4,580,914	388,824	387,498	(1,326)	(0.3)
12	Time-of-Use GPTU	4,681,858	421,973	422,492	519	0.1
13	Energy Intensive EIP	447,618	29,871	32,689	2,819	9.4
14	Total Primary	10,516,047	\$ 920,587	\$ 919,763	\$ (824)	(0.1)
Lighting & Unmetered Class						
15	Metered Lighting GML	12,195	\$ 720	\$ 827	\$ 107	14.9
16	Universal Unmetered Lighting UUL	71,839	4,111	4,728	617	15.0
17	Unmetered GU	93,426	7,772	7,919	147	1.9
18	Total Lighting & Unmetered	177,460	\$ 12,603	\$ 13,474	\$ 871	6.9
Self-generation Class						
19	Small Self-generation GSG-1	-	\$ -	\$ -	\$ -	NA
20	Large Self-generation GSG-2	101,694	10,526	10,526	-	-
21	Total Self-generation	101,694	\$ 10,526	\$ 10,526	\$ -	-
22	Total Full Service	30,328,039	\$ 2,981,062	\$ 2,898,903	\$ (82,160)	(2.8)
ROA SERVICE						
Secondary Class						
23	Energy-only GS	-	\$ -	\$ -	\$ -	NA
24	Demand GSD	-	-	-	-	NA
25	Total Secondary	-	\$ -	\$ -	\$ -	NA
Primary Class						
26	Energy-only GP	-	\$ -	\$ -	\$ -	NA
27	Demand GPD	-	-	-	-	NA
28	Total Primary	-	\$ -	\$ -	\$ -	NA
29	Total ROA Service	-	\$ -	\$ -	\$ -	NA
30	Total Jurisdictional Service	30,328,039	\$ 2,981,062	\$ 2,898,903	\$ (82,160)	(2.8)
31	Plus: Rounding		-	3	3	
32	Total Jurisdictional Base Revenues		\$ 2,981,062	\$ 2,898,906	\$ (82,157)	(2.8)
33	Less: PSCR Factor Revenues		190,772	190,772	-	
34	Less: DR Surcharge		-	59,182	59,182	
35	Less: IRM Surcharge		-	-	-	
36	Total Jurisdictional Base Revenues		\$ 2,790,291	\$ 2,648,952	\$ (141,339)	(5.1)
37	Total Jurisdictional Revenues		2,981,062	2,898,906	(82,157)	(2.8)
38	Plus: ERC Deferral Surcharge		-	-	-	
39	Total Jurisdictional Revenues Requested		\$ 2,981,062	\$ 2,898,906	\$ (82,157)	(2.8)

Schedule F-2.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Present and Proposed Pro Forma Revenues by Rate Schedule
Delivery Revenues

Case No.: U-21389
Exhibit No.: A-16 (AMG-1)
Schedule: F-2.0
Page: 3 of 3
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)
Line			Revenue		Net Increase / (Decrease)	
No.	Description	Sales	Present	Proposed	Revenue	Percent
		MWh	\$000	\$000	\$000	%
	FULL SERVICE					
	Residential Class					
1	Summer On-peak RSP	12,185,915	\$ 910,517	\$ 1,064,279	\$ 153,762	16.9
2	Smart Hours RSH	125,514	9,431	11,015	1,584	16.8
3	Night Time Savers RPM	105,411	7,932	9,263	1,330	16.8
4	Non-Transmitting Meters RSM	142,301	10,474	12,270	1,796	17.1
5	Total Residential Class	12,559,141	\$ 938,355	\$ 1,096,826	\$ 158,471	16.9
	Secondary Class					
6	Energy-only GS	3,996,698	\$ 225,612	\$ 285,467	\$ 59,854	26.5
7	Time-of-Use GSTU	27,489	1,449	1,861	412	28.4
8	Demand GSD	2,949,510	117,772	148,382	30,611	26.0
9	Total Secondary	6,973,697	\$ 344,833	\$ 435,709	\$ 90,877	26.4
	Primary Class					
10	Energy-only GP	805,656	\$ 15,417	\$ 18,593	\$ 3,176	20.6
	Voltage Level 1	15,684	41	51	10	23.9
	Voltage Level 2	31,696	283	337	54	19.1
	Voltage Level 3	758,276	15,094	18,206	3,112	20.6
11	Demand GPD	4,580,914	39,889	47,652	7,763	19.5
12	Time-of-Use GPTU	4,681,858	66,815	79,211	12,396	18.6
13	Energy Intensive EIP	447,618	2,306	2,857	551	23.9
14	Total Primary	10,516,047	\$ 124,427	\$ 148,313	\$ 23,886	19.2
	Lighting & Unmetered Class					
15	Metered Lighting GML	12,195	\$ 825	\$ 1,073	\$ 248	30.1
16	Universal Unmetered Lighting UUL	71,839	24,505	31,764	7,259	29.6
17	Unmetered GU	93,426	2,124	2,565	441	20.8
18	Total Lighting & Unmetered	177,460	\$ 27,454	\$ 35,402	\$ 7,948	29.0
	Self-generation Class					
19	Small Self-generation GSG-1	-	\$ -	\$ -	\$ -	NA
20	Large Self-generation GSG-2	101,694	1,687	2,058	371	22.0
21	Total Self-generation	101,694	\$ 1,687	\$ 2,058	\$ 371	22.0
22	Total Full Service	30,328,039	\$ 1,436,756	\$ 1,718,309	\$ 281,554	19.6
	ROA SERVICE					
	Secondary Class					
23	Energy-only GS	23,026	\$ 1,041	\$ 1,383	\$ 342	32.9
24	Demand GSD	172,462	6,526	8,305	1,780	27.3
25	Total Secondary	195,488	\$ 7,567	\$ 9,688	\$ 2,122	28.0
	Primary Class					
26	Energy-only GP	75,604	\$ 1,347	\$ 1,645	\$ 298	22.1
27	Demand GPD	3,442,823	23,011	27,674	4,663	20.3
28	Total Primary	3,518,427	\$ 24,358	\$ 29,319	\$ 4,961	20.4
29	Total ROA Service	3,713,915	\$ 31,925	\$ 39,008	\$ 7,083	22.2
30	Total Jurisdictional Service	34,041,954	\$ 1,468,680	\$ 1,757,317	\$ 288,637	19.7
31	Plus: Rounding		-	605	605	
32	Total Jurisdictional Base Revenues		\$ 1,468,680	\$ 1,757,922	\$ 289,242	19.7
33	Less: PSCR Factor Revenues		-	-	-	
34	Less: DR Surcharge		-	-	-	
35	Less: IRM Surcharge		-	6,263	6,263	
36	Total Jurisdictional Base Revenues		\$ 1,468,680	\$ 1,751,659	\$ 282,978	19.3
37	Total Jurisdictional Revenues		1,468,680	1,757,922	289,242	19.7
38	Plus: ERC Deferral Surcharge		-	8,558	8,558	
39	Total Jurisdictional Revenues Requested		\$ 1,468,680	\$ 1,766,481	\$ 297,800	20.3

Schedule F-2.1

MICHIGAN PUBLIC SERVICE COMMISSION
 Consumers Energy Company
 Calculation of Rate Design Targets
 (\$000)

Case No.: U-21389
 Exhibit No.: A-16 (AMG-2)
 Schedule: F-2.1
 Page: 1 of 1
 Witness: AMGast
 Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	
Line				Secondary Class		Primary Class			Lighting & Unmetered Class			Self Gen. Class			
No.	Description	Jurisdictional	Residential	GS	GSD	GP	GPD	GPTU	EIP	GML	UUL	GU	GSG-1	GSG-2	Source
Production															
1	Capacity (SRM)	\$ 201,135	\$ 97,930	\$ 29,710	\$ 18,755	\$ 5,112	\$ 21,318	\$ 25,776	\$ 1,896	\$ 1	\$ -	\$ 371	\$ -	\$ 266	
	Capacity Adjustments ⁽¹⁾														
2	DR Revenue Requirement	(46,789)	(22,074)	(6,781)	(4,442)	(1,199)	(5,380)	(6,278)	(498)	(5)	(28)	(102)	-	-	WP-AMG-7
3	Self Generation	-	29	9	6	2	6	8	1	0	-	0	-	(60)	Alloc. Line 1
4	Total Adjustments	(46,789)	(22,045)	(6,772)	(4,437)	(1,198)	(5,374)	(6,271)	(497)	(5)	(28)	(102)	-	(60)	
5	Capacity (SRM) Design Target	\$ 154,347	\$ 75,885	\$ 22,938	\$ 14,319	\$ 3,914	\$ 15,944	\$ 19,505	\$ 1,399	\$ (4)	\$ (28)	\$ 270	\$ -	\$ 206	
Energy ⁽²⁾															
6	Energy Adjustments ⁽¹⁾	\$ 1,968,000	\$ 844,704	\$ 274,142	\$ 196,200	\$ 52,782	\$ 274,007	\$ 293,027	\$ 27,753	\$ 635	\$ 3,646	\$ 5,748	\$ -	\$ (4,645)	
7	Self Generation	-	(1,989)	(645)	(462)	(124)	(645)	(690)	(65)	(1)	(9)	(14)	-	4,645	Alloc. Line 6
8	Total Adjustments	-	(1,989)	(645)	(462)	(124)	(645)	(690)	(65)	(1)	(9)	(14)	-	4,645	
9	Energy Design Target	\$ 1,968,000	\$ 842,715	\$ 273,497	\$ 195,738	\$ 52,658	\$ 273,362	\$ 292,337	\$ 27,687	\$ 634	\$ 3,638	\$ 5,735	\$ -	\$ -	
10	Total Production	\$ 2,122,346	\$ 918,600	\$ 296,435	\$ 210,057	\$ 56,572	\$ 289,306	\$ 311,842	\$ 29,086	\$ 629	\$ 3,609	\$ 6,004	\$ -	\$ 206	
Transmission															
12	Transmission	538,999	248,150	76,014	50,938	14,154	65,598	74,596	6,373	118	653	1,226	-	1,179	
13	Self Generation	-	544	167	112	31	144	164	14	0	1	3	-	(1,179)	Alloc. Line 12
14	Transmission Design Target	\$ 538,999	\$ 248,695	\$ 76,180	\$ 51,049	\$ 14,185	\$ 65,742	\$ 74,759	\$ 6,387	\$ 118	\$ 655	\$ 1,228	\$ -	\$ -	
Delivery															
15	Distribution ⁽³⁾	\$ 1,512,076	\$ 925,194	\$ 246,097	\$ 144,126	\$ 16,945	\$ 70,984	\$ 68,297	\$ 1,450	\$ 1,003	\$ 29,905	\$ 2,481	\$ -	\$ 5,593	
16	Customer	239,577	179,947	37,291	9,297	2,659	4,297	4,269	124	54	1,433	71	-	134	
17	Subtotal Delivery	1,751,653	1,105,141	283,387	153,423	19,604	75,281	72,566	1,574	1,058	31,338	2,553	-	5,727	
	Credit Adjustments ⁽⁴⁾														
18	Substation Credits	\$ -	\$ 953	\$ 254	\$ 148	\$ 3	\$ (570)	\$ (243)	\$ (136)	\$ 1	\$ 31	\$ 3	\$ -	\$ (445)	WP-AMG-12
19	Life Line Credits	(0)	(13,221)	4,059	2,566	560	2,667	2,849	230	11	279	-	-	-	WP-AMG-9
20	Total Adjustments	(0)	(12,268)	4,313	2,714	564	2,098	2,606	94	12	310	3	-	(445)	
21	Total Delivery Design Target	\$ 1,751,653	\$ 1,092,873	\$ 287,700	\$ 156,138	\$ 20,168	\$ 77,379	\$ 75,172	\$ 1,668	\$ 1,070	\$ 31,648	\$ 2,556	\$ -	\$ 5,282	
22	Total Rate Design Target	\$ 4,412,998	\$ 2,260,168	\$ 660,315	\$ 417,243	\$ 90,925	\$ 432,427	\$ 461,774	\$ 37,141	\$ 1,817	\$ 35,912	\$ 9,788	\$ -	\$ 5,488	
23	DR Surcharge	\$ 59,132	\$ 27,919	\$ 8,576	\$ 5,574	\$ 1,517	\$ 6,803	\$ 7,943	\$ 630	\$ 6	\$ 36	\$ 129	\$ -	\$ -	WP-AMG-19
24	IRM Surcharge	\$ 6,269	\$ 3,956	\$ 1,012	\$ 549	\$ 68	\$ 305	\$ 244	\$ 7	\$ 4	\$ 112	\$ 9	\$ -	\$ 3	Exhibit A-90 (AMG-8), page 1
25	Total Revenue Requirement	\$ 4,478,399	\$ 2,292,043	\$ 669,902	\$ 423,367	\$ 92,510	\$ 439,535	\$ 469,961	\$ 37,778	\$ 1,827	\$ 36,060	\$ 9,926	\$ -	\$ 5,491	

Notes

(1) Capacity and energy production costs adjusted to capture elements occurring outside the Cost-of-Service Study.

(2) Includes Non-SRM capacity related costs.

(3) Excludes IRM costs recovered through the IRM surcharge.

(4) Credit adjustments approved in Case No. U-20963 and included in approved settlement for Case No. U-21224.

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Residential Summer On-peak (RSP)

Case No.: U-21389
Exhibit No.: A-16 (AMG-3)
Schedule: F-3.0
Page: 1 of 23
Witness: AMGast
Date: May 2023

		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	On-peak kWh	827,023	MWh	@	0.078133	\$ 64,618	0.098747	\$ 81,666	\$ 17,048
2	Off-peak kWh	3,519,128	MWh	@	0.049341	173,637	0.063696	224,154	50,517
	Capacity								
3	On-peak kWh	827,023	MWh	@	0.048453	40,072	0.010101	8,354	(31,718)
4	Off-peak kWh	3,519,128	MWh	@	0.032567	114,607	0.006790	23,895	(90,713)
	Winter (Oct. - May)								
5	Non-capacity All kWh	7,839,274	MWh	@	0.053169	416,806	0.065273	511,693	94,887
6	Capacity All kWh	7,839,274	MWh	@	0.025374	198,914	0.005290	41,470	(157,444)
	Provisions								
	Device Cycling								
7	AC Cycling	342,162	Bills	@	(8.00)	(2,737)	(8.00)	(2,737)	-
8	Back-Up Generator Pilot	-	Bills	@	(22.40)	-	(12.99)	-	-
9	Electric Water Tank Pilot	24,000	Bills	@	(3.20)	(77)	(1.88)	(45)	32
10	Peak Time Rewards	1,184	MWh	@	(1.00)	(1,184)	(1.00)	(1,184)	-
	Critical Peak Pricing								
11	Critical-peak Charge	490	MWh	@	1.00	490	1.00	490	-
12	Off-peak Credits	24,954	MWh	@	(0.014702)	(367)	(0.019625)	(490)	(123)
13	Annual PSCR Factor	12,185,915	MWh	@	0.005970	72,750	0.005970	72,750	-
14	Total Production					\$ 1,077,529		\$ 960,015	\$ (117,514)
	Transmission								
	Summer (June - Sept.)								
15	On-peak kWh	827,023	MWh	@	0.031798	\$ 26,298	0.033104	\$ 27,378	\$ 1,080
16	Off-peak kWh	3,519,128	MWh	@	0.021372	75,211	0.022250	78,301	3,090
	Winter (Oct. - May)								
17	All kWh	7,839,274	MWh	@	0.016652	130,540	0.017336	135,902	5,362
18	Total Transmission					\$ 232,048		\$ 241,580	\$ 9,532
	Delivery								
19	System Access	19,417,083	Bills	@	8.00	\$ 155,337	8.00	\$ 155,337	\$ -
20	Distribution	12,185,915	MWh	@	0.064152	781,751	0.076455	931,674	149,923
	Provisions								
21	Senior Citizen (RSC)	3,883,861	Bills	@	(4.00)	(15,535)	(4.00)	(15,535)	-
22	Low Income Credit (LIAC)	50,400	Bills	@	(30.00)	(1,512)	(30.00)	(1,512)	-
23	Income Assistance (RIA)	1,190,399	Bills	@	(8.00)	(9,523)	(8.00)	(9,523)	-
24	Total Delivery					\$ 910,517		\$ 1,060,440	\$ 149,923
25	Total Full Service					\$ 2,220,094		\$ 2,262,035	\$ 41,941
	ROA SERVICE								
	Delivery								
26	System Access	-	Bills	@	8.00	\$ -	8.00	\$ -	\$ -
27	Distribution	-	MWh	@	0.064152	-	0.076455	-	-
	Provisions								
28	Senior Citizen (RSC)	-	Bills	@	(4.00)	-	(4.00)	-	-
29	Low Income Credit (LIAC)	-	Bills	@	(30.00)	-	(30.00)	-	-
30	Income Assistance (RIA)	-	Bills	@	(8.00)	-	(8.00)	-	-
31	Total Delivery					\$ -		\$ -	\$ -
32	Total Full & ROA Service					\$ 2,220,094		\$ 2,262,035	\$ 41,941

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION
 Consumers Energy Company
 Present and Proposed Revenue Detail

Case No.: U-21389
 Exhibit No.: A-16 (AMG-3)
 Schedule: F-3.0
 Page: 2 of 23
 Witness: AMGast
 Date: May 2023

Residential Smart Hours (RSH)

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	On-peak kWh	7,567	MWh	@	0.078133	\$ 591	0.098747	\$ 747	\$ 156
2	Off-peak kWh	34,872	MWh	@	0.049341	1,721	0.063696	2,221	501
	Capacity								
3	On-peak kWh	7,567	MWh	@	0.048453	367	0.010101	76	(290)
4	Off-peak kWh	34,872	MWh	@	0.032567	1,136	0.006790	237	(899)
	Winter (Oct. - May)								
	Non-capacity								
5	On-peak kWh	12,107	MWh	@	0.059616	722	0.072052	872	151
6	Off-peak kWh	70,959	MWh	@	0.051905	3,683	0.063934	4,537	854
	Capacity								
7	On-peak kWh	12,107	MWh	@	0.027940	338	0.005825	71	(268)
8	Off-peak kWh	70,959	MWh	@	0.024562	1,743	0.005121	363	(1,380)
	Provisions								
	Device Cycling								
9	AC Cycling	2,895	Bills	@	(8.00)	(23)	(8.00)	(23)	-
10	Back-Up Generator Pilot	-	Bills	@	(22.40)	-	(12.99)	-	-
11	Electric Water Tank Pilot	-	Bills	@	(3.20)	-	(1.88)	-	-
12	Peak Time Rewards	22	MWh	@	(1.00)	(22)	(1.00)	(22)	-
	Critical Peak Pricing								
13	Critical-peak Charge	9	MWh	@	1.00	9	1.00	9	-
14	Off-peak Credits	591	MWh	@	(0.014702)	(9)	(0.019625)	(12)	(3)
15	Annual PSCR Factor	125,514	MWh	@	0.005970	749	0.005970	749	-
16	Total Production					\$ 11,004		\$ 9,826	\$ (1,179)
	Transmission								
	Summer (June - Sept.)								
17	On-peak kWh	7,567	MWh	@	0.031798	\$ 241	0.033104	\$ 250	\$ 10
18	Off-peak kWh	34,872	MWh	@	0.021372	745	0.022250	776	31
	Winter (Oct. - May)								
19	On-peak kWh	12,107	MWh	@	0.018336	222	0.019089	231	9
20	Off-peak kWh	70,959	MWh	@	0.016119	1,144	0.016781	1,191	47
21	Total Transmission					\$ 2,352		\$ 2,448	\$ 97
	Delivery								
22	System Access	198,142	Bills	@	8.00	\$ 1,585	8.00	\$ 1,585	\$ -
23	Distribution	125,514	MWh	@	0.064152	8,052	0.076455	9,596	1,544
	Provisions								
24	Senior Citizen (RSC)	33,104	Bills	@	(4.00)	(132)	(4.00)	(132)	-
25	Low Income Credit (LIAC)	-	Bills	@	(30.00)	-	(30.00)	-	-
26	Income Assistance (RIA)	9,172	Bills	@	(8.00)	(73)	(8.00)	(73)	-
27	Total Delivery					\$ 9,431		\$ 10,976	\$ 1,544
28	Total Full Service					\$ 22,787		\$ 23,250	\$ 462
	ROA SERVICE								
	Delivery								
29	System Access	-	Bills	@	8.00	\$ -	8.00	\$ -	\$ -
30	Distribution	-	MWh	@	0.064152	-	0.076455	-	-
	Provisions								
31	Senior Citizen (RSC)	-	Bills	@	(4.00)	-	(4.00)	-	-
32	Low Income Credit (LIAC)	-	Bills	@	(30.00)	-	(30.00)	-	-
33	Income Assistance (RIA)	-	Bills	@	(8.00)	-	(8.00)	-	-
34	Total Delivery					\$ -		\$ -	\$ -
35	Total Full & ROA Service					\$ 22,787		\$ 23,250	\$ 462

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION
 Consumers Energy Company
 Present and Proposed Revenue Detail
 Residential Nighttime Savers (RPM)

Case No.: U-21389
 Exhibit No.: A-16 (AMG-3)
 Schedule: F-3.0
 Page: 3 of 23
 Witness: AMGast
 Date: May 2023

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line No.	Description	Billing Determinants			Present		Proposed		Net Increase / (Decrease)
		Quantity	Units		Rate	Revenue	Rate	Revenue	
					\$/unit	\$000	\$/unit	\$000	
									\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	On-peak kWh	7,127	MWh	@	0.078133	\$ 557	0.098747	\$ 704	\$ 147
2	Off-peak kWh	14,724	MWh	@	0.055220	813	0.070946	1,045	232
3	Super Off-peak kWh	23,134	MWh	@	0.044158	1,022	0.057238	1,324	303
	Capacity								
4	On-peak kWh	7,127	MWh	@	0.048453	345	0.010101	72	(273)
5	Off-peak kWh	14,724	MWh	@	0.036740	541	0.007660	113	(428)
6	Super Off-peak kWh	23,134	MWh	@	0.022366	517	0.004663	108	(410)
	Winter (Oct. - May)								
	Non-capacity								
7	On-peak kWh	7,932	MWh	@	0.059616	473	0.072052	571	99
8	Off-peak kWh	19,965	MWh	@	0.058664	1,171	0.072238	1,442	271
9	Super Off-peak kWh	32,528	MWh	@	0.045916	1,494	0.056583	1,841	347
	Capacity								
10	On-peak kWh	7,932	MWh	@	0.027940	222	0.005825	46	(175)
11	Off-peak kWh	19,965	MWh	@	0.027016	539	0.005632	112	(427)
12	Super Off-peak kWh	32,528	MWh	@	0.019376	630	0.004040	131	(499)
	Provisions								
	Device Cycling								
13	AC Cycling	3,568	Bills	@	(8.00)	(29)	(8.00)	(29)	-
14	Back-Up Generator Pilot	-	Bills	@	(22.40)	-	(12.99)	-	-
15	Electric Water Tank Pilot	-	Bills	@	(3.20)	-	(1.88)	-	-
16	Peak Time Rewards	36	MWh	@	(1.00)	(36)	(1.00)	(36)	-
	Critical Peak Pricing								
17	Critical-peak Charge	-	MWh	@	1.00	-	1.00	-	-
18	Off-peak Credits	1,153	MWh	@	(0.014702)	(17)	(0.019625)	(23)	(6)
19	Annual PSCR Factor	105,411	MWh	@	0.005970	629	0.005970	629	-
20	Total Production					\$ 8,872		\$ 8,051	\$ (820)
	Transmission								
	Summer (June - Sept.)								
21	On-peak kWh	7,127	MWh	@	0.031798	\$ 227	0.033104	\$ 236	\$ 9
22	Off-peak kWh	14,724	MWh	@	0.024111	355	0.025101	370	15
23	Super Off-peak kWh	23,134	MWh	@	0.014678	340	0.015281	354	14
	Winter (Oct. - May)								
24	On-peak kWh	7,932	MWh	@	0.018336	145	0.019089	151	6
25	Off-peak kWh	19,965	MWh	@	0.017730	354	0.018458	369	15
26	Super Off-peak kWh	32,528	MWh	@	0.012716	414	0.013238	431	17
27	Total Transmission					\$ 1,834		\$ 1,910	\$ 75
	Delivery								
28	System Access	165,280	Bills	@	8.00	\$ 1,322	8.00	\$ 1,322	\$ -
29	Separately Metered EV Charger	300	Bills	@	(8.00)	\$ (2)	(8.00)	\$ (2)	\$ -
30	Distribution	105,411	MWh	@	0.064152	6,762	0.076455	8,059	1,297
	Provisions								
31	Senior Citizen (RSC)	28,759	Bills	@	(4.00)	(115)	(4.00)	(115)	-
32	Low Income Credit (LIAC)	-	Bills	@	(30.00)	-	(30.00)	-	-
33	Income Assistance (RIA)	4,332	Bills	@	(8.00)	(35)	(8.00)	(35)	-
34	Total Delivery					\$ 7,932		\$ 9,229	\$ 1,297
35	Total Full Service					\$ 18,638		\$ 19,190	\$ 552
	ROA SERVICE								
	Delivery								
36	System Access	-	Bills	@	8.00	\$ -	8.00	\$ -	\$ -
37	Distribution	-	MWh	@	0.064152	-	0.076455	-	-
	Provisions								
38	Senior Citizen (RSC)	-	Bills	@	(4.00)	-	(4.00)	-	-
39	Low Income Credit (LIAC)	-	Bills	@	(30.00)	-	(30.00)	-	-
40	Income Assistance (RIA)	-	Bills	@	(8.00)	-	(8.00)	-	-
41	Total Delivery					\$ -		\$ -	\$ -
42	Total Full & ROA Service					\$ 18,638		\$ 19,190	\$ 552

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Residential Non-Transmitting Meters (RSM)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 4 of 23

Witness: AMGast

Date: May 2023

(a)		(b)	(c)	(d)		(e)	(f)	(g)	(h)
Line		Billing Determinants		Present		Proposed		Net Increase	
No.	Description	Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)	
				\$/unit	\$000	\$/unit	\$000	\$000	
FULL SERVICE									
Production									
Summer (June - Sept.)									
Non-capacity									
1	First 600 kWh	25,621	MWh	@	0.053169	\$ 1,362	0.065273	\$ 1,672	\$ 310
2	Excess kWh	18,360	MWh	@	0.078133	1,434	0.098747	1,813	378
Capacity									
3	First 600 kWh	25,621	MWh	@	0.025374	650	0.005290	136	(515)
4	Excess kWh	18,360	MWh	@	0.048453	890	0.010101	185	(704)
Winter (Oct. - May)									
5	Non-capacity All kWh	98,320	MWh	@	0.053169	5,228	0.065273	6,418	1,190
6	Capacity All kWh	98,320	MWh	@	0.025374	2,495	0.005290	520	(1,975)
7	Annual PSCR Factor	142,301	MWh	@	0.005970	850	0.005970	850	-
8	Total Production					\$ 12,908		\$ 11,594	\$ (1,315)
Transmission									
Summer (June - Sept.)									
9	First 600 kWh	25,621	MWh	@	0.016652	\$ 427	0.017336	\$ 444	\$ 18
10	Excess kWh	18,360	MWh	@	0.031798	584	0.033104	608	24
Winter (Oct. - May)									
11	All kWh	98,320	MWh	@	0.016652	1,637	0.017336	1,704	67
12	Total Transmission					\$ 2,648		\$ 2,756	\$ 109
Delivery									
13	System Access	173,693	Bills	@	8.00	\$ 1,390	8.00	\$ 1,390	\$ -
14	Opt-out Fee	101,753	Bills	@	3.00	305	3.00	305	-
15	Distribution	142,301	MWh	@	0.064152	9,129	0.076455	10,880	1,751
Provisions									
16	Senior Citizen (RSC)	6,404	Bills	@	(4.00)	(26)	(4.00)	(26)	-
17	Low Income Credit (LIAC)	-	Bills	@	(30.00)	-	(30.00)	-	-
18	Income Assistance (RIA)	40,465	Bills	@	(8.00)	(324)	(8.00)	(324)	-
19	Total Delivery					\$ 10,474		\$ 12,225	\$ 1,751
20	Total Full Service					\$ 26,030		\$ 26,575	\$ 545
ROA SERVICE									
Delivery									
21	System Access	-	Bills	@	8.00	\$ -	8.00	\$ -	\$ -
22	Opt-out Fee	-	Bills	@	3.00	-	3.00	-	-
23	Distribution	-	MWh	@	0.064152	-	0.076455	-	-
Provisions									
24	Senior Citizen (RSC)	-	Bills	@	(4.00)	-	(4.00)	-	-
25	Low Income Credit (LIAC)	-	Bills	@	(30.00)	-	(30.00)	-	-
26	Income Assistance (RIA)	-	Bills	@	(8.00)	-	(8.00)	-	-
27	Total Delivery					\$ -		\$ -	\$ -
28	Total Full & ROA Service					\$ 26,030		\$ 26,575	\$ 545

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Present and Proposed Revenue Detail

Case No.: U-21389
 Exhibit No.: A-16 (AMG-3)
 Schedule: F-3.0
 Page: 5 of 23
 Witness: AMGast
 Date: May 2023

Secondary Energy-only (GS)

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
1	Non-capacity All kWh	1,444,421	MWh	@	0.056152	\$ 81,107	0.071055	\$ 102,633	\$ 21,526
2	Capacity All kWh	1,444,421	MWh	@	0.029502	42,613	0.007053	10,187	(32,426)
	Winter (Oct. - May)								
3	Non-capacity All kWh	2,552,277	MWh	@	0.054738	139,707	0.066209	168,984	29,277
4	Capacity All kWh	2,552,277	MWh	@	0.020650	52,705	0.004937	12,601	(40,104)
5	Annual PSCR Factor	3,996,698	MWh	@	0.005970	23,860	0.005970	23,860	-
6	Total Production					\$ 339,992		\$ 318,265	\$ (21,726)
	Transmission								
	Summer (June - Sept.)								
7	All kWh	1,444,421	MWh	@	0.021030	\$ 30,376	0.023427	\$ 33,838	\$ 3,462
	Winter (Oct. - May)								
8	All kWh	2,552,277	MWh	@	0.014720	37,570	0.016398	41,852	4,283
9	Total Transmission					\$ 67,946		\$ 75,691	\$ 7,745
	Delivery								
10	System Access	2,353,460	Bills	@	20.00	\$ 47,069	20.00	\$ 47,069	\$ -
11	Distribution	3,996,698	MWh	@	0.044692	178,620	0.059423	237,496	58,875
	Provisions								
12	Education GEI	96,645	MWh	@	(0.000802)	(78)	(0.001010)	(98)	(20)
13	Total Delivery					\$ 225,612		\$ 284,467	\$ 58,855
14	Total Full Service					\$ 633,550		\$ 678,423	\$ 44,874
	ROA SERVICE								
	Delivery								
15	System Access	1,153	Bills	@	20.00	\$ 23	20.00	\$ 23	\$ -
16	Distribution	23,026	MWh	@	0.044692	1,029	0.059423	1,368	339
	Provisions								
17	Education GEI	13,863	MWh	@	(0.000802)	(11)	(0.001010)	(14)	(3)
18	Total Delivery					\$ 1,041		\$ 1,377	\$ 336
19	Total Full & ROA Service					\$ 634,591		\$ 679,801	\$ 45,210

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Secondary Time-of-Use (GSTU)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 6 of 23

Witness: AMGast

Date: May 2023

(a)		(b)		(c)		(d)		(e)		(f)		(g)		(h)	
Line		Billing Determinants			Present		Proposed		Net Increase						
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	Rate	Revenue	/ (Decrease)				
					\$/unit	\$000	\$/unit	\$000	\$/unit	\$000	\$000				
FULL SERVICE															
Production															
Summer (June - Sept.)															
Non-capacity															
1	On-peak kWh	1,793	MWh	@	0.083474	\$ 150	0.103565	\$ 186	\$						36
2	Mid-peak kWh	4,938	MWh	@	0.058660	290	0.074350	367							77
3	Off-peak kWh	5,593	MWh	@	0.045423	254	0.057908	324							70
Capacity															
4	On-peak kWh	1,793	MWh	@	0.035796	64	0.008558	15							(49)
5	Mid-peak kWh	4,938	MWh	@	0.029742	147	0.007111	35							(112)
6	Off-peak kWh	5,593	MWh	@	0.019194	107	0.004589	26							(82)
Winter (Oct. - May)															
Non-capacity															
7	On-peak kWh	7,946	MWh	@	0.060678	482	0.073195	582							99
8	Off-peak kWh	7,219	MWh	@	0.048157	348	0.058461	422							74
Capacity															
9	On-peak kWh	7,946	MWh	@	0.022389	178	0.005353	43							(135)
10	Off-peak kWh	7,219	MWh	@	0.016727	121	0.003999	29							(92)
Provisions															
11	Interruptible (GSI)	-	MWh	@	(0.017799)	-	(0.017673)	-							-
12	Annual PSCR Factor	27,489	MWh	@	0.005970	164	0.005970	164							-
13	Total Production					\$ 2,304		\$ 2,192	\$						(112)
Transmission															
Summer (June - Sept.)															
14	On-peak kWh	1,793	MWh	@	0.025517	\$ 46	0.028425	\$ 51	\$						5
15	Mid-peak kWh	4,938	MWh	@	0.021201	105	0.023618	117							12
16	Off-peak kWh	5,593	MWh	@	0.013682	77	0.015242	85							9
Winter (Oct. - May)															
17	On-peak kWh	7,946	MWh	@	0.015959	127	0.017779	141							14
18	Off-peak kWh	7,219	MWh	@	0.011924	86	0.013283	96							10
19	Total Transmission					\$ 440		\$ 490	\$						50
Delivery															
20	System Access	11,015	Bills	@	20.00	\$ 220	20.00	\$ 220	\$						-
21	Distribution	27,489	MWh	@	0.044692	1,229	0.059423	1,633							405
Provisions															
22	Education GEI	-	MWh	@	(0.000802)	-	(0.001010)	-							-
23	Total Delivery					\$ 1,449		\$ 1,854	\$						405
24	Total Full Service					\$ 4,193		\$ 4,536	\$						343

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Secondary Demand (GSD)

Case No.: U-21389
Exhibit No.: A-16 (AMG-3)
Schedule: F-3.0
Page: 7 of 23
Witness: AMGast
Date: May 2023

		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	Peak kW	2,956	MW	@	11.07	\$ 32,725	19.29	\$ 57,025	\$ 24,300
2	All kWh	1,060,140	MWh	@	0.032069	33,998	0.032115	34,046	49
	Capacity								
3	Peak kW	2,956	MW	@	8.09	23,916	1.88	5,558	(18,358)
4	All kWh	1,060,140	MWh	@	-	-	-	-	-
	Winter (Oct. - May)								
	Non-capacity								
5	Peak kW	5,241	MW	@	5.24	27,463	9.13	47,850	20,387
6	All kWh	1,889,370	MWh	@	0.031261	59,064	0.029925	56,539	(2,524)
	Capacity								
7	Peak kW	5,241	MW	@	7.20	37,735	1.67	8,752	(28,983)
8	All kWh	1,889,370	MWh	@	-	-	-	-	-
9	Power Factor Adjustment					172		296	124
	Provisions								
10	Interruptible (GSI)-Summer	28	MW	@	(7.00)	(196)	(7.00)	(196)	-
11	Interruptible (GSI)-Winter	50	MW	@	(6.00)	(300)	(6.00)	(300)	-
12	Annual PSCR Factor	2,949,510	MWh	@	0.005970	17,609	0.005970	17,609	-
13	Total Production					\$ 232,185		\$ 227,180	\$ (5,005)
	Transmission								
	Summer (June - Sept.)								
14	Peak kW	2,956	MW	@	5.94	\$ 17,560	6.71	\$ 19,836	\$ 2,276
	Winter (Oct. - May)								
15	Peak kW	5,241	MW	@	5.29	27,725	5.96	31,236	3,511
16	Total Transmission					\$ 45,285		\$ 51,073	\$ 5,788
	Delivery								
17	System Access	249,381	Bills	@	30.00	\$ 7,481	30.00	\$ 7,481	\$ -
18	Peak kW	8,197	MW	@	1.00	8,197	1.00	8,197	-
19	Distribution	2,949,510	MWh	@	0.034591	102,027	0.044787	132,100	30,073
20	Power Factor Adjustment					155		198	42
	Provisions								
21	Education GEI	137,614	MWh	@	(0.000647)	(89)	(0.000822)	(113)	(24)
22	Total Delivery					\$ 117,772		\$ 147,863	\$ 30,092
23	Total Full Service					\$ 395,241		\$ 426,116	\$ 30,875
	ROA SERVICE								
	Delivery								
24	System Access	5,542	Bills	@	30.00	\$ 166	30.00	\$ 166	\$ -
25	Peak kW	427	MW	@	1.00	427	1.00	427	-
26	Distribution	172,462	MWh	@	0.034591	5,966	0.044787	7,724	1,758
27	Power Factor Adjustment					10		12	3
	Provisions								
28	Education GEI	66,447	MWh	@	(0.000647)	(43)	(0.000822)	(55)	(12)
29	Total Delivery					\$ 6,526		\$ 8,275	\$ 1,749
30	Total Full & ROA Service					\$ 401,767		\$ 434,391	\$ 32,624

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy-only Voltage Level 1 (GP VL 1)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 8 of 23

Witness: AMGast

Date: May 2023

(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants		Present		Proposed		Net Increase
No.	Description	Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)
				\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE							
	Production							
	Summer (June - Sept.)							
1	Non-capacity All kWh	5,472	MWh	@	0.054929	\$ 301	0.066117 \$ 362	\$ 61
2	Capacity All kWh	5,472	MWh	@	0.025180	138	0.005815 32	(106)
	Winter (Oct. - May)							
3	Non-capacity All kWh	10,212	MWh	@	0.053546	547	0.061608 629	82
4	Capacity All kWh	10,212	MWh	@	0.017625	180	0.004070 42	(138)
5	Power Factor Adjustment					18	16	(2)
6	Annual PSCR Factor	15,684	MWh	@	0.005970	94	0.005970 94	-
7	Total Production					\$ 1,277	\$ 1,174	\$ (102)
	Transmission							
	Summer (June - Sept.)							
8	All kWh	5,472	MWh	@	0.019871	\$ 109	0.021351 \$ 117	\$ 8
	Winter (Oct. - May)							
9	All kWh	10,212	MWh	@	0.013909	142	0.014945 153	11
10	Total Transmission					\$ 251	\$ 269	\$ 19
	Delivery							
11	System Access	67	Bills	@	100.00	\$ 7	100.00 \$ 7	\$ -
12	Distribution	15,684	MWh	@	0.002171	34	0.002777 44	10
13	Substation Ownership Credit	460	MWh	@	(0.001123)	(1)	(0.001376) (1)	(0)
14	Power Factor Adjustment					1	1	0
	Provisions							
15	Education GEI	-	MWh	@	(0.000533)	-	(0.000636) -	-
16	Total Delivery					\$ 41	\$ 50	\$ 10
17	Total Full Service					\$ 1,568	\$ 1,494	\$ (74)
	ROA SERVICE							
	Delivery							
18	System Access	-	Bills	@	100.00	\$ -	100.00 \$ -	\$ -
19	Distribution	-	MWh	@	0.002171	-	0.002777 -	-
20	Substation Ownership Credit	-	MWh	@	(0.001123)	-	(0.001376) -	-
21	Power Factor Adjustment					-	-	-
	Provisions							
22	Education GEI	-	MWh	@	(0.000533)	-	(0.000636) -	-
23	Total Delivery					\$ -	\$ -	\$ -
24	Total Full & ROA Service					\$ 1,568	\$ 1,494	\$ (74)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy-only Voltage Level 2 (GP VL 2)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 9 of 23

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
1	Non-capacity All kWh	9,746	MWh	@	0.055588	\$ 542	0.066884	\$ 652	\$ 110
2	Capacity All kWh	9,746	MWh	@	0.025563	249	0.005899	57	(192)
	Winter (Oct. - May)								
3	Non-capacity All kWh	21,950	MWh	@	0.054189	1,189	0.062323	1,368	179
4	Capacity All kWh	21,950	MWh	@	0.017893	393	0.004129	91	(302)
5	Power Factor Adjustment					62		57	(5)
6	Annual PSCR Factor	31,696	MWh	@	0.005970	189	0.005970	189	-
7	Total Production					\$ 2,625		\$ 2,414	\$ (211)
	Transmission								
	Summer (June - Sept.)								
8	All kWh	9,746	MWh	@	0.020173	\$ 197	0.021661	\$ 211	\$ 15
	Winter (Oct. - May)								
9	All kWh	21,950	MWh	@	0.014120	310	0.015162	333	23
10	Total Transmission					\$ 507		\$ 544	\$ 37
	Delivery								
11	System Access	452	Bills	@	100.00	\$ 45	100.00	\$ 45	\$ -
12	Distribution	31,696	MWh	@	0.007661	243	0.009340	296	53
13	Substation Ownership Credit	7,396	MWh	@	(0.001513)	(11)	(0.001783)	(13)	(2)
14	Power Factor Adjustment					6		7	1
	Provisions								
15	Education GEI	-	MWh	@	(0.000533)	-	(0.000636)	-	-
16	Total Delivery					\$ 283		\$ 335	\$ 53
17	Total Full Service					\$ 3,414		\$ 3,293	\$ (121)
	ROA SERVICE								
	Delivery								
18	System Access	29	Bills	@	100.00	\$ 3	100.00	\$ 3	\$ -
19	Distribution	2,770	MWh	@	0.007661	21	0.009340	26	5
20	Substation Ownership Credit	-	MWh	@	(0.001513)	-	(0.001783)	-	-
21	Power Factor Adjustment					(0)		(0)	(0)
	Provisions								
22	Education GEI	-	MWh	@	(0.000533)	-	(0.000636)	-	-
23	Total Delivery					\$ 24		\$ 29	\$ 5
24	Total Full & ROA Service					\$ 3,438		\$ 3,322	\$ (116)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy-only Voltage Level 3 (GP VL 3)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 10 of 23

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
1	Non-capacity All kWh	256,746	MWh	@	0.056099	\$ 14,403	0.067624	\$ 17,362	\$ 2,959
2	Capacity All kWh	256,746	MWh	@	0.025915	6,654	0.005992	1,538	(5,115)
	Winter (Oct. - May)								
3	Non-capacity All kWh	501,530	MWh	@	0.054687	27,427	0.063013	31,603	4,176
4	Capacity All kWh	501,530	MWh	@	0.018140	9,098	0.004194	2,103	(6,994)
5	Power Factor Adjustment					723		660	(62)
6	Annual PSCR Factor	758,276	MWh	@	0.005970	4,527	0.005970	4,527	-
7	Total Production					\$ 62,831		\$ 57,794	\$ (5,037)
	Transmission								
	Summer (June - Sept.)								
8	All kWh	256,746	MWh	@	0.020451	\$ 5,251	0.022000	\$ 5,648	\$ 398
	Winter (Oct. - May)								
9	All kWh	501,530	MWh	@	0.014315	7,179	0.015399	7,723	544
10	Total Transmission					\$ 12,430		\$ 13,371	\$ 941
	Delivery								
11	System Access	17,224	Bills	@	100.00	\$ 1,722	100.00	\$ 1,722	\$ -
12	Distribution	758,276	MWh	@	0.017474	13,250	0.021460	16,273	3,022
13	Power Factor Adjustment					166		204	38
	Provisions								
14	Education GEI	85,056	MWh	@	(0.000533)	(45)	(0.000636)	(54)	(9)
15	Total Delivery					\$ 15,094		\$ 18,145	\$ 3,052
16	Total Full Service					\$ 90,355		\$ 89,311	\$ (1,044)
	ROA SERVICE								
	Delivery								
17	System Access	674	Bills	@	100.00	\$ 67	100.00	\$ 67	\$ -
18	Distribution	72,834	MWh	@	0.017474	1,273	0.021460	1,563	290
19	Power Factor Adjustment					4		5	1
	Provisions								
20	Education GEI	38,635	MWh	@	(0.000533)	(21)	(0.000636)	(25)	(4)
21	Total Delivery					\$ 1,323		\$ 1,611	\$ 287
22	Total Full & ROA Service					\$ 91,678		\$ 90,921	\$ (757)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Demand Voltage Level 1 (GPD VL 1)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 11 of 23

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	On-peak kW	564	MW	@	11.17	\$ 6,295	16.30	\$ 9,186	\$ 2,891
2	On-peak kWh	74,426	MWh	@	0.042163	3,138	0.044427	3,307	169
3	Off-peak kWh	250,778	MWh	@	0.026469	6,638	0.028381	7,117	479
4	Capacity On-peak kW	564	MW	@	8.94	5,038	1.93	1,088	(3,951)
	Winter (Oct. - May)								
	Non-capacity								
5	On-peak kW	1,216	MW	@	9.65	11,730	14.08	17,115	5,385
6	On-peak kWh	165,191	MWh	@	0.033687	5,565	0.033956	5,609	44
7	Off-peak kWh	543,249	MWh	@	0.028934	15,718	0.029592	16,076	357
8	Capacity On-peak kW	1,216	MW	@	8.29	10,077	1.79	2,176	(7,901)
	Provisions								
9	Economic Development GED	-	MW	@	-	-	-	-	-
	Interruptible GI								
10	Summer Contracted kW	53	MW	@	(7.00)	(372)	(7.00)	(372)	-
11	Winter Contracted kW	106	MW	@	(6.00)	(638)	(6.00)	(638)	-
	Interruptible GI-2								
12	Summer Capacity & Transmission	-	MWh	@	0.031935	-	0.033203	-	-
13	Winter Capacity & Transmission	-	MWh	@	0.029184	-	0.029597	-	-
14	LMP	-	MWh	@	-	-	-	-	-
15	Power Factor Adjustment					(132)		(127)	5
16	Annual PSCR Factor	1,033,644	MWh	@	0.005970	6,171	0.005970	6,171	-
17	Total Production					\$ 69,226		\$ 66,706	\$ (2,520)
	Transmission								
	Summer (June - Sept.)								
18	On-peak kW	564	MW	@	7.38	\$ 4,159	7.93	\$ 4,469	\$ 310
	Winter (Oct. - May)								
19	On-peak kW	1,216	MW	@	6.87	8,351	7.38	8,971	620
20	Total Transmission					\$ 12,510		\$ 13,440	\$ 930
	Delivery								
21	System Access	274	Bills	@	200.00	\$ 55	200.00	\$ 55	\$ -
22	Maximum kW	2,493	MW	@	0.74	1,845	0.95	2,368	524
23	Substation Ownership	46	MW	@	(0.46)	(21)	(0.58)	(27)	(6)
24	Distribution	1,033,644	MWh	@	-	-	-	-	-
25	Power Factor Adjustment					(4)		(5)	(1)
	Provisions								
26	Education GEI	-	MWh	@	(0.000201)	-	(0.000210)	-	-
27	Total Delivery					\$ 1,875		\$ 2,392	\$ 517
28	Total Full Service					\$ 83,611		\$ 82,537	\$ (1,074)
	ROA SERVICE								
	Delivery								
29	System Access	161	Bills	@	200.00	\$ 32	200.00	\$ 32	\$ -
30	Maximum kW	2,197	MW	@	0.74	1,626	0.95	2,087	461
31	Substation Ownership	225	MW	@	(0.46)	(104)	(0.58)	(131)	(27)
32	Distribution	1,130,638	MWh	@	-	-	-	-	-
33	Power Factor Adjustment					(4)		(5)	(1)
	Provisions								
34	Education GEI	3,157	MWh	@	(0.000201)	(1)	(0.000210)	(1)	(0)
35	Total Delivery					\$ 1,550		\$ 1,983	\$ 433
36	Total Full & ROA Service					\$ 85,161		\$ 84,520	\$ (640)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Present and Proposed Revenue Detail

Case No.: U-21389
Exhibit No.: A-16 (AMG-3)
Schedule: F-3.0
Page: 12 of 23
Witness: AMGast
Date: May 2023

Primary Demand Voltage Level 2 (GPD VL 2)

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	On-peak kW	898	MW	@	11.30	\$ 10,149	16.49	\$ 14,810	\$ 4,661
2	On-peak kWh	120,049	MWh	@	0.042669	5,122	0.044942	5,395	273
3	Off-peak kWh	352,713	MWh	@	0.026787	9,448	0.028710	10,126	678
4	Capacity On-peak kW	898	MW	@	9.08	8,155	1.96	1,760	(6,394)
	Winter (Oct. - May)								
	Non-capacity								
5	On-peak kW	1,847	MW	@	9.77	18,045	14.24	26,302	8,256
6	On-peak kWh	248,601	MWh	@	0.034091	8,475	0.034350	8,539	64
7	Off-peak kWh	727,305	MWh	@	0.029281	21,296	0.029935	21,772	476
8	Capacity On-peak kW	1,847	MW	@	8.42	15,552	1.82	3,362	(12,190)
	Provisions								
9	Economic Development GED Interruptible GI	-	MW	@	-	-	-	-	-
10	Summer Contracted kW	32	MW	@	(7.00)	(221)	(7.00)	(221)	-
11	Winter Contracted kW	63	MW	@	(6.00)	(379)	(6.00)	(379)	-
	Interruptible GI-2								
12	Summer Capacity & Transmission	-	MWh	@	0.038803	-	0.037025	-	-
13	Winter Capacity & Transmission	-	MWh	@	0.035411	-	0.033216	-	-
14	LMP	-	MWh	@	-	-	-	-	-
15	Power Factor Adjustment					(129)		(123)	6
16	Annual PSCR Factor	1,448,667	MWh	@	0.005970	8,649	0.005970	8,649	-
17	Total Production					\$ 104,162		\$ 99,991	\$ (4,171)
	Transmission								
	Summer (June - Sept.)								
18	On-peak kW	898	MW	@	7.49	\$ 6,727	8.04	\$ 7,221	\$ 494
	Winter (Oct. - May)								
19	On-peak kW	1,847	MW	@	6.97	12,874	7.49	13,834	960
20	Total Transmission					\$ 19,601		\$ 21,055	\$ 1,454
	Delivery								
21	System Access	701	Bills	@	200.00	\$ 140	200.00	\$ 140	\$ -
22	Maximum kW	3,342	MW	@	2.62	8,755	3.19	10,660	1,905
23	Substation Ownership	314	MW	@	(0.62)	(195)	(0.74)	(232)	(38)
24	Distribution	1,448,667	MWh	@	-	-	-	-	-
25	Power Factor Adjustment					(12)		(14)	(3)
	Provisions								
26	Education GEI	27,849	MWh	@	(0.000201)	(6)	(0.000210)	(6)	(0)
27	Total Delivery					\$ 8,683		\$ 10,547	\$ 1,864
28	Total Full Service					\$ 132,446		\$ 131,594	\$ (852)
	ROA SERVICE								
	Delivery								
29	System Access	450	Bills	@	200.00	\$ 90	200.00	\$ 90	\$ -
30	Maximum kW	2,822	MW	@	2.62	7,393	3.19	9,002	1,608
31	Substation Ownership	342	MW	@	(0.62)	(212)	(0.74)	(253)	(41)
32	Distribution	1,351,423	MWh	@	-	-	-	-	-
33	Power Factor Adjustment					15		18	3
	Provisions								
34	Education GEI	69,123	MWh	@	(0.000201)	(14)	(0.000210)	(15)	(1)
35	Total Delivery					\$ 7,272		\$ 8,842	\$ 1,570
36	Total Full & ROA Service					\$ 139,718		\$ 140,435	\$ 718

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Demand Voltage Level 3 (GPD VL 3)

Case No.: U-21389
Exhibit No.: A-16 (AMG-3)
Schedule: F-3.0
Page: 13 of 23
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	On-peak kW	1,367	MW	@	11.41	\$ 15,598	16.67	\$ 22,788	\$ 7,190
2	On-peak kWh	188,305	MWh	@	0.043061	8,109	0.045440	8,557	448
3	Off-peak kWh	525,139	MWh	@	0.027033	14,196	0.029028	15,244	1,048
4	Capacity On-peak kW	1,367	MW	@	9.20	12,576	1.99	2,720	(9,856)
	Winter (Oct. - May)								
	Non-capacity								
5	On-peak kW	2,641	MW	@	9.86	26,043	14.40	38,034	11,991
6	On-peak kWh	360,950	MWh	@	0.034405	12,418	0.034730	12,536	117
7	Off-peak kWh	1,024,209	MWh	@	0.029550	30,265	0.030267	31,000	734
8	Capacity On-peak kW	2,641	MW	@	8.53	22,530	1.84	4,860	(17,670)
	Provisions								
9	Economic Development GED	-	MW	@	-	-	-	-	-
	Interruptible GI								
10	Summer Contracted kW	5	MW	@	(7.00)	(36)	(7.00)	(36)	-
11	Winter Contracted kW	10	MW	@	(6.00)	(62)	(6.00)	(62)	-
	Interruptible GI-2								
12	Summer Capacity & Transmission	-	MWh	@	0.040422	-	0.037996	-	-
13	Winter Capacity & Transmission	-	MWh	@	0.038216	-	0.035105	-	-
14	LMP	-	MWh	@	-	-	-	-	-
15	Power Factor Adjustment					97		93	(4)
16	Annual PSCR Factor	2,098,603	MWh	@	0.005970	12,529	0.005970	12,529	-
17	Total Production					\$ 154,262		\$ 148,261	\$ (6,001)
	Transmission								
	Summer (June - Sept.)								
18	On-peak kW	1,367	MW	@	7.60	\$ 10,389	8.17	\$ 11,168	\$ 779
	Winter (Oct. - May)								
19	On-peak kW	2,641	MW	@	7.07	18,674	7.60	20,073	1,400
20	Total Transmission					\$ 29,063		\$ 31,242	\$ 2,179
	Delivery								
21	System Access	8,582	Bills	@	200.00	\$ 1,716	200.00	\$ 1,716	-
22	Maximum kW	5,352	MW	@	5.16	27,617	6.13	32,809	5,192
23	Distribution	2,098,603	MWh	@	-	-	-	-	-
24	Power Factor Adjustment					19		23	4
	Provisions								
25	Education GEI	107,908	MWh	@	(0.000201)	(22)	(0.000210)	(23)	(1)
26	Total Delivery					\$ 29,331		\$ 34,525	\$ 5,194
27	Total Full Service					\$ 212,656		\$ 214,029	\$ 1,372
	ROA SERVICE								
	Delivery								
28	System Access	3,439	Bills	@	200.00	\$ 688	200.00	\$ 688	-
29	Maximum kW	2,617	MW	@	5.16	13,502	6.13	16,041	2,538
30	Distribution	960,761	MWh	@	-	-	-	-	-
31	Power Factor Adjustment					30		35	6
	Provisions								
32	Education GEI	153,597	MWh	@	(0.000201)	(31)	(0.000210)	(32)	(1)
33	Total Delivery					\$ 14,189		\$ 16,731	\$ 2,542
34	Total Full & ROA Service					\$ 226,845		\$ 230,760	\$ 3,915

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Time-of-Use Voltage Level 1 (GPTU VL 1)

Case No.: U-21389
 Exhibit No.: A-16 (AMG-3)
 Schedule: F-3.0
 Page: 14 of 23
 Witness: AMGast
 Date: May 2023

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	High-peak kWh	12,200	MWh	@	0.079435	\$ 969	0.096813	\$ 1,181	\$ 212
2	Mid-peak kWh	16,297	MWh	@	0.065988	1,075	0.082384	1,343	267
3	Low-peak kWh	39,282	MWh	@	0.049204	1,933	0.061403	2,412	479
4	Off-peak kWh	85,166	MWh	@	0.041890	3,568	0.052761	4,493	926
	Capacity								
5	High-peak kWh	12,200	MWh	@	0.028718	350	0.006600	81	(270)
6	Mid-peak kWh	16,297	MWh	@	0.027414	447	0.006301	103	(344)
7	Low-peak kWh	39,282	MWh	@	0.022017	865	0.005060	199	(666)
8	Off-peak kWh	85,166	MWh	@	0.014872	1,267	0.003418	291	(975)
	Winter (Oct. - May)								
	Non-capacity								
9	High-peak kWh	23,301	MWh	@	0.059055	1,376	0.069509	1,620	244
10	Mid-peak kWh	30,706	MWh	@	0.053930	1,656	0.064526	1,981	325
11	Off-peak kWh	190,807	MWh	@	0.048735	9,299	0.058218	11,108	1,809
	Capacity								
12	High-peak kWh	23,301	MWh	@	0.018424	429	0.004235	99	(331)
13	Mid-peak kWh	30,706	MWh	@	0.018416	565	0.004233	130	(436)
14	Off-peak kWh	190,807	MWh	@	0.015850	3,024	0.003643	695	(2,329)
15	Power Factor Adjustment					43		42	(2)
	Provisions								
	Interruptible GI								
16	Summer Contracted kW	-	MW	@	-	-	(7.00)	-	-
17	Winter Contracted kW	-	MW	@	-	-	(6.00)	-	-
18	Annual PSCR Factor	397,758	MWh	@	0.005970	2,375	0.005970	2,375	-
19	Total Production					\$ 29,242		\$ 28,152	\$ (1,090)
	Transmission								
	Summer (June - Sept.)								
20	High-peak kWh	12,200	MWh	@	0.023290	\$ 284	0.025348	\$ 309	\$ 25
21	Mid-peak kWh	16,297	MWh	@	0.022233	362	0.024198	394	32
22	Low-peak kWh	39,282	MWh	@	0.017856	701	0.019434	763	62
23	Off-peak kWh	85,166	MWh	@	0.012061	1,027	0.013127	1,118	91
24	Winter (Oct. - May)								
25	High-peak kWh	23,301	MWh	@	0.014942	348	0.016263	379	31
26	Mid-peak kWh	30,706	MWh	@	0.014935	459	0.016256	499	41
27	Off-peak kWh	190,807	MWh	@	0.012854	2,453	0.013991	2,670	217
28	Total Transmission					\$ 5,634		\$ 6,133	\$ 498
	Delivery								
29	System Access	227	Bills	@	200.00	\$ 45	200.00	\$ 45	\$ -
30	Maximum kW	1,164	MW	@	0.74	861	0.95	1,106	244
31	Substation Ownership	207	MW	@	(0.46)	(95)	(0.58)	(120)	(25)
32	Distribution	397,758	MWh	@	-	-	-	-	-
33	Power Factor Adjustment					1		2	0
	Provisions								
34	Education GEI	10,473	MWh	@	(0.000201)	(2)	(0.000210)	(2)	(0)
35	Total Delivery					\$ 811		\$ 1,031	\$ 220
36	Total Full Service					\$ 35,687		\$ 35,315	\$ (372)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Time-of-Use Voltage Level 2 (GPTU VL 2)

Case No.: U-21389
Exhibit No.: A-16 (AMG-3)
Schedule: F-3.0
Page: 15 of 23
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	High-peak kWh	15,606	MWh	@	0.080388	\$ 1,255	0.097936	\$ 1,528	\$ 274
2	Mid-peak kWh	20,947	MWh	@	0.066780	1,399	0.083340	1,746	347
3	Low-peak kWh	56,128	MWh	@	0.049794	2,795	0.062115	3,486	692
4	Off-peak kWh	109,754	MWh	@	0.042393	4,653	0.053373	5,858	1,205
	Capacity								
5	High-peak kWh	15,606	MWh	@	0.029155	455	0.006696	104	(350)
6	Mid-peak kWh	20,947	MWh	@	0.027831	583	0.006392	134	(449)
7	Low-peak kWh	56,128	MWh	@	0.022352	1,255	0.005133	288	(966)
8	Off-peak kWh	109,754	MWh	@	0.015098	1,657	0.003468	381	(1,276)
	Winter (Oct. - May)								
	Non-capacity								
9	High-peak kWh	36,356	MWh	@	0.059764	2,173	0.070315	2,556	384
10	Mid-peak kWh	49,188	MWh	@	0.054577	2,685	0.065275	3,211	526
11	Off-peak kWh	286,729	MWh	@	0.049320	14,141	0.058893	16,886	2,745
	Capacity								
12	High-peak kWh	36,356	MWh	@	0.018704	680	0.004296	156	(524)
13	Mid-peak kWh	49,188	MWh	@	0.018696	920	0.004294	211	(708)
14	Off-peak kWh	286,729	MWh	@	0.016091	4,614	0.003696	1,060	(3,554)
15	Power Factor Adjustment					(17)		(16)	1
	Provisions								
	Interruptible GI								
16	Summer Contracted kW	-	MW	@	-	-	(7.00)	-	-
17	Winter Contracted kW	-	MW	@	-	-	(6.00)	-	-
18	Annual PSCR Factor	574,708	MWh	@	0.005970	3,431	0.005970	3,431	-
19	Total Production					\$ 42,677		\$ 41,021	\$ (1,656)
	Transmission								
	Summer (June - Sept.)								
20	High-peak kWh	15,606	MWh	@	0.023644	\$ 369	0.025716	\$ 401	\$ 32
21	Mid-peak kWh	20,947	MWh	@	0.022571	473	0.024549	514	41
22	Low-peak kWh	56,128	MWh	@	0.018127	1,017	0.019716	1,107	89
23	Off-peak kWh	109,754	MWh	@	0.012244	1,344	0.013317	1,462	118
	Winter (Oct. - May)								
24	High-peak kWh	36,356	MWh	@	0.015169	551	0.016499	600	48
25	Mid-peak kWh	49,188	MWh	@	0.015162	746	0.016492	811	65
26	Off-peak kWh	286,729	MWh	@	0.013049	3,742	0.014194	4,070	328
27	Total Transmission					\$ 8,242		\$ 8,965	\$ 723
	Delivery								
28	System Access	492	Bills	@	200.00	\$ 98	200.00	\$ 98	\$ -
29	Maximum kW	1,610	MW	@	2.62	4,218	3.19	5,135	918
30	Substation Ownership	261	MW	@	(0.62)	(162)	(0.74)	(193)	(31)
31	Distribution	574,708	MWh	@	-	-	-	-	-
32	Power Factor Adjustment					(2)		(2)	(0)
	Provisions								
33	Education GEI	16,903	MWh	@	(0.000201)	(3)	(0.000210)	(4)	(0)
34	Total Delivery					\$ 4,149		\$ 5,035	\$ 886
35	Total Full Service					\$ 55,068		\$ 55,021	\$ (47)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Time-of-Use Voltage Level 3 (GPTU VL 3)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 16 of 23

Witness: AMGast

Date: May 2023

(a)		(b)		(c)		(d)		(e)		(f)		(g)		(h)
Line		Billing Determinants			Present			Proposed			Net Increase			
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	Rate	Revenue	/ (Decrease)			
					\$/unit	\$000	\$/unit	\$000	\$/unit	\$000	\$000			
FULL SERVICE														
Production														
Summer (June - Sept.)														
Non-capacity														
1	High-peak kWh	129,064	MWh	@	0.081127	\$ 10,471	0.099020	\$ 12,780	\$		\$		2,309	
2	Mid-peak kWh	169,998	MWh	@	0.067394	11,457	0.084262	14,324					2,868	
3	Low-peak kWh	393,095	MWh	@	0.050252	19,754	0.062803	24,688					4,934	
4	Off-peak kWh	700,656	MWh	@	0.042782	29,975	0.053964	37,810					7,835	
Capacity														
5	High-peak kWh	129,064	MWh	@	0.029557	3,815	0.006801	878					(2,937)	
6	Mid-peak kWh	169,998	MWh	@	0.028214	4,796	0.006493	1,104					(3,693)	
7	Low-peak kWh	393,095	MWh	@	0.022660	8,908	0.005214	2,050					(6,858)	
8	Off-peak kWh	700,656	MWh	@	0.015306	10,724	0.003522	2,468					(8,257)	
Winter (Oct. - May)														
Non-capacity														
9	High-peak kWh	236,344	MWh	@	0.060313	14,255	0.071094	16,803					2,548	
10	Mid-peak kWh	322,752	MWh	@	0.055079	17,777	0.065997	21,301					3,524	
11	Off-peak kWh	1,757,484	MWh	@	0.049773	87,475	0.059545	104,649					17,174	
Capacity														
12	High-peak kWh	236,344	MWh	@	0.018962	4,482	0.004364	1,031					(3,450)	
13	Mid-peak kWh	322,752	MWh	@	0.018954	6,117	0.004362	1,408					(4,710)	
14	Off-peak kWh	1,757,484	MWh	@	0.016313	28,670	0.003754	6,598					(22,072)	
15	Power Factor Adjustment					608		583					(25)	
Provisions														
Interruptible GI														
16	Summer Contracted kW	-	MW	@	-	-	(7.00)	-					-	
17	Winter Contracted kW	-	MW	@	-	-	(6.00)	-					-	
18	Annual PSCR Factor	3,709,392	MWh	@	0.005970	22,145	0.005970	22,145					-	
19	Total Production					\$ 281,428		\$ 270,618			\$		(10,810)	
Transmission														
Summer (June - Sept.)														
20	High-peak kWh	129,064	MWh	@	0.023970	\$ 3,094	0.026119	\$ 3,371	\$		\$		277	
21	Mid-peak kWh	169,998	MWh	@	0.022882	3,890	0.024934	4,239					349	
22	Low-peak kWh	393,095	MWh	@	0.018377	7,224	0.020025	7,872					648	
23	Off-peak kWh	700,656	MWh	@	0.012413	8,697	0.013526	9,477					780	
Winter (Oct. - May)														
24	High-peak kWh	236,344	MWh	@	0.015378	3,634	0.016757	3,960					326	
25	Mid-peak kWh	322,752	MWh	@	0.015371	4,961	0.016750	5,406					445	
26	Off-peak kWh	1,757,484	MWh	@	0.013229	23,250	0.014416	25,336					2,086	
27	Total Transmission					\$ 54,750		\$ 59,661			\$		4,911	
Delivery														
28	System Access	15,656	Bills	@	200.00	\$ 3,131	200.00	\$ 3,131	\$		\$		-	
29	Maximum kW	11,364	MW	@	5.16	58,639	6.13	69,663					11,023	
30	Distribution	3,709,392	MWh	@	-	-	-	-					-	
31	Power Factor Adjustment					138		164					26	
Provisions														
32	Education GEI	266,636	MWh	@	(0.000201)	(54)	(0.000210)	(56)					(2)	
33	Total Delivery					\$ 61,855		\$ 72,901			\$		11,047	
34	Total Full Service					\$ 398,033		\$ 403,180			\$		5,148	

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy Intensive Level 1 (EIP VL 1)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 17 of 23

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	Critical-peak kWh	1,333	MWh	@	0.106778	\$ 142	0.154881	\$ 206	\$ 64
2	High-peak kWh	5,312	MWh	@	0.071185	378	0.103254	548	170
3	Mid-peak kWh	6,635	MWh	@	0.059191	393	0.088722	589	196
4	Low-peak kWh	44,036	MWh	@	0.045119	1,987	0.067797	2,985	999
5	Off-peak kWh	89,821	MWh	@	0.035526	3,191	0.053839	4,836	1,645
	Capacity								
6	Emergency kWh	-	MWh	@	1.000000	-	1.000000	-	-
7	Critical-peak kWh	1,333	MWh	@	0.014969	20	0.007776	10	(10)
8	High-peak kWh	5,312	MWh	@	0.009979	53	0.005184	28	(25)
9	Mid-peak kWh	6,635	MWh	@	0.009754	65	0.005067	34	(31)
10	Low-peak kWh	44,036	MWh	@	0.008023	353	0.004168	184	(170)
11	Off-peak kWh	89,821	MWh	@	0.005134	461	0.002667	240	(222)
	Winter (Oct. - May)								
	Non-capacity								
12	Critical-peak kWh	1,063	MWh	@	0.077160	82	0.109554	116	34
13	High-peak kWh	13,757	MWh	@	0.051440	708	0.073036	1,005	297
14	Mid-peak kWh	14,749	MWh	@	0.048036	708	0.069404	1,024	315
15	Off-peak kWh	197,697	MWh	@	0.041715	8,247	0.059896	11,841	3,594
	Capacity								
16	Emergency kWh	-	MWh	@	1.000000	-	1.000000	-	-
17	Critical-peak kWh	1,063	MWh	@	0.009690	10	0.005034	5	(5)
18	High-peak kWh	13,757	MWh	@	0.006460	89	0.003356	46	(43)
19	Mid-peak kWh	14,749	MWh	@	0.006373	94	0.003310	49	(45)
20	Off-peak kWh	197,697	MWh	@	0.005585	1,104	0.002901	574	(531)
21	Power Factor Adjustment					7		9	2
22	Interruptible Credit	374,401	MWh	@	-	-	(0.013595)	(5,090)	(5,090)
23	Annual PSCR Factor	374,401	MWh	@	0.005970	2,235	0.005970	2,235	-
24	Total Production					\$ 20,327		\$ 21,474	\$ 1,146
	Transmission								
	Summer (June - Sept.)								
25	Critical-peak kWh	1,333	MWh	@	0.030929	\$ 41	0.035492	\$ 47	\$ 6
26	High-peak kWh	5,312	MWh	@	0.020619	110	0.023661	126	16
27	Mid-peak kWh	6,635	MWh	@	0.020155	134	0.023128	153	20
28	Low-peak kWh	44,036	MWh	@	0.016576	730	0.019022	838	108
29	Off-peak kWh	89,821	MWh	@	0.010608	953	0.012173	1,093	141
	Winter (Oct. - May)								
30	Critical-peak kWh	1,063	MWh	@	0.020022	21	0.022976	24	3
31	High-peak kWh	13,757	MWh	@	0.013348	184	0.015317	211	27
32	Mid-peak kWh	14,749	MWh	@	0.013169	194	0.015111	223	29
33	Off-peak kWh	197,697	MWh	@	0.011540	2,281	0.013242	2,618	336
34	Total Transmission					\$ 4,648		\$ 5,333	\$ 686
	Delivery								
35	System Access	66	Bills	@	200.00	\$ 13	200.00	\$ 13	\$ -
36	Maximum kW	1,163	MW	@	0.74	861	0.95	1,105	244
37	Substation Ownership	-	MW	@	(0.46)	-	(0.58)	-	-
38	Power Factor Adjustment					0		0	0
39	Distribution	374,401	MWh	@	-	-	-	-	-
40	Total Delivery					\$ 874		\$ 1,119	\$ 244
41	Total Full Service					\$ 25,850		\$ 27,926	\$ 2,076

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy Intensive Level 2 (EIP VL 2)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 18 of 23

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	Critical-peak kWh	178	MWh	@	0.108059	\$ 19	0.156678	\$ 28	\$ 9
2	High-peak kWh	703	MWh	@	0.072039	51	0.104452	73	23
3	Mid-peak kWh	908	MWh	@	0.059901	54	0.089751	82	27
4	Low-peak kWh	7,354	MWh	@	0.045660	336	0.068583	504	169
5	Off-peak kWh	15,035	MWh	@	0.035952	541	0.054464	819	278
	Capacity								
6	Emergency kWh	-	MWh	@	1.000000	-	1.000000	-	-
7	Critical-peak kWh	178	MWh	@	0.015197	3	0.007889	1	(1)
8	High-peak kWh	703	MWh	@	0.010131	7	0.005259	4	(3)
9	Mid-peak kWh	908	MWh	@	0.009902	9	0.005140	5	(4)
10	Low-peak kWh	7,354	MWh	@	0.008145	60	0.004228	31	(29)
11	Off-peak kWh	15,035	MWh	@	0.005212	78	0.002706	41	(38)
	Winter (Oct. - May)								
	Non-capacity								
12	Critical-peak kWh	164	MWh	@	0.078086	13	0.110825	18	5
13	High-peak kWh	1,754	MWh	@	0.052057	91	0.073883	130	38
14	Mid-peak kWh	1,913	MWh	@	0.048612	93	0.070209	134	41
15	Off-peak kWh	35,995	MWh	@	0.042216	1,520	0.060591	2,181	661
	Capacity								
16	Emergency kWh	-	MWh	@	1.000000	-	1.000000	-	-
17	Critical-peak kWh	164	MWh	@	0.009837	2	0.005107	1	(1)
18	High-peak kWh	1,754	MWh	@	0.006558	12	0.003405	6	(6)
19	Mid-peak kWh	1,913	MWh	@	0.006470	12	0.003358	6	(6)
20	Off-peak kWh	35,995	MWh	@	0.005670	204	0.002943	106	(98)
21	Power Factor Adjustment					(15)		(21)	(5)
22	Interruptible Credit	64,003	MWh	@	-	-	(0.013595)	(870)	(870)
23	Annual PSCR Factor	64,003	MWh	@	0.005970	382	0.005970	382	-
24	Total Production					\$ 3,471		\$ 3,661	\$ 190
	Transmission								
	Summer (June - Sept.)								
25	Critical-peak kWh	178	MWh	@	0.031399	\$ 6	0.036007	\$ 6	\$ 1
26	High-peak kWh	703	MWh	@	0.020932	15	0.024004	17	2
27	Mid-peak kWh	908	MWh	@	0.020461	19	0.023463	21	3
28	Low-peak kWh	7,354	MWh	@	0.016828	124	0.019298	142	18
29	Off-peak kWh	15,035	MWh	@	0.010769	162	0.012350	186	24
	Winter (Oct. - May)								
30	Critical-peak kWh	164	MWh	@	0.020326	3	0.023309	4	0
31	High-peak kWh	1,754	MWh	@	0.013551	24	0.015539	27	3
32	Mid-peak kWh	1,913	MWh	@	0.013369	26	0.015330	29	4
33	Off-peak kWh	35,995	MWh	@	0.011715	422	0.013434	484	62
34	Total Transmission					\$ 799		\$ 916	\$ 117
	Delivery								
35	System Access	80	Bills	@	200.00	\$ 16	200.00	\$ 16	\$ -
36	Maximum kW	467	MW	@	2.62	1,224	3.19	1,491	266
37	Substation Ownership	187	MW	@	(0.62)	(116)	(0.74)	(138)	(22)
38	Power Factor Adjustment					(6)		(7)	(1)
39	Distribution	64,003	MWh	@	-	-	-	-	-
40	Total Delivery					\$ 1,119		\$ 1,362	\$ 243
41	Total Full Service					\$ 5,389		\$ 5,939	\$ 550

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy Intensive Level 3 (EIP VL 3)

Case No.: U-21389
Exhibit No.: A-16 (AMG-3)
Schedule: F-3.0
Page: 19 of 23
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	Critical-peak kWh	19	MWh	@	0.109052	\$ 2	0.158412	\$ 3	\$ 1
2	High-peak kWh	93	MWh	@	0.072701	7	0.105608	10	3
3	Mid-peak kWh	158	MWh	@	0.060452	10	0.090745	14	5
4	Low-peak kWh	1,679	MWh	@	0.046080	77	0.069343	116	39
5	Off-peak kWh	2,131	MWh	@	0.036283	77	0.055067	117	40
	Capacity								
6	Emergency kWh	-	MWh	@	1.000000	-	1.000000	-	-
7	Critical-peak kWh	19	MWh	@	0.015406	0	0.008012	0	(0)
8	High-peak kWh	93	MWh	@	0.010270	1	0.005342	0	(0)
9	Mid-peak kWh	158	MWh	@	0.010039	2	0.005221	1	(1)
10	Low-peak kWh	1,679	MWh	@	0.008257	14	0.004295	7	(7)
11	Off-peak kWh	2,131	MWh	@	0.005284	11	0.002748	6	(5)
	Winter (Oct. - May)								
	Non-capacity								
12	Critical-peak kWh	9	MWh	@	0.078804	1	0.112052	1	0
13	High-peak kWh	132	MWh	@	0.052536	7	0.074701	10	3
14	Mid-peak kWh	196	MWh	@	0.049059	10	0.070986	14	4
15	Off-peak kWh	4,797	MWh	@	0.042604	204	0.061262	294	90
	Capacity								
16	Emergency kWh	-	MWh	@	1.000000	-	1.000000	-	-
17	Critical-peak kWh	9	MWh	@	0.009973	0	0.005187	0	(0)
18	High-peak kWh	132	MWh	@	0.006649	1	0.003458	0	(0)
19	Mid-peak kWh	196	MWh	@	0.006559	1	0.003411	1	(1)
20	Off-peak kWh	4,797	MWh	@	0.005748	28	0.002989	14	(13)
21	Power Factor Adjustment					(1)		(1)	(0)
22	Interruptible Credit	9,214	MWh	@	-	-	(0.013595)	(125)	(125)
23	Annual PSCR Factor	9,214	MWh	@	0.005970	55	0.005970	55	-
24	Total Production					\$ 507		\$ 538	\$ 32
	Transmission								
	Summer (June - Sept.)								
25	Critical-peak kWh	19	MWh	@	0.031832	\$ 1	0.036571	\$ 1	\$ 0
26	High-peak kWh	93	MWh	@	0.021221	2	0.024380	2	0
27	Mid-peak kWh	158	MWh	@	0.020744	3	0.023831	4	0
28	Low-peak kWh	1,679	MWh	@	0.017060	29	0.019600	33	4
29	Off-peak kWh	2,131	MWh	@	0.010918	23	0.012543	27	3
	Winter (Oct. - May)								
30	Critical-peak kWh	9	MWh	@	0.020607	0	0.023674	0	0
31	High-peak kWh	132	MWh	@	0.013738	2	0.015783	2	0
32	Mid-peak kWh	196	MWh	@	0.013554	3	0.015570	3	0
33	Off-peak kWh	4,797	MWh	@	0.011877	57	0.013645	65	8
34	Total Transmission					\$ 119		\$ 137	\$ 18
	Delivery								
35	System Access	53	Bills	@	200.00	\$ 11	200.00	\$ 11	\$ -
36	Maximum kW	59	MW	@	5.16	303	6.13	360	57
37	Power Factor Adjustment					(1)		(1)	(0)
38	Distribution	9,214	MWh	@	-	-	-	-	-
39	Total Delivery					\$ 313		\$ 369	\$ 57
40	Total Full Service					\$ 939		\$ 1,045	\$ 106

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Metered Lighting (GML)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 20 of 23

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants		Present		Proposed		Net Increase
No.	Description	Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)
				\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE							
	Production							
1	Secondary All kWh	11,686	MWh @	0.045428	\$ 531	0.052717	\$ 616	\$ 85
2	Primary All kWh	509	MWh @	0.022294	11	0.025871	13	2
3	Annual PSCR Factor	12,195	MWh @	0.005970	73	0.005970	73	-
4	Total Production				\$ 615		\$ 702	\$ 87
	Transmission							
5	Secondary All kWh	11,686	MWh @	0.008754	102	0.009919	116	14
6	Primary All kWh	509	MWh @	0.004296	2	0.004868	2	0
7	Total Transmission	12,195			104		118	14
	Delivery							
	<u>Secondary</u>							
8	System Access	4,739	Bills @	10.00	\$ 47	10.00	\$ 47	\$ -
9	Distribution	11,686	MWh @	0.064240	751	0.084486	987	237
	<u>Primary</u>							
10	System Access	110	Bills @	20.00	\$ 2	20.00	\$ 2	\$ -
11	Distribution	509	MWh @	0.048956	25	0.064385	33	8
12	Total Delivery				\$ 825		\$ 1,070	\$ 244
13	Total Full Service				\$ 1,545		\$ 1,890	\$ 345

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Universal Unmetered Light (UUL)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 21 of 23

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
Line	Billing Determinants							Present					Proposed					Net Increase
No.	Description	Customer	Company	Units	Watts	Customer MWh	Company MWh	Production	Delivery	Production	Delivery	Revenue	Production	Delivery	Production	Delivery	Revenue	Net Increase /(Decrease)
								\$/Light	\$/Light	\$/Light	\$/Light	\$000	\$/Light	\$/Light	\$/Light	\$/Light	\$000	\$000
1	15 - 24 Watts	12	-	Lights	20	0	- @	0.35	5.67	0.35	8.67	\$ 0	0.41	7.63	0.41	10.63	\$ 0	\$ 0
2	25 - 34 Watts	-	-	Lights	30	-	- @	0.53	6.11	0.53	9.11	-	0.61	8.28	0.61	11.28	-	-
3	35 - 44 Watts	1,524	31,068	Lights	40	21	430 @	0.71	6.55	0.71	9.55	330	0.82	8.92	0.82	11.92	411	81
4	45 - 54 Watts	1,308	922,740	Lights	50	23	15,986 @	0.89	7.00	0.89	10.00	10,059	1.03	9.57	1.03	12.57	12,563	2,504
5	55 - 64 Watts	1,092	72	Lights	60	23	1 @	1.07	7.44	1.07	10.44	10	1.24	10.21	1.24	13.21	14	3
6	65 - 74 Watts	312	124,344	Lights	70	8	3,025 @	1.25	7.88	1.25	10.88	1,511	1.44	10.86	1.44	13.86	1,906	395
7	75 - 84 Watts	72	2,412	Lights	80	2	67 @	1.43	8.32	1.43	11.32	31	1.65	11.51	1.65	14.51	40	8
8	85 - 94 Watts	-	120,000	Lights	90	-	3,759 @	1.61	8.76	1.61	11.76	1,604	1.86	12.15	1.86	15.15	2,041	437
9	95 - 104 Watts	-	180	Lights	100	-	6 @	1.78	9.20	1.78	12.20	3	2.07	12.80	2.07	15.80	3	1
10	105 - 114 Watts	-	7,092	Lights	110	-	272 @	1.96	9.64	1.96	12.64	104	2.27	13.44	2.27	16.44	133	29
11	115 - 124 Watts	804	648,348	Lights	120	34	27,117 @	2.14	10.08	2.14	13.08	9,878	2.48	14.09	2.48	17.09	12,701	2,824
12	125 - 134 Watts	-	276	Lights	130	-	13 @	2.32	10.53	2.32	13.53	4	2.69	14.73	2.69	17.73	6	1
13	135 - 144 Watts	-	-	Lights	140	-	- @	2.50	10.97	2.50	13.97	-	2.90	15.38	2.90	18.38	-	-
14	145 - 154 Watts	-	84	Lights	150	-	4 @	2.68	11.41	2.68	14.41	1	3.11	16.03	3.11	19.03	2	0
15	155 - 164 Watts	-	-	Lights	160	-	- @	2.86	11.85	2.86	14.85	-	3.31	16.67	3.31	19.67	-	-
16	165 - 174 Watts	1,404	94,668	Lights	170	83	5,616 @	3.04	12.29	3.04	15.29	1,757	3.52	17.32	3.52	20.32	2,286	529
17	175 - 184 Watts	-	-	Lights	180	-	- @	3.22	12.73	3.22	15.73	-	3.73	17.96	3.73	20.96	-	-
18	185 - 194 Watts	-	156	Lights	190	-	10 @	3.40	13.17	3.40	16.17	3	3.94	18.61	3.94	21.61	4	1
19	195 - 204 Watts	-	48	Lights	200	-	3 @	3.58	13.62	3.58	16.62	1	4.14	19.26	4.14	22.26	1	0
20	205 - 214 Watts	504	12,588	Lights	210	37	923 @	3.76	14.06	3.76	17.06	271	4.35	19.90	4.35	22.90	355	84
21	215 - 224 Watts	-	-	Lights	220	-	- @	3.94	14.50	3.94	17.50	-	4.56	20.55	4.56	23.55	-	-
22	225 - 234 Watts	-	-	Lights	230	-	- @	4.12	14.94	4.12	17.94	-	4.77	21.19	4.77	24.19	-	-
23	235 - 244 Watts	-	-	Lights	240	-	- @	4.30	15.38	4.30	18.38	-	4.98	21.84	4.98	24.84	-	-
24	245 - 254 Watts	60	204	Lights	250	5	18 @	4.48	15.82	4.48	18.82	6	5.18	22.49	5.18	25.49	8	2
25	255 - 264 Watts	-	-	Lights	260	-	- @	4.65	16.26	4.65	19.26	-	5.39	23.13	5.39	26.13	-	-
26	265 - 274 Watts	-	-	Lights	270	-	- @	4.83	16.71	4.83	19.71	-	5.60	23.78	5.60	26.78	-	-
27	275 - 284 Watts	432	3,756	Lights	280	42	367 @	5.01	17.15	5.01	20.15	104	5.81	24.42	5.81	27.42	138	34
28	285 - 294 Watts	12	1,020	Lights	290	1	103 @	5.19	17.59	5.19	20.59	27	6.01	25.07	6.01	28.07	35	9
29	295 - 304 Watts	-	-	Lights	300	-	- @	5.37	18.03	5.37	21.03	-	6.22	25.72	6.22	28.72	-	-
30	305 - 314 Watts	108	-	Lights	310	12	- @	5.55	18.47	5.55	21.47	3	6.43	26.36	6.43	29.36	4	1
31	315 - 324 Watts	540	65,916	Lights	320	60	7,371 @	5.73	18.91	5.73	21.91	1,835	6.64	27.01	6.64	30.01	2,434	599
32	325 - 334 Watts	-	-	Lights	330	-	- @	5.91	19.35	5.91	22.35	-	6.85	27.65	6.85	30.65	-	-
33	335 - 344 Watts	-	-	Lights	340	-	- @	6.09	19.80	6.09	22.80	-	7.05	28.30	7.05	31.30	-	-
34	345 - 354 Watts	-	-	Lights	350	-	- @	6.27	20.24	6.27	23.24	-	7.26	28.94	7.26	31.94	-	-
35	355 - 364 Watts	-	-	Lights	360	-	- @	6.45	20.68	6.45	23.68	-	7.47	29.59	7.47	32.59	-	-
36	365 - 374 Watts	-	-	Lights	370	-	- @	6.63	21.12	6.63	24.12	-	7.68	30.24	7.68	33.24	-	-
37	375 - 384 Watts	-	-	Lights	380	-	- @	6.81	21.56	6.81	24.56	-	7.88	30.88	7.88	33.88	-	-
38	385 - 394 Watts	-	-	Lights	390	-	- @	6.99	22.00	6.99	25.00	-	8.09	31.53	8.09	34.53	-	-
39	395 - 404 Watts	-	-	Lights	400	-	- @	7.17	22.44	7.17	25.44	-	8.30	32.17	8.30	35.17	-	-
40	405 - 414 Watts	24	-	Lights	410	3	- @	7.35	22.89	7.35	25.89	1	8.51	32.82	8.51	35.82	1	0
41	415 - 424 Watts	-	-	Lights	420	-	- @	7.52	23.33	7.52	26.33	-	8.72	33.47	8.72	36.47	-	-
42	425 - 434 Watts	-	-	Lights	430	-	- @	7.70	23.77	7.70	26.77	-	8.92	34.11	8.92	37.11	-	-
43	435 - 444 Watts	-	-	Lights	440	-	- @	7.88	24.21	7.88	27.21	-	9.13	34.76	9.13	37.76	-	-
44	445 - 454 Watts	-	-	Lights	450	-	- @	8.06	24.65	8.06	27.65	-	9.34	35.40	9.34	38.40	-	-
45	455 - 464 Watts	852	1,212	Lights	460	137	195 @	8.24	25.09	8.24	28.09	72	9.55	36.05	9.55	39.05	98	25
46	465 - 474 Watts	-	-	Lights	470	-	- @	8.42	25.53	8.42	28.53	-	9.75	36.70	9.75	39.70	-	-
47	475 - 484 Watts	180	35,928	Lights	480	30	6,030 @	8.60	25.98	8.60	28.98	1,356	9.96	37.34	9.96	40.34	1,816	459
48	Conversion Credit	-	129,840	Lights							(6.04)	(784)				(8.35)	(1,084)	(300)
49	Annual PSCR Factor	9,240	2,072,112	Lights		521	71,317	0.005970		0.005970		429	0.005970		0.005970		429	-
50	Total Unmetered LED	9,240	2,072,112	Lights		521	71,317					\$ 28,616					\$ 36,344	\$ 7,728

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Unmetered Service (GU)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 22 of 23

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line	Description	Billing Determinants		Present		Proposed		Net Increase
No.		Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)
				\$/unit	\$000	\$/unit	\$000	\$000
	FULL SERVICE							
	Production							
1	Non-capacity All kWh	93,426	MWh @	0.051503	\$ 4,812	0.061382	\$ 5,735	\$ 923
2	Capacity All kWh	93,426	MWh @	0.013565	1,267	0.002887	270	(998)
3	Annual PSCR Factor	93,426	MWh @	0.005970	558	0.005970	558	-
4	Total Production				\$ 6,637		\$ 6,562	\$ (75)
	Transmission							
5	All kWh	93,426	MWh @	0.012151	1,135	0.013147	1,228	93
	Delivery							
6	System Access	5,536	Bills @	2.00	\$ 11	2.00	\$ 11	\$ -
7	Distribution	93,426	MWh @	0.022612	2,113	0.027235	2,544	432
8	Total Delivery				\$ 2,124		\$ 2,556	\$ 432
9	Total Full Service				\$ 9,896		\$ 10,346	\$ 450

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Large Self-generation (GSG-2)

Case No.: U-21389

Exhibit No.: A-16 (AMG-3)

Schedule: F-3.0

Page: 23 of 23

Witness: AMGast

Date: May 2023

(a)		(b)		(c)		(d)		(e)		(f)		(g)		(h)	
Line		Billing Determinants			Present			Proposed			Net Increase				
No.	Description	Quantity	Units		Rate	Revenue		Rate	Revenue		Rate	Revenue		/ (Decrease)	
					\$/unit	\$000		\$/unit	\$000		\$/unit	\$000		\$000	
STANDBY SERVICE															
Production															
Voltage Level 1															
1	On-peak kW-day	158	MW	@	11.33	\$ 1,785		11.33	\$ 1,785		\$		\$	-	
2	All kWh	46,664	MWh	@	0.047444	2,214		0.047444	2,214					-	
Voltage Level 2															
3	On-peak kW-day	157	MW	@	11.49	1,800		11.49	1,800					-	
4	All kWh	54,438	MWh	@	0.066009	3,593		0.066009	3,593					-	
Voltage Level 3															
5	On-peak kW-day	9	MW	@	11.67	110		11.67	110					-	
6	All kWh	592	MWh	@	0.048878	29		0.048878	29					-	
7	Power Factor Adjustment					129			129					-	
8	Total Production					\$ 9,661			\$ 9,661		\$		\$	-	
Transmission															
Voltage Level 1															
9	On-peak kW-day	158	MW	@	2.65	\$ 418		2.65	\$ 418		\$		\$	-	
Voltage Level 2															
10	On-peak kW-day	157	MW	@	2.69	421		2.69	421					-	
Voltage Level 3															
11	On-peak kW-day	9	MW	@	2.73	26		2.73	26					-	
12	Total Transmission					\$ 865			\$ 865		\$		\$	-	
Delivery															
System Access															
Voltage Level 1															
13	Standby	48	Bills	@	200.00	\$ 10		100.00	\$ 5		\$		\$	(5)	
14	Supplemental	36	Bills	@	100.00	4		100.00	4					-	
15	Maximum kW	839	MW	@	0.74	621		0.95	797					176	
16	Interconnect	355	MW	@	(0.74)	(263)		(0.95)	(337)					(75)	
17	Substation Ownership	512	MW	@	(0.46)	(236)		(0.58)	(297)					(61)	
Voltage Level 2															
18	Standby	24	Bills	@	200.00	\$ 5		100.00	\$ 2		\$		\$	(2)	
19	Supplemental	60	Bills	@	100.00	6		100.00	6					-	
20	Maximum kW	599	MW	@	2.62	1,570		3.19	1,912					342	
21	Substation Ownership	200	MW	@	(0.62)	(124)		(0.74)	(148)					(24)	
Voltage Level 3															
22	Standby	-	Bills	@	200.00	\$ -		100.00	\$ -		\$		\$	-	
23	Supplemental	12	Bills	@	100.00	1		100.00	1					-	
24	Maximum kW	14	MW	@	5.16	72		6.13	86					14	
25	Power Factor Adjustment					22			26					5	
26	Total Delivery					\$ 1,687			\$ 2,056		\$		\$	369	
27	Total Standby Service					\$ 12,214			\$ 12,582		\$		\$	369	

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Summer On-peak RSP

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 1 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	8.89	8.91	0.02	0.2%	178.1	8.80	8.84	0.04	0.5%	176.7
2	50	16.88	17.06	0.18	1.1%	34.1	15.97	16.37	0.40	2.5%	32.7
3	100	25.76	26.12	0.36	1.4%	26.1	23.93	24.73	0.80	3.3%	24.7
4	150	34.64	35.19	0.55	1.6%	23.5	31.90	33.10	1.20	3.8%	22.1
5	200	43.52	44.25	0.73	1.7%	22.1	39.87	41.47	1.60	4.0%	20.7
6	250	52.40	53.31	0.91	1.7%	21.3	47.84	49.83	2.00	4.2%	19.9
7	300	61.28	62.37	1.09	1.8%	20.8	55.80	58.20	2.39	4.3%	19.4
8	350	70.16	71.44	1.28	1.8%	20.4	63.77	66.56	2.79	4.4%	19.0
9	400	79.04	80.50	1.46	1.8%	20.1	71.74	74.93	3.19	4.5%	18.7
10	450	87.92	89.56	1.64	1.9%	19.9	79.71	83.30	3.59	4.5%	18.5
11	500	96.80	98.62	1.82	1.9%	19.7	87.67	91.66	3.99	4.6%	18.3
12	550	105.68	107.69	2.01	1.9%	19.6	95.64	100.03	4.39	4.6%	18.2
13	600	114.56	116.75	2.19	1.9%	19.5	103.61	108.40	4.79	4.6%	18.1
14	650	123.44	125.81	2.37	1.9%	19.4	111.58	116.76	5.19	4.6%	18.0
15	700	132.32	134.87	2.55	1.9%	19.3	119.54	125.13	5.59	4.7%	17.9
16	750	141.20	143.94	2.73	1.9%	19.2	127.51	133.50	5.99	4.7%	17.8
17	800	150.08	153.00	2.92	1.9%	19.1	135.48	141.86	6.38	4.7%	17.7
18	850	158.96	162.06	3.10	1.9%	19.1	143.44	150.23	6.78	4.7%	17.7
19	900	167.84	171.12	3.28	2.0%	19.0	151.41	158.60	7.18	4.7%	17.6
20	950	176.72	180.19	3.46	2.0%	19.0	159.38	166.96	7.58	4.8%	17.6
21	1,000	185.60	189.25	3.65	2.0%	18.9	167.35	175.33	7.98	4.8%	17.5
22	1,500	274.41	279.87	5.47	2.0%	18.7	247.02	258.99	11.97	4.8%	17.3
23	2,000	363.21	370.50	7.29	2.0%	18.5	326.69	342.66	15.96	4.9%	17.1
24	2,500	452.01	461.12	9.12	2.0%	18.4	406.37	426.32	19.95	4.9%	17.1
25	3,000	540.81	551.75	10.94	2.0%	18.4	486.04	509.98	23.94	4.9%	17.0
26	3,500	629.61	642.37	12.76	2.0%	18.4	565.71	593.65	27.93	4.9%	17.0
27	4,000	718.41	733.00	14.58	2.0%	18.3	645.39	677.31	31.92	4.9%	16.9
28	4,500	807.22	823.62	16.41	2.0%	18.3	725.06	760.98	35.91	5.0%	16.9
29	5,000	896.02	914.25	18.23	2.0%	18.3	804.74	844.64	39.90	5.0%	16.9

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Summer On-peak RSP

Senior Citizen Provision RSC

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 2 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	4.89	4.91	0.02	0.4%	98.1	4.80	4.84	0.04	0.8%	96.7
2	50	12.88	13.06	0.18	1.4%	26.1	11.97	12.37	0.40	3.3%	24.7
3	100	21.76	22.12	0.36	1.7%	22.1	19.93	20.73	0.80	4.0%	20.7
4	150	30.64	31.19	0.55	1.8%	20.8	27.90	29.10	1.20	4.3%	19.4
5	200	39.52	40.25	0.73	1.8%	20.1	35.87	37.47	1.60	4.5%	18.7
6	250	48.40	49.31	0.91	1.9%	19.7	43.84	45.83	2.00	4.6%	18.3
7	300	57.28	58.37	1.09	1.9%	19.5	51.80	54.20	2.39	4.6%	18.1
8	350	66.16	67.44	1.28	1.9%	19.3	59.77	62.56	2.79	4.7%	17.9
9	400	75.04	76.50	1.46	1.9%	19.1	67.74	70.93	3.19	4.7%	17.7
10	450	83.92	85.56	1.64	2.0%	19.0	75.71	79.30	3.59	4.7%	17.6
11	500	92.80	94.62	1.82	2.0%	18.9	83.67	87.66	3.99	4.8%	17.5
12	550	101.68	103.69	2.01	2.0%	18.9	91.64	96.03	4.39	4.8%	17.5
13	600	110.56	112.75	2.19	2.0%	18.8	99.61	104.40	4.79	4.8%	17.4
14	650	119.44	121.81	2.37	2.0%	18.7	107.58	112.76	5.19	4.8%	17.3
15	700	128.32	130.87	2.55	2.0%	18.7	115.54	121.13	5.59	4.8%	17.3
16	750	137.20	139.94	2.73	2.0%	18.7	123.51	129.50	5.99	4.8%	17.3
17	800	146.08	149.00	2.92	2.0%	18.6	131.48	137.86	6.38	4.9%	17.2
18	850	154.96	158.06	3.10	2.0%	18.6	139.44	146.23	6.78	4.9%	17.2
19	900	163.84	167.12	3.28	2.0%	18.6	147.41	154.60	7.18	4.9%	17.2
20	950	172.72	176.19	3.46	2.0%	18.5	155.38	162.96	7.58	4.9%	17.2
21	1,000	181.60	185.25	3.65	2.0%	18.5	163.35	171.33	7.98	4.9%	17.1
22	1,500	270.41	275.87	5.47	2.0%	18.4	243.02	254.99	11.97	4.9%	17.0
23	2,000	359.21	366.50	7.29	2.0%	18.3	322.69	338.66	15.96	4.9%	16.9
24	2,500	448.01	457.12	9.12	2.0%	18.3	402.37	422.32	19.95	5.0%	16.9
25	3,000	536.81	547.75	10.94	2.0%	18.3	482.04	505.98	23.94	5.0%	16.9
26	3,500	625.61	638.37	12.76	2.0%	18.2	561.71	589.65	27.93	5.0%	16.8
27	4,000	714.41	729.00	14.58	2.0%	18.2	641.39	673.31	31.92	5.0%	16.8
28	4,500	803.22	819.62	16.41	2.0%	18.2	721.06	756.98	35.91	5.0%	16.8
29	5,000	892.02	910.25	18.23	2.0%	18.2	800.74	840.64	39.90	5.0%	16.8

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Summer On-peak RSP

Income Assistance Provision RIA

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 3 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	0.89	0.91	0.02	2.1%	18.1	0.80	0.84	0.04	5.0%	16.7
2	50	8.88	9.06	0.18	2.1%	18.1	7.97	8.37	0.40	5.0%	16.7
3	100	17.76	18.12	0.36	2.1%	18.1	15.93	16.73	0.80	5.0%	16.7
4	150	26.64	27.19	0.55	2.1%	18.1	23.90	25.10	1.20	5.0%	16.7
5	200	35.52	36.25	0.73	2.1%	18.1	31.87	33.47	1.60	5.0%	16.7
6	250	44.40	45.31	0.91	2.1%	18.1	39.84	41.83	2.00	5.0%	16.7
7	300	53.28	54.37	1.09	2.1%	18.1	47.80	50.20	2.39	5.0%	16.7
8	350	62.16	63.44	1.28	2.1%	18.1	55.77	58.56	2.79	5.0%	16.7
9	400	71.04	72.50	1.46	2.1%	18.1	63.74	66.93	3.19	5.0%	16.7
10	450	79.92	81.56	1.64	2.1%	18.1	71.71	75.30	3.59	5.0%	16.7
11	500	88.80	90.62	1.82	2.1%	18.1	79.67	83.66	3.99	5.0%	16.7
12	550	97.68	99.69	2.01	2.1%	18.1	87.64	92.03	4.39	5.0%	16.7
13	600	106.56	108.75	2.19	2.1%	18.1	95.61	100.40	4.79	5.0%	16.7
14	650	115.44	117.81	2.37	2.1%	18.1	103.58	108.76	5.19	5.0%	16.7
15	700	124.32	126.87	2.55	2.1%	18.1	111.54	117.13	5.59	5.0%	16.7
16	750	133.20	135.94	2.73	2.1%	18.1	119.51	125.50	5.99	5.0%	16.7
17	800	142.08	145.00	2.92	2.1%	18.1	127.48	133.86	6.38	5.0%	16.7
18	850	150.96	154.06	3.10	2.1%	18.1	135.44	142.23	6.78	5.0%	16.7
19	900	159.84	163.12	3.28	2.1%	18.1	143.41	150.60	7.18	5.0%	16.7
20	950	168.72	172.19	3.46	2.1%	18.1	151.38	158.96	7.58	5.0%	16.7
21	1,000	177.60	181.25	3.65	2.1%	18.1	159.35	167.33	7.98	5.0%	16.7
22	1,500	266.41	271.87	5.47	2.1%	18.1	239.02	250.99	11.97	5.0%	16.7
23	2,000	355.21	362.50	7.29	2.1%	18.1	318.69	334.66	15.96	5.0%	16.7
24	2,500	444.01	453.12	9.12	2.1%	18.1	398.37	418.32	19.95	5.0%	16.7
25	3,000	532.81	543.75	10.94	2.1%	18.1	478.04	501.98	23.94	5.0%	16.7
26	3,500	621.61	634.37	12.76	2.1%	18.1	557.71	585.65	27.93	5.0%	16.7
27	4,000	710.41	725.00	14.58	2.1%	18.1	637.39	669.31	31.92	5.0%	16.7
28	4,500	799.22	815.62	16.41	2.1%	18.1	717.06	752.98	35.91	5.0%	16.7
29	5,000	888.02	906.25	18.23	2.1%	18.1	796.74	836.64	39.90	5.0%	16.7

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Smart Hours RSH

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 4 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	8.89	8.91	0.02	0.2%	178.1	8.79	8.83	0.04	0.5%	176.7
2	50	16.87	17.06	0.18	1.1%	34.1	15.93	16.34	0.41	2.6%	32.7
3	100	25.75	26.11	0.37	1.4%	26.1	23.87	24.69	0.82	3.4%	24.7
4	150	34.62	35.17	0.55	1.6%	23.4	31.80	33.03	1.23	3.9%	22.0
5	200	43.49	44.22	0.73	1.7%	22.1	39.74	41.38	1.64	4.1%	20.7
6	250	52.36	53.28	0.92	1.7%	21.3	47.67	49.72	2.05	4.3%	19.9
7	300	61.24	62.33	1.10	1.8%	20.8	55.61	58.07	2.46	4.4%	19.4
8	350	70.11	71.39	1.28	1.8%	20.4	63.54	66.41	2.87	4.5%	19.0
9	400	78.98	80.45	1.47	1.9%	20.1	71.48	74.76	3.28	4.6%	18.7
10	450	87.85	89.50	1.65	1.9%	19.9	79.41	83.10	3.69	4.7%	18.5
11	500	96.73	98.56	1.83	1.9%	19.7	87.34	91.45	4.10	4.7%	18.3
12	550	105.60	107.61	2.01	1.9%	19.6	95.28	99.79	4.51	4.7%	18.1
13	600	114.47	116.67	2.20	1.9%	19.4	103.21	108.14	4.92	4.8%	18.0
14	650	123.34	125.72	2.38	1.9%	19.3	111.15	116.48	5.34	4.8%	17.9
15	700	132.22	134.78	2.56	1.9%	19.3	119.08	124.83	5.75	4.8%	17.8
16	750	141.09	143.83	2.75	1.9%	19.2	127.02	133.17	6.16	4.8%	17.8
17	800	149.96	152.89	2.93	2.0%	19.1	134.95	141.52	6.57	4.9%	17.7
18	850	158.83	161.95	3.11	2.0%	19.1	142.88	149.86	6.98	4.9%	17.6
19	900	167.71	171.00	3.30	2.0%	19.0	150.82	158.21	7.39	4.9%	17.6
20	950	176.58	180.06	3.48	2.0%	19.0	158.75	166.55	7.80	4.9%	17.5
21	1,000	185.45	189.11	3.66	2.0%	18.9	166.69	174.90	8.21	4.9%	17.5
22	1,500	274.18	279.67	5.49	2.0%	18.6	246.03	258.34	12.31	5.0%	17.2
23	2,000	362.90	370.23	7.33	2.0%	18.5	325.38	341.79	16.42	5.0%	17.1
24	2,500	451.63	460.78	9.16	2.0%	18.4	404.72	425.24	20.52	5.1%	17.0
25	3,000	540.35	551.34	10.99	2.0%	18.4	484.06	508.69	24.62	5.1%	17.0
26	3,500	629.08	641.89	12.82	2.0%	18.3	563.41	592.14	28.73	5.1%	16.9
27	4,000	717.80	732.45	14.65	2.0%	18.3	642.75	675.58	32.83	5.1%	16.9
28	4,500	806.53	823.01	16.48	2.0%	18.3	722.10	759.03	36.94	5.1%	16.9
29	5,000	895.25	913.56	18.31	2.0%	18.3	801.44	842.48	41.04	5.1%	16.8

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Smart Hours RSH

Senior Citizen Provision RSC

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 5 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	4.89	4.91	0.02	0.4%	98.1	4.79	4.83	0.04	0.9%	96.7
2	50	12.87	13.06	0.18	1.4%	26.1	11.93	12.34	0.41	3.4%	24.7
3	100	21.75	22.11	0.37	1.7%	22.1	19.87	20.69	0.82	4.1%	20.7
4	150	30.62	31.17	0.55	1.8%	20.8	27.80	29.03	1.23	4.4%	19.4
5	200	39.49	40.22	0.73	1.9%	20.1	35.74	37.38	1.64	4.6%	18.7
6	250	48.36	49.28	0.92	1.9%	19.7	43.67	45.72	2.05	4.7%	18.3
7	300	57.24	58.33	1.10	1.9%	19.4	51.61	54.07	2.46	4.8%	18.0
8	350	66.11	67.39	1.28	1.9%	19.3	59.54	62.41	2.87	4.8%	17.8
9	400	74.98	76.45	1.47	2.0%	19.1	67.48	70.76	3.28	4.9%	17.7
10	450	83.85	85.50	1.65	2.0%	19.0	75.41	79.10	3.69	4.9%	17.6
11	500	92.73	94.56	1.83	2.0%	18.9	83.34	87.45	4.10	4.9%	17.5
12	550	101.60	103.61	2.01	2.0%	18.8	91.28	95.79	4.51	4.9%	17.4
13	600	110.47	112.67	2.20	2.0%	18.8	99.21	104.14	4.92	5.0%	17.4
14	650	119.34	121.72	2.38	2.0%	18.7	107.15	112.48	5.34	5.0%	17.3
15	700	128.22	130.78	2.56	2.0%	18.7	115.08	120.83	5.75	5.0%	17.3
16	750	137.09	139.83	2.75	2.0%	18.6	123.02	129.17	6.16	5.0%	17.2
17	800	145.96	148.89	2.93	2.0%	18.6	130.95	137.52	6.57	5.0%	17.2
18	850	154.83	157.95	3.11	2.0%	18.6	138.88	145.86	6.98	5.0%	17.2
19	900	163.71	167.00	3.30	2.0%	18.6	146.82	154.21	7.39	5.0%	17.1
20	950	172.58	176.06	3.48	2.0%	18.5	154.75	162.55	7.80	5.0%	17.1
21	1,000	181.45	185.11	3.66	2.0%	18.5	162.69	170.90	8.21	5.0%	17.1
22	1,500	270.18	275.67	5.49	2.0%	18.4	242.03	254.34	12.31	5.1%	17.0
23	2,000	358.90	366.23	7.33	2.0%	18.3	321.38	337.79	16.42	5.1%	16.9
24	2,500	447.63	456.78	9.16	2.0%	18.3	400.72	421.24	20.52	5.1%	16.8
25	3,000	536.35	547.34	10.99	2.0%	18.2	480.06	504.69	24.62	5.1%	16.8
26	3,500	625.08	637.89	12.82	2.1%	18.2	559.41	588.14	28.73	5.1%	16.8
27	4,000	713.80	728.45	14.65	2.1%	18.2	638.75	671.58	32.83	5.1%	16.8
28	4,500	802.53	819.01	16.48	2.1%	18.2	718.10	755.03	36.94	5.1%	16.8
29	5,000	891.25	909.56	18.31	2.1%	18.2	797.44	838.48	41.04	5.1%	16.8

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Smart Hours RSH

Income Assistance Provision RIA

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 6 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	0.89	0.91	0.02	2.1%	18.1	0.79	0.83	0.04	5.2%	16.7
2	50	8.87	9.06	0.18	2.1%	18.1	7.93	8.34	0.41	5.2%	16.7
3	100	17.75	18.11	0.37	2.1%	18.1	15.87	16.69	0.82	5.2%	16.7
4	150	26.62	27.17	0.55	2.1%	18.1	23.80	25.03	1.23	5.2%	16.7
5	200	35.49	36.22	0.73	2.1%	18.1	31.74	33.38	1.64	5.2%	16.7
6	250	44.36	45.28	0.92	2.1%	18.1	39.67	41.72	2.05	5.2%	16.7
7	300	53.24	54.33	1.10	2.1%	18.1	47.61	50.07	2.46	5.2%	16.7
8	350	62.11	63.39	1.28	2.1%	18.1	55.54	58.41	2.87	5.2%	16.7
9	400	70.98	72.45	1.47	2.1%	18.1	63.48	66.76	3.28	5.2%	16.7
10	450	79.85	81.50	1.65	2.1%	18.1	71.41	75.10	3.69	5.2%	16.7
11	500	88.73	90.56	1.83	2.1%	18.1	79.34	83.45	4.10	5.2%	16.7
12	550	97.60	99.61	2.01	2.1%	18.1	87.28	91.79	4.51	5.2%	16.7
13	600	106.47	108.67	2.20	2.1%	18.1	95.21	100.14	4.92	5.2%	16.7
14	650	115.34	117.72	2.38	2.1%	18.1	103.15	108.48	5.34	5.2%	16.7
15	700	124.22	126.78	2.56	2.1%	18.1	111.08	116.83	5.75	5.2%	16.7
16	750	133.09	135.83	2.75	2.1%	18.1	119.02	125.17	6.16	5.2%	16.7
17	800	141.96	144.89	2.93	2.1%	18.1	126.95	133.52	6.57	5.2%	16.7
18	850	150.83	153.95	3.11	2.1%	18.1	134.88	141.86	6.98	5.2%	16.7
19	900	159.71	163.00	3.30	2.1%	18.1	142.82	150.21	7.39	5.2%	16.7
20	950	168.58	172.06	3.48	2.1%	18.1	150.75	158.55	7.80	5.2%	16.7
21	1,000	177.45	181.11	3.66	2.1%	18.1	158.69	166.90	8.21	5.2%	16.7
22	1,500	266.18	271.67	5.49	2.1%	18.1	238.03	250.34	12.31	5.2%	16.7
23	2,000	354.90	362.23	7.33	2.1%	18.1	317.38	333.79	16.42	5.2%	16.7
24	2,500	443.63	452.78	9.16	2.1%	18.1	396.72	417.24	20.52	5.2%	16.7
25	3,000	532.35	543.34	10.99	2.1%	18.1	476.06	500.69	24.62	5.2%	16.7
26	3,500	621.08	633.89	12.82	2.1%	18.1	555.41	584.14	28.73	5.2%	16.7
27	4,000	709.80	724.45	14.65	2.1%	18.1	634.75	667.58	32.83	5.2%	16.7
28	4,500	798.53	815.01	16.48	2.1%	18.1	714.10	751.03	36.94	5.2%	16.7
29	5,000	887.25	905.56	18.31	2.1%	18.1	793.44	834.48	41.04	5.2%	16.7

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Nighttime Savers RPM

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 7 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	8.85	8.88	0.03	0.4%	177.7	8.78	8.82	0.05	0.5%	176.5
2	50	16.51	16.83	0.32	1.9%	33.7	15.77	16.24	0.47	3.0%	32.5
3	100	25.02	25.66	0.63	2.5%	25.7	23.55	24.48	0.93	4.0%	24.5
4	150	33.53	34.48	0.95	2.8%	23.0	31.32	32.72	1.40	4.5%	21.8
5	200	42.04	43.31	1.27	3.0%	21.7	39.09	40.96	1.87	4.8%	20.5
6	250	50.56	52.14	1.58	3.1%	20.9	46.87	49.20	2.33	5.0%	19.7
7	300	59.07	60.97	1.90	3.2%	20.3	54.64	57.44	2.80	5.1%	19.1
8	350	67.58	69.79	2.22	3.3%	19.9	62.42	65.68	3.27	5.2%	18.8
9	400	76.09	78.62	2.53	3.3%	19.7	70.19	73.92	3.73	5.3%	18.5
10	450	84.60	87.45	2.85	3.4%	19.4	77.96	82.16	4.20	5.4%	18.3
11	500	93.11	96.28	3.16	3.4%	19.3	85.74	90.40	4.66	5.4%	18.1
12	550	101.62	105.10	3.48	3.4%	19.1	93.51	98.64	5.13	5.5%	17.9
13	600	110.13	113.93	3.80	3.4%	19.0	101.28	106.88	5.60	5.5%	17.8
14	650	118.64	122.76	4.11	3.5%	18.9	109.06	115.12	6.06	5.6%	17.7
15	700	127.16	131.59	4.43	3.5%	18.8	116.83	123.36	6.53	5.6%	17.6
16	750	135.67	140.41	4.75	3.5%	18.7	124.60	131.60	7.00	5.6%	17.5
17	800	144.18	149.24	5.06	3.5%	18.7	132.38	139.84	7.46	5.6%	17.5
18	850	152.69	158.07	5.38	3.5%	18.6	140.15	148.08	7.93	5.7%	17.4
19	900	161.20	166.90	5.70	3.5%	18.5	147.93	156.32	8.40	5.7%	17.4
20	950	169.71	175.72	6.01	3.5%	18.5	155.70	164.56	8.86	5.7%	17.3
21	1,000	178.22	184.55	6.33	3.6%	18.5	163.47	172.80	9.33	5.7%	17.3
22	1,500	263.33	272.83	9.49	3.6%	18.2	241.21	255.20	13.99	5.8%	17.0
23	2,000	348.44	361.10	12.66	3.6%	18.1	318.95	337.60	18.66	5.8%	16.9
24	2,500	433.56	449.38	15.82	3.6%	18.0	396.68	420.00	23.32	5.9%	16.8
25	3,000	518.67	537.65	18.99	3.7%	17.9	474.42	502.40	27.99	5.9%	16.7
26	3,500	603.78	625.93	22.15	3.7%	17.9	552.15	584.81	32.65	5.9%	16.7
27	4,000	688.89	714.20	25.32	3.7%	17.9	629.89	667.21	37.32	5.9%	16.7
28	4,500	774.00	802.48	28.48	3.7%	17.8	707.63	749.61	41.98	5.9%	16.7
29	5,000	859.11	890.76	31.65	3.7%	17.8	785.36	832.01	46.64	5.9%	16.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Nighttime Savers RPM

Senior Citizen Provision RSC

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 8 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	4.85	4.88	0.02	0.5%	97.5	4.78	4.82	0.05	1.0%	96.5
2	50	12.51	12.75	0.24	1.9%	25.5	11.77	12.24	0.47	4.0%	24.5
3	100	21.02	21.50	0.48	2.3%	21.5	19.55	20.48	0.93	4.8%	20.5
4	150	29.53	30.26	0.72	2.4%	20.2	27.32	28.72	1.40	5.1%	19.1
5	200	38.04	39.01	0.96	2.5%	19.5	35.09	36.96	1.87	5.3%	18.5
6	250	46.56	47.76	1.20	2.6%	19.1	42.87	45.20	2.33	5.4%	18.1
7	300	55.07	56.51	1.44	2.6%	18.8	50.64	53.44	2.80	5.5%	17.8
8	350	63.58	65.26	1.68	2.6%	18.6	58.42	61.68	3.27	5.6%	17.6
9	400	72.09	74.01	1.92	2.7%	18.5	66.19	69.92	3.73	5.6%	17.5
10	450	80.60	82.77	2.17	2.7%	18.4	73.96	78.16	4.20	5.7%	17.4
11	500	89.11	91.52	2.41	2.7%	18.3	81.74	86.40	4.66	5.7%	17.3
12	550	97.62	100.27	2.65	2.7%	18.2	89.51	94.64	5.13	5.7%	17.2
13	600	106.13	109.02	2.89	2.7%	18.2	97.28	102.88	5.60	5.8%	17.1
14	650	114.64	117.77	3.13	2.7%	18.1	105.06	111.12	6.06	5.8%	17.1
15	700	123.16	126.52	3.37	2.7%	18.1	112.83	119.36	6.53	5.8%	17.1
16	750	131.67	135.28	3.61	2.7%	18.0	120.60	127.60	7.00	5.8%	17.0
17	800	140.18	144.03	3.85	2.7%	18.0	128.38	135.84	7.46	5.8%	17.0
18	850	148.69	152.78	4.09	2.8%	18.0	136.15	144.08	7.93	5.8%	17.0
19	900	157.20	161.53	4.33	2.8%	17.9	143.93	152.32	8.40	5.8%	16.9
20	950	165.71	170.28	4.57	2.8%	17.9	151.70	160.56	8.86	5.8%	16.9
21	1,000	174.22	179.03	4.81	2.8%	17.9	159.47	168.80	9.33	5.8%	16.9
22	1,500	259.33	266.55	7.22	2.8%	17.8	237.21	251.20	13.99	5.9%	16.7
23	2,000	344.44	354.07	9.62	2.8%	17.7	314.95	333.60	18.66	5.9%	16.7
24	2,500	429.56	441.59	12.03	2.8%	17.7	392.68	416.00	23.32	5.9%	16.6
25	3,000	514.67	529.10	14.44	2.8%	17.6	470.42	498.40	27.99	5.9%	16.6
26	3,500	599.78	616.62	16.84	2.8%	17.6	548.15	580.81	32.65	6.0%	16.6
27	4,000	684.89	704.14	19.25	2.8%	17.6	625.89	663.21	37.32	6.0%	16.6
28	4,500	770.00	791.65	21.66	2.8%	17.6	703.63	745.61	41.98	6.0%	16.6
29	5,000	855.11	879.17	24.06	2.8%	17.6	781.36	828.01	46.64	6.0%	16.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Nighthtime Savers RPM

Income Assistance Provision RIA

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 9 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	0.85	0.88	0.03	3.7%	17.7	0.78	0.82	0.05	6.0%	16.5
2	50	8.51	8.83	0.32	3.7%	17.7	7.77	8.24	0.47	6.0%	16.5
3	100	17.02	17.66	0.63	3.7%	17.7	15.55	16.48	0.93	6.0%	16.5
4	150	25.53	26.48	0.95	3.7%	17.7	23.32	24.72	1.40	6.0%	16.5
5	200	34.04	35.31	1.27	3.7%	17.7	31.09	32.96	1.87	6.0%	16.5
6	250	42.56	44.14	1.58	3.7%	17.7	38.87	41.20	2.33	6.0%	16.5
7	300	51.07	52.97	1.90	3.7%	17.7	46.64	49.44	2.80	6.0%	16.5
8	350	59.58	61.79	2.22	3.7%	17.7	54.42	57.68	3.27	6.0%	16.5
9	400	68.09	70.62	2.53	3.7%	17.7	62.19	65.92	3.73	6.0%	16.5
10	450	76.60	79.45	2.85	3.7%	17.7	69.96	74.16	4.20	6.0%	16.5
11	500	85.11	88.28	3.16	3.7%	17.7	77.74	82.40	4.66	6.0%	16.5
12	550	93.62	97.10	3.48	3.7%	17.7	85.51	90.64	5.13	6.0%	16.5
13	600	102.13	105.93	3.80	3.7%	17.7	93.28	98.88	5.60	6.0%	16.5
14	650	110.64	114.76	4.11	3.7%	17.7	101.06	107.12	6.06	6.0%	16.5
15	700	119.16	123.59	4.43	3.7%	17.7	108.83	115.36	6.53	6.0%	16.5
16	750	127.67	132.41	4.75	3.7%	17.7	116.60	123.60	7.00	6.0%	16.5
17	800	136.18	141.24	5.06	3.7%	17.7	124.38	131.84	7.46	6.0%	16.5
18	850	144.69	150.07	5.38	3.7%	17.7	132.15	140.08	7.93	6.0%	16.5
19	900	153.20	158.90	5.70	3.7%	17.7	139.93	148.32	8.40	6.0%	16.5
20	950	161.71	167.72	6.01	3.7%	17.7	147.70	156.56	8.86	6.0%	16.5
21	1,000	170.22	176.55	6.33	3.7%	17.7	155.47	164.80	9.33	6.0%	16.5
22	1,500	255.33	264.83	9.49	3.7%	17.7	233.21	247.20	13.99	6.0%	16.5
23	2,000	340.44	353.10	12.66	3.7%	17.7	310.95	329.60	18.66	6.0%	16.5
24	2,500	425.56	441.38	15.82	3.7%	17.7	388.68	412.00	23.32	6.0%	16.5
25	3,000	510.67	529.65	18.99	3.7%	17.7	466.42	494.40	27.99	6.0%	16.5
26	3,500	595.78	617.93	22.15	3.7%	17.7	544.15	576.81	32.65	6.0%	16.5
27	4,000	680.89	706.20	25.32	3.7%	17.7	621.89	659.21	37.32	6.0%	16.5
28	4,500	766.00	794.48	28.48	3.7%	17.7	699.63	741.61	41.98	6.0%	16.5
29	5,000	851.11	882.76	31.65	3.7%	17.7	777.36	824.01	46.64	6.0%	16.5

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Non-Transmitting Meters RSM

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 10 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	8.80	8.82	0.02	0.3%	176.4	8.80	8.82	0.02	0.3%	176.4
2	50	15.97	16.22	0.25	1.6%	32.4	15.97	16.22	0.25	1.6%	32.4
3	100	23.93	24.43	0.50	2.1%	24.4	23.93	24.43	0.50	2.1%	24.4
4	150	31.90	32.65	0.75	2.3%	21.8	31.90	32.65	0.75	2.3%	21.8
5	200	39.87	40.87	1.00	2.5%	20.4	39.87	40.87	1.00	2.5%	20.4
6	250	47.84	49.09	1.25	2.6%	19.6	47.84	49.09	1.25	2.6%	19.6
7	300	55.80	57.30	1.50	2.7%	19.1	55.80	57.30	1.50	2.7%	19.1
8	350	63.77	65.52	1.75	2.7%	18.7	63.77	65.52	1.75	2.7%	18.7
9	400	71.74	73.74	2.00	2.8%	18.4	71.74	73.74	2.00	2.8%	18.4
10	450	79.71	81.95	2.25	2.8%	18.2	79.71	81.95	2.25	2.8%	18.2
11	500	87.67	90.17	2.50	2.8%	18.0	87.67	90.17	2.50	2.8%	18.0
12	550	95.64	98.39	2.75	2.9%	17.9	95.64	98.39	2.75	2.9%	17.9
13	600	103.61	106.61	3.00	2.9%	17.8	103.61	106.61	3.00	2.9%	17.8
14	650	114.74	117.53	2.79	2.4%	18.1	111.58	114.82	3.25	2.9%	17.7
15	700	125.86	128.45	2.58	2.1%	18.3	119.54	123.04	3.50	2.9%	17.6
16	750	136.99	139.37	2.38	1.7%	18.6	127.51	131.26	3.75	2.9%	17.5
17	800	148.12	150.29	2.17	1.5%	18.8	135.48	139.48	4.00	3.0%	17.4
18	850	159.24	161.21	1.96	1.2%	19.0	143.44	147.69	4.25	3.0%	17.4
19	900	170.37	172.13	1.76	1.0%	19.1	151.41	155.91	4.50	3.0%	17.3
20	950	181.50	183.05	1.55	0.9%	19.3	159.38	164.13	4.75	3.0%	17.3
21	1,000	192.62	193.97	1.34	0.7%	19.4	167.35	172.34	5.00	3.0%	17.2
22	1,500	303.89	303.16	(0.73)	-0.2%	20.2	247.02	254.52	7.50	3.0%	17.0
23	2,000	415.16	412.36	(2.80)	-0.7%	20.6	326.69	336.69	9.99	3.1%	16.8
24	2,500	526.43	521.56	(4.87)	-0.9%	20.9	406.37	418.86	12.49	3.1%	16.8
25	3,000	637.69	630.76	(6.94)	-1.1%	21.0	486.04	501.03	14.99	3.1%	16.7
26	3,500	748.96	739.96	(9.00)	-1.2%	21.1	565.71	583.20	17.49	3.1%	16.7
27	4,000	860.23	849.16	(11.07)	-1.3%	21.2	645.39	665.38	19.99	3.1%	16.6
28	4,500	971.50	958.35	(13.14)	-1.4%	21.3	725.06	747.55	22.49	3.1%	16.6
29	5,000	1,082.77	1,067.55	(15.21)	-1.4%	21.4	804.74	829.72	24.98	3.1%	16.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Non-Transmitting Meters RSM

Senior Citizen Provision RSC

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 11 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	4.80	4.82	0.02	0.5%	96.4	4.80	4.82	0.02	0.5%	96.4
2	50	11.97	12.22	0.25	2.1%	24.4	11.97	12.22	0.25	2.1%	24.4
3	100	19.93	20.43	0.50	2.5%	20.4	19.93	20.43	0.50	2.5%	20.4
4	150	27.90	28.65	0.75	2.7%	19.1	27.90	28.65	0.75	2.7%	19.1
5	200	35.87	36.87	1.00	2.8%	18.4	35.87	36.87	1.00	2.8%	18.4
6	250	43.84	45.09	1.25	2.8%	18.0	43.84	45.09	1.25	2.8%	18.0
7	300	51.80	53.30	1.50	2.9%	17.8	51.80	53.30	1.50	2.9%	17.8
8	350	59.77	61.52	1.75	2.9%	17.6	59.77	61.52	1.75	2.9%	17.6
9	400	67.74	69.74	2.00	3.0%	17.4	67.74	69.74	2.00	3.0%	17.4
10	450	75.71	77.95	2.25	3.0%	17.3	75.71	77.95	2.25	3.0%	17.3
11	500	83.67	86.17	2.50	3.0%	17.2	83.67	86.17	2.50	3.0%	17.2
12	550	91.64	94.39	2.75	3.0%	17.2	91.64	94.39	2.75	3.0%	17.2
13	600	99.61	102.61	3.00	3.0%	17.1	99.61	102.61	3.00	3.0%	17.1
14	650	110.74	113.53	2.79	2.5%	17.5	107.58	110.82	3.25	3.0%	17.0
15	700	121.86	124.45	2.58	2.1%	17.8	115.54	119.04	3.50	3.0%	17.0
16	750	132.99	135.37	2.38	1.8%	18.0	123.51	127.26	3.75	3.0%	17.0
17	800	144.12	146.29	2.17	1.5%	18.3	131.48	135.48	4.00	3.0%	16.9
18	850	155.24	157.21	1.96	1.3%	18.5	139.44	143.69	4.25	3.0%	16.9
19	900	166.37	168.13	1.76	1.1%	18.7	147.41	151.91	4.50	3.1%	16.9
20	950	177.50	179.05	1.55	0.9%	18.8	155.38	160.13	4.75	3.1%	16.9
21	1,000	188.62	189.97	1.34	0.7%	19.0	163.35	168.34	5.00	3.1%	16.8
22	1,500	299.89	299.16	(0.73)	-0.2%	19.9	243.02	250.52	7.50	3.1%	16.7
23	2,000	411.16	408.36	(2.80)	-0.7%	20.4	322.69	332.69	9.99	3.1%	16.6
24	2,500	522.43	517.56	(4.87)	-0.9%	20.7	402.37	414.86	12.49	3.1%	16.6
25	3,000	633.69	626.76	(6.94)	-1.1%	20.9	482.04	497.03	14.99	3.1%	16.6
26	3,500	744.96	735.96	(9.00)	-1.2%	21.0	561.71	579.20	17.49	3.1%	16.5
27	4,000	856.23	845.16	(11.07)	-1.3%	21.1	641.39	661.38	19.99	3.1%	16.5
28	4,500	967.50	954.35	(13.14)	-1.4%	21.2	721.06	743.55	22.49	3.1%	16.5
29	5,000	1,078.77	1,063.55	(15.21)	-1.4%	21.3	800.74	825.72	24.98	3.1%	16.5

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Non-Transmitting Meters RSM

Income Assistance Provision RIA

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 12 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	4.80	4.82	0.02	0.5%	96.4	4.80	4.82	0.02	0.5%	96.4
2	50	11.97	12.22	0.25	2.1%	24.4	11.97	12.22	0.25	2.1%	24.4
3	100	19.93	20.43	0.50	2.5%	20.4	19.93	20.43	0.50	2.5%	20.4
4	150	27.90	28.65	0.75	2.7%	19.1	27.90	28.65	0.75	2.7%	19.1
5	200	35.87	36.87	1.00	2.8%	18.4	35.87	36.87	1.00	2.8%	18.4
6	250	43.84	45.09	1.25	2.8%	18.0	43.84	45.09	1.25	2.8%	18.0
7	300	51.80	53.30	1.50	2.9%	17.8	51.80	53.30	1.50	2.9%	17.8
8	350	59.77	61.52	1.75	2.9%	17.6	59.77	61.52	1.75	2.9%	17.6
9	400	67.74	69.74	2.00	3.0%	17.4	67.74	69.74	2.00	3.0%	17.4
10	450	75.71	77.95	2.25	3.0%	17.3	75.71	77.95	2.25	3.0%	17.3
11	500	83.67	86.17	2.50	3.0%	17.2	83.67	86.17	2.50	3.0%	17.2
12	550	91.64	94.39	2.75	3.0%	17.2	91.64	94.39	2.75	3.0%	17.2
13	600	99.61	102.61	3.00	3.0%	17.1	99.61	102.61	3.00	3.0%	17.1
14	650	110.74	113.53	2.79	2.5%	17.5	107.58	110.82	3.25	3.0%	17.0
15	700	121.86	124.45	2.58	2.1%	17.8	115.54	119.04	3.50	3.0%	17.0
16	750	132.99	135.37	2.38	1.8%	18.0	123.51	127.26	3.75	3.0%	17.0
17	800	144.12	146.29	2.17	1.5%	18.3	131.48	135.48	4.00	3.0%	16.9
18	850	155.24	157.21	1.96	1.3%	18.5	139.44	143.69	4.25	3.0%	16.9
19	900	166.37	168.13	1.76	1.1%	18.7	147.41	151.91	4.50	3.1%	16.9
20	950	177.50	179.05	1.55	0.9%	18.8	155.38	160.13	4.75	3.1%	16.9
21	1,000	188.62	189.97	1.34	0.7%	19.0	163.35	168.34	5.00	3.1%	16.8
22	1,500	299.89	299.16	(0.73)	-0.2%	19.9	243.02	250.52	7.50	3.1%	16.7
23	2,000	411.16	408.36	(2.80)	-0.7%	20.4	322.69	332.69	9.99	3.1%	16.6
24	2,500	522.43	517.56	(4.87)	-0.9%	20.7	402.37	414.86	12.49	3.1%	16.6
25	3,000	633.69	626.76	(6.94)	-1.1%	20.9	482.04	497.03	14.99	3.1%	16.6
26	3,500	744.96	735.96	(9.00)	-1.2%	21.0	561.71	579.20	17.49	3.1%	16.5
27	4,000	856.23	845.16	(11.07)	-1.3%	21.1	641.39	661.38	19.99	3.1%	16.5
28	4,500	967.50	954.35	(13.14)	-1.4%	21.2	721.06	743.55	22.49	3.1%	16.5
29	5,000	1,078.77	1,063.55	(15.21)	-1.4%	21.3	800.74	825.72	24.98	3.1%	16.5

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Energy-only GS

Case No.: U-21389
Exhibit No.: A-16 (AMG-4)
Schedule F-4.0
Page 13 of 46
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	250	57.84	60.92	3.08	5.3%	24.4	53.70	57.43	3.73	6.9%	23.0
2	500	95.69	101.85	6.16	6.4%	20.4	87.40	94.85	7.45	8.5%	19.0
3	750	133.53	142.77	9.24	6.9%	19.0	121.10	132.28	11.18	9.2%	17.6
4	1,000	171.38	183.69	12.32	7.2%	18.4	154.80	169.70	14.90	9.6%	17.0
5	1,500	247.06	265.54	18.48	7.5%	17.7	222.20	244.55	22.35	10.1%	16.3
6	2,000	322.75	347.39	24.64	7.6%	17.4	289.60	319.41	29.81	10.3%	16.0
7	2,500	398.44	429.24	30.79	7.7%	17.2	357.00	394.26	37.26	10.4%	15.8
8	3,000	474.13	511.08	36.95	7.8%	17.0	424.40	469.11	44.71	10.5%	15.6
9	3,500	549.82	592.93	43.11	7.8%	16.9	491.80	543.96	52.16	10.6%	15.5
10	4,000	625.50	674.78	49.27	7.9%	16.9	559.20	618.81	59.61	10.7%	15.5
11	4,500	701.19	756.62	55.43	7.9%	16.8	626.60	693.66	67.06	10.7%	15.4
12	5,000	776.88	838.47	61.59	7.9%	16.8	694.00	768.52	74.51	10.7%	15.4
13	6,000	928.26	1,002.16	73.91	8.0%	16.7	828.80	918.22	89.42	10.8%	15.3
14	7,000	1,079.63	1,165.86	86.23	8.0%	16.7	963.60	1,067.92	104.32	10.8%	15.3
15	8,000	1,231.01	1,329.55	98.54	8.0%	16.6	1,098.40	1,217.62	119.22	10.9%	15.2
16	9,000	1,382.38	1,493.25	110.86	8.0%	16.6	1,233.20	1,367.33	134.13	10.9%	15.2
17	10,000	1,533.76	1,656.94	123.18	8.0%	16.6	1,368.00	1,517.03	149.03	10.9%	15.2
18	15,000	2,290.64	2,475.41	184.77	8.1%	16.5	2,042.00	2,265.55	223.55	10.9%	15.1
19	20,000	3,047.52	3,293.88	246.36	8.1%	16.5	2,716.00	3,014.06	298.06	11.0%	15.1
20	25,000	3,804.40	4,112.35	307.95	8.1%	16.4	3,390.00	3,762.58	372.57	11.0%	15.1
21	30,000	4,561.28	4,930.82	369.54	8.1%	16.4	4,064.00	4,511.09	447.09	11.0%	15.0
22	35,000	5,318.16	5,749.29	431.13	8.1%	16.4	4,738.00	5,259.61	521.61	11.0%	15.0
23	40,000	6,075.04	6,567.76	492.72	8.1%	16.4	5,412.00	6,008.12	596.12	11.0%	15.0
24	45,000	6,831.92	7,386.23	554.31	8.1%	16.4	6,086.00	6,756.64	670.63	11.0%	15.0
25	50,000	7,588.80	8,204.70	615.90	8.1%	16.4	6,760.00	7,505.15	745.15	11.0%	15.0
26	55,000	8,345.68	9,023.17	677.49	8.1%	16.4	7,434.00	8,253.67	819.66	11.0%	15.0
27	60,000	9,102.56	9,841.64	739.08	8.1%	16.4	8,108.00	9,002.18	894.18	11.0%	15.0
28	65,000	9,859.44	10,660.11	800.67	8.1%	16.4	8,782.00	9,750.70	968.69	11.0%	15.0
29	70,000	10,616.32	11,478.58	862.26	8.1%	16.4	9,456.00	10,499.21	1,043.21	11.0%	15.0

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Energy-only GS

Education Provision GEI

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 14 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	250	57.64	60.67	3.03	5.3%	24.3	53.50	57.17	3.67	6.9%	22.9
2	500	95.29	101.34	6.05	6.4%	20.3	87.00	94.35	7.35	8.4%	18.9
3	750	132.93	142.01	9.08	6.8%	18.9	120.50	131.52	11.02	9.1%	17.5
4	1,000	170.57	182.68	12.11	7.1%	18.3	154.00	168.69	14.69	9.5%	16.9
5	1,500	245.86	264.03	18.16	7.4%	17.6	221.00	243.04	22.04	10.0%	16.2
6	2,000	321.15	345.37	24.22	7.5%	17.3	288.00	317.39	29.39	10.2%	15.9
7	2,500	396.44	426.71	30.27	7.6%	17.1	355.00	391.73	36.74	10.3%	15.7
8	3,000	471.72	508.05	36.33	7.7%	16.9	421.99	466.08	44.08	10.4%	15.5
9	3,500	547.01	589.39	42.38	7.7%	16.8	488.99	540.43	51.43	10.5%	15.4
10	4,000	622.30	670.74	48.44	7.8%	16.8	555.99	614.77	58.78	10.6%	15.4
11	4,500	697.58	752.08	54.49	7.8%	16.7	622.99	689.12	66.13	10.6%	15.3
12	5,000	772.87	833.42	60.55	7.8%	16.7	689.99	763.47	73.47	10.6%	15.3
13	6,000	923.44	996.10	72.66	7.9%	16.6	823.99	912.16	88.17	10.7%	15.2
14	7,000	1,074.02	1,158.79	84.77	7.9%	16.6	957.99	1,060.85	102.87	10.7%	15.2
15	8,000	1,224.59	1,321.47	96.88	7.9%	16.5	1,091.98	1,209.54	117.56	10.8%	15.1
16	9,000	1,375.17	1,484.16	108.99	7.9%	16.5	1,225.98	1,358.24	132.26	10.8%	15.1
17	10,000	1,525.74	1,646.84	121.10	7.9%	16.5	1,359.98	1,506.93	146.95	10.8%	15.1
18	15,000	2,278.61	2,460.26	181.65	8.0%	16.4	2,029.97	2,250.40	220.42	10.9%	15.0
19	20,000	3,031.48	3,273.68	242.20	8.0%	16.4	2,699.96	2,993.86	293.90	10.9%	15.0
20	25,000	3,784.35	4,087.10	302.75	8.0%	16.3	3,369.95	3,737.33	367.37	10.9%	14.9
21	30,000	4,537.22	4,900.52	363.30	8.0%	16.3	4,039.94	4,480.79	440.85	10.9%	14.9
22	35,000	5,290.09	5,713.94	423.85	8.0%	16.3	4,709.93	5,224.26	514.32	10.9%	14.9
23	40,000	6,042.96	6,527.36	484.40	8.0%	16.3	5,379.92	5,967.72	587.80	10.9%	14.9
24	45,000	6,795.83	7,340.78	544.95	8.0%	16.3	6,049.91	6,711.19	661.27	10.9%	14.9
25	50,000	7,548.70	8,154.20	605.50	8.0%	16.3	6,719.90	7,454.65	734.75	10.9%	14.9
26	55,000	8,301.57	8,967.62	666.05	8.0%	16.3	7,389.89	8,198.12	808.22	10.9%	14.9
27	60,000	9,054.44	9,781.04	726.60	8.0%	16.3	8,059.88	8,941.58	881.70	10.9%	14.9
28	65,000	9,807.31	10,594.46	787.15	8.0%	16.3	8,729.87	9,685.05	955.17	10.9%	14.9
29	70,000	10,560.18	11,407.88	847.70	8.0%	16.3	9,399.86	10,428.51	1,028.65	10.9%	14.9

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Demand GSD

Case No.: U-21389
Exhibit No.: A-16 (AMG-4)
Schedule F-4.0
Page 15 of 46
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	97.44	107.30	9.85	10.1%	21.5	87.41	91.67	4.26	4.9%	18.3
2	1,000	164.89	184.59	19.71	12.0%	18.5	144.81	153.34	8.52	5.9%	15.3
3	1,500	232.33	261.89	29.56	12.7%	17.5	202.22	215.00	12.78	6.3%	14.3
4	2,000	299.77	339.19	39.42	13.1%	17.0	259.63	276.67	17.05	6.6%	13.8
5	2,500	367.22	416.49	49.27	13.4%	16.7	317.03	338.34	21.31	6.7%	13.5
6	3,000	434.66	493.78	59.12	13.6%	16.5	374.44	400.01	25.57	6.8%	13.3
7	3,500	502.10	571.08	68.98	13.7%	16.3	431.85	461.68	29.83	6.9%	13.2
8	4,000	569.55	648.38	78.83	13.8%	16.2	489.25	523.34	34.09	7.0%	13.1
9	4,500	636.99	725.67	88.68	13.9%	16.1	546.66	585.01	38.35	7.0%	13.0
10	5,000	704.43	802.97	98.54	14.0%	16.1	604.06	646.68	42.61	7.1%	12.9
11	6,000	839.32	957.56	118.25	14.1%	16.0	718.88	770.02	51.14	7.1%	12.8
12	7,000	974.21	1,112.16	137.95	14.2%	15.9	833.69	893.35	59.66	7.2%	12.8
13	8,000	1,109.09	1,266.75	157.66	14.2%	15.8	948.50	1,016.69	68.18	7.2%	12.7
14	9,000	1,243.98	1,421.35	177.37	14.3%	15.8	1,063.32	1,140.02	76.71	7.2%	12.7
15	10,000	1,378.86	1,575.94	197.08	14.3%	15.8	1,178.13	1,263.36	85.23	7.2%	12.6
16	11,000	1,513.75	1,730.53	216.78	14.3%	15.7	1,292.94	1,386.70	93.75	7.3%	12.6
17	12,000	1,648.64	1,885.13	236.49	14.3%	15.7	1,407.76	1,510.03	102.28	7.3%	12.6
18	13,000	1,783.52	2,039.72	256.20	14.4%	15.7	1,522.57	1,633.37	110.80	7.3%	12.6
19	14,000	1,918.41	2,194.32	275.91	14.4%	15.7	1,637.38	1,756.70	119.32	7.3%	12.5
20	15,000	2,053.30	2,348.91	295.61	14.4%	15.7	1,752.19	1,880.04	127.84	7.3%	12.5
21	16,000	2,188.18	2,503.51	315.32	14.4%	15.6	1,867.01	2,003.37	136.37	7.3%	12.5
22	17,000	2,323.07	2,658.10	335.03	14.4%	15.6	1,981.82	2,126.71	144.89	7.3%	12.5
23	18,000	2,457.96	2,812.69	354.74	14.4%	15.6	2,096.63	2,250.05	153.41	7.3%	12.5
24	19,000	2,592.84	2,967.29	374.44	14.4%	15.6	2,211.45	2,373.38	161.94	7.3%	12.5
25	20,000	2,727.73	3,121.88	394.15	14.4%	15.6	2,326.26	2,496.72	170.46	7.3%	12.5
26	21,000	2,862.62	3,276.48	413.86	14.5%	15.6	2,441.07	2,620.05	178.98	7.3%	12.5
27	22,000	2,997.50	3,431.07	433.57	14.5%	15.6	2,555.89	2,743.39	187.50	7.3%	12.5
28	23,000	3,132.39	3,585.66	453.27	14.5%	15.6	2,670.70	2,866.73	196.03	7.3%	12.5
29	24,000	3,267.28	3,740.26	472.98	14.5%	15.6	2,785.51	2,990.06	204.55	7.3%	12.5

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Demand GSD

Education Provision GEI

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 16 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	97.12	106.00	8.88	9.1%	21.2	87.08	90.37	3.29	3.8%	18.1
2	1,000	164.24	181.99	17.76	10.8%	18.2	144.17	150.74	6.57	4.6%	15.1
3	1,500	231.36	257.99	26.63	11.5%	17.2	201.25	211.10	9.86	4.9%	14.1
4	2,000	298.48	333.99	35.51	11.9%	16.7	258.33	271.47	13.14	5.1%	13.6
5	2,500	365.60	409.99	44.39	12.1%	16.4	315.41	331.84	16.43	5.2%	13.3
6	3,000	432.72	485.98	53.27	12.3%	16.2	372.50	392.21	19.71	5.3%	13.1
7	3,500	499.84	561.98	62.14	12.4%	16.1	429.58	452.58	23.00	5.4%	12.9
8	4,000	566.96	637.98	71.02	12.5%	15.9	486.66	512.95	26.28	5.4%	12.8
9	4,500	634.08	713.98	79.90	12.6%	15.9	543.75	573.31	29.57	5.4%	12.7
10	5,000	701.20	789.97	88.78	12.7%	15.8	600.83	633.68	32.85	5.5%	12.7
11	6,000	835.44	941.97	106.53	12.8%	15.7	715.00	754.42	39.42	5.5%	12.6
12	7,000	969.68	1,093.96	124.29	12.8%	15.6	829.16	875.15	45.99	5.5%	12.5
13	8,000	1,103.92	1,245.96	142.04	12.9%	15.6	943.33	995.89	52.56	5.6%	12.4
14	9,000	1,238.16	1,397.95	159.80	12.9%	15.5	1,057.49	1,116.63	59.13	5.6%	12.4
15	10,000	1,372.39	1,549.95	177.55	12.9%	15.5	1,171.66	1,237.36	65.70	5.6%	12.4
16	11,000	1,506.63	1,701.94	195.31	13.0%	15.5	1,285.83	1,358.10	72.27	5.6%	12.3
17	12,000	1,640.87	1,853.93	213.06	13.0%	15.4	1,399.99	1,478.84	78.84	5.6%	12.3
18	13,000	1,775.11	2,005.93	230.82	13.0%	15.4	1,514.16	1,599.57	85.41	5.6%	12.3
19	14,000	1,909.35	2,157.92	248.57	13.0%	15.4	1,628.32	1,720.31	91.99	5.6%	12.3
20	15,000	2,043.59	2,309.92	266.33	13.0%	15.4	1,742.49	1,841.05	98.56	5.7%	12.3
21	16,000	2,177.83	2,461.91	284.08	13.0%	15.4	1,856.66	1,961.78	105.13	5.7%	12.3
22	17,000	2,312.07	2,613.91	301.84	13.1%	15.4	1,970.82	2,082.52	111.70	5.7%	12.3
23	18,000	2,446.31	2,765.90	319.59	13.1%	15.4	2,084.99	2,203.25	118.27	5.7%	12.2
24	19,000	2,580.55	2,917.90	337.35	13.1%	15.4	2,199.15	2,323.99	124.84	5.7%	12.2
25	20,000	2,714.79	3,069.89	355.10	13.1%	15.3	2,313.32	2,444.73	131.41	5.7%	12.2
26	21,000	2,849.03	3,221.88	372.86	13.1%	15.3	2,427.49	2,565.46	137.98	5.7%	12.2
27	22,000	2,983.27	3,373.88	390.61	13.1%	15.3	2,541.65	2,686.20	144.55	5.7%	12.2
28	23,000	3,117.51	3,525.87	408.37	13.1%	15.3	2,655.82	2,806.94	151.12	5.7%	12.2
29	24,000	3,251.75	3,677.87	426.12	13.1%	15.3	2,769.98	2,927.67	157.69	5.7%	12.2

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Time-of-Use GSTU

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 17 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	103.77	110.56	6.79	6.5%	22.1	86.17	93.97	7.80	9.1%	18.8
2	1,000	187.54	201.12	13.59	7.2%	20.1	152.33	167.94	15.61	10.2%	16.8
3	1,500	271.30	291.68	20.38	7.5%	19.4	218.50	241.91	23.41	10.7%	16.1
4	2,000	355.07	382.25	27.17	7.7%	19.1	284.66	315.87	31.21	11.0%	15.8
5	2,500	438.84	472.81	33.97	7.7%	18.9	350.83	389.84	39.01	11.1%	15.6
6	3,000	522.61	563.37	40.76	7.8%	18.8	417.00	463.81	46.82	11.2%	15.5
7	3,500	606.38	653.93	47.55	7.8%	18.7	483.16	537.78	54.62	11.3%	15.4
8	4,000	690.15	744.49	54.35	7.9%	18.6	549.33	611.75	62.42	11.4%	15.3
9	4,500	773.91	835.05	61.14	7.9%	18.6	615.49	685.72	70.22	11.4%	15.2
10	5,000	857.68	925.61	67.93	7.9%	18.5	681.66	759.69	78.03	11.4%	15.2
11	6,000	1,025.22	1,106.74	81.52	8.0%	18.4	813.99	907.62	93.63	11.5%	15.1
12	7,000	1,192.76	1,287.86	95.10	8.0%	18.4	946.32	1,055.56	109.24	11.5%	15.1
13	8,000	1,360.29	1,468.98	108.69	8.0%	18.4	1,078.65	1,203.50	124.84	11.6%	15.0
14	9,000	1,527.83	1,650.11	122.28	8.0%	18.3	1,210.99	1,351.43	140.45	11.6%	15.0
15	10,000	1,695.37	1,831.23	135.86	8.0%	18.3	1,343.32	1,499.37	156.05	11.6%	15.0
16	11,000	1,862.90	2,012.35	149.45	8.0%	18.3	1,475.65	1,647.31	171.66	11.6%	15.0
17	12,000	2,030.44	2,193.48	163.04	8.0%	18.3	1,607.98	1,795.25	187.26	11.6%	15.0
18	13,000	2,197.98	2,374.60	176.62	8.0%	18.3	1,740.31	1,943.18	202.87	11.7%	14.9
19	14,000	2,365.51	2,555.72	190.21	8.0%	18.3	1,872.65	2,091.12	218.48	11.7%	14.9
20	15,000	2,533.05	2,736.84	203.80	8.0%	18.2	2,004.98	2,239.06	234.08	11.7%	14.9
21	16,000	2,700.58	2,917.97	217.38	8.0%	18.2	2,137.31	2,387.00	249.69	11.7%	14.9
22	17,000	2,868.12	3,099.09	230.97	8.1%	18.2	2,269.64	2,534.93	265.29	11.7%	14.9
23	18,000	3,035.66	3,280.21	244.56	8.1%	18.2	2,401.97	2,682.87	280.90	11.7%	14.9
24	19,000	3,203.19	3,461.34	258.14	8.1%	18.2	2,534.30	2,830.81	296.50	11.7%	14.9
25	20,000	3,370.73	3,642.46	271.73	8.1%	18.2	2,666.64	2,978.74	312.11	11.7%	14.9
26	21,000	3,538.27	3,823.58	285.31	8.1%	18.2	2,798.97	3,126.68	327.71	11.7%	14.9
27	22,000	3,705.80	4,004.71	298.90	8.1%	18.2	2,931.30	3,274.62	343.32	11.7%	14.9
28	23,000	3,873.34	4,185.83	312.49	8.1%	18.2	3,063.63	3,422.56	358.92	11.7%	14.9
29	24,000	4,040.88	4,366.95	326.07	8.1%	18.2	3,195.96	3,570.49	374.53	11.7%	14.9

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Time-of-Use GSTU

Education Provision GEI

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 18 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	103.37	110.06	6.69	6.5%	22.0	85.76	93.46	7.70	9.0%	18.7
2	1,000	186.73	200.11	13.38	7.2%	20.0	151.53	166.93	15.40	10.2%	16.7
3	1,500	270.10	290.17	20.07	7.4%	19.3	217.29	240.39	23.10	10.6%	16.0
4	2,000	353.47	380.23	26.76	7.6%	19.0	283.06	313.85	30.79	10.9%	15.7
5	2,500	436.84	470.28	33.45	7.7%	18.8	348.82	387.32	38.49	11.0%	15.5
6	3,000	520.20	560.34	40.14	7.7%	18.7	414.59	460.78	46.19	11.1%	15.4
7	3,500	603.57	650.40	46.82	7.8%	18.6	480.35	534.25	53.89	11.2%	15.3
8	4,000	686.94	740.45	53.51	7.8%	18.5	546.12	607.71	61.59	11.3%	15.2
9	4,500	770.31	830.51	60.20	7.8%	18.5	611.88	681.17	69.29	11.3%	15.1
10	5,000	853.67	920.56	66.89	7.8%	18.4	677.65	754.64	76.99	11.4%	15.1
11	6,000	1,020.41	1,100.68	80.27	7.9%	18.3	809.18	901.56	92.38	11.4%	15.0
12	7,000	1,187.14	1,280.79	93.65	7.9%	18.3	940.71	1,048.49	107.78	11.5%	15.0
13	8,000	1,353.88	1,460.90	107.03	7.9%	18.3	1,072.24	1,195.42	123.18	11.5%	14.9
14	9,000	1,520.61	1,641.02	120.41	7.9%	18.2	1,203.77	1,342.34	138.58	11.5%	14.9
15	10,000	1,687.35	1,821.13	133.78	7.9%	18.2	1,335.30	1,489.27	153.97	11.5%	14.9
16	11,000	1,854.08	2,001.24	147.16	7.9%	18.2	1,466.83	1,636.20	169.37	11.5%	14.9
17	12,000	2,020.81	2,181.36	160.54	7.9%	18.2	1,598.36	1,783.13	184.77	11.6%	14.9
18	13,000	2,187.55	2,361.47	173.92	8.0%	18.2	1,729.89	1,930.05	200.17	11.6%	14.8
19	14,000	2,354.28	2,541.58	187.30	8.0%	18.2	1,861.42	2,076.98	215.56	11.6%	14.8
20	15,000	2,521.02	2,721.69	200.68	8.0%	18.1	1,992.95	2,223.91	230.96	11.6%	14.8
21	16,000	2,687.75	2,901.81	214.05	8.0%	18.1	2,124.48	2,370.84	246.36	11.6%	14.8
22	17,000	2,854.49	3,081.92	227.43	8.0%	18.1	2,256.01	2,517.76	261.76	11.6%	14.8
23	18,000	3,021.22	3,262.03	240.81	8.0%	18.1	2,387.54	2,664.69	277.15	11.6%	14.8
24	19,000	3,187.96	3,442.15	254.19	8.0%	18.1	2,519.07	2,811.62	292.55	11.6%	14.8
25	20,000	3,354.69	3,622.26	267.57	8.0%	18.1	2,650.60	2,958.54	307.95	11.6%	14.8
26	21,000	3,521.43	3,802.37	280.95	8.0%	18.1	2,782.13	3,105.47	323.35	11.6%	14.8
27	22,000	3,688.16	3,982.49	294.33	8.0%	18.1	2,913.66	3,252.40	338.74	11.6%	14.8
28	23,000	3,854.89	4,162.60	307.70	8.0%	18.1	3,045.19	3,399.33	354.14	11.6%	14.8
29	24,000	4,021.63	4,342.71	321.08	8.0%	18.1	3,176.72	3,546.25	369.54	11.6%	14.8

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Time-of-Use GSTU

Secondary Interruptible Provision

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 19 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	94.87	101.73	6.86	7.2%	20.3	77.27	85.13	7.87	10.2%	17.0
2	1,000	169.74	183.45	13.71	8.1%	18.3	134.53	150.26	15.73	11.7%	15.0
3	1,500	244.61	265.18	20.57	8.4%	17.7	191.80	215.40	23.60	12.3%	14.4
4	2,000	319.47	346.90	27.43	8.6%	17.3	249.07	280.53	31.46	12.6%	14.0
5	2,500	394.34	428.63	34.28	8.7%	17.1	306.33	345.66	39.33	12.8%	13.8
6	3,000	469.21	510.35	41.14	8.8%	17.0	363.60	410.79	47.20	13.0%	13.7
7	3,500	544.08	592.08	48.00	8.8%	16.9	420.86	475.93	55.06	13.1%	13.6
8	4,000	618.95	673.80	54.85	8.9%	16.8	478.13	541.06	62.93	13.2%	13.5
9	4,500	693.82	755.53	61.71	8.9%	16.8	535.40	606.19	70.79	13.2%	13.5
10	5,000	768.69	837.25	68.56	8.9%	16.7	592.66	671.32	78.66	13.3%	13.4
11	6,000	918.42	1,000.70	82.28	9.0%	16.7	707.20	801.59	94.39	13.3%	13.4
12	7,000	1,068.16	1,164.15	95.99	9.0%	16.6	821.73	931.85	110.12	13.4%	13.3
13	8,000	1,217.90	1,327.60	109.70	9.0%	16.6	936.26	1,062.12	125.86	13.4%	13.3
14	9,000	1,367.64	1,491.05	123.42	9.0%	16.6	1,050.79	1,192.38	141.59	13.5%	13.2
15	10,000	1,517.37	1,654.50	137.13	9.0%	16.5	1,165.33	1,322.65	157.32	13.5%	13.2
16	11,000	1,667.11	1,817.95	150.84	9.0%	16.5	1,279.86	1,452.91	173.05	13.5%	13.2
17	12,000	1,816.85	1,981.40	164.55	9.1%	16.5	1,394.39	1,583.17	188.78	13.5%	13.2
18	13,000	1,966.59	2,144.85	178.27	9.1%	16.5	1,508.92	1,713.44	204.51	13.6%	13.2
19	14,000	2,116.32	2,308.30	191.98	9.1%	16.5	1,623.46	1,843.70	220.25	13.6%	13.2
20	15,000	2,266.06	2,471.75	205.69	9.1%	16.5	1,737.99	1,973.97	235.98	13.6%	13.2
21	16,000	2,415.80	2,635.20	219.41	9.1%	16.5	1,852.52	2,104.23	251.71	13.6%	13.2
22	17,000	2,565.53	2,798.65	233.12	9.1%	16.5	1,967.05	2,234.50	267.44	13.6%	13.1
23	18,000	2,715.27	2,962.10	246.83	9.1%	16.5	2,081.59	2,364.76	283.17	13.6%	13.1
24	19,000	2,865.01	3,125.55	260.55	9.1%	16.5	2,196.12	2,495.03	298.91	13.6%	13.1
25	20,000	3,014.75	3,289.00	274.26	9.1%	16.4	2,310.65	2,625.29	314.64	13.6%	13.1
26	21,000	3,164.48	3,452.46	287.97	9.1%	16.4	2,425.19	2,755.55	330.37	13.6%	13.1
27	22,000	3,314.22	3,615.91	301.68	9.1%	16.4	2,539.72	2,885.82	346.10	13.6%	13.1
28	23,000	3,463.96	3,779.36	315.40	9.1%	16.4	2,654.25	3,016.08	361.83	13.6%	13.1
29	24,000	3,613.70	3,942.81	329.11	9.1%	16.4	2,768.78	3,146.35	377.57	13.6%	13.1

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 1

Case No.: U-21389
Exhibit No.: A-16 (AMG-4)
Schedule F-4.0
Page 20 of 46
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	151.08	148.97	(2.10)	-1.4%	29.8	143.63	142.64	(0.98)	-0.7%	28.5
2	1,000	202.15	197.94	(4.21)	-2.1%	19.8	187.25	185.28	(1.97)	-1.1%	18.5
3	1,500	253.23	246.91	(6.31)	-2.5%	16.5	230.88	227.92	(2.95)	-1.3%	15.2
4	2,000	304.30	295.89	(8.42)	-2.8%	14.8	274.50	270.57	(3.94)	-1.4%	13.5
5	2,500	355.38	344.86	(10.52)	-3.0%	13.8	318.13	313.21	(4.92)	-1.5%	12.5
6	3,000	406.45	393.83	(12.62)	-3.1%	13.1	361.75	355.85	(5.90)	-1.6%	11.9
7	4,000	508.60	491.77	(16.83)	-3.3%	12.3	449.00	441.13	(7.87)	-1.8%	11.0
8	5,000	610.76	589.72	(21.04)	-3.4%	11.8	536.26	526.42	(9.84)	-1.8%	10.5
9	6,000	712.91	687.66	(25.25)	-3.5%	11.5	623.51	611.70	(11.81)	-1.9%	10.2
10	7,000	815.06	785.60	(29.46)	-3.6%	11.2	710.76	696.98	(13.78)	-1.9%	10.0
11	8,000	917.21	883.54	(33.66)	-3.7%	11.0	798.01	782.26	(15.74)	-2.0%	9.8
12	9,000	1,019.36	981.49	(37.87)	-3.7%	10.9	885.26	867.55	(17.71)	-2.0%	9.6
13	10,000	1,121.51	1,079.43	(42.08)	-3.8%	10.8	972.51	952.83	(19.68)	-2.0%	9.5
14	15,000	1,632.27	1,569.15	(63.12)	-3.9%	10.5	1,408.77	1,379.25	(29.52)	-2.1%	9.2
15	20,000	2,143.02	2,058.86	(84.16)	-3.9%	10.3	1,845.02	1,805.66	(39.36)	-2.1%	9.0
16	25,000	2,653.78	2,548.58	(105.20)	-4.0%	10.2	2,281.28	2,232.08	(49.20)	-2.2%	8.9
17	30,000	3,164.53	3,038.29	(126.24)	-4.0%	10.1	2,717.53	2,658.49	(59.04)	-2.2%	8.9
18	35,000	3,675.29	3,528.01	(147.28)	-4.0%	10.1	3,153.79	3,084.91	(68.88)	-2.2%	8.8
19	40,000	4,186.04	4,017.72	(168.32)	-4.0%	10.0	3,590.04	3,511.32	(78.72)	-2.2%	8.8
20	45,000	4,696.80	4,507.44	(189.36)	-4.0%	10.0	4,026.30	3,937.74	(88.56)	-2.2%	8.8
21	50,000	5,207.55	4,997.15	(210.40)	-4.0%	10.0	4,462.55	4,364.15	(98.40)	-2.2%	8.7
22	60,000	6,229.06	5,976.58	(252.48)	-4.1%	10.0	5,335.06	5,216.98	(118.08)	-2.2%	8.7
23	70,000	7,250.57	6,956.01	(294.56)	-4.1%	9.9	6,207.57	6,069.81	(137.76)	-2.2%	8.7
24	80,000	8,272.08	7,935.44	(336.64)	-4.1%	9.9	7,080.08	6,922.64	(157.44)	-2.2%	8.7
25	90,000	9,293.59	8,914.87	(378.72)	-4.1%	9.9	7,952.59	7,775.47	(177.12)	-2.2%	8.6
26	100,000	10,315.10	9,894.30	(420.80)	-4.1%	9.9	8,825.10	8,628.30	(196.80)	-2.2%	8.6
27	110,000	11,336.61	10,873.73	(462.88)	-4.1%	9.9	9,697.61	9,481.13	(216.48)	-2.2%	8.6
28	120,000	12,358.12	11,853.16	(504.96)	-4.1%	9.9	10,570.12	10,333.96	(236.16)	-2.2%	8.6
29	130,000	13,379.63	12,832.59	(547.04)	-4.1%	9.9	11,442.63	11,186.79	(255.84)	-2.2%	8.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 1

Education Provision GEI

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 21 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	150.81	148.65	(2.16)	-1.4%	29.7	143.36	142.32	(1.04)	-0.7%	28.5
2	1,000	201.62	197.31	(4.31)	-2.1%	19.7	186.72	184.65	(2.07)	-1.1%	18.5
3	1,500	252.43	245.96	(6.47)	-2.6%	16.4	230.08	226.97	(3.11)	-1.4%	15.1
4	2,000	303.24	294.61	(8.62)	-2.8%	14.7	273.44	269.29	(4.14)	-1.5%	13.5
5	2,500	354.05	343.27	(10.78)	-3.0%	13.7	316.80	311.62	(5.18)	-1.6%	12.5
6	3,000	404.85	391.92	(12.93)	-3.2%	13.1	360.15	353.94	(6.21)	-1.7%	11.8
7	4,000	506.47	489.23	(17.24)	-3.4%	12.2	446.87	438.59	(8.28)	-1.9%	11.0
8	5,000	608.09	586.54	(21.55)	-3.5%	11.7	533.59	523.24	(10.36)	-1.9%	10.5
9	6,000	709.71	683.84	(25.87)	-3.6%	11.4	620.31	607.88	(12.43)	-2.0%	10.1
10	7,000	811.33	781.15	(30.18)	-3.7%	11.2	707.03	692.53	(14.50)	-2.1%	9.9
11	8,000	912.94	878.46	(34.49)	-3.8%	11.0	793.74	777.18	(16.57)	-2.1%	9.7
12	9,000	1,014.56	975.76	(38.80)	-3.8%	10.8	880.46	861.82	(18.64)	-2.1%	9.6
13	10,000	1,116.18	1,073.07	(43.11)	-3.9%	10.7	967.18	946.47	(20.71)	-2.1%	9.5
14	15,000	1,624.27	1,559.61	(64.67)	-4.0%	10.4	1,400.77	1,369.71	(31.07)	-2.2%	9.1
15	20,000	2,132.36	2,046.14	(86.22)	-4.0%	10.2	1,834.36	1,792.94	(41.42)	-2.3%	9.0
16	25,000	2,640.45	2,532.68	(107.78)	-4.1%	10.1	2,267.95	2,216.18	(51.78)	-2.3%	8.9
17	30,000	3,148.54	3,019.21	(129.33)	-4.1%	10.1	2,701.54	2,639.41	(62.13)	-2.3%	8.8
18	35,000	3,656.63	3,505.75	(150.89)	-4.1%	10.0	3,135.13	3,062.65	(72.49)	-2.3%	8.8
19	40,000	4,164.72	3,992.28	(172.44)	-4.1%	10.0	3,568.72	3,485.88	(82.84)	-2.3%	8.7
20	45,000	4,672.81	4,478.82	(194.00)	-4.2%	10.0	4,002.31	3,909.12	(93.20)	-2.3%	8.7
21	50,000	5,180.90	4,965.35	(215.55)	-4.2%	9.9	4,435.90	4,332.35	(103.55)	-2.3%	8.7
22	60,000	6,197.08	5,938.42	(258.66)	-4.2%	9.9	5,303.08	5,178.82	(124.26)	-2.3%	8.6
23	70,000	7,213.26	6,911.49	(301.77)	-4.2%	9.9	6,170.26	6,025.29	(144.97)	-2.3%	8.6
24	80,000	8,229.44	7,884.56	(344.88)	-4.2%	9.9	7,037.44	6,871.76	(165.68)	-2.4%	8.6
25	90,000	9,245.62	8,857.63	(387.99)	-4.2%	9.8	7,904.62	7,718.23	(186.39)	-2.4%	8.6
26	100,000	10,261.80	9,830.70	(431.10)	-4.2%	9.8	8,771.80	8,564.70	(207.10)	-2.4%	8.6
27	110,000	11,277.98	10,803.77	(474.21)	-4.2%	9.8	9,638.98	9,411.17	(227.81)	-2.4%	8.6
28	120,000	12,294.16	11,776.84	(517.32)	-4.2%	9.8	10,506.16	10,257.64	(248.52)	-2.4%	8.5
29	130,000	13,310.34	12,749.91	(560.43)	-4.2%	9.8	11,373.34	11,104.11	(269.23)	-2.4%	8.5

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 2

Case No.: U-21389
Exhibit No.: A-16 (AMG-4)
Schedule F-4.0
Page 22 of 46
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	154.49	152.89	(1.61)	-1.0%	30.6	146.93	146.47	(0.46)	-0.3%	29.3
2	1,000	208.99	205.77	(3.21)	-1.5%	20.6	193.86	192.94	(0.92)	-0.5%	19.3
3	1,500	263.48	258.66	(4.82)	-1.8%	17.2	240.79	239.41	(1.38)	-0.6%	16.0
4	2,000	317.97	311.54	(6.43)	-2.0%	15.6	287.73	285.88	(1.84)	-0.6%	14.3
5	2,500	372.46	364.43	(8.03)	-2.2%	14.6	334.66	332.35	(2.31)	-0.7%	13.3
6	3,000	426.96	417.31	(9.64)	-2.3%	13.9	381.59	378.82	(2.77)	-0.7%	12.6
7	4,000	535.94	523.08	(12.86)	-2.4%	13.1	475.45	471.76	(3.69)	-0.8%	11.8
8	5,000	644.93	628.86	(16.07)	-2.5%	12.6	569.32	564.71	(4.61)	-0.8%	11.3
9	6,000	753.91	734.63	(19.28)	-2.6%	12.2	663.18	657.65	(5.53)	-0.8%	11.0
10	7,000	862.90	840.40	(22.50)	-2.6%	12.0	757.04	750.59	(6.45)	-0.9%	10.7
11	8,000	971.88	946.17	(25.71)	-2.6%	11.8	850.90	843.53	(7.38)	-0.9%	10.5
12	9,000	1,080.87	1,051.94	(28.93)	-2.7%	11.7	944.77	936.47	(8.30)	-0.9%	10.4
13	10,000	1,189.85	1,157.71	(32.14)	-2.7%	11.6	1,038.63	1,029.41	(9.22)	-0.9%	10.3
14	15,000	1,734.78	1,686.57	(48.21)	-2.8%	11.2	1,507.95	1,494.12	(13.83)	-0.9%	10.0
15	20,000	2,279.70	2,215.42	(64.28)	-2.8%	11.1	1,977.26	1,958.82	(18.44)	-0.9%	9.8
16	25,000	2,824.63	2,744.28	(80.35)	-2.8%	11.0	2,446.58	2,423.53	(23.05)	-0.9%	9.7
17	30,000	3,369.55	3,273.13	(96.42)	-2.9%	10.9	2,915.89	2,888.23	(27.66)	-0.9%	9.6
18	35,000	3,914.48	3,801.99	(112.49)	-2.9%	10.9	3,385.21	3,352.94	(32.27)	-1.0%	9.6
19	40,000	4,459.40	4,330.84	(128.56)	-2.9%	10.8	3,854.52	3,817.64	(36.88)	-1.0%	9.5
20	45,000	5,004.33	4,859.70	(144.63)	-2.9%	10.8	4,323.84	4,282.35	(41.49)	-1.0%	9.5
21	50,000	5,549.25	5,388.55	(160.70)	-2.9%	10.8	4,793.15	4,747.05	(46.10)	-1.0%	9.5
22	60,000	6,639.10	6,446.26	(192.84)	-2.9%	10.7	5,731.78	5,676.46	(55.32)	-1.0%	9.5
23	70,000	7,728.95	7,503.97	(224.98)	-2.9%	10.7	6,670.41	6,605.87	(64.54)	-1.0%	9.4
24	80,000	8,818.80	8,561.68	(257.12)	-2.9%	10.7	7,609.04	7,535.28	(73.76)	-1.0%	9.4
25	90,000	9,908.65	9,619.39	(289.26)	-2.9%	10.7	8,547.67	8,464.69	(82.98)	-1.0%	9.4
26	100,000	10,998.50	10,677.10	(321.40)	-2.9%	10.7	9,486.30	9,394.10	(92.20)	-1.0%	9.4
27	110,000	12,088.35	11,734.81	(353.54)	-2.9%	10.7	10,424.93	10,323.51	(101.42)	-1.0%	9.4
28	120,000	13,178.20	12,792.52	(385.68)	-2.9%	10.7	11,363.56	11,252.92	(110.64)	-1.0%	9.4
29	130,000	14,268.05	13,850.23	(417.82)	-2.9%	10.7	12,302.19	12,182.33	(119.86)	-1.0%	9.4

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 2

Education Provision GEI

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 23 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	154.23	152.57	(1.66)	-1.1%	30.5	146.67	146.15	(0.51)	-0.3%	29.2
2	1,000	208.45	205.14	(3.32)	-1.6%	20.5	193.33	192.31	(1.02)	-0.5%	19.2
3	1,500	262.68	257.70	(4.98)	-1.9%	17.2	240.00	238.46	(1.54)	-0.6%	15.9
4	2,000	316.90	310.27	(6.63)	-2.1%	15.5	286.66	284.61	(2.05)	-0.7%	14.2
5	2,500	371.13	362.84	(8.29)	-2.2%	14.5	333.33	330.76	(2.56)	-0.8%	13.2
6	3,000	425.36	415.41	(9.95)	-2.3%	13.8	379.99	376.92	(3.07)	-0.8%	12.6
7	4,000	533.81	520.54	(13.27)	-2.5%	13.0	473.32	469.22	(4.10)	-0.9%	11.7
8	5,000	642.26	625.68	(16.58)	-2.6%	12.5	566.65	561.53	(5.13)	-0.9%	11.2
9	6,000	750.71	730.81	(19.90)	-2.7%	12.2	659.98	653.83	(6.15)	-0.9%	10.9
10	7,000	859.16	835.95	(23.22)	-2.7%	11.9	753.31	746.14	(7.17)	-1.0%	10.7
11	8,000	967.62	941.08	(26.54)	-2.7%	11.8	846.64	838.44	(8.20)	-1.0%	10.5
12	9,000	1,076.07	1,046.22	(29.85)	-2.8%	11.6	939.97	930.75	(9.22)	-1.0%	10.3
13	10,000	1,184.52	1,151.35	(33.17)	-2.8%	11.5	1,033.30	1,023.05	(10.25)	-1.0%	10.2
14	15,000	1,726.78	1,677.03	(49.75)	-2.9%	11.2	1,499.95	1,484.58	(15.38)	-1.0%	9.9
15	20,000	2,269.04	2,202.70	(66.34)	-2.9%	11.0	1,966.60	1,946.10	(20.50)	-1.0%	9.7
16	25,000	2,811.30	2,728.38	(82.92)	-2.9%	10.9	2,433.25	2,407.63	(25.63)	-1.1%	9.6
17	30,000	3,353.56	3,254.05	(99.51)	-3.0%	10.8	2,899.90	2,869.15	(30.75)	-1.1%	9.6
18	35,000	3,895.82	3,779.73	(116.09)	-3.0%	10.8	3,366.55	3,330.68	(35.88)	-1.1%	9.5
19	40,000	4,438.08	4,305.40	(132.68)	-3.0%	10.8	3,833.20	3,792.20	(41.00)	-1.1%	9.5
20	45,000	4,980.34	4,831.08	(149.26)	-3.0%	10.7	4,299.85	4,253.73	(46.12)	-1.1%	9.5
21	50,000	5,522.60	5,356.75	(165.85)	-3.0%	10.7	4,766.50	4,715.25	(51.25)	-1.1%	9.4
22	60,000	6,607.12	6,408.10	(199.02)	-3.0%	10.7	5,699.80	5,638.30	(61.50)	-1.1%	9.4
23	70,000	7,691.64	7,459.45	(232.19)	-3.0%	10.7	6,633.10	6,561.35	(71.75)	-1.1%	9.4
24	80,000	8,776.16	8,510.80	(265.36)	-3.0%	10.6	7,566.40	7,484.40	(82.00)	-1.1%	9.4
25	90,000	9,860.68	9,562.15	(298.53)	-3.0%	10.6	8,499.70	8,407.45	(92.25)	-1.1%	9.3
26	100,000	10,945.20	10,613.50	(331.70)	-3.0%	10.6	9,433.00	9,330.50	(102.50)	-1.1%	9.3
27	110,000	12,029.72	11,664.85	(364.87)	-3.0%	10.6	10,366.30	10,253.55	(112.75)	-1.1%	9.3
28	120,000	13,114.24	12,716.20	(398.04)	-3.0%	10.6	11,299.60	11,176.60	(123.00)	-1.1%	9.3
29	130,000	14,198.76	13,767.55	(431.21)	-3.0%	10.6	12,232.90	12,099.65	(133.25)	-1.1%	9.3

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 3

Case No.: U-21389
Exhibit No.: A-16 (AMG-4)
Schedule F-4.0
Page 24 of 46
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	159.97	159.58	(0.39)	-0.2%	31.9	152.31	153.08	0.77	0.5%	30.6
2	1,000	219.94	219.16	(0.78)	-0.4%	21.9	204.62	206.15	1.54	0.8%	20.6
3	1,500	279.91	278.74	(1.17)	-0.4%	18.6	256.92	259.23	2.30	0.9%	17.3
4	2,000	339.88	338.32	(1.56)	-0.5%	16.9	309.23	312.30	3.07	1.0%	15.6
5	2,500	399.85	397.90	(1.94)	-0.5%	15.9	361.54	365.38	3.84	1.1%	14.6
6	3,000	459.82	457.48	(2.33)	-0.5%	15.2	413.85	418.45	4.60	1.1%	13.9
7	4,000	579.76	576.64	(3.11)	-0.5%	14.4	518.46	524.60	6.14	1.2%	13.1
8	5,000	699.70	695.81	(3.89)	-0.6%	13.9	623.08	630.76	7.67	1.2%	12.6
9	6,000	819.63	814.97	(4.67)	-0.6%	13.6	727.70	736.91	9.21	1.3%	12.3
10	7,000	939.57	934.13	(5.45)	-0.6%	13.3	832.31	843.06	10.75	1.3%	12.0
11	8,000	1,059.51	1,053.29	(6.22)	-0.6%	13.2	936.93	949.21	12.28	1.3%	11.9
12	9,000	1,179.45	1,172.45	(7.00)	-0.6%	13.0	1,041.54	1,055.36	13.82	1.3%	11.7
13	10,000	1,299.39	1,291.61	(7.78)	-0.6%	12.9	1,146.16	1,161.51	15.35	1.3%	11.6
14	15,000	1,899.09	1,887.42	(11.67)	-0.6%	12.6	1,669.24	1,692.27	23.02	1.4%	11.3
15	20,000	2,498.78	2,483.22	(15.56)	-0.6%	12.4	2,192.32	2,223.02	30.70	1.4%	11.1
16	25,000	3,098.48	3,079.03	(19.45)	-0.6%	12.3	2,715.40	2,753.78	38.38	1.4%	11.0
17	30,000	3,698.17	3,674.83	(23.34)	-0.6%	12.2	3,238.48	3,284.53	46.05	1.4%	10.9
18	35,000	4,297.87	4,270.64	(27.23)	-0.6%	12.2	3,761.56	3,815.29	53.72	1.4%	10.9
19	40,000	4,897.56	4,866.44	(31.12)	-0.6%	12.2	4,284.64	4,346.04	61.40	1.4%	10.9
20	45,000	5,497.26	5,462.25	(35.01)	-0.6%	12.1	4,807.72	4,876.80	69.07	1.4%	10.8
21	50,000	6,096.95	6,058.05	(38.90)	-0.6%	12.1	5,330.80	5,407.55	76.75	1.4%	10.8
22	60,000	7,296.34	7,249.66	(46.68)	-0.6%	12.1	6,376.96	6,469.06	92.10	1.4%	10.8
23	70,000	8,495.73	8,441.27	(54.46)	-0.6%	12.1	7,423.12	7,530.57	107.45	1.4%	10.8
24	80,000	9,695.12	9,632.88	(62.24)	-0.6%	12.0	8,469.28	8,592.08	122.80	1.4%	10.7
25	90,000	10,894.51	10,824.49	(70.02)	-0.6%	12.0	9,515.44	9,653.59	138.15	1.5%	10.7
26	100,000	12,093.90	12,016.10	(77.80)	-0.6%	12.0	10,561.60	10,715.10	153.50	1.5%	10.7
27	110,000	13,293.29	13,207.71	(85.58)	-0.6%	12.0	11,607.76	11,776.61	168.85	1.5%	10.7
28	120,000	14,492.68	14,399.32	(93.36)	-0.6%	12.0	12,653.92	12,838.12	184.20	1.5%	10.7
29	130,000	15,692.07	15,590.93	(101.14)	-0.6%	12.0	13,700.08	13,899.63	199.55	1.5%	10.7

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 3

Education Provision GEI

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 25 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	159.70	159.26	(0.44)	-0.3%	31.9	152.04	152.76	0.72	0.5%	30.6
2	1,000	219.41	218.53	(0.88)	-0.4%	21.9	204.08	205.52	1.43	0.7%	20.6
3	1,500	279.11	277.79	(1.32)	-0.5%	18.5	256.12	258.27	2.15	0.8%	17.2
4	2,000	338.81	337.05	(1.76)	-0.5%	16.9	308.17	311.03	2.86	0.9%	15.6
5	2,500	398.52	396.31	(2.20)	-0.6%	15.9	360.21	363.79	3.58	1.0%	14.6
6	3,000	458.22	455.58	(2.64)	-0.6%	15.2	412.25	416.55	4.30	1.0%	13.9
7	4,000	577.62	574.10	(3.52)	-0.6%	14.4	516.33	522.06	5.73	1.1%	13.1
8	5,000	697.03	692.63	(4.40)	-0.6%	13.9	620.42	627.58	7.16	1.2%	12.6
9	6,000	816.44	811.15	(5.29)	-0.6%	13.5	724.50	733.09	8.59	1.2%	12.2
10	7,000	935.84	929.68	(6.17)	-0.7%	13.3	828.58	838.61	10.02	1.2%	12.0
11	8,000	1,055.25	1,048.20	(7.05)	-0.7%	13.1	932.66	944.12	11.46	1.2%	11.8
12	9,000	1,174.65	1,166.73	(7.93)	-0.7%	13.0	1,036.75	1,049.64	12.89	1.2%	11.7
13	10,000	1,294.06	1,285.25	(8.81)	-0.7%	12.9	1,140.83	1,155.15	14.32	1.3%	11.6
14	15,000	1,891.09	1,877.88	(13.21)	-0.7%	12.5	1,661.25	1,682.73	21.48	1.3%	11.2
15	20,000	2,488.12	2,470.50	(17.62)	-0.7%	12.4	2,181.66	2,210.30	28.64	1.3%	11.1
16	25,000	3,085.15	3,063.13	(22.02)	-0.7%	12.3	2,702.08	2,737.88	35.80	1.3%	11.0
17	30,000	3,682.18	3,655.75	(26.43)	-0.7%	12.2	3,222.49	3,265.45	42.96	1.3%	10.9
18	35,000	4,279.21	4,248.38	(30.83)	-0.7%	12.1	3,742.91	3,793.03	50.12	1.3%	10.8
19	40,000	4,876.24	4,841.00	(35.24)	-0.7%	12.1	4,263.32	4,320.60	57.28	1.3%	10.8
20	45,000	5,473.27	5,433.63	(39.64)	-0.7%	12.1	4,783.74	4,848.18	64.44	1.3%	10.8
21	50,000	6,070.30	6,026.25	(44.05)	-0.7%	12.1	5,304.15	5,375.75	71.60	1.3%	10.8
22	60,000	7,264.36	7,211.50	(52.86)	-0.7%	12.0	6,344.98	6,430.90	85.92	1.4%	10.7
23	70,000	8,458.42	8,396.75	(61.67)	-0.7%	12.0	7,385.81	7,486.05	100.24	1.4%	10.7
24	80,000	9,652.48	9,582.00	(70.48)	-0.7%	12.0	8,426.64	8,541.20	114.56	1.4%	10.7
25	90,000	10,846.54	10,767.25	(79.29)	-0.7%	12.0	9,467.47	9,596.35	128.88	1.4%	10.7
26	100,000	12,040.60	11,952.50	(88.10)	-0.7%	12.0	10,508.30	10,651.50	143.20	1.4%	10.7
27	110,000	13,234.66	13,137.75	(96.91)	-0.7%	11.9	11,549.13	11,706.65	157.52	1.4%	10.6
28	120,000	14,428.72	14,323.00	(105.72)	-0.7%	11.9	12,589.96	12,761.80	171.84	1.4%	10.6
29	130,000	15,622.78	15,508.25	(114.53)	-0.7%	11.9	13,630.79	13,816.95	186.16	1.4%	10.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 1

Case No.: U-21389
Exhibit No.: A-16 (AMG-4)
Schedule F-4.0
Page 26 of 46
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	7.50	7.65	0.15	2.0%	7.6	7.11	7.09	(0.03)	-0.4%	7.1
2	110	8.22	8.39	0.16	2.0%	7.6	7.81	7.78	(0.03)	-0.4%	7.1
3	120	8.95	9.13	0.18	2.0%	7.6	8.50	8.47	(0.03)	-0.4%	7.1
4	130	9.68	9.88	0.19	2.0%	7.6	9.19	9.16	(0.03)	-0.4%	7.0
5	140	10.41	10.62	0.21	2.0%	7.6	9.88	9.85	(0.04)	-0.4%	7.0
6	150	11.14	11.37	0.22	2.0%	7.6	10.57	10.53	(0.04)	-0.4%	7.0
7	200	14.79	15.09	0.30	2.0%	7.5	14.03	13.98	(0.05)	-0.4%	7.0
8	250	18.44	18.81	0.37	2.0%	7.5	17.49	17.42	(0.06)	-0.4%	7.0
9	300	22.09	22.54	0.45	2.0%	7.5	20.94	20.87	(0.08)	-0.4%	7.0
10	350	25.73	26.26	0.52	2.0%	7.5	24.40	24.31	(0.09)	-0.4%	6.9
11	400	29.38	29.98	0.60	2.0%	7.5	27.86	27.76	(0.10)	-0.4%	6.9
12	450	33.03	33.70	0.67	2.0%	7.5	31.32	31.20	(0.11)	-0.4%	6.9
13	500	36.68	37.43	0.75	2.0%	7.5	34.77	34.65	(0.13)	-0.4%	6.9
14	600	43.97	44.87	0.90	2.0%	7.5	41.69	41.54	(0.15)	-0.4%	6.9
15	700	51.27	52.32	1.05	2.0%	7.5	48.60	48.43	(0.18)	-0.4%	6.9
16	800	58.56	59.76	1.20	2.0%	7.5	55.52	55.32	(0.20)	-0.4%	6.9
17	900	65.86	67.21	1.35	2.0%	7.5	62.43	62.20	(0.23)	-0.4%	6.9
18	1,000	73.15	74.65	1.50	2.1%	7.5	69.35	69.09	(0.25)	-0.4%	6.9
19	1,500	109.63	111.88	2.25	2.1%	7.5	103.92	103.54	(0.38)	-0.4%	6.9
20	2,000	146.10	149.10	3.00	2.1%	7.5	138.50	137.99	(0.51)	-0.4%	6.9
21	2,500	182.58	186.33	3.75	2.1%	7.5	173.07	172.44	(0.63)	-0.4%	6.9
22	3,000	219.05	223.55	4.50	2.1%	7.5	207.64	206.88	(0.76)	-0.4%	6.9
23	3,500	255.53	260.78	5.25	2.1%	7.5	242.22	241.33	(0.89)	-0.4%	6.9
24	4,000	292.00	298.00	6.00	2.1%	7.5	276.79	275.78	(1.01)	-0.4%	6.9
25	4,500	328.48	335.23	6.75	2.1%	7.4	311.36	310.22	(1.14)	-0.4%	6.9
26	5,000	364.95	372.45	7.50	2.1%	7.4	345.94	344.67	(1.27)	-0.4%	6.9
27	5,500	401.43	409.68	8.25	2.1%	7.4	380.51	379.12	(1.39)	-0.4%	6.9
28	6,000	437.90	446.90	9.00	2.1%	7.4	415.09	413.57	(1.52)	-0.4%	6.9
29	6,500	474.38	484.13	9.75	2.1%	7.4	449.66	448.01	(1.65)	-0.4%	6.9

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 1

Primary Interruptible Provision GI

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 27 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	6.42	6.57	0.15	2.3%	6.6	6.19	6.17	(0.03)	-0.4%	6.2
2	110	7.04	7.21	0.16	2.3%	6.6	6.79	6.76	(0.03)	-0.4%	6.1
3	120	7.66	7.84	0.18	2.3%	6.5	7.39	7.36	(0.03)	-0.4%	6.1
4	130	8.28	8.48	0.19	2.4%	6.5	7.99	7.96	(0.03)	-0.4%	6.1
5	140	8.91	9.12	0.21	2.4%	6.5	8.59	8.55	(0.04)	-0.4%	6.1
6	150	9.53	9.75	0.22	2.4%	6.5	9.19	9.15	(0.04)	-0.4%	6.1
7	200	12.64	12.94	0.30	2.4%	6.5	12.18	12.13	(0.05)	-0.4%	6.1
8	250	15.75	16.12	0.37	2.4%	6.4	15.18	15.12	(0.06)	-0.4%	6.0
9	300	18.86	19.31	0.45	2.4%	6.4	18.18	18.10	(0.08)	-0.4%	6.0
10	350	21.97	22.49	0.52	2.4%	6.4	21.17	21.08	(0.09)	-0.4%	6.0
11	400	25.08	25.68	0.60	2.4%	6.4	24.17	24.07	(0.10)	-0.4%	6.0
12	450	28.18	28.86	0.67	2.4%	6.4	27.17	27.05	(0.11)	-0.4%	6.0
13	500	31.29	32.04	0.75	2.4%	6.4	30.16	30.03	(0.13)	-0.4%	6.0
14	600	37.51	38.41	0.90	2.4%	6.4	36.15	36.00	(0.15)	-0.4%	6.0
15	700	43.73	44.78	1.05	2.4%	6.4	42.15	41.97	(0.18)	-0.4%	6.0
16	800	49.95	51.15	1.20	2.4%	6.4	48.14	47.94	(0.20)	-0.4%	6.0
17	900	56.17	57.52	1.35	2.4%	6.4	54.13	53.90	(0.23)	-0.4%	6.0
18	1,000	62.39	63.89	1.50	2.4%	6.4	60.12	59.87	(0.25)	-0.4%	6.0
19	1,500	93.48	95.73	2.25	2.4%	6.4	90.08	89.70	(0.38)	-0.4%	6.0
20	2,000	124.58	127.58	3.00	2.4%	6.4	120.05	119.54	(0.51)	-0.4%	6.0
21	2,500	155.67	159.42	3.75	2.4%	6.4	150.01	149.37	(0.63)	-0.4%	6.0
22	3,000	186.77	191.26	4.50	2.4%	6.4	179.97	179.21	(0.76)	-0.4%	6.0
23	3,500	217.86	223.11	5.25	2.4%	6.4	209.93	209.04	(0.89)	-0.4%	6.0
24	4,000	248.95	254.95	6.00	2.4%	6.4	239.89	238.88	(1.01)	-0.4%	6.0
25	4,500	280.05	286.80	6.75	2.4%	6.4	269.85	268.71	(1.14)	-0.4%	6.0
26	5,000	311.14	318.64	7.50	2.4%	6.4	299.82	298.55	(1.27)	-0.4%	6.0
27	5,500	342.24	350.49	8.25	2.4%	6.4	329.78	328.38	(1.39)	-0.4%	6.0
28	6,000	373.33	382.33	9.00	2.4%	6.4	359.74	358.22	(1.52)	-0.4%	6.0
29	6,500	404.43	414.17	9.75	2.4%	6.4	389.70	388.05	(1.65)	-0.4%	6.0

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 2

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 28 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	8.85	9.05	0.21	2.3%	9.1	8.32	8.34	0.02	0.2%	8.3
2	110	9.71	9.94	0.23	2.4%	9.0	9.13	9.16	0.02	0.2%	8.3
3	120	10.58	10.82	0.25	2.4%	9.0	9.95	9.97	0.02	0.2%	8.3
4	130	11.44	11.71	0.27	2.4%	9.0	10.76	10.78	0.03	0.2%	8.3
5	140	12.30	12.60	0.29	2.4%	9.0	11.57	11.60	0.03	0.2%	8.3
6	150	13.17	13.48	0.31	2.4%	9.0	12.38	12.41	0.03	0.2%	8.3
7	200	17.49	17.91	0.42	2.4%	9.0	16.44	16.48	0.04	0.2%	8.2
8	250	21.82	22.33	0.52	2.4%	8.9	20.50	20.55	0.05	0.2%	8.2
9	300	26.14	26.76	0.62	2.4%	8.9	24.56	24.62	0.06	0.2%	8.2
10	350	30.46	31.19	0.73	2.4%	8.9	28.62	28.69	0.07	0.2%	8.2
11	400	34.78	35.62	0.83	2.4%	8.9	32.68	32.76	0.08	0.2%	8.2
12	450	39.11	40.04	0.94	2.4%	8.9	36.74	36.83	0.09	0.2%	8.2
13	500	43.43	44.47	1.04	2.4%	8.9	40.80	40.91	0.10	0.2%	8.2
14	600	52.08	53.32	1.25	2.4%	8.9	48.93	49.05	0.12	0.2%	8.2
15	700	60.72	62.18	1.45	2.4%	8.9	57.05	57.19	0.14	0.2%	8.2
16	800	69.37	71.03	1.66	2.4%	8.9	65.17	65.33	0.16	0.2%	8.2
17	900	78.02	79.89	1.87	2.4%	8.9	73.29	73.47	0.18	0.2%	8.2
18	1,000	86.66	88.74	2.08	2.4%	8.9	81.41	81.61	0.20	0.2%	8.2
19	1,500	129.89	133.01	3.12	2.4%	8.9	122.01	122.32	0.30	0.2%	8.2
20	2,000	173.12	177.28	4.16	2.4%	8.9	162.62	163.02	0.40	0.2%	8.2
21	2,500	216.35	221.55	5.20	2.4%	8.9	203.22	203.73	0.50	0.2%	8.1
22	3,000	259.58	265.82	6.23	2.4%	8.9	243.83	244.43	0.60	0.2%	8.1
23	3,500	302.81	310.09	7.27	2.4%	8.9	284.43	285.14	0.70	0.2%	8.1
24	4,000	346.05	354.36	8.31	2.4%	8.9	325.04	325.84	0.80	0.2%	8.1
25	4,500	389.28	398.63	9.35	2.4%	8.9	365.64	366.55	0.90	0.2%	8.1
26	5,000	432.51	442.90	10.39	2.4%	8.9	406.25	407.25	1.01	0.2%	8.1
27	5,500	475.74	487.17	11.43	2.4%	8.9	446.85	447.96	1.11	0.2%	8.1
28	6,000	518.97	531.44	12.47	2.4%	8.9	487.46	488.66	1.21	0.2%	8.1
29	6,500	562.20	575.71	13.51	2.4%	8.9	528.06	529.37	1.31	0.2%	8.1

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 2

Primary Interruptible Provision GI

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 29 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	7.57	7.78	0.21	2.7%	7.8	7.23	7.25	0.02	0.3%	7.2
2	110	8.31	8.53	0.23	2.8%	7.8	7.93	7.95	0.02	0.3%	7.2
3	120	9.04	9.29	0.25	2.8%	7.7	8.63	8.65	0.02	0.3%	7.2
4	130	9.78	10.05	0.27	2.8%	7.7	9.33	9.36	0.03	0.3%	7.2
5	140	10.52	10.81	0.29	2.8%	7.7	10.04	10.06	0.03	0.3%	7.2
6	150	11.25	11.56	0.31	2.8%	7.7	10.74	10.77	0.03	0.3%	7.2
7	200	14.94	15.35	0.42	2.8%	7.7	14.25	14.29	0.04	0.3%	7.1
8	250	18.62	19.14	0.52	2.8%	7.7	17.76	17.81	0.05	0.3%	7.1
9	300	22.30	22.93	0.62	2.8%	7.6	21.28	21.34	0.06	0.3%	7.1
10	350	25.99	26.72	0.73	2.8%	7.6	24.79	24.86	0.07	0.3%	7.1
11	400	29.67	30.50	0.83	2.8%	7.6	28.30	28.38	0.08	0.3%	7.1
12	450	33.36	34.29	0.94	2.8%	7.6	31.82	31.91	0.09	0.3%	7.1
13	500	37.04	38.08	1.04	2.8%	7.6	35.33	35.43	0.10	0.3%	7.1
14	600	44.41	45.66	1.25	2.8%	7.6	42.35	42.47	0.12	0.3%	7.1
15	700	51.78	53.23	1.45	2.8%	7.6	49.38	49.52	0.14	0.3%	7.1
16	800	59.15	60.81	1.66	2.8%	7.6	56.41	56.57	0.16	0.3%	7.1
17	900	66.51	68.38	1.87	2.8%	7.6	63.43	63.61	0.18	0.3%	7.1
18	1,000	73.88	75.96	2.08	2.8%	7.6	70.46	70.66	0.20	0.3%	7.1
19	1,500	110.72	113.84	3.12	2.8%	7.6	105.58	105.89	0.30	0.3%	7.1
20	2,000	147.57	151.72	4.16	2.8%	7.6	140.71	141.11	0.40	0.3%	7.1
21	2,500	184.41	189.60	5.20	2.8%	7.6	175.84	176.34	0.50	0.3%	7.1
22	3,000	221.25	227.48	6.23	2.8%	7.6	210.97	211.57	0.60	0.3%	7.1
23	3,500	258.09	265.36	7.27	2.8%	7.6	246.10	246.80	0.70	0.3%	7.1
24	4,000	294.93	303.24	8.31	2.8%	7.6	281.23	282.03	0.80	0.3%	7.1
25	4,500	331.77	341.12	9.35	2.8%	7.6	316.35	317.26	0.90	0.3%	7.1
26	5,000	368.61	379.01	10.39	2.8%	7.6	351.48	352.49	1.01	0.3%	7.0
27	5,500	405.46	416.89	11.43	2.8%	7.6	386.61	387.72	1.11	0.3%	7.0
28	6,000	442.30	454.77	12.47	2.8%	7.6	421.74	422.94	1.21	0.3%	7.0
29	6,500	479.14	492.65	13.51	2.8%	7.6	456.87	458.17	1.31	0.3%	7.0

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 3

Case No.: U-21389
Exhibit No.: A-16 (AMG-4)
Schedule F-4.0
Page 30 of 46
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	9.35	9.64	0.30	3.2%	9.6	8.81	8.92	0.11	1.2%	8.9
2	110	10.26	10.59	0.33	3.2%	9.6	9.68	9.79	0.12	1.2%	8.9
3	120	11.18	11.53	0.36	3.2%	9.6	10.54	10.66	0.13	1.2%	8.9
4	130	12.09	12.48	0.39	3.2%	9.6	11.40	11.54	0.14	1.2%	8.9
5	140	13.01	13.42	0.42	3.2%	9.6	12.26	12.41	0.15	1.2%	8.9
6	150	13.92	14.37	0.45	3.2%	9.6	13.12	13.28	0.16	1.2%	8.9
7	200	18.49	19.09	0.59	3.2%	9.5	17.43	17.64	0.21	1.2%	8.8
8	250	23.07	23.81	0.74	3.2%	9.5	21.73	22.00	0.27	1.2%	8.8
9	300	27.64	28.53	0.89	3.2%	9.5	26.04	26.36	0.32	1.2%	8.8
10	350	32.21	33.25	1.04	3.2%	9.5	30.35	30.72	0.37	1.2%	8.8
11	400	36.79	37.98	1.19	3.2%	9.5	34.66	35.08	0.42	1.2%	8.8
12	450	41.36	42.70	1.34	3.2%	9.5	38.96	39.44	0.48	1.2%	8.8
13	500	45.93	47.42	1.49	3.2%	9.5	43.27	43.80	0.53	1.2%	8.8
14	600	55.08	56.86	1.78	3.2%	9.5	51.88	52.52	0.64	1.2%	8.8
15	700	64.23	66.31	2.08	3.2%	9.5	60.50	61.24	0.74	1.2%	8.7
16	800	73.37	75.75	2.38	3.2%	9.5	69.11	69.96	0.85	1.2%	8.7
17	900	82.52	85.20	2.68	3.2%	9.5	77.72	78.68	0.96	1.2%	8.7
18	1,000	91.67	94.64	2.97	3.2%	9.5	86.34	87.40	1.06	1.2%	8.7
19	1,500	137.40	141.86	4.46	3.2%	9.5	129.41	131.00	1.59	1.2%	8.7
20	2,000	183.13	189.08	5.95	3.2%	9.5	172.48	174.60	2.12	1.2%	8.7
21	2,500	228.87	236.30	7.43	3.2%	9.5	215.54	218.20	2.66	1.2%	8.7
22	3,000	274.60	283.52	8.92	3.2%	9.5	258.61	261.80	3.19	1.2%	8.7
23	3,500	320.33	330.74	10.41	3.2%	9.4	301.68	305.40	3.72	1.2%	8.7
24	4,000	366.06	377.96	11.90	3.2%	9.4	344.75	349.00	4.25	1.2%	8.7
25	4,500	411.80	425.18	13.38	3.2%	9.4	387.82	392.60	4.78	1.2%	8.7
26	5,000	457.53	472.40	14.87	3.2%	9.4	430.89	436.20	5.31	1.2%	8.7
27	5,500	503.26	519.62	16.36	3.2%	9.4	473.96	479.80	5.84	1.2%	8.7
28	6,000	549.00	566.84	17.84	3.3%	9.4	517.03	523.40	6.37	1.2%	8.7
29	6,500	594.73	614.06	19.33	3.3%	9.4	560.09	567.00	6.91	1.2%	8.7

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 3

Primary Interruptible Provision GI

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 31 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	8.08	8.38	0.30	3.7%	8.4	7.73	7.84	0.11	1.4%	7.8
2	110	8.87	9.20	0.33	3.7%	8.4	8.48	8.60	0.12	1.4%	7.8
3	120	9.66	10.01	0.36	3.7%	8.3	9.24	9.36	0.13	1.4%	7.8
4	130	10.45	10.83	0.39	3.7%	8.3	9.99	10.13	0.14	1.4%	7.8
5	140	11.23	11.65	0.42	3.7%	8.3	10.74	10.89	0.15	1.4%	7.8
6	150	12.02	12.47	0.45	3.7%	8.3	11.49	11.65	0.16	1.4%	7.8
7	200	15.96	16.56	0.59	3.7%	8.3	15.26	15.47	0.21	1.4%	7.7
8	250	19.90	20.65	0.74	3.7%	8.3	19.02	19.29	0.27	1.4%	7.7
9	300	23.84	24.74	0.89	3.7%	8.2	22.79	23.11	0.32	1.4%	7.7
10	350	27.79	28.83	1.04	3.7%	8.2	26.55	26.92	0.37	1.4%	7.7
11	400	31.73	32.92	1.19	3.7%	8.2	30.32	30.74	0.42	1.4%	7.7
12	450	35.67	37.01	1.34	3.8%	8.2	34.08	34.56	0.48	1.4%	7.7
13	500	39.61	41.09	1.49	3.8%	8.2	37.85	38.38	0.53	1.4%	7.7
14	600	47.49	49.27	1.78	3.8%	8.2	45.38	46.01	0.64	1.4%	7.7
15	700	55.37	57.45	2.08	3.8%	8.2	52.91	53.65	0.74	1.4%	7.7
16	800	63.25	65.63	2.38	3.8%	8.2	60.44	61.29	0.85	1.4%	7.7
17	900	71.13	73.81	2.68	3.8%	8.2	67.96	68.92	0.96	1.4%	7.7
18	1,000	79.02	81.99	2.97	3.8%	8.2	75.49	76.56	1.06	1.4%	7.7
19	1,500	118.42	122.88	4.46	3.8%	8.2	113.14	114.73	1.59	1.4%	7.6
20	2,000	157.83	163.78	5.95	3.8%	8.2	150.79	152.91	2.12	1.4%	7.6
21	2,500	197.24	204.67	7.43	3.8%	8.2	188.44	191.09	2.66	1.4%	7.6
22	3,000	236.65	245.57	8.92	3.8%	8.2	226.08	229.27	3.19	1.4%	7.6
23	3,500	276.05	286.46	10.41	3.8%	8.2	263.73	267.45	3.72	1.4%	7.6
24	4,000	315.46	327.36	11.90	3.8%	8.2	301.38	305.63	4.25	1.4%	7.6
25	4,500	354.87	368.25	13.38	3.8%	8.2	339.02	343.80	4.78	1.4%	7.6
26	5,000	394.28	409.15	14.87	3.8%	8.2	376.67	381.98	5.31	1.4%	7.6
27	5,500	433.68	450.04	16.36	3.8%	8.2	414.32	420.16	5.84	1.4%	7.6
28	6,000	473.09	490.93	17.84	3.8%	8.2	451.96	458.34	6.37	1.4%	7.6
29	6,500	512.50	531.83	19.33	3.8%	8.2	489.61	496.52	6.91	1.4%	7.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Time-of-Use GPTU Voltage Level 1

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 32 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	9.19	9.16	(0.02)	-0.3%	9.2	8.44	8.50	0.06	0.7%	8.5
2	110	10.09	10.06	(0.03)	-0.3%	9.1	9.26	9.33	0.06	0.7%	8.5
3	120	10.98	10.95	(0.03)	-0.3%	9.1	10.09	10.16	0.07	0.7%	8.5
4	130	11.88	11.85	(0.03)	-0.3%	9.1	10.91	10.99	0.07	0.7%	8.5
5	140	12.78	12.75	(0.03)	-0.3%	9.1	11.74	11.82	0.08	0.7%	8.4
6	150	13.68	13.64	(0.04)	-0.3%	9.1	12.56	12.65	0.09	0.7%	8.4
7	200	18.17	18.12	(0.05)	-0.3%	9.1	16.68	16.80	0.12	0.7%	8.4
8	250	22.67	22.61	(0.06)	-0.3%	9.0	20.80	20.95	0.14	0.7%	8.4
9	300	27.16	27.09	(0.07)	-0.3%	9.0	24.92	25.09	0.17	0.7%	8.4
10	350	31.65	31.57	(0.09)	-0.3%	9.0	29.04	29.24	0.20	0.7%	8.4
11	400	36.15	36.05	(0.10)	-0.3%	9.0	33.16	33.39	0.23	0.7%	8.3
12	450	40.64	40.53	(0.11)	-0.3%	9.0	37.28	37.54	0.26	0.7%	8.3
13	500	45.13	45.01	(0.12)	-0.3%	9.0	41.40	41.69	0.29	0.7%	8.3
14	600	54.12	53.97	(0.15)	-0.3%	9.0	49.64	49.99	0.35	0.7%	8.3
15	700	63.11	62.93	(0.17)	-0.3%	9.0	57.88	58.29	0.40	0.7%	8.3
16	800	72.10	71.90	(0.20)	-0.3%	9.0	66.13	66.59	0.46	0.7%	8.3
17	900	81.08	80.86	(0.22)	-0.3%	9.0	74.37	74.88	0.52	0.7%	8.3
18	1,000	90.07	89.82	(0.25)	-0.3%	9.0	82.61	83.18	0.58	0.7%	8.3
19	1,500	135.00	134.63	(0.37)	-0.3%	9.0	123.81	124.67	0.86	0.7%	8.3
20	2,000	179.94	179.44	(0.50)	-0.3%	9.0	165.01	166.16	1.15	0.7%	8.3
21	2,500	224.87	224.25	(0.62)	-0.3%	9.0	206.22	207.65	1.44	0.7%	8.3
22	3,000	269.81	269.06	(0.75)	-0.3%	9.0	247.42	249.15	1.73	0.7%	8.3
23	3,500	314.74	313.87	(0.87)	-0.3%	9.0	288.62	290.64	2.01	0.7%	8.3
24	4,000	359.68	358.69	(0.99)	-0.3%	9.0	329.83	332.13	2.30	0.7%	8.3
25	4,500	404.61	403.50	(1.12)	-0.3%	9.0	371.03	373.62	2.59	0.7%	8.3
26	5,000	449.55	448.31	(1.24)	-0.3%	9.0	412.23	415.11	2.88	0.7%	8.3
27	5,500	494.48	493.12	(1.37)	-0.3%	9.0	453.44	456.60	3.16	0.7%	8.3
28	6,000	539.42	537.93	(1.49)	-0.3%	9.0	494.64	498.09	3.45	0.7%	8.3
29	6,500	584.35	582.74	(1.62)	-0.3%	9.0	535.84	539.58	3.74	0.7%	8.3

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Time-of-Use GPTU Voltage Level 2

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 33 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	9.74	9.96	0.21	2.2%	10.0	9.06	9.21	0.15	1.6%	9.2
2	110	10.70	10.93	0.23	2.2%	9.9	9.94	10.11	0.16	1.6%	9.2
3	120	11.65	11.91	0.25	2.2%	9.9	10.83	11.01	0.18	1.6%	9.2
4	130	12.61	12.88	0.27	2.2%	9.9	11.71	11.91	0.19	1.7%	9.2
5	140	13.56	13.86	0.29	2.2%	9.9	12.60	12.81	0.21	1.7%	9.1
6	150	14.52	14.83	0.32	2.2%	9.9	13.49	13.71	0.22	1.7%	9.1
7	200	19.29	19.71	0.42	2.2%	9.9	17.91	18.21	0.30	1.7%	9.1
8	250	24.06	24.59	0.53	2.2%	9.8	22.34	22.72	0.37	1.7%	9.1
9	300	28.83	29.47	0.63	2.2%	9.8	26.77	27.22	0.45	1.7%	9.1
10	350	33.61	34.34	0.74	2.2%	9.8	31.20	31.72	0.52	1.7%	9.1
11	400	38.38	39.22	0.84	2.2%	9.8	35.63	36.22	0.60	1.7%	9.1
12	450	43.15	44.10	0.95	2.2%	9.8	40.06	40.73	0.67	1.7%	9.1
13	500	47.92	48.98	1.05	2.2%	9.8	44.49	45.23	0.74	1.7%	9.0
14	600	57.47	58.73	1.26	2.2%	9.8	53.34	54.24	0.89	1.7%	9.0
15	700	67.01	68.49	1.47	2.2%	9.8	62.20	63.24	1.04	1.7%	9.0
16	800	76.56	78.24	1.68	2.2%	9.8	71.06	72.25	1.19	1.7%	9.0
17	900	86.10	88.00	1.89	2.2%	9.8	79.92	81.26	1.34	1.7%	9.0
18	1,000	95.65	97.75	2.10	2.2%	9.8	88.77	90.26	1.49	1.7%	9.0
19	1,500	143.37	146.53	3.16	2.2%	9.8	133.06	135.29	2.23	1.7%	9.0
20	2,000	191.09	195.30	4.21	2.2%	9.8	177.35	180.32	2.98	1.7%	9.0
21	2,500	238.82	244.08	5.26	2.2%	9.8	221.63	225.35	3.72	1.7%	9.0
22	3,000	286.54	292.86	6.31	2.2%	9.8	265.92	270.39	4.47	1.7%	9.0
23	3,500	334.26	341.63	7.37	2.2%	9.8	310.20	315.42	5.21	1.7%	9.0
24	4,000	381.99	390.41	8.42	2.2%	9.8	354.49	360.45	5.96	1.7%	9.0
25	4,500	429.71	439.18	9.47	2.2%	9.8	398.78	405.48	6.70	1.7%	9.0
26	5,000	477.44	487.96	10.52	2.2%	9.8	443.06	450.51	7.44	1.7%	9.0
27	5,500	525.16	536.74	11.58	2.2%	9.8	487.35	495.54	8.19	1.7%	9.0
28	6,000	572.88	585.51	12.63	2.2%	9.8	531.64	540.57	8.93	1.7%	9.0
29	6,500	620.61	634.29	13.68	2.2%	9.8	575.92	585.60	9.68	1.7%	9.0

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Time-of-Use GPTU Voltage Level 3

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 34 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	10.81	11.15	0.34	3.1%	11.1	10.02	10.32	0.30	3.0%	10.3
2	110	11.87	12.24	0.37	3.1%	11.1	11.00	11.33	0.33	3.0%	10.3
3	120	12.93	13.34	0.40	3.1%	11.1	11.98	12.35	0.37	3.0%	10.3
4	130	13.99	14.43	0.44	3.1%	11.1	12.96	13.36	0.40	3.1%	10.3
5	140	15.05	15.53	0.47	3.1%	11.1	13.94	14.37	0.43	3.1%	10.3
6	150	16.12	16.62	0.51	3.1%	11.1	14.93	15.38	0.46	3.1%	10.3
7	200	21.42	22.09	0.67	3.1%	11.0	19.84	20.44	0.61	3.1%	10.2
8	250	26.73	27.57	0.84	3.2%	11.0	24.74	25.50	0.76	3.1%	10.2
9	300	32.03	33.04	1.01	3.2%	11.0	29.65	30.57	0.91	3.1%	10.2
10	350	37.34	38.52	1.18	3.2%	11.0	34.56	35.63	1.06	3.1%	10.2
11	400	42.64	43.99	1.35	3.2%	11.0	39.47	40.69	1.22	3.1%	10.2
12	450	47.95	49.46	1.52	3.2%	11.0	44.38	45.75	1.37	3.1%	10.2
13	500	53.25	54.94	1.69	3.2%	11.0	49.29	50.81	1.52	3.1%	10.2
14	600	63.86	65.88	2.02	3.2%	11.0	59.11	60.93	1.83	3.1%	10.2
15	700	74.47	76.83	2.36	3.2%	11.0	68.92	71.05	2.13	3.1%	10.2
16	800	85.08	87.78	2.70	3.2%	11.0	78.74	81.17	2.43	3.1%	10.1
17	900	95.69	98.73	3.04	3.2%	11.0	88.56	91.30	2.74	3.1%	10.1
18	1,000	106.30	109.67	3.37	3.2%	11.0	98.38	101.42	3.04	3.1%	10.1
19	1,500	159.35	164.41	5.06	3.2%	11.0	147.46	152.03	4.56	3.1%	10.1
20	2,000	212.40	219.15	6.75	3.2%	11.0	196.55	202.63	6.08	3.1%	10.1
21	2,500	265.45	273.88	8.43	3.2%	11.0	245.64	253.24	7.60	3.1%	10.1
22	3,000	318.50	328.62	10.12	3.2%	11.0	294.73	303.85	9.13	3.1%	10.1
23	3,500	371.55	383.36	11.80	3.2%	11.0	343.81	354.46	10.65	3.1%	10.1
24	4,000	424.60	438.09	13.49	3.2%	11.0	392.90	405.07	12.17	3.1%	10.1
25	4,500	477.65	492.83	15.18	3.2%	11.0	441.99	455.68	13.69	3.1%	10.1
26	5,000	530.70	547.57	16.86	3.2%	11.0	491.08	506.28	15.21	3.1%	10.1
27	5,500	583.75	602.30	18.55	3.2%	11.0	540.16	556.89	16.73	3.1%	10.1
28	6,000	636.80	657.04	20.24	3.2%	11.0	589.25	607.50	18.25	3.1%	10.1
29	6,500	689.85	711.78	21.92	3.2%	11.0	638.34	658.11	19.77	3.1%	10.1

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy Intensive EIP Voltage Level 1

Case No.: U-21389
Exhibit No.: A-16 (AMG-4)
Schedule F-4.0
Page 35 of 46
Witness: AMGast
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	6.94	7.85	0.91	13.2%	7.9	6.46	7.08	0.61	9.5%	7.1
2	110	7.61	8.62	1.00	13.2%	7.8	7.09	7.77	0.67	9.5%	7.1
3	120	8.28	9.38	1.10	13.2%	7.8	7.72	8.45	0.74	9.5%	7.0
4	130	8.96	10.15	1.19	13.2%	7.8	8.34	9.14	0.80	9.6%	7.0
5	140	9.63	10.91	1.28	13.3%	7.8	8.97	9.83	0.86	9.6%	7.0
6	150	10.31	11.68	1.37	13.3%	7.8	9.60	10.52	0.92	9.6%	7.0
7	200	13.67	15.50	1.83	13.3%	7.8	12.73	13.96	1.23	9.6%	7.0
8	250	17.04	19.33	2.28	13.4%	7.7	15.86	17.39	1.53	9.7%	7.0
9	300	20.41	23.15	2.74	13.4%	7.7	18.99	20.83	1.84	9.7%	6.9
10	350	23.78	26.98	3.19	13.4%	7.7	22.12	24.27	2.15	9.7%	6.9
11	400	27.15	30.80	3.65	13.4%	7.7	25.26	27.71	2.45	9.7%	6.9
12	450	30.52	34.63	4.11	13.5%	7.7	28.39	31.15	2.76	9.7%	6.9
13	500	33.89	38.45	4.56	13.5%	7.7	31.52	34.59	3.07	9.7%	6.9
14	600	40.62	46.10	5.48	13.5%	7.7	37.78	41.47	3.68	9.7%	6.9
15	700	47.36	53.75	6.39	13.5%	7.7	44.05	48.34	4.29	9.7%	6.9
16	800	54.10	61.40	7.30	13.5%	7.7	50.31	55.22	4.91	9.8%	6.9
17	900	60.84	69.05	8.21	13.5%	7.7	56.58	62.10	5.52	9.8%	6.9
18	1,000	67.57	76.70	9.13	13.5%	7.7	62.84	68.98	6.13	9.8%	6.9
19	1,500	101.26	114.95	13.69	13.5%	7.7	94.16	103.36	9.20	9.8%	6.9
20	2,000	134.95	153.20	18.25	13.5%	7.7	125.48	137.75	12.27	9.8%	6.9
21	2,500	168.64	191.45	22.81	13.5%	7.7	156.80	172.14	15.34	9.8%	6.9
22	3,000	202.32	229.70	27.38	13.5%	7.7	188.12	206.53	18.40	9.8%	6.9
23	3,500	236.01	267.95	31.94	13.5%	7.7	219.44	240.92	21.47	9.8%	6.9
24	4,000	269.70	306.20	36.50	13.5%	7.7	250.76	275.30	24.54	9.8%	6.9
25	4,500	303.39	344.45	41.06	13.5%	7.7	282.08	309.69	27.61	9.8%	6.9
26	5,000	337.07	382.70	45.63	13.5%	7.7	313.41	344.08	30.67	9.8%	6.9
27	5,500	370.76	420.95	50.19	13.5%	7.7	344.73	378.47	33.74	9.8%	6.9
28	6,000	404.45	459.20	54.75	13.5%	7.7	376.05	412.86	36.81	9.8%	6.9
29	6,500	438.14	497.45	59.31	13.5%	7.7	407.37	447.24	39.88	9.8%	6.9

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy Intensive EIP Voltage Level 2

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 36 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	8.63	9.90	1.27	14.7%	9.9	8.23	9.22	0.99	12.0%	9.2
2	110	9.48	10.87	1.40	14.7%	9.9	9.03	10.12	1.08	12.0%	9.2
3	120	10.32	11.84	1.52	14.7%	9.9	9.84	11.02	1.18	12.0%	9.2
4	130	11.16	12.81	1.65	14.8%	9.9	10.64	11.92	1.28	12.1%	9.2
5	140	12.01	13.78	1.78	14.8%	9.8	11.44	12.82	1.38	12.1%	9.2
6	150	12.85	14.75	1.90	14.8%	9.8	12.24	13.72	1.48	12.1%	9.1
7	200	17.07	19.60	2.54	14.9%	9.8	16.26	18.23	1.97	12.1%	9.1
8	250	21.28	24.45	3.17	14.9%	9.8	20.27	22.74	2.47	12.2%	9.1
9	300	25.50	29.30	3.80	14.9%	9.8	24.29	27.25	2.96	12.2%	9.1
10	350	29.72	34.15	4.44	14.9%	9.8	28.30	31.75	3.45	12.2%	9.1
11	400	33.93	39.01	5.07	15.0%	9.8	32.32	36.26	3.94	12.2%	9.1
12	450	38.15	43.86	5.71	15.0%	9.7	36.33	40.77	4.44	12.2%	9.1
13	500	42.37	48.71	6.34	15.0%	9.7	40.35	45.28	4.93	12.2%	9.1
14	600	50.80	58.41	7.61	15.0%	9.7	48.38	54.29	5.92	12.2%	9.0
15	700	59.23	68.11	8.88	15.0%	9.7	56.41	63.31	6.90	12.2%	9.0
16	800	67.66	77.81	10.15	15.0%	9.7	64.44	72.32	7.89	12.2%	9.0
17	900	76.10	87.51	11.41	15.0%	9.7	72.47	81.34	8.88	12.2%	9.0
18	1,000	84.53	97.21	12.68	15.0%	9.7	80.49	90.36	9.86	12.3%	9.0
19	1,500	126.70	145.72	19.02	15.0%	9.7	120.64	135.43	14.79	12.3%	9.0
20	2,000	168.86	194.23	25.37	15.0%	9.7	160.79	180.51	19.72	12.3%	9.0
21	2,500	211.03	242.73	31.71	15.0%	9.7	200.94	225.59	24.65	12.3%	9.0
22	3,000	253.19	291.24	38.05	15.0%	9.7	241.08	270.67	29.58	12.3%	9.0
23	3,500	295.36	339.75	44.39	15.0%	9.7	281.23	315.75	34.51	12.3%	9.0
24	4,000	337.52	388.25	50.73	15.0%	9.7	321.38	360.82	39.45	12.3%	9.0
25	4,500	379.69	436.76	57.07	15.0%	9.7	361.53	405.90	44.38	12.3%	9.0
26	5,000	421.85	485.27	63.41	15.0%	9.7	401.67	450.98	49.31	12.3%	9.0
27	5,500	464.02	533.77	69.76	15.0%	9.7	441.82	496.06	54.24	12.3%	9.0
28	6,000	506.18	582.28	76.10	15.0%	9.7	481.97	541.14	59.17	12.3%	9.0
29	6,500	548.35	630.79	82.44	15.0%	9.7	522.12	586.21	64.10	12.3%	9.0

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy Intensive EIP Voltage Level 3

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 37 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	10.62	11.62	1.00	9.4%	11.6	9.66	10.87	1.22	12.6%	10.9
2	110	11.67	12.77	1.10	9.4%	11.6	10.60	11.94	1.34	12.6%	10.9
3	120	12.71	13.91	1.20	9.4%	11.6	11.55	13.01	1.46	12.7%	10.8
4	130	13.75	15.05	1.30	9.5%	11.6	12.49	14.08	1.58	12.7%	10.8
5	140	14.79	16.19	1.40	9.5%	11.6	13.44	15.14	1.71	12.7%	10.8
6	150	15.84	17.34	1.50	9.5%	11.6	14.38	16.21	1.83	12.7%	10.8
7	200	21.05	23.05	2.00	9.5%	11.5	19.11	21.55	2.44	12.8%	10.8
8	250	26.26	28.76	2.50	9.5%	11.5	23.84	26.89	3.05	12.8%	10.8
9	300	31.47	34.47	3.00	9.5%	11.5	28.57	32.22	3.66	12.8%	10.7
10	350	36.69	40.19	3.50	9.5%	11.5	33.29	37.56	4.27	12.8%	10.7
11	400	41.90	45.90	4.00	9.5%	11.5	38.02	42.90	4.87	12.8%	10.7
12	450	47.11	51.61	4.50	9.6%	11.5	42.75	48.23	5.48	12.8%	10.7
13	500	52.32	57.32	5.00	9.6%	11.5	47.48	53.57	6.09	12.8%	10.7
14	600	62.75	68.75	6.00	9.6%	11.5	56.93	64.24	7.31	12.8%	10.7
15	700	73.17	80.17	7.00	9.6%	11.5	66.39	74.92	8.53	12.9%	10.7
16	800	83.60	91.59	8.00	9.6%	11.4	75.84	85.59	9.75	12.9%	10.7
17	900	94.02	103.02	9.00	9.6%	11.4	85.30	96.27	10.97	12.9%	10.7
18	1,000	104.44	114.44	10.00	9.6%	11.4	94.75	106.94	12.19	12.9%	10.7
19	1,500	156.57	171.56	15.00	9.6%	11.4	142.03	160.31	18.28	12.9%	10.7
20	2,000	208.69	228.69	20.00	9.6%	11.4	189.31	213.68	24.37	12.9%	10.7
21	2,500	260.81	285.81	25.00	9.6%	11.4	236.59	267.05	30.47	12.9%	10.7
22	3,000	312.93	342.93	30.00	9.6%	11.4	283.86	320.42	36.56	12.9%	10.7
23	3,500	365.05	400.05	35.00	9.6%	11.4	331.14	373.79	42.66	12.9%	10.7
24	4,000	417.18	457.17	40.00	9.6%	11.4	378.42	427.17	48.75	12.9%	10.7
25	4,500	469.30	514.29	45.00	9.6%	11.4	425.69	480.54	54.84	12.9%	10.7
26	5,000	521.42	571.42	50.00	9.6%	11.4	472.97	533.91	60.94	12.9%	10.7
27	5,500	573.54	628.54	55.00	9.6%	11.4	520.25	587.28	67.03	12.9%	10.7
28	6,000	625.66	685.66	60.00	9.6%	11.4	567.52	640.65	73.12	12.9%	10.7
29	6,500	677.78	742.78	65.00	9.6%	11.4	614.80	694.02	79.22	12.9%	10.7

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Energy-only GS

Retail Open Access

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 38 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	250	31.17	35.01	3.83	12.3%	14.0	31.17	35.01	3.83	12.3%	14.0
2	500	42.35	50.01	7.67	18.1%	10.0	42.35	50.01	7.67	18.1%	10.0
3	750	53.52	65.02	11.50	21.5%	8.7	53.52	65.02	11.50	21.5%	8.7
4	1,000	64.69	80.02	15.33	23.7%	8.0	64.69	80.02	15.33	23.7%	8.0
5	1,500	87.04	110.03	23.00	26.4%	7.3	87.04	110.03	23.00	26.4%	7.3
6	2,000	109.38	140.05	30.66	28.0%	7.0	109.38	140.05	30.66	28.0%	7.0
7	2,500	131.73	170.06	38.33	29.1%	6.8	131.73	170.06	38.33	29.1%	6.8
8	3,000	154.08	200.07	45.99	29.9%	6.7	154.08	200.07	45.99	29.9%	6.7
9	3,500	176.42	230.08	53.66	30.4%	6.6	176.42	230.08	53.66	30.4%	6.6
10	4,000	198.77	260.09	61.32	30.9%	6.5	198.77	260.09	61.32	30.9%	6.5
11	4,500	221.11	290.10	68.99	31.2%	6.4	221.11	290.10	68.99	31.2%	6.4
12	5,000	243.46	320.12	76.66	31.5%	6.4	243.46	320.12	76.66	31.5%	6.4
13	6,000	288.15	380.14	91.99	31.9%	6.3	288.15	380.14	91.99	31.9%	6.3
14	7,000	332.84	440.16	107.32	32.2%	6.3	332.84	440.16	107.32	32.2%	6.3
15	8,000	377.54	500.18	122.65	32.5%	6.3	377.54	500.18	122.65	32.5%	6.3
16	9,000	422.23	560.21	137.98	32.7%	6.2	422.23	560.21	137.98	32.7%	6.2
17	10,000	466.92	620.23	153.31	32.8%	6.2	466.92	620.23	153.31	32.8%	6.2
18	15,000	690.38	920.35	229.97	33.3%	6.1	690.38	920.35	229.97	33.3%	6.1
19	20,000	913.84	1,220.46	306.62	33.6%	6.1	913.84	1,220.46	306.62	33.6%	6.1
20	25,000	1,137.30	1,520.58	383.28	33.7%	6.1	1,137.30	1,520.58	383.28	33.7%	6.1
21	30,000	1,360.76	1,820.69	459.93	33.8%	6.1	1,360.76	1,820.69	459.93	33.8%	6.1
22	35,000	1,584.22	2,120.81	536.59	33.9%	6.1	1,584.22	2,120.81	536.59	33.9%	6.1
23	40,000	1,807.68	2,420.92	613.24	33.9%	6.1	1,807.68	2,420.92	613.24	33.9%	6.1
24	45,000	2,031.14	2,721.04	689.90	34.0%	6.0	2,031.14	2,721.04	689.90	34.0%	6.0
25	50,000	2,254.60	3,021.15	766.55	34.0%	6.0	2,254.60	3,021.15	766.55	34.0%	6.0
26	55,000	2,478.06	3,321.27	843.21	34.0%	6.0	2,478.06	3,321.27	843.21	34.0%	6.0
27	60,000	2,701.52	3,621.38	919.86	34.0%	6.0	2,701.52	3,621.38	919.86	34.0%	6.0
28	65,000	2,924.98	3,921.50	996.52	34.1%	6.0	2,924.98	3,921.50	996.52	34.1%	6.0
29	70,000	3,148.44	4,221.61	1,073.17	34.1%	6.0	3,148.44	4,221.61	1,073.17	34.1%	6.0

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Demand GSD

Retail Open Access

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 39 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	48.55	53.86	5.31	10.9%	10.8	48.55	53.86	5.31	10.9%	10.8
2	1,000	67.10	77.71	10.62	15.8%	7.8	67.10	77.71	10.62	15.8%	7.8
3	1,500	85.65	101.57	15.93	18.6%	6.8	85.65	101.57	15.93	18.6%	6.8
4	2,000	104.19	125.43	21.23	20.4%	6.3	104.19	125.43	21.23	20.4%	6.3
5	2,500	122.74	149.28	26.54	21.6%	6.0	122.74	149.28	26.54	21.6%	6.0
6	3,000	141.29	173.14	31.85	22.5%	5.8	141.29	173.14	31.85	22.5%	5.8
7	3,500	159.84	197.00	37.16	23.2%	5.6	159.84	197.00	37.16	23.2%	5.6
8	4,000	178.39	220.86	42.47	23.8%	5.5	178.39	220.86	42.47	23.8%	5.5
9	4,500	196.94	244.71	47.78	24.3%	5.4	196.94	244.71	47.78	24.3%	5.4
10	5,000	215.48	268.57	53.09	24.6%	5.4	215.48	268.57	53.09	24.6%	5.4
11	6,000	252.58	316.28	63.70	25.2%	5.3	252.58	316.28	63.70	25.2%	5.3
12	7,000	289.68	364.00	74.32	25.7%	5.2	289.68	364.00	74.32	25.7%	5.2
13	8,000	326.78	411.71	84.94	26.0%	5.1	326.78	411.71	84.94	26.0%	5.1
14	9,000	363.87	459.43	95.55	26.3%	5.1	363.87	459.43	95.55	26.3%	5.1
15	10,000	400.97	507.14	106.17	26.5%	5.1	400.97	507.14	106.17	26.5%	5.1
16	11,000	438.07	554.85	116.79	26.7%	5.0	438.07	554.85	116.79	26.7%	5.0
17	12,000	475.16	602.57	127.40	26.8%	5.0	475.16	602.57	127.40	26.8%	5.0
18	13,000	512.26	650.28	138.02	26.9%	5.0	512.26	650.28	138.02	26.9%	5.0
19	14,000	549.36	698.00	148.64	27.1%	5.0	549.36	698.00	148.64	27.1%	5.0
20	15,000	586.45	745.71	159.26	27.2%	5.0	586.45	745.71	159.26	27.2%	5.0
21	16,000	623.55	793.42	169.87	27.2%	5.0	623.55	793.42	169.87	27.2%	5.0
22	17,000	660.65	841.14	180.49	27.3%	4.9	660.65	841.14	180.49	27.3%	4.9
23	18,000	697.75	888.85	191.11	27.4%	4.9	697.75	888.85	191.11	27.4%	4.9
24	19,000	734.84	936.57	201.72	27.5%	4.9	734.84	936.57	201.72	27.5%	4.9
25	20,000	771.94	984.28	212.34	27.5%	4.9	771.94	984.28	212.34	27.5%	4.9
26	21,000	809.04	1,031.99	222.96	27.6%	4.9	809.04	1,031.99	222.96	27.6%	4.9
27	22,000	846.13	1,079.71	233.57	27.6%	4.9	846.13	1,079.71	233.57	27.6%	4.9
28	23,000	883.23	1,127.42	244.19	27.6%	4.9	883.23	1,127.42	244.19	27.6%	4.9
29	24,000	920.33	1,175.14	254.81	27.7%	4.9	920.33	1,175.14	254.81	27.7%	4.9

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Time-of-Use GSTU

Retail Open Access

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 40 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	42.35	50.01	7.67	18.1%	10.0	42.35	50.01	7.67	18.1%	10.0
2	1,000	64.69	80.02	15.33	23.7%	8.0	64.69	80.02	15.33	23.7%	8.0
3	1,500	87.04	110.03	23.00	26.4%	7.3	87.04	110.03	23.00	26.4%	7.3
4	2,000	109.38	140.05	30.66	28.0%	7.0	109.38	140.05	30.66	28.0%	7.0
5	2,500	131.73	170.06	38.33	29.1%	6.8	131.73	170.06	38.33	29.1%	6.8
6	3,000	154.08	200.07	45.99	29.9%	6.7	154.08	200.07	45.99	29.9%	6.7
7	3,500	176.42	230.08	53.66	30.4%	6.6	176.42	230.08	53.66	30.4%	6.6
8	4,000	198.77	260.09	61.32	30.9%	6.5	198.77	260.09	61.32	30.9%	6.5
9	4,500	221.11	290.10	68.99	31.2%	6.4	221.11	290.10	68.99	31.2%	6.4
10	5,000	243.46	320.12	76.66	31.5%	6.4	243.46	320.12	76.66	31.5%	6.4
11	6,000	288.15	380.14	91.99	31.9%	6.3	288.15	380.14	91.99	31.9%	6.3
12	7,000	332.84	440.16	107.32	32.2%	6.3	332.84	440.16	107.32	32.2%	6.3
13	8,000	377.54	500.18	122.65	32.5%	6.3	377.54	500.18	122.65	32.5%	6.3
14	9,000	422.23	560.21	137.98	32.7%	6.2	422.23	560.21	137.98	32.7%	6.2
15	10,000	466.92	620.23	153.31	32.8%	6.2	466.92	620.23	153.31	32.8%	6.2
16	11,000	511.61	680.25	168.64	33.0%	6.2	511.61	680.25	168.64	33.0%	6.2
17	12,000	556.30	740.28	183.97	33.1%	6.2	556.30	740.28	183.97	33.1%	6.2
18	13,000	601.00	800.30	199.30	33.2%	6.2	601.00	800.30	199.30	33.2%	6.2
19	14,000	645.69	860.32	214.63	33.2%	6.1	645.69	860.32	214.63	33.2%	6.1
20	15,000	690.38	920.35	229.97	33.3%	6.1	690.38	920.35	229.97	33.3%	6.1
21	16,000	735.07	980.37	245.30	33.4%	6.1	735.07	980.37	245.30	33.4%	6.1
22	17,000	779.76	1,040.39	260.63	33.4%	6.1	779.76	1,040.39	260.63	33.4%	6.1
23	18,000	824.46	1,100.41	275.96	33.5%	6.1	824.46	1,100.41	275.96	33.5%	6.1
24	19,000	869.15	1,160.44	291.29	33.5%	6.1	869.15	1,160.44	291.29	33.5%	6.1
25	20,000	913.84	1,220.46	306.62	33.6%	6.1	913.84	1,220.46	306.62	33.6%	6.1
26	21,000	958.53	1,280.48	321.95	33.6%	6.1	958.53	1,280.48	321.95	33.6%	6.1
27	22,000	1,003.22	1,340.51	337.28	33.6%	6.1	1,003.22	1,340.51	337.28	33.6%	6.1
28	23,000	1,047.92	1,400.53	352.61	33.6%	6.1	1,047.92	1,400.53	352.61	33.6%	6.1
29	24,000	1,092.61	1,460.55	367.94	33.7%	6.1	1,092.61	1,460.55	367.94	33.7%	6.1

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 1

Retail Open Access

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 41 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	101.09	101.41	0.32	0.3%	20.3	101.09	101.41	0.32	0.3%	20.3
2	1,000	102.17	102.81	0.64	0.6%	10.3	102.17	102.81	0.64	0.6%	10.3
3	1,500	103.26	104.22	0.96	0.9%	6.9	103.26	104.22	0.96	0.9%	6.9
4	2,000	104.34	105.63	1.29	1.2%	5.3	104.34	105.63	1.29	1.2%	5.3
5	2,500	105.43	107.04	1.61	1.5%	4.3	105.43	107.04	1.61	1.5%	4.3
6	3,000	106.51	108.44	1.93	1.8%	3.6	106.51	108.44	1.93	1.8%	3.6
7	4,000	108.68	111.26	2.57	2.4%	2.8	108.68	111.26	2.57	2.4%	2.8
8	5,000	110.86	114.07	3.21	2.9%	2.3	110.86	114.07	3.21	2.9%	2.3
9	6,000	113.03	116.88	3.86	3.4%	1.9	113.03	116.88	3.86	3.4%	1.9
10	7,000	115.20	119.70	4.50	3.9%	1.7	115.20	119.70	4.50	3.9%	1.7
11	8,000	117.37	122.51	5.14	4.4%	1.5	117.37	122.51	5.14	4.4%	1.5
12	9,000	119.54	125.33	5.79	4.8%	1.4	119.54	125.33	5.79	4.8%	1.4
13	10,000	121.71	128.14	6.43	5.3%	1.3	121.71	128.14	6.43	5.3%	1.3
14	15,000	132.57	142.21	9.65	7.3%	0.9	132.57	142.21	9.65	7.3%	0.9
15	20,000	143.42	156.28	12.86	9.0%	0.8	143.42	156.28	12.86	9.0%	0.8
16	25,000	154.28	170.35	16.08	10.4%	0.7	154.28	170.35	16.08	10.4%	0.7
17	30,000	165.13	184.42	19.29	11.7%	0.6	165.13	184.42	19.29	11.7%	0.6
18	35,000	175.99	198.49	22.51	12.8%	0.6	175.99	198.49	22.51	12.8%	0.6
19	40,000	186.84	212.56	25.72	13.8%	0.5	186.84	212.56	25.72	13.8%	0.5
20	45,000	197.70	226.63	28.94	14.6%	0.5	197.70	226.63	28.94	14.6%	0.5
21	50,000	208.55	240.70	32.15	15.4%	0.5	208.55	240.70	32.15	15.4%	0.5
22	60,000	230.26	268.84	38.58	16.8%	0.4	230.26	268.84	38.58	16.8%	0.4
23	70,000	251.97	296.98	45.01	17.9%	0.4	251.97	296.98	45.01	17.9%	0.4
24	80,000	273.68	325.12	51.44	18.8%	0.4	273.68	325.12	51.44	18.8%	0.4
25	90,000	295.39	353.26	57.87	19.6%	0.4	295.39	353.26	57.87	19.6%	0.4
26	100,000	317.10	381.40	64.30	20.3%	0.4	317.10	381.40	64.30	20.3%	0.4
27	110,000	338.81	409.54	70.73	20.9%	0.4	338.81	409.54	70.73	20.9%	0.4
28	120,000	360.52	437.68	77.16	21.4%	0.4	360.52	437.68	77.16	21.4%	0.4
29	130,000	382.23	465.82	83.59	21.9%	0.4	382.23	465.82	83.59	21.9%	0.4

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 2

Retail Open Access

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 42 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	103.83	104.73	0.90	0.9%	20.9	103.83	104.73	0.90	0.9%	20.9
2	1,000	107.66	109.46	1.80	1.7%	10.9	107.66	109.46	1.80	1.7%	10.9
3	1,500	111.49	114.19	2.70	2.4%	7.6	111.49	114.19	2.70	2.4%	7.6
4	2,000	115.32	118.92	3.60	3.1%	5.9	115.32	118.92	3.60	3.1%	5.9
5	2,500	119.15	123.65	4.50	3.8%	4.9	119.15	123.65	4.50	3.8%	4.9
6	3,000	122.98	128.38	5.40	4.4%	4.3	122.98	128.38	5.40	4.4%	4.3
7	4,000	130.64	137.84	7.20	5.5%	3.4	130.64	137.84	7.20	5.5%	3.4
8	5,000	138.31	147.30	9.00	6.5%	2.9	138.31	147.30	9.00	6.5%	2.9
9	6,000	145.97	156.76	10.79	7.4%	2.6	145.97	156.76	10.79	7.4%	2.6
10	7,000	153.63	166.22	12.59	8.2%	2.4	153.63	166.22	12.59	8.2%	2.4
11	8,000	161.29	175.68	14.39	8.9%	2.2	161.29	175.68	14.39	8.9%	2.2
12	9,000	168.95	185.14	16.19	9.6%	2.1	168.95	185.14	16.19	9.6%	2.1
13	10,000	176.61	194.60	17.99	10.2%	1.9	176.61	194.60	17.99	10.2%	1.9
14	15,000	214.92	241.90	26.99	12.6%	1.6	214.92	241.90	26.99	12.6%	1.6
15	20,000	253.22	289.20	35.98	14.2%	1.4	253.22	289.20	35.98	14.2%	1.4
16	25,000	291.53	336.50	44.98	15.4%	1.3	291.53	336.50	44.98	15.4%	1.3
17	30,000	329.83	383.80	53.97	16.4%	1.3	329.83	383.80	53.97	16.4%	1.3
18	35,000	368.14	431.10	62.97	17.1%	1.2	368.14	431.10	62.97	17.1%	1.2
19	40,000	406.44	478.40	71.96	17.7%	1.2	406.44	478.40	71.96	17.7%	1.2
20	45,000	444.75	525.70	80.96	18.2%	1.2	444.75	525.70	80.96	18.2%	1.2
21	50,000	483.05	573.00	89.95	18.6%	1.1	483.05	573.00	89.95	18.6%	1.1
22	60,000	559.66	667.60	107.94	19.3%	1.1	559.66	667.60	107.94	19.3%	1.1
23	70,000	636.27	762.20	125.93	19.8%	1.1	636.27	762.20	125.93	19.8%	1.1
24	80,000	712.88	856.80	143.92	20.2%	1.1	712.88	856.80	143.92	20.2%	1.1
25	90,000	789.49	951.40	161.91	20.5%	1.1	789.49	951.40	161.91	20.5%	1.1
26	100,000	866.10	1,046.00	179.90	20.8%	1.0	866.10	1,046.00	179.90	20.8%	1.0
27	110,000	942.71	1,140.60	197.89	21.0%	1.0	942.71	1,140.60	197.89	21.0%	1.0
28	120,000	1,019.32	1,235.20	215.88	21.2%	1.0	1,019.32	1,235.20	215.88	21.2%	1.0
29	130,000	1,095.93	1,329.80	233.87	21.3%	1.0	1,095.93	1,329.80	233.87	21.3%	1.0

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 3

Retail Open Access

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 43 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	108.74	110.83	2.09	1.9%	22.2	108.74	110.83	2.09	1.9%	22.2
2	1,000	117.47	121.66	4.18	3.6%	12.2	117.47	121.66	4.18	3.6%	12.2
3	1,500	126.21	132.49	6.27	5.0%	8.8	126.21	132.49	6.27	5.0%	8.8
4	2,000	134.95	143.31	8.37	6.2%	7.2	134.95	143.31	8.37	6.2%	7.2
5	2,500	143.69	154.14	10.46	7.3%	6.2	143.69	154.14	10.46	7.3%	6.2
6	3,000	152.42	164.97	12.55	8.2%	5.5	152.42	164.97	12.55	8.2%	5.5
7	4,000	169.90	186.63	16.73	9.8%	4.7	169.90	186.63	16.73	9.8%	4.7
8	5,000	187.37	208.29	20.92	11.2%	4.2	187.37	208.29	20.92	11.2%	4.2
9	6,000	204.84	229.94	25.10	12.3%	3.8	204.84	229.94	25.10	12.3%	3.8
10	7,000	222.32	251.60	29.28	13.2%	3.6	222.32	251.60	29.28	13.2%	3.6
11	8,000	239.79	273.26	33.46	14.0%	3.4	239.79	273.26	33.46	14.0%	3.4
12	9,000	257.27	294.91	37.65	14.6%	3.3	257.27	294.91	37.65	14.6%	3.3
13	10,000	274.74	316.57	41.83	15.2%	3.2	274.74	316.57	41.83	15.2%	3.2
14	15,000	362.11	424.86	62.74	17.3%	2.8	362.11	424.86	62.74	17.3%	2.8
15	20,000	449.48	533.14	83.66	18.6%	2.7	449.48	533.14	83.66	18.6%	2.7
16	25,000	536.85	641.43	104.58	19.5%	2.6	536.85	641.43	104.58	19.5%	2.6
17	30,000	624.22	749.71	125.49	20.1%	2.5	624.22	749.71	125.49	20.1%	2.5
18	35,000	711.59	858.00	146.41	20.6%	2.5	711.59	858.00	146.41	20.6%	2.5
19	40,000	798.96	966.28	167.32	20.9%	2.4	798.96	966.28	167.32	20.9%	2.4
20	45,000	886.33	1,074.57	188.24	21.2%	2.4	886.33	1,074.57	188.24	21.2%	2.4
21	50,000	973.70	1,182.85	209.15	21.5%	2.4	973.70	1,182.85	209.15	21.5%	2.4
22	60,000	1,148.44	1,399.42	250.98	21.9%	2.3	1,148.44	1,399.42	250.98	21.9%	2.3
23	70,000	1,323.18	1,615.99	292.81	22.1%	2.3	1,323.18	1,615.99	292.81	22.1%	2.3
24	80,000	1,497.92	1,832.56	334.64	22.3%	2.3	1,497.92	1,832.56	334.64	22.3%	2.3
25	90,000	1,672.66	2,049.13	376.47	22.5%	2.3	1,672.66	2,049.13	376.47	22.5%	2.3
26	100,000	1,847.40	2,265.70	418.30	22.6%	2.3	1,847.40	2,265.70	418.30	22.6%	2.3
27	110,000	2,022.14	2,482.27	460.13	22.8%	2.3	2,022.14	2,482.27	460.13	22.8%	2.3
28	120,000	2,196.88	2,698.84	501.96	22.8%	2.2	2,196.88	2,698.84	501.96	22.8%	2.2
29	130,000	2,371.62	2,915.41	543.79	22.9%	2.2	2,371.62	2,915.41	543.79	22.9%	2.2

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 1

Retail Open Access

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 44 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	0.34	0.39	0.04	12.6%	0.4	0.34	0.39	0.04	12.6%	0.4
2	110	0.36	0.41	0.05	13.4%	0.4	0.36	0.41	0.05	13.4%	0.4
3	120	0.37	0.43	0.05	14.0%	0.4	0.37	0.43	0.05	14.0%	0.4
4	130	0.39	0.44	0.06	14.6%	0.3	0.39	0.44	0.06	14.6%	0.3
5	140	0.40	0.46	0.06	15.2%	0.3	0.40	0.46	0.06	15.2%	0.3
6	150	0.42	0.48	0.07	15.7%	0.3	0.42	0.48	0.07	15.7%	0.3
7	200	0.49	0.58	0.09	17.8%	0.3	0.49	0.58	0.09	17.8%	0.3
8	250	0.56	0.67	0.11	19.4%	0.3	0.56	0.67	0.11	19.4%	0.3
9	300	0.63	0.76	0.13	20.6%	0.3	0.63	0.76	0.13	20.6%	0.3
10	350	0.70	0.86	0.15	21.6%	0.2	0.70	0.86	0.15	21.6%	0.2
11	400	0.78	0.95	0.17	22.4%	0.2	0.78	0.95	0.17	22.4%	0.2
12	450	0.85	1.05	0.20	23.1%	0.2	0.85	1.05	0.20	23.1%	0.2
13	500	0.92	1.14	0.22	23.6%	0.2	0.92	1.14	0.22	23.6%	0.2
14	600	1.07	1.33	0.26	24.5%	0.2	1.07	1.33	0.26	24.5%	0.2
15	700	1.21	1.51	0.30	25.2%	0.2	1.21	1.51	0.30	25.2%	0.2
16	800	1.35	1.70	0.35	25.7%	0.2	1.35	1.70	0.35	25.7%	0.2
17	900	1.50	1.89	0.39	26.2%	0.2	1.50	1.89	0.39	26.2%	0.2
18	1,000	1.64	2.08	0.44	26.5%	0.2	1.64	2.08	0.44	26.5%	0.2
19	1,500	2.36	3.02	0.65	27.6%	0.2	2.36	3.02	0.65	27.6%	0.2
20	2,000	3.09	3.96	0.87	28.2%	0.2	3.09	3.96	0.87	28.2%	0.2
21	2,500	3.81	4.89	1.09	28.6%	0.2	3.81	4.89	1.09	28.6%	0.2
22	3,000	4.53	5.83	1.31	28.8%	0.2	4.53	5.83	1.31	28.8%	0.2
23	3,500	5.25	6.77	1.52	29.0%	0.2	5.25	6.77	1.52	29.0%	0.2
24	4,000	5.97	7.71	1.74	29.2%	0.2	5.97	7.71	1.74	29.2%	0.2
25	4,500	6.69	8.65	1.96	29.3%	0.2	6.69	8.65	1.96	29.3%	0.2
26	5,000	7.41	9.59	2.18	29.4%	0.2	7.41	9.59	2.18	29.4%	0.2
27	5,500	8.13	10.53	2.39	29.4%	0.2	8.13	10.53	2.39	29.4%	0.2
28	6,000	8.86	11.47	2.61	29.5%	0.2	8.86	11.47	2.61	29.5%	0.2
29	6,500	9.58	12.41	2.83	29.6%	0.2	9.58	12.41	2.83	29.6%	0.2

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 2

Retail Open Access

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 45 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	0.75	0.88	0.13	17.1%	0.9	0.75	0.88	0.13	17.1%	0.9
2	110	0.80	0.94	0.14	17.5%	0.9	0.80	0.94	0.14	17.5%	0.9
3	120	0.86	1.01	0.15	17.9%	0.8	0.86	1.01	0.15	17.9%	0.8
4	130	0.91	1.08	0.17	18.2%	0.8	0.91	1.08	0.17	18.2%	0.8
5	140	0.97	1.15	0.18	18.5%	0.8	0.97	1.15	0.18	18.5%	0.8
6	150	1.02	1.21	0.19	18.7%	0.8	1.02	1.21	0.19	18.7%	0.8
7	200	1.30	1.55	0.26	19.7%	0.8	1.30	1.55	0.26	19.7%	0.8
8	250	1.57	1.89	0.32	20.3%	0.8	1.57	1.89	0.32	20.3%	0.8
9	300	1.85	2.23	0.38	20.8%	0.7	1.85	2.23	0.38	20.8%	0.7
10	350	2.12	2.57	0.45	21.1%	0.7	2.12	2.57	0.45	21.1%	0.7
11	400	2.39	2.91	0.51	21.4%	0.7	2.39	2.91	0.51	21.4%	0.7
12	450	2.67	3.24	0.58	21.6%	0.7	2.67	3.24	0.58	21.6%	0.7
13	500	2.94	3.58	0.64	21.7%	0.7	2.94	3.58	0.64	21.7%	0.7
14	600	3.49	4.26	0.77	22.0%	0.7	3.49	4.26	0.77	22.0%	0.7
15	700	4.04	4.93	0.89	22.2%	0.7	4.04	4.93	0.89	22.2%	0.7
16	800	4.59	5.61	1.02	22.3%	0.7	4.59	5.61	1.02	22.3%	0.7
17	900	5.14	6.29	1.15	22.4%	0.7	5.14	6.29	1.15	22.4%	0.7
18	1,000	5.68	6.96	1.28	22.5%	0.7	5.68	6.96	1.28	22.5%	0.7
19	1,500	8.43	10.34	1.92	22.8%	0.7	8.43	10.34	1.92	22.8%	0.7
20	2,000	11.17	13.73	2.56	22.9%	0.7	11.17	13.73	2.56	22.9%	0.7
21	2,500	13.91	17.11	3.20	23.0%	0.7	13.91	17.11	3.20	23.0%	0.7
22	3,000	16.65	20.49	3.83	23.0%	0.7	16.65	20.49	3.83	23.0%	0.7
23	3,500	19.40	23.87	4.47	23.1%	0.7	19.40	23.87	4.47	23.1%	0.7
24	4,000	22.14	27.25	5.11	23.1%	0.7	22.14	27.25	5.11	23.1%	0.7
25	4,500	24.88	30.63	5.75	23.1%	0.7	24.88	30.63	5.75	23.1%	0.7
26	5,000	27.62	34.02	6.39	23.1%	0.7	27.62	34.02	6.39	23.1%	0.7
27	5,500	30.37	37.40	7.03	23.2%	0.7	30.37	37.40	7.03	23.2%	0.7
28	6,000	33.11	40.78	7.67	23.2%	0.7	33.11	40.78	7.67	23.2%	0.7
29	6,500	35.85	44.16	8.31	23.2%	0.7	35.85	44.16	8.31	23.2%	0.7

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 3

Retail Open Access

Case No.: U-21389

Exhibit No.: A-16 (AMG-4)

Schedule F-4.0

Page 46 of 46

Witness: AMGast

Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	1.60	1.88	0.28	17.3%	1.9	1.60	1.88	0.28	17.3%	1.9
2	110	1.74	2.05	0.30	17.5%	1.9	1.74	2.05	0.30	17.5%	1.9
3	120	1.88	2.21	0.33	17.7%	1.8	1.88	2.21	0.33	17.7%	1.8
4	130	2.02	2.38	0.36	17.8%	1.8	2.02	2.38	0.36	17.8%	1.8
5	140	2.16	2.55	0.39	18.0%	1.8	2.16	2.55	0.39	18.0%	1.8
6	150	2.30	2.72	0.42	18.1%	1.8	2.30	2.72	0.42	18.1%	1.8
7	200	3.00	3.56	0.55	18.5%	1.8	3.00	3.56	0.55	18.5%	1.8
8	250	3.70	4.39	0.69	18.7%	1.8	3.70	4.39	0.69	18.7%	1.8
9	300	4.40	5.23	0.83	18.9%	1.7	4.40	5.23	0.83	18.9%	1.7
10	350	5.10	6.07	0.97	19.0%	1.7	5.10	6.07	0.97	19.0%	1.7
11	400	5.80	6.91	1.11	19.1%	1.7	5.80	6.91	1.11	19.1%	1.7
12	450	6.50	7.75	1.25	19.2%	1.7	6.50	7.75	1.25	19.2%	1.7
13	500	7.20	8.59	1.39	19.2%	1.7	7.20	8.59	1.39	19.2%	1.7
14	600	8.60	10.27	1.66	19.3%	1.7	8.60	10.27	1.66	19.3%	1.7
15	700	10.00	11.94	1.94	19.4%	1.7	10.00	11.94	1.94	19.4%	1.7
16	800	11.40	13.62	2.22	19.4%	1.7	11.40	13.62	2.22	19.4%	1.7
17	900	12.80	15.30	2.49	19.5%	1.7	12.80	15.30	2.49	19.5%	1.7
18	1,000	14.21	16.98	2.77	19.5%	1.7	14.21	16.98	2.77	19.5%	1.7
19	1,500	21.21	25.37	4.16	19.6%	1.7	21.21	25.37	4.16	19.6%	1.7
20	2,000	28.21	33.75	5.54	19.7%	1.7	28.21	33.75	5.54	19.7%	1.7
21	2,500	35.21	42.14	6.93	19.7%	1.7	35.21	42.14	6.93	19.7%	1.7
22	3,000	42.22	50.53	8.32	19.7%	1.7	42.22	50.53	8.32	19.7%	1.7
23	3,500	49.22	58.92	9.70	19.7%	1.7	49.22	58.92	9.70	19.7%	1.7
24	4,000	56.22	67.31	11.09	19.7%	1.7	56.22	67.31	11.09	19.7%	1.7
25	4,500	63.22	75.70	12.47	19.7%	1.7	63.22	75.70	12.47	19.7%	1.7
26	5,000	70.23	84.08	13.86	19.7%	1.7	70.23	84.08	13.86	19.7%	1.7
27	5,500	77.23	92.47	15.24	19.7%	1.7	77.23	92.47	15.24	19.7%	1.7
28	6,000	84.23	100.86	16.63	19.7%	1.7	84.23	100.86	16.63	19.7%	1.7
29	6,500	91.23	109.25	18.02	19.7%	1.7	91.23	109.25	18.02	19.7%	1.7

EXHIBIT A-87 (AMG-5)
IS CONFIDENTIAL AND BEING FILED
UNDER SEAL WITH THE MPSC

	(a)	(b)	(c)	(d)	(e)	(f)
Line	Program plus	Financial	Reconciled			
No.	Description	Cust. Credits ⁽¹⁾	Incentive ⁽¹⁾	Balance ⁽¹⁾	Total	Units
FULL SERVICE						
<u>Residential Class</u>						
1	Summer On-peak RSP	0.002426	0.000174	(0.000377)	0.002223	per kWh
2	Smart Hours RSH	0.002426	0.000174	(0.000377)	0.002223	per kWh
3	Night Time Savers RPM	0.002426	0.000174	(0.000377)	0.002223	per kWh
4	Non-Transmitting Meters RSM	0.002426	0.000174	(0.000377)	0.002223	per kWh
<u>Secondary Class</u>						
5	Energy-only GS	0.002326	0.000167	(0.000362)	0.002131	per kWh
6	Time-of-Use GSTU	0.002326	0.000167	(0.000362)	0.002131	per kWh
7	Demand GSD	0.75	0.05	(0.12)	0.68	per Peak kW
<u>Primary Class</u>						
Energy-only GP						
8	Voltage Level 1	0.002011	0.000144	(0.000313)	0.001842	per kWh
9	Voltage Level 2	0.002034	0.000146	(0.000317)	0.001863	per kWh
10	Voltage Level 3	0.002057	0.000147	(0.000320)	0.001884	per kWh
Demand GPD						
11	Voltage Level 1	0.85	0.06	(0.13)	0.78	per On-Peak kW
12	Voltage Level 2	0.86	0.06	(0.13)	0.79	per On-Peak kW
13	Voltage Level 3	0.88	0.06	(0.13)	0.81	per On-Peak kW
Time-of-Use GPTU						
14	Voltage Level 1	0.001816	0.000130	(0.000282)	0.001664	per kWh
15	Voltage Level 2	0.001837	0.000132	(0.000285)	0.001684	per kWh
16	Voltage Level 3	0.001857	0.000133	(0.000288)	0.001702	per kWh
Energy Intensive EIP						
17	Voltage Level 1	0.001532	0.000110	(0.000238)	0.001404	per kWh
18	Voltage Level 2	0.001550	0.000111	(0.000241)	0.001420	per kWh
19	Voltage Level 3	0.001567	0.000113	(0.000243)	0.001437	per kWh
<u>Lighting & Unmetered Class</u>						
20	Metered Lighting GML	0.000556	0.000040	(0.000086)	0.000510	per kWh
21	Universal Unmetered Lighting UUL	0.000544	0.000039	(0.000085)	0.000498	per kWh
22	Unmetered GU	0.001504	0.000108	(0.000234)	0.001378	per kWh
ROA SERVICE						
<u>Secondary Class</u>						
23	Energy-only GS	NA	NA	NA	NA	
24	Demand GSD	NA	NA	NA	NA	
<u>Primary Class</u>						
Energy-only GP						
25	Voltage Level 1	NA	NA	NA	NA	
26	Voltage Level 2	NA	NA	NA	NA	
27	Voltage Level 3	NA	NA	NA	NA	
Demand GPD						
28	Voltage Level 1	NA	NA	NA	NA	
29	Voltage Level 2	NA	NA	NA	NA	
30	Voltage Level 3	NA	NA	NA	NA	

(a)					(e)				
Line No.	Description	Test-year	ERC	ERC Deferral	Line No.	Description	Test-year	ERC	ERC Deferral
		Sales	Deferral ⁽¹⁾	Surcharge			Sales	Deferral ⁽¹⁾	Surcharge
		(MWh)	(\$000)	(\$ / kWh)			(MWh)	(\$000)	(\$ / kWh)
FULL SERVICE					ROA SERVICE				
1	Residential Class	12,559,141	\$ 5,400	0.000430	1	Residential Class	-	\$ -	0.000430
	<u>Secondary Class</u>					<u>Secondary Class</u>			
2	Energy-only GS	3,996,698	1,367	0.000342	2	Energy-only GS	23,026	8	0.000342
3	Time-of-Use GSTU	27,489	9	0.000342	3	Time-of-Use GSTU	-	-	0.000342
4	Demand GSD	2,949,510	708	0.000240	4	Demand GSD	172,462	41	0.000240
	<u>Primary Class</u>					<u>Primary Class</u>			
	Energy-only GP					Energy-only GP			
5	Voltage Level 1	15,684	0	0.000021	5	Voltage Level 1	-	-	0.000021
6	Voltage Level 2	31,696	2	0.000068	6	Voltage Level 2	2,770	0	0.000068
7	Voltage Level 3	758,276	85	0.000112	7	Voltage Level 3	72,834	8	0.000112
	Demand GPD					Demand GPD			
8	Voltage Level 1	1,033,644	16	0.000015	8	Voltage Level 1	1,130,638	17	0.000015
9	Voltage Level 2	1,448,667	71	0.000049	9	Voltage Level 2	1,351,423	66	0.000049
10	Voltage Level 3	2,098,603	168	0.000080	10	Voltage Level 3	960,761	77	0.000080
	Time-of-Use GPTU								
11	Voltage Level 1	397,758	6	0.000015	11	Total ROA Service		\$ 218	
12	Voltage Level 2	574,708	28	0.000049					
13	Voltage Level 3	3,709,392	297	0.000080	12	Total Surcharge Revenue Collected		\$ 8,558	
	Energy Intensive EIP				13	Rounding		(3)	
14	Voltage Level 1	374,401	6	0.000015	14	Total Surcharge Revenue Target		\$ 8,556	
15	Voltage Level 2	64,003	3	0.000049					
16	Voltage Level 3	9,214	1	0.000080					
	<u>Lighting & Unmetered Class</u>								
17	Metered Lighting GML	12,195	5	0.000424					
18	Universal Unmetered Lighting UUL	71,839	153	0.002131					
19	Unmetered GU	93,426	12	0.000133					
	<u>Self-generation Class</u>								
20	Large Self-generation GSG-2								
21	Voltage Level 1	46,664	1	0.000015					
22	Voltage Level 2	54,438	3	0.000049					
23	Voltage Level 3	592	0	0.000080					
24	Total Full Service		\$ 8,341						

(1) A-81 (EAD-5), line 15

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Investment Recovery Mechanism (IRM) Surcharge

Year 1 - 12 months ending February 28, 2025

Case No.: U-21389
Exhibit No.: A-90 (AMG-8)
Page: 1 of 3
Witness: AMGast
Date: May 2023

(a)					(e)				
(b)					(f)				
(c)					(g)				
(d)					(h)				
Line	Test-year	IRM	IRM		Line	Test-year	IRM	IRM	
No.	Description	Sales	Revenue ⁽¹⁾	Surcharge	No.	Description	Sales	Revenue ⁽¹⁾	Surcharge
		(MWh)	(\$000)	(\$ / kWh)			(MWh)	(\$000)	(\$ / kWh)
FULL SERVICE					ROA SERVICE				
1	Residential Class	12,559,141	\$ 3,956	0.000315	1	Residential Class	-	\$ -	0.000315
	<u>Secondary Class</u>					<u>Secondary Class</u>			
2	Energy-only GS	3,996,698	999	0.000250	2	Energy-only GS	23,026	6	0.000250
3	Time-of-Use GSTU	27,489	7	0.000250	3	Time-of-Use GSTU	-	-	0.000250
4	Demand GSD	2,949,510	519	0.000176	4	Demand GSD	172,462	30	0.000176
	<u>Primary Class</u>					<u>Primary Class</u>			
	Energy-only GP					Energy-only GP			
5	Voltage Level 1	15,684	0	0.000015	5	Voltage Level 1	-	-	0.000015
6	Voltage Level 2	31,696	2	0.000049	6	Voltage Level 2	2,770	0	0.000049
7	Voltage Level 3	758,276	61	0.000080	7	Voltage Level 3	72,834	6	0.000080
	Demand GPD					Demand GPD			
8	Voltage Level 1	1,033,644	11	0.000011	8	Voltage Level 1	1,130,638	12	0.000011
9	Voltage Level 2	1,448,667	52	0.000036	9	Voltage Level 2	1,351,423	49	0.000036
10	Voltage Level 3	2,098,603	124	0.000059	10	Voltage Level 3	960,761	57	0.000059
	Time-of-Use GPTU								
11	Voltage Level 1	397,758	4	0.000011	11	Total ROA Service		\$ 160	
12	Voltage Level 2	574,708	21	0.000036					
13	Voltage Level 3	3,709,392	219	0.000059	12	Total Surcharge Revenue Collected		\$ 6,269	
	Energy Intensive EIP				13	Rounding		(6)	
14	Voltage Level 1	374,401	4	0.000011	14	Total Surcharge Revenue Target		\$ 6,263	
15	Voltage Level 2	64,003	2	0.000036					
16	Voltage Level 3	9,214	1	0.000059					
	<u>Lighting & Unmetered Class</u>								
17	Metered Lighting GML	12,195	4	0.000310					
18	Universal Unmetered Lighting UUL	71,839	112	0.001560					
19	Unmetered GU	93,426	9	0.000098					
	<u>Self-generation Class</u>								
20	Large Self-generation GSG-2								
21	Voltage Level 1	46,664	1	0.000011					
22	Voltage Level 2	54,438	2	0.000036					
23	Voltage Level 3	592	0	0.000059					
24	Total Full Service		\$ 6,109						

(1) A-81 (EAD-5), line 15

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Investment Recovery Mechanism (IRM) Surcharge

Year 2 - 12 months ending February 28, 2026

Case No.: U-21389

Exhibit No.: A-90 (AMG-8)

Page: 2 of 3

Witness: AMGast

Date: May 2023

(a)					(e)				
Line No.	Description	Test-year	IRM	IRM	Line No.	Description	Test-year	IRM	IRM
		Sales	Revenue ⁽¹⁾	Surcharge			Sales	Revenue ⁽¹⁾	Surcharge
		(MWh)	(\$000)	(\$ / kWh)			(MWh)	(\$000)	(\$ / kWh)
FULL SERVICE					ROA SERVICE				
1	Residential Class	12,559,141	\$ 11,203	0.000892	1	Residential Class	-	\$ -	0.000892
	<u>Secondary Class</u>					<u>Secondary Class</u>			
2	Energy-only GS	3,996,698	2,834	0.000709	2	Energy-only GS	23,026	16	0.000709
3	Time-of-Use GSTU	27,489	19	0.000709	3	Time-of-Use GSTU	-	-	0.000709
4	Demand GSD	2,949,510	1,469	0.000498	4	Demand GSD	172,462	86	0.000498
	<u>Primary Class</u>					<u>Primary Class</u>			
	Energy-only GP					Energy-only GP			
5	Voltage Level 1	15,684	1	0.000044	5	Voltage Level 1	-	-	0.000044
6	Voltage Level 2	31,696	5	0.000143	6	Voltage Level 2	2,770	0	0.000143
7	Voltage Level 3	758,276	178	0.000235	7	Voltage Level 3	72,834	17	0.000235
	Demand GPD					Demand GPD			
8	Voltage Level 1	1,033,644	32	0.000031	8	Voltage Level 1	1,130,638	35	0.000031
9	Voltage Level 2	1,448,667	145	0.000100	9	Voltage Level 2	1,351,423	135	0.000100
10	Voltage Level 3	2,098,603	346	0.000165	10	Voltage Level 3	960,761	159	0.000165
	Time-of-Use GPTU								
11	Voltage Level 1	397,758	12	0.000031	11	Total ROA Service		\$ 448	
12	Voltage Level 2	574,708	57	0.000100					
13	Voltage Level 3	3,709,392	612	0.000165	12	Total Surcharge Revenue Collected		\$ 17,742	
	Energy Intensive EIP				13	Rounding		5.56	
14	Voltage Level 1	374,401	12	0.000031	14	Total Surcharge Revenue Target		\$ 17,748	
15	Voltage Level 2	64,003	6	0.000100					
16	Voltage Level 3	9,214	2	0.000165					
	<u>Lighting & Unmetered Class</u>								
17	Metered Lighting GML	12,195	11	0.000879					
18	Universal Unmetered Lighting UUL	71,839	318	0.004420					
19	Unmetered GU	93,426	26	0.000277					
	<u>Self-generation Class</u>								
20	Large Self-generation GSG-2								
21	Voltage Level 1	46,664	1	0.000031					
22	Voltage Level 2	54,438	5	0.000100					
23	Voltage Level 3	592	0	0.000165					
24	Total Full Service		\$ 17,294						

(1) A-81 (EAD-5), line 15

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Investment Recovery Mechanism (IRM) Surcharge

Year 3 - 12 months ending February 28, 2027

Case No.: U-21389

Exhibit No.: A-90 (AMG-8)

Page: 3 of 3

Witness: AMGast

Date: May 2023

(a)		(b)		(c)		(d)		(e)		(f)		(g)		(h)	
Line		Test-year	IRM	IRM	Line		Test-year	IRM	IRM	Line		Test-year	IRM	IRM	
No.	Description	Sales	Revenue ⁽¹⁾	Surcharge	No.	Description	Sales	Revenue ⁽¹⁾	Surcharge	No.	Description	Sales	Revenue ⁽¹⁾	Surcharge	
		(MWh)	(\$000)	(\$ / kWh)			(MWh)	(\$000)	(\$ / kWh)			(MWh)	(\$000)	(\$ / kWh)	
	FULL SERVICE					ROA SERVICE									
1	Residential Class	12,559,141	\$ 18,424	0.001467	1	Residential Class	-	\$ -	0.001467						
	<u>Secondary Class</u>					<u>Secondary Class</u>									
2	Energy-only GS	3,996,698	4,668	0.001168	2	Energy-only GS	23,026	27	0.001168						
3	Time-of-Use GSTU	27,489	32	0.001168	3	Time-of-Use GSTU	-	-	0.001168						
4	Demand GSD	2,949,510	2,416	0.000819	4	Demand GSD	172,462	141	0.000819						
	<u>Primary Class</u>					<u>Primary Class</u>									
	Energy-only GP					Energy-only GP									
5	Voltage Level 1	15,684	1	0.000072	5	Voltage Level 1	-	-	0.000072						
6	Voltage Level 2	31,696	7	0.000233	6	Voltage Level 2	2,770	1	0.000233						
7	Voltage Level 3	758,276	291	0.000384	7	Voltage Level 3	72,834	28	0.000384						
	Demand GPD					Demand GPD									
8	Voltage Level 1	1,033,644	53	0.000051	8	Voltage Level 1	1,130,638	58	0.000051						
9	Voltage Level 2	1,448,667	239	0.000165	9	Voltage Level 2	1,351,423	223	0.000165						
10	Voltage Level 3	2,098,603	571	0.000272	10	Voltage Level 3	960,761	261	0.000272						
	Time-of-Use GPTU														
11	Voltage Level 1	397,758	20	0.000051	11	Total ROA Service		\$ 739							
12	Voltage Level 2	574,708	95	0.000165											
13	Voltage Level 3	3,709,392	1,009	0.000272	12	Total Surcharge Revenue Collected		\$ 29,192							
	Energy Intensive EIP				13	Rounding		16							
14	Voltage Level 1	374,401	19	0.000051	14	Total Surcharge Revenue Target		\$ 29,208							
15	Voltage Level 2	64,003	11	0.000165											
16	Voltage Level 3	9,214	3	0.000272											
	<u>Lighting & Unmetered Class</u>														
17	Metered Lighting GML	12,195	18	0.001446											
18	Universal Unmetered Lighting UUL	71,839	523	0.007274											
19	Unmetered GU	93,426	43	0.000456											
	<u>Self-generation Class</u>														
	Large Self-generation GSG-2														
20	Voltage Level 1	46,664	2	0.000051											
21	Voltage Level 2	54,438	9	0.000165											
22	Voltage Level 3	592	0	0.000272											
23															
24	Total Full Service		\$ 28,453												

(1) A-81 (EAD-5), line 15

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electric utility and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

KENDRA K. GROB

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of Actual and Projected Employee Benefits O&M Expenses

For the Years 2022, 2023, 2024 and Test Year Twelve Months ending February 28, 2025

(\$000)

Case No.: U-21389

Exhibit No.: A-91 (KKG-1)

Page: 1 of 2

Witness: KKGrob

Date: May 2023

Line No.	(a) Description	(b)	(c)	(d)	(e)
		Actual 12 mos. ended 12/31/2022	Projected 12 mos. Ending 12/31/2023	Projected 12 mos. Ending 12/31/2024	Projected 12 mos. Ending 2/28/25
1	Pension Plans A/B	(\$29,860)	(\$41,041)	(\$43,416)	(\$43,327)
2	Defined Company Contribution Plan	14,105	14,512	14,831	14,883
3	401 (k) Employees' Savings Plan	12,974	13,343	13,637	13,684
4	Active Health Care/Life Insurance/LTD	26,332	26,442	27,667	27,881
5	Retiree Health Care and Life Insurance	(69,030)	(42,416)	(39,229)	(39,608)
6	Other Benefits	3,279	3,407	3,482	3,494
7	Total Expense	(\$42,200)	(\$25,753)	(\$23,028)	(\$22,993)

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	2022 Actual	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2022	Merit & Inflation 12 Mos Ending Dec 31, 2023	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2023	Merit & Inflation 12 Mos Ending Dec 31, 2024	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2024	Merit & Inflation 2 Mos Ending 2/28/2025	Other Adjustments	Projected O&M 12 Mos Ending 2/28/2025
			(c) * Inflation Rate			(e) * Inflation Rate		(g) * Inflation Rate		(b) + (d) + (f) + (h) + (i)
1	Pension Plans A/B	\$ (29,860)	\$ (29,860)	\$ (1,165)	\$ (31,025)	\$ (683)	\$ (31,707)	\$ (111)	(\$11,509)	\$ (43,327)
2	Defined Company Contribution Plan	\$ 14,105	\$ 14,105	\$ 550	\$ 14,655	\$ 322	\$ 14,978	\$ 52	\$ (147)	\$ 14,883
3	401 (k) Employees' Savings Plan	\$ 12,974	\$ 12,974	\$ 506	\$ 13,480	\$ 297	\$ 13,777	\$ 48	\$ (141)	\$ 13,684
4	Active Health Care/Life Insurance/LTD	\$ 26,332	\$ 26,332	\$ 1,027	\$ 27,359	\$ 602	\$ 27,961	\$ 98	\$ (178)	\$ 27,881
5	Retiree Health Care and Life Insurance	\$ (69,030)	\$ (69,030)	\$ (2,692)	\$ (71,722)	\$ (1,578)	\$ (73,300)	\$ (257)	\$ 33,949	\$ (39,608)
6	Other Benefits	\$ 3,279	\$ 3,279	\$ 128	\$ 3,407	\$ 75	\$ 3,482	\$ 12	\$ -	\$ 3,494
	Other Benefits non labor	2,073	2,073	81	2,154	47	2,201	8	-	2,209
	Other Benefits labor	1,206	1,206	47	1,253	28	1,281	4	-	1,285
7	Total Employee Benefits O&M Expenses	\$ (42,200)	\$ (42,200)	\$ (1,646)	\$ (43,846)	\$ (965)	\$ (44,810)	\$ (157)	\$ 21,974	\$ (22,993)

*Pension and Retiree HC are based on AON actuarial studies.

**DCCP, 401k, and Active HC are based on projected participation by employees

Notes

	2023	2024	2025
8 Annual merit increase (testimony of Company witness Amy M. Conrad)			
Annual merit increase	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated inflation rate	3.9%	2.2%	0.4%
9 Annual inflation rates per WP-JCA-51			
Annual inflation rates	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated inflation rate	3.9%	2.2%	0.4%

CMS Energy - Pension Plan A
ASC 715 Pension Expense Estimates (\$ millions)
Minimum Required Contributions - Baseline Scenario

Prepared on January 18, 2023

	2022	2023	2024	2025	2026	2027	2028	2029	2030
Funding Target	\$ 1,034.2	\$ 1,026.6	\$ 1,078.2	\$ 1,084.3	\$ 1,124.2	\$ 1,160.0	\$ 1,195.1	\$ 1,226.2	\$ 1,254.4
Value of Plan Assets	\$ 1,985.3	\$ 1,797.3	\$ 1,806.4	\$ 1,684.8	\$ 1,719.6	\$ 1,755.0	\$ 1,791.1	\$ 1,826.2	\$ 1,860.7
Funding Balance	\$ 1,116.6	\$ 878.3	\$ 872.3	\$ 817.4	\$ 793.4	\$ 764.1	\$ 731.3	\$ 699.5	\$ 668.9
Funded %	84.0%	89.5%	86.6%	80.0%	82.4%	85.4%	88.7%	91.9%	95.0%
Effective Interest Rate	5.46%	5.29%	5.16%	5.38%	5.32%	5.30%	5.30%	5.30%	5.30%
Contribution by Plan Year									
Utility	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0
Nonutility	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0
At-Risk?	No	No	No	No	No	No	No	No	No
Benefit Restrictions?	No	No	No	No	No	No	No	No	No
Participant Count	3,004	2,799	2,726	2,659	2,586	2,520	2,459	2,399	2,343
PBGC Liability	\$ 1,413.3	\$ 1,304.9	\$ 1,109.9	\$ 1,016.6	\$ 1,050.8	\$ 1,086.3	\$ 1,121.8	\$ 1,154.5	\$ 1,184.0
Market Value of Assets	\$ 2,100.0	\$ 1,633.9	\$ 1,670.9	\$ 1,710.0	\$ 1,747.2	\$ 1,784.8	\$ 1,822.0	\$ 1,857.8	\$ 1,892.9
PBGC Flat Rate Premium	\$ 0.3	\$ 0.3	\$ 0.3	\$ 0.3	\$ 0.3	\$ 0.2	\$ 0.2	\$ 0.2	\$ 0.2
PBGC Variable Rate Premium	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total PBGC Premium	\$ 0.3	\$ 0.3	\$ 0.3	\$ 0.3	\$ 0.3	\$ 0.2	\$ 0.2	\$ 0.2	\$ 0.2
Projected Benefit Obligation	\$ 1,872.0	\$ 1,238.2	\$ 1,249.0	\$ 1,260.0	\$ 1,265.9	\$ 1,268.5	\$ 1,267.0	\$ 1,259.5	\$ 1,247.1
Market Value of Assets	2,100.0	1,633.9	1,670.9	1,710.0	1,747.2	1,784.8	1,822.0	1,857.8	1,892.9
Funded Status	\$ (228.0)	\$ (395.7)	\$ (421.9)	\$ (450.0)	\$ (481.3)	\$ (516.3)	\$ (555.0)	\$ (598.3)	\$ (645.8)
ASC 715 Funded %	112.2%	132.0%	133.8%	135.7%	138.0%	140.7%	143.8%	147.5%	151.8%
ASC 715 Accounting Expense									
Utility	\$ (6.6)	\$ (32.6)	\$ (35.5)	\$ (36.7)	\$ (34.9)	\$ (30.4)	\$ (34.3)	\$ (40.6)	\$ (45.5)
Nonutility	(0.2)	(0.9)	(1.0)	(1.0)	(0.9)	(0.8)	(0.9)	(1.1)	(1.2)
Total ASC 715 Accounting Expense	\$ (6.8)	\$ (33.5)	\$ (36.5)	\$ (37.7)	\$ (35.8)	\$ (31.2)	\$ (35.2)	\$ (41.7)	\$ (46.7)
2022 Settlement Accounting Nonutility	\$ 0.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2020 Settlement Accounting Utility Amortization1	\$ 5.3	\$ 5.3	\$ 5.3	\$ 5.3	\$ 5.3	\$ 5.3	\$ 5.3	\$ 1.5	\$ 0.0
2021 Settlement Accounting Utility Amortization2	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	0.1
2022 Settlement Accounting Utility Amortization3	0.9	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.0
Total Expense	2.6	(22.9)	(25.9)	(27.1)	(25.2)	(20.6)	(24.6)	(34.9)	(44.6)
Components of Total Expense									
Service Cost	\$ 37.0	\$ 27.9	\$ 26.0	\$ 25.0	\$ 23.7	\$ 22.4	\$ 20.7	\$ 19.4	\$ 18.7
Interest Cost	\$ 55.2	\$ 61.4	\$ 62.0	\$ 62.4	\$ 62.7	\$ 62.7	\$ 62.5	\$ 62.1	\$ 61.3
Expected Return on Assets	(119.2)	(127.1)	(128.0)	(128.8)	(126.7)	(126.7)	(128.8)	(130.9)	(133.2)
Amortization of Outstanding Components	20.2	4.3	4.3	3.7	4.5	10.4	10.4	7.7	6.5
Total ASC 715 Accounting Expense	(6.8)	(33.5)	(36.5)	(37.7)	(35.8)	(31.2)	(35.2)	(41.7)	(46.7)
2022 Settlement Accounting Nonutility	\$ 0.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2020 Settlement Accounting Utility Amortization1	\$ 5.3	\$ 5.3	\$ 5.3	\$ 5.3	\$ 5.3	\$ 5.3	\$ 5.3	\$ 1.5	\$ 0.0
2021 Settlement Accounting Utility Amortization2	\$ 2.6	\$ 2.6	\$ 2.6	\$ 2.6	\$ 2.6	\$ 2.6	\$ 2.6	\$ 2.6	\$ 0.1
2022 Settlement Accounting Utility Amortization3	0.9	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.0
Total Expense	2.6	(22.9)	(25.9)	(27.1)	(25.2)	(20.6)	(24.6)	(34.9)	(44.6)
Market Related Value of Assets	\$ 1,928.2	\$ 1,804.2	\$ 1,827.3	\$ 1,828.8	\$ 1,801.6	\$ 1,801.9	\$ 1,833.3	\$ 1,864.7	\$ 1,897.4
Assumptions									
Discount Rate	3.02% & 3.76%	5.24%	5.24%	5.24%	5.24%	5.24%	5.24%	5.24%	5.24%
Expected Return on Assets	6.50%	7.20%	7.20%	7.20%	7.20%	7.20%	7.20%	7.20%	7.20%
Salary Increases	3.60%	3.55%	3.55%	3.55%	3.55%	3.55%	3.55%	3.55%	3.55%

1 The information related to the 2020 Settlement Accounting Utility Amortization was provided directly by CMS and was not calculated or reviewed by Aon. The amortization was set up as a regulatory asset for the utility portion of the 2020 settlement loss of \$35.5 million at August 31, 2020. As provided by CMS, the amortizations starts in September 2020 over service life of 8.55 years. Additionally, the settlement attributable to the utility portion on December 31, 2020 is \$9.8 million, which is also amortized over service of 8.55 years beginning in January 2021. Nonutility portions of the settlement loss are \$0.9 million and \$0.3 million for August 31, 2020 and December 31, 2020 respectively.

2 The information related to the 2021 Settlement Accounting Utility Amortization was provided directly by CMS and was not calculated or reviewed by Aon. The amortization was set up as a regulatory asset for the utility portion of the 2021 settlement loss of \$17.6 million at October 31, 2021. As provided by CMS, the amortizations starts in November 2021 over service life of 8.19 years. Additionally, the settlement attributable to the utility portion on December 31, 2021 is \$3.3 million, which is also amortized over service of 8.19 years beginning in January 2022. Nonutility portions of the settlement loss are \$0.5 million and \$0.1 million for October 31, 2021 and December 31, 2021 respectively.

3 The information related to the 2022 Settlement Accounting Utility Amortization was provided directly by CMS and was not calculated or reviewed by Aon. The amortizations were set up as regulatory assets for the utility portions of the 2022 settlement losses. The settlements attributable to the utility portions are \$3.9 million, \$3.7 million, \$11.5 million, and \$2.5 million as of March 31, 2022, June 30, 2022, September 30, 2022, and December 31, 2022 respectively. As provided by CMS, the amortizations are amortized over service life of 8.11 years beginning in April 2022, July 2022, October 2022, and January 2023 respectively. Nonutility portions of the settlement loss are \$0.1 million, \$0.1 million, \$0.3 million, and \$0.1 million for March 31, 2022, June 30, 2022, September 30, 2022, and December 31, 2022 respectively.

Prepared on January 18, 2023

[illegible]

2023-2030 projections reflect the following:

- January 1, 2022 census data.
- PBO discount rate of 5.24% (Plan A) and 5.14% (Plan B) in fiscal 2023+, based on December 31, 2022 yield curve analysis.
- Service Cost effective interest rate of 5.27% (Plan A) and 5.28% (Plan B) in fiscal 2023+, based on December 31, 2022 yield curve analysis.
- Interest Cost effective interest rate of 5.12% (Plan A) and 5.06% (Plan B) in fiscal 2023+, based on December 31, 2022 yield curve analysis.
- Expense is allocated to Utility/Non Utility based on PBO using the allocation percentages for 2023 expense.
- Contributions are allocated to Utility/Non Utility based on the most recent allocation percentages provided by CMS for the December 2020 contributions.
- December 31, 2022 market assets provided by CMS, with expected returns thereafter.
- Expected and actual asset returns of 7.20% in fiscal 2023+.
- The plans will waive Funding Balance when necessary, in order to maintain a Funded Percentage of at least 80%.
- Other provisions, assumptions and methods are the same as those used for the December 31, 2022 ASC 715 disclosures.

CMS Energy

ASC 715 OPEB Expense Estimates (\$ millions)

Prepared on January 18, 2023

	2022	2023	2024	2025	2026	2027	2028	2029
Funded Status, January 1								
Accumulated Postretirement Benefit Obligation	\$ (1,165.8)	\$ (889.4)	\$ (891.1)	\$ (890.8)	\$ (889.0)	\$ (885.4)	\$ (880.2)	(873.30)
Plan Assets at Fair Value	\$ 1,787.4	\$ 1,446.5	\$ 1,493.9	\$ 1,542.8	\$ 1,594.1	\$ 1,647.6	\$ 1,703.8	1,762.90
Funded Status	\$ 621.6	\$ 557.1	\$ 602.8	\$ 652.0	\$ 705.1	\$ 762.2	\$ 823.6	889.60
ASC 715 Accounting Expense								
Utility	\$ (112.8)	\$ (70.2)	\$ (64.8)	\$ (69.0)	\$ (72.3)	\$ (62.8)	\$ (55.6)	(61.20)
Nonutility	\$ (7.5)	\$ (6.0)	\$ (5.9)	\$ (6.1)	\$ (6.4)	\$ (6.3)	\$ (6.2)	(6.60)
Total	\$ (120.3)	\$ (76.2)	\$ (70.7)	\$ (75.1)	\$ (78.7)	\$ (69.1)	\$ (61.8)	(67.80)
Components of Total Expense								
Service Cost	\$ 17.3	\$ 11.6	\$ 11.3	\$ 11.0	\$ 10.7	\$ 10.5	\$ 10.2	10.00
Interest Cost	\$ 27.9	\$ 44.0	\$ 44.0	\$ 44.0	\$ 43.8	\$ 43.6	\$ 43.3	43.00
Expected Return on Assets	\$ (114.6)	\$ (102.3)	\$ (105.5)	\$ (109.0)	\$ (112.6)	\$ (116.5)	\$ (120.4)	(124.80)
Amortization of Net (Gain) or Loss	\$ 0.6	\$ 11.7	\$ 10.3	\$ 8.9	\$ 7.6	\$ 6.3	\$ 5.1	4.00
Amortization of Prior Service Cost	\$ (51.5)	\$ (41.2)	\$ (30.8)	\$ (30.0)	\$ (28.2)	\$ (13.0)	\$ -	-
Total Expense	\$ (120.3)	\$ (76.2)	\$ (70.7)	\$ (75.1)	\$ (78.7)	\$ (69.1)	\$ (61.8)	(67.80)
Assumptions								
APBO Discount Rate	2.99%	5.21%	5.21%	5.21%	5.21%	5.21%	5.21%	5.21%
Service Cost Effective Interest Rate	3.23%	5.31%	5.31%	5.31%	5.31%	5.31%	5.31%	5.31%
Interest Cost Effective Interest Rate	2.45%	5.10%	5.10%	5.10%	5.10%	5.10%	5.10%	5.10%
Expected Return on Assets	6.50%	7.20%	7.20%	7.20%	7.20%	7.20%	7.20%	7.20%
Trend Rate—Initial Pre-65	6.00%	6.25%	6.00%	6.00%	5.75%	5.50%	5.50%	5.25%
Trend Rate—Initial Post-65	6.50%	6.50%	6.25%	6.25%	6.00%	5.75%	5.50%	5.25%
Trend Rate—Ultimate	4.75%	4.75%	4.75%	4.75%	4.75%	4.75%	4.75%	4.75%
Trend Rate—Ultimate Year Pre-65	2028	2032	2032	2032	2032	2032	2032	2032
Trend Rate—Ultimate Year Post-65	2028	2032	2032	2032	2032	2032	2032	2032
Expected Contribution	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	\$ 0.0	0

2023-2029 expense projections reflect the following:

- January 1, 2022 census data.
- APBO discount rate of 5.21% in fiscal 2023+, based on December 31, 2022 yield curve.
- Service Cost effective interest rate of 5.31% in fiscal 2023+, based on December 31, 2022 yield curve.
- Interest Cost effective interest rate of 5.10% in fiscal 2023+, based on December 31, 2022 yield curve.
- December 31, 2022 market assets provided by CMS.
- Other provisions, assumptions and methods are the same as those used for December 31, 2022 ASC 715 disclosures.

EXHIBIT A-94 (KKG-4)
IS CONFIDENTIAL AND BEING FILED
UNDER SEAL WITH THE MPSC

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-21389

EXHIBITS

OF

QUENTIN A. GUINN

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Summary of Actual and Projected Capital Expenditures
 Operations Support
 (\$000)

Case No.: U-21389
Exhibit No.: A-12 (QAG-1)
Schedule: B-5.5
Page: 1 of 2
Witness: QAGuinn
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)
		Historical Year	Projected Bridge Period			Projected Test Year
Line No	Description	12 Mos Ended 12/31/2022	12 Mos Ended 12/31/2023	2 Mos Ending 2/29/2024	14 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025
1	Asset Preservation	27,318	21,621	3,397	25,017	35,083
2	Other Equipment	862	671	-	671	690
3	Total Capital	28,181	22,292	3,397	25,688	35,773
			Projected			
		2 Mos Ending 2/28/2023	12 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025	26 Mos Ending 2/28/2025	
4	Asset Preservation	1,719	23,298	35,083	60,101	
5	Other Equipment	-	671	690	1,361	
6	Total Capital	1,719	23,969	35,773	61,462	

Schedule: B-5.5

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Actual and Projected Capital Expenditures
 Operations Support
 (\$000)

Case No.: U-21389
 Exhibit No.: A-12 (QAG-1)
 Schedule: B-5.5
 Page: 2 of 2
 Witness: QAGuinn
 Date: May 2023

Line No	(a) Description	(b)	(c)	(d)	(e)	(f)
		Historical Year	Projected Bridge Period			Projected Test Year
		12 Mos Ended 12/31/2022	12 Mos Ended 12/31/2023	2 Mos Ending 2/29/2024	14 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025
1	Asset Preservation	27,318	21,621	3,397	25,017	35,083
	Contractor	22,901	17,640	2,771	20,412	28,624
	Labor	1,393	1,094	172	1,265	1,775
	Materials	1,841	1,673	263	1,936	2,715
	Business Expenses	74	67	10	77	108
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	1,109	1,147	180	1,327	1,862
2	Other Equipment	862	671	-	671	690
	Contractor	26	20	-	20	21
	Labor	-	-	-	-	-
	Materials	836	651	-	651	669
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-	-
3	Total Capital	28,181	22,292	3,397	25,688	35,773

	Projected			
	2 Mos Ending 2/28/2023	12 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025	26 Mos Ending 2/28/2025
4	Asset Preservation	1,719	23,298	35,083
	Contractor	1,403	19,009	28,624
	Labor	87	1,178	1,775
	Materials	133	1,803	2,715
	Business Expenses	5	72	108
	Contingency	-	-	-
	Other (Loadings, Chargebacks)	91	1,236	1,862
5	Other Equipment	-	671	690
	Contractor	-	20	21
	Labor	-	-	-
	Materials	-	651	669
	Business Expenses	-	-	-
	Contingency	-	-	-
	Other (Loadings, Chargebacks)	-	-	-
6	Total Capital	1,719	23,969	35,773

61,462

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Summary of Actual & Projected O&M Expenses
 Operations Support
 (\$000)

Case No.: U-21389
 Exhibit No.: A-95 (QAG-2)
 Page: 1 of 2
 Witness: QAGuinn
 Date: May 2023

(a)		(b)	(c)
Line No.	Description	2022 Actual	Projected 12-Mos Ending 2/28/2025
1	Facilities	14,134	12,762
	Labor	3,958	4,501
	Non-Labor Other	10,177	8,261
2	Real Estate	2,613	2,351
	Labor	1,803	1,866
	Non-Labor Other	810	485
3	Supply Chain	904	1,073
	Labor	90	83
	Non-Labor Other	813	990
4	Total Operations Support of Area O&M Expenses	\$ 17,651	\$ 16,186
	Labor	5,761	6,367
	Non-Labor Other	10,987	8,746

Michigan Public Service Commission
Consumers Energy Company
Detailed List of Projected Electric Capital Expenditures
Operations Support
(000's)

Case No: U-21389
Exhibit No: A-96 (QAG-3)
Page: 1 of 3
Witness: QAGuinn
Date: May 2023

Line No		Historical Year	Projected Bridge Period			Projected Test Year
		12 Mos Ended 12/31/2022	12 Mos Ended 12/31/2023	2 Mos Ending 2/29/2024	14 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025
		<u>Electric</u>	<u>Electric</u>	<u>Electric</u>	<u>Electric</u>	<u>Electric</u>
1	Other Equipment	862	671	-	671	690
2	Wellness Equip	98	51	-	51	52
3	Computer Equipment	18	17	-	17	17
4	Print Equipment	-	43	-	43	44
5	Real Estate Tools & Equipment	18	10	-	10	10
6	Supply Chain Tools & Equipment	727	507	-	507	522
7	Facilities Tools	-	43	-	43	43
8	Emergent Repairs	1,523	1,932	152	2,084	1,856
9	Asset Preservation - Unplanned Repair	1,523	1,932	152	2,084	1,856
10	Asset Replacement	16,020	9,949	958	10,906	10,998
11	Statewide Paving	3,755	1,998	107	2,105	1,327
12	Statewide Roofing	6,533	1,576	295	1,872	3,366
13	Statewide Mechanical/Electrical	3,846	5,107	342	5,449	3,975
14	Statewide Elevators	1,318	1,091	30	1,121	365
15	Marshall Roadway Paving	26	40	171	211	1,829
16	Furniture	541	136	12	147	136
17	New Construction	3,054	3,760	684	4,443	8,539
18	Lansing Service Center	311	3,146	523	3,669	6,402
19	Hastings Service Center	3	543	37	580	815
20	Greenville Cold Storage Building	-	-	27	27	289
21	Ludington Cold Storage Bldg	-	50	96	146	1,032
23	Hardy Headquarter Building	2,740	21	-	21	-
24	Renovations	5,017	5,980	1,603	7,583	13,690
25	Kalamazoo Service Center	182	848	452	1,300	5,513
26	Parnall Renovations (P1-3)	2,715	214	-	214	-
27	Jackson Dispatch	292	543	-	543	-
29	Midland Building	85	-	-	-	-
30	Grand Rapids LVD-UG	1,545	1,386	-	1,386	-
31	Control/Dispatch Center	81	1,551	903	2,454	5,500
32	South Monroe Interior Renovation	5	200	248	448	2,677
33	EV Infrastructure	113	1,238	-	1,238	-
34	Miscellaneous	1,705	-	-	-	-
35	Total Capital	28,181	22,292	3,397	25,688	35,773

New Construction & Renovations - Current Construction Cost

	Location	Type of Operations	Electric Allocation	Gas Allocation	Total Historical Cost 2022	Allocated Electric Cost 2022	Allocated Gas Cost 2022	Total Plan Cost 2023	Allocated Electric Cost 2023	Allocated Gas Cost 2023	Total Plan Cost 2024	Allocated Electric Cost 2024	Allocated Gas Cost 2024	Total Plan Cost 2025	Allocated Electric Cost 2025	Allocated Gas Cost 2025	Total Plan Cost 2026	Allocated Electric Cost 2026	Allocated Gas Cost 2026	Total Plan Cost 2027	Allocated Electric Cost 2027	Allocated Gas Cost 2027	Total Electric Cost	Total Gas Cost	Total Cost
New Construction	Lansing SC	Common	54%	46%	\$572,250	\$310,674	\$261,575	\$5,797,757	\$3,145,863	\$2,651,894	\$11,260,988	\$6,110,212	\$5,150,778	\$17,548,383	\$9,521,742	\$8,026,621	\$11,601,997	\$6,294,918	\$5,306,479	\$0	\$0	\$0	\$25,983,408	\$21,397,343	\$46,780,750
New Construction	Hastings SC	Common	54%	46%	\$5,304	\$2,280	\$2,425	\$1,000,000	\$542,600	\$457,400	\$800,000	\$434,080	\$365,740	\$9,000,000	\$4,888,400	\$4,116,000	\$6,200,000	\$3,364,120	\$2,835,880	\$0	\$0	\$0	\$9,277,080	\$7,778,224	\$17,005,304
New Construction	Greenville Cold Storage	Common	54%	46%	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$583,094	\$366,701	\$366,701	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$316,387	\$266,701	\$583,094
New Construction	Ludington Cold Storage	Electric	100%	0%	\$0	\$0	\$0	\$50,000	\$50,000	\$0	\$1,119,655	\$1,119,655	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1,269,655	\$0	\$1,269,655
New Construction	Hardy Headquarters	Generation	100%	0%	\$2,740,458	\$2,740,458	\$0	\$21,135	\$21,135	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$2,761,594	\$0	\$2,761,594
Renovations	Kalamazoo SC	Common	54%	46%	\$334,472	\$181,581	\$152,881	\$1,362,989	\$648,078	\$514,911	\$9,716,090	\$5,271,905	\$4,444,145	\$14,891,512	\$8,080,134	\$6,813,178	\$7,338,061	\$3,981,632	\$3,356,428	\$51,271	\$27,851	\$23,435	\$18,391,200	\$15,503,157	\$33,894,357
Renovations	Parnall Renovations (P1-3)	Corporate	54%	46%	\$5,000,228	\$2,714,624	\$2,285,604	\$394,893	\$214,249	\$180,624	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$2,928,891	\$2,466,228	\$5,395,121
Renovations	Jackson Dispatch	Common	54%	46%	\$537,536	\$293,828	\$245,708	\$1,000,000	\$542,600	\$457,400	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$834,428	\$703,108	\$1,537,536
Renovations	Midland Building	Electric	54%	46%	\$156,830	\$85,143	\$71,687	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$85,143	\$71,687	\$156,830
Renovations	Grand Rapids LVD-UG	Electric	100%	0%	\$1,545,013	\$1,545,013	\$0	\$1,386,430	\$1,386,430	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$2,931,443	\$0	\$2,931,443
Renovations	Control/Dispatch Center	Corporate	100%	0%	\$80,871	\$80,871	\$0	\$1,551,138	\$1,551,138	\$0	\$5,417,740	\$5,417,740	\$0	\$5,909,820	\$5,909,820	\$0	\$1,921,902	\$1,921,902	\$0	\$0	\$0	\$0	\$14,881,471	\$0	\$14,881,471
Renovations	South Monroe Interior Renovation	Electric	100%	0%	\$5,185	\$5,185	\$0	\$200,000	\$200,000	\$0	\$2,900,000	\$2,900,000	\$0	\$300,000	\$300,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3,405,185	\$0	\$3,405,185
Renovations	EY Infrastructure	Common	Varies based on site location		\$153,721	\$112,661	\$21,061	\$1,864,471	\$1,237,833	\$626,639	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1,930,494	\$647,658	\$1,998,152

Location		Historical Cost 2022	Plan Cost 2023	Plan Cost 2024	Plan Cost 2025	Plan Cost 2026	Plan Cost 2027	Total Cost
Lansing SC	Master Planning							\$0
	Programming	\$161,054						\$161,054
	Land Acquisition							\$0
	Engineering	\$411,196	\$1,950,000	\$100,000	\$100,000			\$2,461,196
	Construction		\$3,847,757	\$13,160,988	\$17,448,382	\$9,426,397		\$41,883,505
	Furnishings					\$1,750,000		\$1,750,000
	Commissioning					\$425,000		\$425,000
Hastings SC	Master Planning	\$5,304						\$5,304
	Programming		\$25,000					\$25,000
	Land Acquisition		\$975,000					\$975,000
	Engineering			\$800,000				\$800,000
	Construction				\$9,000,000	\$5,130,000		\$14,130,000
	Furnishings					\$750,000		\$750,000
	Commissioning					\$240,000		\$240,000
Greenville Cold Storage	Master Planning							\$0
	Programming							\$0
	Land Acquisition							\$0
	Engineering							\$0
	Construction				\$568,204			\$568,204
	Furnishings				\$14,888			\$14,888
	Commissioning							\$0
Ludington Cold Storage	Master Planning							\$0
	Programming							\$0
	Land Acquisition							\$0
	Engineering		\$50,000					\$50,000
	Construction			\$1,119,655	\$85,000			\$1,204,655
	Furnishings				\$15,000			\$15,000
	Commissioning							\$0
Hardy Headquarters	Master Planning							\$0
	Programming							\$0
	Land Acquisition							\$0
	Engineering							\$0
	Construction	\$2,587,058	\$24,135					\$2,608,058
	Furnishings	\$84,078						\$84,078
	Commissioning	\$69,313						\$69,313
Kalamazoo SC	Master Planning	\$334,472						\$334,472
	Programming							\$0
	Land Acquisition							\$0
	Engineering		\$1,562,989					\$1,562,989
	Construction		\$9,166,090	\$14,541,512	\$6,988,061			\$30,695,663
	Furnishings		\$500,000	\$250,000	\$250,000			\$1,000,000
	Commissioning		\$50,000	\$100,000	\$100,000	\$51,271		\$301,271
Parnall Renovations (P1-3)	Master Planning							\$0
	Programming							\$0
	Land Acquisition							\$0
	Engineering							\$0
	Construction	\$4,363,440	\$284,893					\$4,648,333
	Furnishings	\$636,783						\$636,783
	Commissioning		\$130,000					\$130,000
Jackson Dispatch	Master Planning							\$0
	Programming							\$0
	Land Acquisition							\$0
	Engineering	\$75,000						\$75,000
	Construction	\$801,147	\$970,000					\$1,771,147
	Furnishings	\$161,389						\$161,389
	Commissioning		\$30,000					\$30,000
Midland Building	Master Planning							\$0
	Programming	\$18,130						\$18,130
	Land Acquisition							\$0
	Engineering	\$138,700						\$138,700
	Construction							\$0
	Furnishings							\$0
	Commissioning							\$0
Grand Rapids LVD-UG	Master Planning							\$0
	Programming							\$0
	Land Acquisition							\$0
	Engineering	\$64,000						\$64,000
	Construction	\$1,360,110	\$1,386,430					\$2,746,540
	Furnishings	\$120,906						\$120,906
	Commissioning							\$0
Control/Dispatch Center	Master Planning	\$80,871						\$80,871
	Programming		\$65,000					\$65,000
	Land Acquisition							\$0
	Engineering				\$57,546			\$57,546
	Construction	\$393,480	\$5,287,746	\$3,437,388	\$1,711,802			\$11,464,500
	Furnishings		\$105,000	\$2,415,000	\$105,000			\$2,625,000
	Commissioning							\$0
South Monroe Interior Renovation	Master Planning							\$0
	Programming	\$5,185						\$5,185
	Land Acquisition							\$0
	Engineering		\$200,000					\$200,000
	Construction			\$2,900,000	\$162,500			\$3,062,500
	Furnishings				\$75,000			\$75,000
	Commissioning				\$62,500			\$62,500
EY Infrastructure	Master Planning							\$0
	Programming							\$0
	Land Acquisition							\$0
	Engineering	\$6,500	\$85,725					\$92,225
	Construction	\$127,221	\$1,773,246					\$1,898,467
	Furnishings							\$0
	Commissioning							\$0
Location		Historical Cost 2022	Plan Cost 2023	Plan Cost 2024	Plan Cost 2025	Plan Cost 2026	Plan Cost 2027	Total Cost

Lansing/Hastings/Kalamazoo Service Centers - Current Construction Cost

	Location	Type of Operations	Electric Allocation	Gas Allocation	Total Historical Cost 2022	Allocated Electric Cost 2022	Allocated Gas Cost 2022	Total Plan Cost 2023	Allocated Electric Cost 2023	Allocated Gas Cost 2023	Total Plan Cost 2024	Allocated Electric Cost 2024	Allocated Gas Cost 2024	Total Plan Cost 2025	Allocated Electric Cost 2025	Allocated Gas Cost 2025	Total Plan Cost 2026	Allocated Electric Cost 2026	Allocated Gas Cost 2026	Total Plan Cost 2027	Allocated Electric Cost 2027	Allocated Gas Cost 2027	Total Electric Cost	Total Gas Cost	Total Cost
New Construction	Lansing SC	Common	54%	46%	\$572,250	\$310,674	\$261,575	\$5,797,757	\$3,145,863	\$2,651,894	\$11,260,988	\$6,110,212	\$5,150,776	\$17,548,363	\$9,521,742	\$8,026,621	\$11,601,397	\$6,294,918	\$5,306,479	\$0	\$0	\$0	\$25,383,409	\$21,397,345	\$46,780,755
New Construction	Hastings SC	Common	54%	46%	\$5,304	\$2,880	\$2,425	\$1,000,000	\$542,600	\$457,400	\$800,000	\$434,080	\$365,920	\$9,000,000	\$4,883,400	\$4,116,600	\$6,200,000	\$3,364,120	\$2,835,880	\$0	\$0	\$0	\$9,227,080	\$7,778,225	\$17,005,304
Renovations	Kalamazoo SC	Common	54%	46%	\$334,472	\$181,585	\$152,887	\$1,562,989	\$848,078	\$714,911	\$9,716,090	\$5,271,950	\$4,444,140	\$14,891,512	\$8,080,134	\$6,811,378	\$7,338,061	\$3,981,632	\$3,356,429	\$51,273	\$27,821	\$23,452	\$18,391,200	\$15,503,197	\$33,894,397

	Location		Total Historical Cost 2022	Total Plan Cost 2023	Total Plan Cost 2024	Total Plan Cost 2025	Total Plan Cost 2026	Total Plan Cost 2027	Total Cost
New Construction	Lansing SC	Master Planning							\$0
		Programming	\$161,054						\$161,054
		Land Acquisition							\$0
		Engineering	\$411,196	\$1,950,000	\$100,000	\$100,000			\$2,561,196
		Construction		\$3,847,757	\$11,160,988	\$17,448,363	\$9,426,397		\$41,883,505
		Furnishings					\$1,750,000		\$1,750,000
		Commissioning					\$425,000		\$425,000
New Construction	Hastings SC	Master Planning	\$5,304						\$5,304
		Programming		\$25,000					\$25,000
		Land Acquisition		\$975,000					\$975,000
		Engineering			\$800,000				\$800,000
		Construction				\$9,000,000	\$5,130,000		\$14,130,000
		Furnishings					\$750,000		\$750,000
		Commissioning					\$320,000		\$320,000
	Kalamazoo SC	Master Planning							\$0
		Programming	\$334,472						\$334,472
		Land Acquisition							\$0
		Engineering		\$1,562,989					\$1,562,989
		Construction			\$9,166,090	\$14,541,512	\$6,988,061		\$30,695,663
		Furnishings			\$500,000	\$250,000	\$250,000		\$1,000,000
		Commissioning			\$50,000	\$100,000	\$100,000	\$51,273	\$301,273
	Location		Total Plan Cost 2022	Total Plan Cost 2023	Total Plan Cost 2024	Total Plan Cost 2025	Total Plan Cost 2026	Total Plan Cost 2027	Total Cost

Lansing/Hastings/Kalamazoo Service Centers - Construction Cost As Filed in U-20963

Location	Existing Floor Area	Number of Employees	Type of Operations	Electric Allocation	Gas Allocation	Projected Total Plan Cost 2020	Projected Electric Cost 2020	Projected Gas Cost 2020	Projected Total Plan Cost 2021	Projected Electric Cost 2021	Projected Gas Cost 2021	Projected Total Plan Cost 2022	Projected Electric Cost 2022	Projected Gas Cost 2022	Projected Total Plan Cost 2023	Projected Electric Cost 2023	Projected Gas Cost 2023	Projected Total Cost
Lansing SC	150,594 SF	412	Electric Operations, Gas Operations	54%	46%	\$1,782,689	\$966,574	\$816,115	\$4,000,000	\$2,168,800	\$1,831,200	\$25,000,000	\$13,555,000	\$11,445,000	\$24,000,000	\$13,012,800	\$10,987,200	\$54,782,689
Kalamazoo SC	140,884 SF	248	Electric Operations, Gas Operations	54%	46%	\$1,618	\$877	\$741	\$2,063,673	\$1,118,924	\$944,749	\$28,637,665	\$15,527,342	\$13,110,323	\$21,307,751	\$11,553,063	\$9,754,688	\$52,010,707
Hastings SC	12,317 SF	44	Electric Operations, Gas Operations	54%	46%	\$4,574	\$2,480	\$2,094	\$1,000,000	\$542,200	\$457,800	\$10,500,000	\$5,693,100	\$4,806,900	\$9,000,000	\$4,879,800	\$4,120,200	\$20,504,574

Location	Item	Total Plan Cost 2020	Total Plan Cost 2021	Total Plan Cost 2022	Total Plan Cost 2023	Total Cost
Lansing SC	Engineering		\$1,500,000			
	Land Acquisition	\$1,782,689				
	Construction		\$2,500,000	\$25,000,000	\$20,750,000	
	Furnishings				\$3,000,000	
	Commissioning				\$250,000	\$54,782,689
Kalamazoo SC	Engineering	\$1,618	\$2,063,673			
	Land Acquisition					
	Construction			\$28,637,665	\$18,057,751	
	Furnishings				\$3,000,000	
	Commissioning				\$250,000	\$52,010,707
Hastings SC	Engineering	\$4,574				
	Land Acquisition		\$1,000,000			
	Construction			\$10,500,000	\$7,400,000	
	Furnishings				\$1,500,000	
	Commissioning				\$100,000	\$20,504,574

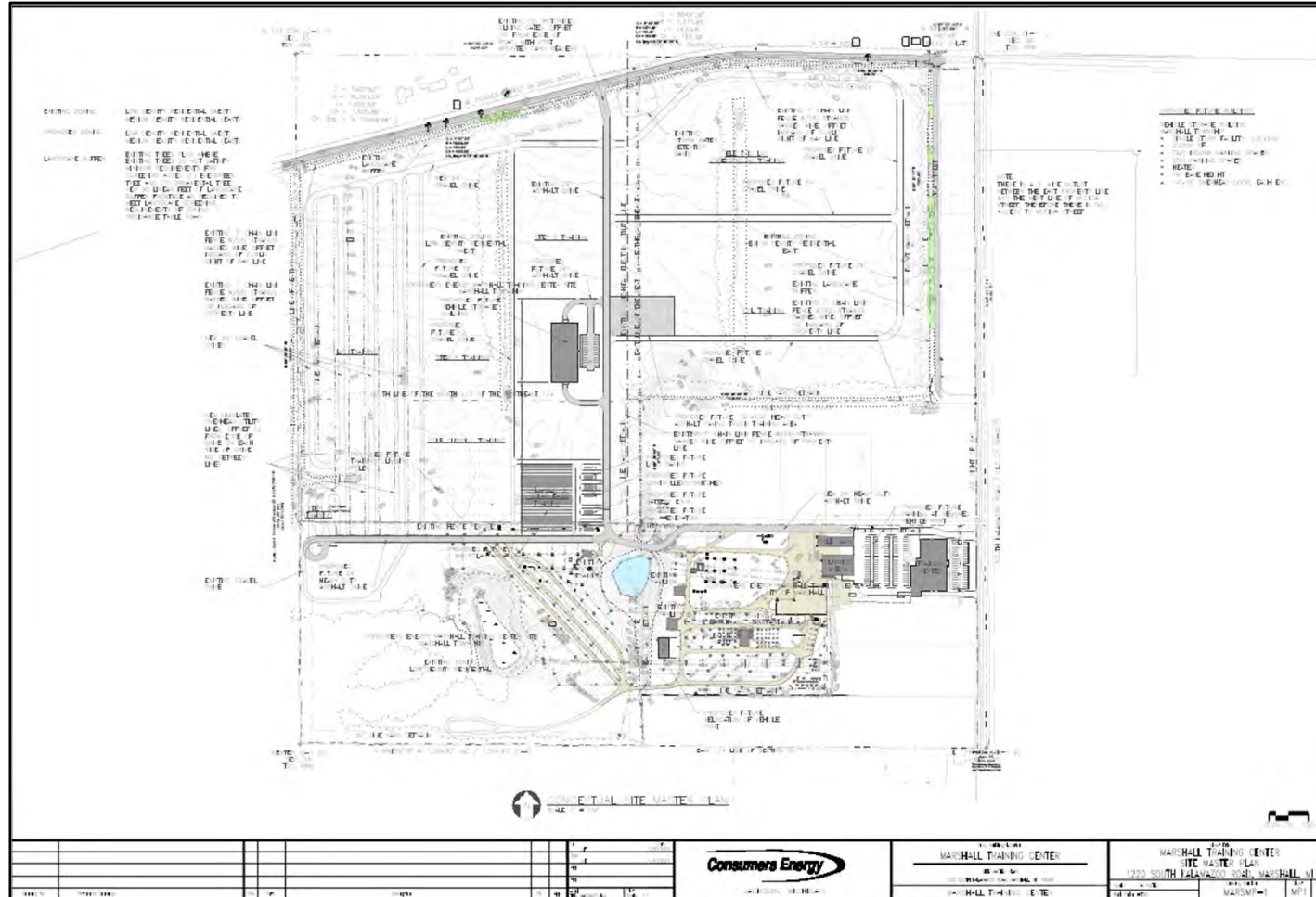
Marshall Roadway Paving

Plan Summary



[illegible]

Marshall Roadway Paving: Conceptual Site Master Plan Including Gravel Roadways



Lansing Service Center

Plan Summary



Lansing Service Center – Health Assessment

Facility Assessment - Lansing Service Center					
Location	Lansing	Maximum Possible Points (80)	Servicable > 48 Points	Poor < 48 Points	Good > 64 Points
Zone	MMOM				
Safety					
Compatible Use	0	5	3	2	4
Site Conditions	3	5	3	2	4
Traffic/Site Flow	1	5	3	2	4
Environmental Hazard	1	5	3	2	4
Quality					
Branding	2	5	3	2	4
Workplace Efficiency	3	5	3	2	4
Years since last major systems upgrade	0	5	3	2	4
Cost					
Facility Operating Costs	4	5	3	2	4
Non Facility Cost Impact	4	5	3	2	4
Space Optimization	3	5	3	2	4
Delivery					
Drive Time	3	5	3	2	4
Customer Response	4	5	3	2	4
Sustainability	1	5	3	2	4
Morale					
Employee Pride	2	5	3	2	4
Retention	3	5	3	2	4
Wellness	5	5	3	2	4
Total					
	39	80	48	32	64

[illegible]

Lansing Service Center – Health Assessment Scoring Criteria

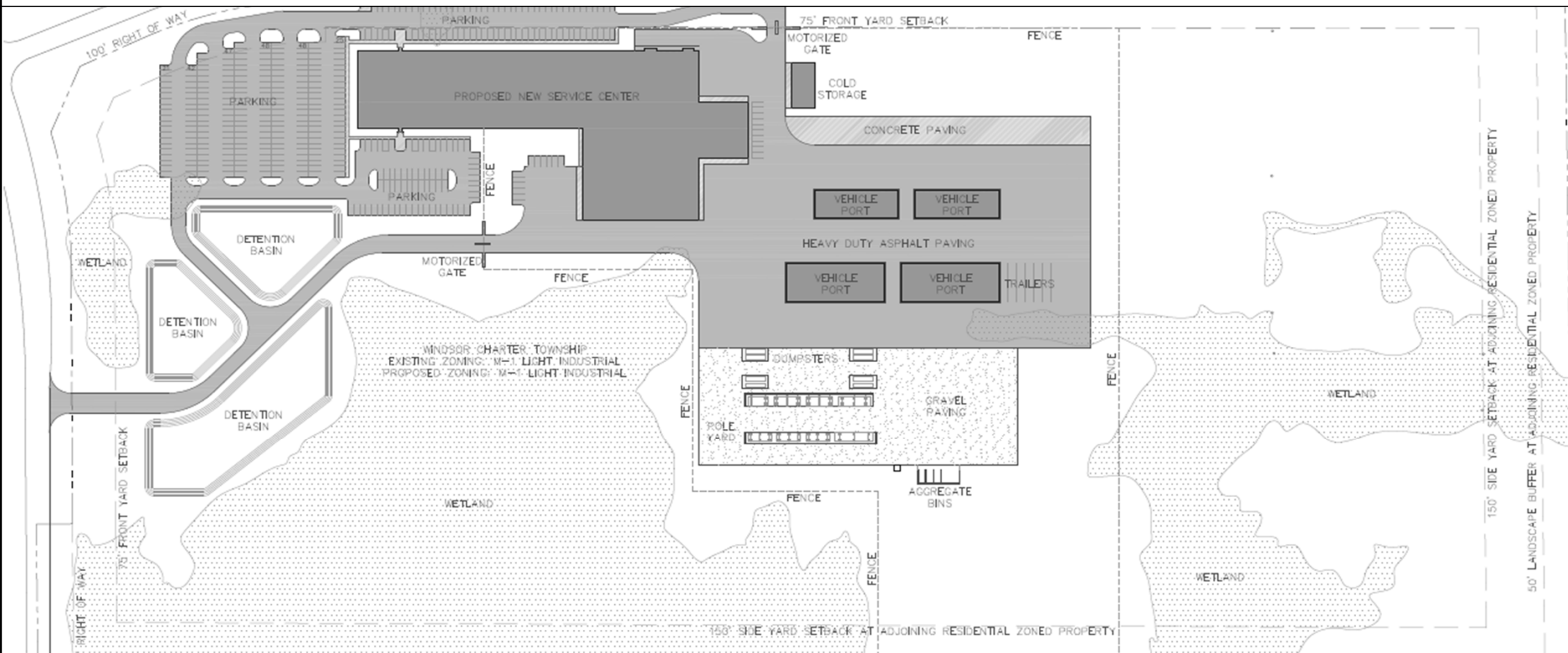
Cost (15 Points)				Score 0 Does not satisfy criteria	Score 1 Does not satisfy most listed criteria	Score 2 Does not satisfy some listed criteria	Score 3 Satisfies minimal criteria	Score 4 Satisfies most listed criteria	Score 5 Satisfies all listed criteria
		Pro	Con						
Facility Operating Costs	O&M costs to operate and maintain facility	Cost per square foot for facility less than statewide average cost per square foot (\$5.31)	Cost per square foot for facility more than statewide average cost per square foot (\$5.31)	costs greater than \$10.00 per square foot	costs between \$8.00 to \$10.00 per square foot	costs between \$7.25 to \$7.99 per square foot	costs between \$7.00 to \$7.24 per square foot	costs between \$5.75 to \$6.99 per square foot	Combined operating costs less than \$5.75 per square foot
Non Facility Cost Impact	O&M cost impacts to operations	Building correct size, centrally located within customer base area, able to accommodate necessary materials, able to maneuver onsite	Building too large, eccentrically located within customer base area, unable to accommodate necessary materials, difficult to maneuver onsite	Building eccentrically located within service territory, inadequate space	Building eccentrically located within service territory, underutilized space	Building eccentrically located within service territory	Building centrally located within service territory	Building accommodates operational needs, centrally located within service territory	Building accommodates operational need, space utilized efficiently, centrally located within service territory
Space Optimization	Reduce excess square footage	Space fully utilized, space utilized efficiently	Space under utilized, space utilized inefficiently, indoor parking area	Building vacant	Building partially occupied or partially vacant	Building occupied inefficiently, includes IPA	Building fully occupied, includes IPA	Building fully occupied, no IPA	occupied, no IPA, includes cold storage

Delivery (15 Points)				Score 0 Does not satisfy criteria	Score 1 Does not satisfy most listed criteria	Score 2 Does not satisfy some listed criteria	Score 3 Satisfies minimal criteria	Score 4 Satisfies most listed criteria	Score 5 Satisfies all listed criteria
		Pro	Con						
Drive Time	Service territory area of proper size	Centrally located within service area(s), drive time balanced across territory and within targeted limits	Eccentrically located within service area(s), drive time unbalanced across territory and/or exceeds targeted limits	service territory and size of service territory allows drive time >1:20 Gas and >1:30 Electric	service territory and size of service territory allows drive time < 1:20 Gas and <1:30 Electric	service territory and size of service territory allows drive time < 1:10 Gas and <1:15 Electric	service territory and size of service territory allows drive time < 1:00 Gas and <1:00 Electric	service territory and size of service territory allows drive time < 0:50 Gas and <0:45 Electric	service territory and size of service territory allows drive time < 0:40 Gas and <0:30 Electric
Customer Response	Meets needs of customers	Customer response time metrics fall below targets	Customer response times exceed targets	response times meet target <90% of time, Electric CAIDI >215 minutes	response times meet target >90% of time, Electric CAIDI <215 minutes	response times meet target >95% of time, Electric CAIDI <195 minutes	response times meet target >97% of time, Electric CAIDI <180 minutes	response times meet target >98% of time, Electric CAIDI <165 minutes	response times meet target >99% of time, Electric CAIDI <150 minutes
Sustainability	Facility incorporates reduced energy utilization and minimizes environmental impact	Facility constructed in accordance with recent Energy Codes, includes onsite stormwater treatment. Green building certification	Facility constructed with no insulation, utilizes energy intensive equipment, no onsite storm water treatment	Minimal roofing insulation	Upgraded roofing insulation	Upgraded roofing insulation and windows	Upgraded roofing insulation, windows and building envelope	insulation, windows, and building envelope with onsite storm water treatment	LEED Certified (or higher LEED rated) facility

Lansing Service Center – Health Assessment Scoring Criteria

Morale (15 Points)				Score 0 Does not satisfy criteria	Score 1 Does not satisfy most listed criteria	Score 2 Does not satisfy some listed criteria	Score 3 Satisfies minimal criteria	Score 4 Satisfies most listed criteria	Score 5 Satisfies all listed criteria
Employee Wellness		Pro	Con						
Employee Pride	Employees are engaged	Employee engagement scores > company average	Employee engagement scores < company average	Deteriorated facility	Deteriorated or dated facility with limited amenities	Dated facility with minimal amenities	Dated facility with full amenities and onsite parking	Modern facility with full amenities	Modern facility with full amenities, onsite parking
Retention	Facilities located in a desirable area, space is attractive to Millennials and supports millennial work styles	Downtown location, local amenities available, space supports varied work styles	Remote location, few local amenities, fixed work space	Remote location, no local amenities	some amenities available within 20 minutes driving distance	Remote location, amenities available within 20 minutes driving distance	Remote location, amenities available within 15 minutes driving distance	Diverse location, amenities available within 10 minutes driving distance	Diverse location, amenities available within walking distance
Wellness	Wellness facilities	Wellness room, space for trainer, drinking fountain, shower facilities nearby, space for wellness classes, well maintained exercise equipment, mirrors, lockers	Wellness room combined with non-compatible uses (i.e.: conference room), aged exercise equipment, poorly maintained exercise equipment, remote shower facilities	No shower, drinking fountain, Wellness or Nursing Mother Room, no additional space available	Shower and drinking fountain, no Wellness or Nursing Mother Room, no additional space available	Wellness Room or Nursing Mother Room, shower and drinking fountain, no additional space available	Nursing Mother Room, shower and drinking fountain, with space available to add Wellness or NMR	Wellness Room and Nursing Mother room, shower and drinking fountain, with space available for Trainer	Wellness room, Trainer room, Nursing Mother Room, shower and drinking fountain nearby

Lansing Service Center – Conceptual Site Plan



Lansing Service Center – Conceptual Site Plan



Lansing Service Center – Alternatives Matrix

						
Alternative Scenario Evaluation Matrix - Lansing Service Center Major Systems Upgrade						
	Weight	Build on New Site Score	Renov. Existing Score	Do Nothing Score	Score	COMMENTS
Section 1 - Affordable (50%)						
Operating cost per square foot below statewide average	40%	4.0	3.0	2.0		Current facility operating cost for Lansing is \$6.97 per square foot, versus statewide average operating cost of \$5.31 per square foot.
Construction cost	10%	1.0	4.0	5.0		
Section 1 - Scenario Average Weighted Score Total (Best = 5):		1.7	1.6	1.3	0.0	
Section 2 - Safe (25%)						
Hazardous Material Abatement (asbestos, lead paint)	5.0%	4.0	3.0	1.0		Current location is 3' below FEMA 100 year flood elev. Current location is former Manufactured Gas Plant site
Site located outside high crime area	5.0%	4.0	1.0	1.0		
Pedestrian traffic route separated from truck traffic route	2.5%	4.0	1.0	1.0		
Site located outside defined FEMA 100 year flood elevation	10.0%	4.0	1.0	1.0		
Site soils not impacted by previous use (Manufactured Gas Plant)	2.5%	4.0	1.0	1.0		
Section 2 - Scenario Average Score Total (Best = 5):		1.0	0.4	0.3	0.0	
Section 3 - Reliable (17.5%)						
Overall score above 72 as measured by Facilities Scorecard	2.5%	4.0	3.0	1.0		Current overall facility score for Lansing is 39
Building size matches operational needs	5.0%	4.0	2.0	2.0		Building fully occupied currently (pre COVID-19)
Improve CAIDI Score > 3%	4.0%	4.0	1.0	1.0		
On Premise start times improved by 5 to 10 minutes	5.0%	4.0	2.0	1.0		Improved material flow and vehicle loading
Drive through vehicle repair bays	2.5%	4.0	2.0	2.0		Improved vehicle repair time, reduced vehicle damage
Section 3 - Scenario Weighted Score Total (Best = 5):		0.6	0.3	0.2	0.0	
Section 4 - Clean (7.5%)						
LEED V4.1 Certification	2.5%	4.0	3.0	1.0		Stormwater currently direct discharge to Grand River
Stormwater detention and treatment	2.5%	4.0	1.0	1.0		
Onsite solar power generation	1.0%	4.0	1.0	1.0		
Section 4 - Scenario Weighted Score Total (Best = 5):		0.2	0.1	0.1	0.0	
TOTAL WEIGHTED VALUE SCORE: (BEST = 5)						
	100%	3.5	2.4	1.8	0.0	

Lansing Service Center – Alternatives Matrix Criteria

Safe
Hazardous Material Abatement (asbestos, lead paint)
Site located outside high crime area
Pedestrian traffic route separated from truck traffic route
Site located outside defined FEMA 100 year flood elevation
Site soils not impacted by previous use
Affordable
Operating cost per square foot below statewide average
Construction cost
Reliable
Overall score above 72 as measured by Facilities Scorecard
Building size matches operational needs
Improve CAIDI Score > 3%
On Premise start times improved by 5 to 10 minutes
Drive through vehicle repair bays
Clean
LEED V4.1 Certification
Stormwater detention and treatment
Onsite solar power generation
Non-Monetary Considerations
Other Risks & Barriers
Regulatory
Technical/Engineering
Aesthetics
People
Process
Technology
Customer Requirements
Real Estate
Municipality Requirements

New Lansing Service Center – Potential Locations

Property Details	Site A	Site B	Site C
Site Address	6964 N. Canal Rd. Dimondale, MI 48821	401 N. Verlinden Ave. Lansing, MI 48915	S. Waverly Rd & W. Miller Rd. Lansing, MI 48911
County	Eaton	Ingham	Ingham
Municipality	Windsor Township	City of Lansing	City of Lansing
Acreage	77.67	57	50

New Lansing Service Center – Potential Locations

Site Characteristics	Site A	Site B	Site C
Site Address	6964 N. Canal Rd. Dimondale, MI 48821	401 N. Verlinden Ave. Lansing, MI 48915	S. Waverly Rd. & W. Miller Rd. Lansing, MI 48911
Internal CE Security Assessment	Low Risk	High Risk	Medium Risk
Distance to Highways	Less than 1 mile to I-96	¼ Mile from M43; 2 miles to I-496	1.5 miles to I-96
Zoning	✓ M1 Industrial	✓ Industrial	✗ Residential
3 Phase	✓ Yes	✓ Yes	✗ No
Natural Gas	✓ Yes	✓ Yes	✓ Yes
Water	✗ No; water main extension pending	✓ Yes	✓ Yes
Sewer	✓ Yes	✓ Yes	✓ Yes
Fiber	✗ No	✗ No	✗ No
Wetlands Identified	Yes; 60 usable acres	No; 57 usable acres	Yes; 40 usable acres
Known Environmental Contamination	✓ No	✗ Yes	✓ No

New Lansing Service Center – Potential Locations – Gas / Electric Customer Response Time Comparisons

Electric work order x minutes					
Number of Work Orders and Time					
Verlinden Site					
Option	Electric				
10 minutes	17,984	179,840			
20 minutes	63,792	1,275,840			
30 minutes	15,949	478,470			
40 minutes	972	38,880			
50 minutes					
Total	98,697	1,973,030	Best Electric but worst Gas		
Waverly Site					
Option	Electric				
10 minutes	7,505	75,050			
20 minutes	52,207	1,044,140			
30 minutes	31,058	931,740			
40 minutes	7,841	313,640			
50 minutes	86	4,300			
Total	98,697	2,368,870	Worst Electric but best Gas		
Canal Site					
Option	Electric				
10 minutes	8,464	84,640			
20 minutes	56,349	1,126,980			
30 minutes	28,126	843,780			
40 minutes	5,742	229,680			
50 minutes	16	800			
Total	98,697	2,285,880	Mid for both Electric and Gas		

Gas work order x minutes					
Number of Work Orders and Time					
Verlinden Site					
Option	Gas				
10 minutes	41,742	417,420			
20 minutes	62,020	1,240,400			
30 minutes	22,865	685,950			
40 minutes	1,939	77,560			
50 minutes					
Total	128,566	2,421,330	Worst Gas but best Electric		
Waverly Site					
Option	Gas				
10 minutes	26,982	269,820			
20 minutes	77,463	1,549,260			
30 minutes	19,380	581,400			
40 minutes	4,741	189,640			
50 minutes					
Total	128,566	2,590,120	Best Gas but worst Electric		
Canal Site					
Option	Gas				
10 minutes	21,801	218,010			
20 minutes	85,674	1,713,480			
30 minutes	17,453	523,590			
40 minutes	3,638	145,520			
50 minutes					
Total	128,566	2,600,600	Mid for both Gas and Electric		

New Lansing Service Center – Potential Locations within City of Lansing

401 N. Verlinden Ave. Lansing, MI 48915

Positives:

- Low purchase price
- Potential tax incentives available
- Utilities available

Limitations:

- Environmental concerns
- Unknown abandon utilities and tunnels under surface
- LBWL substation on site along with HVD line but unable to serve ourselves with electric
- Fiber would need to be ran to site
- Medium-risk security assessment needed

S. Waverly Rd. & W. Miller Rd. Lansing, MI 48911

Positives:

- Potential assistance with site preparation
- Potential assistance with demo of asbestos

Limitations:

- Wetlands and possibly not enough usable property
- Potentially unable to have a drive on S. Waverly due to hill and high speeds coming northbound on Waverly - both drives would have to be off of W. Miller Rd
- Length of time to potentially acquire site
- Demo of houses with asbestos
- Probably unable to serve ourselves with electric
- Fiber is currently some distance away – would need to be ran to site
- Currently zoned residential and would need to be rezoned to light industrial
- Medium-risk security assessment needed
- Unknown purchase price

New Lansing Service Center – Potential Locations within City of Lansing

South of Grand River, Ave.

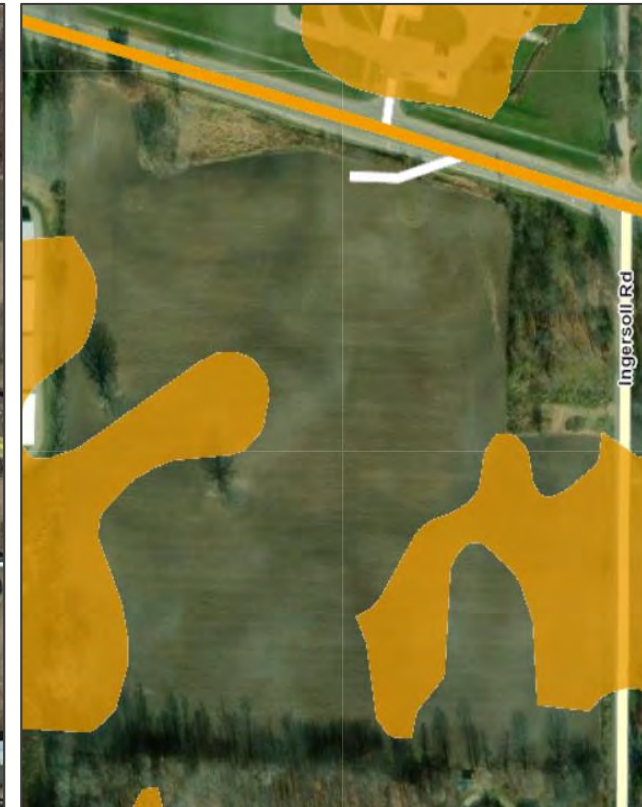
Positives:

- Utilities available (Water & Sewer)

Limitations:

- Fuel station across the street
- Possible wetlands
- Further north
- Lots of traffic on road

*No other data was collected regarding this location



New Lansing Service Center Potential Location 401 N. Verlinden Ave. Lansing - Site Characteristics 57 acres

Response Time:	• 87% of reported outages are reached in 30 minutes timeframe • 100% of reported outages are reached within 45 minutes
Ease of Travel:	CE Operations: • Difficulty with turning and maneuvering trucks and trailers through city streets and blocks • Travel impediment due to foot and bike traffic, CATA bus stops, railroad crossing and street parking in residential areas • Difficult to access highways through residential areas: 2 miles to I-496
Security:	• CE Security Risk Assessment Rating – HIGH • Crimes occurring within 1-mile radius of parcel within 12/26/19-01/23/20: Assault – 15; Weapon Offenses – 6; Larceny – 6; Vandalism – 4; Burglary – 4; Motor Vehicle Theft – 3; Robbery – 1 • 74 sex offenders in registry within 1-mile radius
Wetlands identified:	• No identified wetlands on the parcel
Environmental concerns:	• Significant concern regarding existing contamination and abandoned underground utility facilities on the property. Further due diligence would be required if site is selected to evaluate impact to service center construction and operations.

New Lansing Service Center Potential Location

S. Waverly & W. Miller Rd, Lansing - Site Characteristics

33 usable acres

Response Time:	88% of reported outages are reached in the 30 minutes 100% of reported outages are reached within 50 minutes
Ease of Travel:	CE Operations: - Difficulty with turning and maneuvering trucks and trailers through city streets and blocks - Travel impediment due to foot and bike traffic, CATA bus stops and street parking in residential areas - Difficult to access highways through residential areas: 1.5 miles to I-96
Security:	CE Security Risk Assessment Rating – MEDIUM - Crimes occurring within 1-mile radius of parcel within 12/26/19-01/23/20: Assault – 6; Vandalism – 1; Larceny – 1; Motor Vehicle Theft – 1 16 sex offenders in registry within 1-mile radius
Wetlands identified:	- The State's National Wetland Inventory map indicates 17 acres of regulated wetlands of the 50-acre parcel - CE would not be able to use 17 acres without costly wetland mitigation. Would limit or possibly eliminate future expansion on site.
Environmental concerns:	- LEAP reports asbestos abatement needed prior to demolition of existing houses on parcel - CE Environmental reports existing contaminated site on the adjacent property to the east



New Lansing Service Center Potential Location

6964 N. Canal Rd, Dimondale - Site Characteristics

60 usable acres

Response Time:	• 91% of reported outages are reached in the 30 minute timeframe • 100% of reported outages are reached within 45 minutes
Ease of Travel:	CE Operations: • Located in the industrial area of Windsor Commerce Parkway. Roads and intersections provide for wider turns for maneuvering trucks and trailers • Access to highways through commercial areas: 0.75 miles to I-96; 1.35 miles to I-69 • Limited foot and bike traffic and no parking on roads
Security:	• CE Security Risk Assessment Rating – LOW • Crimes occurring within 1-mile radius of parcel within 12/26/19-01/23/20: Assault – 0 • 4 sex offenders in registry within 1-mile radius • High police presence due to Michigan State Police within 0.6 miles from parcel
Wetlands identified:	• The State's National Wetland Inventory map indicates 7.29 acres of regulated wetlands on the 77.67-acre parcel • Property size allows for CE use to be designed to avoid wetland areas and allow for future expansion.
Environmental concerns:	• No known environmental concerns

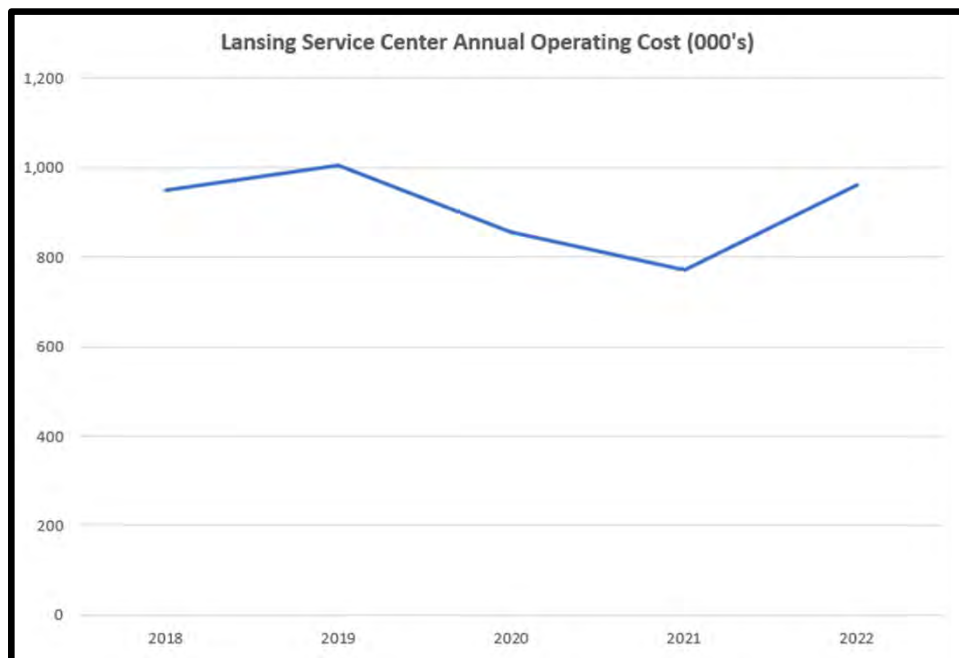
New Lansing Service Center – Gas Benefits

New Lansing Service Center - Gas Customer Benefits					
Gas work order x minutes					
Number of Work Orders x Travel Time					
Canal - New Site	Gas	WO x Min.	Willow - Current Site	Gas	Adjust.
10 minutes	21,801	218,010	10 minutes	25,726	257,260
20 minutes	85,674	1,713,480	20 minutes	62,260	1,245,200
30 minutes	17,453	523,590	30 minutes	30,560	916,800
40 minutes	3,638	145,520	40 minutes	9,489	379,560
50 minutes			50 minutes	531	26,550
Total	128,566	2,600,600	Total	128,566	2,825,370
Difference:		224,770	Minutes reduction in travel time to customer sites compared to Willow Road		
		8.0%	Reduction in travel time to customer sites compared to Willow Road		
Cost Savings:		\$309,059	Productivity recaptured annually based on \$2.75 per minute per employee		
Gas Response Time Improvement:		224,770			

New Lansing Service Center – Electric Benefits

New Lansing Service Center - Electric Customer Benefits					
Electric work order x minutes					
Number of Work Orders x Travel Time					
Canal - New Site	Electric	WO x Min.	Willow - Current Site	Electric	WO x Min.
10 minutes	8,464	84,640	10 minutes	2,142	21,420
20 minutes	56,349	1,126,980	20 minutes	58,784	1,175,680
30 minutes	28,126	843,780	30 minutes	31,496	944,880
40 minutes	5,742	229,680	40 minutes	6,239	249,560
50 minutes	16	800	50 minutes	36	1,800
Total	98,697	2,285,880	Total	98,697	2,393,340
2018 Electric Outages		50,135	Customers		
2019 Electric Outages		56,670	Customers		
Total Electric Outages		106,805	Customers		
Avg Annual Outages		53,403	Customers		
Difference:		107,460	Minutes reduction in travel time to customer sites compared to Willow Road		
		4.5%	Reduction in travel time to customer sites compared to Willow Road		
Cost Savings:		\$185,906	Productivity recaptured annually based on \$3.46 per minute per employee		
Electric CAIDI Improvement:		53,730	Annual reduction in travel minutes		
		4,478	Monthly reduction in travel minutes		

Lansing Service Center – Operating Costs



Service Center	Existing Building Area	Existing Building 2022 Annual Operating Cost	Existing Building Cost Per Square Foot	New Building Estimated Cost Per Square Foot	New Building Area - Current Conceptual	New Building Estimated Annual Cost
Lansing	150,594 SF	\$960,864	\$6.38	\$5.75	125,000 SF	\$718,750

Hastings Service Center

Plan Summary



Hastings Service Center – Health Assessment

Facility Assessment - Hastings Service Center					
Location	Hastings	Maximum Possible Points (80)	Serviceable > 48 Points	Poor < 48 Points	Good > 64 Points
Zone	SWOM				
Safety					
Compatible Use	4	5	3	2	4
Site Conditions	4	5	3	2	4
Traffic/Site Flow	2	5	3	2	4
Environmental Hazard	3	5	3	2	4
Quality					
Branding	3	5	3	2	4
Workplace Efficiency	3	5	3	2	4
Years since last major systems upgrade	1	5	3	2	4
Cost					
Facility Operating Costs	0	5	3	2	4
Non Facility Cost Impact	4	5	3	2	4
Space Optimization	2	5	3	2	4
Delivery					
Drive Time	4	5	3	2	4
Customer Response	3	5	3	2	4
Sustainability	3	5	3	2	4
Morale					
Employee Pride	1	5	3	2	4
Retention	3	5	3	2	4
Wellness	1	5	3	2	4
Total	41	80	48	32	64

[illegible]

Hastings Service Center – Health Assessment Scoring Criteria

Cost (15 Points)				Score 0 Does not satisfy criteria	Score 1 Does not satisfy most listed criteria	Score 2 Does not satisfy some listed criteria	Score 3 Satisfies minimal criteria	Score 4 Satisfies most listed criteria	Score 5 Satisfies all listed criteria
		Pro	Con						
Facility Operating Costs	O&M costs to operate and maintain facility	Cost per square foot for facility less than statewide average cost per square foot (\$5.31)	Cost per square foot for facility more than statewide average cost per square foot (\$5.31)	costs greater than \$10.00 per square foot	costs between \$8.00 to \$10.00 per square foot	costs between \$7.25 to \$7.99 per square foot	costs between \$7.00 to \$7.24 per square foot	costs between \$5.75 to \$6.99 per square foot	Combined operating costs less than \$5.75 per square foot
Non Facility Cost Impact	O&M cost impacts to operations	Building correct size, centrally located within customer base area, able to accommodate necessary materials, able to maneuver onsite	Building too large, eccentrically located within customer base area, unable to accommodate necessary materials, difficult to maneuver onsite	Building eccentrically located within service territory, inadequate space	Building eccentrically located within service territory, underutilized space	Building eccentrically located within service territory	Building centrally located within service territory	Building accommodates operational needs, centrally located within service territory	Building accommodates operational need, space utilized efficiently, centrally located within service territory
Space Optimization	Reduce excess square footage	Space fully utilized, space utilized efficiently	Space under utilized, space utilized inefficiently, indoor parking area	Building vacant	Building partially occupied or partially vacant	Building occupied inefficiently, includes IPA	Building fully occupied, includes IPA	Building fully occupied, no IPA	occupied, no IPA, includes cold storage

Delivery (15 Points)				Score 0 Does not satisfy criteria	Score 1 Does not satisfy most listed criteria	Score 2 Does not satisfy some listed criteria	Score 3 Satisfies minimal criteria	Score 4 Satisfies most listed criteria	Score 5 Satisfies all listed criteria
		Pro	Con						
Drive Time	Service territory area of proper size	Centrally located within service area(s), drive time balanced across territory and within targeted limits	Eccentrically located within service area(s), drive time unbalanced across territory and/or exceeds targeted limits	service territory and size of service territory allows drive time >1:20 Gas and >1:30 Electric	service territory and size of service territory allows drive time < 1:20 Gas and <1:30 Electric	service territory and size of service territory allows drive time < 1:10 Gas and <1:15 Electric	service territory and size of service territory allows drive time < 1:00 Gas and <1:00 Electric	service territory and size of service territory allows drive time < 0:50 Gas and <0:45 Electric	service territory and size of service territory allows drive time < 0:40 Gas and <0:30 Electric
Customer Response	Meets needs of customers	Customer response time metrics fall below targets	Customer response times exceed targets	response times meet target <90% of time, Electric CAIDI >215 minutes	response times meet target >90% of time, Electric CAIDI <215 minutes	response times meet target >95% of time, Electric CAIDI <195 minutes	response times meet target >97% of time, Electric CAIDI <180 minutes	response times meet target >98% of time, Electric CAIDI <165 minutes	response times meet target >99% of time, Electric CAIDI <150 minutes
Sustainability	Facility incorporates reduced energy utilization and minimizes environmental impact	Facility constructed in accordance with recent Energy Codes, includes onsite stormwater treatment. Green building certification	Facility constructed with no insulation, utilizes energy intensive equipment, no onsite storm water treatment	Minimal roofing insulation	Upgraded roofing insulation	Upgraded roofing insulation and windows	Upgraded roofing insulation, windows and building envelope	insulation, windows, and building envelope with onsite storm water treatment	LEED Certified (or higher LEED rated) facility

Hastings Service Center – Health Assessment Scoring Criteria

Morale (15 Points)				Score 0 Does not satisfy criteria	Score 1 Does not satisfy most listed criteria	Score 2 Does not satisfy some listed criteria	Score 3 Satisfies minimal criteria	Score 4 Satisfies most listed criteria	Score 5 Satisfies all listed criteria
Employee Wellness		Pro	Con						
Employee Pride	Employees are engaged	Employee engagement scores > company average	Employee engagement scores < company average	Deteriorated facility	Deteriorated or dated facility with limited amenities	Dated facility with minimal amenities	Dated facility with full amenities and onsite parking	Modern facility with full amenities	Modern facility with full amenities, onsite parking
Retention	Facilities located in a desirable area, space is attractive to Millennials and supports millennial work styles	Downtown location, local amenities available, space supports varied work styles	Remote location, few local amenities, fixed work space	Remote location, no local amenities	some amenities available within 20 minutes driving distance	Remote location, amenities available within 20 minutes driving distance	Remote location, amenities available within 15 minutes driving distance	Diverse location, amenities available within 10 minutes driving distance	Diverse location, amenities available within walking distance
Wellness	Wellness facilities	Wellness room, space for trainer, drinking fountain, shower facilities nearby, space for wellness classes, well maintained exercise equipment, mirrors, lockers	Wellness room combined with non-compatible uses (i.e.: conference room), aged exercise equipment, poorly maintained exercise equipment, remote shower facilities	No shower, drinking fountain, Wellness or Nursing Mother Room, no additional space available	Shower and drinking fountain, no Wellness or Nursing Mother Room, no additional space available	Wellness Room or Nursing Mother Room, shower and drinking fountain, no additional space available	Nursing Mother Room, shower and drinking fountain, with space available to add Wellness or NMR	Wellness Room and Nursing Mother room, shower and drinking fountain, with space available for Trainer	Wellness room, Trainer room, Nursing Mother Room, shower and drinking fountain nearby

GREEN STREET

HASTINGS AVENUE

FINAL SITE PLAN

Service Center

Office

Restroom

Parking

Green Street

Hastings Avenue

Final Site Plan

Hastings Service Center – Alternatives Matrix

						
Alternative Scenario Evaluation Matrix - Hastings Service Center Major Systems Upgrade						
	Weight	Build on New Site Score	Build on Ex. Site Score	Renov. Existing Score	Do Nothing Score	COMMENTS
Section 1 - Affordable (50%)						
Operating cost per square foot below statewide average	40%	4.0	4.0	2.0	1.0	Current facility operating cost for Hastings is \$13.85 per square foot, versus statewide average operating cost of \$5.31 per square foot.
Construction cost	10%	2.0	2.0	3.0	4.0	
Section 1 - Scenario Average Weighted Score Total (Best = 5):		1.8	1.8	1.1	0.8	
Section 2 - Safe (12.5%)						
Hazardous Material Abatement (asbestos, lead paint)	2.5%	4.00	4.0	3.0	2.0	
Reduced site congestion and improved vehicular traffic flow	10%	4.00	3.00	1.00	1.00	
Section 2 - Scenario Average Score Total (Best = 5):		0.5	0.4	0.2	0.2	
Section 3 - Reliable (25%)						
Overall score above 72 as measured by Facilities Scorecard	5%	4.0	4.0	3.0	1.0	Current overall facility score for Hastings is 41
On Premise start times improved by 5 to 10 minutes	10%	4.0	4.0	2.0	1.0	Improved material flow and vehicle loading
Drive through vehicle repair bays	5%	4.0	4.0	3.0	3.0	Improved vehicle repair time, reduced vehicle damage
Temporary relocation of operations	5%	4.0	1.0	2.0	4.0	
Section 3 - Scenario Weighted Score Total (Best = 5):		1.0	0.9	0.6	0.5	
Section 4 - Clean (12.5%)						
LEED V4.1 Certification	2.5%	4.0	4.0	2.0	1.0	
Stormwater detention and treatment	2.5%	4.0	4.0	3.0	1.0	Stormwater currently discharged to ditch
Onsite solar power generation	2.5%	4.0	4.0	1.0	1.0	
Municipal Utilities Available	5%	1.0	4.0	4.0	4.0	Available parcels are located outside Hastings urban area, private well and septic will be required
Section 3 - Scenario Weighted Score Total (Best = 5):		0.4	0.5	0.4	0.3	
TOTAL WEIGHTED VALUE SCORE: (BEST = 5)						
	100%	3.7	3.6	2.2	1.7	

Hastings Service Center – Alternatives Matrix Criteria

Safe
Hazardous Material Abatement (asbestos, lead paint)
Reduced site congestion and improved vehicular traffic flow
Affordable
Operating cost per square foot below statewide average
Construction cost
Reliable
Overall score above 72 as measured by Facilities Scorecard
On Premise start times improved by 5 to 10 minutes
Temporary relocation of operations
Drive through vehicle repair bays
Clean
LEED V4.1 Certification
Stormwater detention and treatment
Onsite solar power generation
Municipal Utilities Available
Non-Monetary Considerations
Other Risks & Barriers
Regulatory
Technical/Engineering
Aesthetics
People
Process
Technology
Customer Requirements
Real Estate
Municipality Requirements

Gas Call Drive Time Summary

- **For the current Hastings Service Center and the proposed site at Vacant Land M-179**
 - Within 20 minutes we are able to reach 50% of gas customers and in 40 minutes we are able to reach 100% of gas customers.
- **For the proposed site at Vacant Land Enterprise Drive**
 - Within 25 minutes we are able to reach 50% of gas customers and in 45 minutes we are able to reach 100% of gas customers.

Electric Outage Drive Time Summary

- **For the current Hastings Service Center and the proposed site at Vacant Land M-179**
 - Within 15 minutes we are able to reach 50% of electric customers and in 30 minutes we are able to reach 100% of electric customers.
- **For the proposed site at Vacant Land Enterprise Drive**
 - Within 20 minutes we are able to reach 50% of electric customers and in 30 minutes we are able to reach 100% of electric customers.

Summary Site Comparison

Vacant Land: Enterprise Dr

- Public water and septic
- Would add 5 minutes to both electric and gas drive times

Vacant Land: M-179

- Private water and septic
- Drive times would remain similar to what is being experienced at the current location

Hastings Service Center Site Options Comparison

Vacant Land: Enterprise Dr, Hastings, MI 49058

Characteristics:

- Property – 18 acres
- Municipal Water – Available at the street
- Municipal Sewer – Available at the street
- Fiber – Available
- Wetlands – No mitigation required
- 3-Phase – Available at the street
- Natural Gas – Available at the street

Vacant Land: M-179, Hastings, MI 49058

Characteristics:

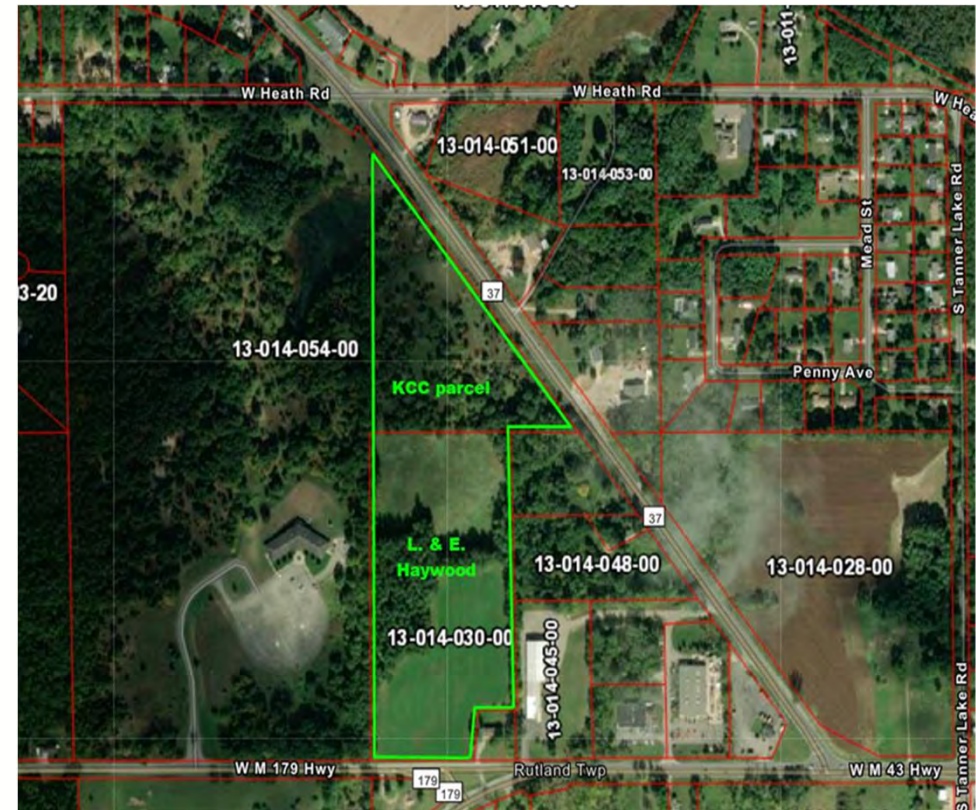
- Property – 30 Acres
- Private Water System – New system needed
- Private Septic System – New system needed
- Fiber – Available
- Wetlands – No mitigation required
- 3-Phase – Available at the street
- Natural Gas – Available at the street

Vacant Land: M-179, Hastings, MI 49058

Location Description: Just west of the corner of M-43 and M-37, on M-179 at M-43. Just East of Kellogg Community College parcel.

Access: Main entrance off M-37

Owner: 2 property owners have been approached one being Kellogg Community College



Vacant Land: Enterprise Dr, Hastings, MI 49058

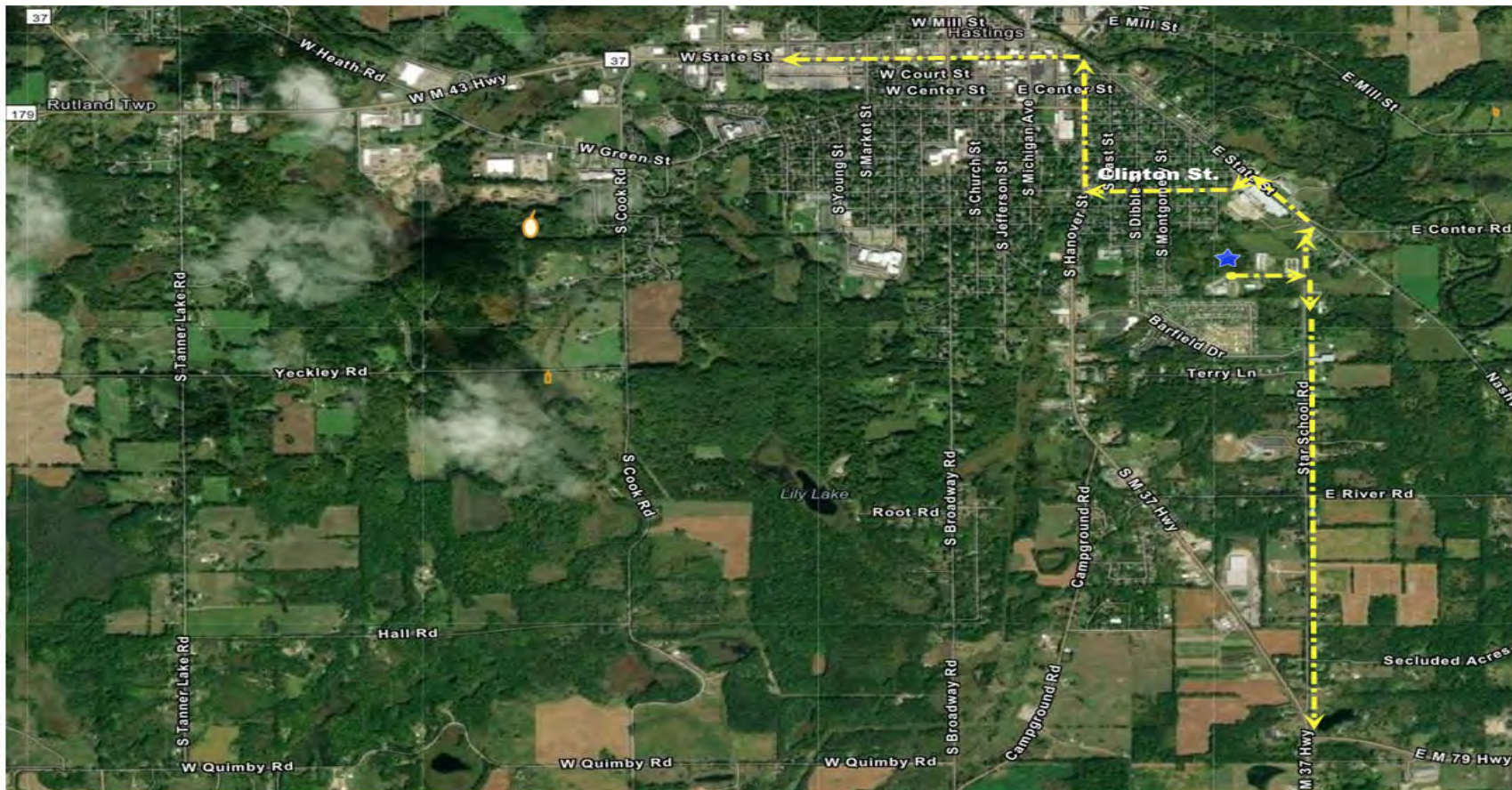
Location Description: Southeast in the City just off E State St and Star School Rd in the Industrial Park

Access: There could be multiple drives on E. Enterprise Dr. Northeast parcel has access to Star School Rd.

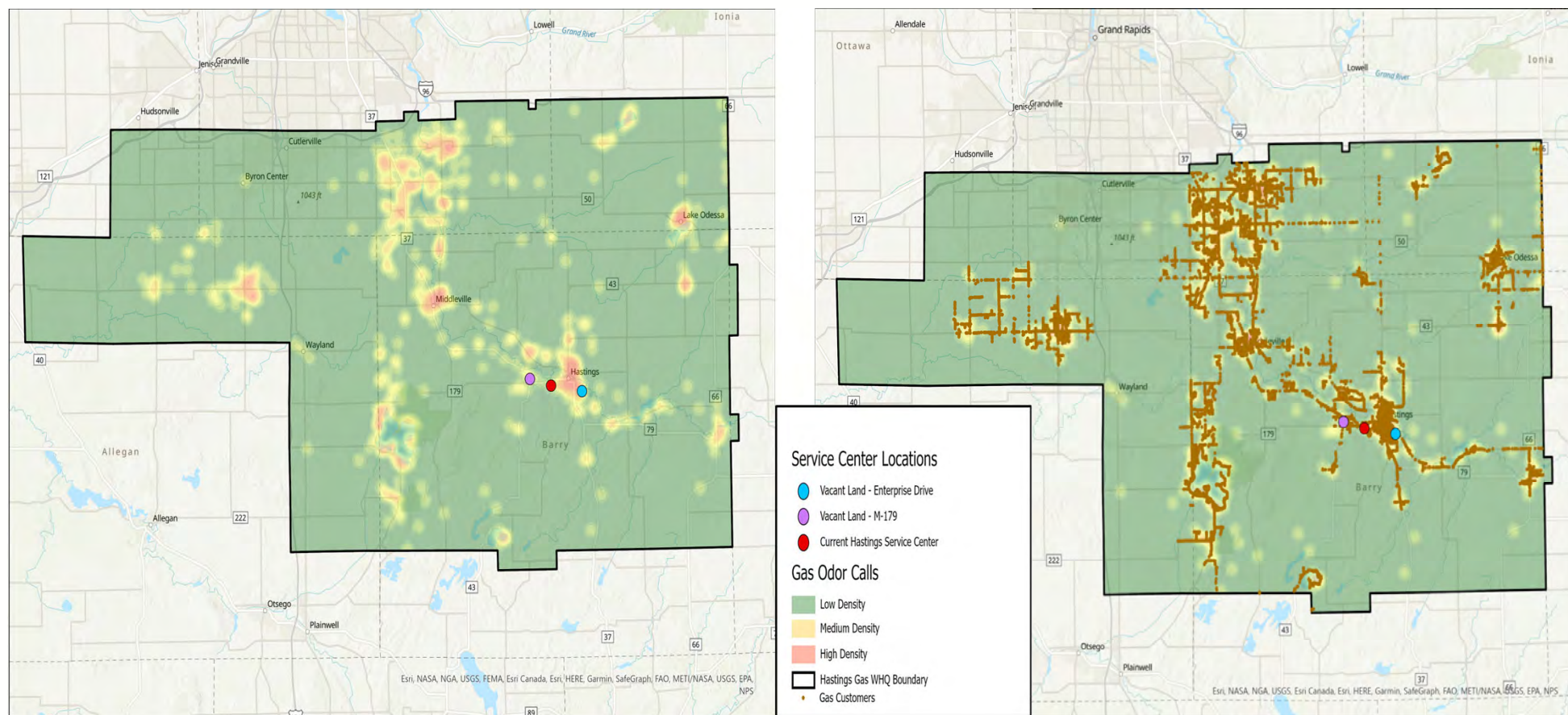
Owner: City of Hastings, parcels are currently for sale.



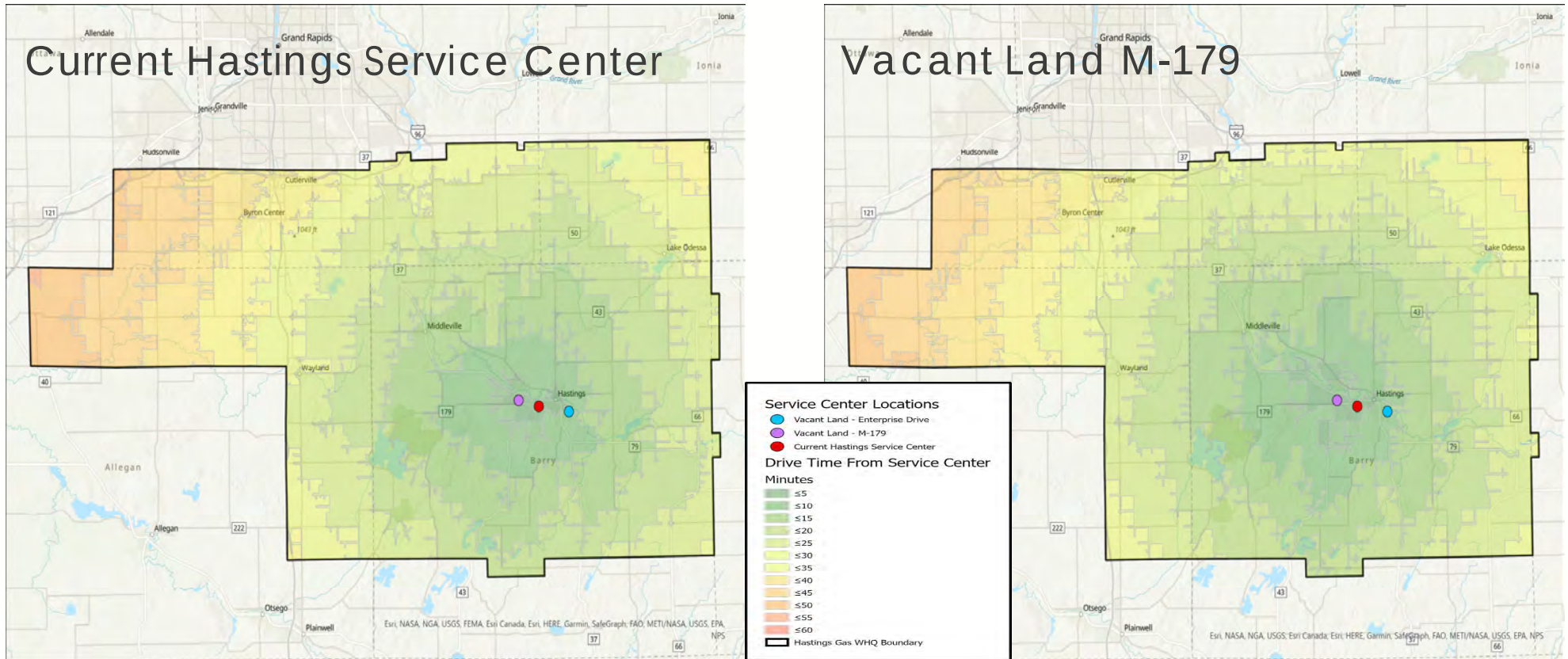
Class A Road Access Vacant Land Enterprise Dr.



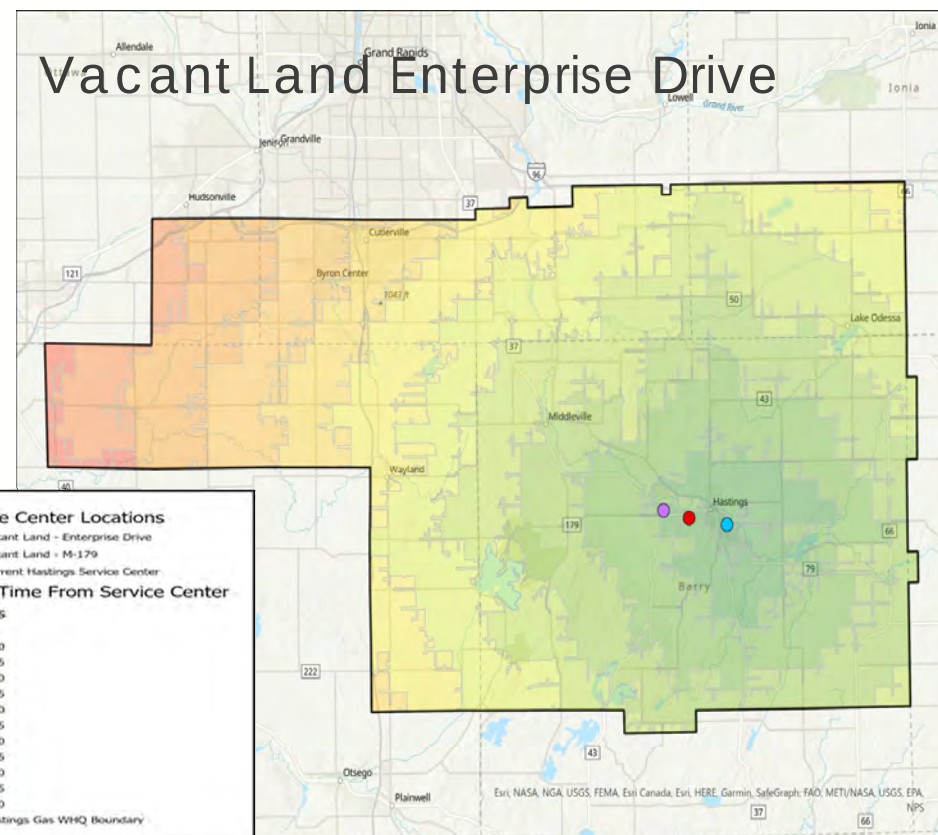
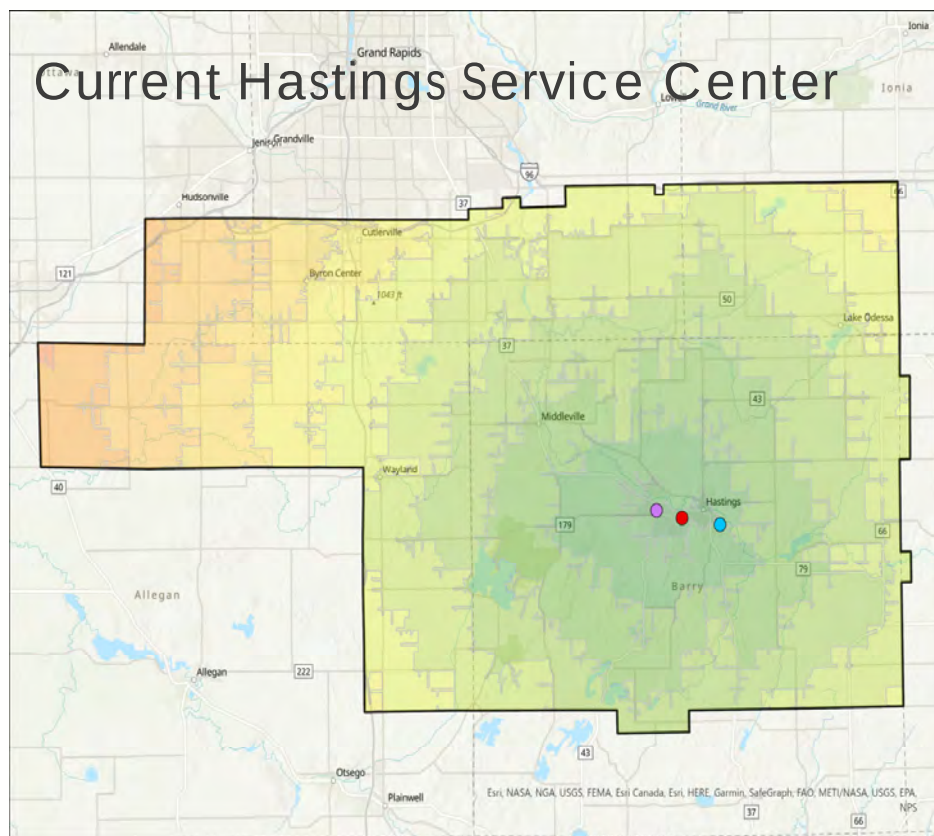
2019-2020 Density of Gas Odor Calls and Customer Data



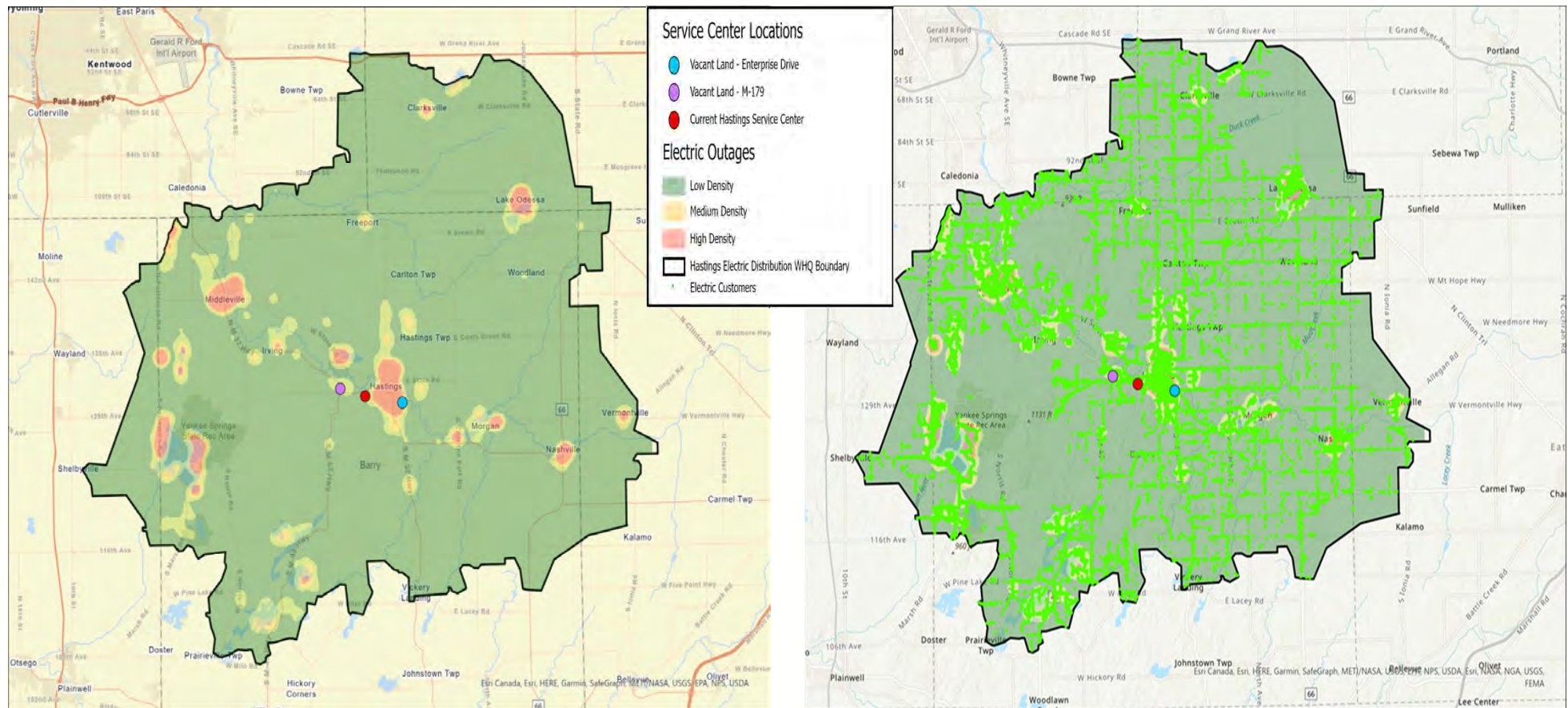
Gas Call Drive Times



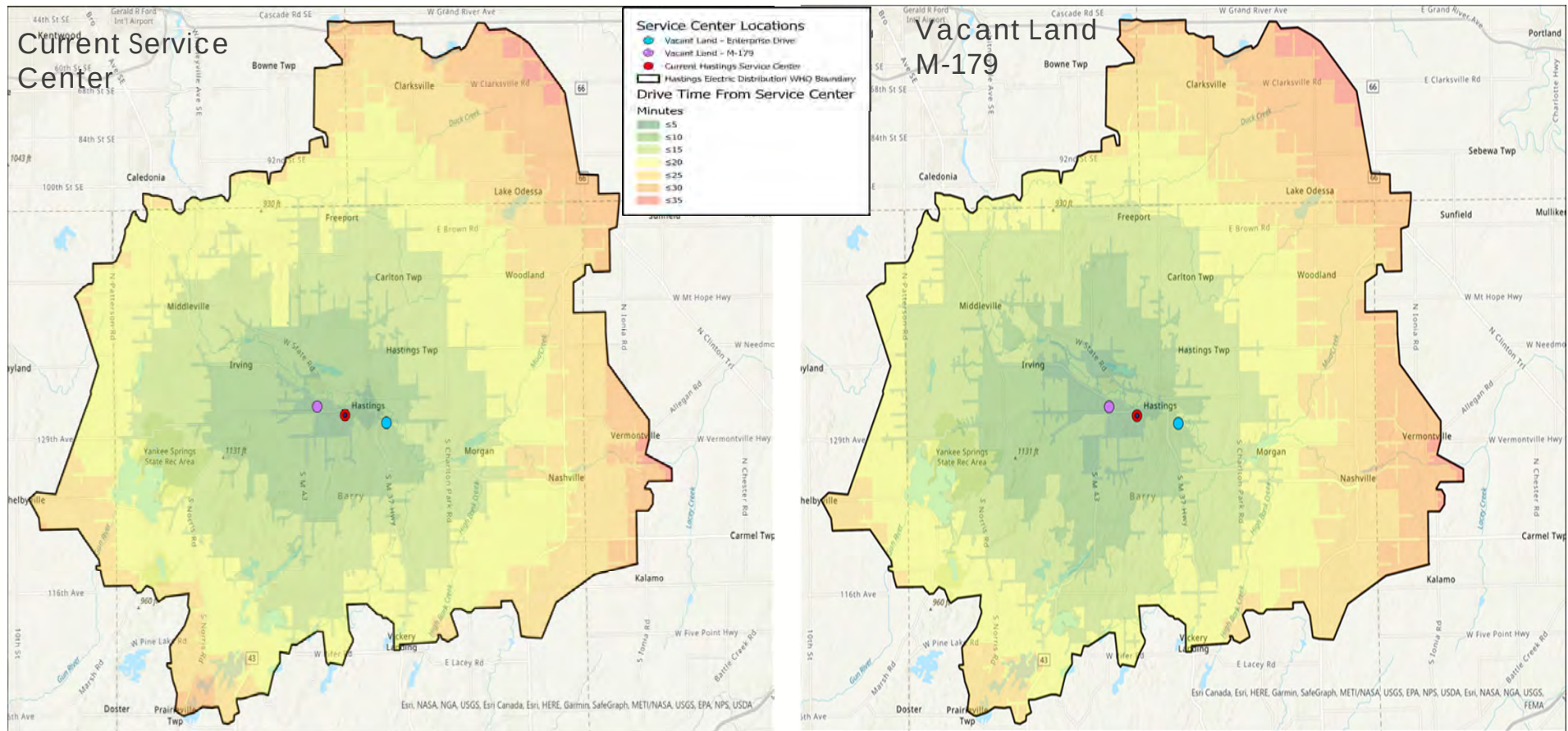
Gas Call Drive Times



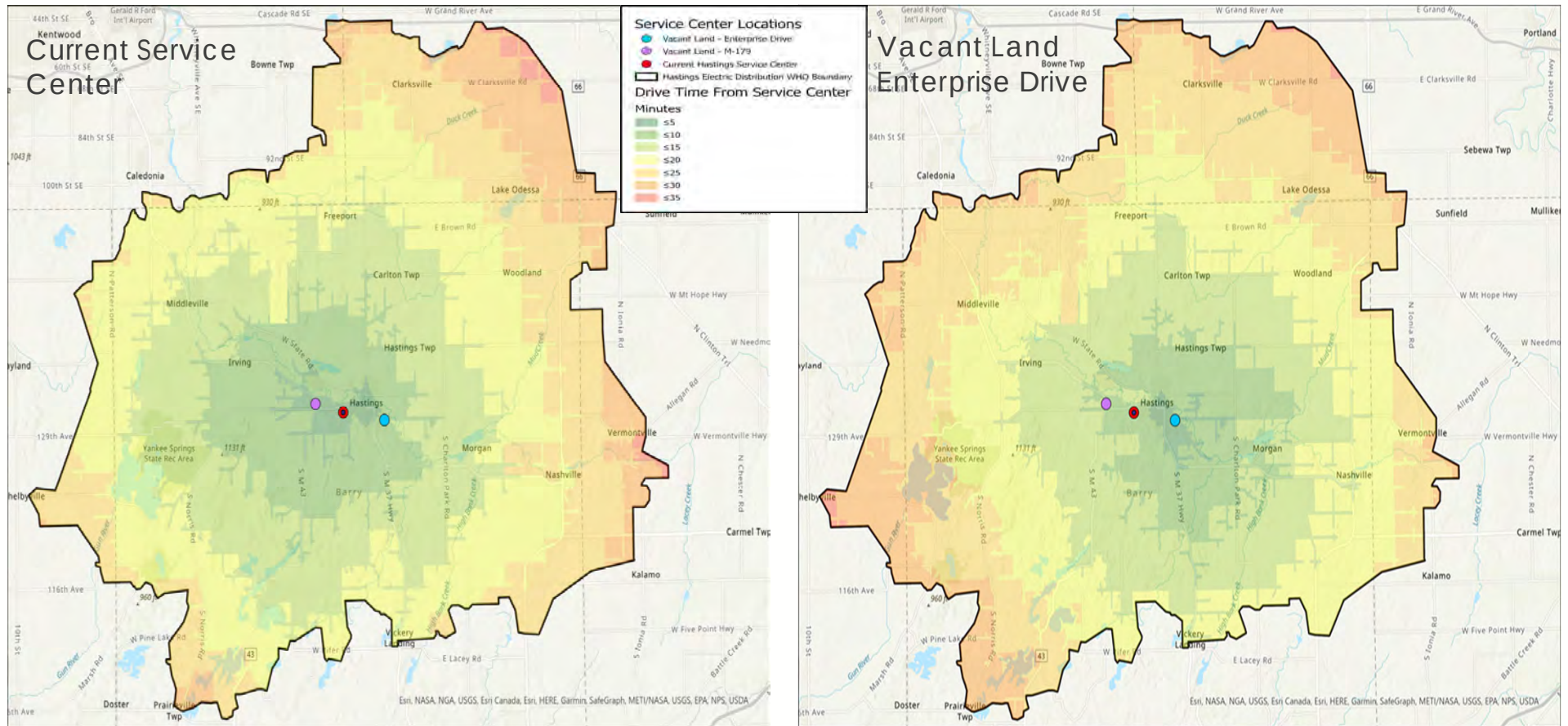
2019-2020 Density of Electric Outages and Customer Data



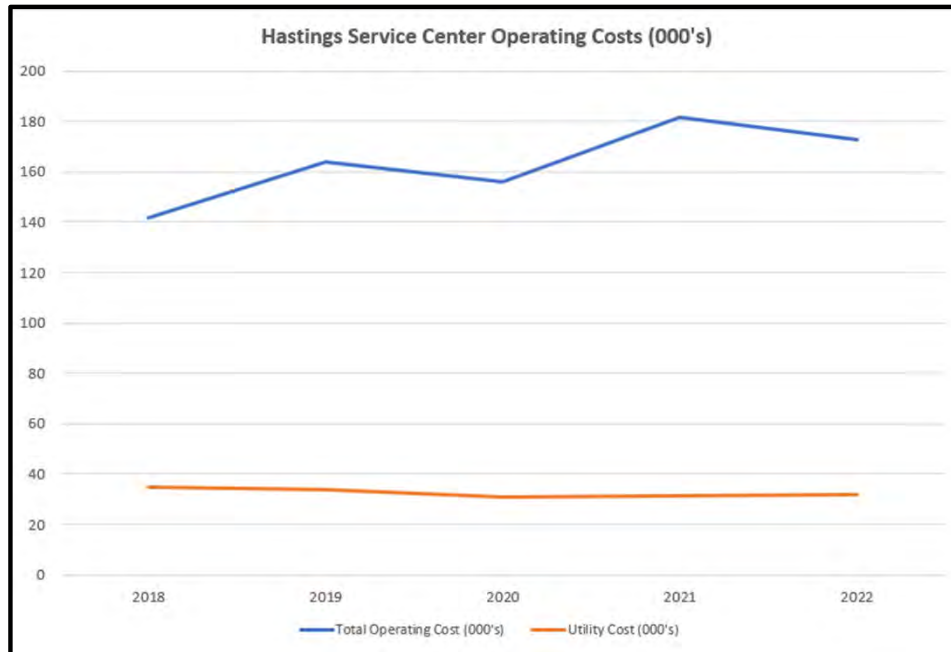
Electric Call Drive Times



Electric Call Drive Times



Hastings Service Center – Operating Costs



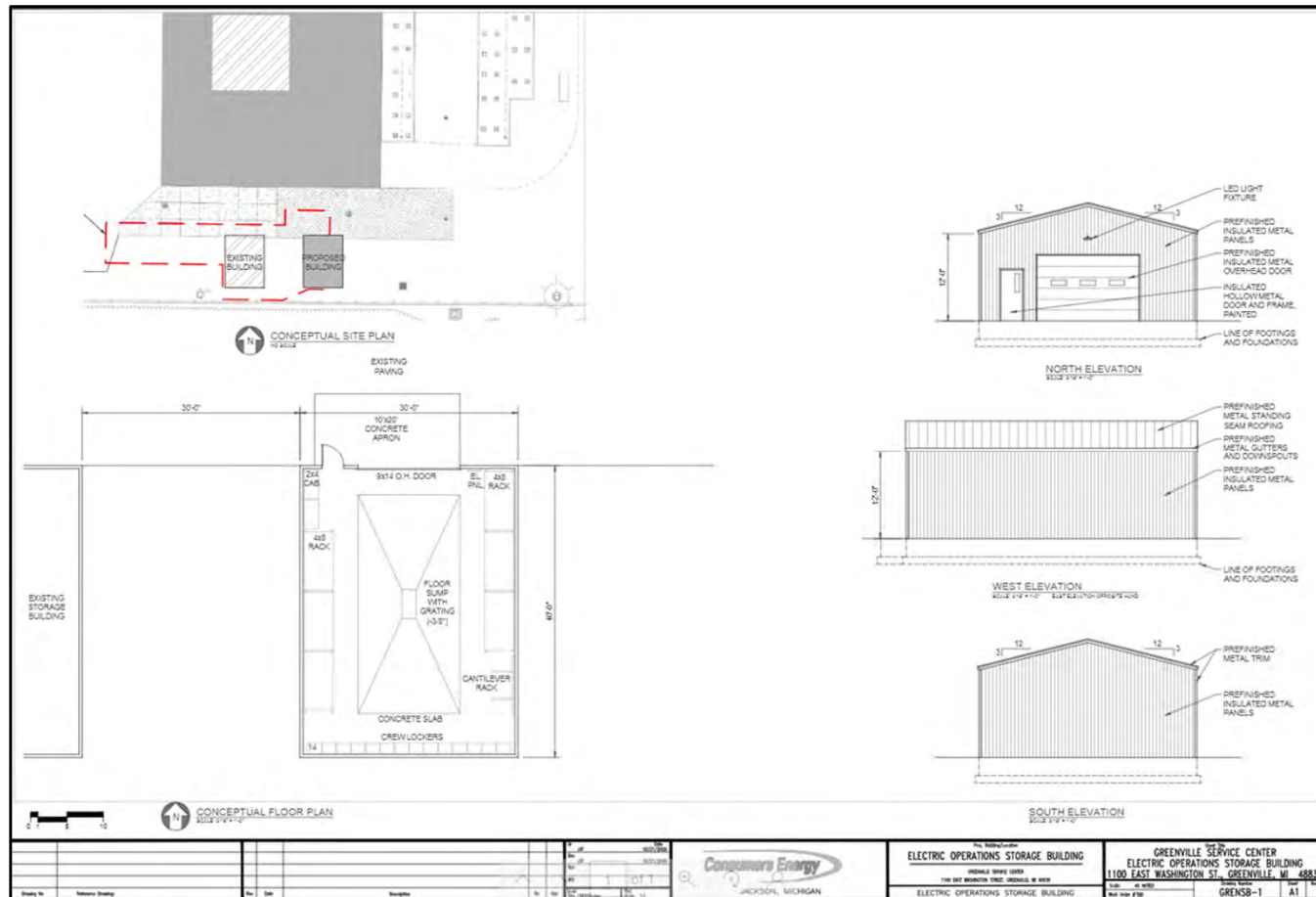
Service Center	Existing Building Area	Existing Building 2022 Annual Operating Cost	Existing Building Cost Per Square Foot	New Building Estimated Cost Per Square Foot	New Building Area - Current Conceptual	New Building Estimated Annual Cost
Hastings	12,317 SF	\$172,514	\$14.01	\$9.70	23,500 SF	\$227,950

Greenville Storage Building

Plan Summary



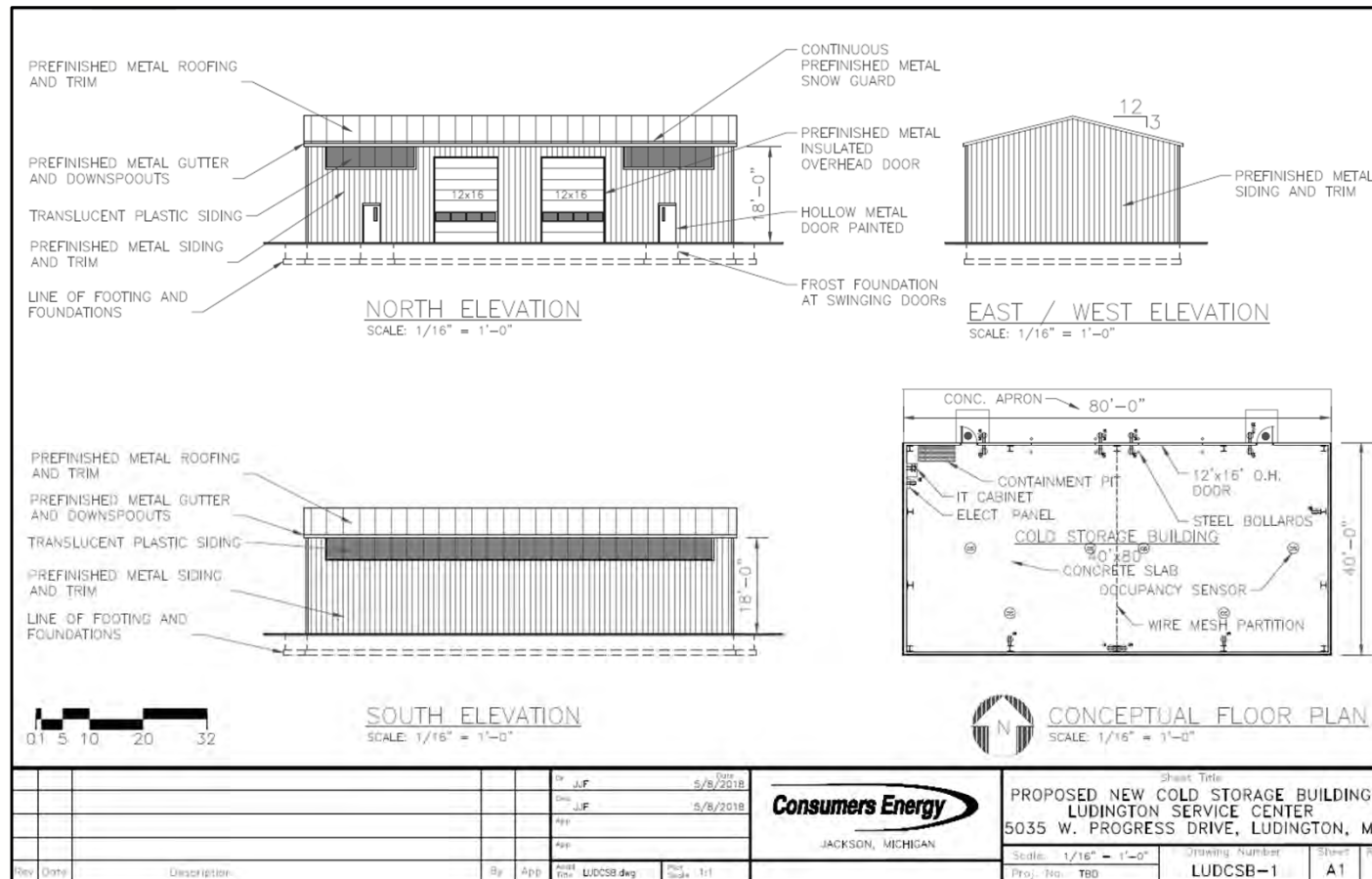
Greenville Storage Building – Conceptual Plan



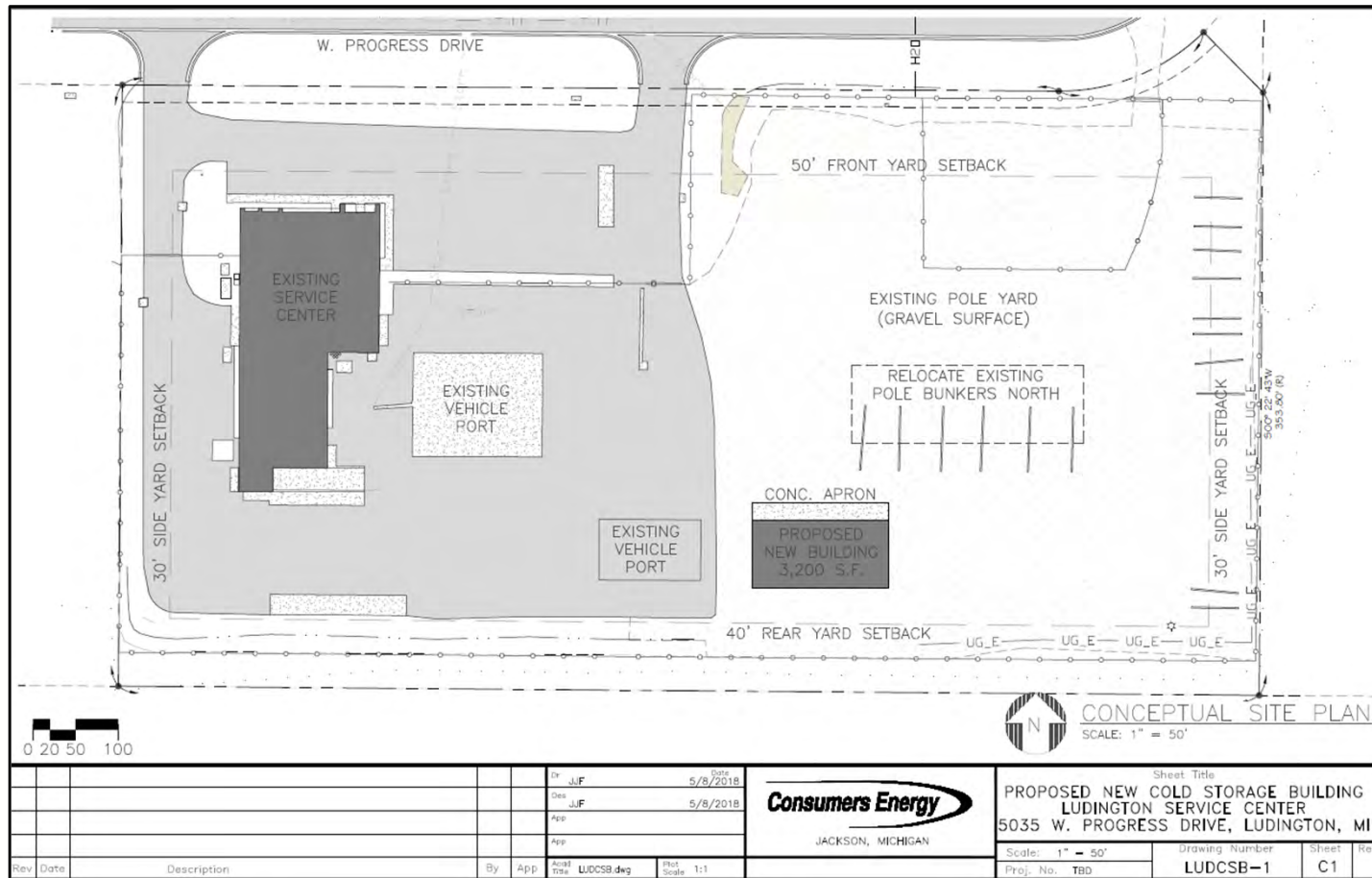
Ludington Storage Building

Plan Summary





Ludington Storage Building – Conceptual Plan



Hardy Headquarter Building

Plan Summary



ZONING DATA

EXISTING ZONING: PROPOSED ZONING:	#1 TBD #1 LIGHT INDUSTRIAL
MINIMUM PARCEL SIZE: PROPOSED PARCEL SIZE:	2 ACRES 106.93 ACRES (PARCEL #16-28-300-04) 8.35 ACRES PORTION OF PARCEL UTILIZED 89.22 ACRES (PARCEL #16-28-405-04) ADJACENT
MINIMUM LOT WIDTH: PROPOSED LOT WIDTH:	200 FEET #0 FEET
REQUIRED FRONT YARD SETBACK: PROPOSED FRONT YARD SETBACK:	75 FEET 157 FEET 36TH STREET / 111 FEET CRISTON HARDY DRIVE
REQUIRED FRONT PARKING SETBACK: PROPOSED FRONT PARKING SETBACK:	85 FEET (75 FEET + 10 FEET) 98 FEET
REQUIRED SIDE YARD SETBACK: PROPOSED SIDE YARD SETBACK:	25 FEET / 50 FEET ABUTTING RESIDENTIAL 123 FEET
REQUIRED SIDE PARKING SETBACK: PROPOSED SIDE PARKING SETBACK:	5 FEET / 20 FEET ABUTTING RESIDENTIAL 50 FEET
REQUIRED REAR YARD SETBACK: PROPOSED REAR YARD SETBACK:	50 FEET / 100 FEET ABUTTING RESIDENTIAL N/A FEET (NO REAR YARD)
MAXIMUM BUILDING SIZE: PROPOSED BUILDING SIZE:	107,000 SQUARE FEET 9,140 SQUARE FEET (4,048 SF OFFICE / 4,732 SF WAREHOUSE)

Hardy Headquarter – HQ Building, Vehicle Port & Solar Array



Kalamazoo Service Center

Plan Summary



Kalamazoo Service Center – Health Assessment

Facility Assessment - Kalamazoo Service Center					
Location	Kalamazoo	Maximum Possible Points (80)	Servicable > 48 Points	Poor < 48 Points	Good > 64 Points
Zone	SWOM				
Safety					
Compatible Use	4	5	3	2	4
Site Conditions	5	5	3	2	4
Traffic/Site Flow	3	5	3	2	4
Environmental Hazard	0	5	3	2	4
Quality					
Branding	3	5	3	2	4
Workplace Efficiency	3	5	3	2	4
Years since last major systems upgrade	0	5	3	2	4
Cost					
Facility Operating Costs	5	5	3	2	4
Non Facility Cost Impact	4	5	3	2	4
Space Optimization	1	5	3	2	4
Delivery					
Drive Time	3	5	3	2	4
Customer Response	4	5	3	2	4
Sustainability	1	5	3	2	4
Morale					
Employee Pride	1	5	3	2	4
Retention	4	5	3	2	4
Wellness	5	5	3	2	4
Total		80	48	32	64

[illegible]

Kalamazoo Service Center – Health Assessment Scoring Criteria

Cost (15 Points)				Score 0 Does not satisfy criteria	Score 1 Does not satisfy most listed criteria	Score 2 Does not satisfy some listed criteria	Score 3 Satisfies minimal criteria	Score 4 Satisfies most listed criteria	Score 5 Satisfies all listed criteria
		Pro	Con						
Facility Operating Costs	O&M costs to operate and maintain facility	Cost per square foot for facility less than statewide average cost per square foot (\$5.31)	Cost per square foot for facility more than statewide average cost per square foot (\$5.31)	costs greater than \$10.00 per square foot	costs between \$8.00 to \$10.00 per square foot	costs between \$7.25 to \$7.99 per square foot	costs between \$7.00 to \$7.24 per square foot	costs between \$5.75 to \$6.99 per square foot	Combined operating costs less than \$5.75 per square foot
Non Facility Cost Impact	O&M cost impacts to operations	Building correct size, centrally located within customer base area, able to accommodate necessary materials, able to maneuver onsite	Building too large, eccentrically located within customer base area, unable to accommodate necessary materials, difficult to maneuver onsite	Building eccentrically located within service territory, inadequate space	Building eccentrically located within service territory, underutilized space	Building eccentrically located within service territory	Building centrally located within service territory	Building accommodates operational needs, centrally located within service territory	Building accommodates operational need, space utilized efficiently, centrally located within service territory
Space Optimization	Reduce excess square footage	Space fully utilized, space utilized efficiently	Space under utilized, space utilized inefficiently, indoor parking area	Building vacant	Building partially occupied or partially vacant	Building occupied inefficiently, includes IPA	Building fully occupied, includes IPA	Building fully occupied, no IPA	occupied, no IPA, includes cold storage

Delivery (15 Points)				Score 0 Does not satisfy criteria	Score 1 Does not satisfy most listed criteria	Score 2 Does not satisfy some listed criteria	Score 3 Satisfies minimal criteria	Score 4 Satisfies most listed criteria	Score 5 Satisfies all listed criteria
		Pro	Con						
Drive Time	Service territory area of proper size	Centrally located within service area(s), drive time balanced across territory and within targeted limits	Eccentrically located within service area(s), drive time unbalanced across territory and/or exceeds targeted limits	service territory and size of service territory allows drive time >1:20 Gas and >1:30 Electric	service territory and size of service territory allows drive time < 1:20 Gas and <1:30 Electric	service territory and size of service territory allows drive time < 1:10 Gas and <1:15 Electric	service territory and size of service territory allows drive time < 1:00 Gas and <1:00 Electric	service territory and size of service territory allows drive time < 0:50 Gas and <0:45 Electric	service territory and size of service territory allows drive time < 0:40 Gas and <0:30 Electric
Customer Response	Meets needs of customers	Customer response time metrics fall below targets	Customer response times exceed targets	response times meet target <90% of time, Electric CAIDI >215 minutes	response times meet target >90% of time, Electric CAIDI <215 minutes	response times meet target >95% of time, Electric CAIDI <195 minutes	response times meet target >97% of time, Electric CAIDI <180 minutes	response times meet target >98% of time, Electric CAIDI <165 minutes	response times meet target >99% of time, Electric CAIDI <150 minutes
Sustainability	Facility incorporates reduced energy utilization and minimizes environmental impact	Facility constructed in accordance with recent Energy Codes, includes onsite stormwater treatment. Green building certification	Facility constructed with no insulation, utilizes energy intensive equipment, no onsite storm water treatment	Minimal roofing insulation	Upgraded roofing insulation	Upgraded roofing insulation and windows	Upgraded roofing insulation, windows and building envelope	insulation, windows, and building envelope with onsite storm water treatment	LEED Certified (or higher LEED rated) facility

Kalamazoo Service Center – Health Assessment Scoring Criteria

Morale (15 Points)				Score 0 Does not satisfy criteria	Score 1 Does not satisfy most listed criteria	Score 2 Does not satisfy some listed criteria	Score 3 Satisfies minimal criteria	Score 4 Satisfies most listed criteria	Score 5 Satisfies all listed criteria
Employee Wellness		Pro	Con						
Employee Pride	Employees are engaged	Employee engagement scores > company average	Employee engagement scores < company average	Deteriorated facility	Deteriorated or dated facility with limited amenities	Dated facility with minimal amenities	Dated facility with full amenities and onsite parking	Modern facility with full amenities	Modern facility with full amenities, onsite parking
Retention	Facilities located in a desirable area, space is attractive to Millennials and supports millennial work styles	Downtown location, local amenities available, space supports varied work styles	Remote location, few local amenities, fixed work space	Remote location, no local amenities	some amenities available within 20 minutes driving distance	Remote location, amenities available within 20 minutes driving distance	Remote location, amenities available within 15 minutes driving distance	Diverse location, amenities available within 10 minutes driving distance	Diverse location, amenities available within walking distance
Wellness	Wellness facilities	Wellness room, space for trainer, drinking fountain, shower facilities nearby, space for wellness classes, well maintained exercise equipment, mirrors, lockers	Wellness room combined with non-compatible uses (i.e.: conference room), aged exercise equipment, poorly maintained exercise equipment, remote shower facilities	No shower, drinking fountain, Wellness or Nursing Mother Room, no additional space available	Shower and drinking fountain, no Wellness or Nursing Mother Room, no additional space available	Wellness Room or Nursing Mother Room, shower and drinking fountain, no additional space available	Nursing Mother Room, shower and drinking fountain, with space available to add Wellness or NMR	Wellness Room and Nursing Mother room, shower and drinking fountain, with space available for Trainer	Wellness room, Trainer room, Nursing Mother Room, shower and drinking fountain nearby

Kalamazoo Service Center First Floor: Existing & Conceptual Plan



Existing



Conceptual Plan

Conceptual Plan

SHEET NO. 101			SHEET NO. 102			SHEET NO. 103			SHEET NO. 104			SHEET NO. 105			SHEET NO. 106			SHEET NO. 107			SHEET NO. 108			SHEET NO. 109			SHEET NO. 110			SHEET NO. 111			SHEET NO. 112			SHEET NO. 113			SHEET NO. 114			SHEET NO. 115			SHEET NO. 116			SHEET NO. 117			SHEET NO. 118			SHEET NO. 119			SHEET NO. 120			SHEET NO. 121			SHEET NO. 122			SHEET NO. 123			SHEET NO. 124			SHEET NO. 125			SHEET NO. 126			SHEET NO. 127			SHEET NO. 128			SHEET NO. 129			SHEET NO. 130			SHEET NO. 131			SHEET NO. 132			SHEET NO. 133			SHEET NO. 134			SHEET NO. 135			SHEET NO. 136			SHEET NO. 137			SHEET NO. 138			SHEET NO. 139			SHEET NO. 140			SHEET NO. 141			SHEET NO. 142			SHEET NO. 143			SHEET NO. 144			SHEET NO. 145			SHEET NO. 146			SHEET NO. 147			SHEET NO. 148			SHEET NO. 149			SHEET NO. 150			SHEET NO. 151			SHEET NO. 152			SHEET NO. 153			SHEET NO. 154			SHEET NO. 155			SHEET NO. 156			SHEET NO. 157			SHEET NO. 158			SHEET NO. 159			SHEET NO. 160			SHEET NO. 161			SHEET NO. 162			SHEET NO. 163			SHEET NO. 164			SHEET NO. 165			SHEET NO. 166			SHEET NO. 167			SHEET NO. 168			SHEET NO. 169			SHEET NO. 170			SHEET NO. 171			SHEET NO. 172			SHEET NO. 173			SHEET NO. 174			SHEET NO. 175			SHEET NO. 176			SHEET NO. 177			SHEET NO. 178			SHEET NO. 179			SHEET NO. 180			SHEET NO. 181			SHEET NO. 182			SHEET NO. 183			SHEET NO. 184			SHEET NO. 185			SHEET NO. 186			SHEET NO. 187			SHEET NO. 188			SHEET NO. 189			SHEET NO. 190			SHEET NO. 191			SHEET NO. 192			SHEET NO. 193			SHEET NO. 194			SHEET NO. 195			SHEET NO. 196			SHEET NO. 197			SHEET NO. 198			SHEET NO. 199			SHEET NO. 200			SHEET NO. 201			SHEET NO. 202			SHEET NO. 203			SHEET NO. 204		
---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--	---------------	--	--

Kalamazoo Service Center: Renovation Expected Energy Usage Reductions (Alternative #1)

CONSUMERS KALAMAZOO SERVICE CENTER

2500 EAST CORK STREET

BUILDING AREAS BY TYPE

	ALT 1	ALT 2	ALT 3	ALT 4	ALT 5	ALT 6
OFFICE	58,000	58,000	45,000	45,000	47,200	44,000
STORAGE	48,500	48,500	48,500	48,500	48,500	27,200
INDOOR PARKING	20,200	20,200	20,200	20,200	47,800	30,200
AUTO MAINTENANCE	20,400	20,400	20,400	21,500	46,800	23,100
TOTAL	147,100	147,100	134,100	135,200	190,300	124,500

ESTIMATED ENERGY USE - CONCEPT LEVEL

	ALT 1	ALT 2	ALT 3	ALT 4	ALT 5	ALT 6
Total Area (SF)	147,100	147,100	134,100	135,200	190,300	124,500
Baseline EUI, CBECS Weighted (kBtu/SF)	42	42	41	41	43	44
Estimated Overall Reduction	10%	20%	25%	30%	35%	45%
Reduced EUI (kBtu/SF)	38	33	30	28	28	24
Estimated Annual Energy Use (kBtu)	5,524,470	4,910,640	4,086,975	3,851,470	5,302,635	3,033,800

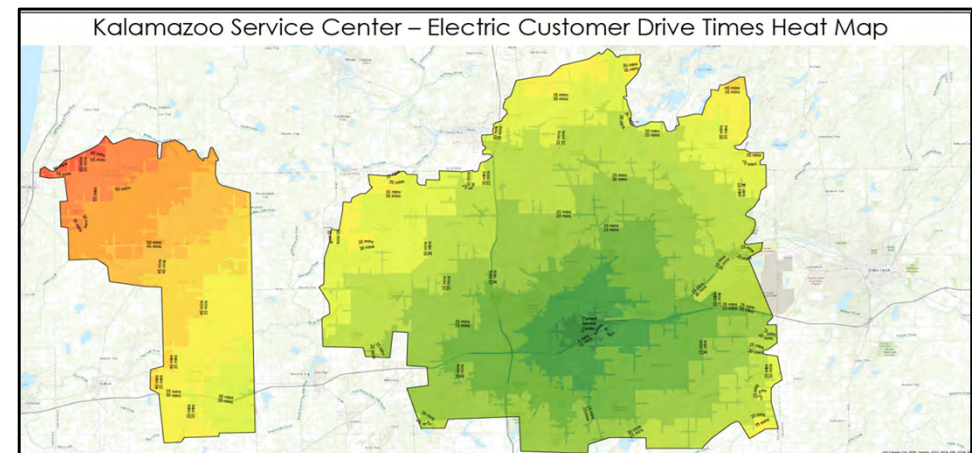
Kalamazoo Service Center – Electric Customer Drive Times

Kalamazoo Service Center – Electric Customer Drive Times

Eastern Electric			
Drive Time (minutes)	Count of Customers	Percentage	Cumulative Percentage
5	1,450	1.1%	1.1%
10	20,759	16.0%	17.1%
15	46,583	35.9%	53.1%
20	40,146	31.0%	84.0%
25	8,651	6.7%	90.7%
30	6,174	4.8%	95.5%
35	1,953	1.5%	97.0%
40	934	0.7%	97.7%
45	646	0.5%	98.2%
50	1,493	1.2%	99.4%
55	605	0.5%	99.8%
60	229	0.2%	100.0%
Grand Total	129,623	100%	

Western Electric			
Drive Time (minutes)	Count of Customers	Percentage	Cumulative Percentage
5	4,202	3.2%	3.2%
10	33,706	26.0%	29.2%
15	39,159	30.2%	59.5%
20	28,523	22.0%	81.5%
25	13,224	10.2%	91.7%
30	5,375	4.1%	95.8%
35	2,721	2.1%	97.9%
40	1,906	1.5%	99.4%
45	606	0.5%	99.8%
50	201	0.2%	100.0%
Grand Total	129,623	100.00%	

Northern Electric			
Drive Time (minutes)	Count of Customers	Percentage	Cumulative Percentage
5	2,597	2.0%	2.0%
10	18,922	14.6%	16.6%
15	35,461	27.4%	44.0%
20	44,914	34.6%	78.6%
25	15,626	12.1%	90.7%
30	5,527	4.3%	94.9%
35	4,444	3.4%	98.4%
40	1,688	1.3%	99.7%
45	441	0.3%	100.0%
50	3	0.0%	
Grand Total	129,623	100%	



Kalamazoo Service Center – Gas Customer Drive Times

Kalamazoo Service Center – Gas Customer Drive Times

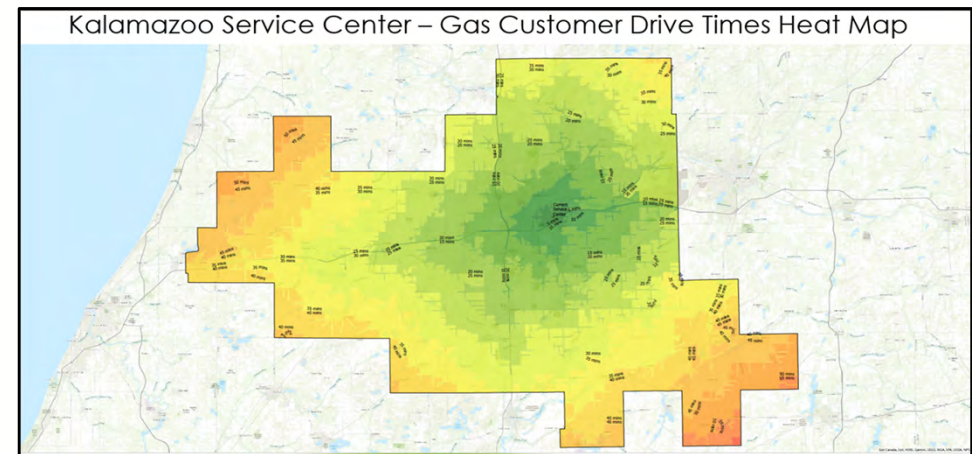
Current Gas			
Drive Time (minutes)	Count of Customers		
5	5,113	5.1%	5.1%
10	21,900	21.7%	26.8%
15	36,026	35.7%	62.5%
20	19,219	19.1%	81.5%
25	7,601	7.5%	89.1%
30	3,731	3.7%	92.8%
35	4,713	4.7%	97.4%
40	1,608	1.6%	99.0%
45	749	0.7%	99.8%
50	162	0.2%	99.9%
55	59	0.1%	100.0%
Grand Total	100,881	100%	

Northern Gas			
Drive Time (minutes)	Count of Customers		
5	2,379	2.4%	2.4%
10	10,292	10.2%	12.6%
15	20,680	20.5%	33.1%
20	33,090	32.8%	65.9%
25	15,911	15.8%	81.6%
30	9,315	9.2%	90.9%
35	3,191	3.2%	94.0%
40	3,866	3.8%	97.9%
45	666	0.7%	98.5%
50	710	0.7%	99.2%
55	306	0.3%	99.5%
60	374	0.4%	99.9%
65	101	0.1%	100.0%
Grand Total	100,881	100%	

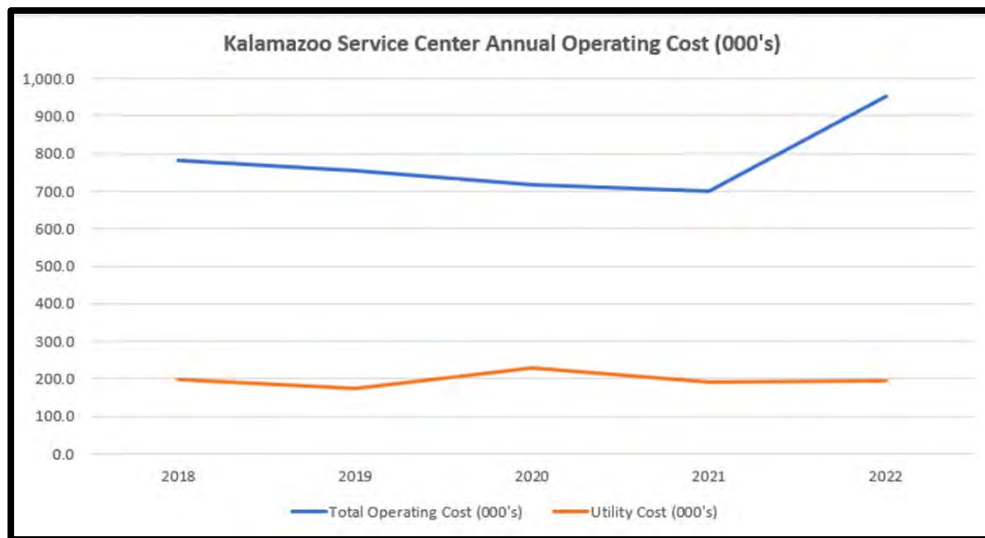
Western Gas			
Drive Time (minutes)	Count of Customers		
5	2,857	2.8%	2.8%
10	23,143	22.9%	25.8%
15	30,990	30.7%	56.5%
20	23,967	23.8%	80.2%
25	9,378	9.3%	89.5%
30	5,814	5.8%	95.3%
35	2,483	2.5%	97.8%
40	1,096	1.1%	98.9%
45	582	0.6%	99.4%
50	341	0.3%	99.8%
55	230	0.2%	100.0%
Grand Total	100,881	100%	

Eastern Gas			
Drive Time (minutes)	Count of Customers		
5	1,285	1.3%	1.3%
10	16,770	16.6%	17.9%
15	32,817	32.5%	50.4%
20	29,244	29.0%	79.4%
25	7,816	7.7%	87.2%
30	6,199	6.1%	93.3%
35	2,469	2.4%	95.8%
40	3,025	3.0%	98.8%
45	1,197	1.2%	99.9%
50	59	0.1%	100.0%
Grand Total	100,881	100%	

Southern Gas			
Drive Time (minutes)	Count of Customers		
5	5,349	5.3%	5.3%
10	20,447	20.3%	25.6%
15	30,252	30.0%	55.6%
20	23,832	23.6%	79.2%
25	9,218	9.1%	88.3%
30	5,877	5.8%	94.1%
35	3,262	3.2%	97.4%
40	1,943	1.9%	99.3%
45	311	0.3%	99.6%
50	341	0.3%	100.0%
55	49	0.0%	100.0%
Grand Total	100,881		



Kalamazoo Service Center – Operating Costs

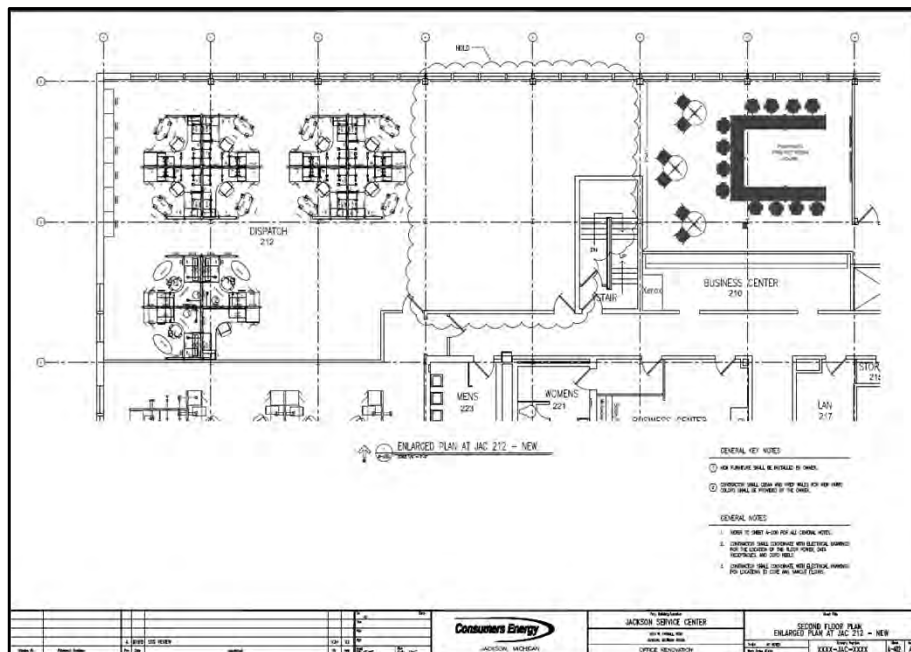
[illegible]

Jackson Dispatch

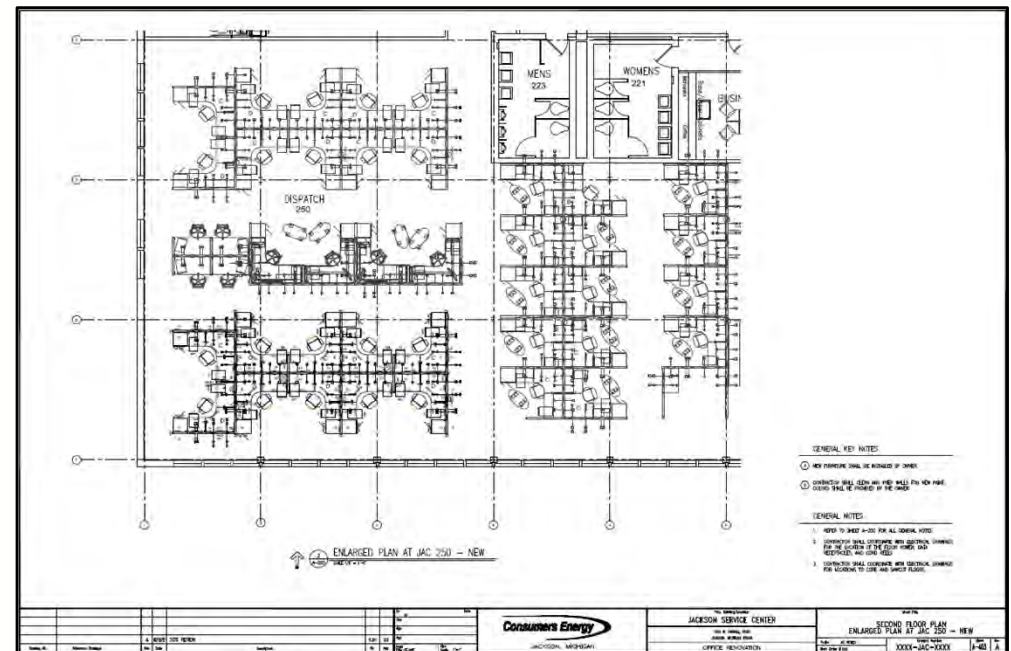
Plan Summary



Jackson Dispatch: Rooms 212 & 250 Conceptual Plans



Room 212



Room 250

Parnal 1-3

Plan Summary



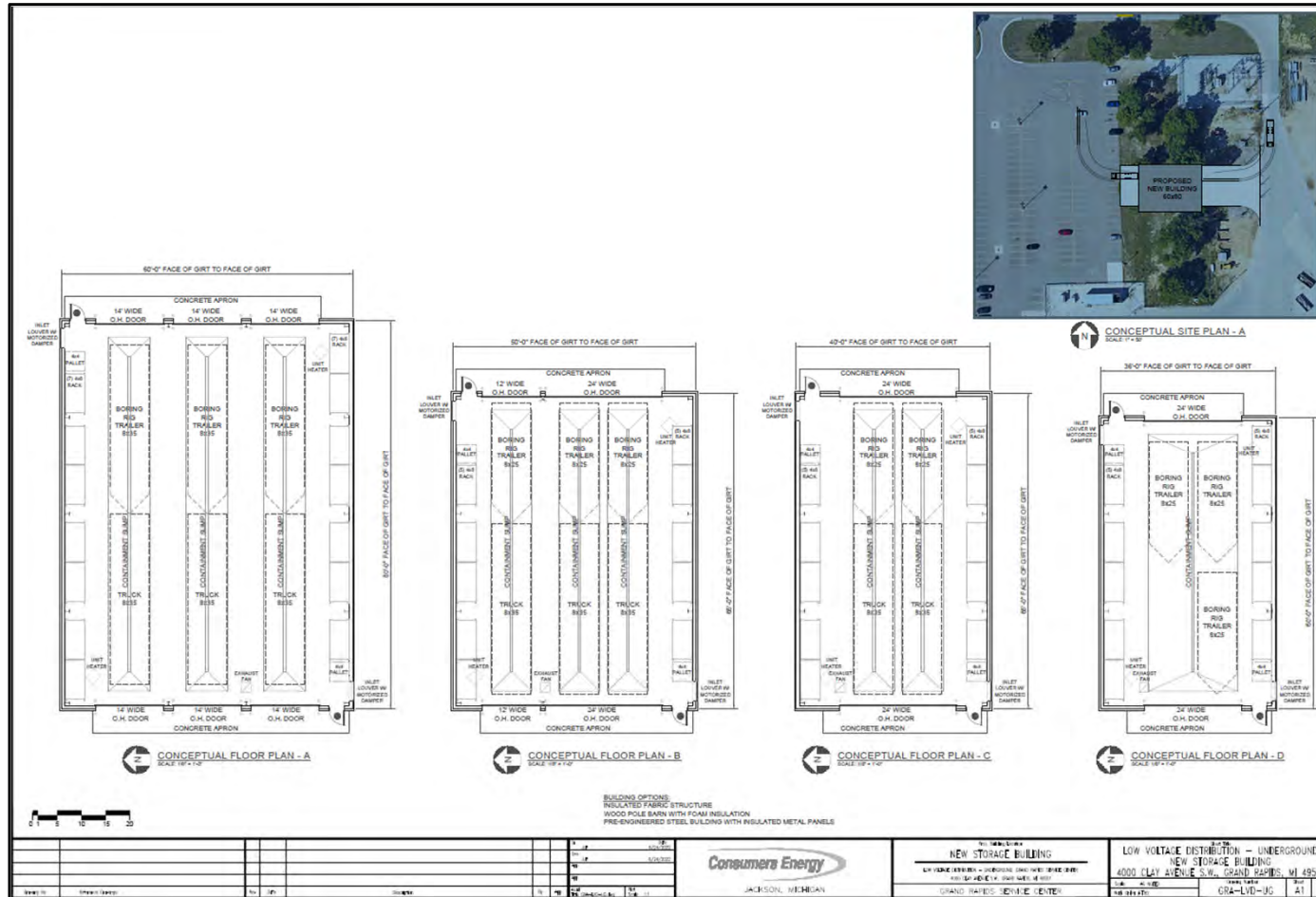
Architectural floor plan of the Parnall 13 Conceptual Plan, 3rd Floor South. The plan shows a large open office area with numerous desks and chairs, a central corridor, and various support spaces including a lounge, meeting rooms, and restrooms. A north arrow and scale bar are provided at the bottom left. A title block at the bottom right identifies the project as 'PARNALL 13 RENOVATIONS' and includes the 'Consumers Energy' logo.

Grand Rapids LVD UG

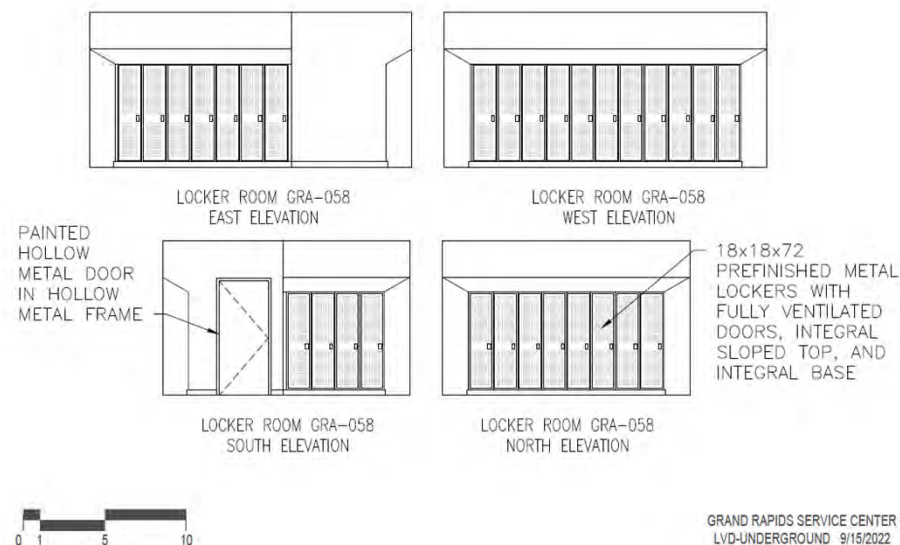
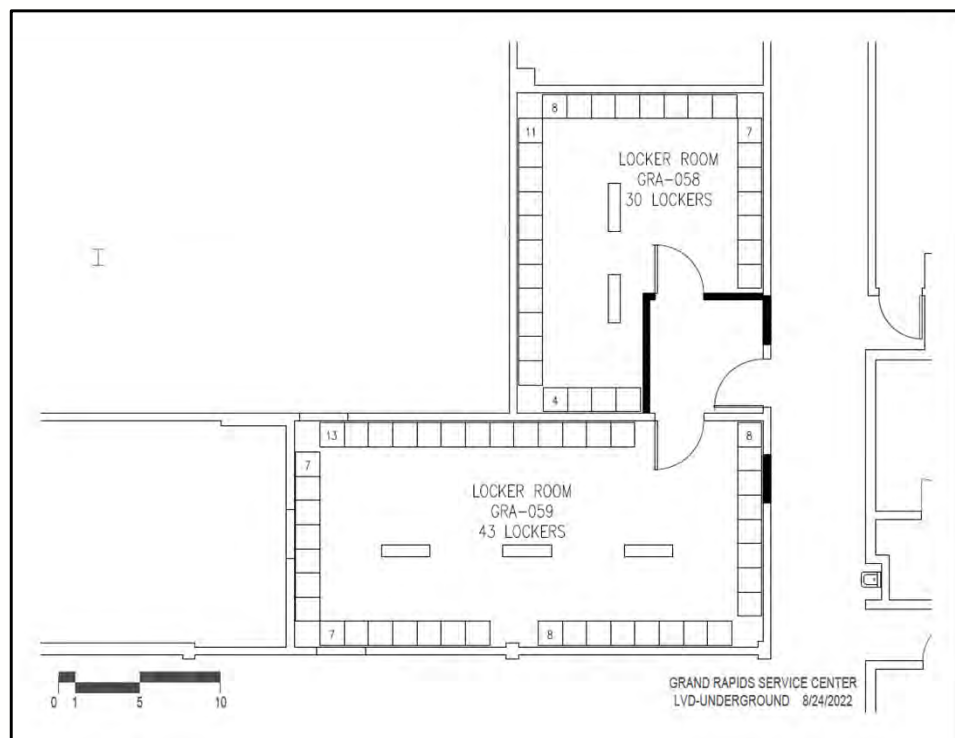
Plan Summary



Grand Rapids LVD UG: Conceptual Floor Plan & Site Plan



Grand Rapids LVD UG: Lockers Plan



Control/Dispatch Consolidation

Plan Summary



Control/Dispatch Consolidation: Alternatives Considered

Alternative 1: New Control Center Building (Case No. U-20963)

\$100.4 Million

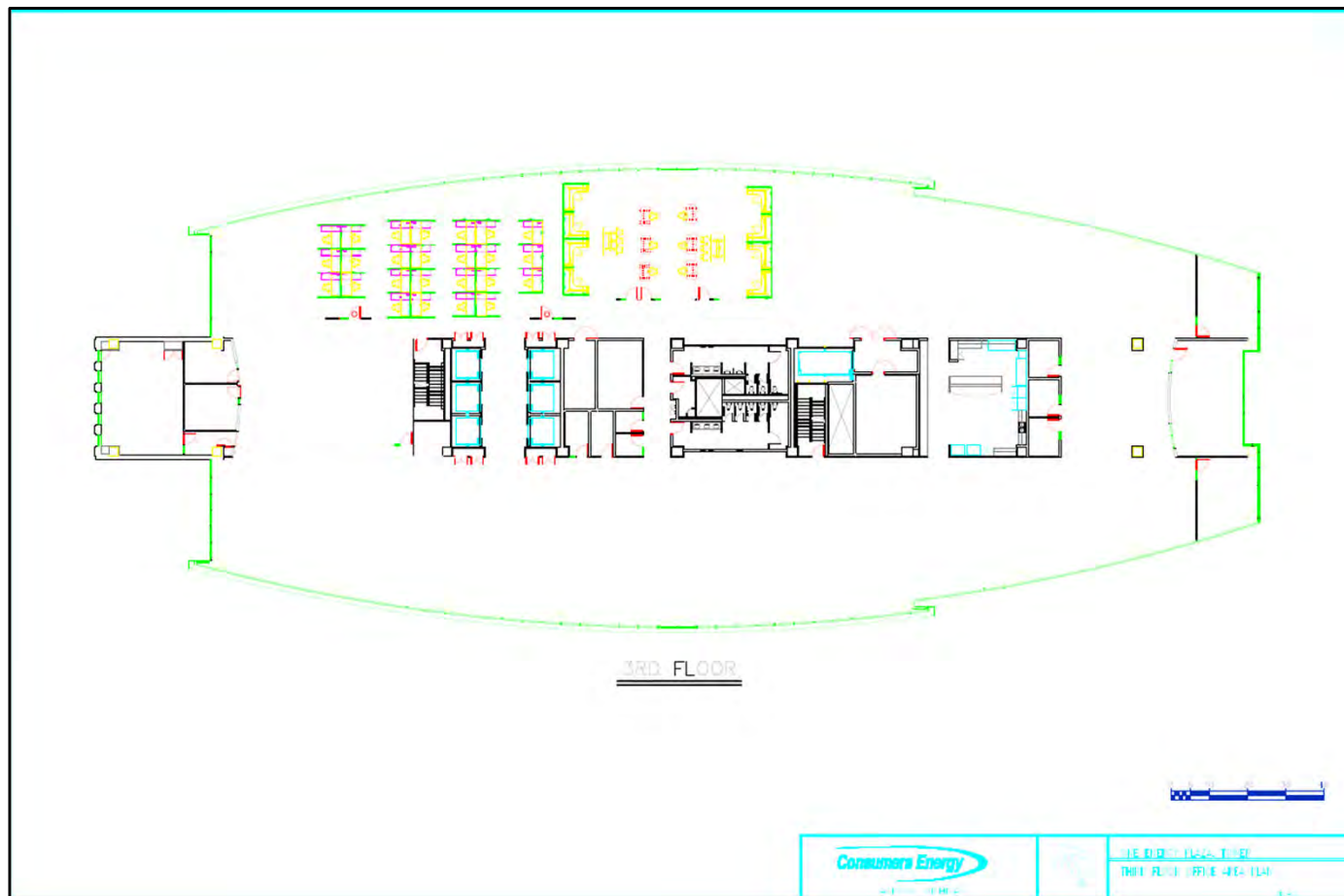
Item	2021 - Total Plan	2021 - Electric Only	2022 - Total Plan	2022 - Electric Only	2023 - Total Plan	2023 - Electric Only	2024 - Total Plan	2024 - Electric Only	Subtotals
Budget per year	\$1,000,000		\$28,764,660		\$38,268,692		\$32,383,096		\$100,416,448
Design Engineering	\$850,000		\$1,000,000		\$250,000		\$250,000		\$2,350,000
Fiber optic service			\$5,000,000						\$5,000,000
Site infrastructure			\$4,764,660				\$1,000,000		\$5,764,660
Building construction			\$9,500,000		\$22,018,692		\$5,133,096		\$36,651,788
Commissioning					\$1,000,000		\$500,000		\$1,500,000
Furnishings					\$3,500,000		\$500,000		\$4,000,000
IT equipment and systems							\$15,000,000		\$15,000,000
Loadings/AFUDC	\$150,000		\$8,500,000		\$11,500,000		\$10,000,000		\$30,150,000
Subtotals	\$1,000,000	\$840,000	\$28,764,660	\$24,162,314	\$38,268,692	\$32,145,701	\$32,383,096	\$27,201,801	\$100,416,448

Alternative 2: Use OEP Floors 3 & 4 and JSC Backup Site (Current Case)

\$14.8 Million

Location		Historical Cost 2022	Plan Cost 2023	Plan Cost 2024	Plan Cost 2025	Plan Cost 2026	Plan Cost 2027	Total Cost
Control/Dispatch Center	Master Planning	\$80,871						\$80,871
	Programming		\$65,000					\$65,000
	Land Acquisition							\$0
	Engineering		\$353,460		\$57,540			\$411,000
	Construction		\$1,027,678	\$5,287,740	\$3,437,280	\$1,711,802		\$11,464,500
	Furnishings		\$105,000		\$2,415,000	\$105,000		\$2,625,000
	Commissioning			\$130,000		\$105,100		\$235,100

Control/Dispatch Consolidation: OEP 3rd Floor Conceptual Plan



Control/Dispatch Consolidation: OEP 4th Floor Conceptual Plan



South Monroe Service Center Building Renovations Rough Order of Magnitude (ROM) Rev (0)

This is a conceptual estimate with an expected accuracy range of -30%/+50%.

SCOPE: Renovation of 3,830 FT2 of office and mezzanine spaces to include architectural/mechanical/electrical upgrades.
Ventilation improvements of the existing internal parking area (IPA) and the addition of an attached 1,300 FT2 storage building.
Total affected space 11,027 FT2.

	Anticipated Construction Cost	\$1,839,567.86			
	Construction Contingency	6.8%			
			Cost	Contingency	Total Cost
5101500	Material Stock Issues	\$	105.74	\$ 7.19	\$ 112.93
5101700	NonStock Material (FURNISHINGS)	\$	79,859.62	\$ 5,430.45	\$ 85,290.07
5101700	NonStock Material (IT EQPT)	\$	44,612.00	\$ 3,033.62	\$ 47,645.62
5101750	N-Slk Mtl Nonstorerm	\$	-	\$ -	\$ -
5200800	Lodging	\$	-	\$ -	\$ -
5201200	General Travel-Other	\$	1,327.73	\$ 90.29	\$ 1,418.02
5201600	Meals & Ent-50%	\$	-	\$ -	\$ -
5202800	Other Business Expenses	\$	3,245.56	\$ 220.70	\$ 3,466.26
5210000	Fees-Regulatory/Permit	\$	-	\$ -	\$ -
5227000	Indirect Lbr Cr/Adj (FEDT)	\$	49,175.22	\$ 3,343.91	\$ 52,519.13
5360000	IT Tech Chgbk-Labor	\$	34,531.06	\$ 2,348.11	\$ 36,879.17
5361000	IT SDvlp Chgbk-Labor	\$	2,278.33	\$ 154.93	\$ 2,433.26
5373000	IS&T Tech Chgbk-Contractor	\$	3,156.63	\$ 214.65	\$ 3,371.28
5374000	IT SDvlp Chgbk-Contractor	\$	279.98	\$ 19.04	\$ 299.02
5380000	Lab Services Chargeback-Labor	\$	1,200.00	\$ 81.60	\$ 1,281.60
5380659	Lab Chargeback-LaborNDT	\$	-	\$ -	\$ -
5390000	Lab Srvc Chgbk-NLbr	\$	600.00	\$ 40.80	\$ 640.80
5390659	Lab Chargeback-NonLabor NDT	\$	-	\$ -	\$ -
5503500	Other Outside Services - CONSTRUCTION: RENOVATION	\$	1,839,567.86	\$ 125,090.61	\$ 1,964,658.47
5503500	Other Outside Services - DESIGN FEE - WHITMAN	\$	126,200.00	\$ 8,581.60	\$ 134,781.60
5503500	Other Outside Services - FURNISHINGS INSTALL - MARKXMODA	\$	9,140.38	\$ 621.55	\$ 9,761.93
5503500	Other Outside Services - IT MGMT: CSM	\$	106,897.00	\$ 7,269.00	\$ 114,166.00
5503500	Other Outside Services - IT CABLING: CPS	\$	15,850.00	\$ 1,077.80	\$ 16,927.80
5503500	Other Outside Services - AV: BLUEWATER	\$	49,649.99	\$ 3,376.20	\$ 53,026.19
5503500	Other Outside Services - SHELVING/INSTALL	\$	28,615.38	\$ 1,945.85	\$ 30,561.23
5504100	BTS O/S Managed Services	\$	-	\$ -	\$ -
5508000	Contractor Costs-Other	\$	12,740.56	\$ 866.36	\$ 13,606.92
5509000	Contractor Costs - Labor (Pro-Unlimited)	\$	141,561.82	\$ 9,626.20	\$ 151,188.02
5890000	Other Nonlabor Expense	\$	-	\$ -	\$ -
5894000	Sales/Salvage	\$	-	\$ -	\$ -
A1000	OM&C Direct Labor	\$	4,240.00	\$ 288.32	\$ 4,528.32
A3000	Non Exempt Direct Labor	\$	-	\$ -	\$ -
A3500	Exempt Direct Labor (CE Facilities Staff)	\$	28,145.08	\$ 1,913.87	\$ 30,058.95
M1034	Real Estate Services - Labor	\$	-	\$ -	\$ -
M1061	ENVIRONMENTAL LABOR	\$	-	\$ -	\$ -
M1090	Facil Eng/Proj Labor	\$	-	\$ -	\$ -
M1516	Environ Compl - Labor	\$	-	\$ -	\$ -
M1520	Facil Eng Proj Cap L	\$	-	\$ -	\$ -
M2034	Real Estate - Non Lbr	\$	-	\$ -	\$ -
M2090	FACIL ENG NLABOR	\$	-	\$ -	\$ -
M2516	Environ Compl - NLbr	\$	-	\$ -	\$ -
M2520	Fac Eng Proj Cap Nlb	\$	-	\$ -	\$ -
	Project Direct Totals:	\$	2,582,979.95	\$ 175,642.64	\$ 2,758,622.58
5231300	OPEB Expense Capitalized				1,917.60
5250000	Pwr Plt Capitalized AFUDC				45,325.00
5251000	Pwr Plt Capitalized Pension				8,507.65
5252000	PP Capitalz Oth Cost				20,733.28
5253000	PP Capitalz Adm&Gen				7,930.19
	Corporate Overheads:				84,413.72
O & M Costs				Total Capital Project Costs:	2,843,036
IT					
5101700	NonStock Material			\$	2,500.00
5360000	IT Tech Chgbk-Labor			\$	1,500.00
5361000	IT SDvlp Chgbk-Labor			\$	-
5373000	IS&T Tech Chgbk-Contractor			\$	-
5374000	IT SDvlp Chgbk-Contractor			\$	-
5503500	Other Outside Services -			\$	2,500.00
5504100	BTS O/S Managed Services -			\$	-
	IT O & M:	\$	6,500.00		
O & M: 35687754					
5101500	Material Stock Issues			\$	-
5101750	N-Slk Mtl Nonstorerm			\$	-
5503500	Other Outside Services - Remove Existing Furnishings			\$	1,800.00
A3500	Exempt Direct Labor			\$	-
	O & M:	\$	1,800.00		
	Total Project Costs:		2,851,336		
	CAPITAL COSTS				
	CONSTRUCTION:	\$	1,779,321.16		
	COR:	\$	60,246.70		
	DESIGN SERVICES:	\$	126,200.00		
	IT:	\$	257,254.99		
	FURNISHINGS:	\$	89,000.00		
	EQUIPMENT:	\$	28,615.38		
	CONTRACT CONST MGMT:	\$	154,302.38		
	CE STAFFING:	\$	82,760.29		
	CE MISC COSTS:	\$	5,279.04		
	CONTINGENCY:	\$	175,642.64		
	OVERHEADS:	\$	84,413.72		
	TOTAL CAPITAL COSTS	\$	2,843,036.30		
	O & M COSTS:				
	CONSTRUCTION:	\$	-		
	MISCELLANEOUS:				
	FURNISHINGS:	\$	1,800.00		
	IT:	\$	6,500.00		
	TOTAL O & M COSTS:	\$	8,300.00		
	TOTAL PROJECT COSTS:	\$	2,851,336.30		

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for the)
the generation and distribution of)
electricity and for other relief.)

Case No. U-21389

EXHIBITS

OF

CULLEN M. HALE

ON BEHALF OF

CONSUMERS ENERGY COMPANY

May 2023

Schedule: B-5.6

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of Actual and Projected Capital Expenditures

Customer Experience and Operations

(\$000)

Case No.: U-21389

Exhibit No.: A-12 (CMH-1)

Schedule: B-5.6

Page: 1 of 2

Witness: CMHale

Date: May 2023

Line No	(a) Description	(b)	(c)	(d)	(e)	(f)
		Historical Year	Projected Bridge Period			Projected Test Year
		12 Mos Ended 12/31/2022	12 Mos Ended 12/31/2023	2 Mos Ending 2/29/2024	14 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025
1	Business Customer Care	-	-	-	-	-
	Contractor					
	Labor					
	Materials					
	Business Expenses					
	Contingency					
	Other (Loadings, Chargebacks)					
2	Customer Contact Center	3	-	-	-	-
	Contractor					
	Labor	3				
	Materials					
	Business Expenses					
	Contingency					
	Other (Loadings, Chargebacks)	1				
3	Digital Customer Operations	49	310	-	310	-
	Contractor	30	310		310	
	Labor					
	Materials					
	Business Expenses					
	Contingency					
	Other (Loadings, Chargebacks)	19				
4	Total Capital	52	310	-	310	-

Schedule: B-5.6

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Actual and Projected Capital Expenditures

(\$000)

Case No.: U-21389

Exhibit No.: A-12 (CMH-1)

Schedule: B-5.6

Page: 2 of 2

Witness: CMHale

Date: May 2023

	Projected			
	2 Mos Ending 2/28/2023	12 Mos Ending 2/29/2024	12 Mos Ending 2/28/2025	26 Mos Ending 2/28/2025
5 Business Customer Care	-	-	-	-
Contractor				
Labor				
Materials				
Business Expenses				
Contingency				
Other (Loadings, Chargebacks)				
6 Customer Contact Center	-	-	-	-
Contractor				
Labor				
Materials				
Business Expenses				
Contingency				
Other (Loadings, Chargebacks)				
7 Digital Customer Operations	-	310	-	310
Contractor		310		310
Labor				
Materials				
Business Expenses				
Contingency				
Other (Loadings, Chargebacks)				
8 Total Capital	-	310	-	310

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Summary of Actual & Projected O&M Expenses
 Customer Experience and Operations
 (\$000)

Case No.: U-21389
 Exhibit No.: A-109 (CMH-2)
 Page: 1 of 4
 Witness: CMHale
 Date: May 2023

(a)		(b)	(c)
Line No.	Description	2022 Actual	Projected 12-Mos Ending 2/28/2025
1	Billing & Payment	18,212	18,301
	Labor	2,640	2,604
	Material	761	769
	Contractor	9,299	9,345
	Non-Labor Overheads	0	0
	Non-Labor Other	5,513	5,583
2	Customer Interactions	25,769	26,544
	Labor	18,115	18,744
	Material	30	0
	Contractor	6,616	6,137
	Non-Labor Overheads	0	0
	Non-Labor Other	1,009	1,664
4	Customer Experience & Ops O&M	\$ 43,982	\$ 44,845
	Labor	20,755	21,348
	Material	791	769
	Contractor	15,914	15,482
	Non-Labor Overheads	0	0
	Non-Labor Other	6,521	7,247

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Summary of Actual & Projected O&M Expenses
 Customer Experience and Operations
 (\$000)

Case No.: U-21389
 Exhibit No.: A-109 (CMH-2)
 Page: 2 of 4
 Witness: CMHale
 Date: May 2023

(a)		(b)	(c)
Line No.	Description	2022 Actual	Projected 12-Mos Ending 2/28/2025
1	Customer Billing	8,784	9,089
	Labor	2,250	2,185
	Material	758	769
	Contractor	498	576
	Non-Labor Overheads		
	Non-Labor Other	5,278	5,559
2	Customer Payment Programs	9,429	9,212
	Labor	390	419
	Material	3	
	Contractor	8,801	8,769
	Non-Labor Overheads		
	Non-Labor Other	235	24
3	Billing & Payment O&M	\$ 18,212	\$ 18,301
	Labor	2,640	2,604
	Material	761	769
	Contractor	9,299	9,345
	Non-Labor Overheads	0	0
	Non-Labor Other	5,513	5,583

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Summary of Actual & Projected O&M Expenses
Customer Experience and Operations
(\$000)

Case No.: U-21389
Exhibit No.: A-109 (CMH-2)
Page: 3 of 4
Witness: CMHale
Date: May 2023

(a)		(b)	(c)
Line No.	Description	2022 Actual	Projected 12-Mos Ending 2/28/2025
1	Analytics & Outreach	470	446
	Labor	205	109
	Material	8	
	Contractor	248	333
	Non-Labor Overheads		
	Non-Labor Other	9	4
2	Business Customer Care	2,110	2,256
	Labor	1,902	1,936
	Material	2	
	Contractor	152	269
	Non-Labor Overheads		
	Non-Labor Other	54	51
3	Credit & Assistance	2,942	3,605
	Labor	1,758	1,889
	Material	1	
	Contractor	985	1,363
	Non-Labor Overheads		
	Non-Labor Other	198	353
4	Customer Contact Center	16,349	16,626
	Labor	12,935	13,258
	Material	16	
	Contractor	2,821	2,215
	Non-Labor Overheads		
	Non-Labor Other	576	1,152
5	Digital Customer Operations	3,899	3,611
	Labor	1,315	1,550
	Material	3	
	Contractor	2,409	1,957
	Non-Labor Overheads		
	Non-Labor Other	172	104
6	Customer Interactions O&M	\$ 25,769	\$ 26,544
	Labor	18,115	18,744
	Material	30	0
	Contractor	6,616	6,137
	Non-Labor Overheads	0	0
	Non-Labor Other	1,009	1,664

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Summary of O&M Expenses Projected Using Inflation
Customer Experience and Operations
(\$000)

Case No.: U-21389
Exhibit No.: A-109 (CMH-2)
Page: 4 of 4
Witness: CMHale
Date: May 2023

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	2022 Actual	Base O&M for Merit & Inflation 12 Mos Ended Dec 31, 2022	Merit & Inflation 12 Mos Ending Dec 31, 2023	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2023	Merit & Inflation 12 Mos Ending Dec 31, 2024	Base O&M for Merit & Inflation 12 Mos Ending Dec 31, 2024	Merit & Inflation 2 Mos Ending 2/28/2025	Other Adjustments	Projected O&M 12 Mos Ending 2/28/2025
			(c) * Inflation Rate			(e) * Inflation Rate		(g) * Inflation Rate		(b) + (d) + (f) + (h) + (i)
1	Customer Billing	18,212	18,212	710	18,923	416	19,339	68	-1,106	18,301
	Labor	2,640	2,640	103	2,743	60	2,803	10	-209	2,604
	Material	761	761	30	791	17	808	3	-42	769
	Contractor	9,299	9,299	363	9,661	213	9,874	35	-563	9,345
	Non-Labor Overheads	0	0	0	0	0	0	0	0	0
	Non-Labor Other	5,513	5,513	215	5,728	126	5,854	20	-291	5,583
2	Customer Interactions	25,769	25,769	1,005	26,774	589	27,363	96	-915	26,544
	Labor	18,115	18,115	706	18,821	414	19,236	67	-559	18,744
	Material	30	30	1	31	1	32	0	-32	0
	Contractor	6,616	6,616	258	6,874	151	7,025	25	-913	6,137
	Non-Labor Overheads	0	0	0	0	0	0	0	0	0
	Non-Labor Other	1,009	1,009	39	1,048	23	1,071	4	589	1,664
3	Customer Experience & Ops O&M	\$ 43,982	\$ 43,982	\$ 1,715	\$ 45,697	\$ 1,005	\$ 46,702	\$ 163	\$ (2,020)	\$ 44,845
	Labor	20,755	20,755	809	21,564	474	22,039	77	-768	21,348
	Material	791	791	31	822	18	840	3	-74	769
	Contractor	15,914	15,914	621	16,535	364	16,899	59	-1,476	15,482
	Non-Labor Overheads	0	0	0	0	0	0	0	0	0
	Non-Labor Other	6,521	6,521	254	6,776	149	6,925	24	298	7,247

Notes			
	2023	2024	2025
4 Annual merit increase			
Annual merit increase	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated merit increase	3.9%	2.2%	0.4%
Annual inflation rates per WP-JCA-51			
Annual inflation rates	3.9%	2.2%	2.1%
Number of months in the period	12	12	2
Pro-rated inflation rate	3.9%	2.2%	0.4%